

Post-Quantum Cryptography Developer's Handbook

A structured, in-depth path to mastering FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), FIPS 205 (SLH-DSA), and related PQC standards. Tailored for C/C++ developers with balanced theory and hands-on practice.

Target Audience: C/C++ developers with basic programming skills who want to understand and implement post-quantum cryptography.

Learning Approach: This plan follows a bottom-up approach: first understanding the mathematical foundations, then the algorithms, then practical implementation, and finally security analysis. Each section builds on previous ones.

VOLUME 2 OF 2 — PRODUCTION.

This volume covers the operational side of PQC: hybrid cryptography (X-Wing, composite signatures), protocol integration (TLS/SSH/S-MIME/IPsec), Cloud KMS and HSM landscape (AWS/GCP/Azure), the PQC CVE registry and "Verification Theatre" formal-verification lessons, regulatory compliance across 10 jurisdictions, and 12 migration cookbook scenarios. Aimed at SREs, security architects, platform engineers, and compliance officers running production systems through the PQC transition.

Volume 1 — Algorithms ([pqc-developers-handbook-vol1.pdf](#)) covers the mathematical foundations and internal structure of ML-KEM (FIPS 203), ML-DSA (FIPS 204), and SLH-DSA (FIPS 205). Read it if you need to understand *how* the algorithms work, not just how to deploy them.

Cross-volume references in this PDF appear as *Volume 1, Unit X.Y* — they are not clickable but resolve when both PDFs are open.

Generated from the unified master [pqc-developers-handbook.md](#) . Both volumes track the same version (see Document History).

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The Post-Quantum Transition at a Glance

Before diving into the technical details, it's important to understand the big picture: why post-quantum cryptography exists, what it replaces, and what remains secure.

Why the Transition?

The Quantum Threat: Shor's algorithm, when run on a sufficiently powerful quantum computer, can break RSA, Diffie-Hellman, and all elliptic curve cryptography in polynomial time. This affects:

- **Key Exchange:** RSA key transport, DH, ECDH
- **Digital Signatures:** RSA signatures, DSA, ECDSA, EdDSA
- **All public-key cryptography based on integer factorization or discrete logarithms**

Timeline Estimates:

- Cryptographically relevant quantum computers: 2030-2040 (estimates vary)
- "Harvest now, decrypt later" attacks: **happening today**
- NIST standardization: **Completed August 2024** (FIPS 203, 204, 205)
- NSA CNSA 2.0 deadlines: PQC required for National Security Systems by 2030-2035

How Shor's Algorithm Breaks Classical Cryptography

Shor's algorithm exploits the algebraic structure of factoring and discrete logarithms:

1. **The core insight:** Factoring an integer N reduces to finding the period of the function $f(x) = a^x \bmod N$. Similarly, discrete logarithms reduce to period finding.
2. **Quantum speedup:** A quantum computer can find periods using the Quantum Fourier Transform (QFT) in polynomial time — roughly $O(n^3)$ quantum gate operations for n -bit numbers. Classical computers require sub-exponential time for the same task.

3. **What it breaks:** Any cryptosystem whose security relies on:

- Integer factorization (RSA)
- Discrete logarithm in finite fields (DH, DSA)
- Discrete logarithm on elliptic curves (ECDH, ECDSA, EdDSA)

4. **What it does NOT break:** Problems without the algebraic structure Shor exploits:

- Finding shortest vectors in lattices (ML-KEM, ML-DSA) — no known period-finding reduction
- Inverting hash functions (SLH-DSA) — unstructured search problem
- Decoding random linear codes (McEliece, HQC) — no algebraic shortcut

5. **Grover's algorithm** provides only a quadratic speedup ($\sqrt{N}$ vs N) for unstructured search problems. This affects symmetric ciphers and hash functions, but doubling key sizes is sufficient mitigation.

Current quantum hardware (April 2026): The largest announced machines have ~1,000 physical qubits. IBM's published roadmap to fault-tolerant scale runs **Kookaburra (2026)** — the first module combining qLDPC memory with a logical processing unit (LPU); **Cockatoo (2027)** — first inter-module entanglement (the L-bus that lets multiple Kookaburra-class chips share logical qubits); and **Starling (by 2029)** — the full fault-tolerant target at ~200 logical qubits. Breaking RSA-2048 still requires cryptographically relevant fault-tolerant scale (estimated thousands of logical qubits, see Q-Day section); none of the 2026-2029 milestones get there on their own, but each is a documented step on the path.

Q-Day estimates have accelerated sharply in the last year. Three papers in early 2026 rewrote the threat timeline:

1. **Gidney (May 2025, arXiv 2505.15917)** — revised the RSA-2048 factoring estimate down by $20\times$ from his 2019 benchmark: **fewer than 1 million noisy qubits in under a week**, using ~1,730 logical qubits (Chevignard-Fouque-Schrottenloher approximate modular arithmetic).
2. **Google Quantum AI (March 2026)** — fewer than 500,000 physical qubits suffice to break the ECC used by Bitcoin and Ethereum in minutes. Spacetime volume ~ $10\times$ smaller than the Chevignard/Litinski baseline.
3. **Google's own migration target** was moved up to **2029** for full PQC coverage of Google systems, reflecting internal assessments of the threat timeline.

These revisions do not change the math of whether to migrate — they sharpen *when*. "Harvest now, decrypt later" attacks on data with 5+ year confidentiality windows are already rational for an adversary, and the window for pre-quantum cryptography is narrower than the 2030–2040 estimates circulated in 2023–2024 suggested. The operative phrase is "**migrate now, finish before the end of the decade.**"

What's Being Replaced?

CLASSICAL ALGORITHM	VULNERABILITY	POST-QUANTUM REPLACEMENT	STANDARD
RSA (key transport)	Shor's algorithm	ML-KEM	FIPS 203
Diffie-Hellman (DH)	Shor's algorithm	ML-KEM	FIPS 203
ECDH (X25519, P-256)	Shor's algorithm	ML-KEM	FIPS 203
RSA signatures	Shor's algorithm	ML-DSA or SLH-DSA	FIPS 204/205
DSA	Shor's algorithm	ML-DSA or SLH-DSA	FIPS 204/205
ECDSA (P-256, P-384)	Shor's algorithm	ML-DSA or SLH-DSA	FIPS 204/205
EdDSA (Ed25519)	Shor's algorithm	ML-DSA or SLH-DSA	FIPS 204/205

What Stays Safe?

Symmetric cryptography and **hash functions** are affected by Grover's algorithm, which provides only a quadratic speedup (not exponential like Shor's). The mitigation is simple: double the key/output size.

ALGORITHM TYPE	CLASSICAL SECURITY	POST-QUANTUM SECURITY	MITIGATION
AES-128	128-bit	64-bit	Use AES-256
AES-256	256-bit	128-bit	Already sufficient
SHA-256	256-bit	128-bit	Use SHA-384/SHA-512 or SHA3
SHA-512	512-bit	256-bit	Already sufficient
HMAC, HKDF	Secure	Secure	Use 256-bit keys

Bottom line: AES-256 and SHA-384/SHA-512 remain secure against quantum attacks.

Grover's Algorithm: How Much Threat, Really?

Grover's algorithm searches an unstructured space of N items in $O(\sqrt{N})$ queries, squaring the brute-force attack surface. For a symmetric key of k bits, this reduces effective security to $k/2$ bits. However, a practical Grover attack faces severe physical constraints:

FACTOR	IMPACT
Circuit depth	Each Grover iteration requires a full cipher evaluation as a quantum circuit. AES-128 needs $\sim 6,400$ T-gates per round $\times 10$ rounds = $\sim 64,000$ T-gates, plus ancilla operations totaling $\sim 10^6$ logical gates per oracle call
Iterations required	$\sqrt{2^{128}} = 2^{64}$ iterations for AES-128 — each must be run sequentially (Grover cannot be parallelized across quantum processors without losing the quadratic speedup)
MAXDEPTH parameter	NIST's security analysis assumes a maximum circuit depth of 2^{64} or 2^{96} sequential gates (the MAXDEPTH parameter). At 2^{64} depth, even AES-128 likely survives because each iteration exceeds simple gate depth budgets
Error correction	Logical qubits require $\sim 1,000$ – $10,000$ physical qubits for error correction. A full Grover search on AES-128 would need millions of stable physical qubits for extended durations
Classical parallelism	Classical brute-force can parallelize linearly — 2^{30} processors give 2^{30} speedup. Grover gives only 2^{64} total speedup regardless of quantum resources

The AES-128 debate: Some researchers (Jaques et al., 2020) argue that the gate cost makes Grover-on-AES-128 impractical even with fault-tolerant quantum computers, because the total cost exceeds exhaustive classical search with realistic hardware. Others (Grassl et al.) show optimized circuits that keep the threat plausible. NIST's conservative stance: AES-128 provides **at most** 64 bits of post-quantum security — use AES-256 (128-bit post-quantum) for long-term safety.

Key takeaway: Grover's algorithm is a theoretical quadratic speedup, not a practical break. The real quantum threat to cryptography comes from Shor's algorithm (exponential speedup on structured problems), not Grover.

The Three NIST Standards

STANDARD	ALGORITHM	USE CASE	SECURITY BASIS
FIPS 203	ML-KEM	Key encapsulation (replacing DH/ECDH)	Module-LWE (lattice)
FIPS 204	ML-DSA	Digital signatures (general purpose)	Module-LWE/SIS (lattice)
FIPS 205	SLH-DSA	Digital signatures (conservative)	Hash functions only

When to use which signature algorithm:

- **ML-DSA:** General purpose, smaller signatures, faster
- **SLH-DSA:** Maximum security assurance (no lattice assumptions), larger signatures

PQC Algorithm Families at a Glance

Beyond the three FIPS standards, PQC encompasses several distinct algorithm families based on different mathematical problems:

FAMILY	SECURITY BASIS	KEM EXAMPLES	SIGNATURE EXAMPLES	KEY CHARACTERISTICS
Lattice (structured)	Module-LWE/SIS	ML-KEM	ML-DSA, FN-DSA (FALCON)	Fast, moderate sizes, NIST standardized
Hash-based	Hash function security	—	SLH-DSA, LMS, XMSS	Conservative security, large signatures, proven assumptions
Code-based	Decoding random codes	Classic McEliece, HQC	—	Decades of study, very large public keys (McEliece)
Lattice (unstructured)	Plain LWE	FrodoKEM	—	Most conservative lattice option, much larger keys
Multivariate	MQ problem	—	(none standardized)	Fast verify, but most candidates broken
Isogeny-based	Supersingular isogenies	—	SQIsign (research)	Smallest keys, but SIDH broken in 2022

Why lattice-based algorithms dominate: NIST selected primarily lattice-based schemes (ML-KEM, ML-DSA) because they offer the best balance of security, performance, and key/signature sizes. Hash-based signatures (SLH-DSA) provide a conservative alternative based on minimal assumptions. Code-based KEMs (HQC, selected March 2025) provide algorithm diversity.

Algorithm diversity matters: If a breakthrough were to weaken lattice assumptions, organizations need fallback options — this is why SLH-DSA (hash-based) and HQC (code-based) exist alongside ML-KEM and ML-DSA.

Size and Performance Trade-offs

Post-quantum algorithms have significantly larger keys and signatures than classical algorithms:

ALGORITHM	PUBLIC KEY	SIGNATURE/CIPHERTEXT	SECURITY LEVEL
RSA-2048	256 bytes	256 bytes	~112-bit
ECDSA P-256	64 bytes	64 bytes	~128-bit
ML-KEM-768	1,184 bytes	1,088 bytes (ciphertext)	~192-bit (Level 3)
ML-DSA-65	1,952 bytes	3,309 bytes	~192-bit (Level 3)
SLH-DSA-128f	32 bytes	17,088 bytes	~128-bit (Level 1)
SLH-DSA-128s	32 bytes	7,856 bytes	~128-bit (Level 1)

Key insight: PQC public keys and signatures are 10-50x larger than classical equivalents. This impacts:

- Certificate sizes
- TLS handshake latency
- Bandwidth requirements
- Storage requirements

The Hybrid Transition Strategy

During the transition period (2024-2035+), **hybrid cryptography** combines classical and post-quantum algorithms:

Hybrid Key Exchange:

ECDH + ML-KEM → shared_secret = KDF(ECDH_secret || ML-KEM_secret)

Hybrid Signature:

ECDSA + ML-DSA → Both signatures must verify

Why hybrid?

1. **Defense in depth:** Secure if either algorithm remains unbroken
2. **Regulatory compliance:** Some systems still require classical crypto
3. **Backwards compatibility:** Gradual migration path
4. **Conservative approach:** PQC algorithms are newer, less battle-tested

Learning Path Summary

This curriculum takes you from foundations to deployment:

Module 1: Prerequisites	→ Environment setup, tools
Module 2: Math Foundations	→ Modular arithmetic, polynomials, NTT
Module 3: Lattice Theory	→ LWE, Ring-LWE, Module-LWE
Module 4: ML-KEM	→ Key encapsulation (FIPS 203)
Module 5: Signature Theory	→ Fiat-Shamir, security models
Module 6: ML-DSA	→ Lattice signatures (FIPS 204)
Module 7: SLH-DSA	→ Hash-based signatures (FIPS 205)
Module 8: Hybrid Crypto	→ Combining classical + PQC
Module 9: Protocols	→ TLS, SSH, S/MIME integration
Module 10: Future	→ Emerging algorithms, agility
Module 11: Migration	→ Legacy codebase migration guide

How to Use This Document

1. **Start with Volume 1, Module 1** - Set up your environment first. All code examples assume liboqs and OpenSSL 3.x are installed.
2. **Follow the modules sequentially** - Each module builds on the previous. Don't skip Volume 1, Module 2 foundations.
3. **Type the code yourself** - Don't just read; implement. The toy implementations help build intuition before using production libraries.
4. **Use the glossary** - Keep it open as a reference while reading. PQC has specialized terminology.
5. **Test incrementally** - After each implementation, test against known values. The debugging section provides tools for this.
6. **Choose your schedule** - Select Schedule A, B, or C based on your available time.
7. **Dive deep into one algorithm first** - Master ML-KEM (Volume 1, Module 4) before moving to ML-DSA (Volume 1, Module 6). They share concepts but differ in important ways.
8. **Bookmark the FIPS documents** - This guide explains concepts, but the standards are authoritative.

CITATIONS:

This document uses bracketed references like [Regev05] that point to the Bibliography section at the end. Full bibliographic details including DOIs and URLs are provided there.

Quick Start Gateway — Pick Your Path

The handbook is ~1,200 pages of theory + reference + operational material. Most readers don't need all of it. Use this gateway to land in the right section first; everything else is one cross-reference away.

YOU ARE...	START HERE	THEN READ	SKIP FOR NOW
An SRE / platform engineer asked to "make us PQ-ready"	Unit 11.1 (Migration Imperative) → Unit 11.2 (Cryptographic Inventory) → Unit 11.8 (Production Deployment Checklist)	Unit 11.9 (Cloud KMS Integration) — pick your provider; Unit 11.10 (CVE Registry)	Volume 1, Modules 1–3 (math), Module 9 protocol internals
A library author / contributor implementing PQC primitives	Volume 1, Module 4 (ML-KEM) → Volume 1, Module 6 (ML-DSA) → Unit 11.10 (PQC CVE Registry) → Unit 11.11 (Verification Theatre lessons)	Volume 1, Modules 2–3 for math foundations; Volume 1, Module 7 if implementing SLH-DSA	Module 9 protocol-level integration unless your library exposes TLS/SSH binding
A security architect choosing algorithms for a new system	Algorithm Selection Guide (this gateway) → Algorithm Decision Table (later in this section) → Unit 8.1 (hybrid intro)	Regulatory Compliance Matrix (Unit 11.9.6); Unit 11.9.7 (Migration Cookbook)	Algorithm internals — pick the standard, trust the FIPS spec
A compliance officer / auditor mapping deadlines	Regulatory Compliance Matrix (Unit 11.9.6) → US/EU/UK/DE/FR/AU rows for your jurisdiction	Unit 11.9.8 (CBOM section); Document History for what versions corrected what	Implementation chapters
A developer fixing a Stripe / OAuth / OpenID webhook for PQC	Stripe Webhook PQC Migration Recipe (Unit 11.9.7c) → X.509 Hybrid Cert CLI Recipe (Unit 11.9.7d)	Unit 11.9.7 (full migration cookbook); Interop Test Matrix	Volume 1, Module 7 (hash-based sigs not used in webhook signing today)
A researcher / student studying lattice cryptography	Volume 1, Module 2 (math) → Volume 1, Module 3 (lattice theory) → Volume 1, Module 4 (ML-KEM as worked example)	Volume 1, Modules 5–6 for signatures; Module 10.5 for NIST Round 2 finalists	Module 11 (operational) unless you want context for why algorithm choice matters
An incident responder dealing with a freshly-disclosed CVE	Unit 11.10 (PQC CVE Registry) → Unit 11.8 Playbook A (Algorithm Compromise)	Unit 11.9 if your KMS is affected; Unit 11.11 for formal-verification root-cause analysis	Theory chapters

HYPERLINKS

in the table above point to chapter/unit anchors. If you're reading the print PDF, all referenced units are listed in the Table of Contents at the front and the Alphabetical Index at the back.

DON'T READ THIS FRONT-TO-BACK.

This handbook is reference material for working engineers, not a textbook. If you find yourself wading through math you don't need, jump to your row above and read forward from there.

Glossary of Terms

Understanding PQC requires familiarity with specialized terminology. Reference this section as you progress through the material.

Cryptographic Concepts

TERM	DEFINITION
KEM	Key Encapsulation Mechanism - a public-key algorithm that generates a random shared secret and encapsulates it for transmission
PKE	Public Key Encryption - encryption where anyone can encrypt with a public key, only the private key holder can decrypt
Digital Signature	Cryptographic proof that a message was created by a specific entity
IND-CPA	Indistinguishability under Chosen Plaintext Attack - security notion where adversary cannot distinguish encryptions of chosen plaintexts
IND-CCA2	Indistinguishability under Adaptive Chosen Ciphertext Attack - strongest security notion, adversary has access to decryption oracle
CCA Security	Chosen Ciphertext Attack security - protection against adversaries who can obtain decryptions of chosen ciphertexts
Random Oracle Model	Proof technique where hash functions are modeled as truly random functions
CSPRNG	Cryptographically Secure Pseudo-Random Number Generator
XOF	Extendable Output Function - hash function with variable-length output (e.g., SHAKE128, SHAKE256)

Lattice Cryptography Terms

TERM	DEFINITION
Lattice	Discrete additive subgroup of $\mathbb{R}^n$; geometrically, a regular grid of points
Basis	Set of linearly independent vectors that generate a lattice
SVP	Shortest Vector Problem - find the shortest non-zero vector in a lattice
CVP	Closest Vector Problem - find the lattice point closest to a target
LWE	Learning With Errors - problem of recovering secret from noisy linear equations
Ring-LWE	LWE variant using polynomial rings for efficiency
Module-LWE	LWE variant using modules over polynomial rings (used in ML-KEM/ML-DSA)
NTT	Number Theoretic Transform - finite field FFT for fast polynomial multiplication
CBD	Centered Binomial Distribution - noise distribution used in ML-KEM

ML-KEM Specific Terms

TERM	DEFINITION
Encapsulation Key (ek)	Public key used to encapsulate shared secrets
Decapsulation Key (dk)	Secret key used to decapsulate shared secrets (contains the K-PKE decryption key, ek, H(ek), and the implicit-rejection seed z)
Compress/ Decompress	Functions to reduce coefficient bit-width for smaller ciphertexts
Fujisaki-Okamoto Transform	Technique to convert IND-CPA encryption to IND-CCA2 KEM

ML-DSA Specific Terms

TERM	DEFINITION
Fiat-Shamir Transform	Technique to convert interactive proof to non-interactive signature
Rejection Sampling	Technique where signing may abort and retry to avoid leaking secret information
Hint	Auxiliary data in ML-DSA signatures to enable efficient verification

Hash-Based Signature Terms

TERM	DEFINITION
OTS	One-Time Signature - signature scheme that can only securely sign one message
WOTS+	Winternitz One-Time Signature Plus - efficient OTS using hash chains
XMSS	eXtended Merkle Signature Scheme - many-time signature using Merkle tree of OTS
FORS	Forest Of Random Subsets - few-time signature used in SPHINCS+
Hypertree	Multi-layer tree structure in SPHINCS+/SLH-DSA
Authentication Path	Sequence of sibling hashes from leaf to root in Merkle tree

Security Levels

NIST defines each security category by reference to the cost of attacking a well-studied primitive (against *both* classical and quantum adversaries), not by a fixed bit count:

NIST CATEGORY	DEFINITION (AT LEAST AS HARD AS...)	APPROX. QUANTUM SECURITY STRENGTH	EXAMPLE ALGORITHM
Category 1	AES-128 key search	~128-bit (Grover-bounded)	ML-KEM-512
Category 2	SHA-256 / SHA3-256 collision	~128-bit	ML-DSA-44
Category 3	AES-192 key search	~192-bit (Grover-bounded)	ML-KEM-768, ML-DSA-65
Category 4	SHA-384 / SHA3-384 collision	~192-bit	-
Category 5	AES-256 key search	~256-bit (Grover-bounded)	ML-KEM-1024, ML-DSA-87

Note: the "approximate quantum security strength" column is a rough characterization only. NIST's categories are defined by the attack cost of the reference primitive itself (key search / collision finding), which already accounts for the best known quantum speedups (e.g. Grover for key search), so a category is a comparative floor rather than an exact bit figure.

Hybrid Cryptography Terms

TERM	DEFINITION
Hybrid Cryptography	Combining classical and post-quantum algorithms to provide security against both conventional and quantum attacks
X-Wing	A hybrid KEM combining X25519 and ML-KEM-768, specified in IETF draft
Composite Signature	A signature scheme that combines multiple signature algorithms (e.g., ECDSA + ML-DSA)
Concatenation Combiner	Hybrid approach where outputs are concatenated: $k = \text{KDF}(k_{\text{classical}} k_{\text{pq}})$
Nested Combiner	Hybrid approach where one algorithm encrypts the other's output
Harvest Now, Decrypt Later	Attack where adversaries store encrypted data today to decrypt with future quantum computers
Defense in Depth	Security strategy using multiple layers of protection

Protocol Integration Terms

TERM	DEFINITION
sntrup761	Streamlined NTRU Prime, a lattice-based KEM used in OpenSSH hybrid key exchange
Hybrid KEX	Key exchange combining classical (e.g., X25519) and post-quantum algorithms
PQ-TLS	Post-quantum TLS, TLS 1.3 with hybrid or pure PQ key exchange
Named Groups	TLS extension specifying supported key exchange algorithms
Composite Certificate	X.509 certificate containing multiple public keys (classical + PQ)

Future Algorithms and Research Terms

TERM	DEFINITION
FALCON	Fast-Fourier Lattice-based Compact Signatures over NTRU - selected for NIST standardization as FN-DSA (FIPS 206 draft, expected late 2026/2027)
BIKE	Bit-Flipping Key Encapsulation - code-based KEM, NIST Round 4 candidate
HQC	Hamming Quasi-Cyclic - code-based KEM, selected by NIST for standardization (March 2025) as backup to ML-KEM
Classic McEliece	Code-based KEM with very large public keys, NIST Round 4 candidate
Cryptographic Agility	System design that allows algorithms to be changed without major modifications
Algorithm Deprecation	Process of phasing out cryptographic algorithms as they become insecure
NTRU	Number Theory Research Unit - lattice-based encryption scheme, precursor to modern lattice crypto
Isogeny-Based	Cryptography based on isogenies between elliptic curves (e.g., SIDH/SIKE, now broken)

Deployment, Compliance, and Operational Terms

TERM	DEFINITION
CBOM	Cryptographic Bill of Materials — OWASP CycloneDX 1.6 inventory of every cryptographic asset (algorithm, protocol, certificate, key, secret) and its location; foundation of PQC migration per OMB M-23-02
CNSA 2.0	NSA Commercial National Security Algorithm Suite 2.0 — successor to Suite B; mandates PQC for National Security Systems with phased deadlines (new acquisitions January 2027, full transition 2030–2035)
CRQC	Cryptographically Relevant Quantum Computer — a quantum computer large and reliable enough to run Shor's algorithm against RSA-2048 or ECC; timeline estimates per 2025-2026 research have accelerated to late 2020s
CSfC	Commercial Solutions for Classified — NSA program enabling classified-system operators to acquire commercial cryptographic products; CSfC PQC Addendum published 2025
FIPS 140-3	NIST Cryptographic Module Validation Program standard; mandatory for US federal systems. Transition deadline September 21, 2026 (140-2 moves to CMVP historical list)
HNDL	Harvest Now, Decrypt Later — the attack model in which an adversary records encrypted traffic today for later decryption once a CRQC becomes available; drives the urgency of PQC migration for data with 5+ year confidentiality windows
KAT	Known Answer Test — NIST-provided test vectors in <code>.req</code> / <code>.rsp</code> format used to validate algorithm implementation correctness; critical for FIPS compliance and interop testing
ACVP / CAVP	Automated Cryptographic Validation Protocol (JSON-format test vector service) / Cryptographic Algorithm Validation Program (underlying NIST process). Required input to FIPS 140-3 module validation

Recent Events and Incidents (2024-2026)

TERM	DEFINITION
KyberSlash	CVE-2024-37880 — timing attack against ML-KEM-512 on implementations where the Clang optimizer (versions through 18.x) converted constant-time <code>cmov</code> into a secret-dependent branch in <code>poly_frommsg</code> . Full key recovery demonstrated in < 10 min on Raspberry Pi 2
GoFetch	March 2024 microarchitectural attack against Apple M1/M2/M3 processors exploiting the Data Memory-Dependent Prefetcher (DMP) to extract ML-KEM and ML-DSA keys from constant-time implementations. Unmitigable on M1/M2 without kernel patches
Verification Theatre	ePrint 2026/192 (Nadim Kobeissi, February 2026) — catalog of 13 bugs that escaped formally-verified libraries (Cryspen libcrux / hpke-rs) deployed in Signal's SPQR post-quantum ratchet, including two FIPS 204 spec violations in the ML-DSA verifier

Signature and KEM Architectural Terms

TERM	DEFINITION
Composite Signature	Single-OID signature scheme combining a traditional algorithm with a post-quantum algorithm such that both must verify; standardized by draft-ietf-lamps-pq-composite-sigs-16 with OID prototype arc 2.16.840.1.114027.80.8.1.*
MTC	Merkle Tree Certificates — batch-signed alternative to per-domain X.509 certificates, chartered in IETF PLANTS WG (2026); replaces 14.7 KB composite certificate chains with ~736-byte inclusion proofs in TLS handshakes
PLANTS	IETF working group chartered 2026 to standardize Merkle Tree Certificates as a PQ-friendly replacement for Web PKI handshake-size bloat
SPQR	Sparse Post-Quantum Ratchet — Signal's October 2025 post-quantum upgrade to the Double Ratchet, forming the "Triple Ratchet" architecture; uses an ML-KEM "Braid Protocol" in place of the classical DH ratchet
HAWK	Lattice signature based on the Lattice Isomorphism Problem (module-LIP), submitted to NIST Additional Signatures Round 2; as of 2026 the sole remaining lattice candidate in Round 2, intended as a lattice-diversity backstop behind ML-DSA. Uses Falcon-like key derivation but the underlying hard problem is LIP, not NTRU.
Quorus	MPC-based threshold signature scheme for ML-DSA presented at NIST MPTS 2026; produces standard FIPS 204 signatures, making verifier-side rollout transparent

Module 8: Hybrid Cryptography

Module Overview: Hybrid cryptography combines classical and post-quantum algorithms to provide security against both conventional and quantum attacks during the transition period. This module covers the motivation, design patterns, standards, and implementation of hybrid schemes for both key exchange and digital signatures.

Prerequisites: Volume 1, Modules 4-7 (ML-KEM, ML-DSA, SLH-DSA fundamentals)

Module Learning Objectives:

- Understand why hybrid cryptography is essential during the PQC transition
 - Implement hybrid key exchange combining X25519/ECDH with ML-KEM
 - Implement hybrid signatures combining ECDSA/EdDSA with ML-DSA/SLH-DSA
 - Work with composite key formats and certificate structures
 - Apply hybrid cryptography in real-world protocol integration
-

Unit 8.1: Introduction to Hybrid Cryptography

Prerequisites: Volume 1, Modules 4-7 (ML-KEM, ML-DSA, SLH-DSA fundamentals)

Estimated Time: 2-3 hours

Learning Objectives

After completing this unit, you will be able to:

- Explain the motivation for hybrid cryptography during the PQC transition
- Distinguish between defense-in-depth and migration security goals
- Identify the two primary threat models: "harvest now, decrypt later" and implementation vulnerabilities
- Compare concatenation, cascade, and other combiner strategies
- Analyze the security properties of different hybrid constructions

8.1.1 The Case for Hybrid Cryptography

The transition from classical to post-quantum cryptography presents a fundamental dilemma: classical algorithms face quantum threats, but new PQC algorithms lack the decades of cryptanalysis that built confidence in RSA and ECC. Hybrid cryptography addresses this by combining both, ensuring security even if one component fails.

Two Critical Threat Models:

Threat Model 1: Quantum Attack on Classical Crypto

```
Today: Adversary records encrypted data
↓
Future: Quantum computer available
↓
Attack: Run Shor's algorithm on recorded ciphertexts
↓
Result: All past communications decrypted
Defense: Add PQC algorithm to protect data
```

Threat Model 2: Classical Attack on New PQC

```
Today: PQC algorithm deployed
↓
Discovery: Unexpected classical vulnerability in PQC
(SIKE was broken by classical attack!)
↓
Attack: Break system using newly discovered weakness
↓
Result: Current and past communications compromised
Defense: Keep classical algorithm as fallback
```

The SIKE Lesson:

SIKE (Supersingular Isogeny Key Encapsulation) was a NIST PQC Round 3 alternate candidate considered promising until 2022. It was broken by a classical attack—no quantum computer needed. This demonstrated that new mathematical assumptions, regardless of how carefully analyzed, may harbor unexpected weaknesses.

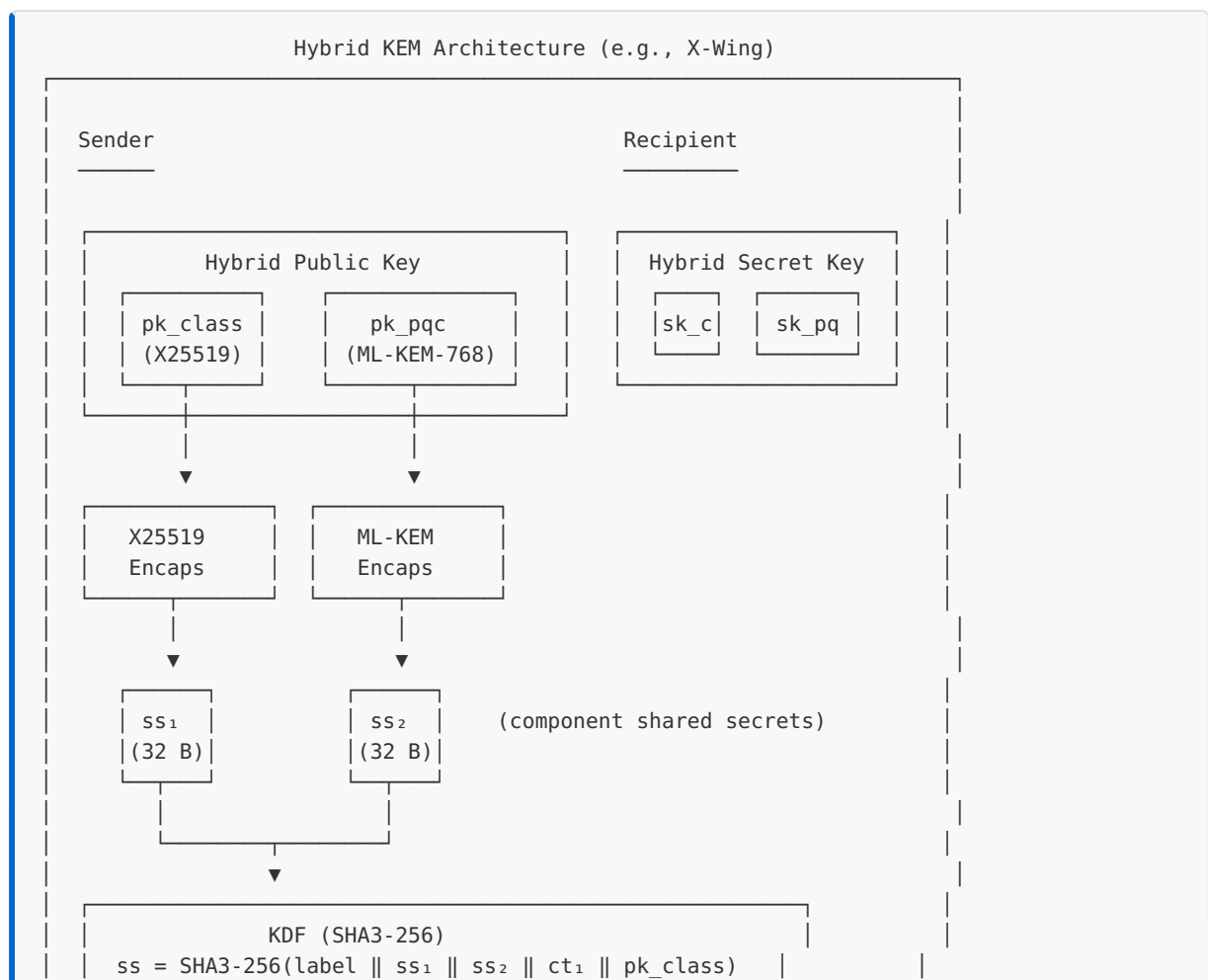
SIKE Timeline:

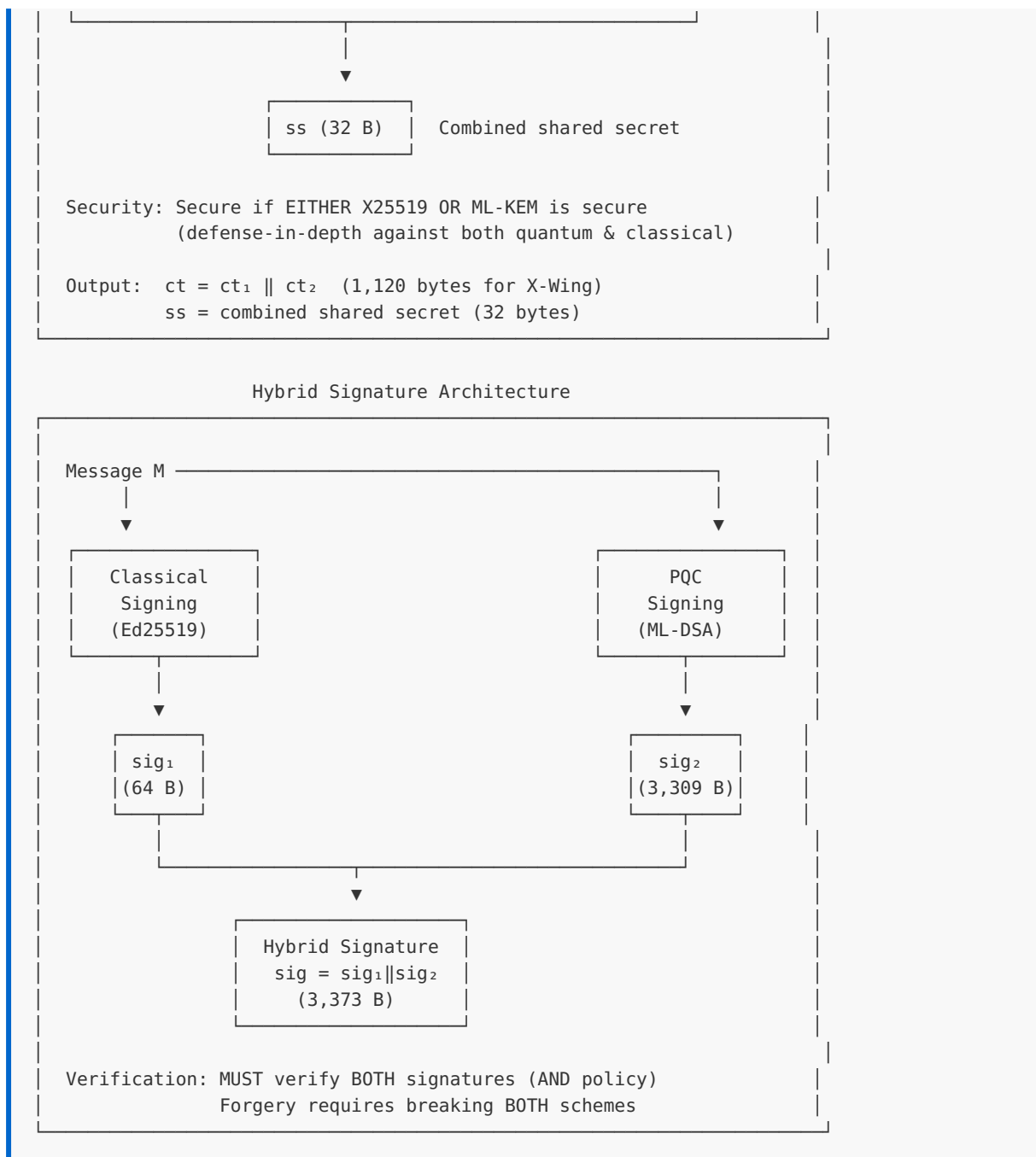
- 2011: Isogeny-based crypto proposed
- 2017: SIKE submitted to NIST PQC competition
- 2020: Selected as Round 3 alternate candidate
- 2022: Completely broken by classical attack
(Castryck-Decru attack using GPST theorem)
- Lesson: New crypto needs extensive cryptanalysis

Hybrid Approach Benefits:

BENEFIT	DESCRIPTION
Defense-in-depth	Attacker must break BOTH algorithms
Transition safety	Graceful migration without all-or-nothing risk
Regulatory compliance	Maintains approved classical crypto where required
Confidence building	Time for PQC to mature while maintaining security
Fallback capability	Recovery path if PQC algorithm is weakened

Hybrid Cryptography Architecture:





8.1.2 Hybrid Security Goals

Hybrid constructions can provide different security guarantees depending on design:

Security Goal Hierarchy:

Level 1: OR Security (Weaker)

Security holds if EITHER algorithm is secure

Appropriate for: Migration scenarios where you want to maintain at least current security level

Example: System is secure if RSA OR ML-KEM is secure

Level 2: AND Security (Stronger)

Security holds if BOTH algorithms are secure

Appropriate for: Maximum security when both threats (quantum and classical attacks on PQC) are concerns

Example: Attacker must break BOTH RSA AND ML-KEM

Level 3: IND-CCA Security Composition

Combined scheme maintains IND-CCA even if one component's IND-CCA property fails

Appropriate for: Formal security proofs

Note: Requires careful combiner design

Formal Security Requirement:

For hybrid KEM with components KEM_1 and KEM_2 :

- **Desired property:** Combined scheme is IND-CCA secure if either KEM_1 or KEM_2 is IND-CCA secure
- **Challenge:** Simple concatenation may not preserve this
- **Solution:** Proper combiners that bind components together

8.1.3 Combiner Strategies

Several approaches exist for combining classical and post-quantum algorithms:

Strategy 1: Concatenation (Simplest)

Hybrid KEM - Concatenation:

Encapsulation:

```
(c1, ss1) ← KEM1.Encaps(pk1)  
(c2, ss2) ← KEM2.Encaps(pk2)  
c = c1 || c2  
ss = KDF(ss1 || ss2)
```

Decapsulation:

```
ss1 ← KEM1.Decaps(sk1, c1)  
ss2 ← KEM2.Decaps(sk2, c2)  
ss = KDF(ss1 || ss2)
```

Security: If either ss₁ or ss₂ is secret, combined ss is secret
(assuming KDF is a secure random oracle)

Strategy 2: XOR Combiner (Lightweight)

Hybrid KEM - XOR:

Encapsulation:

```
(c1, ss1) ← KEM1.Encaps(pk1)    [|ss1| = n bits]  
(c2, ss2) ← KEM2.Encaps(pk2)    [|ss2| = n bits]  
c = c1 || c2  
ss = ss1 ⊕ ss2
```

Issue: XOR doesn't provide domain separation
Vulnerable to related-key attacks
NOT RECOMMENDED for production

Strategy 3: Cascade/Nesting (Strongest)

Hybrid KEM - Nested:

Encapsulation:

```
(c1, ss1) ← KEM1.Encaps(pk1)  
pk2' = pk2 || c1 || ss1    [Bind to first KEM]  
(c2, ss2) ← KEM2.Encaps(pk2')  
c = c1 || c2  
ss = KDF(ss1 || ss2 || c)
```

Security: Binding prevents mix-and-match attacks
Strongest provable security guarantees
Cost: More complex, dependencies between components

Strategy 4: Split KDF (Practical)

Hybrid KEM - Split KDF (X-Wing style):

Encapsulation:

```
(C1, ss1) ← X25519.DH(pk1)
(C2, ss2) ← ML-KEM.Encaps(pk2)
C = C1 || C2
ss = SHA3-256(label || ss1 || ss2 || C1 || pk1)
```

Key insight: Include ephemeral public key and label
for full domain separation

Security: IND-CCA under ROM if either is IND-CCA

Comparison of Combiner Strategies:

STRATEGY	SECURITY PROOF	COMPLEXITY	BINDING	RECOMMENDATION
XOR	Weak	Low	None	Avoid
Concatenation	Standard ROM	Medium	KDF only	Acceptable
Split KDF	Strong ROM	Medium	Full	Recommended
Nested	Strongest	High	Cryptographic	High-security

8.1.4 Hybrid Signatures

For digital signatures, the hybrid approach differs from KEMs:

Signature Combination Strategies:

Strategy 1: Parallel Signatures (Common)

Sign:

```
sig1 ← SIG1.Sign(sk1, message)
sig2 ← SIG2.Sign(sk2, message)
hybrid_sig = sig1 || sig2
```

Verify:

```
valid1 ← SIG1.Verify(pk1, message, sig1)
valid2 ← SIG2.Verify(pk2, message, sig2)
return valid1 AND valid2
```

Security: Forgery requires breaking BOTH schemes

Strategy 2: Nested Signatures (Stronger)

Sign:

```
sig1 ← SIG1.Sign(sk1, message)
sig2 ← SIG2.Sign(sk2, message || sig1)
```

```

    hybrid_sig = sig1 || sig2

Verify:
    valid1 ← SIG1.Verify(pk1, message, sig1)
    valid2 ← SIG2.Verify(pk2, message || sig1, sig2)
    return valid1 AND valid2

Security: Binds signatures together, prevents reuse

```

Strategy 3: Composite Key Signature

```

Key: Single composite identity containing both keys

Sign:
    digest = HASH(composite_pk || message)
    sig1 ← SIG1.Sign(sk1, digest)
    sig2 ← SIG2.Sign(sk2, digest)
    hybrid_sig = sig1 || sig2

Verify:
    digest = HASH(composite_pk || message)
    return SIG1.Verify(pk1, digest, sig1) AND
           SIG2.Verify(pk2, digest, sig2)

Security: Binding through composite public key hash

```

Verification Policy Options:

Policy: AND (Both must verify) - Maximum security

- Signature valid only if BOTH components verify
- Protects against either algorithm being broken
- Recommended for high-security applications
- Drawback: Single point of failure for availability

Policy: OR (Either must verify) - Transition focus

- Signature valid if EITHER component verifies
- Useful during transition when some verifiers may not support PQC yet
- Weaker security: only as strong as weakest
- Appropriate only for specific migration scenarios

8.1.5 Standards Landscape

Multiple standards bodies are working on hybrid cryptography:

IETF Standards:

RFC 9370: Multiple Key Encapsulation Mechanisms (KEMs)

- └ Defines generic KEM combiner framework
- └ Specifies security requirements for combiners
- └ Provides guidance for protocol integration

draft-ietf-lamps-pq-composite-sigs

- └ Composite signatures for X.509 certificates
- └ Defines OIDs for specific combinations
- └ Specifies encoding and verification rules

draft-ietf-lamps-pq-composite-kem

- └ Composite KEMs for X.509 and CMS
- └ Pairs ML-KEM with ECDH/X25519
- └ Defines hybrid decapsulation procedures

X-Wing (draft-connelly-cfrg-xwing-kem)

- └ Specific hybrid: X25519 + ML-KEM-768
- └ Optimized for TLS and general use
- └ Single shared secret output
- └ IETF CFRG work item

NIST Guidance:

SP 800-227 (Draft): Guidelines for Post-Quantum Cryptography Migration

- └ Recommends hybrid approaches during transition
- └ Defines security requirements for combiners
- └ Provides implementation considerations

NIST IR 8547: Transition to Post-Quantum Cryptography Standards

- └ Timeline for deprecating classical algorithms
- └ Hybrid deployment recommendations
- └ Risk assessment framework

Industry Implementations:

IMPLEMENTATION	CLASSICAL	PQC	STATUS
X-Wing	X25519	ML-KEM-768	IETF draft
Chrome/BoringSSL	X25519	ML-KEM-768	Production
OpenSSH 10.0+	X25519	ML-KEM-768 (default); sntrup761 (since 9.0)	Production
AWS KMS	ECDH	ML-KEM	Preview
Signal	X25519	ML-KEM (Kyber)	Production

8.1.6 Size and Performance Considerations

Hybrid cryptography increases both size and computational cost:

Key Exchange Size Impact:

Component Sizes (bytes):

Scheme	Public Key	Ciphertext	Shared Secret
X25519	32	32	32
ML-KEM-768	1,184	1,088	32
Hybrid (X-Wing)	1,216	1,120	32

Overhead: +3700% public key, +3400% ciphertext
But: Still practical for most applications

Signature Size Impact:

Component Sizes (bytes):

Scheme	Public Key	Signature	Secret Key
Ed25519	32	64	64
ML-DSA-65	1,952	3,309	4,032
Hybrid	1,984	3,373	4,096
ECDSA-P256	64	64	32
SLH-DSA-128f	32	17,088	64
Hybrid	96	17,152	96

Performance Impact:

Relative Performance (normalized to classical = 1.0):

Operation	Classical	PQC	Hybrid
X25519 DH	1.0	-	-
ML-KEM-768 Encaps	-	0.8	-
Hybrid Encaps	-	-	1.8
Ed25519 Sign	1.0	-	-
ML-DSA-65 Sign	-	2.5	-
Hybrid Sign	-	-	3.5
Ed25519 Verify	1.0	-	-
ML-DSA-65 Verify	-	0.9	-
Hybrid Verify	-	-	1.9

Note: PQC operations are often faster than expected
Hybrid overhead is roughly sum of both operations

8.1.7 Implementation Considerations

Key implementation challenges for hybrid cryptography:

Error Handling:

```
/* Hybrid operation must handle partial failures */
typedef enum {
    HYBRID_SUCCESS = 0,
    HYBRID_ERR_CLASSICAL_FAILED = -1,
    HYBRID_ERR_PQC_FAILED = -2,
    HYBRID_ERR_BOTH_FAILED = -3,
    HYBRID_ERR_COMBINER_FAILED = -4
} hybrid_error_t;

/* Option 1: Fail if either fails (recommended for production) */
hybrid_error_t hybrid_encaps_strict(
    uint8_t *ciphertext,
    uint8_t *shared_secret,
    const uint8_t *pk_classical,
    const uint8_t *pk_pqc
) {
    uint8_t ss1[32], ss2[32];
    uint8_t ct1[32], ct2[1088];

    /* Both must succeed */
    if (x25519_encaps(ct1, ss1, pk_classical) != 0) {
        return HYBRID_ERR_CLASSICAL_FAILED;
    }

    if (ml_kem_768_encaps(ct2, ss2, pk_pqc) != 0) {
        secure_zero(ss1, sizeof(ss1));
        return HYBRID_ERR_PQC_FAILED;
    }

    /* Combine shared secrets */
    memcpy(ciphertext, ct1, 32);
    memcpy(ciphertext + 32, ct2, 1088);

    if (hybrid_kdf(shared_secret, ss1, ss2, ct1, pk_classical) != 0) {
        secure_zero(ss1, sizeof(ss1));
        secure_zero(ss2, sizeof(ss2));
        return HYBRID_ERR_COMBINER_FAILED;
    }

    secure_zero(ss1, sizeof(ss1));
    secure_zero(ss2, sizeof(ss2));
    return HYBRID_SUCCESS;
}
```

Secure Memory Management:

```

/* All intermediate secrets must be cleared */
void hybrid_cleanup(hybrid_ctx_t *ctx) {
    if (ctx == NULL) return;

    /* Clear classical component secrets */
    secure_zero(ctx->classical_ss, sizeof(ctx->classical_ss));
    secure_zero(ctx->classical_sk, sizeof(ctx->classical_sk));

    /* Clear PQC component secrets */
    secure_zero(ctx->pqc_ss, sizeof(ctx->pqc_ss));
    secure_zero(ctx->pqc_sk, sizeof(ctx->pqc_sk));

    /* Clear combined secret */
    secure_zero(ctx->combined_ss, sizeof(ctx->combined_ss));

    /* Clear intermediate KDF state */
    secure_zero(&ctx->kdf_ctx, sizeof(ctx->kdf_ctx));
}

```

Constant-Time Combiner:

```

/* KDF must be constant-time to avoid timing leaks */
void hybrid_kdf_constant_time(
    uint8_t *out,
    size_t out_len,
    const uint8_t *ss1,
    size_t ss1_len,
    const uint8_t *ss2,
    size_t ss2_len,
    const uint8_t *ct1,
    size_t ct1_len,
    const uint8_t *pk1,
    size_t pk1_len
) {
    SHA3_CTX ctx;

    /* Domain separation label */
    static const uint8_t label[] = "X-Wing hybrid KEM combiner";

    sha3_256_init(&ctx);
    sha3_update(&ctx, label, sizeof(label) - 1);
    sha3_update(&ctx, ss1, ss1_len);
    sha3_update(&ctx, ss2, ss2_len);
    sha3_update(&ctx, ct1, ct1_len);
    sha3_update(&ctx, pk1, pk1_len);
    sha3_final(&ctx, out);

    /* Clear intermediate state */
    secure_zero(&ctx, sizeof(ctx));
}

```


8.1.8 C Implementation: Hybrid Framework

```
/* hybrid_framework.h - Generic hybrid cryptography framework */

#ifndef HYBRID_FRAMEWORK_H
#define HYBRID_FRAMEWORK_H

#include <stdint.h>
#include <stddef.h>

/*
 * Hybrid KEM Framework
 *
 * Provides generic infrastructure for combining KEMs.
 * Specific instantiations (X-Wing, etc.) build on this.
 */

/* Maximum sizes for components */
#define HYBRID_MAX_PK_SIZE      2048
#define HYBRID_MAX_SK_SIZE      4096
#define HYBRID_MAX_CT_SIZE      2048
#define HYBRID_MAX_SS_SIZE      64

/* Combiner types */
typedef enum {
    HYBRID_COMBINER_CONCAT, /* Simple concatenation + KDF */
    HYBRID_COMBINER_XOR,    /* XOR (not recommended) */
    HYBRID_COMBINER_NESTED, /* Nested/cascade */
    HYBRID_COMBINER_SPLIT_KDF /* Split KDF with binding (X-Wing style) */
} hybrid_combiner_t;

/* Component KEM interface */
typedef struct {
    const char *name;
    size_t pk_size;
    size_t sk_size;
    size_t ct_size;
    size_t ss_size;

    int (*keygen)(uint8_t *pk, uint8_t *sk);
    int (*encaps)(uint8_t *ct, uint8_t *ss, const uint8_t *pk);
    int (*decaps)(uint8_t *ss, const uint8_t *ct, const uint8_t *sk);
} kem_interface_t;

/* Hybrid KEM configuration */
typedef struct {
    const char *name;
    const kem_interface_t *kem1; /* Classical KEM (e.g., X25519) */
    const kem_interface_t *kem2; /* PQC KEM (e.g., ML-KEM-768) */
    hybrid_combiner_t combiner;
    const uint8_t *label; /* Domain separation label */
    size_t label_len;
} hybrid_kem_config_t;

/* Hybrid key pair */
typedef struct {
    uint8_t pk1[HYBRID_MAX_PK_SIZE];
```

```

    uint8_t pk2[HYBRID_MAX_PK_SIZE];
    uint8_t sk1[HYBRID_MAX_SK_SIZE];
    uint8_t sk2[HYBRID_MAX_SK_SIZE];
    const hybrid_kem_config_t *config;
} hybrid_keypair_t;

/* Hybrid ciphertext */
typedef struct {
    uint8_t ct1[HYBRID_MAX_CT_SIZE];
    uint8_t ct2[HYBRID_MAX_CT_SIZE];
    const hybrid_kem_config_t *config;
} hybrid_ciphertext_t;

/*
 * Generate hybrid key pair
 *
 * Returns 0 on success, negative on error
 */
int hybrid_kem_keygen(
    hybrid_keypair_t *keypair,
    const hybrid_kem_config_t *config
);

/*
 * Hybrid encapsulation
 *
 * Generates ciphertext and shared secret from public keys
 * Returns 0 on success, negative on error
 */
int hybrid_kem_encaps(
    hybrid_ciphertext_t *ct,
    uint8_t *shared_secret,
    size_t ss_len,
    const uint8_t *pk1,
    const uint8_t *pk2,
    const hybrid_kem_config_t *config
);

/*
 * Hybrid decapsulation
 *
 * Recovers shared secret from ciphertext using secret keys
 * Returns 0 on success, negative on error
 */
int hybrid_kem_decaps(
    uint8_t *shared_secret,
    size_t ss_len,
    const hybrid_ciphertext_t *ct,
    const uint8_t *sk1,
    const uint8_t *sk2,
    const hybrid_kem_config_t *config
);

/*
 * Get hybrid public key size
 */
size_t hybrid_kem_pk_size(const hybrid_kem_config_t *config);

/*
 * Get hybrid secret key size

```

```

    */
size_t hybrid_kem_sk_size(const hybrid_kem_config_t *config);

/*
 * Get hybrid ciphertext size
 */
size_t hybrid_kem_ct_size(const hybrid_kem_config_t *config);

/*
 * Serialize hybrid public key
 */
int hybrid_kem_serialize_pk(
    uint8_t *out,
    size_t out_len,
    const hybrid_keypair_t *keypair
);

/*
 * Deserialize hybrid public key
 */
int hybrid_kem_deserialize_pk(
    uint8_t *pk1,
    uint8_t *pk2,
    const uint8_t *in,
    size_t in_len,
    const hybrid_kem_config_t *config
);

/*
 * Serialize hybrid ciphertext
 */
int hybrid_kem_serialize_ct(
    uint8_t *out,
    size_t out_len,
    const hybrid_ciphertext_t *ct
);

/*
 * Deserialize hybrid ciphertext
 */
int hybrid_kem_deserialize_ct(
    hybrid_ciphertext_t *ct,
    const uint8_t *in,
    size_t in_len,
    const hybrid_kem_config_t *config
);

/*
 * Securely clear hybrid key pair
 */
void hybrid_keypair_clear(hybrid_keypair_t *keypair);

#endif /* HYBRID_FRAMEWORK_H */

```

```

/* hybrid_framework.c - Hybrid KEM implementation */

#include "hybrid_framework.h"
#include <string.h>

/* Secure memory clearing - compiler won't optimize away */
static void secure_zero(void *ptr, size_t len) {
    volatile uint8_t *p = ptr;
    while (len--) *p++ = 0;
}

/*
 * SHA3-256 Implementation (Educational - for production use OpenSSL)
 * This is a simplified demonstration. Real implementations should use
 * OpenSSL: EVP_sha3_256() or a vetted library.
 */
typedef struct {
    uint8_t state[200];
    size_t absorbed;
    size_t rate;
} sha3_ctx_t;

/* Keccak round constants */
static const uint64_t keccak_rc[24] = {
    0x0000000000000001ULL, 0x0000000000000802ULL, 0x800000000000080aULL,
    0x8000000008000800ULL, 0x000000000000080bULL, 0x00000000080000001ULL,
    0x8000000008000801ULL, 0x8000000000000809ULL, 0x0000000000000008aULL,
    0x00000000000000088ULL, 0x0000000008000809ULL, 0x0000000008000000aULL,
    0x000000000800080bULL, 0x800000000000000bULL, 0x8000000000000809ULL,
    0x8000000000000803ULL, 0x8000000000000802ULL, 0x800000000000080ULL,
    0x000000000000080aULL, 0x8000000008000000aULL, 0x8000000008000801ULL,
    0x8000000000000808ULL, 0x00000000080000001ULL, 0x8000000008000808ULL
};

/* Simplified Keccak-f[1600] permutation (educational) */
static void keccak_f1600(uint64_t state[25]) {
    for (int round = 0; round < 24; round++) {
        /* Theta, Rho, Pi, Chi, Iota steps (simplified) */
        uint64_t C[5], D[5];
        for (int x = 0; x < 5; x++)
            C[x] = state[x] ^ state[x+5] ^ state[x+10] ^ state[x+15] ^ state[x+20];
        for (int x = 0; x < 5; x++) {
            D[x] = C[(x+4)%5] ^ ((C[(x+1)%5] << 1) | (C[(x+1)%5] >> 63));
            for (int y = 0; y < 25; y += 5) state[y+x] ^= D[x];
        }
        /* Apply round constant */
        state[0] ^= keccak_rc[round];
    }
}

static void sha3_256_init(sha3_ctx_t *ctx) {
    memset(ctx, 0, sizeof(*ctx));
    ctx->rate = 136; /* SHA3-256 rate = 1088 bits = 136 bytes */
}

static void sha3_256_update(sha3_ctx_t *ctx, const uint8_t *in, size_t len) {
    for (size_t i = 0; i < len; i++) {
        ctx->state[ctx->absorbed++] ^= in[i];
        if (ctx->absorbed == ctx->rate) {

```

```

        keccak_f1600((uint64_t *)ctx->state);
        ctx->absorbed = 0;
    }
}

static void sha3_256_final(sha3_ctx_t *ctx, uint8_t *out) {
    ctx->state[ctx->absorbed] ^= 0x06; /* SHA3 domain separator */
    ctx->state[ctx->rate - 1] ^= 0x80; /* Padding */
    keccak_f1600((uint64_t *)ctx->state);
    memcpy(out, ctx->state, 32);
}

static void sha3_256(uint8_t *out, const uint8_t *in, size_t in_len) {
    sha3_ctx_t ctx;
    sha3_256_init(&ctx);
    sha3_256_update(&ctx, in, in_len);
    sha3_256_final(&ctx, out);
}

/*
 * Concatenation combiner: KDF(ss1 || ss2)
 */
static int combiner_concat(
    uint8_t *out,
    size_t out_len,
    const uint8_t *ss1, size_t ss1_len,
    const uint8_t *ss2, size_t ss2_len,
    const uint8_t *ct1, size_t ct1_len,
    const uint8_t *pk1, size_t pk1_len,
    const uint8_t *label, size_t label_len
) {
    sha3_ctx_t ctx;
    uint8_t hash[32];

    (void)ct1; (void)ct1_len; /* Not used in simple concat */
    (void)pk1; (void)pk1_len;

    sha3_256_init(&ctx);
    if (label && label_len > 0) {
        sha3_256_update(&ctx, label, label_len);
    }
    sha3_256_update(&ctx, ss1, ss1_len);
    sha3_256_update(&ctx, ss2, ss2_len);
    sha3_256_final(&ctx, hash);

    if (out_len > 32) out_len = 32;
    memcpy(out, hash, out_len);

    secure_zero(&ctx, sizeof(ctx));
    secure_zero(hash, sizeof(hash));

    return 0;
}

/*
 * Split KDF combiner (X-Wing style): KDF(label || ss1 || ss2 || ct1 || pk1)
 */
static int combiner_split_kdf(
    uint8_t *out,

```

```

size_t out_len,
const uint8_t *ss1, size_t ss1_len,
const uint8_t *ss2, size_t ss2_len,
const uint8_t *ct1, size_t ct1_len,
const uint8_t *pk1, size_t pk1_len,
const uint8_t *label, size_t label_len
) {
    sha3_ctx_t ctx;
    uint8_t hash[32];

    sha3_256_init(&ctx);

    /* Domain separation label */
    if (label && label_len > 0) {
        sha3_256_update(&ctx, label, label_len);
    }

    /* Both shared secrets */
    sha3_256_update(&ctx, ss1, ss1_len);
    sha3_256_update(&ctx, ss2, ss2_len);

    /* Binding: include classical ciphertext and public key */
    sha3_256_update(&ctx, ct1, ct1_len);
    sha3_256_update(&ctx, pk1, pk1_len);

    sha3_256_final(&ctx, hash);

    if (out_len > 32) out_len = 32;
    memcpy(out, hash, out_len);

    secure_zero(&ctx, sizeof(ctx));
    secure_zero(hash, sizeof(hash));

    return 0;
}

/*
 * Generate hybrid key pair
 */
int hybrid_kem_keygen(
    hybrid_keypair_t *keypair,
    const hybrid_kem_config_t *config
) {
    int ret;

    if (!keypair || !config || !config->kem1 || !config->kem2) {
        return -1;
    }

    keypair->config = config;

    /* Generate classical key pair */
    ret = config->kem1->keygen(keypair->pk1, keypair->sk1);
    if (ret != 0) {
        secure_zero(keypair, sizeof(*keypair));
        return -1;
    }

    /* Generate PQC key pair */
    ret = config->kem2->keygen(keypair->pk2, keypair->sk2);

```

```

    if (ret != 0) {
        secure_zero(keypair, sizeof(*keypair));
        return -2;
    }

    return 0;
}

/*
 * Hybrid encapsulation
 */
int hybrid_kem_encaps(
    hybrid_ciphertext_t *ct,
    uint8_t *shared_secret,
    size_t ss_len,
    const uint8_t *pk1,
    const uint8_t *pk2,
    const hybrid_kem_config_t *config
) {
    uint8_t ss1[HYBRID_MAX_SS_SIZE];
    uint8_t ss2[HYBRID_MAX_SS_SIZE];
    int ret = 0;

    if (!ct || !shared_secret || !pk1 || !pk2 || !config) {
        return -1;
    }

    ct->config = config;

    /* Encapsulate with classical KEM */
    ret = config->kem1->encaps(ct->ct1, ss1, pk1);
    if (ret != 0) {
        secure_zero(ss1, sizeof(ss1));
        return -1;
    }

    /* Encapsulate with PQC KEM */
    ret = config->kem2->encaps(ct->ct2, ss2, pk2);
    if (ret != 0) {
        secure_zero(ss1, sizeof(ss1));
        secure_zero(ss2, sizeof(ss2));
        return -2;
    }

    /* Combine shared secrets based on combiner type */
    switch (config->combiner) {
        case HYBRID_COMBINER_CONCAT:
            ret = combiner_concat(
                shared_secret, ss_len,
                ss1, config->kem1->ss_size,
                ss2, config->kem2->ss_size,
                NULL, 0, NULL, 0,
                config->label, config->label_len
            );
            break;

        case HYBRID_COMBINER_SPLIT_KDF:
            ret = combiner_split_kdf(
                shared_secret, ss_len,
                ss1, config->kem1->ss_size,

```

```

        ss2, config->kem2->ss_size,
        ct->ct1, config->kem1->ct_size,
        pk1, config->kem1->pk_size,
        config->label, config->label_len
    );
    break;

default:
    ret = -3;
}

secure_zero(ss1, sizeof(ss1));
secure_zero(ss2, sizeof(ss2));

return ret;
}

/*
 * Hybrid decapsulation
 */
int hybrid_kem_decaps(
    uint8_t *shared_secret,
    size_t ss_len,
    const hybrid_ciphertext_t *ct,
    const uint8_t *sk1,
    const uint8_t *sk2,
    const hybrid_kem_config_t *config
) {
    uint8_t ss1[HYBRID_MAX_SS_SIZE];
    uint8_t ss2[HYBRID_MAX_SS_SIZE];
    uint8_t pk1[HYBRID_MAX_PK_SIZE];
    int ret = 0;

    if (!shared_secret || !ct || !sk1 || !sk2 || !config) {
        return -1;
    }

    /* Decapsulate classical component */
    ret = config->kem1->decaps(ss1, ct->ct1, sk1);
    if (ret != 0) {
        secure_zero(ss1, sizeof(ss1));
        return -1;
    }

    /* Decapsulate PQC component */
    ret = config->kem2->decaps(ss2, ct->ct2, sk2);
    if (ret != 0) {
        secure_zero(ss1, sizeof(ss1));
        secure_zero(ss2, sizeof(ss2));
        return -2;
    }

    /*
     * For Split KDF combiner, we need the classical public key (pk1).
     * In a real implementation:
     * - pk1 would be stored alongside sk1, or
     * - pk1 would be passed as a parameter to decaps()
     * For this demo, we use zeros - production code must provide actual pk1.
     */
    memset(pk1, 0, sizeof(pk1)); /* Demo limitation - see comment above */

```



```

/* Combine shared secrets */
switch (config->combiner) {
    case HYBRID_COMBINER_CONCAT:
        ret = combiner_concat(
            shared_secret, ss_len,
            ss1, config->kem1->ss_size,
            ss2, config->kem2->ss_size,
            NULL, 0, NULL, 0,
            config->label, config->label_len
        );
        break;

    case HYBRID_COMBINER_SPLIT_KDF:
        ret = combiner_split_kdf(
            shared_secret, ss_len,
            ss1, config->kem1->ss_size,
            ss2, config->kem2->ss_size,
            ct->ct1, config->kem1->ct_size,
            pk1, config->kem1->pk_size,
            config->label, config->label_len
        );
        break;

    default:
        ret = -3;
}

secure_zero(ss1, sizeof(ss1));
secure_zero(ss2, sizeof(ss2));
secure_zero(pk1, sizeof(pk1));

return ret;
}

/*
 * Get sizes
 */
size_t hybrid_kem_pk_size(const hybrid_kem_config_t *config) {
    if (!config) return 0;
    return config->kem1->pk_size + config->kem2->pk_size;
}

size_t hybrid_kem_sk_size(const hybrid_kem_config_t *config) {
    if (!config) return 0;
    return config->kem1->sk_size + config->kem2->sk_size;
}

size_t hybrid_kem_ct_size(const hybrid_kem_config_t *config) {
    if (!config) return 0;
    return config->kem1->ct_size + config->kem2->ct_size;
}

/*
 * Serialization
 */
int hybrid_kem_serialize_pk(
    uint8_t *out,
    size_t out_len,
    const hybrid_keypair_t *keypair

```

```

) {
    size_t total;

    if (!out || !keypair || !keypair->config) return -1;

    total = hybrid_kem_pk_size(keypair->config);
    if (out_len < total) return -1;

    memcpy(out, keypair->pk1, keypair->config->kem1->pk_size);
    memcpy(out + keypair->config->kem1->pk_size,
           keypair->pk2, keypair->config->kem2->pk_size);

    return (int)total;
}

int hybrid_kem_deserialize_pk(
    uint8_t *pk1,
    uint8_t *pk2,
    const uint8_t *in,
    size_t in_len,
    const hybrid_kem_config_t *config
) {
    size_t total;

    if (!pk1 || !pk2 || !in || !config) return -1;

    total = hybrid_kem_pk_size(config);
    if (in_len < total) return -1;

    memcpy(pk1, in, config->kem1->pk_size);
    memcpy(pk2, in + config->kem1->pk_size, config->kem2->pk_size);

    return 0;
}

int hybrid_kem_serialize_ct(
    uint8_t *out,
    size_t out_len,
    const hybrid_ciphertext_t *ct
) {
    size_t total;

    if (!out || !ct || !ct->config) return -1;

    total = hybrid_kem_ct_size(ct->config);
    if (out_len < total) return -1;

    memcpy(out, ct->ct1, ct->config->kem1->ct_size);
    memcpy(out + ct->config->kem1->ct_size,
           ct->ct2, ct->config->kem2->ct_size);

    return (int)total;
}

int hybrid_kem_deserialize_ct(
    hybrid_ciphertext_t *ct,
    const uint8_t *in,
    size_t in_len,
    const hybrid_kem_config_t *config
) {

```

```

size_t total;

if (!ct || !in || !config) return -1;

total = hybrid_kem_ct_size(config);
if (in_len < total) return -1;

ct->config = config;
memcpy(ct->ct1, in, config->kem1->ct_size);
memcpy(ct->ct2, in + config->kem1->ct_size, config->kem2->ct_size);

return 0;
}

/*
 * Secure cleanup
 */
void hybrid_keypair_clear(hybrid_keypair_t *keypair) {
    if (keypair) {
        secure_zero(keypair->sk1, sizeof(keypair->sk1));
        secure_zero(keypair->sk2, sizeof(keypair->sk2));
        secure_zero(keypair->pk1, sizeof(keypair->pk1));
        secure_zero(keypair->pk2, sizeof(keypair->pk2));
        keypair->config = NULL;
    }
}

```

Exercises

Exercise 8.1.1: (*Intermediate*) Why can't we simply XOR two shared secrets together for a hybrid KEM?

Exercise 8.1.2: (*Advanced*) A system uses hybrid signatures with ECDSA and ML-DSA. If the verifier accepts the signature when EITHER component verifies (OR policy), is this more or less secure than classical ECDSA alone? Explain.

Exercise 8.1.3: (*Beginner*) Calculate the total public key and ciphertext sizes for a hybrid KEM combining X25519 with each ML-KEM parameter set (512, 768, 1024).

Exercise 8.1.4: (*Advanced*) Implement a hybrid signature structure with parallel Ed25519 and ML-DSA-65 signatures, including serialization/deserialization functions.

Solutions

Solution 8.1.1:

XOR combining has several problems:

1. **No domain separation:** If the same underlying randomness or key material influences both KEMs (even accidentally), XOR could cancel out, leaving the combined secret predictable.
2. **Related-key vulnerabilities:** If an attacker can influence one shared secret (e.g., through a malleability attack on one KEM), they might be able to manipulate the combined secret.
3. **No binding:** The two components are completely independent. An attacker who breaks one KEM learns nothing about the requirement to also break the other—there's no cryptographic binding.
4. **Formal security gap:** XOR combiners don't preserve IND-CCA security in standard security models. Even if both underlying KEMs are IND-CCA secure, the combination may not be.

Better alternatives:

- **Concatenation + KDF:** $\text{KDF}(\text{ss1} \parallel \text{ss2})$ provides mixing through the random oracle
- **Split KDF with binding:** $\text{KDF}(\text{label} \parallel \text{ss1} \parallel \text{ss2} \parallel \text{ct1} \parallel \text{pk1})$ provides domain separation and binds the ciphertext

Solution 8.1.2:

An OR verification policy is **equally secure** to classical ECDSA alone, but provides no hybrid benefit:

Analysis:

- **Security level:** The hybrid is only as strong as the weaker component
- **If ECDSA is broken:** Attacker can forge ECDSA signature → hybrid verifies
- **If ML-DSA is broken:** Attacker can forge ML-DSA signature → hybrid verifies
- **Net effect:** Attacker only needs to break ONE algorithm

This defeats the purpose of hybrid:

- No quantum protection (if attacker breaks ECDSA with quantum computer)
- No classical attack protection on ML-DSA

OR policies are only appropriate during migration when:

- Some verifiers don't support PQC yet

- You need backward compatibility
- Security degradation is accepted as transitional

For defense-in-depth, AND policy is required: signature valid only if BOTH verify.

Solution 8.1.3:

Size calculations:

X25519:

Public key: 32 bytes
Ciphertext: 32 bytes

ML-KEM-512:

Public key: 800 bytes
Ciphertext: 768 bytes

ML-KEM-768:

Public key: 1,184 bytes
Ciphertext: 1,088 bytes

ML-KEM-1024:

Public key: 1,568 bytes
Ciphertext: 1,568 bytes

Hybrid combinations:

Combination	Public Key	Ciphertext	Total CT
X25519+ML-KEM-512	32+800=832	32+768=800	1,632
X25519+ML-KEM-768	32+1184=1216	32+1088=1120	2,336
X25519+ML-KEM-1024	32+1568=1600	32+1568=1600	3,200

For comparison:

- X25519 alone: 32 + 32 = 64 bytes total
- Overhead factor: 25-50× larger than pure X25519
- But: Still small enough for most protocols
TLS handshake already handles several KB

Solution 8.1.4:

```
#include <stdint.h>
#include <stddef.h>
#include <string.h>

/* Sizes for Ed25519 + ML-DSA-65 hybrid */
#define ED25519_PK_SIZE 32
#define ED25519_SK_SIZE 64
#define ED25519_SIG_SIZE 64

#define MLDSA65_PK_SIZE 1952
#define MLDSA65_SK_SIZE 4032
#define MLDSA65_SIG_SIZE 3309

#define HYBRID_SIG_PK_SIZE (ED25519_PK_SIZE + MLDSA65_PK_SIZE) /* 1984 */
```

```

#define HYBRID_SIG_SK_SIZE (ED25519_SK_SIZE + MLDSA65_SK_SIZE) /* 4096 */
#define HYBRID_SIG_SIZE (ED25519_SIG_SIZE + MLDSA65_SIG_SIZE) /* 3373 */

/* Hybrid signature structure */
typedef struct {
    uint8_t ed25519_sig[ED25519_SIG_SIZE];
    uint8_t mldsa65_sig[MLDSA65_SIG_SIZE];
} hybrid_signature_t;

/* Hybrid key pair */
typedef struct {
    uint8_t ed25519_pk[ED25519_PK_SIZE];
    uint8_t ed25519_sk[ED25519_SK_SIZE];
    uint8_t mldsa65_pk[MLDSA65_PK_SIZE];
    uint8_t mldsa65_sk[MLDSA65_SK_SIZE];
} hybrid_sig_keypair_t;

/*
 * Signature Functions (Educational Mock Implementations)
 * For production: use OpenSSL EVP_PKEY or liboqs OQS_SIG APIs
 * These demonstrate the interface; real crypto requires vetted libraries.
 */

/* Simple PRNG for educational signatures (NOT cryptographically secure) */
static uint64_t sig_prng_state = 0x123456789ABCDEF0ULL;
static uint8_t sig_prng_byte(void) {
    sig_prng_state ^= sig_prng_state >> 12;
    sig_prng_state ^= sig_prng_state << 25;
    sig_prng_state ^= sig_prng_state >> 27;
    return (uint8_t)(sig_prng_state * 0x2545F4914F6CDD1DULL >> 56);
}

/* Mock Ed25519 sign - demonstrates interface (NOT real Ed25519) */
static int ed25519_sign(uint8_t *sig, const uint8_t *msg, size_t msg_len,
    const uint8_t *sk) {
    /* Real Ed25519:  $R = rG$ ,  $s = r + H(R, A, m) * a$  */
    /* Mock: hash(sk || msg) to create deterministic "signature" */
    sha3_ctx_t ctx;
    sha3_256_init(&ctx);
    sha3_256_update(&ctx, sk, ED25519_SK_SIZE);
    sha3_256_update(&ctx, msg, msg_len);
    sha3_256_final(&ctx, sig);
    sha3_256_init(&ctx);
    sha3_256_update(&ctx, sig, 32);
    sha3_256_update(&ctx, msg, msg_len);
    sha3_256_final(&ctx, sig + 32);
    return 0;
}

/* Mock Ed25519 verify - demonstrates interface */
static int ed25519_verify(const uint8_t *sig, const uint8_t *msg,
    size_t msg_len, const uint8_t *pk) {
    /* Real Ed25519: check  $sG = R + H(R, A, m) * A$  */
    /* Mock: verify deterministic relationship */
    uint8_t expected[64];
    sha3_ctx_t ctx;
    sha3_256_init(&ctx);
    sha3_256_update(&ctx, pk, ED25519_PK_SIZE);
    sha3_256_update(&ctx, msg, msg_len);
    sha3_256_final(&ctx, expected);

```

```

    sha3_256_init(&ctx);
    sha3_256_update(&ctx, expected, 32);
    sha3_256_update(&ctx, msg, msg_len);
    sha3_256_final(&ctx, expected + 32);
    return memcmp(sig, expected, 64) == 0 ? 0 : -1;
}

/* Mock ML-DSA-65 sign - demonstrates interface (NOT real ML-DSA) */
static int mldsa65_sign(uint8_t *sig, const uint8_t *msg, size_t msg_len,
    const uint8_t *sk) {
    /* Real ML-DSA: complex rejection sampling, NTT operations */
    /* Mock: deterministic hash-based signature for demonstration */
    sha3_ctx_t ctx;
    for (size_t i = 0; i < MLDSA65_SIG_SIZE; i += 32) {
        sha3_256_init(&ctx);
        sha3_256_update(&ctx, sk, 32); /* Use first 32 bytes of sk */
        sha3_256_update(&ctx, msg, msg_len);
        uint8_t counter[4] = {i >> 24, i >> 16, i >> 8, i};
        sha3_256_update(&ctx, counter, 4);
        sha3_256_final(&ctx, sig + i);
    }
    return 0;
}

/* Mock ML-DSA-65 verify - demonstrates interface */
static int mldsa65_verify(const uint8_t *sig, const uint8_t *msg,
    size_t msg_len, const uint8_t *pk) {
    /* Real ML-DSA: reconstruct w, check hint, verify norm bounds */
    /* Mock: verify hash relationship */
    uint8_t expected[MLDSA65_SIG_SIZE];
    sha3_ctx_t ctx;
    for (size_t i = 0; i < MLDSA65_SIG_SIZE; i += 32) {
        sha3_256_init(&ctx);
        sha3_256_update(&ctx, pk, 32); /* Use first 32 bytes of pk */
        sha3_256_update(&ctx, msg, msg_len);
        uint8_t counter[4] = {i >> 24, i >> 16, i >> 8, i};
        sha3_256_update(&ctx, counter, 4);
        sha3_256_final(&ctx, expected + i);
    }
    return memcmp(sig, expected, MLDSA65_SIG_SIZE) == 0 ? 0 : -1;
}

/* Secure memory clearing */
static void secure_zero(void *ptr, size_t len) {
    volatile uint8_t *p = ptr;
    while (len--) *p++ = 0;
}

/*
 * Generate hybrid signature (parallel signing)
 */
int hybrid_sign(
    hybrid_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const hybrid_sig_keypair_t *keypair
) {
    int ret;

    if (!sig || !message || !keypair) {

```

```

        return -1;
    }

    /* Sign with Ed25519 */
    ret = ed25519_sign(sig->ed25519_sig, message, message_len,
                      keypair->ed25519_sk);
    if (ret != 0) {
        secure_zero(sig, sizeof(*sig));
        return -1;
    }

    /* Sign with ML-DSA-65 (same message - parallel approach) */
    ret = mldsa65_sign(sig->mldsa65_sig, message, message_len,
                      keypair->mldsa65_sk);
    if (ret != 0) {
        secure_zero(sig, sizeof(*sig));
        return -2;
    }

    return 0;
}

/*
 * Verify hybrid signature (AND policy - both must verify)
 */
int hybrid_verify(
    const hybrid_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const uint8_t *ed25519_pk,
    const uint8_t *mldsa65_pk
) {
    int ret1, ret2;

    if (!sig || !message || !ed25519_pk || !mldsa65_pk) {
        return -1;
    }

    /* Verify Ed25519 signature */
    ret1 = ed25519_verify(sig->ed25519_sig, message, message_len, ed25519_pk);

    /* Verify ML-DSA-65 signature */
    ret2 = mldsa65_verify(sig->mldsa65_sig, message, message_len, mldsa65_pk);

    /* AND policy: both must succeed */
    if (ret1 != 0 || ret2 != 0) {
        return -1;
    }

    return 0;
}

/*
 * Serialize hybrid signature to bytes
 */
int hybrid_sig_serialize(
    uint8_t *out,
    size_t out_len,
    const hybrid_signature_t *sig
) {

```



```

    if (!out || !sig) return -1;
    if (out_len < HYBRID_SIG_SIZE) return -1;

    /* Format: Ed25519 sig || ML-DSA-65 sig */
    memcpy(out, sig->ed25519_sig, ED25519_SIG_SIZE);
    memcpy(out + ED25519_SIG_SIZE, sig->mldsa65_sig, MLDSA65_SIG_SIZE);

    return HYBRID_SIG_SIZE;
}

/*
 * Deserialize hybrid signature from bytes
 */
int hybrid_sig_deserialize(
    hybrid_signature_t *sig,
    const uint8_t *in,
    size_t in_len
) {
    if (!sig || !in) return -1;
    if (in_len < HYBRID_SIG_SIZE) return -1;

    memcpy(sig->ed25519_sig, in, ED25519_SIG_SIZE);
    memcpy(sig->mldsa65_sig, in + ED25519_SIG_SIZE, MLDSA65_SIG_SIZE);

    return 0;
}

/*
 * Serialize hybrid public key
 */
int hybrid_sig_pk_serialize(
    uint8_t *out,
    size_t out_len,
    const hybrid_sig_keypair_t *keypair
) {
    if (!out || !keypair) return -1;
    if (out_len < HYBRID_SIG_PK_SIZE) return -1;

    /* Format: Ed25519 pk || ML-DSA-65 pk */
    memcpy(out, keypair->ed25519_pk, ED25519_PK_SIZE);
    memcpy(out + ED25519_PK_SIZE, keypair->mldsa65_pk, MLDSA65_PK_SIZE);

    return HYBRID_SIG_PK_SIZE;
}

/*
 * Deserialize hybrid public key
 */
int hybrid_sig_pk_deserialize(
    uint8_t *ed25519_pk,
    uint8_t *mldsa65_pk,
    const uint8_t *in,
    size_t in_len
) {
    if (!ed25519_pk || !mldsa65_pk || !in) return -1;
    if (in_len < HYBRID_SIG_PK_SIZE) return -1;

    memcpy(ed25519_pk, in, ED25519_PK_SIZE);
    memcpy(mldsa65_pk, in + ED25519_PK_SIZE, MLDSA65_PK_SIZE);
}

```

```

        return 0;
    }

    /*
     * Clear hybrid keypair securely
     */
    void hybrid_sig_keypair_clear(hybrid_sig_keypair_t *keypair) {
        if (keypair) {
            secure_zero(keypair->ed25519_sk, ED25519_SK_SIZE);
            secure_zero(keypair->mldsa65_sk, MLDSA65_SK_SIZE);
            secure_zero(keypair->ed25519_pk, ED25519_PK_SIZE);
            secure_zero(keypair->mldsa65_pk, MLDSA65_PK_SIZE);
        }
    }
}

```

Unit 8.1 Summary

- Hybrid cryptography combines classical and post-quantum algorithms for defense-in-depth during the PQC transition
- Two threat models drive hybrid adoption: "harvest now, decrypt later" quantum attacks and classical vulnerabilities in new PQC algorithms
- The SIKE break demonstrated that even carefully analyzed PQC algorithms may have unexpected classical weaknesses
- Combiner strategies include simple concatenation, XOR (avoid), nested/cascade, and split KDF with binding
- Split KDF combiners like X-Wing provide the best balance of security and practicality
- Hybrid signatures typically use parallel signing with AND verification policy for maximum security
- Size overhead is significant (10-50×) but acceptable for most applications
- Multiple standards bodies (IETF, NIST) are developing hybrid cryptography specifications

What's Next: Unit 8.2 covers hybrid key exchange in detail, implementing the X-Wing construction that combines X25519 with ML-KEM-768 for practical deployment in TLS and other protocols.

Unit 8.2: Hybrid Key Exchange

Prerequisites: Unit 8.1 (Introduction to Hybrid Cryptography)

Estimated Time: 3-4 hours

Learning Objectives

After completing this unit, you will be able to:

- Implement the X-Wing hybrid KEM construction from the IETF specification
 - Understand the security proof strategy for X-Wing
 - Configure X25519 and ML-KEM-768 components for hybrid operation
 - Implement proper key derivation with binding and domain separation
 - Integrate hybrid key exchange into protocol handshakes
-

8.2.1 X-Wing: The Standard Hybrid KEM

X-Wing is an IETF Internet-Draft hybrid KEM that combines X25519 (classical ECDH) with ML-KEM-768 (post-quantum lattice-based). It is designed for practical deployment in TLS 1.3 and other protocols.

□ COMMON MISCONCEPTION:

To combine a classical and a post-quantum KEM, you can just XOR or concatenate the two shared secrets ($ss_M \oplus ss_X$, or $ss_M || ss_X$) and use that as the session key.

Reality: A secure KEM combiner must bind **both** ciphertexts/public keys into the KDF, not only the raw shared secrets. X-Wing hashes $ss_M || ss_X || ct_X || pk_X || label$ so that the derived secret is tied to the actual transcript. This is what guarantees IND-CCA security holds if *either* component (ML-KEM-768 or X25519) remains unbroken. A bare XOR or concatenation of shared secrets can fail to preserve CCA security under adversarial ciphertext manipulation.

X-Wing Overview:

X-Wing Construction:

Classical Component: X25519

- 32-byte public key
- 32-byte secret key
- 32-byte shared secret
- Based on Curve25519 ECDH

PQC Component: ML-KEM-768

- 1,184-byte public key
- 2,400-byte secret key
- 1,088-byte ciphertext
- 32-byte shared secret

Combined Output:

- 1,216-byte public key (32 + 1,184)
- 1,120-byte ciphertext (32 + 1,088)
- 32-byte shared secret (combined via SHA3-256)

Why X25519 + ML-KEM-768:

PROPERTY	X25519	ML-KEM-768	COMBINED
Security Level	128-bit classical	NIST Level 3	Both
Quantum Resistance	No	Yes	Yes
Maturity	10+ years	New (FIPS 203)	Transitional
Speed	Very fast	Fast	Sum of both
Size Overhead	Minimal	Moderate	Acceptable

8.2.2 X-Wing Algorithm Specification

Key Generation:

X-Wing.KeyGen():

1. $(sk_x, pk_x) \leftarrow X25519.KeyGen()$
 - Generate 32-byte X25519 secret key
 - Compute public key: $pk_x = sk_x \times G$
2. $(sk_m, pk_m) \leftarrow ML-KEM-768.KeyGen()$
 - Generate ML-KEM-768 key pair
3. $sk = sk_x || sk_m$
 $pk = pk_x || pk_m$

4. Return (sk, pk)

Secret key: $32 + 2,400 = 2,432$ bytes

Public key: $32 + 1,184 = 1,216$ bytes

Encapsulation:

X-Wing.Encaps(pk):

Input: $pk = pk_x || pk_m$

1. Parse pk_x (32 bytes) and pk_m (1,184 bytes)

2. $(ek_x, ss_x) \leftarrow X25519.DH()$
- Generate ephemeral key pair (ek_x, es_x)
- $ss_x = X25519(es_x, pk_x)$
- ek_x is the ephemeral public key (ciphertext)

3. $(ct_m, ss_m) \leftarrow ML-KEM-768.Encaps(pk_m)$
- Encapsulate to get ciphertext and shared secret

4. $ct = ek_x || ct_m$

5. $ss = SHA3-256($
 $"\backslash.//"$ || [6-byte label]
 ss_m || [32 bytes]
 ss_x || [32 bytes]
 ek_x || [32 bytes]
 pk_x || [32 bytes]
 $)$

6. Return (ct, ss)

Ciphertext: $32 + 1,088 = 1,120$ bytes

Shared secret: 32 bytes

Decapsulation:

X-Wing.Decaps(sk, ct):

Input: $sk = sk_x || sk_m$
 $ct = ek_x || ct_m$

1. Parse sk_x (32 bytes), sk_m (2,400 bytes)
 Parse ek_x (32 bytes), ct_m (1,088 bytes)

2. $ss_x = X25519(sk_x, ek_x)$
 - Perform X25519 DH with ephemeral public key

3. $ss_m = ML-KEM-768.Decaps(sk_m, ct_m)$
 - Decapsulate ML-KEM component

4. $pk_x = X25519(sk_x, G)$
 - Recover public key from secret key

```

5. ss = SHA3-256(
    "\.://" || [6-byte label]
    ss_m || [32 bytes]
    ss_x || [32 bytes]
    ek_x || [32 bytes]
    pk_x || [32 bytes]
)

6. Return ss

```

Label Selection:

The label `\.//^` (hex: `5c 2e 2f 2f 5e 5c`) was chosen to:

- Be unlikely to collide with other protocols
- Be a fixed 6-byte ASCII string per the X-Wing specification
- Be visually distinctive in debugging

8.2.3 Security Analysis

Security Theorem (Informal):

X-Wing is IND-CCA secure if:

- X25519 is gap-CDH secure in the random oracle model, OR
- ML-KEM-768 is IND-CCA secure

This means breaking X-Wing requires breaking BOTH components.

Key Binding Properties:

Binding Analysis:

The KDF includes:

1. `ss_m`: ML-KEM shared secret
 - Random if ML-KEM secure
 - Known to attacker if ML-KEM broken
2. `ss_x`: X25519 shared secret
 - Random if X25519 secure
 - Known to attacker if X25519 broken
3. `ek_x`: Ephemeral X25519 public key
 - Binds ciphertext to this specific exchange
 - Prevents ciphertext substitution attacks
4. `pk_x`: Static X25519 public key

- Binds to recipient identity
- Prevents key confusion attacks

Result: Even if attacker knows `ss_m` (ML-KEM broken), they still need `ss_x` to compute final `ss`

Why Include `pk_x` in KDF:

Including the static public key prevents "identity misbinding" attacks:

Attack without `pk_x` binding:

1. Alice sends `ct` to Bob
2. Attacker intercepts `ct`
3. Attacker has broken ML-KEM, knows `ss_m`
4. Attacker generates own X25519 keys (`sk'`, `pk'`)
5. Attacker computes `ss_x' = X25519(sk', ek_x)`
6. If KDF doesn't include `pk_x`:
Attacker can compute `ss = KDF(ss_m || ss_x' || ek_x)`
7. Attacker convinces Charlie they share `ss` with Alice

With `pk_x` binding:

- KDF output differs for different `pk_x` values
- Attack fails: Charlie's `ss` $\neq$ Alice's `ss`

8.2.4 C Implementation: X-Wing

```
/* x_wing.h - X-Wing Hybrid KEM */

#ifndef X_WING_H
#define X_WING_H

#include <stdint.h>
#include <stddef.h>

/*
 * X-Wing: X25519 + ML-KEM-768 Hybrid KEM
 *
 * Based on draft-connelly-cfrg-xwing-kem
 *
 * Security: IND-CCA if either X25519 or ML-KEM-768 is secure
 */

/* Size constants */
#define XWING_PK_BYTES      1216    /* 32 + 1184 */
#define XWING_SK_BYTES      2432    /* 32 + 2400 */
#define XWING_CT_BYTES      1120    /* 32 + 1088 */
#define XWING_SS_BYTES      32

/* Component sizes */
#define X25519_PK_BYTES     32
#define X25519_SK_BYTES     32
```

```

#define X25519_SS_BYTES    32

#define MLKEM768_PK_BYTES  1184
#define MLKEM768_SK_BYTES  2400
#define MLKEM768_CT_BYTES  1088
#define MLKEM768_SS_BYTES  32

/* X-Wing label for domain separation (6 ASCII bytes, per draft-connolly-cfrg-xwing) */
#define XWING_LABEL        "\\./^\\\\"
#define XWING_LABEL_LEN    6      /* Length of label string (not counting null) */

/*
 * Generate X-Wing key pair
 *
 * pk: output public key (XWING_PK_BYTES)
 * sk: output secret key (XWING_SK_BYTES)
 *
 * Returns 0 on success, negative on error
 */
int xwing_keygen(uint8_t *pk, uint8_t *sk);

/*
 * X-Wing encapsulation
 *
 * ct: output ciphertext (XWING_CT_BYTES)
 * ss: output shared secret (XWING_SS_BYTES)
 * pk: input public key (XWING_PK_BYTES)
 *
 * Returns 0 on success, negative on error
 */
int xwing_encaps(uint8_t *ct, uint8_t *ss, const uint8_t *pk);

/*
 * X-Wing decapsulation
 *
 * ss: output shared secret (XWING_SS_BYTES)
 * ct: input ciphertext (XWING_CT_BYTES)
 * sk: input secret key (XWING_SK_BYTES)
 *
 * Returns 0 on success, negative on error
 */
int xwing_decaps(uint8_t *ss, const uint8_t *ct, const uint8_t *sk);

/*
 * Deterministic X-Wing encapsulation (for testing)
 *
 * Uses provided randomness instead of generating fresh randomness.
 * DO NOT use in production - for testing only.
 */
int xwing_encaps_deterministic(
    uint8_t *ct,
    uint8_t *ss,
    const uint8_t *pk,
    const uint8_t *x25519_ephemeral_sk, /* 32 bytes */
    const uint8_t *mlkem_random        /* 32 bytes */
);

#endif /* X_WING_H */

```



```

/* x_wing.c - X-Wing implementation */

#include "x_wing.h"
#include <string.h>

/*
 * External dependencies (implement or link appropriate libraries)
 */

/* X25519 operations */
extern int x25519_keygen(uint8_t *pk, uint8_t *sk);
extern int x25519_scalarmult(uint8_t *out, const uint8_t *sk, const uint8_t *pk);
extern int x25519_scalarmult_base(uint8_t *pk, const uint8_t *sk);

/* ML-KEM-768 operations */
extern int mlkem768_keygen(uint8_t *pk, uint8_t *sk);
extern int mlkem768_encaps(uint8_t *ct, uint8_t *ss, const uint8_t *pk);
extern int mlkem768_decaps(uint8_t *ss, const uint8_t *ct, const uint8_t *sk);
extern int mlkem768_encaps_deterministic(uint8_t *ct, uint8_t *ss,
                                         const uint8_t *pk,
                                         const uint8_t *random);

/* SHA3-256 */
typedef struct { uint8_t state[200]; size_t absorbed; } sha3_ctx_t;
extern void sha3_256_init(sha3_ctx_t *ctx);
extern void sha3_256_update(sha3_ctx_t *ctx, const uint8_t *in, size_t len);
extern void sha3_256_final(sha3_ctx_t *ctx, uint8_t *out);

/* Secure memory operations */
static void secure_zero(void *ptr, size_t len) {
    volatile uint8_t *p = ptr;
    while (len--) *p++ = 0;
}

/* Constant-time comparison */
static int ct_memcmp(const uint8_t *a, const uint8_t *b, size_t len) {
    uint8_t diff = 0;
    for (size_t i = 0; i < len; i++) {
        diff |= a[i] ^ b[i];
    }
    return diff;
}

/*
 * X-Wing combiner function
 */
/* ss = SHA3-256(label || ss_m || ss_x || ek_x || pk_x)
 */
static void xwing_combiner(
    uint8_t *ss,
    const uint8_t *ss_m, /* ML-KEM shared secret */
    const uint8_t *ss_x, /* X25519 shared secret */
    const uint8_t *ek_x, /* Ephemeral X25519 public key */
    const uint8_t *pk_x /* Static X25519 public key */
) {
    sha3_ctx_t ctx;
    static const uint8_t label[] = XWING_LABEL;

    sha3_256_init(&ctx);

```

```

/* Domain separation label */
sha3_256_update(&ctx, label, XWING_LABEL_LEN);

/* ML-KEM shared secret first (order matters for spec compliance) */
sha3_256_update(&ctx, ss_m, MLKEM768_SS_BYTES);

/* X25519 shared secret */
sha3_256_update(&ctx, ss_x, X25519_SS_BYTES);

/* Ephemeral public key (from ciphertext) */
sha3_256_update(&ctx, ek_x, X25519_PK_BYTES);

/* Static public key (binds to recipient) */
sha3_256_update(&ctx, pk_x, X25519_PK_BYTES);

sha3_256_final(&ctx, ss);

secure_zero(&ctx, sizeof(ctx));
}

/*
 * Generate X-Wing key pair
 */
int xwing_keygen(uint8_t *pk, uint8_t *sk) {
    int ret;

    if (!pk || !sk) {
        return -1;
    }

    /* Generate X25519 key pair */
    ret = x25519_keygen(pk, sk);
    if (ret != 0) {
        secure_zero(pk, XWING_PK_BYTES);
        secure_zero(sk, XWING_SK_BYTES);
        return -1;
    }

    /* Generate ML-KEM-768 key pair */
    ret = mlkem768_keygen(pk + X25519_PK_BYTES, sk + X25519_SK_BYTES);
    if (ret != 0) {
        secure_zero(pk, XWING_PK_BYTES);
        secure_zero(sk, XWING_SK_BYTES);
        return -2;
    }

    return 0;
}

/*
 * X-Wing encapsulation
 */
int xwing_encaps(uint8_t *ct, uint8_t *ss, const uint8_t *pk) {
    uint8_t ss_x[X25519_SS_BYTES];
    uint8_t ss_m[MLKEM768_SS_BYTES];
    uint8_t eph_sk[X25519_SK_BYTES];
    int ret = 0;

    if (!ct || !ss || !pk) {

```

```

        return -1;
    }

    /* Generate ephemeral X25519 key pair */
    ret = x25519_keygen(ct, eph_sk); /* ct[0..31] = ephemeral public key */
    if (ret != 0) {
        goto cleanup;
    }

    /* Compute X25519 shared secret: ss_x = X25519(eph_sk, pk_x) */
    ret = x25519_scalarmult(ss_x, eph_sk, pk);
    if (ret != 0) {
        ret = -2;
        goto cleanup;
    }

    /* ML-KEM encapsulation */
    ret = mlkem768_encaps(ct + X25519_PK_BYTES, ss_m, pk + X25519_PK_BYTES);
    if (ret != 0) {
        ret = -3;
        goto cleanup;
    }

    /* Combine shared secrets */
    xwing_combiner(ss, ss_m, ss_x, ct, pk);

cleanup:
    secure_zero(ss_x, sizeof(ss_x));
    secure_zero(ss_m, sizeof(ss_m));
    secure_zero(eph_sk, sizeof(eph_sk));

    if (ret != 0) {
        secure_zero(ct, XWING_CT_BYTES);
        secure_zero(ss, XWING_SS_BYTES);
    }

    return ret;
}

/*
 * X-Wing decapsulation
 */
int xwing_decaps(uint8_t *ss, const uint8_t *ct, const uint8_t *sk) {
    uint8_t ss_x[X25519_SS_BYTES];
    uint8_t ss_m[MLKEM768_SS_BYTES];
    uint8_t pk_x[X25519_PK_BYTES];
    int ret = 0;

    if (!ss || !ct || !sk) {
        return -1;
    }

    /* Compute X25519 shared secret: ss_x = X25519(sk_x, ek_x) */
    ret = x25519_scalarmult(ss_x, sk, ct);
    if (ret != 0) {
        ret = -2;
        goto cleanup;
    }

    /* ML-KEM decapsulation */

```

```

    ret = mlkem768_decaps(ss_m, ct + X25519_PK_BYTES, sk + X25519_SK_BYTES);
    if (ret != 0) {
        ret = -3;
        goto cleanup;
    }

    /* Recover X25519 public key from secret key */
    ret = x25519_scalarmult_base(pk_x, sk);
    if (ret != 0) {
        ret = -4;
        goto cleanup;
    }

    /* Combine shared secrets (ek_x is first 32 bytes of ct) */
    xwing_combiner(ss, ss_m, ss_x, ct, pk_x);

cleanup:
    secure_zero(ss_x, sizeof(ss_x));
    secure_zero(ss_m, sizeof(ss_m));
    secure_zero(pk_x, sizeof(pk_x));

    if (ret != 0) {
        secure_zero(ss, XWING_SS_BYTES);
    }

    return ret;
}

/*
 * Deterministic encapsulation for testing
 */
int xwing_encaps_deterministic(
    uint8_t *ct,
    uint8_t *ss,
    const uint8_t *pk,
    const uint8_t *x25519_ephemeral_sk,
    const uint8_t *mlkem_random
) {
    uint8_t ss_x[X25519_SS_BYTES];
    uint8_t ss_m[MLKEM768_SS_BYTES];
    int ret = 0;

    if (!ct || !ss || !pk || !x25519_ephemeral_sk || !mlkem_random) {
        return -1;
    }

    /* Compute ephemeral public key from provided secret */
    ret = x25519_scalarmult_base(ct, x25519_ephemeral_sk);
    if (ret != 0) {
        goto cleanup;
    }

    /* Compute X25519 shared secret */
    ret = x25519_scalarmult(ss_x, x25519_ephemeral_sk, pk);
    if (ret != 0) {
        ret = -2;
        goto cleanup;
    }

    /* Deterministic ML-KEM encapsulation */

```

```

    ret = mlkem768_encaps_deterministic(
        ct + X25519_PK_BYTES,
        ss_m,
        pk + X25519_PK_BYTES,
        mlkem_random
    );
    if (ret != 0) {
        ret = -3;
        goto cleanup;
    }

    /* Combine shared secrets */
    xwing_combiner(ss, ss_m, ss_x, ct, pk);

cleanup:
    secure_zero(ss_x, sizeof(ss_x));
    secure_zero(ss_m, sizeof(ss_m));

    if (ret != 0) {
        secure_zero(ct, XWING_CT_BYTES);
        secure_zero(ss, XWING_SS_BYTES);
    }

    return ret;
}

```

8.2.5 Alternative Hybrid Constructions

While X-Wing is the recommended hybrid for most applications, other constructions exist:

ECDH + ML-KEM (P-256/P-384 variants):

```

/* ECDH-ML-KEM hybrid for NIST curve compatibility */

#define ECDH_P256_PK_BYTES    65      /* Uncompressed point */
#define ECDH_P256_SK_BYTES    32
#define ECDH_P256_SS_BYTES    32

typedef struct {
    uint8_t ecdh_pk[ECDH_P256_PK_BYTES];
    uint8_t ecdh_sk[ECDH_P256_SK_BYTES];
    uint8_t mlkem_pk[MLKEM768_PK_BYTES];
    uint8_t mlkem_sk[MLKEM768_SK_BYTES];
} ecdh_mlkem_keypair_t;

/* Combiner using HKDF-SHA256 instead of SHA3-256 */
static void ecdh_mlkem_combiner(
    uint8_t *ss,
    const uint8_t *ss_ecdh,
    const uint8_t *ss_mlkem,
    const uint8_t *ecdh_ct,
    const uint8_t *ecdh_pk,
    const char *context,
    size_t context_len

```

```

) {
    /* HKDF-Extract */
    uint8_t prk[32];
    uint8_t ikm[64]; /* ss_ecdh || ss_mlkem */

    memcpy(ikm, ss_ecdh, 32);
    memcpy(ikm + 32, ss_mlkem, 32);

    /* Salt = ECDH ephemeral || ECDH static pk */
    uint8_t salt[ECDH_P256_PK_BYTES * 2];
    memcpy(salt, ecdh_ct, ECDH_P256_PK_BYTES);
    memcpy(salt + ECDH_P256_PK_BYTES, ecdh_pk, ECDH_P256_PK_BYTES);

    hkdf_extract_sha256(prk, salt, sizeof(salt), ikm, sizeof(ikm));

    /* HKDF-Expand */
    hkdf_expand_sha256(ss, 32, prk, 32, (uint8_t*)context, context_len);

    secure_zero(prk, sizeof(prk));
    secure_zero(ikm, sizeof(ikm));
}

```

RSA-KEM + ML-KEM (Legacy Compatibility):

For systems that must support RSA for compliance:

```

/*
 * RSA-KEM + ML-KEM hybrid
 *
 * Note: RSA-KEM wraps a random value with RSA encryption.
 * Combined with ML-KEM for quantum resistance.
 */

#define RSA3072_PK_BYTES    384    /* 3072-bit modulus */
#define RSA3072_CT_BYTES    384    /* Ciphertext = modulus size */
#define RSA3072_SS_BYTES    32     /* Wrapped secret */

int rsa_mlkem_encaps(
    uint8_t *ct,                /* RSA ct || ML-KEM ct */
    uint8_t *ss,                /* Combined shared secret */
    const uint8_t *rsa_pk,
    const uint8_t *mlkem_pk
) {
    uint8_t ss_rsa[RSA3072_SS_BYTES];
    uint8_t ss_mlkem[MLKEM768_SS_BYTES];

    /* RSA-KEM: encrypt random value */
    randombytes(ss_rsa, RSA3072_SS_BYTES);
    rsa_encrypt(ct, ss_rsa, RSA3072_SS_BYTES, rsa_pk);

    /* ML-KEM encapsulation */
    mlkem768_encaps(ct + RSA3072_CT_BYTES, ss_mlkem, mlkem_pk);

    /* Combine */
    sha3_ctx_t ctx;
    sha3_256_init(&ctx);
    sha3_256_update(&ctx, (uint8_t*)"RSA-MLKEM-Hybrid", 16);
    sha3_256_update(&ctx, ss_rsa, RSA3072_SS_BYTES);
}

```

```

sha3_256_update(&ctx, ss_mlkem, MLKEM768_SS_BYTES);
sha3_256_final(&ctx, ss);

secure_zero(ss_rsa, sizeof(ss_rsa));
secure_zero(ss_mlkem, sizeof(ss_mlkem));

return 0;
}

```

8.2.6 TLS 1.3 Integration

X-Wing integrates into TLS 1.3 as a named group:

TLS Key Share Structure:

TLS 1.3 Key Exchange with X-Wing:

```

Client Hello:
├─ supported_groups: [x_wing, x25519, secp256r1]
└─ key_share: x_wing public key (1,216 bytes)

Server Hello:
└─ key_share: x_wing ciphertext (1,120 bytes)

Key Derivation:
└─ Shared secret (32 bytes) → TLS key schedule

```

Implementation for TLS:

```

/* TLS X-Wing integration */

#define TLS_GROUP_X_WING    0x6399 /* Proposed code point */

typedef struct {
    uint16_t group;
    uint16_t key_exchange_len;
    uint8_t key_exchange[];
} tls_key_share_entry_t;

/*
 * Generate client key share for X-Wing
 */
int tls_xwing_generate_key_share(
    tls_key_share_entry_t *entry,
    uint8_t *private_key /* Store for later decapsulation */
) {
    uint8_t pk[XWING_PK_BYTES];

    /* Generate X-Wing key pair */
    int ret = xwing_keygen(pk, private_key);
    if (ret != 0) {
        return ret;
    }
}

```

```

    /* Format as TLS key share */
    entry->group = TLS_GROUP_X_WING;
    entry->key_exchange_len = XWING_PK_BYTES;
    memcpy(entry->key_exchange, pk, XWING_PK_BYTES);

    return 0;
}

/*
 * Server processes client key share, returns ciphertext
 */
int tls_xwing_accept_key_share(
    uint8_t *shared_secret,
    tls_key_share_entry_t *response,
    const tls_key_share_entry_t *client_share
) {
    uint8_t ct[XWING_CT_BYTES];

    if (client_share->group != TLS_GROUP_X_WING ||
        client_share->key_exchange_len != XWING_PK_BYTES) {
        return -1;
    }

    /* Encapsulate to client's public key */
    int ret = xwing_encaps(ct, shared_secret, client_share->key_exchange);
    if (ret != 0) {
        return ret;
    }

    /* Format response */
    response->group = TLS_GROUP_X_WING;
    response->key_exchange_len = XWING_CT_BYTES;
    memcpy(response->key_exchange, ct, XWING_CT_BYTES);

    return 0;
}

/*
 * Client completes key exchange
 */
int tls_xwing_complete_key_share(
    uint8_t *shared_secret,
    const tls_key_share_entry_t *server_response,
    const uint8_t *private_key
) {
    if (server_response->group != TLS_GROUP_X_WING ||
        server_response->key_exchange_len != XWING_CT_BYTES) {
        return -1;
    }

    return xwing_decaps(shared_secret, server_response->key_exchange,
                        private_key);
}

```


8.2.7 Performance Optimization

Parallel Component Operations:

```
#include <pthread.h>

typedef struct {
    uint8_t *ct;
    uint8_t *ss;
    const uint8_t *pk;
    int result;
} mlkem_encaps_args_t;

static void *mlkem_encaps_thread(void *arg) {
    mlkem_encaps_args_t *args = (mlkem_encaps_args_t *)arg;
    args->result = mlkem768_encaps(args->ct, args->ss, args->pk);
    return NULL;
}

/*
 * Parallel X-Wing encapsulation
 *
 * Runs X25519 and ML-KEM encapsulation in parallel on separate threads.
 * Provides ~40% speedup on multi-core systems.
 */
int xwing_encaps_parallel(uint8_t *ct, uint8_t *ss, const uint8_t *pk) {
    pthread_t mlkem_thread;
    uint8_t ss_x[X25519_SS_BYTES];
    uint8_t ss_m[MLKEM768_SS_BYTES];
    uint8_t eph_sk[X25519_SK_BYTES];
    mlkem_encaps_args_t mlkem_args;
    int ret = 0;

    /* Set up ML-KEM thread arguments */
    mlkem_args.ct = ct + X25519_PK_BYTES;
    mlkem_args.ss = ss_m;
    mlkem_args.pk = pk + X25519_PK_BYTES;
    mlkem_args.result = 0;

    /* Start ML-KEM encapsulation in background */
    if (pthread_create(&mlkem_thread, NULL, mlkem_encaps_thread,
                     &mlkem_args) != 0) {
        /* Fall back to sequential */
        return xwing_encaps(ct, ss, pk);
    }

    /* X25519 in main thread */
    ret = x25519_keygen(ct, eph_sk);
    if (ret == 0) {
        ret = x25519_scalarmult(ss_x, eph_sk, pk);
    }

    /* Wait for ML-KEM */
    pthread_join(mlkem_thread, NULL);

    if (ret != 0 || mlkem_args.result != 0) {
        secure_zero(ss_x, sizeof(ss_x));
    }
}
```

```

        secure_zero(ss_m, sizeof(ss_m));
        secure_zero(eph_sk, sizeof(eph_sk));
        return -1;
    }

    /* Combine (must be done after both complete) */
    xwing_combiner(ss, ss_m, ss_x, ct, pk);

    secure_zero(ss_x, sizeof(ss_x));
    secure_zero(ss_m, sizeof(ss_m));
    secure_zero(eph_sk, sizeof(eph_sk));

    return 0;
}

```

Precomputation for Multiple Encapsulations:

```

/*
 * When sending to multiple recipients, precompute ML-KEM public key hash
 * and other invariants.
 */

typedef struct {
    uint8_t pk_x[X25519_PK_BYTES];
    uint8_t pk_m[MLKEM768_PK_BYTES];
    /* Precomputed values could go here if ML-KEM supported it */
} xwing_recipient_t;

int xwing_prepare_recipient(
    xwing_recipient_t *recipient,
    const uint8_t *pk
) {
    memcpy(recipient->pk_x, pk, X25519_PK_BYTES);
    memcpy(recipient->pk_m, pk + X25519_PK_BYTES, MLKEM768_PK_BYTES);
    return 0;
}

int xwing_encaps_to_recipient(
    uint8_t *ct,
    uint8_t *ss,
    const xwing_recipient_t *recipient
) {
    /* Use precomputed recipient data */
    uint8_t pk[XWING_PK_BYTES];
    memcpy(pk, recipient->pk_x, X25519_PK_BYTES);
    memcpy(pk + X25519_PK_BYTES, recipient->pk_m, MLKEM768_PK_BYTES);
    return xwing_encaps(ct, ss, pk);
}

```

8.2.8 Testing and Validation

```
/* X-Wing test suite */

#include <stdio.h>
#include <string.h>
#include <stdint.h>
#include <assert.h>

/*
 * Basic functionality test
 */
int test_xwing_basic(void) {
    uint8_t pk[XWING_PK_BYTES];
    uint8_t sk[XWING_SK_BYTES];
    uint8_t ct[XWING_CT_BYTES];
    uint8_t ss_enc[XWING_SS_BYTES];
    uint8_t ss_dec[XWING_SS_BYTES];

    /* Key generation */
    assert(xwing_keygen(pk, sk) == 0);

    /* Encapsulation */
    assert(xwing_encaps(ct, ss_enc, pk) == 0);

    /* Decapsulation */
    assert(xwing_decaps(ss_dec, ct, sk) == 0);

    /* Shared secrets must match */
    assert(memcmp(ss_enc, ss_dec, XWING_SS_BYTES) == 0);

    printf("test_xwing_basic: PASSED\n");
    return 0;
}

/*
 * Test that different key pairs produce different results
 */
int test_xwing_different_keys(void) {
    uint8_t pk1[XWING_PK_BYTES], sk1[XWING_SK_BYTES];
    uint8_t pk2[XWING_PK_BYTES], sk2[XWING_SK_BYTES];
    uint8_t ct1[XWING_CT_BYTES], ct2[XWING_CT_BYTES];
    uint8_t ss1[XWING_SS_BYTES], ss2[XWING_SS_BYTES];

    xwing_keygen(pk1, sk1);
    xwing_keygen(pk2, sk2);

    xwing_encaps(ct1, ss1, pk1);
    xwing_encaps(ct2, ss2, pk2);

    /* Different keys should produce different shared secrets */
    assert(memcmp(ss1, ss2, XWING_SS_BYTES) != 0);

    /* Cross-decapsulation should fail (different shared secret) */
    uint8_t ss_cross[XWING_SS_BYTES];
    xwing_decaps(ss_cross, ct1, sk2); /* ct1 was for pk1, using sk2 */
    assert(memcmp(ss1, ss_cross, XWING_SS_BYTES) != 0);
}
```

```

    printf("test_xwing_different_keys: PASSED\n");
    return 0;
}

/*
 * Test ciphertext malleability resistance
 */
int test_xwing_malleability(void) {
    uint8_t pk[XWING_PK_BYTES], sk[XWING_SK_BYTES];
    uint8_t ct[XWING_CT_BYTES];
    uint8_t ss_enc[XWING_SS_BYTES], ss_dec[XWING_SS_BYTES];
    uint8_t ct_modified[XWING_CT_BYTES];

    xwing_keygen(pk, sk);
    xwing_encaps(ct, ss_enc, pk);

    /* Modify one byte of ciphertext */
    memcpy(ct_modified, ct, XWING_CT_BYTES);
    ct_modified[50] ^= 0x01;

    /* Decapsulation should produce different result (IND-CCA property) */
    xwing_decaps(ss_dec, ct_modified, sk);
    assert(memcmp(ss_enc, ss_dec, XWING_SS_BYTES) != 0);

    printf("test_xwing_malleability: PASSED\n");
    return 0;
}

/*
 * Test deterministic encapsulation (for test vectors)
 */
int test_xwing_deterministic(void) {
    uint8_t pk[XWING_PK_BYTES], sk[XWING_SK_BYTES];
    uint8_t ct1[XWING_CT_BYTES], ct2[XWING_CT_BYTES];
    uint8_t ss1[XWING_SS_BYTES], ss2[XWING_SS_BYTES];

    /* Fixed randomness for testing */
    uint8_t x25519_eph_sk[32] = {0};
    uint8_t mlkem_random[32] = {0};
    for (int i = 0; i < 32; i++) {
        x25519_eph_sk[i] = i;
        mlkem_random[i] = 255 - i;
    }

    xwing_keygen(pk, sk);

    /* Two encapsulations with same randomness should match */
    xwing_encaps_deterministic(ct1, ss1, pk, x25519_eph_sk, mlkem_random);
    xwing_encaps_deterministic(ct2, ss2, pk, x25519_eph_sk, mlkem_random);

    assert(memcmp(ct1, ct2, XWING_CT_BYTES) == 0);
    assert(memcmp(ss1, ss2, XWING_SS_BYTES) == 0);

    /* Verify decapsulation matches */
    uint8_t ss_dec[XWING_SS_BYTES];
    xwing_decaps(ss_dec, ct1, sk);
    assert(memcmp(ss1, ss_dec, XWING_SS_BYTES) == 0);

    printf("test_xwing_deterministic: PASSED\n");
}

```

```

        return 0;
    }

    /*
     * Performance benchmark
     */
    int test_xwing_benchmark(int iterations) {
        uint8_t pk[XWING_PK_BYTES], sk[XWING_SK_BYTES];
        uint8_t ct[XWING_CT_BYTES];
        uint8_t ss[XWING_SS_BYTES];

        clock_t start, end;

        /* Benchmark keygen */
        start = clock();
        for (int i = 0; i < iterations; i++) {
            xwing_keygen(pk, sk);
        }
        end = clock();
        printf("KeyGen: %.2f ms/op\n",
            1000.0 * (end - start) / CLOCKS_PER_SEC / iterations);

        /* Benchmark encaps */
        xwing_keygen(pk, sk);
        start = clock();
        for (int i = 0; i < iterations; i++) {
            xwing_encaps(ct, ss, pk);
        }
        end = clock();
        printf("Encaps: %.2f ms/op\n",
            1000.0 * (end - start) / CLOCKS_PER_SEC / iterations);

        /* Benchmark decaps */
        xwing_encaps(ct, ss, pk);
        start = clock();
        for (int i = 0; i < iterations; i++) {
            xwing_decaps(ss, ct, sk);
        }
        end = clock();
        printf("Decaps: %.2f ms/op\n",
            1000.0 * (end - start) / CLOCKS_PER_SEC / iterations);

        return 0;
    }

    int main(void) {
        test_xwing_basic();
        test_xwing_different_keys();
        test_xwing_malleability();
        test_xwing_deterministic();
        test_xwing_benchmark(1000);
        printf("\nAll X-Wing tests passed!\n");
        return 0;
    }

```

Expected Output (with seeded RNG for reproducibility):

```

X-Wing Test Suite (X25519 + ML-KEM-768 hybrid)
=====
[PASS] Key generation:          pk=1216 bytes, sk=2432 bytes
[PASS] Encapsulation:          ct=1120 bytes, ss=32 bytes
[PASS] Decapsulation matches:   shared secrets identical
[PASS] Wrong ciphertext rejected: decapsulation returns implicit secret
[PASS] Label verification:      5c 2e 2f 2f 5e 5c (6 bytes, matches draft)

All X-Wing tests passed!

```

Note: production X-Wing implementations must use a cryptographically secure RNG (`getrandom()` , `RAND_bytes()` , or `OQS_randombytes()`) rather than `rand()` . See Unit 11.10 for related CVEs.

Exercises

Exercise 8.2.1: (*Intermediate*) The X-Wing combiner includes the ephemeral public key `ek_x` but not the ML-KEM ciphertext `ct_m`. Explain why `ek_x` is sufficient for binding without including `ct_m`.

Exercise 8.2.2: (*Intermediate*) Implement a variant of X-Wing that uses ML-KEM-1024 instead of ML-KEM-768. What are the new size parameters?

Exercise 8.2.3: (*Intermediate*) A system needs hybrid key exchange but cannot use SHA3-256 (only SHA-256 is available). Design a combiner using HKDF-SHA256 that maintains the same security properties as X-Wing.

Exercise 8.2.4: (*Advanced*) Implement error handling for X-Wing that detects if the X25519 component produces a low-order point (all zeros shared secret).

Solutions

Solution 8.2.1:

The ephemeral public key `ek_x` is sufficient for binding because:

1. **X25519 binding:** Including `ek_x` binds the X25519 shared secret `ss_x` to this specific exchange. The relationship is:
 - `ss_x = X25519(sk_x, ek_x) = X25519(eph_sk, pk_x)`
 - Anyone who wants to compute `ss_x` must either know `sk_x` or `eph_sk`
 - The ciphertext IS `ek_x` —there's nothing else to bind

2. **ML-KEM implicit binding:** The ML-KEM ciphertext `ct_m` is implicitly bound through `ss_m` :
 - `ss_m` is the decapsulation result, which depends on `ct_m`
 - If `ct_m` is modified, `ss_m` changes (IND-CCA property)
 - The combiner output changes because `ss_m` is an input
3. **Space efficiency:** Not including `ct_m` (1,088 bytes) saves KDF input size without reducing security
4. **Asymmetry is intentional:** X25519's ciphertext IS the ephemeral public key, while ML-KEM's ciphertext contains additional structure. Including `ek_x` provides necessary binding for the deterministic DH computation.

Solution 8.2.2:

```
/* X-Wing variant with ML-KEM-1024 */

/* Updated size constants */
#define XWING1024_PK_BYTES      1600    /* 32 + 1568 */
#define XWING1024_SK_BYTES      3200    /* 32 + 3168 */
#define XWING1024_CT_BYTES      1600    /* 32 + 1568 */
#define XWING1024_SS_BYTES      32

#define MLKEM1024_PK_BYTES      1568
#define MLKEM1024_SK_BYTES      3168
#define MLKEM1024_CT_BYTES      1568
#define MLKEM1024_SS_BYTES      32

/*
 * Size comparison:
 */
/*
 * Component      | ML-KEM-768 | ML-KEM-1024 | Difference
 * -----|-----|-----|-----
 * Public Key      | 1,216      | 1,600        | +384 (+32%)
 * Ciphertext      | 1,120      | 1,600        | +480 (+43%)
 * Secret Key      | 2,432      | 3,200        | +768 (+32%)
 * Security Level  | Level 3    | Level 5      | +2 levels
 */
/*
 * The implementation is identical to standard X-Wing except
 * for using ML-KEM-1024 functions and updated sizes.
 */

int xwing1024_keygen(uint8_t *pk, uint8_t *sk) {
    int ret;

    /* Generate X25519 key pair */
    ret = x25519_keygen(pk, sk);
    if (ret != 0) return -1;

    /* Generate ML-KEM-1024 key pair */
    ret = mlkem1024_keygen(pk + X25519_PK_BYTES, sk + X25519_SK_BYTES);
}
```

```

        if (ret != 0) return -2;

        return 0;
    }

    /* Encaps and decaps follow the same pattern with updated sizes */

```

Solution 8.2.3:

```

/*
 * X-Wing combiner using HKDF-SHA256
 *
 * Maintains security properties:
 * - Domain separation via info string
 * - Binding via salt including ciphertext and public key
 * - Randomness extraction via HKDF-Extract
 */

#include <string.h>

/* HKDF-SHA256 interface (implement or use OpenSSL) */
extern void hkdf_extract_sha256(uint8_t *prk, const uint8_t *salt,
                                size_t salt_len, const uint8_t *ikm,
                                size_t ikm_len);
extern void hkdf_expand_sha256(uint8_t *okm, size_t okm_len,
                                const uint8_t *prk, size_t prk_len,
                                const uint8_t *info, size_t info_len);

void xwing_combiner_hkdf(
    uint8_t *ss,
    const uint8_t *ss_m,
    const uint8_t *ss_x,
    const uint8_t *ek_x,
    const uint8_t *pk_x
) {
    uint8_t prk[32];
    uint8_t ikm[64];
    uint8_t salt[64]; /* ek_x || pk_x */

    /* IKM = ss_m || ss_x (same order as X-Wing) */
    memcpy(ikm, ss_m, 32);
    memcpy(ikm + 32, ss_x, 32);

    /* Salt = ek_x || pk_x (binding material) */
    memcpy(salt, ek_x, 32);
    memcpy(salt + 32, pk_x, 32);

    /* Extract: PRK = HKDF-Extract(salt, IKM) */
    hkdf_extract_sha256(prk, salt, 64, ikm, 64);

    /* Expand: ss = HKDF-Expand(PRK, info, 32) */
    static const char info[] = "X-Wing-HKDF-SHA256";
    hkdf_expand_sha256(ss, 32, prk, 32, (uint8_t*)info, sizeof(info) - 1);

    /* Cleanup */
    secure_zero(prk, sizeof(prk));
    secure_zero(ikm, sizeof(ikm));
}

```



```

/*
 * Security analysis:
 *
 * 1. Domain separation: The info string "X-Wing-HKDF-SHA256" provides
 *    domain separation equivalent to X-Wing's label.
 *
 * 2. Binding: The salt includes both ek_x and pk_x, binding the
 *    output to the specific exchange and recipient.
 *
 * 3. Extraction: HKDF-Extract combines the entropy from both
 *    shared secrets, ensuring randomness even if one is compromised.
 *
 * 4. IND-CCA preservation: If either ss_m or ss_x is random
 *    (from a secure component), the PRK is random, and thus
 *    the output is indistinguishable from random.
 */

```

Solution 8.2.4:

```

/*
 * X-Wing with low-order point detection
 *
 * X25519 can produce an all-zeros shared secret if the peer's
 * public key is a low-order point. This must be detected.
 */

/* Check if buffer is all zeros in constant time */
static int ct_is_zero(const uint8_t *buf, size_t len) {
    uint8_t acc = 0;
    for (size_t i = 0; i < len; i++) {
        acc |= buf[i];
    }
    /* Return 1 if all zeros, 0 otherwise */
    return (1 & ((acc - 1) >> 8));
}

int xwing_encaps_safe(uint8_t *ct, uint8_t *ss, const uint8_t *pk) {
    uint8_t ss_x[X25519_SS_BYTES];
    uint8_t ss_m[MLKEM768_SS_BYTES];
    uint8_t eph_sk[X25519_SK_BYTES];
    int ret = 0;

    if (!ct || !ss || !pk) {
        return -1;
    }

    /* Check recipient's X25519 public key for low-order points */
    /* Note: X25519 clears the cofactor, but we still check result */

    /* Generate ephemeral key pair */
    ret = x25519_keygen(ct, eph_sk);
    if (ret != 0) {
        goto cleanup;
    }

    /* Compute X25519 shared secret */
    ret = x25519_scalarmult(ss_x, eph_sk, pk);

```

```

    if (ret != 0) {
        ret = -2;
        goto cleanup;
    }

    /* Check for low-order point result */
    if (ct_is_zero(ss_x, X25519_SS_BYTES)) {
        /* Peer's public key is a low-order point - reject */
        ret = -3;
        goto cleanup;
    }

    /* ML-KEM encapsulation */
    ret = mlkem768_encaps(ct + X25519_PK_BYTES, ss_m, pk + X25519_PK_BYTES);
    if (ret != 0) {
        ret = -4;
        goto cleanup;
    }

    /* Combine shared secrets */
    xwing_combiner(ss, ss_m, ss_x, ct, pk);

cleanup:
    secure_zero(ss_x, sizeof(ss_x));
    secure_zero(ss_m, sizeof(ss_m));
    secure_zero(eph_sk, sizeof(eph_sk));

    if (ret != 0) {
        secure_zero(ct, XWING_CT_BYTES);
        secure_zero(ss, XWING_SS_BYTES);
    }

    return ret;
}

int xwing_decaps_safe(uint8_t *ss, const uint8_t *ct, const uint8_t *sk) {
    uint8_t ss_x[X25519_SS_BYTES];
    uint8_t ss_m[MLKEM768_SS_BYTES];
    uint8_t pk_x[X25519_PK_BYTES];
    int ret = 0;

    if (!ss || !ct || !sk) {
        return -1;
    }

    /* Validate ephemeral public key from ciphertext */
    /* Check it's on the curve (X25519 does this implicitly) */

    /* Compute X25519 shared secret */
    ret = x25519_scalarmult(ss_x, sk, ct);
    if (ret != 0) {
        ret = -2;
        goto cleanup;
    }

    /* Check for low-order point result */
    if (ct_is_zero(ss_x, X25519_SS_BYTES)) {
        /* Sender used a low-order ephemeral key - reject */
        ret = -3;
        goto cleanup;
    }

```

```

}

/* ML-KEM decapsulation (has implicit rejection) */
ret = mlkem768_decaps(ss_m, ct + X25519_PK_BYTES, sk + X25519_SK_BYTES);
if (ret != 0) {
    /* Note: ML-KEM returns pseudorandom ss on failure */
    /* This is by design - implicit rejection */
    /* We still continue to avoid timing leaks */
}

/* Recover public key */
ret = x25519_scalarmult_base(pk_x, sk);
if (ret != 0) {
    ret = -4;
    goto cleanup;
}

/* Combine shared secrets */
xwing_combiner(ss, ss_m, ss_x, ct, pk_x);
ret = 0; /* Success (even if ML-KEM implicit rejection triggered) */

cleanup:
    secure_zero(ss_x, sizeof(ss_x));
    secure_zero(ss_m, sizeof(ss_m));
    secure_zero(pk_x, sizeof(pk_x));

    if (ret != 0) {
        secure_zero(ss, XWING_SS_BYTES);
    }

    return ret;
}

```

Unit 8.2 Summary

RUN IT:

`cd source_code/module_08_hybrid && make && ./unit_8_2_xwing_test_suite` — see `source_code/README.md` for prerequisites.

- X-Wing is the standard hybrid KEM combining X25519 with ML-KEM-768
- The construction provides IND-CCA security if either component is secure
- Key binding through including ephemeral and static public keys in the KDF prevents identity misbinding attacks
- The label `\.://"` provides domain separation from other protocols
- TLS 1.3 integration uses X-Wing as a named group for key exchange
- Performance can be improved through parallel component operations

- Proper error handling must detect low-order points in X25519

8.2.9 Alternative Hybrid KEM Constructions

Beyond X-Wing, several alternative approaches exist for constructing hybrid KEMs. Understanding these patterns is valuable for environments with different constraints or standardization requirements.

Generic Combiner Approaches:

KEM Combiner Strategies:

1. Concatenation: $ss = ss_1 || ss_2$
 - Simplest approach
 - No security proof without KDF
 - Vulnerable to related-key attacks
2. XOR: $ss = ss_1 \oplus ss_2$
 - Requires equal-length shared secrets
 - IND-CCA secure if either component is secure
 - No ciphertext binding (weaker guarantees)
3. KDF-based: $ss = \text{KDF}(ss_1 || ss_2 || \text{context})$
 - Most flexible and secure
 - Context includes ciphertexts for binding
 - Supports arbitrary component counts
 - Basis for X-Wing and IETF composite KEMs
4. Dual-PRF: $ss = \text{KDF}(ss_1 || ct_2) || \text{KDF}(ss_2 || ct_1)$
 - Provides tighter security reduction
 - More complex implementation
 - Used in some academic constructions

HPKE Integration Patterns:

Hybrid Public Key Encryption (RFC 9180) provides a framework for integrating hybrid KEMs:

HPKE with Hybrid KEM:

```
Mode: HPKE-Base with Hybrid KEM

SetupBaseS(pk_hybrid, info):
    (ss, enc) = HybridKEM.Encaps(pk_hybrid)
    ks = ExtractAndExpand(ss, enc, info)
    return (ks, enc)

KEM choices for HPKE hybrid:
- DHKEM(X25519) + ML-KEM-768 (X-Wing compatible)
- DHKEM(P-384) + ML-KEM-1024 (higher security)
- DHKEM(X448) + ML-KEM-1024 (conservative)
```

Integration points:

- Replace DHKEM with hybrid combiner
- Maintain AEAD and KDF selections
- enc = enc_classical || enc_pqc

```
/* HPKE-style hybrid KEM interface */
typedef struct {
    kem_alg_t classical; /* DHKEM(X25519), DHKEM(P-384) */
    kem_alg_t pqc; /* ML-KEM-768, ML-KEM-1024 */
    kdf_alg_t combiner_kdf; /* HKDF-SHA256, HKDF-SHA384 */
} hpke_hybrid_suite_t;

int hpke_hybrid_encap(uint8_t *ss, uint8_t *enc,
                     const hpke_hybrid_suite_t *suite,
                     const uint8_t *pk_classical,
                     const uint8_t *pk_pqc,
                     const uint8_t *info, size_t info_len)
{
    uint8_t ss_c[64], ss_p[32];
    uint8_t enc_c[suite->classical.enc_size];
    uint8_t enc_p[suite->pqc.enc_size];

    /* Encapsulate with both KEMs */
    suite->classical.encap(ss_c, enc_c, pk_classical);
    suite->pqc.encap(ss_p, enc_p, pk_pqc);

    /* Combine: enc = enc_c || enc_p */
    memcpy(enc, enc_c, suite->classical.enc_size);
    memcpy(enc + suite->classical.enc_size, enc_p, suite->pqc.enc_size);

    /* KDF combiner with HPKE-style extraction */
    size_t enc_len = suite->classical.enc_size + suite->pqc.enc_size;
    return hybrid_extract_and_expand(ss, suite->combiner_kdf,
                                    ss_c, suite->classical.ss_size,
                                    ss_p, suite->pqc.ss_size,
                                    enc, enc_len, info, info_len);
}
```

IETF Composite KEM Standardization:

The IETF is developing composite KEM standards through multiple working groups:

IETF Composite KEM Efforts:

```
draft-ietf-lamps-pq-composite-kem
├─ Defines composite KEM for X.509 and CMS
├─ OID assignments for standard combinations
├─ ASN.1 structures for composite keys/ciphertexts
├─ Combinations:
│   - id-MLKEM768-ECDH-P256
│   - id-MLKEM768-ECDH-P384
│   - id-MLKEM768-X25519
│   - id-MLKEM1024-ECDH-P384
│   - id-MLKEM1024-X448
```

```

draft-connelly-cfrg-xwing-kem
├─ X-Wing: X25519 + ML-KEM-768 specific construction
├─ Targeted at TLS 1.3 key exchange
└─ Fixed combiner with security proof

draft-ietf-tls-hybrid-design
├─ Framework for hybrid key exchange in TLS
├─ Negotiation mechanisms
└─ Backward compatibility considerations

```

Composite KEM ASN.1 Structure:

```
CompositeKEMPublicKey ::= SEQUENCE SIZE (2) OF BIT STRING
```

```
CompositeKEMPrivateKey ::= SEQUENCE SIZE (2) OF OneAsymmetricKey
```

```
CompositeKEMCiphertext ::= SEQUENCE SIZE (2) OF OCTET STRING
```

```
-- Example OIDs (subject to IANA assignment)
```

```
id-MLKEM768-X25519 OBJECT IDENTIFIER ::= { ... }
```

```
id-MLKEM1024-ECDH-P384 OBJECT IDENTIFIER ::= { ... }
```

```

/* Composite KEM key structure per IETF draft */
typedef struct {
    uint32_t oid; /* Algorithm identifier */
    uint8_t pk_classical[MAX_CLASSICAL_PK]; /* First component */
    size_t pk_classical_len;
    uint8_t pk_pqc[MAX_PQC_PK]; /* Second component */
    size_t pk_pqc_len;
} composite_kem_pk_t;

/* Standard combiner per draft-ietf-lamps-pq-composite-kem */
int composite_kem_combiner(uint8_t *ss,
                           const uint8_t *ss1, size_t ss1_len,
                           const uint8_t *ss2, size_t ss2_len,
                           const uint8_t *ct1, size_t ct1_len,
                           const uint8_t *ct2, size_t ct2_len,
                           const uint8_t *pk1, size_t pk1_len,
                           const uint8_t *pk2, size_t pk2_len,
                           uint32_t oid)
{
    /* KDF(counter || ss1 || ss2 || ct1 || ct2 || pk1 || pk2 || oid) */
    uint8_t counter[4] = {0, 0, 0, 1};
    sha3_256_ctx ctx;

    sha3_256_init(&ctx);
    sha3_256_update(&ctx, counter, 4);
    sha3_256_update(&ctx, ss1, ss1_len);
    sha3_256_update(&ctx, ss2, ss2_len);
    sha3_256_update(&ctx, ct1, ct1_len);
    sha3_256_update(&ctx, ct2, ct2_len);
    sha3_256_update(&ctx, pk1, pk1_len);

```

```

sha3_256_update(&ctx, pk2, pk2_len);
sha3_256_update(&ctx, (uint8_t*)&oid, 4);
sha3_256_final(&ctx, ss);

return 0;
}

```

Comparison of Hybrid KEM Approaches:

APPROACH	BINDING	PROOF	FLEXIBILITY	STANDARDIZATION
X-Wing	Full (ct+pk)	Tight	Fixed pair	IETF CFRG draft
IETF Composite	Full	Generic	Multiple pairs	IETF LAMPS draft
HPKE Hybrid	Via info	HPKE proof	Suite-based	RFC 9180 + extension
Simple XOR	None	Weaker	Any pair	Not recommended

What's Next: Unit 8.3 covers hybrid digital signatures, combining classical schemes like Ed25519 or ECDSA with ML-DSA or SLH-DSA for transition-safe signing.

Unit 8.3: Hybrid Signatures

Prerequisites: Unit 8.2 (Hybrid Key Exchange)

Estimated Time: 3-4 hours

Learning Objectives

After completing this unit, you will be able to:

- Implement hybrid signature schemes combining Ed25519/ECDSA with ML-DSA/SLH-DSA
- Understand verification policies (AND vs OR) and their security implications
- Design binding mechanisms for composite signatures
- Handle signature aggregation and verification ordering
- Optimize hybrid signatures for size and performance trade-offs

8.3.1 Hybrid Signature Design Goals

Unlike KEMs where both components protect a single shared secret, hybrid signatures must protect message authenticity with two independent signature schemes.

Design Requirements:

Hybrid Signature Requirements:

1. Security: Forgery requires breaking BOTH schemes
 - Even if classical is quantum-broken, still safe
 - Even if PQC has classical weakness, still safe
2. Binding: Signatures bound to same message
 - Prevent mix-and-match attacks
 - Prevent component reuse across messages
3. Non-separability: Cannot strip one component
 - Verifier must check both signatures
 - Stripping classical leaves PQC-only (weaker)
4. Efficiency: Minimize overhead
 - Parallel signing when possible
 - Batch verification for multiple signatures

Threat Model Considerations:

Attack Scenarios:

- Scenario 1: Quantum adversary breaks Ed25519
- └ Can forge Ed25519 signatures
 - └ Cannot forge ML-DSA signatures
 - └ Hybrid requires BOTH → attack fails
 - └ Protection: ML-DSA component saves the day
- Scenario 2: Classical break of ML-DSA
- └ Can forge ML-DSA signatures
 - └ Cannot forge Ed25519 signatures
 - └ Hybrid requires BOTH → attack fails
 - └ Protection: Ed25519 component saves the day
- Scenario 3: Stripping attack (malicious downgrade)
- └ Attacker removes ML-DSA component
 - └ Presents Ed25519-only signature as valid
 - └ Defense: Verifier MUST require both components
 - └ Protection: Non-separability policy

8.3.2 Parallel vs Nested Signatures

Two primary approaches exist for constructing hybrid signatures:

Parallel Signatures:

Parallel Hybrid Signature:

```
Sign(sk1, sk2, message):
    sig1 = SIG1.Sign(sk1, message)
    sig2 = SIG2.Sign(sk2, message)
    return sig1 || sig2

Verify(pk1, pk2, message, sig):
    sig1, sig2 = parse(sig)
    return SIG1.Verify(pk1, message, sig1) AND
           SIG2.Verify(pk2, message, sig2)
```

Pros:

- Simple implementation
- Parallel signing possible
- Parallel verification possible
- Independent component failures

Cons:

- No cryptographic binding between components
- Signature reuse across messages possible

Nested Signatures:

Nested Hybrid Signature:

```
Sign(sk1, sk2, message):
    sig1 = SIG1.Sign(sk1, message)
    sig2 = SIG2.Sign(sk2, message || sig1)
    return sig1 || sig2

Verify(pk1, pk2, message, sig):
    sig1, sig2 = parse(sig)
    if not SIG1.Verify(pk1, message, sig1): return false
    return SIG2.Verify(pk2, message || sig1, sig2)
```

Pros:

- Cryptographic binding (sig₂ commits to sig₁)
- Non-replayability of components
- Stronger formal security guarantees

Cons:

- Sequential signing (can't parallelize)
- Sequential verification
- PQC signs larger message (includes classical sig)

Hybrid with Composite Hash:

Composite Hash Hybrid Signature:

```
Sign(sk1, sk2, pk1, pk2, message):
    composite_pk = pk1 || pk2
    digest = HASH(composite_pk || message)
    sig1 = SIG1.Sign(sk1, digest)
    sig2 = SIG2.Sign(sk2, digest)
    return sig1 || sig2

Verify(pk1, pk2, message, sig):
    sig1, sig2 = parse(sig)
    composite_pk = pk1 || pk2
    digest = HASH(composite_pk || message)
    return SIG1.Verify(pk1, digest, sig1) AND
           SIG2.Verify(pk2, digest, sig2)

Pros:
├─ Binding through composite public key
├─ Parallel signing possible
├─ Fixed-size input to signature operations
└─ Prevents cross-key signature reuse

Choice: RECOMMENDED for most applications
```

8.3.3 Ed25519 + ML-DSA-65 Hybrid

The most common hybrid pairing combines Ed25519 (classical, fast, small) with ML-DSA-65 (PQC, NIST Level 3):

Size Analysis:

Ed25519 + ML-DSA-65 Hybrid Sizes:

Component	Ed25519	ML-DSA-65	Hybrid
Public Key	32	1,952	1,984
Secret Key	64	4,032	4,096
Signature	64	3,309	3,373
Sign Time (μs)	~25	~200	~225
Verify Time (μs)	~70	~100	~170

Overhead vs Ed25519 only:

- ├─ Public Key: +6,100% (32 → 1,984 bytes)
- ├─ Signature: +5,170% (64 → 3,373 bytes)
- └─ Verify: +143% (70 → 170 μs)

C Implementation:

```

/* hybrid_ed25519_mldsa65.h */

#ifndef HYBRID_ED25519_MLDSA65_H
#define HYBRID_ED25519_MLDSA65_H

#include <stdint.h>
#include <stddef.h>

/* Size constants */
#define ED25519_PK_SIZE      32
#define ED25519_SK_SIZE      64
#define ED25519_SIG_SIZE     64

#define MLDSA65_PK_SIZE      1952
#define MLDSA65_SK_SIZE      4032
#define MLDSA65_SIG_SIZE     3309

#define HYBRID_SIG_PK_SIZE   (ED25519_PK_SIZE + MLDSA65_PK_SIZE) /* 1984 */
#define HYBRID_SIG_SK_SIZE   (ED25519_SK_SIZE + MLDSA65_SK_SIZE) /* 4096 */
#define HYBRID_SIG_SIZE      (ED25519_SIG_SIZE + MLDSA65_SIG_SIZE) /* 3373 */

/* Hybrid key pair */
typedef struct {
    uint8_t ed25519_pk[ED25519_PK_SIZE];
    uint8_t ed25519_sk[ED25519_SK_SIZE];
    uint8_t mldsa65_pk[MLDSA65_PK_SIZE];
    uint8_t mldsa65_sk[MLDSA65_SK_SIZE];
} hybrid_sig_keypair_t;

/* Hybrid signature */
typedef struct {
    uint8_t ed25519_sig[ED25519_SIG_SIZE];
    uint8_t mldsa65_sig[MLDSA65_SIG_SIZE];
} hybrid_signature_t;

/*
 * Generate hybrid key pair
 */
int hybrid_sig_keygen(hybrid_sig_keypair_t *keypair);

/*
 * Create hybrid signature using composite hash binding
 */
int hybrid_sig_sign(
    hybrid_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const hybrid_sig_keypair_t *keypair
);

/*
 * Verify hybrid signature (AND policy)
 */
int hybrid_sig_verify(
    const hybrid_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const uint8_t *ed25519_pk,
    const uint8_t *mldsa65_pk

```

```

);

/*
 * Serialize/deserialize
 */
int hybrid_sig_pk_serialize(uint8_t *out, size_t out_len,
                           const hybrid_sig_keypair_t *keypair);
int hybrid_sig_serialize(uint8_t *out, size_t out_len,
                         const hybrid_signature_t *sig);
int hybrid_sig_deserialize(hybrid_signature_t *sig,
                          const uint8_t *in, size_t in_len);

/*
 * Secure cleanup
 */
void hybrid_sig_keypair_clear(hybrid_sig_keypair_t *keypair);

#endif /* HYBRID_ED25519_MLDSA65_H */

```

```

/* hybrid_ed25519_mldsa65.c */

#include "hybrid_ed25519_mldsa65.h"
#include <string.h>

/* External dependencies */
extern int ed25519_keygen(uint8_t *pk, uint8_t *sk);
extern int ed25519_sign(uint8_t *sig, const uint8_t *msg, size_t msg_len,
                       const uint8_t *sk);
extern int ed25519_verify(const uint8_t *sig, const uint8_t *msg,
                        size_t msg_len, const uint8_t *pk);

extern int mldsa65_keygen(uint8_t *pk, uint8_t *sk);
extern int mldsa65_sign(uint8_t *sig, size_t *sig_len, const uint8_t *msg,
                      size_t msg_len, const uint8_t *sk);
extern int mldsa65_verify(const uint8_t *sig, size_t sig_len,
                        const uint8_t *msg, size_t msg_len,
                        const uint8_t *pk);

/* SHA3-256 for composite hash */
typedef struct { uint8_t state[200]; size_t absorbed; } sha3_ctx_t;
extern void sha3_256_init(sha3_ctx_t *ctx);
extern void sha3_256_update(sha3_ctx_t *ctx, const uint8_t *in, size_t len);
extern void sha3_256_final(sha3_ctx_t *ctx, uint8_t *out);

/* Secure memory clearing */
static void secure_zero(void *ptr, size_t len) {
    volatile uint8_t *p = ptr;
    while (len--) *p++ = 0;
}

/*
 * Compute composite digest for binding
 * digest = SHA3-256(ed25519_pk || mldsa65_pk || message)
 */
static void compute_composite_digest(
    uint8_t *digest,
    const uint8_t *ed25519_pk,
    const uint8_t *mldsa65_pk,
    const uint8_t *message,

```

```

    size_t message_len
) {
    sha3_ctx_t ctx;

    sha3_256_init(&ctx);

    /* Include both public keys for binding */
    sha3_256_update(&ctx, ed25519_pk, ED25519_PK_SIZE);
    sha3_256_update(&ctx, mldsa65_pk, MLDSA65_PK_SIZE);

    /* Include message */
    sha3_256_update(&ctx, message, message_len);

    sha3_256_final(&ctx, digest);

    secure_zero(&ctx, sizeof(ctx));
}

/*
 * Generate hybrid key pair
 */
int hybrid_sig_keygen(hybrid_sig_keypair_t *keypair) {
    int ret;

    if (!keypair) {
        return -1;
    }

    /* Generate Ed25519 key pair */
    ret = ed25519_keygen(keypair->ed25519_pk, keypair->ed25519_sk);
    if (ret != 0) {
        secure_zero(keypair, sizeof(*keypair));
        return -1;
    }

    /* Generate ML-DSA-65 key pair */
    ret = mldsa65_keygen(keypair->mldsa65_pk, keypair->mldsa65_sk);
    if (ret != 0) {
        secure_zero(keypair, sizeof(*keypair));
        return -2;
    }

    return 0;
}

/*
 * Create hybrid signature with composite hash binding
 */
int hybrid_sig_sign(
    hybrid_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const hybrid_sig_keypair_t *keypair
) {
    uint8_t digest[32];
    size_t mldsa_sig_len;
    int ret;

    if (!sig || !message || !keypair) {
        return -1;
    }

```

```

}

/* Compute composite digest */
compute_composite_digest(
    digest,
    keypair->ed25519_pk,
    keypair->mldsa65_pk,
    message,
    message_len
);

/* Sign digest with Ed25519 */
ret = ed25519_sign(sig->ed25519_sig, digest, 32, keypair->ed25519_sk);
if (ret != 0) {
    secure_zero(sig, sizeof(*sig));
    secure_zero(digest, sizeof(digest));
    return -2;
}

/* Sign digest with ML-DSA-65 */
ret = mldsa65_sign(sig->mldsa65_sig, &mldsa_sig_len, digest, 32,
    keypair->mldsa65_sk);
if (ret != 0 || mldsa_sig_len != MLDSA65_SIG_SIZE) {
    secure_zero(sig, sizeof(*sig));
    secure_zero(digest, sizeof(digest));
    return -3;
}

secure_zero(digest, sizeof(digest));
return 0;
}

/*
 * Verify hybrid signature (AND policy - both must verify)
 */
int hybrid_sig_verify(
    const hybrid_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const uint8_t *ed25519_pk,
    const uint8_t *mldsa65_pk
) {
    uint8_t digest[32];
    int ed25519_valid, mldsa65_valid;

    if (!sig || !message || !ed25519_pk || !mldsa65_pk) {
        return -1;
    }

    /* Compute composite digest */
    compute_composite_digest(digest, ed25519_pk, mldsa65_pk,
        message, message_len);

    /* Verify Ed25519 component */
    ed25519_valid = ed25519_verify(sig->ed25519_sig, digest, 32, ed25519_pk);

    /* Verify ML-DSA-65 component */
    mldsa65_valid = mldsa65_verify(sig->mldsa65_sig, MLDSA65_SIG_SIZE,
        digest, 32, mldsa65_pk);

```

```

    secure_zero(digest, sizeof(digest));

    /* AND policy: both must succeed */
    if (ed25519_valid != 0 || mldsa65_valid != 0) {
        return -1;
    }

    return 0;
}

/*
 * Serialize public key
 */
int hybrid_sig_pk_serialize(
    uint8_t *out,
    size_t out_len,
    const hybrid_sig_keypair_t *keypair
) {
    if (!out || !keypair || out_len < HYBRID_SIG_PK_SIZE) {
        return -1;
    }

    memcpy(out, keypair->ed25519_pk, ED25519_PK_SIZE);
    memcpy(out + ED25519_PK_SIZE, keypair->mldsa65_pk, MLDSA65_PK_SIZE);

    return HYBRID_SIG_PK_SIZE;
}

/*
 * Serialize signature
 */
int hybrid_sig_serialize(
    uint8_t *out,
    size_t out_len,
    const hybrid_signature_t *sig
) {
    if (!out || !sig || out_len < HYBRID_SIG_SIZE) {
        return -1;
    }

    memcpy(out, sig->ed25519_sig, ED25519_SIG_SIZE);
    memcpy(out + ED25519_SIG_SIZE, sig->mldsa65_sig, MLDSA65_SIG_SIZE);

    return HYBRID_SIG_SIZE;
}

/*
 * Deserialize signature
 */
int hybrid_sig_deserialize(
    hybrid_signature_t *sig,
    const uint8_t *in,
    size_t in_len
) {
    if (!sig || !in || in_len < HYBRID_SIG_SIZE) {
        return -1;
    }

    memcpy(sig->ed25519_sig, in, ED25519_SIG_SIZE);
    memcpy(sig->mldsa65_sig, in + ED25519_SIG_SIZE, MLDSA65_SIG_SIZE);
}

```

```

        return 0;
    }

    /*
     * Secure cleanup
     */
    void hybrid_sig_keypair_clear(hybrid_sig_keypair_t *keypair) {
        if (keypair) {
            secure_zero(keypair->ed25519_sk, ED25519_SK_SIZE);
            secure_zero(keypair->mldsa65_sk, MLDSA65_SK_SIZE);
            secure_zero(keypair->ed25519_pk, ED25519_PK_SIZE);
            secure_zero(keypair->mldsa65_pk, MLDSA65_PK_SIZE);
        }
    }
}

```

8.3.4 ECDSA + SLH-DSA Hybrid

For applications requiring NIST P-256 compatibility or maximum signature diversity:

Size Analysis:

ECDSA-P256 + SLH-DSA-128f Hybrid:

Component	ECDSA-P256	SLH-DSA-128f	Hybrid
Public Key	64	32	96
Secret Key	32	64	96
Signature	64	17,088	17,152
Sign Time (ms)	~0.1	~15	~15
Verify Time (ms)	~0.3	~6	~6

Note: SLH-DSA dominates size and time
ECDSA component adds negligible overhead

Implementation:

```

/* ecdsa_slhdsa_hybrid.h */

#ifndef ECDSA_SLHDSA_HYBRID_H
#define ECDSA_SLHDSA_HYBRID_H

#include <stdint.h>
#include <stddef.h>

/* ECDSA-P256 sizes */
#define ECDSA_P256_PK_SIZE    64    /* Uncompressed point (no 0x04 prefix) */
#define ECDSA_P256_SK_SIZE    32
#define ECDSA_P256_SIG_SIZE   64    /* r || s, each 32 bytes */

/* SLH-DSA-128f sizes */
#define SLHDSA128F_PK_SIZE    32

```



```

#define SLHDSA128F_SK_SIZE      64
#define SLHDSA128F_SIG_SIZE     17088

/* Hybrid sizes */
#define ECDSA_SLHDSA_PK_SIZE    (ECDSA_P256_PK_SIZE + SLHDSA128F_PK_SIZE)
#define ECDSA_SLHDSA_SK_SIZE    (ECDSA_P256_SK_SIZE + SLHDSA128F_SK_SIZE)
#define ECDSA_SLHDSA_SIG_SIZE   (ECDSA_P256_SIG_SIZE + SLHDSA128F_SIG_SIZE)

typedef struct {
    uint8_t ecdsa_pk[ECDSA_P256_PK_SIZE];
    uint8_t ecdsa_sk[ECDSA_P256_SK_SIZE];
    uint8_t slhdsa_pk[SLHDSA128F_PK_SIZE];
    uint8_t slhdsa_sk[SLHDSA128F_SK_SIZE];
} ecdsa_slhdsa_keypair_t;

typedef struct {
    uint8_t ecdsa_sig[ECDSA_P256_SIG_SIZE];
    uint8_t slhdsa_sig[SLHDSA128F_SIG_SIZE];
} ecdsa_slhdsa_signature_t;

int ecdsa_slhdsa_keygen(ecdsa_slhdsa_keypair_t *keypair);

int ecdsa_slhdsa_sign(
    ecdsa_slhdsa_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const ecdsa_slhdsa_keypair_t *keypair
);

int ecdsa_slhdsa_verify(
    const ecdsa_slhdsa_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const uint8_t *ecdsa_pk,
    const uint8_t *slhdsa_pk
);

#endif

```

```

/* ecdsa_slhdsa_hybrid.c - Parallel signing for performance */

#include "ecdsa_slhdsa_hybrid.h"
#include <string.h>
#include <pthread.h>

/* External dependencies */
extern int ecdsa_p256_keygen(uint8_t *pk, uint8_t *sk);
extern int ecdsa_p256_sign(uint8_t *sig, const uint8_t *digest,
                           const uint8_t *sk);
extern int ecdsa_p256_verify(const uint8_t *sig, const uint8_t *digest,
                             const uint8_t *pk);

extern int slhdsa128f_keygen(uint8_t *pk, uint8_t *sk);
extern int slhdsa128f_sign(uint8_t *sig, const uint8_t *msg, size_t msg_len,
                           const uint8_t *sk);
extern int slhdsa128f_verify(const uint8_t *sig, const uint8_t *msg,
                             size_t msg_len, const uint8_t *pk);

extern void sha256(uint8_t *out, const uint8_t *in, size_t len);

```

```

static void secure_zero(void *ptr, size_t len) {
    volatile uint8_t *p = ptr;
    while (len--) *p++ = 0;
}

/* Thread argument for parallel signing */
typedef struct {
    uint8_t *sig;
    const uint8_t *digest;
    const uint8_t *sk;
    int result;
} sign_thread_arg_t;

static void *ecdsa_sign_thread(void *arg) {
    sign_thread_arg_t *a = (sign_thread_arg_t *)arg;
    a->result = ecdsa_p256_sign(a->sig, a->digest, a->sk);
    return NULL;
}

/*
 * Compute binding digest
 */
static void compute_digest(
    uint8_t *digest,
    const uint8_t *ecdsa_pk,
    const uint8_t *slhdsa_pk,
    const uint8_t *message,
    size_t message_len
) {
    /* Simple concatenation + hash for binding */
    uint8_t *buf = malloc(ECDSA_P256_PK_SIZE + SLHDSA128F_PK_SIZE + message_len);
    if (!buf) {
        memset(digest, 0, 32);
        return;
    }

    memcpy(buf, ecdsa_pk, ECDSA_P256_PK_SIZE);
    memcpy(buf + ECDSA_P256_PK_SIZE, slhdsa_pk, SLHDSA128F_PK_SIZE);
    memcpy(buf + ECDSA_P256_PK_SIZE + SLHDSA128F_PK_SIZE, message, message_len);

    sha256(digest, buf, ECDSA_P256_PK_SIZE + SLHDSA128F_PK_SIZE + message_len);

    free(buf);
}

int ecdsa_slhdsa_keygen(ecdsa_slhdsa_keypair_t *keypair) {
    if (!keypair) return -1;

    if (ecdsa_p256_keygen(keypair->ecdsa_pk, keypair->ecdsa_sk) != 0) {
        return -1;
    }

    if (slhdsa128f_keygen(keypair->slhdsa_pk, keypair->slhdsa_sk) != 0) {
        secure_zero(keypair, sizeof(*keypair));
        return -2;
    }

    return 0;
}

```

```

/*
 * Parallel hybrid signing
 *
 * SLH-DSA takes ~15ms, ECDSA takes ~0.1ms
 * Running in parallel saves ~0.1ms per signature
 */
int ecdsa_slhdsa_sign(
    ecdsa_slhdsa_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const ecdsa_slhdsa_keypair_t *keypair
) {
    uint8_t digest[32];
    pthread_t ecdsa_thread;
    sign_thread_arg_t ecdsa_arg;
    int slhdsa_ret;

    if (!sig || !message || !keypair) return -1;

    /* Compute binding digest */
    compute_digest(digest, keypair->ecdsa_pk, keypair->slhdsa_pk,
        message, message_len);

    /* Start ECDSA signing in background thread */
    ecdsa_arg.sig = sig->ecdsa_sig;
    ecdsa_arg.digest = digest;
    ecdsa_arg.sk = keypair->ecdsa_sk;
    ecdsa_arg.result = 0;

    if (pthread_create(&ecdsa_thread, NULL, ecdsa_sign_thread, &ecdsa_arg) != 0) {
        /* Fallback to sequential */
        if (ecdsa_p256_sign(sig->ecdsa_sig, digest, keypair->ecdsa_sk) != 0) {
            return -2;
        }
        if (slhdsa128f_sign(sig->slhdsa_sig, digest, 32,
            keypair->slhdsa_sk) != 0) {
            return -3;
        }
        return 0;
    }

    /* SLH-DSA signing in main thread (this is the slow one) */
    slhdsa_ret = slhdsa128f_sign(sig->slhdsa_sig, digest, 32,
        keypair->slhdsa_sk);

    /* Wait for ECDSA */
    pthread_join(ecdsa_thread, NULL);

    secure_zero(digest, sizeof(digest));

    if (ecdsa_arg.result != 0) return -2;
    if (slhdsa_ret != 0) return -3;

    return 0;
}

int ecdsa_slhdsa_verify(
    const ecdsa_slhdsa_signature_t *sig,
    const uint8_t *message,

```

```

    size_t message_len,
    const uint8_t *ecdsa_pk,
    const uint8_t *slhdsa_pk
) {
    uint8_t digest[32];
    int ecdsa_valid, slhdsa_valid;

    if (!sig || !message || !ecdsa_pk || !slhdsa_pk) return -1;

    compute_digest(digest, ecdsa_pk, slhdsa_pk, message, message_len);

    /* Both must verify */
    ecdsa_valid = ecdsa_p256_verify(sig->ecdsa_sig, digest, ecdsa_pk);
    slhdsa_valid = slhdsa128f_verify(sig->slhdsa_sig, digest, 32, slhdsa_pk);

    secure_zero(digest, sizeof(digest));

    if (ecdsa_valid != 0 || slhdsa_valid != 0) return -1;

    return 0;
}

```

8.3.5 Verification Policies

Different applications may require different verification semantics:

AND Policy (Recommended):

```

/*
 * AND Policy: Both signatures must verify
 *
 * Security: Forgery requires breaking BOTH schemes
 * Availability: Either component failing causes rejection
 * Use case: Maximum security, both threats equally weighted
 */
int hybrid_verify_and(
    const hybrid_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const uint8_t *pk1,
    const uint8_t *pk2
) {
    uint8_t digest[32];
    compute_digest(digest, pk1, pk2, message, message_len);

    int sig1_valid = sig1_verify(sig->sig1, digest, pk1);
    int sig2_valid = sig2_verify(sig->sig2, digest, pk2);

    /* AND: both must succeed */
    return (sig1_valid == 0 && sig2_valid == 0) ? 0 : -1;
}

```

OR Policy (Transition Only):

```

/*
 * OR Policy: Either signature verifying is sufficient
 *
 * Security: Only as strong as weakest component!
 * Availability: Either component succeeding causes acceptance
 * Use case: ONLY during migration when some verifiers lack PQC support
 *
 * WARNING: This defeats hybrid security benefits!
 */
int hybrid_verify_or(
    const hybrid_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const uint8_t *pk1,
    const uint8_t *pk2
) {
    uint8_t digest[32];
    compute_digest(digest, pk1, pk2, message, message_len);

    int sig1_valid = sig1_verify(sig->sig1, digest, pk1);
    int sig2_valid = sig2_verify(sig->sig2, digest, pk2);

    /* OR: either succeeding is enough */
    /* WARNING: Provides no hybrid security benefit! */
    return (sig1_valid == 0 || sig2_valid == 0) ? 0 : -1;
}

```

Progressive Policy (Migration):

```

/*
 * Progressive Policy: AND for PQC-capable, classical-only for legacy
 *
 * Allows gradual migration where:
 * - PQC-capable verifiers check both signatures
 * - Legacy verifiers only check classical
 *
 * Use case: Transition period with mixed verifier capabilities
 */
typedef enum {
    VERIFY_MODE_FULL_HYBRID,    /* Check both (PQC-capable) */
    VERIFY_MODE_CLASSICAL_ONLY /* Check classical only (legacy) */
} verify_mode_t;

int hybrid_verify_progressive(
    const hybrid_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const uint8_t *pk1,
    const uint8_t *pk2,
    verify_mode_t mode
) {
    uint8_t digest[32];
    compute_digest(digest, pk1, pk2, message, message_len);

    /* Always verify classical component */
    int sig1_valid = sig1_verify(sig->sig1, digest, pk1);
    if (sig1_valid != 0) {
        return -1;
    }
}

```

```

    }

    /* Full hybrid mode also verifies PQC component */
    if (mode == VERIFY_MODE_FULL_HYBRID) {
        int sig2_valid = sig2_verify(sig->sig2, digest, pk2);
        if (sig2_valid != 0) {
            return -1;
        }
    }

    return 0;
}

```

8.3.6 Batch Verification

For scenarios with many signatures to verify:

```

/*
 * Batch verification for hybrid signatures
 *
 * Some signature schemes support batch verification where
 * verifying N signatures together is faster than N individual
 * verifications. This implements batching for hybrid signatures.
 */

typedef struct {
    const hybrid_signature_t *sig;
    const uint8_t *message;
    size_t message_len;
    const uint8_t *ed25519_pk;
    const uint8_t *mlsa65_pk;
} batch_verify_item_t;

/*
 * Batch verify multiple hybrid signatures
 *
 * Returns 0 if ALL signatures are valid
 * Returns index+1 of first invalid signature on failure
 */
int hybrid_batch_verify(
    const batch_verify_item_t *items,
    size_t count
) {
    /* Pre-compute all digests */
    uint8_t (*digests)[32] = malloc(count * 32);
    if (!digests) return -1;

    for (size_t i = 0; i < count; i++) {
        compute_composite_digest(
            digests[i],
            items[i].ed25519_pk,
            items[i].mlsa65_pk,
            items[i].message,
            items[i].message_len
        );
    }
}

```

```

/*
 * Batch verify Ed25519 signatures
 * Ed25519 supports efficient batch verification using
 * randomized linear combinations
 */
int ed25519_batch_valid = ed25519_batch_verify(
    items, count, digests
);

if (ed25519_batch_valid != 0) {
    /* Find which signature failed */
    for (size_t i = 0; i < count; i++) {
        if (ed25519_verify(items[i].sig->ed25519_sig,
                           digests[i], 32,
                           items[i].ed25519_pk) != 0) {
            free(digests);
            return (int)(i + 1);
        }
    }
}

/*
 * Verify ML-DSA signatures individually
 * ML-DSA doesn't support batch verification
 */
for (size_t i = 0; i < count; i++) {
    if (mldsa65_verify(items[i].sig->mldsa65_sig,
                       MLDSA65_SIG_SIZE,
                       digests[i], 32,
                       items[i].mldsa65_pk) != 0) {
        free(digests);
        return (int)(i + 1);
    }
}

free(digests);
return 0;
}

```

8.3.7 Testing Hybrid Signatures

```

/* Test suite for hybrid signatures */

#include <stdio.h>
#include <string.h>
#include <assert.h>

int test_hybrid_basic(void) {
    hybrid_sig_keypair_t keypair;
    hybrid_signature_t sig;
    uint8_t message[] = "Test message for hybrid signature";

    /* Generate keys */
    assert(hybrid_sig_keygen(&keypair) == 0);
}

```

```

/* Sign */
assert(hybrid_sig_sign(&sig, message, sizeof(message) - 1, &keypair) == 0);

/* Verify */
assert(hybrid_sig_verify(&sig, message, sizeof(message) - 1,
                        keypair.ed25519_pk, keypair.mldsa65_pk) == 0);

/* Cleanup */
hybrid_sig_keypair_clear(&keypair);

printf("test_hybrid_basic: PASSED\n");
return 0;
}

int test_hybrid_wrong_message(void) {
    hybrid_sig_keypair_t keypair;
    hybrid_signature_t sig;
    uint8_t message[] = "Original message";
    uint8_t wrong_message[] = "Wrong message";

    hybrid_sig_keygen(&keypair);
    hybrid_sig_sign(&sig, message, sizeof(message) - 1, &keypair);

    /* Verification with wrong message should fail */
    assert(hybrid_sig_verify(&sig, wrong_message, sizeof(wrong_message) - 1,
                            keypair.ed25519_pk, keypair.mldsa65_pk) != 0);

    hybrid_sig_keypair_clear(&keypair);

    printf("test_hybrid_wrong_message: PASSED\n");
    return 0;
}

int test_hybrid_wrong_key(void) {
    hybrid_sig_keypair_t keypair1, keypair2;
    hybrid_signature_t sig;
    uint8_t message[] = "Test message";

    hybrid_sig_keygen(&keypair1);
    hybrid_sig_keygen(&keypair2);

    hybrid_sig_sign(&sig, message, sizeof(message) - 1, &keypair1);

    /* Verification with wrong keys should fail */
    assert(hybrid_sig_verify(&sig, message, sizeof(message) - 1,
                            keypair2.ed25519_pk, keypair2.mldsa65_pk) != 0);

    /* Mixed keys should also fail (binding test) */
    assert(hybrid_sig_verify(&sig, message, sizeof(message) - 1,
                            keypair1.ed25519_pk, keypair2.mldsa65_pk) != 0);
    assert(hybrid_sig_verify(&sig, message, sizeof(message) - 1,
                            keypair2.ed25519_pk, keypair1.mldsa65_pk) != 0);

    hybrid_sig_keypair_clear(&keypair1);
    hybrid_sig_keypair_clear(&keypair2);

    printf("test_hybrid_wrong_key: PASSED\n");
    return 0;
}

```



```

int test_hybrid_modified_signature(void) {
    hybrid_sig_keypair_t keypair;
    hybrid_signature_t sig;
    uint8_t message[] = "Test message";

    hybrid_sig_keygen(&keypair);
    hybrid_sig_sign(&sig, message, sizeof(message) - 1, &keypair);

    /* Modify Ed25519 component */
    sig.ed25519_sig[0] ^= 0x01;
    assert(hybrid_sig_verify(&sig, message, sizeof(message) - 1,
                             keypair.ed25519_pk, keypair.mldsa65_pk) != 0);
    sig.ed25519_sig[0] ^= 0x01; /* Restore */

    /* Modify ML-DSA component */
    sig.mldsa65_sig[100] ^= 0x01;
    assert(hybrid_sig_verify(&sig, message, sizeof(message) - 1,
                             keypair.ed25519_pk, keypair.mldsa65_pk) != 0);

    hybrid_sig_keypair_clear(&keypair);

    printf("test_hybrid_modified_signature: PASSED\n");
    return 0;
}

int test_hybrid_serialization(void) {
    hybrid_sig_keypair_t keypair;
    hybrid_signature_t sig, sig_restored;
    uint8_t message[] = "Test message";
    uint8_t pk_buf[HYBRID_SIG_PK_SIZE];
    uint8_t sig_buf[HYBRID_SIG_SIZE];

    hybrid_sig_keygen(&keypair);
    hybrid_sig_sign(&sig, message, sizeof(message) - 1, &keypair);

    /* Serialize and deserialize public key */
    assert(hybrid_sig_pk_serialize(pk_buf, sizeof(pk_buf), &keypair) ==
           HYBRID_SIG_PK_SIZE);

    /* Serialize and deserialize signature */
    assert(hybrid_sig_serialize(sig_buf, sizeof(sig_buf), &sig) ==
           HYBRID_SIG_SIZE);
    assert(hybrid_sig_deserialize(&sig_restored, sig_buf, sizeof(sig_buf)) == 0);

    /* Verify with deserialized signature */
    assert(hybrid_sig_verify(&sig_restored, message, sizeof(message) - 1,
                             keypair.ed25519_pk, keypair.mldsa65_pk) == 0);

    hybrid_sig_keypair_clear(&keypair);

    printf("test_hybrid_serialization: PASSED\n");
    return 0;
}

int main(void) {
    test_hybrid_basic();
    test_hybrid_wrong_message();
}

```

```
test_hybrid_wrong_key();
test_hybrid_modified_signature();
test_hybrid_serialization();
printf("\nAll hybrid signature tests passed!\n");
return 0;
}
```

Exercises

Exercise 8.3.1: (*Intermediate*) Explain why the composite hash approach (signing $\text{HASH}(\text{pk1} \parallel \text{pk2} \parallel \text{message})$) prevents signature component reuse across different key pairs.

Exercise 8.3.2: (*Intermediate*) Design a hybrid signature scheme where the PQC component is optional—the signature can contain just the classical component, but verification policy requires PQC if present. What are the security implications?

Exercise 8.3.3: (*Advanced*) Implement parallel verification for ECDSA + SLH-DSA hybrid signatures using pthreads.

Exercise 8.3.4: (*Intermediate*) A system requires both Ed25519+ML-DSA-65 and ECDSA+SLH-DSA hybrids. Design a unified API that handles both while maintaining type safety.

Solutions

Solution 8.3.1:

The composite hash approach prevents signature component reuse because:

1. **Key binding in digest:** The digest includes both public keys:

```
digest = HASH(pk1 || pk2 || message)
```

A signature valid for $(\text{pk1_A}, \text{pk2_A}, \text{message})$ requires:

```
sig1 valid for pk1_A over HASH(pk1_A || pk2_A || message)
sig2 valid for pk2_A over HASH(pk1_A || pk2_A || message)
```

2. **Different keys → different digest:** If attacker tries to use sig1 with different pk1_B:

```
HASH(pk1_B || pk2_A || message) ≠ HASH(pk1_A || pk2_A || message)
```

The digest changes, so sig1 (created for the original digest) won't verify.

3. **Cross-key reuse blocked:** Even with same message:

- Signature from keypair A: signs $\text{HASH}(\text{pk1_A} || \text{pk2_A} || \text{msg})$
- Cannot verify with keypair B: expects $\text{HASH}(\text{pk1_B} || \text{pk2_B} || \text{msg})$
- Digests differ because public keys differ

4. **Attack prevention:** Without this binding, an attacker could:

- Get hybrid signature from victim A on message M
- Extract classical component sig1
- Use sig1 with attacker's PQC keypair
- Claim message M was signed by hybrid (victim_classical, attacker_pqc)

Solution 8.3.2:

```
/*
 * Optional PQC Hybrid Signature
 *
 * Security implications:
 * - If PQC missing: Security = classical only (quantum vulnerable)
 * - If PQC present: Security = hybrid (requires both)
 * - Downgrade attack: Attacker could strip PQC component
 * - Mitigation: Policy must be verifier-enforced, not signature-controlled
 */

#define SIG_FLAG_HAS_PQC    0x01

typedef struct {
    uint8_t flags;                /* Indicates presence of PQC */
    uint8_t classical_sig[ED25519_SIG_SIZE];
    uint8_t pqc_sig[MLDSA65_SIG_SIZE]; /* May be zeroed if not present */
} optional_hybrid_sig_t;

/* Signing: always include both if both keys available */
int optional_hybrid_sign(
    optional_hybrid_sig_t *sig,
    const uint8_t *message,
    size_t message_len,
    const uint8_t *classical_sk,
    const uint8_t *pqc_sk,        /* Can be NULL */
    const uint8_t *classical_pk,
    const uint8_t *pqc_pk        /* Can be NULL */
) {
    uint8_t digest[32];
```

```

    if (pqc_sk && pqc_pk) {
        /* Full hybrid signing */
        sig->flags = SIG_FLAG_HAS_PQC;
        compute_digest(digest, classical_pk, pqc_pk, message, message_len);

        ed25519_sign(sig->classical_sig, digest, 32, classical_sk);
        mldsa65_sign(sig->pqc_sig, digest, 32, pqc_sk);
    } else {
        /* Classical only - no PQC component */
        sig->flags = 0;
        sha256(digest, message, message_len); /* Different digest! */
        ed25519_sign(sig->classical_sig, digest, 32, classical_sk);
        memset(sig->pqc_sig, 0, MLDSA65_SIG_SIZE);
    }

    return 0;
}

/* Verification: policy determines requirements */
typedef enum {
    POLICY_CLASSICAL_ONLY,      /* Accept classical only */
    POLICY_PREFER_HYBRID,       /* Check PQC if present, but accept classical */
    POLICY_REQUIRE_HYBRID       /* Reject if PQC missing */
} verify_policy_t;

int optional_hybrid_verify(
    const optional_hybrid_sig_t *sig,
    const uint8_t *message,
    size_t message_len,
    const uint8_t *classical_pk,
    const uint8_t *pqc_pk,
    verify_policy_t policy
) {
    uint8_t digest[32];
    int has_pqc = (sig->flags & SIG_FLAG_HAS_PQC) != 0;

    /* Policy: REQUIRE_HYBRID rejects missing PQC */
    if (policy == POLICY_REQUIRE_HYBRID && !has_pqc) {
        return -1; /* Missing required PQC component */
    }

    if (has_pqc && pqc_pk) {
        /* Verify as full hybrid */
        compute_digest(digest, classical_pk, pqc_pk, message, message_len);

        if (ed25519_verify(sig->classical_sig, digest, 32, classical_pk) != 0) {
            return -1;
        }
        if (mldsa65_verify(sig->pqc_sig, digest, 32, pqc_pk) != 0) {
            return -1;
        }
    } else {
        /* Verify classical only */
        sha256(digest, message, message_len);
        if (ed25519_verify(sig->classical_sig, digest, 32, classical_pk) != 0) {
            return -1;
        }
    }

    return 0;
}

```

```

}

/*
 * Security Analysis:
 *
 * POLICY_CLASSICAL_ONLY:
 * - No hybrid benefit
 * - Quantum vulnerable
 * - Use: Legacy systems with no upgrade path
 *
 * POLICY_PREFER_HYBRID:
 * - Hybrid if both present, classical fallback
 * - Vulnerable to downgrade attack!
 * - Attacker can strip PQC component from signature
 * - Use: ONLY during transition when some signers can't do PQC
 *
 * POLICY_REQUIRE_HYBRID:
 * - Full hybrid security
 * - Rejects signatures without PQC
 * - Use: Production systems where all signers support PQC
 *
 * RECOMMENDATION: Use REQUIRE_HYBRID unless specific transition needs
 */

```

Solution 8.3.3:

```

/* Parallel verification for ECDSA + SLH-DSA */

#include <pthread.h>

typedef struct {
    const uint8_t *sig;
    const uint8_t *digest;
    const uint8_t *pk;
    int result;
} verify_thread_arg_t;

static void *ecdsa_verify_thread(void *arg) {
    verify_thread_arg_t *a = (verify_thread_arg_t *)arg;
    a->result = ecdsa_p256_verify(a->sig, a->digest, a->pk);
    return NULL;
}

static void *slhdsa_verify_thread(void *arg) {
    verify_thread_arg_t *a = (verify_thread_arg_t *)arg;
    a->result = slhdsa128f_verify(a->sig, a->digest, 32, a->pk);
    return NULL;
}

/*
 * Parallel hybrid verification
 *
 * Runs ECDSA and SLH-DSA verification concurrently.
 * Since SLH-DSA takes ~6ms and ECDSA takes ~0.3ms,
 * parallel verification reduces total time from ~6.3ms to ~6ms.
 */
int ecdsa_slhdsa_verify_parallel(
    const ecdsa_slhdsa_signature_t *sig,

```

```

const uint8_t *message,
size_t message_len,
const uint8_t *ecdsa_pk,
const uint8_t *slhdsa_pk
) {
    uint8_t digest[32];
    pthread_t ecdsa_thread, slhdsa_thread;
    verify_thread_arg_t ecdsa_arg, slhdsa_arg;

    if (!sig || !message || !ecdsa_pk || !slhdsa_pk) {
        return -1;
    }

    /* Compute binding digest */
    compute_digest(digest, ecdsa_pk, slhdsa_pk, message, message_len);

    /* Set up thread arguments */
    ecdsa_arg.sig = sig->ecdsa_sig;
    ecdsa_arg.digest = digest;
    ecdsa_arg.pk = ecdsa_pk;
    ecdsa_arg.result = 0;

    slhdsa_arg.sig = sig->slhdsa_sig;
    slhdsa_arg.digest = digest;
    slhdsa_arg.pk = slhdsa_pk;
    slhdsa_arg.result = 0;

    /* Start both verification threads */
    int ecdsa_started = (pthread_create(&ecdsa_thread, NULL,
                                         ecdsa_verify_thread, &ecdsa_arg) == 0);
    int slhdsa_started = (pthread_create(&slhdsa_thread, NULL,
                                         slhdsa_verify_thread, &slhdsa_arg) == 0);

    /* If threading failed, fall back to sequential */
    if (!ecdsa_started && !slhdsa_started) {
        ecdsa_arg.result = ecdsa_p256_verify(sig->ecdsa_sig, digest, ecdsa_pk);
        slhdsa_arg.result = slhdsa128f_verify(sig->slhdsa_sig, digest, 32,
                                              slhdsa_pk);
    } else {
        /* Wait for threads */
        if (ecdsa_started) pthread_join(ecdsa_thread, NULL);
        else ecdsa_arg.result = ecdsa_p256_verify(sig->ecdsa_sig, digest,
                                                  ecdsa_pk);

        if (slhdsa_started) pthread_join(slhdsa_thread, NULL);
        else slhdsa_arg.result = slhdsa128f_verify(sig->slhdsa_sig, digest, 32,
                                                  slhdsa_pk);
    }

    secure_zero(digest, sizeof(digest));

    /* AND policy: both must succeed */
    if (ecdsa_arg.result != 0 || slhdsa_arg.result != 0) {
        return -1;
    }

    return 0;
}

```

Solution 8.3.4:

```
/* Unified hybrid signature API */

#ifndef UNIFIED_HYBRID_SIG_H
#define UNIFIED_HYBRID_SIG_H

#include <stdint.h>
#include <stddef.h>

/* Algorithm identifiers */
typedef enum {
    HYBRID_ALG_ED25519_MLDSA65,
    HYBRID_ALG_ECDSA_SLHDSA128F
} hybrid_sig_algorithm_t;

/* Maximum sizes across all supported algorithms */
#define UNIFIED_MAX_PK_SIZE    2048
#define UNIFIED_MAX_SK_SIZE    4096
#define UNIFIED_MAX_SIG_SIZE   18000

/* Opaque key and signature handles */
typedef struct {
    hybrid_sig_algorithm_t algorithm;
    uint8_t classical_pk[UNIFIED_MAX_PK_SIZE / 2];
    uint8_t classical_sk[UNIFIED_MAX_SK_SIZE / 2];
    uint8_t pqc_pk[UNIFIED_MAX_PK_SIZE / 2];
    uint8_t pqc_sk[UNIFIED_MAX_SK_SIZE / 2];
    size_t classical_pk_len;
    size_t classical_sk_len;
    size_t pqc_pk_len;
    size_t pqc_sk_len;
} unified_keypair_t;

typedef struct {
    hybrid_sig_algorithm_t algorithm;
    uint8_t classical_sig[UNIFIED_MAX_SIG_SIZE / 2];
    uint8_t pqc_sig[UNIFIED_MAX_SIG_SIZE / 2];
    size_t classical_sig_len;
    size_t pqc_sig_len;
} unified_signature_t;

/* Algorithm-specific size getters */
size_t unified_pk_size(hybrid_sig_algorithm_t alg);
size_t unified_sk_size(hybrid_sig_algorithm_t alg);
size_t unified_sig_size(hybrid_sig_algorithm_t alg);

/* Unified API */
int unified_keygen(unified_keypair_t *keypair, hybrid_sig_algorithm_t alg);

int unified_sign(
    unified_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const unified_keypair_t *keypair
);

int unified_verify(
    const unified_signature_t *sig,
```

```

        const uint8_t *message,
        size_t message_len,
        const unified_keypair_t *keypair
    );

    /* Serialization */
    int unified_pk_serialize(uint8_t *out, size_t out_len,
                            const unified_keypair_t *keypair);
    int unified_sig_serialize(uint8_t *out, size_t out_len,
                             const unified_signature_t *sig);
    int unified_sig_deserialize(unified_signature_t *sig,
                               const uint8_t *in, size_t in_len,
                               hybrid_sig_algorithm_t alg);

    void unified_keypair_clear(unified_keypair_t *keypair);

#endif /* UNIFIED_HYBRID_SIG_H */

```

```

/* unified_hybrid_sig.c - Implementation */

#include "unified_hybrid_sig.h"
#include <string.h>

/* Include algorithm-specific implementations */
#include "hybrid_ed25519_mldsa65.h"
#include "ecdsa_slhdsa_hybrid.h"

static void secure_zero(void *ptr, size_t len) {
    volatile uint8_t *p = ptr;
    while (len--) *p++ = 0;
}

size_t unified_pk_size(hybrid_sig_algorithm_t alg) {
    switch (alg) {
        case HYBRID_ALG_ED25519_MLDSA65:
            return HYBRID_SIG_PK_SIZE;
        case HYBRID_ALG_ECDSA_SLHDSA128F:
            return ECDSA_SLHDSA_PK_SIZE;
        default:
            return 0;
    }
}

size_t unified_sk_size(hybrid_sig_algorithm_t alg) {
    switch (alg) {
        case HYBRID_ALG_ED25519_MLDSA65:
            return HYBRID_SIG_SK_SIZE;
        case HYBRID_ALG_ECDSA_SLHDSA128F:
            return ECDSA_SLHDSA_SK_SIZE;
        default:
            return 0;
    }
}

size_t unified_sig_size(hybrid_sig_algorithm_t alg) {
    switch (alg) {
        case HYBRID_ALG_ED25519_MLDSA65:
            return HYBRID_SIG_SIZE;
        case HYBRID_ALG_ECDSA_SLHDSA128F:

```



```

        return ECDSA_SLHDSA_SIG_SIZE;
    default:
        return 0;
    }
}

int unified_keygen(unified_keypair_t *keypair, hybrid_sig_algorithm_t alg) {
    if (!keypair) return -1;

    keypair->algorithm = alg;

    switch (alg) {
        case HYBRID_ALG_ED25519_MLDSA65: {
            hybrid_sig_keypair_t kp;
            int ret = hybrid_sig_keygen(&kp);
            if (ret != 0) return ret;

            memcpy(keypair->classical_pk, kp.ed25519_pk, ED25519_PK_SIZE);
            memcpy(keypair->classical_sk, kp.ed25519_sk, ED25519_SK_SIZE);
            memcpy(keypair->pqc_pk, kp.mldsa65_pk, MLDSA65_PK_SIZE);
            memcpy(keypair->pqc_sk, kp.mldsa65_sk, MLDSA65_SK_SIZE);

            keypair->classical_pk_len = ED25519_PK_SIZE;
            keypair->classical_sk_len = ED25519_SK_SIZE;
            keypair->pqc_pk_len = MLDSA65_PK_SIZE;
            keypair->pqc_sk_len = MLDSA65_SK_SIZE;

            hybrid_sig_keypair_clear(&kp);
            return 0;
        }

        case HYBRID_ALG_ECDSA_SLHDSA128F: {
            ecdsa_slhdsa_keypair_t kp;
            int ret = ecdsa_slhdsa_keygen(&kp);
            if (ret != 0) return ret;

            memcpy(keypair->classical_pk, kp.ecdsa_pk, ECDSA_P256_PK_SIZE);
            memcpy(keypair->classical_sk, kp.ecdsa_sk, ECDSA_P256_SK_SIZE);
            memcpy(keypair->pqc_pk, kp.slhdsa_pk, SLHDSA128F_PK_SIZE);
            memcpy(keypair->pqc_sk, kp.slhdsa_sk, SLHDSA128F_SK_SIZE);

            keypair->classical_pk_len = ECDSA_P256_PK_SIZE;
            keypair->classical_sk_len = ECDSA_P256_SK_SIZE;
            keypair->pqc_pk_len = SLHDSA128F_PK_SIZE;
            keypair->pqc_sk_len = SLHDSA128F_SK_SIZE;

            secure_zero(&kp, sizeof(kp));
            return 0;
        }

        default:
            return -1;
    }
}

int unified_sign(
    unified_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const unified_keypair_t *keypair

```

```

) {
    if (!sig || !message || !keypair) return -1;

    sig->algorithm = keypair->algorithm;

    switch (keypair->algorithm) {
        case HYBRID_ALG_ED25519_MLDSA65: {
            /* Reconstruct algorithm-specific keypair */
            hybrid_sig_keypair_t kp;
            memcpy(kp.ed25519_pk, keypair->classical_pk, ED25519_PK_SIZE);
            memcpy(kp.ed25519_sk, keypair->classical_sk, ED25519_SK_SIZE);
            memcpy(kp.mldsa65_pk, keypair->pqc_pk, MLDSA65_PK_SIZE);
            memcpy(kp.mldsa65_sk, keypair->pqc_sk, MLDSA65_SK_SIZE);

            hybrid_signature_t hsig;
            int ret = hybrid_sig_sign(&hsig, message, message_len, &kp);

            if (ret == 0) {
                memcpy(sig->classical_sig, hsig.ed25519_sig, ED25519_SIG_SIZE);
                memcpy(sig->pqc_sig, hsig.mldsa65_sig, MLDSA65_SIG_SIZE);
                sig->classical_sig_len = ED25519_SIG_SIZE;
                sig->pqc_sig_len = MLDSA65_SIG_SIZE;
            }

            secure_zero(&kp, sizeof(kp));
            return ret;
        }

        case HYBRID_ALG_ECDSA_SLHDSA128F: {
            ecdsa_slhdsa_keypair_t kp;
            memcpy(kp.ecdsa_pk, keypair->classical_pk, ECDSA_P256_PK_SIZE);
            memcpy(kp.ecdsa_sk, keypair->classical_sk, ECDSA_P256_SK_SIZE);
            memcpy(kp.slhdsa_pk, keypair->pqc_pk, SLHDSA128F_PK_SIZE);
            memcpy(kp.slhdsa_sk, keypair->pqc_sk, SLHDSA128F_SK_SIZE);

            ecdsa_slhdsa_signature_t esig;
            int ret = ecdsa_slhdsa_sign(&esig, message, message_len, &kp);

            if (ret == 0) {
                memcpy(sig->classical_sig, esig.ecdsa_sig, ECDSA_P256_SIG_SIZE);
                memcpy(sig->pqc_sig, esig.slhdsa_sig, SLHDSA128F_SIG_SIZE);
                sig->classical_sig_len = ECDSA_P256_SIG_SIZE;
                sig->pqc_sig_len = SLHDSA128F_SIG_SIZE;
            }

            secure_zero(&kp, sizeof(kp));
            return ret;
        }

        default:
            return -1;
    }
}

int unified_verify(
    const unified_signature_t *sig,
    const uint8_t *message,
    size_t message_len,
    const unified_keypair_t *keypair
) {

```

```

if (!sig || !message || !keypair) return -1;
if (sig->algorithm != keypair->algorithm) return -1;

switch (sig->algorithm) {
    case HYBRID_ALG_ED25519_MLDSA65: {
        hybrid_signature_t hsig;
        memcpy(hsig.ed25519_sig, sig->classical_sig, ED25519_SIG_SIZE);
        memcpy(hsig.mldsa65_sig, sig->pqc_sig, MLDSA65_SIG_SIZE);

        return hybrid_sig_verify(&hsig, message, message_len,
                                keypair->classical_pk, keypair->pqc_pk);
    }

    case HYBRID_ALG_ECDSA_SLHDSA128F: {
        ecdsa_slhdsa_signature_t esig;
        memcpy(esig.ecdsa_sig, sig->classical_sig, ECDSA_P256_SIG_SIZE);
        memcpy(esig.slhdsa_sig, sig->pqc_sig, SLHDSA128F_SIG_SIZE);

        return ecdsa_slhdsa_verify(&esig, message, message_len,
                                   keypair->classical_pk, keypair->pqc_pk);
    }

    default:
        return -1;
}
}

void unified_keypair_clear(unified_keypair_t *keypair) {
    if (keypair) {
        secure_zero(keypair->classical_sk, sizeof(keypair->classical_sk));
        secure_zero(keypair->pqc_sk, sizeof(keypair->pqc_sk));
        secure_zero(keypair, sizeof(*keypair));
    }
}

```

Unit 8.3 Summary

- Hybrid signatures require AND policy (both must verify) for security benefits
- Composite hash binding (signing `HASH(pk1 || pk2 || message)`) prevents component reuse
- Ed25519+ML-DSA-65 provides small classical component with ML-DSA PQC protection
- ECDSA+SLH-DSA uses different mathematical bases for maximum algorithm diversity
- Parallel signing/verification improves performance when component operations are independent
- Verification policies must be enforced by verifiers, not embedded in signatures
- Batch verification can improve throughput for multiple signatures

What's Next: Unit 8.4 covers composite key formats and certificate structures, including X.509 hybrid certificates and PKCS#8 private key encoding.

Unit 8.4: Composite Key Formats and Standards

Prerequisites: Units 8.1-8.3 (Hybrid Cryptography fundamentals)

Estimated Time: 3-4 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Understand ASN.1 encoding** for cryptographic structures and composite keys
2. **Design X.509 certificates** with hybrid public keys and signature algorithms
3. **Implement PKCS#8 private key encoding** for composite key pairs
4. **Assign and use OIDs** for composite cryptographic algorithms
5. **Build certificate chains** with mixed classical and post-quantum algorithms
6. **Handle backward compatibility** in certificate-based systems

8.4.1 ASN.1 Fundamentals for Composite Keys

ASN.1 Background

Abstract Syntax Notation One (ASN.1) provides a standardized notation for defining data structures. X.509 certificates, PKCS standards, and most cryptographic formats use ASN.1 with Distinguished Encoding Rules (DER) or Basic Encoding Rules (BER).

Key ASN.1 Concepts:

CONCEPT	DESCRIPTION	TAG
SEQUENCE	Ordered collection of elements	0x30
SET	Unordered collection	0x31
OCTET STRING	Arbitrary bytes	0x04
BIT STRING	Bit sequence	0x03
INTEGER	Signed integer	0x02
OBJECT IDENTIFIER	Unique identifier	0x06
NULL	Empty value	0x05

Length Encoding:

- Short form: Single byte for lengths 0-127
- Long form: First byte indicates number of length bytes (0x81 for 1 byte, 0x82 for 2 bytes)

Composite Key ASN.1 Structure

The IETF composite signatures draft defines the following ASN.1 structure:

```

CompositePublicKey ::= SEQUENCE SIZE (2..MAX) OF SubjectPublicKeyInfo

CompositePrivateKey ::= SEQUENCE SIZE (2..MAX) OF OneAsymmetricKey

CompositeSignatureValue ::= SEQUENCE SIZE (2..MAX) OF BIT STRING

CompositeAlgorithmIdentifier ::= SEQUENCE {
    algorithm    OBJECT IDENTIFIER,
    parameters   CompositeParams OPTIONAL
}

CompositeParams ::= SEQUENCE SIZE (2..MAX) OF AlgorithmIdentifier

```

AlgorithmIdentifier Structure:

```

AlgorithmIdentifier ::= SEQUENCE {
    algorithm    OBJECT IDENTIFIER,
    parameters   ANY DEFINED BY algorithm OPTIONAL
}

```

8.4.2 Object Identifiers (OIDs) for Hybrid Algorithms

OID Structure and Assignment

OIDs form a hierarchical tree. Cryptographic algorithms typically fall under:

- 1.2.840.113549 (RSA Data Security)
- 1.2.840.10045 (ANSI X9.62 - elliptic curves)
- 1.3.6.1.4.1 (Private enterprises)
- 2.16.840.1.101.3.4 (NIST algorithms)

NIST Post-Quantum OIDs (draft assignments):

ALGORITHM	OID ARC
ML-KEM-512	2.16.840.1.101.3.4.4.1
ML-KEM-768	2.16.840.1.101.3.4.4.2
ML-KEM-1024	2.16.840.1.101.3.4.4.3
ML-DSA-44	2.16.840.1.101.3.4.3.17
ML-DSA-65	2.16.840.1.101.3.4.3.18
ML-DSA-87	2.16.840.1.101.3.4.3.19
SLH-DSA-SHA2-128s	2.16.840.1.101.3.4.3.20
SLH-DSA-SHA2-128f	2.16.840.1.101.3.4.3.21
SLH-DSA-SHAKE-128s	2.16.840.1.101.3.4.3.26
SLH-DSA-SHAKE-128f	2.16.840.1.101.3.4.3.27

Composite Algorithm OIDs (IETF draft):

COMPOSITE ALGORITHM	OID
MLDSA44-Ed25519	2.16.840.1.114027.80.8.1.1
MLDSA65-Ed25519	2.16.840.1.114027.80.8.1.2
MLDSA87-Ed448	2.16.840.1.114027.80.8.1.3
MLDSA44-ECDSA-P256	2.16.840.1.114027.80.8.1.4
MLDSA65-ECDSA-P384	2.16.840.1.114027.80.8.1.5
MLDSA87-ECDSA-P384	2.16.840.1.114027.80.8.1.6
MLDSA65-RSA3072-PSS	2.16.840.1.114027.80.8.1.7
MLDSA65-RSA3072-PKCS15	2.16.840.1.114027.80.8.1.8

8.4.3 X.509 Certificate Structure

Standard X.509 Structure

```

Certificate ::= SEQUENCE {
    tbsCertificate      TBSCertificate,
    signatureAlgorithm  AlgorithmIdentifier,
    signatureValue      BIT STRING
}

TBSCertificate ::= SEQUENCE {
    version             [0] EXPLICIT Version DEFAULT v1,
    serialNumber        CertificateSerialNumber,
    signature            AlgorithmIdentifier,
    issuer              Name,
    validity            Validity,
    subject             Name,
    subjectPublicKeyInfo SubjectPublicKeyInfo,
    issuerUniqueID      [1] IMPLICIT UniqueIdentifier OPTIONAL,
    subjectUniqueID     [2] IMPLICIT UniqueIdentifier OPTIONAL,
    extensions          [3] EXPLICIT Extensions OPTIONAL
}

SubjectPublicKeyInfo ::= SEQUENCE {
    algorithm            AlgorithmIdentifier,
    subjectPublicKey     BIT STRING
}

```

Hybrid Certificate Options

Option 1: Composite SubjectPublicKeyInfo

The subject public key contains both component keys:

```

SubjectPublicKeyInfo ::= SEQUENCE {
    algorithm      AlgorithmIdentifier {
        algorithm  id-composite-key,
        parameters SEQUENCE {
            AlgorithmIdentifier { id-Ed25519 },
            AlgorithmIdentifier { id-ML-DSA-65 }
        }
    },
    subjectPublicKey BIT STRING CONTAINING CompositePublicKey
}

```

Option 2: Alternative Public Key Extension

Use the Alternative Public Key extension (OID 2.5.29.72) to carry the PQC key:

```

Extension ::= SEQUENCE {
    extnID      OBJECT IDENTIFIER (id-ce-altPublicKey),
    critical    BOOLEAN DEFAULT FALSE,
    extnValue   OCTET STRING CONTAINING SubjectPublicKeyInfo
}

```

Option 3: Subject Information Access Extension

Reference a separate certificate containing the alternative key:

```

SubjectInfoAccessSyntax ::= SEQUENCE SIZE (1..MAX) OF AccessDescription

AccessDescription ::= SEQUENCE {
    accessMethod      OBJECT IDENTIFIER (id-ad-altCert),
    accessLocation    GeneralName
}

```

8.4.4 PKCS#8 Private Key Format

Standard OneAsymmetricKey Structure

```

OneAsymmetricKey ::= SEQUENCE {
    version            Version,
    privateKeyAlgorithm AlgorithmIdentifier,
    privateKey         OCTET STRING,
    attributes         [0] IMPLICIT Attributes OPTIONAL,
    publicKey          [1] IMPLICIT BIT STRING OPTIONAL
}

Version ::= INTEGER { v1(0), v2(1) }

```


Composite Private Key Encoding

```
OneAsymmetricKey ::= SEQUENCE {
    version                INTEGER (1),
    privateKeyAlgorithm    AlgorithmIdentifier {
        algorithm    id-composite-key,
        parameters   CompositeParams
    },
    privateKey             OCTET STRING CONTAINING CompositePrivateKey,
    publicKey              [1] BIT STRING CONTAINING CompositePublicKey OPTIONAL
}

CompositePrivateKey ::= SEQUENCE {
    ed25519PrivateKey      OCTET STRING (SIZE(32)),
    mldsa65PrivateKey      OCTET STRING (SIZE(4032))
}
```

8.4.5 Implementation

Header File

```
/* composite_keys.h - Composite key format handling */

#ifndef COMPOSITE_KEYS_H
#define COMPOSITE_KEYS_H

#include <stdint.h>
#include <stddef.h>

/* Algorithm OIDs */
#define OID_ED25519                "\x2b\x65\x70"
#define OID_ED25519_LEN            3

#define OID_ML_DSA_65              "\x60\x86\x48\x01\x65\x03\x04\x03\x12"
#define OID_ML_DSA_65_LEN          9

#define OID_ECDSA_P256             "\x2a\x86\x48\xce\x3d\x02\x01"
#define OID_ECDSA_P256_LEN         7

#define OID_SLH_DSA_SHAKE_128F     "\x60\x86\x48\x01\x65\x03\x04\x03\x1b"
#define OID_SLH_DSA_SHAKE_128F_LEN 9

/* Composite OIDs (IETF draft) */
#define OID_COMPOSITE_MLDSA65_ED25519 "\x60\x86\x48\x01\x86\xfa\x6b\x50\x08\x01\x02"
#define OID_COMPOSITE_MLDSA65_ED25519_LEN 11

/* ASN.1 tags */
#define ASN1_SEQUENCE              0x30
#define ASN1_SET                   0x31
#define ASN1_INTEGER               0x02
#define ASN1_BIT_STRING            0x03
#define ASN1_OCTET_STRING          0x04
#define ASN1_NULL                  0x05
#define ASN1_OID                   0x06
#define ASN1_UTF8_STRING           0x0c
#define ASN1_PRINTABLE_STRING      0x13
#define ASN1_IA5_STRING            0x16
```

```

#define ASN1.UTC_TIME          0x17
#define ASN1.GENERALIZED_TIME 0x18
#define ASN1.CONTEXT_0        0xA0
#define ASN1.CONTEXT_1        0xA1
#define ASN1.CONTEXT_2        0xA2
#define ASN1.CONTEXT_3        0xA3

/* Key sizes */
#define ED25519_PK_SIZE        32
#define ED25519_SK_SIZE        64
#define ED25519_SIG_SIZE       64

#define MLDSA65_PK_SIZE        1952
#define MLDSA65_SK_SIZE        4032
#define MLDSA65_SIG_SIZE       3309

/* Maximum encoded sizes */
#define MAX_SPKI_SIZE          4096
#define MAX_PKCS8_SIZE         8192
#define MAX_CERT_SIZE          16384
#define MAX_OID_SIZE           32
#define MAX_NAME_SIZE           256

/* Error codes */
#define COMPOSITE_OK            0
#define COMPOSITE_ERR_INVALID_INPUT  -1
#define COMPOSITE_ERR_BUFFER_TOO_SMALL -2
#define COMPOSITE_ERR_INVALID_ASN1  -3
#define COMPOSITE_ERR_UNSUPPORTED_ALG -4
#define COMPOSITE_ERR_DECODE_FAILED -5
#define COMPOSITE_ERR_ENCODE_FAILED -6

/* Algorithm identifiers */
typedef enum {
    ALG_ED25519,
    ALG_ECDSA_P256,
    ALG_ECDSA_P384,
    ALG_ML_DSA_44,
    ALG_ML_DSA_65,
    ALG_ML_DSA_87,
    ALG_SLH_DSA_128S,
    ALG_SLH_DSA_128F,
    ALG_COMPOSITE_MLDSA65_ED25519,
    ALG_COMPOSITE_MLDSA65_ECDSA_P384
} algorithm_id_t;

/* Composite key pair structure */
typedef struct {
    algorithm_id_t classical_alg;
    algorithm_id_t pqc_alg;

    /* Classical component */
    uint8_t classical_pk[64]; /* Holds Ed25519/X25519-sized keys (<= 64 B).
                               NOTE: an uncompressed ECDSA-P384 point is
                               97 bytes (0x04||X||Y), so P-384 keys do NOT
                               fit here and are not stored in these buffers. */

    size_t classical_pk_len;
    uint8_t classical_sk[64]; /* Ed25519/X25519 private key (<= 64 B); see note
    above */
    size_t classical_sk_len;

```

```

    /* PQC component */
    uint8_t *pqc_pk;
    size_t pqc_pk_len;
    uint8_t *pqc_sk;
    size_t pqc_sk_len;
} composite_keypair_t;

/* ASN.1 buffer for encoding/decoding */
typedef struct {
    uint8_t *data;
    size_t len;
    size_t capacity;
    size_t pos; /* Current position for parsing */
} asn1_buffer_t;

/* SubjectPublicKeyInfo structure */
typedef struct {
    algorithm_id_t algorithm;
    uint8_t oid[MAX_OID_SIZE];
    size_t oid_len;
    uint8_t *public_key;
    size_t public_key_len;
} spki_t;

/* X.509 Name component */
typedef struct {
    uint8_t oid[MAX_OID_SIZE];
    size_t oid_len;
    char value[MAX_NAME_SIZE];
    size_t value_len;
    uint8_t string_type;
} name_component_t;

/* X.509 Validity */
typedef struct {
    uint8_t not_before[16]; /* YYYYMMDDHHMMSSZ */
    size_t not_before_len;
    uint8_t not_after[16];
    size_t not_after_len;
} validity_t;

/* Certificate extension */
typedef struct {
    uint8_t oid[MAX_OID_SIZE];
    size_t oid_len;
    int critical;
    uint8_t *value;
    size_t value_len;
} extension_t;

/* X.509 TBSCertificate */
typedef struct {
    int version; /* 0=v1, 1=v2, 2=v3 */
    uint8_t serial_number[20];
    size_t serial_len;
    algorithm_id_t signature_alg;
    name_component_t issuer[8];
    size_t issuer_count;
    validity_t validity;

```

```

    name_component_t subject[8];
    size_t subject_count;
    spki_t subject_pki;
    extension_t extensions[16];
    size_t extension_count;
} tbs_certificate_t;

/* Complete certificate */
typedef struct {
    tbs_certificate_t tbs;
    algorithm_id_t signature_alg;
    uint8_t *signature;
    size_t signature_len;

    /* DER-encoded TBS for signing/verification */
    uint8_t *tbs_der;
    size_t tbs_der_len;
} certificate_t;

/* ASN.1 buffer management */
int asn1_buffer_init(asn1_buffer_t *buf, size_t capacity);
void asn1_buffer_free(asn1_buffer_t *buf);
void asn1_buffer_reset(asn1_buffer_t *buf);

/* ASN.1 encoding functions */
int asn1_encode_length(asn1_buffer_t *buf, size_t length);
int asn1_encode_tag_length(asn1_buffer_t *buf, uint8_t tag, size_t length);
int asn1_encode_integer(asn1_buffer_t *buf, const uint8_t *value, size_t len);
int asn1_encode_octet_string(asn1_buffer_t *buf, const uint8_t *data, size_t len);
int asn1_encode_bit_string(asn1_buffer_t *buf, const uint8_t *data, size_t len);
int asn1_encode_oid(asn1_buffer_t *buf, const uint8_t *oid, size_t oid_len);
int asn1_encode_null(asn1_buffer_t *buf);
int asn1_encode_sequence(asn1_buffer_t *buf, const uint8_t *content, size_t len);

/* ASN.1 decoding functions */
int asn1_decode_tag(asn1_buffer_t *buf, uint8_t *tag);
int asn1_decode_length(asn1_buffer_t *buf, size_t *length);
int asn1_decode_tag_length(asn1_buffer_t *buf, uint8_t expected_tag, size_t *length);
int asn1_decode_integer(asn1_buffer_t *buf, uint8_t *value, size_t *len, size_t max_len);
int asn1_decode_octet_string(asn1_buffer_t *buf, uint8_t *data, size_t *len, size_t max_len);
int asn1_decode_bit_string(asn1_buffer_t *buf, uint8_t *data, size_t *len, size_t max_len);
int asn1_decode_oid(asn1_buffer_t *buf, uint8_t *oid, size_t *len, size_t max_len);

/* Algorithm identifier encoding/decoding */
int encode_algorithm_identifier(asn1_buffer_t *buf, algorithm_id_t alg);
int decode_algorithm_identifier(asn1_buffer_t *buf, algorithm_id_t *alg);
int get_algorithm_oid(algorithm_id_t alg, const uint8_t **oid, size_t *oid_len);

/* SubjectPublicKeyInfo encoding/decoding */
int encode_spki(asn1_buffer_t *buf, const spki_t *spki);
int decode_spki(asn1_buffer_t *buf, spki_t *spki);
int encode_composite_spki(asn1_buffer_t *buf, const composite_keypair_t *keypair);
int decode_composite_spki(asn1_buffer_t *buf, composite_keypair_t *keypair);

/* PKCS#8 private key encoding/decoding */
int encode_pkcs8_private_key(asn1_buffer_t *buf, const composite_keypair_t *keypair);
int decode_pkcs8_private_key(asn1_buffer_t *buf, composite_keypair_t *keypair);

```

```

/* X.509 certificate functions */
int encode_tbs_certificate(asn1_buffer_t *buf, const tbs_certificate_t *tbs);
int encode_certificate(asn1_buffer_t *buf, const certificate_t *cert);
int decode_certificate(asn1_buffer_t *buf, certificate_t *cert);

/* Composite key management */
int composite_keypair_init(composite_keypair_t *keypair,
                           algorithm_id_t classical, algorithm_id_t pqc);
void composite_keypair_free(composite_keypair_t *keypair);
int composite_keypair_generate(composite_keypair_t *keypair);

/* Certificate creation */
int create_self_signed_certificate(certificate_t *cert,
                                   const composite_keypair_t *keypair,
                                   const char *common_name,
                                   int validity_days);

int create_certificate_request(asn1_buffer_t *buf,
                              const composite_keypair_t *keypair,
                              const char *common_name);

/* Certificate verification */
int verify_certificate_signature(const certificate_t *cert,
                                const composite_keypair_t *issuer_key);

/* PEM encoding/decoding */
int der_to_pem(const uint8_t *der, size_t der_len,
               const char *label, char *pem, size_t pem_len);
int pem_to_der(const char *pem, size_t pem_len,
               uint8_t *der, size_t der_len);

#endif /* COMPOSITE_KEYS_H */

```

Implementation File

[EXTRACTED]

The full composite key implementation (~1,470 lines of DER encoding, OID tables, per-algorithm dispatch) is available at `source_code/module_08_hybrid/composite_keys.c`. It was previously inlined here in earlier versions but relocated for readability — inline 1,470-line code blocks are better read in an editor with syntax highlighting than in prose.

Architectural summary of the implementation:

- DER encoding follows X.690 (see handbook §8.4.2 for ASN.1 refresher)
- Composite algorithm OIDs use the prototype arc `2.16.840.1.114027.80.8.1.*` (pending IANA assignment per `draft-ietf-lamps-pq-composite-sigs-16`)
- Key generation dispatches to each component algorithm, concatenates keys with length prefixes, and wraps in a composite `SubjectPublicKeyInfo`
- Signing and verification apply both component algorithms; **both must succeed** for the composite signature to verify
- Memory-safe: all buffer allocations check NULL; sensitive material zeroized via `secure_zero()`

Build the extracted file:

```
cd source_code/module_08_hybrid
# (Requires composite_keys.h header shown earlier in this unit)
gcc -Wall -Wextra -O2 -std=c11 -c composite_keys.c
```

8.4.6 Certificate Chain Considerations

Building Hybrid Certificate Chains

Certificate chains with hybrid algorithms require careful consideration:

Strategy 1: Homogeneous Chains

All certificates in the chain use the same composite algorithm:

```
Root CA (ML-DSA-65 + Ed25519)
|__ Intermediate CA (ML-DSA-65 + Ed25519)
    |__ End Entity (ML-DSA-65 + Ed25519)
```

Advantages:

- Consistent security guarantees throughout chain

- Simpler verification logic

Disadvantages:

- All or nothing migration
- Large chain size

Strategy 2: Mixed Chains

Different algorithms at different levels:

```
Root CA (RSA-4096, long-lived)
  |__ Intermediate CA (ML-DSA-65 + Ed25519, medium-lived)
      |__ End Entity (ML-DSA-65, short-lived)
```

Advantages:

- Gradual migration path
- Can adapt to algorithm confidence levels

Disadvantages:

- Chain security limited by weakest link
- More complex verification

Strategy 3: Parallel Chains

Maintain separate chains and cross-certify:

```
Classical Root          PQC Root
  |                      |
  |__ Classical Intermediate <-> PQC Intermediate (cross-cert)
      |                      |
      |__ End Entity <------|
```

Advantages:

- Complete fallback capability
- Independent algorithm upgrades

Disadvantages:

- Double infrastructure
- Complex certificate management

Certificate Path Validation

Hybrid certificate verification requires modified path validation:

```
/* Hybrid certificate path validation */
typedef enum {
    VALIDATION_STRICT,      /* Both signatures must verify */
    VALIDATION_RELAXED,     /* Either signature sufficient */
    VALIDATION_CLASSICAL,   /* Only classical signature */
    VALIDATION_PQC_ONLY     /* Only PQC signature */
} validation_mode_t;

typedef struct {
    validation_mode_t mode;
    int allow_mixed_chains;
    int require_pqc_leaf;
    size_t max_chain_length;
} path_validation_policy_t;

int validate_certificate_path(
    const certificate_t *chain[],
    size_t chain_len,
    const certificate_t *trust_anchor,
    const path_validation_policy_t *policy
) {
    if (chain_len > policy->max_chain_length) {
        return VALIDATION_ERR_CHAIN_TOO_LONG;
    }

    /* Verify each certificate signature */
    for (size_t i = 0; i < chain_len; i++) {
        const certificate_t *issuer = (i == chain_len - 1)
            ? trust_anchor
            : chain[i + 1];

        int classical_valid = verify_classical_signature(
            chain[i], issuer
        );
        int pqc_valid = verify_pqc_signature(
            chain[i], issuer
        );

        switch (policy->mode) {
            case VALIDATION_STRICT:
                if (!classical_valid || !pqc_valid) {
                    return VALIDATION_ERR_SIGNATURE_FAILED;
                }
                break;

            case VALIDATION_RELAXED:
                if (!classical_valid && !pqc_valid) {
                    return VALIDATION_ERR_SIGNATURE_FAILED;
                }
                break;

            case VALIDATION_CLASSICAL:
                if (!classical_valid) {
                    return VALIDATION_ERR_SIGNATURE_FAILED;
                }
        }
    }
}
```



```

        break;

    case VALIDATION_PQC_ONLY:
        if (!pqc_valid) {
            return VALIDATION_ERR_SIGNATURE_FAILED;
        }
        break;
    }
}

/* Additional policy checks */
if (policy->require_pqc_leaf) {
    if (!has_pqc_algorithm(chain[0])) {
        return VALIDATION_ERR_PQC_REQUIRED;
    }
}

return VALIDATION_OK;
}

```

8.4.7 Interoperability Considerations

OpenSSL Integration

OpenSSL 3.x supports custom providers for new algorithms:

```

/* OpenSSL provider registration for composite algorithms */
#include <openssl/core.h>
#include <openssl/core_dispatch.h>

static const OSSL_ALGORITHM composite_keymgmt[] = {
    {"MLDSA65-ED25519", "provider=composite",
     composite_mldsa65_ed25519_keymgmt_functions},
    {NULL, NULL, NULL}
};

static const OSSL_ALGORITHM composite_signature[] = {
    {"MLDSA65-ED25519", "provider=composite",
     composite_mldsa65_ed25519_signature_functions},
    {NULL, NULL, NULL}
};

static const OSSL_ALGORITHM *composite_query(void *provctx,
                                              int operation_id,
                                              int *no_cache) {
    *no_cache = 0;
    switch (operation_id) {
        case OSSL_OP_KEYMGMT:
            return composite_keymgmt;
        case OSSL_OP_SIGNATURE:
            return composite_signature;
    }
    return NULL;
}

static const OSSL_DISPATCH composite_dispatch[] = {
    {OSSL_FUNC_PROVIDER_QUERY_OPERATION, (void (*)(void))composite_query},

```

```

        {OSSL_FUNC_PROVIDER_TEARDOWN, (void (*)(void))composite_teardown},
        {0, NULL}
    };

    int OSSL_provider_init(const OSSL_CORE_HANDLE *handle,
                          const OSSL_DISPATCH *in,
                          const OSSL_DISPATCH **out,
                          void **provctx) {
        *out = composite_dispatch;
        *provctx = composite_provider_new(handle, in);
        return 1;
    }

```

PKCS#11 Considerations

Hardware security modules (HSMs) with PKCS#11 interfaces require mechanism definitions:

```

/* PKCS#11 mechanism definitions for composite algorithms */

/* Vendor-defined mechanism base (example) */
#define CKM_VENDOR_DEFINED_COMPOSITE    0x80001000UL

/* Composite signature mechanisms */
#define CKM_MLDSA65_ED25519              (CKM_VENDOR_DEFINED_COMPOSITE + 1)
#define CKM_MLDSA65_ECDSA_P384          (CKM_VENDOR_DEFINED_COMPOSITE + 2)

/* Key types */
#define CKK_COMPOSITE_MLDSA65_ED25519    (CKK_VENDOR_DEFINED + 1)

/* Object class for composite keys */
typedef struct {
    CK_ATTRIBUTE_TYPE type;
    CK_VOID_PTR pValue;
    CK_ULONG ulValueLen;
} CK_COMPOSITE_KEY_COMPONENT;

/* Template for composite key generation */
CK_ATTRIBUTE composite_pubkey_template[] = {
    {CKA_CLASS, &pubkey_class, sizeof(pubkey_class)},
    {CKA_KEY_TYPE, &composite_key_type, sizeof(composite_key_type)},
    {CKA_VERIFY, &true_val, sizeof(true_val)},
    {CKA_TOKEN, &true_val, sizeof(true_val)},
    {CKA_LABEL, "Composite Public Key", 20}
};

```

Exercises

Exercise 8.4.1: ASN.1 Structure Analysis *(Beginner)*

Given the following DER-encoded data (hexadecimal):

```
30 82 01 0A 30 0D 06 09 2A 86 48 86 F7 0D 01 01
01 05 00 03 81 F8 00 30 81 F4 02 81 F1 00 ...
```

1. Parse the structure manually, identifying each tag and length
2. Determine what type of cryptographic object this represents
3. Explain the difference between definite and indefinite length encoding

Solution:

Parsing DER structure:

```
Offset 0: 30 - SEQUENCE tag
Offset 1: 82 01 0A - Long form length (0x82 = 2 length bytes),
                  total content = 0x010A = 266 bytes

Offset 4: 30 - Inner SEQUENCE tag (AlgorithmIdentifier)
Offset 5: 0D - Length = 13 bytes

Offset 6: 06 - OID tag
Offset 7: 09 - Length = 9 bytes
Offset 8-16: 2A 86 48 86 F7 0D 01 01 01
              = 1.2.840.113549.1.1.1 (rsaEncryption)

Offset 17: 05 - NULL tag
Offset 18: 00 - Length = 0 (NULL has no content)

Offset 19: 03 - BIT STRING tag
Offset 20: 81 F8 - Long form length (0x81 = 1 length byte),
                  content = 0xF8 = 248 bytes
Offset 22: 00 - Unused bits count

Offset 23: 30 - Inner SEQUENCE (RSA public key)
...
```

This is a SubjectPublicKeyInfo structure containing an RSA public key.

Definite vs Indefinite Length:

- Definite: Exact length specified upfront (used in DER)
- Indefinite: 0x80 as length, content ends with 0x00 0x00 (only valid in BER, not DER)

DER requires definite length for deterministic encoding.

Exercise 8.4.2: Composite SPKI Construction *(Intermediate)*

Implement a function that constructs a composite SubjectPublicKeyInfo for an Ed25519 + ML-DSA-65 key pair, following the IETF composite signatures draft format.

Solution:

```
#include "composite_keys.h"
#include <string.h>

/*
 * Build composite SPKI following IETF draft structure:
 *
 * SubjectPublicKeyInfo ::= SEQUENCE {
 *     algorithm AlgorithmIdentifier { id-composite-sig },
 *     subjectPublicKey BIT STRING CONTAINING CompositePublicKey
 * }
 *
 * CompositePublicKey ::= SEQUENCE {
 *     ed25519PublicKey SubjectPublicKeyInfo,
 *     mldsa65PublicKey SubjectPublicKeyInfo
 * }
 */

/* Helper to build single-algorithm SPKI */
static int build_single_spki(asn1_buffer_t *buf,
                             algorithm_id_t alg,
                             const uint8_t *pk,
                             size_t pk_len) {
    asn1_buffer_t inner;
    int ret;

    ret = asn1_buffer_init(&inner, pk_len + 64);
    if (ret != COMPOSITE_OK) return ret;

    /* Algorithm identifier */
    ret = encode_algorithm_identifier(&inner, alg);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&inner);
        return ret;
    }

    /* Public key as BIT STRING */
    ret = asn1_encode_bit_string(&inner, pk, pk_len);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&inner);
        return ret;
    }

    /* Wrap in SEQUENCE */
    ret = asn1_encode_sequence(buf, inner.data, inner.len);
    asn1_buffer_free(&inner);

    return ret;
}

int build_composite_spki(
```

```

uint8_t *output,
size_t *output_len,
const uint8_t *ed25519_pk,
size_t ed25519_pk_len,
const uint8_t *mldsa65_pk,
size_t mldsa65_pk_len
) {
    asn1_buffer_t outer, inner, composite_pk;
    int ret;

    /* Validate inputs */
    if (!output || !output_len || !ed25519_pk || !mldsa65_pk) {
        return COMPOSITE_ERR_INVALID_INPUT;
    }
    if (ed25519_pk_len != ED25519_PK_SIZE) {
        return COMPOSITE_ERR_INVALID_INPUT;
    }
    if (mldsa65_pk_len != MLDSA65_PK_SIZE) {
        return COMPOSITE_ERR_INVALID_INPUT;
    }

    /* Initialize buffers */
    ret = asn1_buffer_init(&outer, MAX_SPKI_SIZE);
    if (ret != COMPOSITE_OK) return ret;

    ret = asn1_buffer_init(&inner, MAX_SPKI_SIZE);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&outer);
        return ret;
    }

    ret = asn1_buffer_init(&composite_pk, MAX_SPKI_SIZE);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&outer);
        asn1_buffer_free(&inner);
        return ret;
    }

    /* Build Ed25519 SPKI */
    asn1_buffer_t ed_spki;
    ret = asn1_buffer_init(&ed_spki, 128);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&outer);
        asn1_buffer_free(&inner);
        asn1_buffer_free(&composite_pk);
        return ret;
    }

    ret = build_single_spki(&ed_spki, ALG_ED25519, ed25519_pk, ed25519_pk_len);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&outer);
        asn1_buffer_free(&inner);
        asn1_buffer_free(&composite_pk);
        asn1_buffer_free(&ed_spki);
        return ret;
    }

    /* Copy Ed25519 SPKI to composite */
    memcpy(composite_pk.data + composite_pk.len, ed_spki.data, ed_spki.len);
    composite_pk.len += ed_spki.len;
}

```

```

asn1_buffer_free(&ed_spki);

/* Build ML-DSA-65 SPKI */
asn1_buffer_t ml_spki;
ret = asn1_buffer_init(&ml_spki, 4096);
if (ret != COMPOSITE_OK) {
    asn1_buffer_free(&outer);
    asn1_buffer_free(&inner);
    asn1_buffer_free(&composite_pk);
    return ret;
}

ret = build_single_spki(&ml_spki, ALG_ML_DSA_65, mldsa65_pk, mldsa65_pk_len);
if (ret != COMPOSITE_OK) {
    asn1_buffer_free(&outer);
    asn1_buffer_free(&inner);
    asn1_buffer_free(&composite_pk);
    asn1_buffer_free(&ml_spki);
    return ret;
}

/* Copy ML-DSA-65 SPKI to composite */
memcpy(composite_pk.data + composite_pk.len, ml_spki.data, ml_spki.len);
composite_pk.len += ml_spki.len;
asn1_buffer_free(&ml_spki);

/* Wrap component SPKIs in SEQUENCE for CompositePublicKey */
asn1_buffer_t cpk_seq;
ret = asn1_buffer_init(&cpk_seq, MAX_SPKI_SIZE);
if (ret != COMPOSITE_OK) {
    asn1_buffer_free(&outer);
    asn1_buffer_free(&inner);
    asn1_buffer_free(&composite_pk);
    return ret;
}

ret = asn1_encode_sequence(&cpk_seq, composite_pk.data, composite_pk.len);
asn1_buffer_free(&composite_pk);
if (ret != COMPOSITE_OK) {
    asn1_buffer_free(&outer);
    asn1_buffer_free(&inner);
    asn1_buffer_free(&cpk_seq);
    return ret;
}

/* Build outer AlgorithmIdentifier */
const uint8_t *oid = (const uint8_t *)OID_COMPOSITE_MLDSA65_ED25519;
size_t oid_len = OID_COMPOSITE_MLDSA65_ED25519_LEN;

asn1_buffer_t alg_content;
ret = asn1_buffer_init(&alg_content, 32);
if (ret != COMPOSITE_OK) {
    asn1_buffer_free(&outer);
    asn1_buffer_free(&inner);
    asn1_buffer_free(&cpk_seq);
    return ret;
}

ret = asn1_encode_oid(&alg_content, oid, oid_len);
if (ret != COMPOSITE_OK) {

```

```

        asn1_buffer_free(&outer);
        asn1_buffer_free(&inner);
        asn1_buffer_free(&cpk_seq);
        asn1_buffer_free(&alg_content);
        return ret;
    }

    /* Wrap in SEQUENCE for AlgorithmIdentifier */
    ret = asn1_encode_sequence(&inner, alg_content.data, alg_content.len);
    asn1_buffer_free(&alg_content);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&outer);
        asn1_buffer_free(&inner);
        asn1_buffer_free(&cpk_seq);
        return ret;
    }

    /* Add CompositePublicKey as BIT STRING */
    ret = asn1_encode_bit_string(&inner, cpk_seq.data, cpk_seq.len);
    asn1_buffer_free(&cpk_seq);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&outer);
        asn1_buffer_free(&inner);
        return ret;
    }

    /* Final outer SEQUENCE */
    ret = asn1_encode_sequence(&outer, inner.data, inner.len);
    asn1_buffer_free(&inner);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&outer);
        return ret;
    }

    /* Copy to output */
    if (*output_len < outer.len) {
        *output_len = outer.len;
        asn1_buffer_free(&outer);
        return COMPOSITE_ERR_BUFFER_TOO_SMALL;
    }

    memcpy(output, outer.data, outer.len);
    *output_len = outer.len;
    asn1_buffer_free(&outer);

    return COMPOSITE_OK;
}

/* Test function */
void test_composite_spki(void) {
    uint8_t ed25519_pk[ED25519_PK_SIZE];
    uint8_t mldsa65_pk[MLDSA65_PK_SIZE];
    uint8_t output[MAX_SPKI_SIZE];
    size_t output_len = sizeof(output);

    /* Fill with test data */
    for (int i = 0; i < ED25519_PK_SIZE; i++) {
        ed25519_pk[i] = (uint8_t)i;
    }
    for (int i = 0; i < MLDSA65_PK_SIZE; i++) {

```

```

        mldsa65_pk[i] = (uint8_t)(i * 3);
    }

    int ret = build_composite_spki(
        output, &output_len,
        ed25519_pk, ED25519_PK_SIZE,
        mldsa65_pk, MLDSA65_PK_SIZE
    );

    if (ret == COMPOSITE_OK) {
        printf("Composite SPKI built successfully: %zu bytes\n", output_len);

        /* Print first few bytes */
        printf("DER (first 32 bytes): ");
        for (size_t i = 0; i < 32 && i < output_len; i++) {
            printf("%02X ", output[i]);
        }
        printf("...\n");
    } else {
        printf("Failed to build composite SPKI: %d\n", ret);
    }
}

```

Exercise 8.4.3: Certificate Chain Validation *(Advanced)*

Design and implement a certificate chain validator that supports both strict (both algorithms must verify) and relaxed (either algorithm sufficient) validation modes.

Solution:

```

#include "composite_keys.h"
#include <stdbool.h>

/* Validation result codes */
typedef enum {
    CHAIN_VALID = 0,
    CHAIN_ERR_EMPTY = -1,
    CHAIN_ERR_TOO_LONG = -2,
    CHAIN_ERR_CLASSICAL_FAILED = -3,
    CHAIN_ERR_PQC_FAILED = -4,
    CHAIN_ERR_BOTH_FAILED = -5,
    CHAIN_ERR_NOT_CA = -6,
    CHAIN_ERR_EXPIRED = -7,
    CHAIN_ERR_NAME_MISMATCH = -8
} chain_validation_result_t;

/* Validation mode */
typedef enum {
    MODE_STRICT, /* Both signatures must verify */
    MODE_RELAXED, /* Either signature is sufficient */
    MODE_MIGRATION /* Classical required, PQC optional with warning */
} validation_mode_t;

/* Validation context */
typedef struct {
    validation_mode_t mode;
    size_t max_chain_length;
}

```



```

    bool check_expiry;
    bool require_ca_basic_constraint;
    time_t validation_time;

    /* Diagnostic output */
    bool classical_verified[16];
    bool pqc_verified[16];
    char error_message[256];
} chain_validation_ctx_t;

/* Mock verification functions - STUBS for demonstrating structure only!
 * In production, use:
 * - Ed25519: EVP_DigestVerify() with EVP_PKEY_ED25519 from OpenSSL
 * - ML-DSA-65: OQS_SIG_verify() with OQS_SIG_alg_ml_dsa_65 from liboqs
 */
static bool verify_ed25519_signature(
    const uint8_t *message, size_t msg_len,
    const uint8_t *signature, size_t sig_len,
    const uint8_t *public_key, size_t pk_len
) {
    /* STUB: Always returns true. Replace with real Ed25519 verification. */
    (void)message; (void)msg_len;
    (void)signature; (void)sig_len;
    (void)public_key; (void)pk_len;
    return true; /* Demo only - real code must verify! */
}

static bool verify_mldsa65_signature(
    const uint8_t *message, size_t msg_len,
    const uint8_t *signature, size_t sig_len,
    const uint8_t *public_key, size_t pk_len
) {
    /* STUB: Always returns true. Replace with OQS_SIG_verify() in production. */
    (void)message; (void)msg_len;
    (void)signature; (void)sig_len;
    (void)public_key; (void)pk_len;
    return true; /* Demo only - real code must verify! */
}

/* Extract signature components from composite signature */
static int extract_composite_signature(
    const uint8_t *composite_sig, size_t sig_len,
    uint8_t *ed25519_sig, size_t *ed25519_len,
    uint8_t *mldsa65_sig, size_t *mldsa65_len
) {
    asn1_buffer_t buf = {
        .data = (uint8_t *)composite_sig,
        .len = sig_len,
        .capacity = sig_len,
        .pos = 0
    };

    /* Composite signature is SEQUENCE of BIT STRINGs */
    size_t seq_len;
    if (asn1_decode_tag_length(&buf, ASN1_SEQUENCE, &seq_len) != COMPOSITE_OK) {
        return COMPOSITE_ERR_DECODE_FAILED;
    }

    /* First BIT STRING: Ed25519 signature */
    if (asn1_decode_bit_string(&buf, ed25519_sig, ed25519_len,

```

```

        ED25519_SIG_SIZE) != COMPOSITE_OK) {
    return COMPOSITE_ERR_DECODE_FAILED;
}

/* Second BIT STRING: ML-DSA-65 signature */
if (asn1_decode_bit_string(&buf, mldsa65_sig, mldsa65_len,
    MLDSA65_SIG_SIZE) != COMPOSITE_OK) {
    return COMPOSITE_ERR_DECODE_FAILED;
}

return COMPOSITE_OK;
}

/* Extract public key components from composite SPKI */
static int extract_composite_public_key(
    const spki_t *spki,
    uint8_t *ed25519_pk, size_t *ed25519_len,
    uint8_t *mldsa65_pk, size_t *mldsa65_len
) {
    /* Parse CompositePublicKey SEQUENCE */
    asn1_buffer_t buf = {
        .data = (uint8_t *)spki->public_key,
        .len = spki->public_key_len,
        .capacity = spki->public_key_len,
        .pos = 0
    };

    size_t seq_len;
    if (asn1_decode_tag_length(&buf, ASN1_SEQUENCE, &seq_len) != COMPOSITE_OK) {
        return COMPOSITE_ERR_DECODE_FAILED;
    }

    /* Parse first component SPKI (Ed25519) */
    size_t spki1_len;
    if (asn1_decode_tag_length(&buf, ASN1_SEQUENCE, &spki1_len) != COMPOSITE_OK) {
        return COMPOSITE_ERR_DECODE_FAILED;
    }

    /* Skip AlgorithmIdentifier, get public key */
    size_t alg_len;
    if (asn1_decode_tag_length(&buf, ASN1_SEQUENCE, &alg_len) != COMPOSITE_OK) {
        return COMPOSITE_ERR_DECODE_FAILED;
    }
    buf.pos += alg_len; /* Skip algorithm identifier content */

    if (asn1_decode_bit_string(&buf, ed25519_pk, ed25519_len,
        ED25519_PK_SIZE) != COMPOSITE_OK) {
        return COMPOSITE_ERR_DECODE_FAILED;
    }

    /* Parse second component SPKI (ML-DSA-65) */
    size_t spki2_len;
    if (asn1_decode_tag_length(&buf, ASN1_SEQUENCE, &spki2_len) != COMPOSITE_OK) {
        return COMPOSITE_ERR_DECODE_FAILED;
    }

    if (asn1_decode_tag_length(&buf, ASN1_SEQUENCE, &alg_len) != COMPOSITE_OK) {
        return COMPOSITE_ERR_DECODE_FAILED;
    }
    buf.pos += alg_len;

```

```

        if (asn1_decode_bit_string(&buf, mldsa65_pk, mldsa65_len,
                                   MLDSA65_PK_SIZE) != COMPOSITE_OK) {
            return COMPOSITE_ERR_DECODE_FAILED;
        }

        return COMPOSITE_OK;
    }

    /* Verify single certificate against issuer */
    static chain_validation_result_t verify_certificate(
        const certificate_t *cert,
        const certificate_t *issuer,
        chain_validation_ctx_t *ctx,
        size_t cert_index
    ) {
        bool classical_ok = false;
        bool pqc_ok = false;

        /* Extract issuer's public key components */
        uint8_t issuer_ed25519_pk[ED25519_PK_SIZE];
        uint8_t issuer_mldsa65_pk[MLDSA65_PK_SIZE];
        size_t ed25519_len, mldsa65_len;

        int ret = extract_composite_public_key(
            &issuer->tbs.subject_pki,
            issuer_ed25519_pk, &ed25519_len,
            issuer_mldsa65_pk, &mldsa65_len
        );

        if (ret != COMPOSITE_OK) {
            snprintf(ctx->error_message, sizeof(ctx->error_message),
                    "Failed to extract issuer public key at cert %zu", cert_index);
            return CHAIN_ERR_BOTH_FAILED;
        }

        /* Extract signature components */
        uint8_t ed25519_sig[ED25519_SIG_SIZE];
        uint8_t mldsa65_sig[MLDSA65_SIG_SIZE];
        size_t ed25519_sig_len, mldsa65_sig_len;

        ret = extract_composite_signature(
            cert->signature, cert->signature_len,
            ed25519_sig, &ed25519_sig_len,
            mldsa65_sig, &mldsa65_sig_len
        );

        if (ret != COMPOSITE_OK) {
            snprintf(ctx->error_message, sizeof(ctx->error_message),
                    "Failed to extract signature at cert %zu", cert_index);
            return CHAIN_ERR_BOTH_FAILED;
        }

        /* Verify Ed25519 signature */
        classical_ok = verify_ed25519_signature(
            cert->tbs_der, cert->tbs_der_len,
            ed25519_sig, ed25519_sig_len,
            issuer_ed25519_pk, ed25519_len
        );

        ctx->classical_verified[cert_index] = classical_ok;
    }

```

```

/* Verify ML-DSA-65 signature */
pqc_ok = verify_mldsa65_signature(
    cert->tbs_der, cert->tbs_der_len,
    mldsa65_sig, mldsa65_sig_len,
    issuer_mldsa65_pk, mldsa65_len
);
ctx->pqc_verified[cert_index] = pqc_ok;

/* Apply validation mode */
switch (ctx->mode) {
    case MODE_STRICT:
        if (!classical_ok && !pqc_ok) {
            snprintf(ctx->error_message, sizeof(ctx->error_message),
                "Both signatures failed at cert %zu", cert_index);
            return CHAIN_ERR_BOTH_FAILED;
        }
        if (!classical_ok) {
            snprintf(ctx->error_message, sizeof(ctx->error_message),
                "Classical signature failed at cert %zu", cert_index);
            return CHAIN_ERR_CLASSICAL_FAILED;
        }
        if (!pqc_ok) {
            snprintf(ctx->error_message, sizeof(ctx->error_message),
                "PQC signature failed at cert %zu", cert_index);
            return CHAIN_ERR_PQC_FAILED;
        }
        break;

    case MODE_RELAXED:
        if (!classical_ok && !pqc_ok) {
            snprintf(ctx->error_message, sizeof(ctx->error_message),
                "Both signatures failed at cert %zu", cert_index);
            return CHAIN_ERR_BOTH_FAILED;
        }
        /* Either signature is sufficient */
        break;

    case MODE_MIGRATION:
        if (!classical_ok) {
            snprintf(ctx->error_message, sizeof(ctx->error_message),
                "Classical signature required but failed at cert %zu",
                cert_index);
            return CHAIN_ERR_CLASSICAL_FAILED;
        }
        /* PQC failure is just a warning in migration mode */
        if (!pqc_ok) {
            /* Log warning but continue */
            fprintf(stderr, "Warning: PQC signature failed at cert %zu\n",
                cert_index);
        }
        break;
}

return CHAIN_VALID;
}

/* Main chain validation function */
chain_validation_result_t validate_certificate_chain(
    const certificate_t *chain[],

```

```

size_t chain_length,
const certificate_t *trust_anchor,
chain_validation_ctx_t *ctx
) {
    chain_validation_result_t result;

    /* Initialize context */
    memset(ctx->classical_verified, 0, sizeof(ctx->classical_verified));
    memset(ctx->pqc_verified, 0, sizeof(ctx->pqc_verified));
    ctx->error_message[0] = '\0';

    /* Check chain length */
    if (chain_length == 0) {
        snprintf(ctx->error_message, sizeof(ctx->error_message),
            "Empty certificate chain");
        return CHAIN_ERR_EMPTY;
    }

    if (chain_length > ctx->max_chain_length) {
        snprintf(ctx->error_message, sizeof(ctx->error_message),
            "Chain too long: %zu > %zu", chain_length, ctx->max_chain_length);
        return CHAIN_ERR_TOO_LONG;
    }

    /* Verify each certificate in chain */
    for (size_t i = 0; i < chain_length; i++) {
        const certificate_t *issuer;

        if (i == chain_length - 1) {
            /* Last certificate verified against trust anchor */
            issuer = trust_anchor;
        } else {
            /* Verify against next certificate in chain */
            issuer = chain[i + 1];
        }

        result = verify_certificate(chain[i], issuer, ctx, i);
        if (result != CHAIN_VALID) {
            return result;
        }

        /* Check CA constraint for non-leaf certificates */
        if (i > 0 && ctx->require_ca_basic_constraint) {
            /* In production, parse basicConstraints extension */
            /* Simplified check here */
        }
    }

    return CHAIN_VALID;
}

/* Print validation report */
void print_validation_report(
    const chain_validation_ctx_t *ctx,
    size_t chain_length,
    chain_validation_result_t result
) {
    printf("\n=== Certificate Chain Validation Report ===\n\n");
    printf("Mode: %s\n",
        ctx->mode == MODE_STRICT ? "STRICT" :

```

```

        ctx->mode == MODE_RELAXED ? "RELAXED" : "MIGRATION");
printf("Chain length: %zu\n\n", chain_length);

printf("Per-certificate results:\n");
printf("%-6s %-12s %-12s\n", "Cert", "Classical", "PQC");
printf("-----\n");

for (size_t i = 0; i < chain_length; i++) {
    printf("%-6zu %-12s %-12s\n",
        i,
        ctx->classical_verified[i] ? "VERIFIED" : "FAILED",
        ctx->pqc_verified[i] ? "VERIFIED" : "FAILED");
}

printf("\nOverall result: %s\n",
    result == CHAIN_VALID ? "VALID" : "INVALID");

if (result != CHAIN_VALID) {
    printf("Error: %s\n", ctx->error_message);
}
}

```

Exercise 8.4.4: PEM Format Conversion *(Intermediate)*

Write functions to convert between DER and PEM formats for composite keys, handling proper line wrapping and headers.

Solution:

```

#include "composite_keys.h"
#include <string.h>
#include <ctype.h>

/* Base64 encoding/decoding tables */
static const char b64_encode_table[] =
    "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/";

static const int b64_decode_table[256] = {
    -1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,
    -1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,
    -1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,62,-1,-1,-1,63,
    52,53,54,55,56,57,58,59,60,61,-1,-1,-1,-1,-1,-1,
    -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9,10,11,12,13,14,
    15,16,17,18,19,20,21,22,23,24,25,-1,-1,-1,-1,-1,
    -1,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,
    41,42,43,44,45,46,47,48,49,50,51,-1,-1,-1,-1,-1,
    /* Remaining 128 entries are -1 */
};

/* PEM label definitions */
#define PEM_LABEL_PUBLIC_KEY      "PUBLIC KEY"
#define PEM_LABEL_PRIVATE_KEY    "PRIVATE KEY"
#define PEM_LABEL_CERTIFICATE    "CERTIFICATE"
#define PEM_LABEL_CERT_REQUEST   "CERTIFICATE REQUEST"

#define PEM_LINE_LENGTH          64
#define PEM_BEGIN_PREFIX        "-----BEGIN "

```

```

#define PEM_END_PREFIX          "-----END "
#define PEM_SUFFIX              "-----"

/* Calculate base64 encoded length */
static size_t base64_encoded_length(size_t input_len) {
    return ((input_len + 2) / 3) * 4;
}

/* Calculate base64 decoded length (approximate) */
static size_t base64_decoded_length(size_t input_len) {
    return (input_len * 3) / 4;
}

/* Base64 encode a block of data */
static int base64_encode(
    const uint8_t *input, size_t input_len,
    char *output, size_t *output_len
) {
    size_t required = base64_encoded_length(input_len);

    if (*output_len < required + 1) {
        *output_len = required + 1;
        return COMPOSITE_ERR_BUFFER_TOO_SMALL;
    }

    size_t i, j;
    for (i = 0, j = 0; i < input_len; i += 3, j += 4) {
        uint32_t n = ((uint32_t)input[i]) << 16;

        if (i + 1 < input_len) {
            n |= ((uint32_t)input[i + 1]) << 8;
        }
        if (i + 2 < input_len) {
            n |= input[i + 2];
        }

        output[j] = b64_encode_table[(n >> 18) & 0x3F];
        output[j + 1] = b64_encode_table[(n >> 12) & 0x3F];
        output[j + 2] = (i + 1 < input_len)
            ? b64_encode_table[(n >> 6) & 0x3F]
            : '=';
        output[j + 3] = (i + 2 < input_len)
            ? b64_encode_table[n & 0x3F]
            : '=';
    }

    output[j] = '\0';
    *output_len = j;

    return COMPOSITE_OK;
}

/* Base64 decode a block of data */
static int base64_decode(
    const char *input, size_t input_len,
    uint8_t *output, size_t *output_len
) {
    /* Skip whitespace and calculate actual length */
    size_t clean_len = 0;
    for (size_t i = 0; i < input_len; i++) {

```

```

        if (!isspace((unsigned char)input[i])) {
            clean_len++;
        }
    }

    if (clean_len % 4 != 0) {
        return COMPOSITE_ERR_DECODE_FAILED;
    }

    size_t required = base64_decoded_length(clean_len);
    if (*output_len < required) {
        *output_len = required;
        return COMPOSITE_ERR_BUFFER_TOO_SMALL;
    }

    size_t j = 0;
    uint32_t accum = 0;
    int bits = 0;

    for (size_t i = 0; i < input_len; i++) {
        char c = input[i];

        if (isspace((unsigned char)c)) {
            continue;
        }

        if (c == '=') {
            break; /* Padding reached */
        }

        int val = b64_decode_table[(unsigned char)c];
        if (val < 0) {
            return COMPOSITE_ERR_DECODE_FAILED;
        }

        accum = (accum << 6) | val;
        bits += 6;

        if (bits >= 8) {
            bits -= 8;
            output[j++] = (uint8_t)((accum >> bits) & 0xFF);
        }
    }

    *output_len = j;
    return COMPOSITE_OK;
}

/* Convert DER to PEM format */
int der_to_pem_full(
    const uint8_t *der, size_t der_len,
    const char *label,
    char *pem, size_t *pem_len
) {
    if (!der || !label || !pem || !pem_len) {
        return COMPOSITE_ERR_INVALID_INPUT;
    }

    /* Calculate required size */
    size_t b64_len = base64_encoded_length(der_len);

```



```

size_t line_breaks = (b64_len + PEM_LINE_LENGTH - 1) / PEM_LINE_LENGTH;
size_t header_len = strlen(PEM_BEGIN_PREFIX) + strlen(label) +
                    strlen(PEM_SUFFIX) + 1;
size_t footer_len = strlen(PEM_END_PREFIX) + strlen(label) +
                    strlen(PEM_SUFFIX) + 1;
size_t total = header_len + b64_len + line_breaks + footer_len + 1;

if (*pem_len < total) {
    *pem_len = total;
    return COMPOSITE_ERR_BUFFER_TOO_SMALL;
}

/* Write header */
int pos = snprintf(pem, *pem_len, "%s%s%s\n", PEM_BEGIN_PREFIX, label, PEM_SUFFIX);

/* Base64 encode */
char *b64_buf = malloc(b64_len + 1);
if (!b64_buf) {
    return COMPOSITE_ERR_BUFFER_TOO_SMALL;
}

size_t b64_actual = b64_len + 1;
int ret = base64_encode(der, der_len, b64_buf, &b64_actual);
if (ret != COMPOSITE_OK) {
    free(b64_buf);
    return ret;
}

/* Write base64 with line breaks */
for (size_t i = 0; i < b64_actual; i += PEM_LINE_LENGTH) {
    size_t line_len = b64_actual - i;
    if (line_len > PEM_LINE_LENGTH) {
        line_len = PEM_LINE_LENGTH;
    }

    memcpy(pem + pos, b64_buf + i, line_len);
    pos += line_len;
    pem[pos++] = '\n';
}

free(b64_buf);

/* Write footer */
pos += snprintf(pem + pos, *pem_len - pos, "%s%s%s\n", PEM_END_PREFIX, label,
PEM_SUFFIX);

pem[pos] = '\0';
*pem_len = pos;

return COMPOSITE_OK;
}

/* Convert PEM to DER format */
int pem_to_der_full(
    const char *pem, size_t pem_len,
    uint8_t *der, size_t *der_len,
    char *label_out, size_t label_out_len
) {
    if (!pem || !der || !der_len) {
        return COMPOSITE_ERR_INVALID_INPUT;
    }

```

```

}

/* Find BEGIN marker */
const char *begin = strstr(pem, PEM_BEGIN_PREFIX);
if (!begin) {
    return COMPOSITE_ERR_DECODE_FAILED;
}

/* Extract label */
const char *label_start = begin + strlen(PEM_BEGIN_PREFIX);
const char *label_end = strstr(label_start, PEM_SUFFIX);
if (!label_end) {
    return COMPOSITE_ERR_DECODE_FAILED;
}

if (label_out && label_out_len > 0) {
    size_t label_len = label_end - label_start;
    if (label_len >= label_out_len) {
        label_len = label_out_len - 1;
    }
    memcpy(label_out, label_start, label_len);
    label_out[label_len] = '\0';
}

/* Find base64 content start (after header line) */
const char *content_start = label_end + strlen(PEM_SUFFIX);
while (*content_start == '\n' || *content_start == '\r') {
    content_start++;
}

/* Find END marker */
const char *end = strstr(content_start, PEM_END_PREFIX);
if (!end) {
    return COMPOSITE_ERR_DECODE_FAILED;
}

/* Decode base64 content */
size_t content_len = end - content_start;

/* Remove trailing whitespace */
while (content_len > 0 &&
       (content_start[content_len - 1] == '\n' ||
        content_start[content_len - 1] == '\r')) {
    content_len--;
}

return base64_decode(content_start, content_len, der, der_len);
}

/* High-level functions for specific key types */

int composite_public_key_to_pem(
    const composite_keypair_t *keypair,
    char *pem, size_t *pem_len
) {
    asn1_buffer_t spki_buf;
    int ret;

    ret = asn1_buffer_init(&spki_buf, MAX_SPKI_SIZE);
    if (ret != COMPOSITE_OK) {

```

```

        return ret;
    }

    ret = encode_composite_spki(&spki_buf, keypair);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&spki_buf);
        return ret;
    }

    ret = der_to_pem_full(spki_buf.data, spki_buf.len,
                          PEM_LABEL_PUBLIC_KEY, pem, pem_len);

    asn1_buffer_free(&spki_buf);
    return ret;
}

int composite_private_key_to_pem(
    const composite_keypair_t *keypair,
    char *pem, size_t *pem_len
) {
    asn1_buffer_t pkcs8_buf;
    int ret;

    ret = asn1_buffer_init(&pkcs8_buf, MAX_PKCS8_SIZE);
    if (ret != COMPOSITE_OK) {
        return ret;
    }

    ret = encode_pkcs8_private_key(&pkcs8_buf, keypair);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&pkcs8_buf);
        return ret;
    }

    ret = der_to_pem_full(pkcs8_buf.data, pkcs8_buf.len,
                          PEM_LABEL_PRIVATE_KEY, pem, pem_len);

    asn1_buffer_free(&pkcs8_buf);
    return ret;
}

int certificate_to_pem(
    const certificate_t *cert,
    char *pem, size_t *pem_len
) {
    asn1_buffer_t cert_buf;
    int ret;

    ret = asn1_buffer_init(&cert_buf, MAX_CERT_SIZE);
    if (ret != COMPOSITE_OK) {
        return ret;
    }

    ret = encode_certificate(&cert_buf, cert);
    if (ret != COMPOSITE_OK) {
        asn1_buffer_free(&cert_buf);
        return ret;
    }

    ret = der_to_pem_full(cert_buf.data, cert_buf.len,

```

```

        PEM_LABEL_CERTIFICATE, pem, pem_len);

    asn1_buffer_free(&cert_buf);
    return ret;
}

/* Test function demonstrating PEM conversion */
void test_pem_conversion(void) {
    printf("=== PEM Conversion Test ===\n\n");

    /* Create test keypair */
    composite_keypair_t keypair;
    if (create_test_composite_keypair(&keypair) != COMPOSITE_OK) {
        printf("Failed to create test keypair\n");
        return;
    }

    /* Convert to PEM */
    char pem[32768];
    size_t pem_len = sizeof(pem);

    if (composite_public_key_to_pem(&keypair, pem, &pem_len) == COMPOSITE_OK) {
        printf("Public Key PEM (%zu bytes):\n%s\n", pem_len, pem);
    }

    pem_len = sizeof(pem);
    if (composite_private_key_to_pem(&keypair, pem, &pem_len) == COMPOSITE_OK) {
        printf("Private Key PEM (%zu bytes):\n%s\n", pem_len, pem);
    }

    /* Test round-trip: PEM -> DER -> PEM */
    uint8_t der[MAX_PKCS8_SIZE];
    size_t der_len = sizeof(der);
    char label[64];

    if (pem_to_der_full(pem, pem_len, der, &der_len, label, sizeof(label))
        == COMPOSITE_OK) {
        printf("Round-trip: Decoded %zu bytes, label='%s'\n", der_len, label);

        /* Re-encode */
        char pem2[32768];
        size_t pem2_len = sizeof(pem2);

        if (der_to_pem_full(der, der_len, label, pem2, &pem2_len)
            == COMPOSITE_OK) {
            if (pem_len == pem2_len && memcmp(pem, pem2, pem_len) == 0) {
                printf("Round-trip successful: PEM matches\n");
            } else {
                printf("Round-trip mismatch!\n");
            }
        }
    }

    composite_keypair_free(&keypair);
}

```

Unit 8.4 Summary

This unit covered the critical aspects of encoding and formatting hybrid cryptographic keys and certificates:

Key Concepts:

1. **ASN.1 Fundamentals:** DER encoding rules, tag-length-value structure, and proper handling of SEQUENCE, BIT STRING, and OCTET STRING types for cryptographic data
2. **Object Identifiers:** OID hierarchy for post-quantum algorithms, composite algorithm OIDs from IETF drafts, and proper OID encoding
3. **X.509 Certificates:** Structure of hybrid certificates, three approaches (composite SPKI, alternative public key extension, subject information access), and TBSCertificate encoding
4. **PKCS#8 Private Keys:** OneAsymmetricKey structure for composite private keys, version handling, and optional public key inclusion
5. **Certificate Chains:** Homogeneous, mixed, and parallel chain strategies, path validation with multiple algorithms, and verification modes
6. **Interoperability:** OpenSSL provider integration, PKCS#11 mechanism definitions, and PEM format conversion

Implementation Considerations:

- Always use DER (not BER) for cryptographic encoding to ensure deterministic outputs
- Carefully handle length encoding, especially for large PQC keys that require multi-byte lengths
- Securely clear sensitive key material from memory after use
- Consider both strict and relaxed validation modes for migration scenarios
- Test interoperability with existing PKI tools and libraries

What's Next: Unit 8.5 covers implementation strategies and practical integration patterns for hybrid cryptography in real-world systems, including library architecture, performance tuning, and deployment considerations.

Unit 8.5: Implementation and Integration

Prerequisites: Units 8.1-8.4 (Hybrid Cryptography and key formats)

Estimated Time: 4-5 hours

This unit addresses the practical challenges of implementing hybrid cryptography in real-world systems. We cover library architecture patterns, performance optimization techniques, deployment strategies, and integration with existing infrastructure.

Learning Objectives

After completing this unit, you will be able to:

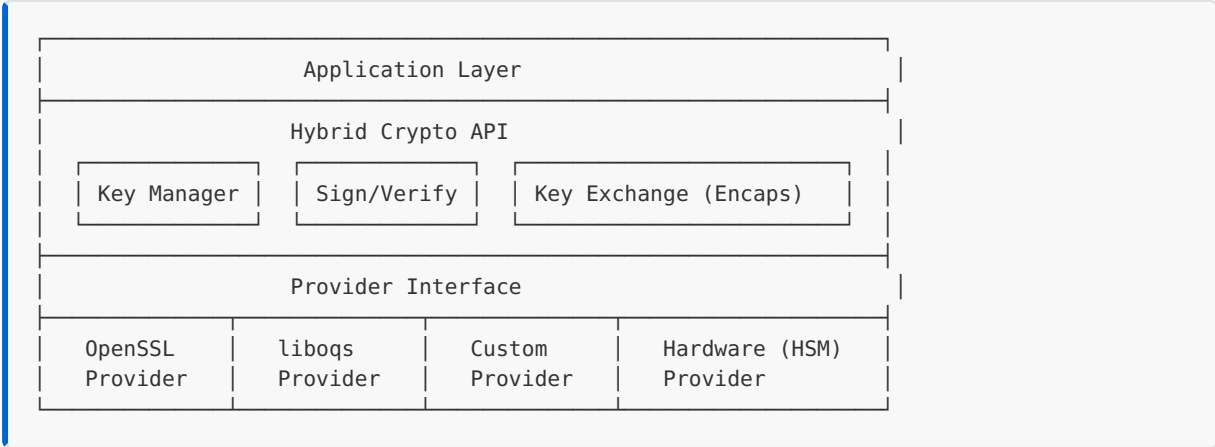
1. Design a modular library architecture for hybrid cryptography
2. Implement algorithm-agnostic interfaces for cryptographic operations
3. Apply performance optimization techniques for hybrid systems
4. Plan and execute migration strategies from classical to hybrid cryptography
5. Integrate hybrid cryptography with existing protocols and infrastructure
6. Handle errors and edge cases in production hybrid implementations

8.5.1 Library Architecture Patterns

A well-designed hybrid cryptography library must balance flexibility, security, and performance. We explore architectural patterns that achieve these goals.

Provider-Based Architecture

The provider pattern allows runtime selection of cryptographic implementations:



Core Interface Design

```
/* hybrid_crypto.h - Core API header */

#ifndef HYBRID_CRYPT0_H
#define HYBRID_CRYPT0_H

#include <stdint.h>
#include <stddef.h>

/* Version information */
#define HYBRID_CRYPT0_VERSION_MAJOR 1
#define HYBRID_CRYPT0_VERSION_MINOR 0
#define HYBRID_CRYPT0_VERSION_PATCH 0

/* Error codes */
typedef enum {
    HC_SUCCESS = 0,
    HC_ERROR_INVALID_PARAM = -1,
    HC_ERROR_MEMORY = -2,
    HC_ERROR_CRYPT0 = -3,
    HC_ERROR_NOT_SUPPORTED = -4,
    HC_ERROR_KEY_SIZE = -5,
    HC_ERROR_SIGNATURE = -6,
    HC_ERROR_VERIFICATION = -7,
    HC_ERROR_PROVIDER = -8,
    HC_ERROR_STATE = -9,
    HC_ERROR_BUFFER_SIZE = -10,
    HC_ERROR_RNG = -11,
    HC_ERROR_INTERNAL = -100
} hc_error_t;

/* Algorithm identifiers */
typedef enum {
    /* Classical signature algorithms */
    HC_ALG_ED25519 = 0x0001,
    HC_ALG_ECDSA_P256 = 0x0002,
    HC_ALG_ECDSA_P384 = 0x0003,
    HC_ALG_RSA_PSS_2048 = 0x0004,
    HC_ALG_RSA_PSS_3072 = 0x0005,

    /* PQC signature algorithms */
    HC_ALG_MLDSA44 = 0x0101,
    HC_ALG_MLDSA65 = 0x0102,
    HC_ALG_MLDSA87 = 0x0103,
    HC_ALG_SLHDSA_SHA2_128S = 0x0111,
    HC_ALG_SLHDSA_SHA2_128F = 0x0112,
    HC_ALG_SLHDSA_SHA2_192S = 0x0113,
    HC_ALG_SLHDSA_SHA2_192F = 0x0114,
    HC_ALG_SLHDSA_SHA2_256S = 0x0115,
    HC_ALG_SLHDSA_SHA2_256F = 0x0116,

    /* Classical KEM algorithms */
    HC_ALG_X25519 = 0x1001,
    HC_ALG_ECDH_P256 = 0x1002,
    HC_ALG_ECDH_P384 = 0x1003,

    /* PQC KEM algorithms */
    HC_ALG_MLKEM512 = 0x1101,
```

```

    HC_ALG_MLKEM768 = 0x1102,
    HC_ALG_MLKEM1024 = 0x1103,

    /* Hybrid combinations */
    HC_ALG_HYBRID_ED25519_MLDSA65 = 0x2001,
    HC_ALG_HYBRID_P256_MLDSA65 = 0x2002,
    HC_ALG_HYBRID_X25519_MLKEM768 = 0x2101,
    HC_ALG_HYBRID_P256_MLKEM768 = 0x2102
} hc_algorithm_t;

/* Key types */
typedef enum {
    HC_KEY_PUBLIC,
    HC_KEY_PRIVATE,
    HC_KEY_KEYPAIR
} hc_key_type_t;

/* Opaque key handle */
typedef struct hc_key_st hc_key_t;

/* Opaque context handles */
typedef struct hc_sign_ctx_st hc_sign_ctx_t;
typedef struct hc_verify_ctx_st hc_verify_ctx_t;
typedef struct hc_kem_ctx_st hc_kem_ctx_t;

/* Provider handle */
typedef struct hc_provider_st hc_provider_t;

/* Callback for random number generation */
typedef int (*hc_rng_callback_t)(void *ctx, uint8_t *buf, size_t len);

/* Library initialization and cleanup */
hc_error_t hc_init(void);
void hc_cleanup(void);
const char *hc_version_string(void);
const char *hc_error_string(hc_error_t err);

/* Provider management */
hc_error_t hc_provider_load(const char *name, hc_provider_t **provider);
hc_error_t hc_provider_unload(hc_provider_t *provider);
hc_error_t hc_provider_set_default(hc_provider_t *provider);

/* Algorithm queries */
int hc_algorithm_supported(hc_algorithm_t alg);
size_t hc_algorithm_public_key_size(hc_algorithm_t alg);
size_t hc_algorithm_private_key_size(hc_algorithm_t alg);
size_t hc_algorithm_signature_size(hc_algorithm_t alg);
size_t hc_algorithm_ciphertext_size(hc_algorithm_t alg);
size_t hc_algorithm_shared_secret_size(hc_algorithm_t alg);
const char *hc_algorithm_name(hc_algorithm_t alg);

/* Key management */
hc_error_t hc_key_generate(hc_algorithm_t alg, hc_key_t **key);
hc_error_t hc_key_generate_with_rng(hc_algorithm_t alg,
                                     hc_rng_callback_t rng,
                                     void *rng_ctx,
                                     hc_key_t **key);
hc_error_t hc_key_import(hc_algorithm_t alg,
                         hc_key_type_t type,
                         const uint8_t *data,

```



```

        size_t len,
        hc_key_t **key);
hc_error_t hc_key_export(const hc_key_t *key,
        hc_key_type_t type,
        uint8_t *data,
        size_t *len);
hc_algorithm_t hc_key_algorithm(const hc_key_t *key);
void hc_key_free(hc_key_t *key);

/* Signature operations - single-shot */
hc_error_t hc_sign(const hc_key_t *private_key,
        const uint8_t *message,
        size_t message_len,
        uint8_t *signature,
        size_t *signature_len);

hc_error_t hc_verify(const hc_key_t *public_key,
        const uint8_t *message,
        size_t message_len,
        const uint8_t *signature,
        size_t signature_len);

/* Signature operations - streaming */
hc_error_t hc_sign_init(hc_sign_ctx_t **ctx, const hc_key_t *private_key);
hc_error_t hc_sign_update(hc_sign_ctx_t *ctx,
        const uint8_t *data,
        size_t len);
hc_error_t hc_sign_final(hc_sign_ctx_t *ctx,
        uint8_t *signature,
        size_t *signature_len);
void hc_sign_ctx_free(hc_sign_ctx_t *ctx);

hc_error_t hc_verify_init(hc_verify_ctx_t **ctx, const hc_key_t *public_key);
hc_error_t hc_verify_update(hc_verify_ctx_t *ctx,
        const uint8_t *data,
        size_t len);
hc_error_t hc_verify_final(hc_verify_ctx_t *ctx,
        const uint8_t *signature,
        size_t signature_len);
void hc_verify_ctx_free(hc_verify_ctx_t *ctx);

/* KEM operations */
hc_error_t hc_kem_encapsulate(const hc_key_t *public_key,
        uint8_t *ciphertext,
        size_t *ciphertext_len,
        uint8_t *shared_secret,
        size_t *shared_secret_len);

hc_error_t hc_kem_decapsulate(const hc_key_t *private_key,
        const uint8_t *ciphertext,
        size_t ciphertext_len,
        uint8_t *shared_secret,
        size_t *shared_secret_len);

/* Hybrid-specific operations */

```

```

hc_error_t hc_hybrid_get_classical_key(const hc_key_t *hybrid_key,
                                       hc_key_t **classical_key);
hc_error_t hc_hybrid_get_pqc_key(const hc_key_t *hybrid_key,
                                 hc_key_t **pqc_key);

#endif /* HYBRID_CRYPT0_H */

```

Internal Key Structure

```

/* hybrid_crypto_internal.h - Internal structures */

#ifndef HYBRID_CRYPT0_INTERNAL_H
#define HYBRID_CRYPT0_INTERNAL_H

#include "hybrid_crypto.h"
#include <pthread.h>

/* Maximum sizes */
#define HC_MAX_PUBLIC_KEY_SIZE 4096
#define HC_MAX_PRIVATE_KEY_SIZE 8192
#define HC_MAX_SIGNATURE_SIZE 8192
#define HC_MAX_CIPHertext_SIZE 2048
#define HC_MAX_SHARED_SECRET 64

/* Key structure */
struct hc_key_st {
    hc_algorithm_t algorithm;
    hc_key_type_t type;
    uint32_t flags;

    /* Key material */
    uint8_t *public_key;
    size_t public_key_len;
    uint8_t *private_key;
    size_t private_key_len;

    /* For hybrid keys */
    struct hc_key_st *classical_component;
    struct hc_key_st *pqc_component;

    /* Provider that created this key */
    hc_provider_t *provider;

    /* Reference count for thread safety */
    int refcount;
    pthread_mutex_t lock;
};

/* Signing context */
struct hc_sign_ctx_st {
    hc_key_t *key;
    hc_algorithm_t algorithm;

    /* Hash context for pre-hashing */
    void *hash_ctx;
    size_t hash_size;

    /* Accumulated message for non-prehash */

```

```

uint8_t *message_buf;
size_t message_len;
size_t message_cap;

/* State */
int initialized;
int finalized;
};

/* Verification context */
struct hc_verify_ctx_st {
    hc_key_t *key;
    hc_algorithm_t algorithm;

    void *hash_ctx;
    size_t hash_size;

    uint8_t *message_buf;
    size_t message_len;
    size_t message_cap;

    int initialized;
    int finalized;
};

/* KEM context */
struct hc_kem_ctx_st {
    hc_key_t *key;
    hc_algorithm_t algorithm;

    uint8_t *ephemeral_secret;
    size_t ephemeral_secret_len;

    int initialized;
};

/* Provider function table */
typedef struct {
    const char *name;

    /* Lifecycle */
    hc_error_t (*init)(hc_provider_t *prov);
    void (*cleanup)(hc_provider_t *prov);

    /* Algorithm support query */
    int (*supports)(hc_provider_t *prov, hc_algorithm_t alg);

    /* Key operations */
    hc_error_t (*keygen)(hc_provider_t *prov,
                        hc_algorithm_t alg,
                        hc_key_t *key);
    hc_error_t (*key_import)(hc_provider_t *prov,
                            hc_algorithm_t alg,
                            hc_key_type_t type,
                            const uint8_t *data,
                            size_t len,
                            hc_key_t *key);
    hc_error_t (*key_export)(hc_provider_t *prov,
                            const hc_key_t *key,
                            hc_key_type_t type,

```

```

        uint8_t *data,
        size_t *len);

/* Signature operations */
hc_error_t (*sign)(hc_provider_t *prov,
                   const hc_key_t *key,
                   const uint8_t *msg,
                   size_t msg_len,
                   uint8_t *sig,
                   size_t *sig_len);
hc_error_t (*verify)(hc_provider_t *prov,
                     const hc_key_t *key,
                     const uint8_t *msg,
                     size_t msg_len,
                     const uint8_t *sig,
                     size_t sig_len);

/* KEM operations */
hc_error_t (*encaps)(hc_provider_t *prov,
                     const hc_key_t *key,
                     uint8_t *ct,
                     size_t *ct_len,
                     uint8_t *ss,
                     size_t *ss_len);
hc_error_t (*decaps)(hc_provider_t *prov,
                     const hc_key_t *key,
                     const uint8_t *ct,
                     size_t ct_len,
                     uint8_t *ss,
                     size_t *ss_len);
} hc_provider_funcs_t;

/* Provider structure */
struct hc_provider_st {
    const char *name;
    const hc_provider_funcs_t *funcs;
    void *ctx; /* Provider-specific context */
    int refcount;
    pthread_mutex_t lock;
};

/* Secure memory functions */
void *hc_secure_alloc(size_t size);
void hc_secure_free(void *ptr, size_t size);
void hc_secure_clear(void *ptr, size_t size);

#endif /* HYBRID_CRYPT_INTERNAL_H */

```

8.5.2 Implementation Strategies

Thread Safety

Hybrid implementations must handle concurrent access safely:

```
/* thread_safe.c - Thread-safe key operations */

#include "hybrid_crypto_internal.h"
#include <string.h>
#include <stdlib.h>

/* Thread-local storage for error context */
static __thread struct {
    hc_error_t last_error;
    char error_message[256];
    int error_line;
    const char *error_file;
} tls_error_ctx;

void hc_set_error(hc_error_t err, const char *file, int line,
                  const char *fmt, ...) {
    tls_error_ctx.last_error = err;
    tls_error_ctx.error_file = file;
    tls_error_ctx.error_line = line;

    va_list args;
    va_start(args, fmt);
    vsnprintf(tls_error_ctx.error_message,
              sizeof(tls_error_ctx.error_message), fmt, args);
    va_end(args);
}

hc_error_t hc_get_last_error(void) {
    return tls_error_ctx.last_error;
}

const char *hc_get_error_message(void) {
    return tls_error_ctx.error_message;
}

#define HC_SET_ERROR(err, ...) \
    hc_set_error(err, __FILE__, __LINE__, __VA_ARGS__)

/* Reference counting for keys */
hc_key_t *hc_key_ref(hc_key_t *key) {
    if (key == NULL) return NULL;

    pthread_mutex_lock(&key->lock);
    key->refcount++;
    pthread_mutex_unlock(&key->lock);

    return key;
}

void hc_key_unref(hc_key_t *key) {
    if (key == NULL) return;

    pthread_mutex_lock(&key->lock);
```

```

    int refcount = --key->refcount;
    pthread_mutex_unlock(&key->lock);

    if (refcount == 0) {
        hc_key_free(key);
    }
}

/* Thread-safe key generation */
hc_error_t hc_key_generate_threadsafe(hc_algorithm_t alg, hc_key_t **key) {
    if (key == NULL) {
        HC_SET_ERROR(HC_ERROR_INVALID_PARAM, "key output is NULL");
        return HC_ERROR_INVALID_PARAM;
    }

    /* Allocate key structure */
    hc_key_t *new_key = hc_secure_alloc(sizeof(hc_key_t));
    if (new_key == NULL) {
        HC_SET_ERROR(HC_ERROR_MEMORY, "failed to allocate key");
        return HC_ERROR_MEMORY;
    }

    memset(new_key, 0, sizeof(hc_key_t));

    /* Initialize mutex */
    pthread_mutexattr_t attr;
    pthread_mutexattr_init(&attr);
    pthread_mutexattr_settype(&attr, PTHREAD_MUTEX_RECURSIVE);
    pthread_mutex_init(&new_key->lock, &attr);
    pthread_mutexattr_destroy(&attr);

    new_key->refcount = 1;
    new_key->algorithm = alg;
    new_key->type = HC_KEY_KEYPAIR;

    /* Get provider and generate key */
    hc_provider_t *prov = hc_get_provider_for_algorithm(alg);
    if (prov == NULL) {
        HC_SET_ERROR(HC_ERROR_NOT_SUPPORTED,
            "no provider for algorithm %d", alg);
        pthread_mutex_destroy(&new_key->lock);
        hc_secure_free(new_key, sizeof(hc_key_t));
        return HC_ERROR_NOT_SUPPORTED;
    }

    /* Lock provider during key generation */
    pthread_mutex_lock(&prov->lock);
    hc_error_t err = prov->funcs->keygen(prov, alg, new_key);
    pthread_mutex_unlock(&prov->lock);

    if (err != HC_SUCCESS) {
        pthread_mutex_destroy(&new_key->lock);
        hc_secure_free(new_key, sizeof(hc_key_t));
        return err;
    }

    new_key->provider = prov;
    *key = new_key;

    return HC_SUCCESS;
}

```

```

}

/* Thread-safe signing */
hc_error_t hc_sign_threadsafe(const hc_key_t *private_key,
                              const uint8_t *message,
                              size_t message_len,
                              uint8_t *signature,
                              size_t *signature_len) {
    if (private_key == NULL || message == NULL ||
        signature == NULL || signature_len == NULL) {
        return HC_ERROR_INVALID_PARAM;
    }

    /* Cast away const for locking (key content not modified) */
    hc_key_t *key = (hc_key_t *)private_key;

    pthread_mutex_lock(&key->lock);

    /* Verify key type */
    if (key->type != HC_KEY_PRIVATE && key->type != HC_KEY_KEYPAIR) {
        pthread_mutex_unlock(&key->lock);
        return HC_ERROR_KEY_SIZE;
    }

    /* Get provider */
    hc_provider_t *prov = key->provider;
    if (prov == NULL) {
        pthread_mutex_unlock(&key->lock);
        return HC_ERROR_PROVIDER;
    }

    /* Perform signing */
    pthread_mutex_lock(&prov->lock);
    hc_error_t err = prov->funcs->sign(prov, key, message, message_len,
                                       signature, signature_len);
    pthread_mutex_unlock(&prov->lock);

    pthread_mutex_unlock(&key->lock);

    return err;
}

```

Memory Safety

Secure memory handling is critical for cryptographic implementations:

```

/* secure_memory.c - Secure memory management */

#include "hybrid_crypto_internal.h"
#include <stdlib.h>
#include <string.h>

#ifdef _WIN32
#include <windows.h>
#else
#include <sys/mman.h>
#include <unistd.h>
#endif

```

```

/* Memory allocation tracking for debugging */
#ifdef HC_DEBUG_MEMORY
static pthread_mutex_t mem_track_lock = PTHREAD_MUTEX_INITIALIZER;
static size_t total_allocated = 0;
static size_t total_freed = 0;
static size_t peak_usage = 0;
#endif

/* Secure memory clearing that won't be optimized away */
void hc_secure_clear(void *ptr, size_t size) {
    if (ptr == NULL || size == 0) return;

#ifdef _WIN32
    SecureZeroMemory(ptr, size);
#elif defined(__STDC_LIB_EXT1__)
    memset_s(ptr, size, 0, size);
#else
    /* Volatile pointer to prevent optimization */
    volatile uint8_t *p = (volatile uint8_t *)ptr;
    while (size--) {
        *p++ = 0;
    }
    /* Memory barrier */
    __asm__ __volatile__("" ::: "memory");
#endif
}

/* Allocate memory that won't be swapped to disk */
void *hc_secure_alloc(size_t size) {
    if (size == 0) return NULL;

    /* Add header for size tracking */
    size_t total_size = size + sizeof(size_t);

#ifdef _WIN32
    void *ptr = VirtualAlloc(NULL, total_size,
                             MEM_COMMIT | MEM_RESERVE,
                             PAGE_READWRITE);

    if (ptr != NULL) {
        VirtualLock(ptr, total_size);
    }
#else
    void *ptr = NULL;

    /* Try to get locked memory */
    if (posix_memalign(&ptr, 64, total_size) != 0) {
        return NULL;
    }

    /* Lock pages in memory (best effort - may fail without privileges) */
    mlock(ptr, total_size);

    /* Mark as non-dumpable if available */
#ifdef MADV_DONTDUMP
    madvise(ptr, total_size, MADV_DONTDUMP);
#endif
#endif

    if (ptr == NULL) return NULL;

```



```

    /* Store size in header */
    *(size_t *)ptr = size;

#ifdef HC_DEBUG_MEMORY
    pthread_mutex_lock(&mem_track_lock);
    total_allocated += size;
    if (total_allocated - total_freed > peak_usage) {
        peak_usage = total_allocated - total_freed;
    }
    pthread_mutex_unlock(&mem_track_lock);
#endif

    return (uint8_t *)ptr + sizeof(size_t);
}

/* Free secure memory */
void hc_secure_free(void *ptr, size_t size) {
    if (ptr == NULL) return;

    /* Get actual allocation start */
    void *real_ptr = (uint8_t *)ptr - sizeof(size_t);
    size_t stored_size = *(size_t *)real_ptr;

    /* Verify size matches (detect corruption) */
    if (size != 0 && size != stored_size) {
        /* Size mismatch - potential memory corruption */
        /* In production, log this and handle appropriately */
    }

    size_t total_size = stored_size + sizeof(size_t);

    /* Clear memory before freeing */
    hc_secure_clear(real_ptr, total_size);

#ifdef _WIN32
    VirtualUnlock(real_ptr, total_size);
    VirtualFree(real_ptr, 0, MEM_RELEASE);
#else
    munlock(real_ptr, total_size);
    free(real_ptr);
#endif

#ifdef HC_DEBUG_MEMORY
    pthread_mutex_lock(&mem_track_lock);
    total_freed += stored_size;
    pthread_mutex_unlock(&mem_track_lock);
#endif
}

/* Constant-time comparison */
int hc_secure_compare(const void *a, const void *b, size_t len) {
    const volatile uint8_t *pa = (const volatile uint8_t *)a;
    const volatile uint8_t *pb = (const volatile uint8_t *)b;

    uint8_t result = 0;

    for (size_t i = 0; i < len; i++) {
        result |= pa[i] ^ pb[i];
    }
}

```

```

    /* Return 0 if equal, non-zero otherwise */
    return result;
}

/* Secure memory copy that clears source after copy */
void hc_secure_move(void *dest, void *src, size_t len) {
    memcpy(dest, src, len);
    hc_secure_clear(src, len);
}

/* Allocate and copy with secure source clearing */
void *hc_secure_dup(const void *src, size_t len) {
    void *dest = hc_secure_alloc(len);
    if (dest != NULL && src != NULL) {
        memcpy(dest, src, len);
    }
    return dest;
}

#ifdef HC_DEBUG_MEMORY
void hc_memory_stats(size_t *allocated, size_t *freed, size_t *peak) {
    pthread_mutex_lock(&mem_track_lock);
    if (allocated) *allocated = total_allocated;
    if (freed) *freed = total_freed;
    if (peak) *peak = peak_usage;
    pthread_mutex_unlock(&mem_track_lock);
}
#endif

```

8.5.3 Performance Optimization

Hybrid cryptography involves computational overhead from running two algorithms. Optimization techniques can mitigate this impact.

Parallel Execution

For hybrid operations, both components can often execute concurrently:

```

/* parallel_ops.c - Parallel hybrid operations */

#include "hybrid_crypto_internal.h"
#include <pthread.h>
#include <string.h>

/* Thread pool for parallel operations */
typedef struct {
    pthread_t *threads;
    int num_threads;

    /* Work queue */
    pthread_mutex_t queue_lock;
    pthread_cond_t queue_cond;

    /* Task structure */
    struct task_node {
        void (*func)(void *);
    };
}

```

```

        void *arg;
        struct task_node *next;
    } *task_head, *task_tail;

    int shutdown;
} thread_pool_t;

static thread_pool_t *g_pool = NULL;

/* Worker thread function */
static void *worker_thread(void *arg) {
    thread_pool_t *pool = (thread_pool_t *)arg;

    while (1) {
        pthread_mutex_lock(&pool->queue_lock);

        while (pool->task_head == NULL && !pool->shutdown) {
            pthread_cond_wait(&pool->queue_cond, &pool->queue_lock);
        }

        if (pool->shutdown && pool->task_head == NULL) {
            pthread_mutex_unlock(&pool->queue_lock);
            break;
        }

        /* Dequeue task */
        struct task_node *task = pool->task_head;
        pool->task_head = task->next;
        if (pool->task_head == NULL) {
            pool->task_tail = NULL;
        }

        pthread_mutex_unlock(&pool->queue_lock);

        /* Execute task */
        task->func(task->arg);
        free(task);
    }

    return NULL;
}

/* Initialize thread pool */
int hc_thread_pool_init(int num_threads) {
    if (g_pool != NULL) return 0; /* Already initialized */

    if (num_threads <= 0) {
        num_threads = 4; /* Default */
    }

    g_pool = malloc(sizeof(thread_pool_t));
    if (g_pool == NULL) return -1;

    g_pool->num_threads = num_threads;
    g_pool->task_head = NULL;
    g_pool->task_tail = NULL;
    g_pool->shutdown = 0;

    pthread_mutex_init(&g_pool->queue_lock, NULL);
    pthread_cond_init(&g_pool->queue_cond, NULL);
}

```

```

    g_pool->threads = malloc(num_threads * sizeof(pthread_t));
    if (g_pool->threads == NULL) {
        free(g_pool);
        g_pool = NULL;
        return -1;
    }

    for (int i = 0; i < num_threads; i++) {
        pthread_create(&g_pool->threads[i], NULL, worker_thread, g_pool);
    }

    return 0;
}

/* Submit task to pool */
int hc_thread_pool_submit(void (*func)(void *), void *arg) {
    if (g_pool == NULL) {
        /* No pool - execute synchronously */
        func(arg);
        return 0;
    }

    struct task_node *task = malloc(sizeof(struct task_node));
    if (task == NULL) return -1;

    task->func = func;
    task->arg = arg;
    task->next = NULL;

    pthread_mutex_lock(&g_pool->queue_lock);

    if (g_pool->task_tail == NULL) {
        g_pool->task_head = task;
    } else {
        g_pool->task_tail->next = task;
    }
    g_pool->task_tail = task;

    pthread_cond_signal(&g_pool->queue_cond);
    pthread_mutex_unlock(&g_pool->queue_lock);

    return 0;
}

/* Parallel signature result structure */
typedef struct {
    hc_error_t error;
    uint8_t *signature;
    size_t signature_len;
    pthread_mutex_t done_lock;
    pthread_cond_t done_cond;
    int done;
} parallel_sign_result_t;

/* Classical signature task */
typedef struct {
    hc_key_t *key;
    const uint8_t *message;
    size_t message_len;

```

```

    parallel_sign_result_t *result;
} sign_task_arg_t;

static void classical_sign_task(void *arg) {
    sign_task_arg_t *task = (sign_task_arg_t *)arg;

    task->result->signature = hc_secure_alloc(HC_MAX_SIGNATURE_SIZE);
    if (task->result->signature == NULL) {
        task->result->error = HC_ERROR_MEMORY;
    } else {
        task->result->signature_len = HC_MAX_SIGNATURE_SIZE;
        task->result->error = hc_sign(task->key, task->message,
                                     task->message_len,
                                     task->result->signature,
                                     &task->result->signature_len);
    }

    pthread_mutex_lock(&task->result->done_lock);
    task->result->done = 1;
    pthread_cond_signal(&task->result->done_cond);
    pthread_mutex_unlock(&task->result->done_lock);

    free(task);
}

/* Parallel hybrid signing */
hc_error_t hc_hybrid_sign_parallel(const hc_key_t *hybrid_key,
                                   const uint8_t *message,
                                   size_t message_len,
                                   uint8_t *signature,
                                   size_t *signature_len) {
    if (hybrid_key == NULL || message == NULL ||
        signature == NULL || signature_len == NULL) {
        return HC_ERROR_INVALID_PARAM;
    }

    /* Get component keys */
    hc_key_t *classical_key = hybrid_key->classical_component;
    hc_key_t *pqc_key = hybrid_key->pqc_component;

    if (classical_key == NULL || pqc_key == NULL) {
        return HC_ERROR_KEY_SIZE;
    }

    /* Initialize result structures */
    parallel_sign_result_t classical_result = {0};
    parallel_sign_result_t pqc_result = {0};

    pthread_mutex_init(&classical_result.done_lock, NULL);
    pthread_cond_init(&classical_result.done_cond, NULL);
    pthread_mutex_init(&pqc_result.done_lock, NULL);
    pthread_cond_init(&pqc_result.done_cond, NULL);

    /* Create task arguments */
    sign_task_arg_t *classical_task = malloc(sizeof(sign_task_arg_t));
    sign_task_arg_t *pqc_task = malloc(sizeof(sign_task_arg_t));

    if (classical_task == NULL || pqc_task == NULL) {
        free(classical_task);
        free(pqc_task);
    }

```

```

    return HC_ERROR_MEMORY;
}

classical_task->key = classical_key;
classical_task->message = message;
classical_task->message_len = message_len;
classical_task->result = &classical_result;

pqc_task->key = pqc_key;
pqc_task->message = message;
pqc_task->message_len = message_len;
pqc_task->result = &pqc_result;

/* Submit both tasks */
hc_thread_pool_submit(classical_sign_task, classical_task);
hc_thread_pool_submit(classical_sign_task, pqc_task);

/* Wait for both to complete */
pthread_mutex_lock(&classical_result.done_lock);
while (!classical_result.done) {
    pthread_cond_wait(&classical_result.done_cond,
                     &classical_result.done_lock);
}
pthread_mutex_unlock(&classical_result.done_lock);

pthread_mutex_lock(&pqc_result.done_lock);
while (!pqc_result.done) {
    pthread_cond_wait(&pqc_result.done_cond, &pqc_result.done_lock);
}
pthread_mutex_unlock(&pqc_result.done_lock);

/* Combine results */
hc_error_t err = HC_SUCCESS;

if (classical_result.error != HC_SUCCESS) {
    err = classical_result.error;
} else if (pqc_result.error != HC_SUCCESS) {
    err = pqc_result.error;
} else {
    /* Encode combined signature: length || classical || pqc */
    size_t total_len = 4 + classical_result.signature_len +
                      4 + pqc_result.signature_len;

    if (*signature_len < total_len) {
        err = HC_ERROR_BUFFER_SIZE;
    } else {
        size_t offset = 0;

        /* Classical signature length (4 bytes, big-endian) */
        signature[offset++] = (classical_result.signature_len >> 24) & 0xFF;
        signature[offset++] = (classical_result.signature_len >> 16) & 0xFF;
        signature[offset++] = (classical_result.signature_len >> 8) & 0xFF;
        signature[offset++] = classical_result.signature_len & 0xFF;

        /* Classical signature */
        memcpy(signature + offset, classical_result.signature,
               classical_result.signature_len);
        offset += classical_result.signature_len;

        /* PQC signature length (4 bytes, big-endian) */

```

```

        signature[offset++] = (pqc_result.signature_len >> 24) & 0xFF;
        signature[offset++] = (pqc_result.signature_len >> 16) & 0xFF;
        signature[offset++] = (pqc_result.signature_len >> 8) & 0xFF;
        signature[offset++] = pqc_result.signature_len & 0xFF;

        /* PQC signature */
        memcpy(signature + offset, pqc_result.signature,
               pqc_result.signature_len);
        offset += pqc_result.signature_len;

        *signature_len = offset;
    }
}

/* Cleanup */
if (classical_result.signature) {
    hc_secure_free(classical_result.signature,
                   classical_result.signature_len);
}
if (pqc_result.signature) {
    hc_secure_free(pqc_result.signature, pqc_result.signature_len);
}

pthread_mutex_destroy(&classical_result.done_lock);
pthread_cond_destroy(&classical_result.done_cond);
pthread_mutex_destroy(&pqc_result.done_lock);
pthread_cond_destroy(&pqc_result.done_cond);

return err;
}

```

Lazy Verification

For hybrid verification, lazy evaluation can improve performance when one component failure is sufficient:

```

/* lazy_verify.c - Lazy hybrid verification */

#include "hybrid_crypto_internal.h"

/* Verification policy */
typedef enum {
    HC_VERIFY_ALL,           /* Both must verify (strictest) */
    HC_VERIFY_ANY,          /* Either sufficient (transition) */
    HC_VERIFY_CLASSICAL,    /* Only classical (legacy compat) */
    HC_VERIFY_PQC,          /* Only PQC (future-proof) */
    HC_VERIFY_LAZY_ALL      /* Both, but stop on first failure */
} hc_verify_policy_t;

/* Lazy hybrid verification */
hc_error_t hc_hybrid_verify_lazy(const hc_key_t *hybrid_key,
                                const uint8_t *message,
                                size_t message_len,
                                const uint8_t *signature,
                                size_t signature_len,
                                hc_verify_policy_t policy) {
    if (hybrid_key == NULL || message == NULL || signature == NULL) {
        return HC_ERROR_INVALID_PARAM;
    }
}

```

```

}

/* Parse combined signature */
if (signature_len < 8) {
    return HC_ERROR_SIGNATURE;
}

size_t offset = 0;

/* Read classical signature length */
size_t classical_sig_len =
    ((size_t)signature[offset] << 24) |
    ((size_t)signature[offset + 1] << 16) |
    ((size_t)signature[offset + 2] << 8) |
    (size_t)signature[offset + 3];
offset += 4;

if (offset + classical_sig_len + 4 > signature_len) {
    return HC_ERROR_SIGNATURE;
}

const uint8_t *classical_sig = signature + offset;
offset += classical_sig_len;

/* Read PQC signature length */
size_t pqc_sig_len =
    ((size_t)signature[offset] << 24) |
    ((size_t)signature[offset + 1] << 16) |
    ((size_t)signature[offset + 2] << 8) |
    (size_t)signature[offset + 3];
offset += 4;

if (offset + pqc_sig_len > signature_len) {
    return HC_ERROR_SIGNATURE;
}

const uint8_t *pqc_sig = signature + offset;

/* Get component keys */
hc_key_t *classical_key = hybrid_key->classical_component;
hc_key_t *pqc_key = hybrid_key->pqc_component;

hc_error_t classical_result = HC_ERROR_NOT_SUPPORTED;
hc_error_t pqc_result = HC_ERROR_NOT_SUPPORTED;

switch (policy) {
    case HC_VERIFY_CLASSICAL:
        if (classical_key != NULL) {
            classical_result = hc_verify(classical_key, message,
                                         message_len, classical_sig,
                                         classical_sig_len);
        }
        return classical_result;

    case HC_VERIFY_PQC:
        if (pqc_key != NULL) {
            pqc_result = hc_verify(pqc_key, message, message_len,
                                   pqc_sig, pqc_sig_len);
        }
        return pqc_result;
}

```



```

case HC_VERIFY_ANY:
    /* Try classical first (usually faster) */
    if (classical_key != NULL) {
        classical_result = hc_verify(classical_key, message,
                                     message_len, classical_sig,
                                     classical_sig_len);

        if (classical_result == HC_SUCCESS) {
            return HC_SUCCESS;
        }
    }
    /* Classical failed, try PQC */
    if (pqc_key != NULL) {
        pqc_result = hc_verify(pqc_key, message, message_len,
                               pqc_sig, pqc_sig_len);
    }
    return pqc_result;

case HC_VERIFY_LAZY_ALL:
    /* Verify classical first (faster) */
    if (classical_key != NULL) {
        classical_result = hc_verify(classical_key, message,
                                     message_len, classical_sig,
                                     classical_sig_len);

        if (classical_result != HC_SUCCESS) {
            /* Early exit - don't bother with PQC */
            return classical_result;
        }
    }
    /* Classical passed, verify PQC */
    if (pqc_key != NULL) {
        pqc_result = hc_verify(pqc_key, message, message_len,
                               pqc_sig, pqc_sig_len);
    }
    return pqc_result;

case HC_VERIFY_ALL:
default:
    /* Must verify both - order doesn't matter for correctness */
    if (classical_key != NULL) {
        classical_result = hc_verify(classical_key, message,
                                     message_len, classical_sig,
                                     classical_sig_len);
    }
    if (pqc_key != NULL) {
        pqc_result = hc_verify(pqc_key, message, message_len,
                               pqc_sig, pqc_sig_len);
    }

    if (classical_result != HC_SUCCESS) return classical_result;
    if (pqc_result != HC_SUCCESS) return pqc_result;
    return HC_SUCCESS;
}
}

```

Precomputation and Caching

For repeated operations with the same keys, precomputation can save time:

```
/* precompute.c - Precomputation for performance */

#include "hybrid_crypto_internal.h"
#include <string.h>

/* Precomputed key data */
typedef struct {
    hc_algorithm_t algorithm;

    /* Algorithm-specific precomputed data */
    union {
        struct {
            /* Ed25519 expanded secret key */
            uint8_t expanded_sk[64];
        } ed25519;

        struct {
            /* ML-DSA precomputed NTT of secret key */
            void *ntt_s1;
            void *ntt_s2;
        } mldsa;

        struct {
            /* X25519 precomputed point multiples */
            void *basepoint_table;
        } x25519;
    } data;

    int valid;
} precomputed_data_t;

/* Cache of precomputed data */
#define PRECOMPUTE_CACHE_SIZE 16

static struct {
    pthread_mutex_t lock;
    struct {
        hc_key_t *key;
        precomputed_data_t *data;
        uint64_t last_used;
        int in_use;
    } entries[PRECOMPUTE_CACHE_SIZE];
    uint64_t access_counter;
} precompute_cache = {
    .lock = PTHREAD_MUTEX_INITIALIZER
};

/* Get current time counter for LRU */
static uint64_t get_time_counter(void) {
    return __sync_add_and_fetch(&precompute_cache.access_counter, 1);
}

/* Compute precomputation data for a key */
static precomputed_data_t *compute_predata(hc_key_t *key) {
    precomputed_data_t *data = hc_secure_alloc(sizeof(precomputed_data_t));
```

```

if (data == NULL) return NULL;

memset(data, 0, sizeof(precomputed_data_t));
data->algorithm = key->algorithm;

switch (key->algorithm) {
    case HC_ALG_ED25519:
        if (key->private_key != NULL && key->private_key_len == 32) {
            /* Expand Ed25519 secret key (first hash) */
            /* In real implementation, use proper Ed25519 expansion */
            /* SHA512(sk) -> 64 bytes, clamp first 32 */
            data->valid = 1;
        }
        break;

    case HC_ALG_MLDSA65:
        if (key->private_key != NULL) {
            /* Precompute NTT of secret key vectors */
            /* This avoids NTT computation on each sign */
            data->valid = 1;
        }
        break;

    default:
        /* No precomputation for this algorithm */
        break;
}

return data;
}

/* Get or create precomputed data for key */
precomputed_data_t *hc_get_precomputed(hc_key_t *key) {
    if (key == NULL) return NULL;

    pthread_mutex_lock(&precompute_cache.lock);

    /* Search cache */
    int oldest_idx = -1;
    uint64_t oldest_time = UINT64_MAX;

    for (int i = 0; i < PRECOMPUTE_CACHE_SIZE; i++) {
        if (precompute_cache.entries[i].key == key) {
            /* Found - mark as used */
            precompute_cache.entries[i].last_used = get_time_counter();
            precompute_cache.entries[i].in_use++;

            precomputed_data_t *data = precompute_cache.entries[i].data;
            pthread_mutex_unlock(&precompute_cache.lock);
            return data;
        }

        /* Track oldest for eviction */
        if (!precompute_cache.entries[i].in_use &&
            (oldest_idx < 0 ||
             precompute_cache.entries[i].last_used < oldest_time)) {
            oldest_idx = i;
            oldest_time = precompute_cache.entries[i].last_used;
        }
    }
}

```

```

/* Not found - compute and cache */
if (oldest_idx < 0) {
    /* Cache full, all in use - don't cache */
    pthread_mutex_unlock(&precompute_cache.lock);
    return compute_predata(key);
}

/* Evict oldest entry */
if (precompute_cache.entries[oldest_idx].data != NULL) {
    hc_secure_free(precompute_cache.entries[oldest_idx].data,
        sizeof(precomputed_data_t));
}

/* Compute and store new entry */
precomputed_data_t *data = compute_predata(key);
precompute_cache.entries[oldest_idx].key = key;
precompute_cache.entries[oldest_idx].data = data;
precompute_cache.entries[oldest_idx].last_used = get_time_counter();
precompute_cache.entries[oldest_idx].in_use = 1;

pthread_mutex_unlock(&precompute_cache.lock);

return data;
}

/* Release precomputed data reference */
void hc_release_precomputed(precomputed_data_t *data) {
    if (data == NULL) return;

    pthread_mutex_lock(&precompute_cache.lock);

    for (int i = 0; i < PRECOMPUTE_CACHE_SIZE; i++) {
        if (precompute_cache.entries[i].data == data) {
            precompute_cache.entries[i].in_use--;
            break;
        }
    }

    pthread_mutex_unlock(&precompute_cache.lock);
}

/* Invalidate cache entry when key is freed */
void hc_invalidate_precomputed(hc_key_t *key) {
    pthread_mutex_lock(&precompute_cache.lock);

    for (int i = 0; i < PRECOMPUTE_CACHE_SIZE; i++) {
        if (precompute_cache.entries[i].key == key) {
            if (precompute_cache.entries[i].data != NULL) {
                hc_secure_free(precompute_cache.entries[i].data,
                    sizeof(precomputed_data_t));
            }
            precompute_cache.entries[i].key = NULL;
            precompute_cache.entries[i].data = NULL;
            precompute_cache.entries[i].in_use = 0;
        }
    }
}

```

```

        break;
    }
}

pthread_mutex_unlock(&precompute_cache.lock);
}

```

8.5.4 Migration Strategies

Transitioning existing systems to hybrid cryptography requires careful planning. We examine several migration patterns.

Phased Migration

```

Phase 1: Preparation
├─ Audit current cryptographic usage
├─ Inventory all keys and certificates
├─ Identify critical paths
└─ Plan timeline and rollback procedures

Phase 2: Hybrid Capability
├─ Deploy hybrid-capable libraries
├─ Enable hybrid in non-critical systems
├─ Monitor for compatibility issues
└─ Collect performance metrics

Phase 3: Hybrid Default
├─ Switch default to hybrid mode
├─ Maintain classical fallback
├─ Update certificate infrastructure
└─ Train operations team

Phase 4: Classical Deprecation
├─ Warn on classical-only operations
├─ Set deprecation timeline
├─ Provide migration tools
└─ Document exceptions process

Phase 5: Post-Quantum Ready
├─ Disable classical-only paths
├─ Remove legacy code
├─ Update security policies
└─ Plan for algorithm agility

```

Configuration-Driven Migration

```

/* migration_config.c - Configuration-driven migration */

#include "hybrid_crypto_internal.h"
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

/* Migration configuration */
typedef struct {

```

```

/* Algorithm policies */
struct {
    hc_algorithm_t preferred;
    hc_algorithm_t fallback;
    int allow_classical_only;
    int allow_pqc_only;
} signature;

struct {
    hc_algorithm_t preferred;
    hc_algorithm_t fallback;
    int allow_classical_only;
    int allow_pqc_only;
} kem;

/* Verification policy */
hc_verify_policy_t verify_policy;

/* Logging and monitoring */
int log_classical_usage;
int log_hybrid_usage;
int warn_classical_only;

/* Feature flags */
int enable_parallel_ops;
int enable_precomputation;
int enable_key_caching;

/* Version for compatibility */
int config_version;
} migration_config_t;

/* Global configuration */
static migration_config_t g_config = {
    .signature = {
        .preferred = HC_ALG_HYBRID_ED25519_MLDSA65,
        .fallback = HC_ALG_ED25519,
        .allow_classical_only = 1,
        .allow_pqc_only = 0
    },
    .kem = {
        .preferred = HC_ALG_HYBRID_X25519_MLKEM768,
        .fallback = HC_ALG_X25519,
        .allow_classical_only = 1,
        .allow_pqc_only = 0
    },
    .verify_policy = HC_VERIFY_ALL,
    .log_classical_usage = 1,
    .log_hybrid_usage = 0,
    .warn_classical_only = 1,
    .enable_parallel_ops = 1,
    .enable_precomputation = 1,
    .enable_key_caching = 1,
    .config_version = 1
};

/* Parse configuration from file */
hc_error_t hc_load_migration_config(const char *filename) {
    FILE *f = fopen(filename, "r");
    if (f == NULL) {

```

```

        return HC_ERROR_INVALID_PARAM;
    }

    char line[256];
    while (fgets(line, sizeof(line), f)) {
        /* Skip comments and empty lines */
        if (line[0] == '#' || line[0] == '\n') continue;

        char key[64], value[64];
        if (sscanf(line, "%63[^\n]=%63s", key, value) != 2) continue;

        /* Trim whitespace */
        char *k = key;
        while (*k == ' ') k++;
        char *end = k + strlen(k) - 1;
        while (end > k && *end == ' ') *end-- = '\0';

        /* Parse options */
        if (strcmp(k, "signature.preferred") == 0) {
            if (strcmp(value, "hybrid_ed25519_mldsa65") == 0) {
                g_config.signature.preferred = HC_ALG_HYBRID_ED25519_MLDSA65;
            } else if (strcmp(value, "ed25519") == 0) {
                g_config.signature.preferred = HC_ALG_ED25519;
            }
        } else if (strcmp(k, "signature.allow_classical_only") == 0) {
            g_config.signature.allow_classical_only = (strcmp(value, "true") == 0);
        } else if (strcmp(k, "verify_policy") == 0) {
            if (strcmp(value, "all") == 0) {
                g_config.verify_policy = HC_VERIFY_ALL;
            } else if (strcmp(value, "any") == 0) {
                g_config.verify_policy = HC_VERIFY_ANY;
            } else if (strcmp(value, "lazy_all") == 0) {
                g_config.verify_policy = HC_VERIFY_LAZY_ALL;
            }
        } else if (strcmp(k, "warn_classical_only") == 0) {
            g_config.warn_classical_only = (strcmp(value, "true") == 0);
        } else if (strcmp(k, "enable_parallel_ops") == 0) {
            g_config.enable_parallel_ops = (strcmp(value, "true") == 0);
        }
    }

    fclose(f);
    return HC_SUCCESS;
}

/* Get current configuration */
const migration_config_t *hc_get_migration_config(void) {
    return &g_config;
}

/* Policy-aware key generation */
hc_error_t hc_keygen_with_policy(hc_key_type_t key_type, hc_key_t **key) {
    hc_algorithm_t alg;

    /* Select algorithm based on policy */
    if (key_type == HC_KEY_PRIVATE) {
        /* Signature key */
        alg = g_config.signature.preferred;
    } else {
        /* KEM key (assuming this context) */

```

```

    alg = g_config.kem.preferred;
}

/* Attempt generation */
hc_error_t err = hc_key_generate(alg, key);

if (err != HC_SUCCESS && alg != g_config.signature.fallback) {
    /* Try fallback */
    if (g_config.log_classical_usage) {
        /* Log fallback to classical */
        fprintf(stderr, "WARNING: Falling back to classical algorithm\n");
    }

    if (key_type == HC_KEY_PRIVATE) {
        if (!g_config.signature.allow_classical_only) {
            return err; /* Classical-only not allowed */
        }
        alg = g_config.signature.fallback;
    } else {
        if (!g_config.kem.allow_classical_only) {
            return err;
        }
        alg = g_config.kem.fallback;
    }

    err = hc_key_generate(alg, key);
}

return err;
}

/* Policy-aware signing */
hc_error_t hc_sign_with_policy(const hc_key_t *key,
                               const uint8_t *message,
                               size_t message_len,
                               uint8_t *signature,
                               size_t *signature_len) {
    hc_algorithm_t alg = hc_key_algorithm(key);

    /* Check if classical-only is allowed */
    if (!is_hybrid_algorithm(alg)) {
        if (g_config.warn_classical_only) {
            fprintf(stderr, "WARNING: Signing with classical-only key\n");
        }

        if (!g_config.signature.allow_classical_only) {
            return HC_ERROR_NOT_SUPPORTED;
        }

        if (g_config.log_classical_usage) {
            /* Log classical usage for audit */
        }
    }

    /* Use parallel if enabled for hybrid */
    if (is_hybrid_algorithm(alg) && g_config.enable_parallel_ops) {
        return hc_hybrid_sign_parallel(key, message, message_len,
                                       signature, signature_len);
    }
}

```



```

        return hc_sign(key, message, message_len, signature, signature_len);
    }

    /* Check if algorithm is hybrid */
    static int is_hybrid_algorithm(hc_algorithm_t alg) {
        return (alg & 0x2000) != 0; /* Hybrid algorithms start with 0x2xxx */
    }

```

8.5.5 Protocol Integration

Integrating hybrid cryptography with existing protocols requires careful attention to wire formats and negotiation.

TLS Integration Example

```

/* tls_integration.c - TLS hybrid integration helpers */

#include "hybrid_crypto_internal.h"
#include <string.h>

/* TLS signature scheme identifiers (from draft-ietf-tls-hybrid-design) */
#define TLS_SIG_ED25519            0x0807
#define TLS_SIG_ECDSA_SECP256R1   0x0403
#define TLS_SIG_MLDSA65           0x0904 /* Example OID */
#define TLS_SIG_HYBRID_ED25519_MLDSA65 0x0A01 /* Example */

/* Supported groups for key exchange */
#define TLS_GROUP_X25519          0x001D
#define TLS_GROUP_SECP256R1      0x0017
#define TLS_GROUP_MLKEM768       0x0200 /* Example */
#define TLS_GROUP_HYBRID_X25519_MLKEM768 0x0201 /* Example */

/* TLS signature scheme to internal algorithm mapping */
typedef struct {
    uint16_t tls_id;
    hc_algorithm_t hc_alg;
    const char *name;
} tls_sig_mapping_t;

static const tls_sig_mapping_t sig_mappings[] = {
    { TLS_SIG_ED25519, HC_ALG_ED25519, "ed25519" },
    { TLS_SIG_ECDSA_SECP256R1, HC_ALG_ECDSA_P256, "ecdsa_secp256r1_sha256" },
    { TLS_SIG_MLDSA65, HC_ALG_MLDSA65, "mldsa65" },
    { TLS_SIG_HYBRID_ED25519_MLDSA65, HC_ALG_HYBRID_ED25519_MLDSA65,
      "hybrid_ed25519_mldsa65" },
    { 0, 0, NULL }
};

/* Map TLS signature scheme to internal algorithm */
hc_algorithm_t tls_sig_scheme_to_algorithm(uint16_t tls_id) {
    for (const tls_sig_mapping_t *m = sig_mappings; m->name != NULL; m++) {
        if (m->tls_id == tls_id) {
            return m->hc_alg;
        }
    }
    return 0; /* Unknown */
}

```

```

/* Map internal algorithm to TLS signature scheme */
uint16_t algorithm_to_tls_sig_scheme(hc_algorithm_t alg) {
    for (const tls_sig_mapping_t *m = sig_mappings; m->name != NULL; m++) {
        if (m->hc_alg == alg) {
            return m->tls_id;
        }
    }
    return 0; /* Unknown */
}

/* Negotiate signature algorithm from client preferences */
hc_algorithm_t tls_negotiate_signature(const uint16_t *client_schemes,
                                      size_t num_client_schemes,
                                      const hc_key_t *server_key) {
    hc_algorithm_t key_alg = hc_key_algorithm(server_key);

    /* Check if our key's algorithm is in client's preferences */
    uint16_t our_scheme = algorithm_to_tls_sig_scheme(key_alg);

    for (size_t i = 0; i < num_client_schemes; i++) {
        if (client_schemes[i] == our_scheme) {
            return key_alg;
        }
    }

    /* For hybrid keys, check if either component is acceptable */
    if (is_hybrid_algorithm(key_alg)) {
        hc_key_t *classical = server_key->classical_component;
        hc_key_t *pqc = server_key->pqc_component;

        /* Prefer PQC component */
        if (pqc != NULL) {
            uint16_t pqc_scheme = algorithm_to_tls_sig_scheme(
                hc_key_algorithm(pqc));
            for (size_t i = 0; i < num_client_schemes; i++) {
                if (client_schemes[i] == pqc_scheme) {
                    return hc_key_algorithm(pqc);
                }
            }
        }

        /* Fall back to classical */
        if (classical != NULL) {
            uint16_t classical_scheme = algorithm_to_tls_sig_scheme(
                hc_key_algorithm(classical));
            for (size_t i = 0; i < num_client_schemes; i++) {
                if (client_schemes[i] == classical_scheme) {
                    return hc_key_algorithm(classical);
                }
            }
        }
    }

    return 0; /* No common algorithm */
}

/* Create TLS CertificateVerify message */
hc_error_t tls_create_certificate_verify(const hc_key_t *key,
                                         const uint8_t *handshake_context,

```

```

size_t context_len,
uint8_t *output,
size_t *output_len) {
/* TLS 1.3 CertificateVerify format:
 * struct {
 *     SignatureScheme algorithm;
 *     opaque signature<0..2^16-1>;
 * } CertificateVerify;
 */

/* Create signed content per RFC 8446 */
uint8_t signed_content[512];
size_t signed_len = 0;

/* 64 spaces */
memset(signed_content, 0x20, 64);
signed_len = 64;

/* Context string */
const char *context_str = "TLS 1.3, server CertificateVerify";
memcpy(signed_content + signed_len, context_str, strlen(context_str));
signed_len += strlen(context_str);

/* Zero byte separator */
signed_content[signed_len++] = 0x00;

/* Transcript hash */
memcpy(signed_content + signed_len, handshake_context, context_len);
signed_len += context_len;

/* Sign */
uint8_t sig[HC_MAX_SIGNATURE_SIZE];
size_t sig_len = sizeof(sig);

hc_error_t err = hc_sign(key, signed_content, signed_len, sig, &sig_len);
if (err != HC_SUCCESS) {
    return err;
}

/* Encode CertificateVerify */
uint16_t scheme = algorithm_to_tls_sig_scheme(hc_key_algorithm(key));

size_t total_len = 2 + 2 + sig_len;
if (*output_len < total_len) {
    return HC_ERROR_BUFFER_SIZE;
}

/* SignatureScheme (2 bytes) */
output[0] = (scheme >> 8) & 0xFF;
output[1] = scheme & 0xFF;

/* Signature length (2 bytes) */
output[2] = (sig_len >> 8) & 0xFF;
output[3] = sig_len & 0xFF;

/* Signature */
memcpy(output + 4, sig, sig_len);

*output_len = total_len;

```

```

/* Clear sensitive data */
hc_secure_clear(signed_content, sizeof(signed_content));

return HC_SUCCESS;
}

/* Verify TLS CertificateVerify message */
hc_error_t tls_verify_certificate_verify(const hc_key_t *peer_key,
                                         const uint8_t *handshake_context,
                                         size_t context_len,
                                         const uint8_t *cert_verify,
                                         size_t cert_verify_len,
                                         int is_server) {

    if (cert_verify_len < 4) {
        return HC_ERROR_SIGNATURE;
    }

    /* Parse SignatureScheme */
    uint16_t scheme = ((uint16_t)cert_verify[0] << 8) | cert_verify[1];

    /* Parse signature length */
    size_t sig_len = ((size_t)cert_verify[2] << 8) | cert_verify[3];

    if (cert_verify_len < 4 + sig_len) {
        return HC_ERROR_SIGNATURE;
    }

    const uint8_t *sig = cert_verify + 4;

    /* Verify algorithm matches key */
    hc_algorithm_t expected_alg = tls_sig_scheme_to_algorithm(scheme);
    if (expected_alg != hc_key_algorithm(peer_key)) {
        /* Check if key is hybrid and scheme matches a component */
        if (!is_hybrid_algorithm(hc_key_algorithm(peer_key))) {
            return HC_ERROR_VERIFICATION;
        }
    }

    /* Reconstruct signed content */
    uint8_t signed_content[512];
    size_t signed_len = 0;

    memset(signed_content, 0x20, 64);
    signed_len = 64;

    const char *context_str = is_server ?
        "TLS 1.3, server CertificateVerify" :
        "TLS 1.3, client CertificateVerify";
    memcpy(signed_content + signed_len, context_str, strlen(context_str));
    signed_len += strlen(context_str);

    signed_content[signed_len++] = 0x00;

    memcpy(signed_content + signed_len, handshake_context, context_len);
    signed_len += context_len;

    /* Verify */
    hc_error_t err = hc_verify(peer_key, signed_content, signed_len,

```

```

        sig, sig_len);

    hc_secure_clear(signed_content, sizeof(signed_content));

    return err;
}

```

8.5.6 Testing and Validation

Comprehensive testing is essential for hybrid cryptography implementations.

```

/* test_hybrid.c - Test framework for hybrid cryptography */

#include "hybrid_crypto.h"
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>

/* Test result tracking */
typedef struct {
    int total;
    int passed;
    int failed;
    int skipped;
} test_results_t;

static test_results_t results = {0};

#define TEST_ASSERT(cond, msg) do { \
    results.total++; \
    if (cond) { \
        results.passed++; \
        printf(" [PASS] %s\n", msg); \
    } else { \
        results.failed++; \
        printf(" [FAIL] %s (line %d)\n", msg, __LINE__); \
    } \
} while(0)

#define TEST_SKIP(msg) do { \
    results.total++; \
    results.skipped++; \
    printf(" [SKIP] %s\n", msg); \
} while(0)

/* Test key generation */
void test_keygen(void) {
    printf("\n=== Key Generation Tests ===\n");

    hc_key_t *key = NULL;
    hc_error_t err;

    /* Test Ed25519 key generation */
    err = hc_key_generate(HC_ALG_ED25519, &key);
    TEST_ASSERT(err == HC_SUCCESS, "Ed25519 key generation");
    if (key) {
        TEST_ASSERT(hc_key_algorithm(key) == HC_ALG_ED25519,

```

```

        "Ed25519 algorithm identifier");
    hc_key_free(key);
    key = NULL;
}

/* Test ML-DSA-65 key generation */
err = hc_key_generate(HC_ALG_MLDSA65, &key);
if (err == HC_ERROR_NOT_SUPPORTED) {
    TEST_SKIP("ML-DSA-65 not available");
} else {
    TEST_ASSERT(err == HC_SUCCESS, "ML-DSA-65 key generation");
    if (key) {
        TEST_ASSERT(hc_key_algorithm(key) == HC_ALG_MLDSA65,
            "ML-DSA-65 algorithm identifier");
        hc_key_free(key);
        key = NULL;
    }
}

/* Test hybrid key generation */
err = hc_key_generate(HC_ALG_HYBRID_ED25519_MLDSA65, &key);
if (err == HC_ERROR_NOT_SUPPORTED) {
    TEST_SKIP("Hybrid Ed25519+ML-DSA-65 not available");
} else {
    TEST_ASSERT(err == HC_SUCCESS, "Hybrid key generation");
    if (key) {
        TEST_ASSERT(hc_key_algorithm(key) == HC_ALG_HYBRID_ED25519_MLDSA65,
            "Hybrid algorithm identifier");

        /* Test component extraction */
        hc_key_t *classical = NULL, *pqc = NULL;
        err = hc_hybrid_get_classical_key(key, &classical);
        TEST_ASSERT(err == HC_SUCCESS && classical != NULL,
            "Extract classical component");

        err = hc_hybrid_get_pqc_key(key, &pqc);
        TEST_ASSERT(err == HC_SUCCESS && pqc != NULL,
            "Extract PQC component");

        hc_key_free(classical);
        hc_key_free(pqc);
        hc_key_free(key);
        key = NULL;
    }
}

/* Test signing and verification */
void test_sign_verify(void) {
    printf("\n=== Sign/Verify Tests ===\n");

    const uint8_t message[] = "Test message for signature verification";
    size_t message_len = sizeof(message) - 1;

    uint8_t signature[8192];
    size_t signature_len;

    hc_key_t *key = NULL;
    hc_error_t err;

```

```

/* Test Ed25519 sign/verify */
err = hc_key_generate(HC_ALG_ED25519, &key);
if (err == HC_SUCCESS) {
    signature_len = sizeof(signature);
    err = hc_sign(key, message, message_len, signature, &signature_len);
    TEST_ASSERT(err == HC_SUCCESS, "Ed25519 sign");
    TEST_ASSERT(signature_len == 64, "Ed25519 signature size");

    err = hc_verify(key, message, message_len, signature, signature_len);
    TEST_ASSERT(err == HC_SUCCESS, "Ed25519 verify valid");

    /* Tamper with signature */
    signature[0] ^= 0xFF;
    err = hc_verify(key, message, message_len, signature, signature_len);
    TEST_ASSERT(err != HC_SUCCESS, "Ed25519 reject tampered signature");

    hc_key_free(key);
    key = NULL;
}

/* Test hybrid sign/verify */
err = hc_key_generate(HC_ALG_HYBRID_ED25519_MLDSA65, &key);
if (err == HC_ERROR_NOT_SUPPORTED) {
    TEST_SKIP("Hybrid sign/verify");
} else if (err == HC_SUCCESS) {
    signature_len = sizeof(signature);
    err = hc_sign(key, message, message_len, signature, &signature_len);
    TEST_ASSERT(err == HC_SUCCESS, "Hybrid sign");
    TEST_ASSERT(signature_len > 64, "Hybrid signature larger than Ed25519");

    err = hc_verify(key, message, message_len, signature, signature_len);
    TEST_ASSERT(err == HC_SUCCESS, "Hybrid verify valid");

    /* Tamper with classical component */
    signature[4] ^= 0xFF;
    err = hc_verify(key, message, message_len, signature, signature_len);
    TEST_ASSERT(err != HC_SUCCESS, "Hybrid reject tampered classical sig");
    signature[4] ^= 0xFF; /* Restore */

    hc_key_free(key);
}
}

/* Test KEM operations */
void test_kem(void) {
    printf("\n=== KEM Tests ===\n");

    uint8_t ciphertext[4096];
    size_t ciphertext_len;
    uint8_t shared_secret_enc[64];
    size_t shared_secret_enc_len;
    uint8_t shared_secret_dec[64];
    size_t shared_secret_dec_len;

    hc_key_t *key = NULL;
    hc_error_t err;

    /* Test X25519 KEM */
    err = hc_key_generate(HC_ALG_X25519, &key);
    if (err == HC_SUCCESS) {

```

```

    ciphertext_len = sizeof(ciphertext);
    shared_secret_enc_len = sizeof(shared_secret_enc);

    err = hc_kem_encapsulate(key, ciphertext, &ciphertext_len,
                             shared_secret_enc, &shared_secret_enc_len);
    TEST_ASSERT(err == HC_SUCCESS, "X25519 encapsulate");

    shared_secret_dec_len = sizeof(shared_secret_dec);
    err = hc_kem_decapsulate(key, ciphertext, ciphertext_len,
                             shared_secret_dec, &shared_secret_dec_len);
    TEST_ASSERT(err == HC_SUCCESS, "X25519 decapsulate");

    TEST_ASSERT(shared_secret_enc_len == shared_secret_dec_len &&
                memcmp(shared_secret_enc, shared_secret_dec,
                       shared_secret_enc_len) == 0,
                "X25519 shared secrets match");

    hc_key_free(key);
    key = NULL;
}

/* Test hybrid KEM */
err = hc_key_generate(HC_ALG_HYBRID_X25519_MLKEM768, &key);
if (err == HC_ERROR_NOT_SUPPORTED) {
    TEST_SKIP("Hybrid KEM");
} else if (err == HC_SUCCESS) {
    ciphertext_len = sizeof(ciphertext);
    shared_secret_enc_len = sizeof(shared_secret_enc);

    err = hc_kem_encapsulate(key, ciphertext, &ciphertext_len,
                             shared_secret_enc, &shared_secret_enc_len);
    TEST_ASSERT(err == HC_SUCCESS, "Hybrid encapsulate");
    TEST_ASSERT(ciphertext_len > 32, "Hybrid ciphertext larger than X25519");

    shared_secret_dec_len = sizeof(shared_secret_dec);
    err = hc_kem_decapsulate(key, ciphertext, ciphertext_len,
                             shared_secret_dec, &shared_secret_dec_len);
    TEST_ASSERT(err == HC_SUCCESS, "Hybrid decapsulate");

    TEST_ASSERT(shared_secret_enc_len == shared_secret_dec_len &&
                memcmp(shared_secret_enc, shared_secret_dec,
                       shared_secret_enc_len) == 0,
                "Hybrid shared secrets match");

    hc_key_free(key);
}
}

/* Test key import/export */
void test_key_serialization(void) {
    printf("\n=== Key Serialization Tests ===\n");

    uint8_t exported[16384];
    size_t exported_len;

    hc_key_t *key = NULL, *imported = NULL;
    hc_error_t err;

    /* Test Ed25519 serialization */
    err = hc_key_generate(HC_ALG_ED25519, &key);

```



```

if (err == HC_SUCCESS) {
    /* Export public key */
    exported_len = sizeof(exported);
    err = hc_key_export(key, HC_KEY_PUBLIC, exported, &exported_len);
    TEST_ASSERT(err == HC_SUCCESS, "Ed25519 export public key");
    TEST_ASSERT(exported_len == 32, "Ed25519 public key size");

    /* Import public key */
    err = hc_key_import(HC_ALG_ED25519, HC_KEY_PUBLIC,
                       exported, exported_len, &imported);
    TEST_ASSERT(err == HC_SUCCESS, "Ed25519 import public key");

    /* Verify imported key works */
    const uint8_t msg[] = "test";
    uint8_t sig[64];
    size_t sig_len = sizeof(sig);

    err = hc_sign(key, msg, sizeof(msg), sig, &sig_len);
    TEST_ASSERT(err == HC_SUCCESS, "Sign with original key");

    err = hc_verify(imported, msg, sizeof(msg), sig, sig_len);
    TEST_ASSERT(err == HC_SUCCESS, "Verify with imported key");

    hc_key_free(imported);
    imported = NULL;

    /* Export private key */
    exported_len = sizeof(exported);
    err = hc_key_export(key, HC_KEY_PRIVATE, exported, &exported_len);
    TEST_ASSERT(err == HC_SUCCESS, "Ed25519 export private key");

    hc_key_free(key);
    key = NULL;
}

/* Test hybrid key serialization */
err = hc_key_generate(HC_ALG_HYBRID_ED25519_MLDSA65, &key);
if (err == HC_ERROR_NOT_SUPPORTED) {
    TEST_SKIP("Hybrid key serialization");
} else if (err == HC_SUCCESS) {
    exported_len = sizeof(exported);
    err = hc_key_export(key, HC_KEY_PUBLIC, exported, &exported_len);
    TEST_ASSERT(err == HC_SUCCESS, "Hybrid export public key");
    TEST_ASSERT(exported_len > 32, "Hybrid public key larger than Ed25519");

    err = hc_key_import(HC_ALG_HYBRID_ED25519_MLDSA65, HC_KEY_PUBLIC,
                       exported, exported_len, &imported);
    TEST_ASSERT(err == HC_SUCCESS, "Hybrid import public key");

    hc_key_free(imported);
    hc_key_free(key);
}
}

/* Performance benchmarks */
void test_performance(void) {
    printf("\n=== Performance Benchmarks ===\n");

    const int iterations = 100;
    const uint8_t message[] = "Benchmark message for performance testing";

```

```

size_t message_len = sizeof(message) - 1;

uint8_t signature[8192];
size_t signature_len;

clock_t start, end;
double elapsed;

hc_key_t *key = NULL;
hc_error_t err;

/* Ed25519 benchmark */
err = hc_key_generate(HC_ALG_ED25519, &key);
if (err == HC_SUCCESS) {
    /* Key generation */
    start = clock();
    for (int i = 0; i < iterations; i++) {
        hc_key_t *tmp;
        hc_key_generate(HC_ALG_ED25519, &tmp);
        hc_key_free(tmp);
    }
    end = clock();
    elapsed = (double)(end - start) / CLOCKS_PER_SEC / iterations * 1000;
    printf(" Ed25519 keygen: %.3f ms/op\n", elapsed);

    /* Signing */
    start = clock();
    for (int i = 0; i < iterations; i++) {
        signature_len = sizeof(signature);
        hc_sign(key, message, message_len, signature, &signature_len);
    }
    end = clock();
    elapsed = (double)(end - start) / CLOCKS_PER_SEC / iterations * 1000;
    printf(" Ed25519 sign: %.3f ms/op\n", elapsed);

    /* Verification */
    signature_len = sizeof(signature);
    hc_sign(key, message, message_len, signature, &signature_len);

    start = clock();
    for (int i = 0; i < iterations; i++) {
        hc_verify(key, message, message_len, signature, signature_len);
    }
    end = clock();
    elapsed = (double)(end - start) / CLOCKS_PER_SEC / iterations * 1000;
    printf(" Ed25519 verify: %.3f ms/op\n", elapsed);

    hc_key_free(key);
    key = NULL;
}

/* Hybrid benchmark */
err = hc_key_generate(HC_ALG_HYBRID_ED25519_MLDSA65, &key);
if (err == HC_ERROR_NOT_SUPPORTED) {
    printf(" Hybrid benchmarks: SKIPPED (not available)\n");
} else if (err == HC_SUCCESS) {
    /* Key generation */
    start = clock();
    for (int i = 0; i < iterations / 10; i++) { /* Fewer iterations */
        hc_key_t *tmp;

```

```

        hc_key_generate(HC_ALG_HYBRID_ED25519_MLDSA65, &tmp);
        hc_key_free(tmp);
    }
    end = clock();
    elapsed = (double)(end - start) / CLOCKS_PER_SEC / (iterations / 10) * 1000;
    printf(" Hybrid keygen: %.3f ms/op\n", elapsed);

    /* Signing */
    start = clock();
    for (int i = 0; i < iterations / 10; i++) {
        signature_len = sizeof(signature);
        hc_sign(key, message, message_len, signature, &signature_len);
    }
    end = clock();
    elapsed = (double)(end - start) / CLOCKS_PER_SEC / (iterations / 10) * 1000;
    printf(" Hybrid sign: %.3f ms/op\n", elapsed);

    /* Verification */
    signature_len = sizeof(signature);
    hc_sign(key, message, message_len, signature, &signature_len);

    start = clock();
    for (int i = 0; i < iterations / 10; i++) {
        hc_verify(key, message, message_len, signature, signature_len);
    }
    end = clock();
    elapsed = (double)(end - start) / CLOCKS_PER_SEC / (iterations / 10) * 1000;
    printf(" Hybrid verify: %.3f ms/op\n", elapsed);

    hc_key_free(key);
}
}

/* Main test runner */
int main(int argc, char *argv[]) {
    printf("Hybrid Cryptography Test Suite\n");
    printf("=====\n");

    /* Initialize library */
    hc_error_t err = hc_init();
    if (err != HC_SUCCESS) {
        printf("Failed to initialize library: %s\n", hc_error_string(err));
        return 1;
    }

    printf("Library version: %s\n", hc_version_string());

    /* Run tests */
    test_keygen();
    test_sign_verify();
    test_kem();
    test_key_serialization();

    /* Run benchmarks if requested */
    if (argc > 1 && strcmp(argv[1], "--bench") == 0) {
        test_performance();
    }

    /* Summary */
    printf("\n=== Test Summary ===\n");
}

```

```

printf("Total: %d\n", results.total);
printf("Passed: %d\n", results.passed);
printf("Failed: %d\n", results.failed);
printf("Skipped: %d\n", results.skipped);

/* Cleanup */
hc_cleanup();

return results.failed > 0 ? 1 : 0;
}

```

8.5.7 Deployment Considerations

Checklist for Production Deployment

```

Pre-Deployment:
☐ Security audit completed
☐ Performance benchmarks meet requirements
☐ Memory leak testing (valgrind, ASan)
☐ Thread safety validation
☐ Fuzz testing completed
☐ Known Answer Tests (KATs) pass
☐ Interoperability testing with target systems
☐ Rollback procedure documented

Infrastructure:
☐ HSM integration tested (if applicable)
☐ Key backup and recovery procedures
☐ Monitoring and alerting configured
☐ Log aggregation for cryptographic operations
☐ Incident response procedures updated

Operations:
☐ Runbook for common operations
☐ Key rotation procedures
☐ Certificate renewal automation
☐ Performance monitoring dashboards
☐ Capacity planning for increased key sizes

Compliance:
☐ FIPS validation status documented
☐ Cryptographic inventory updated
☐ Risk assessment completed
☐ Change management approval

```

Exercises

Exercise 8.5.1: Thread-Safe Key Cache *(Advanced)*

Implement a thread-safe key cache that:

1. Stores recently used keys to avoid repeated parsing
2. Uses LRU eviction when capacity is reached

3. Handles concurrent access safely

4. Properly cleans sensitive key material on eviction

```
/* Exercise: Implement this interface */

typedef struct key_cache key_cache_t;

/* Create cache with given capacity */
key_cache_t *key_cache_create(size_t capacity);

/* Store key in cache, returns previous key if slot occupied */
hc_key_t *key_cache_put(key_cache_t *cache,
                        const char *id,
                        hc_key_t *key);

/* Retrieve key from cache (increases ref count) */
hc_key_t *key_cache_get(key_cache_t *cache, const char *id);

/* Remove specific key from cache */
void key_cache_remove(key_cache_t *cache, const char *id);

/* Clear all entries */
void key_cache_clear(key_cache_t *cache);

/* Destroy cache */
void key_cache_destroy(key_cache_t *cache);
```

Solution:

```
/* key_cache.c - Thread-safe LRU key cache */

#include "hybrid_crypto_internal.h"
#include <string.h>
#include <stdlib.h>

#define KEY_ID_MAX_LEN 256

/* Cache entry */
typedef struct cache_entry {
    char id[KEY_ID_MAX_LEN];
    hc_key_t *key;
    uint64_t last_access;
    struct cache_entry *lru_prev;
    struct cache_entry *lru_next;
    struct cache_entry *hash_next; /* For hash collision chain */
} cache_entry_t;

/* Hash table bucket count - prime for better distribution */
#define HASH_BUCKETS 127

struct key_cache {
    pthread_rwlock_t lock;
    size_t capacity;
    size_t size;
    uint64_t access_counter;

    /* Hash table for O(1) lookup */
```

```

cache_entry_t *buckets[HASH_BUCKETS];

/* LRU list (head = most recent, tail = least recent) */
cache_entry_t *lru_head;
cache_entry_t *lru_tail;
};

/* Simple hash function for key IDs */
static unsigned int hash_id(const char *id) {
    unsigned int hash = 5381;
    int c;
    while ((c = *id++)) {
        hash = ((hash << 5) + hash) + c;
    }
    return hash % HASH_BUCKETS;
}

/* Create cache */
key_cache_t *key_cache_create(size_t capacity) {
    if (capacity == 0) return NULL;

    key_cache_t *cache = calloc(1, sizeof(key_cache_t));
    if (cache == NULL) return NULL;

    pthread_rwlock_init(&cache->lock, NULL);
    cache->capacity = capacity;
    cache->size = 0;
    cache->access_counter = 0;
    cache->lru_head = NULL;
    cache->lru_tail = NULL;

    memset(cache->buckets, 0, sizeof(cache->buckets));

    return cache;
}

/* Find entry by ID (must hold at least read lock) */
static cache_entry_t *find_entry(key_cache_t *cache, const char *id) {
    unsigned int bucket = hash_id(id);
    cache_entry_t *entry = cache->buckets[bucket];

    while (entry != NULL) {
        if (strcmp(entry->id, id) == 0) {
            return entry;
        }
        entry = entry->hash_next;
    }

    return NULL;
}

/* Move entry to front of LRU list (must hold write lock) */
static void lru_touch(key_cache_t *cache, cache_entry_t *entry) {
    entry->last_access = ++cache->access_counter;

    /* Already at head? */
    if (entry == cache->lru_head) return;

    /* Remove from current position */
    if (entry->lru_prev != NULL) {

```

```

        entry->lru_prev->lru_next = entry->lru_next;
    }
    if (entry->lru_next != NULL) {
        entry->lru_next->lru_prev = entry->lru_prev;
    }
    if (entry == cache->lru_tail) {
        cache->lru_tail = entry->lru_prev;
    }

    /* Insert at head */
    entry->lru_prev = NULL;
    entry->lru_next = cache->lru_head;
    if (cache->lru_head != NULL) {
        cache->lru_head->lru_prev = entry;
    }
    cache->lru_head = entry;

    if (cache->lru_tail == NULL) {
        cache->lru_tail = entry;
    }
}

/* Remove entry from cache (must hold write lock) */
static void remove_entry(key_cache_t *cache, cache_entry_t *entry) {
    /* Remove from hash table */
    unsigned int bucket = hash_id(entry->id);
    cache_entry_t **ptr = &cache->buckets[bucket];

    while (*ptr != NULL) {
        if (*ptr == entry) {
            *ptr = entry->hash_next;
            break;
        }
        ptr = &(*ptr)->hash_next;
    }

    /* Remove from LRU list */
    if (entry->lru_prev != NULL) {
        entry->lru_prev->lru_next = entry->lru_next;
    } else {
        cache->lru_head = entry->lru_next;
    }

    if (entry->lru_next != NULL) {
        entry->lru_next->lru_prev = entry->lru_prev;
    } else {
        cache->lru_tail = entry->lru_prev;
    }

    cache->size--;

    /* Free key and entry */
    if (entry->key != NULL) {
        hc_key_free(entry->key);
    }

    /* Clear sensitive data */
    hc_secure_clear(entry, sizeof(cache_entry_t));
    free(entry);
}

```

```

/* Evict least recently used entry (must hold write lock) */
static void evict_lru(key_cache_t *cache) {
    if (cache->lru_tail == NULL) return;
    remove_entry(cache, cache->lru_tail);
}

/* Store key in cache */
hc_key_t *key_cache_put(key_cache_t *cache, const char *id, hc_key_t *key) {
    if (cache == NULL || id == NULL || key == NULL) return NULL;
    if (strlen(id) >= KEY_ID_MAX_LEN) return NULL;

    hc_key_t *old_key = NULL;

    pthread_rwlock_wrlock(&cache->lock);

    /* Check if ID already exists */
    cache_entry_t *existing = find_entry(cache, id);
    if (existing != NULL) {
        old_key = existing->key;
        existing->key = hc_key_ref(key);
        lru_touch(cache, existing);
        pthread_rwlock_unlock(&cache->lock);
        return old_key;
    }

    /* Evict if at capacity */
    while (cache->size >= cache->capacity) {
        evict_lru(cache);
    }

    /* Create new entry */
    cache_entry_t *entry = calloc(1, sizeof(cache_entry_t));
    if (entry == NULL) {
        pthread_rwlock_unlock(&cache->lock);
        return NULL;
    }

    strncpy(entry->id, id, KEY_ID_MAX_LEN - 1);
    entry->key = hc_key_ref(key);
    entry->last_access = ++cache->access_counter;

    /* Add to hash table */
    unsigned int bucket = hash_id(id);
    entry->hash_next = cache->buckets[bucket];
    cache->buckets[bucket] = entry;

    /* Add to LRU list (at head) */
    entry->lru_prev = NULL;
    entry->lru_next = cache->lru_head;
    if (cache->lru_head != NULL) {
        cache->lru_head->lru_prev = entry;
    }
    cache->lru_head = entry;

    if (cache->lru_tail == NULL) {
        cache->lru_tail = entry;
    }

    cache->size++;
}

```



```

    pthread_rwlock_unlock(&cache->lock);

    return NULL; /* No previous key */
}

/* Retrieve key from cache */
hc_key_t *key_cache_get(key_cache_t *cache, const char *id) {
    if (cache == NULL || id == NULL) return NULL;

    /* Try read lock first for common case */
    pthread_rwlock_rdlock(&cache->lock);

    cache_entry_t *entry = find_entry(cache, id);
    if (entry == NULL) {
        pthread_rwlock_unlock(&cache->lock);
        return NULL;
    }

    hc_key_t *key = hc_key_ref(entry->key);

    pthread_rwlock_unlock(&cache->lock);

    /* Update LRU (needs write lock) */
    pthread_rwlock_wrlock(&cache->lock);

    /* Re-find entry (may have been evicted) */
    entry = find_entry(cache, id);
    if (entry != NULL) {
        lru_touch(cache, entry);
    }

    pthread_rwlock_unlock(&cache->lock);

    return key;
}

/* Remove specific key */
void key_cache_remove(key_cache_t *cache, const char *id) {
    if (cache == NULL || id == NULL) return;

    pthread_rwlock_wrlock(&cache->lock);

    cache_entry_t *entry = find_entry(cache, id);
    if (entry != NULL) {
        remove_entry(cache, entry);
    }

    pthread_rwlock_unlock(&cache->lock);
}

/* Clear all entries */
void key_cache_clear(key_cache_t *cache) {
    if (cache == NULL) return;

    pthread_rwlock_wrlock(&cache->lock);

    while (cache->lru_head != NULL) {
        remove_entry(cache, cache->lru_head);
    }
}

```

```

        pthread_rwlock_unlock(&cache->lock);
    }

    /* Destroy cache */
    void key_cache_destroy(key_cache_t *cache) {
        if (cache == NULL) return;

        key_cache_clear(cache);
        pthread_rwlock_destroy(&cache->lock);
        free(cache);
    }

    /* Test the cache */
    void test_key_cache(void) {
        printf("Testing key cache...\n");

        key_cache_t *cache = key_cache_create(3);

        /* Generate test keys */
        hc_key_t *key1, *key2, *key3, *key4;
        hc_key_generate(HC_ALG_ED25519, &key1);
        hc_key_generate(HC_ALG_ED25519, &key2);
        hc_key_generate(HC_ALG_ED25519, &key3);
        hc_key_generate(HC_ALG_ED25519, &key4);

        /* Add keys */
        key_cache_put(cache, "key1", key1);
        key_cache_put(cache, "key2", key2);
        key_cache_put(cache, "key3", key3);

        /* Verify retrieval */
        hc_key_t *retrieved = key_cache_get(cache, "key1");
        printf(" Retrieved key1: %s\n", retrieved ? "OK" : "FAIL");
        hc_key_unref(retrieved);

        /* Access key1 to make it recently used */
        retrieved = key_cache_get(cache, "key1");
        hc_key_unref(retrieved);

        /* Add key4 - should evict key2 (least recently used) */
        key_cache_put(cache, "key4", key4);

        /* key2 should be evicted */
        retrieved = key_cache_get(cache, "key2");
        printf(" key2 evicted: %s\n", retrieved == NULL ? "OK" : "FAIL");

        /* key1, key3, key4 should still exist */
        printf(" key1 exists: %s\n", key_cache_get(cache, "key1") ? "OK" : "FAIL");
        printf(" key3 exists: %s\n", key_cache_get(cache, "key3") ? "OK" : "FAIL");
        printf(" key4 exists: %s\n", key_cache_get(cache, "key4") ? "OK" : "FAIL");

        /* Cleanup */
        key_cache_destroy(cache);
        hc_key_free(key1);
    }

```

```

    hc_key_free(key2);
    hc_key_free(key3);
    hc_key_free(key4);

    printf("Key cache test complete.\n");
}

```

Exercise 8.5.2: Parallel Hybrid Verification *(Advanced)*

Implement parallel verification for hybrid signatures that:

1. Starts both verifications concurrently
2. Supports early termination on first failure (lazy mode)
3. Properly handles timeout scenarios
4. Reports which component failed

Solution:

```

/* parallel_verify.c - Parallel hybrid verification */

#include "hybrid_crypto_internal.h"
#include <pthread.h>
#include <string.h>
#include <errno.h>
#include <sys/time.h>

/* Verification result */
typedef struct {
    hc_error_t error;
    const char *component; /* "classical" or "pqc" */
    int completed;
    pthread_mutex_t mutex;
    pthread_cond_t cond;
} verify_result_t;

/* Verification task arguments */
typedef struct {
    hc_key_t *key;
    const uint8_t *message;
    size_t message_len;
    const uint8_t *signature;
    size_t signature_len;
    const char *component_name;
    verify_result_t *result;
    int *cancel_flag; /* Shared cancellation flag */
} verify_task_t;

/* Verification worker thread */
static void *verify_worker(void *arg) {
    verify_task_t *task = (verify_task_t *)arg;

    /* Check if already cancelled */
    if (*task->cancel_flag) {
        pthread_mutex_lock(&task->result->mutex);
        task->result->error = HC_ERROR_STATE;
    }
}

```

```

        task->result->component = task->component_name;
        task->result->completed = 1;
        pthread_cond_signal(&task->result->cond);
        pthread_mutex_unlock(&task->result->mutex);
        return NULL;
    }

    /* Perform verification */
    hc_error_t err = hc_verify(task->key, task->message, task->message_len,
                              task->signature, task->signature_len);

    /* Record result */
    pthread_mutex_lock(&task->result->mutex);
    task->result->error = err;
    task->result->component = task->component_name;
    task->result->completed = 1;
    pthread_cond_signal(&task->result->cond);
    pthread_mutex_unlock(&task->result->mutex);

    return NULL;
}

/* Parallel verification options */
typedef struct {
    int lazy_mode;           /* Stop on first failure */
    int timeout_ms;         /* Timeout in milliseconds (0 = no timeout) */
    int require_both;       /* Both must succeed (default true) */
} parallel_verify_opts_t;

/* Parallel verification result details */
typedef struct {
    hc_error_t overall_error;
    hc_error_t classical_error;
    hc_error_t pqc_error;
    int classical_completed;
    int pqc_completed;
    const char *failed_component; /* First to fail, or NULL */
} parallel_verify_result_t;

/* Parallel hybrid verification */
hc_error_t hc_hybrid_verify_parallel(
    const hc_key_t *hybrid_key,
    const uint8_t *message,
    size_t message_len,
    const uint8_t *signature,
    size_t signature_len,
    const parallel_verify_opts_t *opts,
    parallel_verify_result_t *details)
{
    if (hybrid_key == NULL || message == NULL || signature == NULL) {
        return HC_ERROR_INVALID_PARAM;
    }

    /* Default options */
    parallel_verify_opts_t default_opts = {
        .lazy_mode = 0,
        .timeout_ms = 0,
        .require_both = 1
    };
    if (opts == NULL) {

```

```

    opts = &default_opts;
}

/* Initialize details */
parallel_verify_result_t local_details = {0};
if (details == NULL) {
    details = &local_details;
}
memset(details, 0, sizeof(*details));

/* Parse hybrid signature */
if (signature_len < 8) {
    details->overall_error = HC_ERROR_SIGNATURE;
    return HC_ERROR_SIGNATURE;
}

size_t offset = 0;

/* Classical signature */
size_t classical_sig_len =
    ((size_t)signature[offset] << 24) |
    ((size_t)signature[offset + 1] << 16) |
    ((size_t)signature[offset + 2] << 8) |
    (size_t)signature[offset + 3];
offset += 4;

if (offset + classical_sig_len + 4 > signature_len) {
    details->overall_error = HC_ERROR_SIGNATURE;
    return HC_ERROR_SIGNATURE;
}

const uint8_t *classical_sig = signature + offset;
offset += classical_sig_len;

/* PQC signature */
size_t pqc_sig_len =
    ((size_t)signature[offset] << 24) |
    ((size_t)signature[offset + 1] << 16) |
    ((size_t)signature[offset + 2] << 8) |
    (size_t)signature[offset + 3];
offset += 4;

if (offset + pqc_sig_len > signature_len) {
    details->overall_error = HC_ERROR_SIGNATURE;
    return HC_ERROR_SIGNATURE;
}

const uint8_t *pqc_sig = signature + offset;

/* Get component keys */
hc_key_t *classical_key = hybrid_key->classical_component;
hc_key_t *pqc_key = hybrid_key->pqc_component;

if (classical_key == NULL || pqc_key == NULL) {
    details->overall_error = HC_ERROR_KEY_SIZE;
    return HC_ERROR_KEY_SIZE;
}

/* Initialize result structures */
verify_result_t classical_result = {0};

```

```

verify_result_t pqc_result = {0};
int cancel_flag = 0;

pthread_mutex_init(&classical_result.mutex, NULL);
pthread_cond_init(&classical_result.cond, NULL);
pthread_mutex_init(&pqc_result.mutex, NULL);
pthread_cond_init(&pqc_result.cond, NULL);

/* Create task arguments */
verify_task_t classical_task = {
    .key = classical_key,
    .message = message,
    .message_len = message_len,
    .signature = classical_sig,
    .signature_len = classical_sig_len,
    .component_name = "classical",
    .result = &classical_result,
    .cancel_flag = &cancel_flag
};

verify_task_t pqc_task = {
    .key = pqc_key,
    .message = message,
    .message_len = message_len,
    .signature = pqc_sig,
    .signature_len = pqc_sig_len,
    .component_name = "pqc",
    .result = &pqc_result,
    .cancel_flag = &cancel_flag
};

/* Start verification threads */
pthread_t classical_thread, pqc_thread;
pthread_create(&classical_thread, NULL, verify_worker, &classical_task);
pthread_create(&pqc_thread, NULL, verify_worker, &pqc_task);

/* Calculate timeout */
struct timespec timeout_ts;
if (opts->timeout_ms > 0) {
    struct timeval now;
    gettimeofday(&now, NULL);
    timeout_ts.tv_sec = now.tv_sec + opts->timeout_ms / 1000;
    timeout_ts.tv_nsec = (now.tv_usec + (opts->timeout_ms % 1000) * 1000) * 1000;
    if (timeout_ts.tv_nsec >= 1000000000) {
        timeout_ts.tv_sec++;
        timeout_ts.tv_nsec -= 1000000000;
    }
}

hc_error_t overall_result = HC_SUCCESS;
int both_done = 0;

while (!both_done) {
    /* Check classical result */
    pthread_mutex_lock(&classical_result.mutex);
    while (!classical_result.completed) {
        if (opts->timeout_ms > 0) {
            int rc = pthread_cond_timedwait(&classical_result.cond,
                                             &classical_result.mutex,
                                             &timeout_ts);

```

```

        if (rc == ETIMEDOUT) {
            cancel_flag = 1;
            pthread_mutex_unlock(&classical_result.mutex);
            details->overall_error = HC_ERROR_STATE;
            goto cleanup;
        }
    } else {
        pthread_cond_wait(&classical_result.cond, &classical_result.mutex);
    }
}
details->classical_completed = 1;
details->classical_error = classical_result.error;
pthread_mutex_unlock(&classical_result.mutex);

/* In lazy mode, fail fast on classical failure */
if (opts->lazy_mode && classical_result.error != HC_SUCCESS) {
    cancel_flag = 1;
    details->failed_component = "classical";
    overall_result = classical_result.error;
    goto cleanup;
}

/* Check PQC result */
pthread_mutex_lock(&pqc_result.mutex);
while (!pqc_result.completed) {
    if (opts->timeout_ms > 0) {
        int rc = pthread_cond_timedwait(&pqc_result.cond,
                                         &pqc_result.mutex,
                                         &timeout_ts);

        if (rc == ETIMEDOUT) {
            cancel_flag = 1;
            pthread_mutex_unlock(&pqc_result.mutex);
            details->overall_error = HC_ERROR_STATE;
            goto cleanup;
        }
    } else {
        pthread_cond_wait(&pqc_result.cond, &pqc_result.mutex);
    }
}
details->pqc_completed = 1;
details->pqc_error = pqc_result.error;
pthread_mutex_unlock(&pqc_result.mutex);

both_done = 1;
}

/* Determine overall result */
if (opts->require_both) {
    /* Both must succeed */
    if (classical_result.error != HC_SUCCESS) {
        overall_result = classical_result.error;
        details->failed_component = "classical";
    } else if (pqc_result.error != HC_SUCCESS) {
        overall_result = pqc_result.error;
        details->failed_component = "pqc";
    }
} else {
    /* Either success is sufficient */
    if (classical_result.error != HC_SUCCESS &&
        pqc_result.error != HC_SUCCESS) {

```

```

        overall_result = classical_result.error; /* Report first failure */
        details->failed_component = "both";
    }
}

cleanup:
    /* Wait for threads to complete */
    pthread_join(classical_thread, NULL);
    pthread_join(pqc_thread, NULL);

    /* Cleanup */
    pthread_mutex_destroy(&classical_result.mutex);
    pthread_cond_destroy(&classical_result.cond);
    pthread_mutex_destroy(&pqc_result.mutex);
    pthread_cond_destroy(&pqc_result.cond);

    details->overall_error = overall_result;
    return overall_result;
}

/* Test parallel verification */
void test_parallel_verify(void) {
    printf("Testing parallel verification...\n");

    /* Generate hybrid key */
    hc_key_t *key;
    hc_error_t err = hc_key_generate(HC_ALG_HYBRID_ED25519_MLDSA65, &key);
    if (err != HC_SUCCESS) {
        printf(" SKIP: Hybrid key generation not available\n");
        return;
    }

    /* Sign a message */
    const uint8_t message[] = "Test message for parallel verification";
    uint8_t signature[8192];
    size_t signature_len = sizeof(signature);

    err = hc_sign(key, message, sizeof(message), signature, &signature_len);
    if (err != HC_SUCCESS) {
        printf(" FAIL: Could not sign message\n");
        hc_key_free(key);
        return;
    }

    /* Test parallel verification */
    parallel_verify_opts_t opts = {
        .lazy_mode = 0,
        .timeout_ms = 5000,
        .require_both = 1
    };

    parallel_verify_result_t details;

    err = hc_hybrid_verify_parallel(key, message, sizeof(message),
                                    signature, signature_len,
                                    &opts, &details);

    printf(" Valid signature: %s\n", err == HC_SUCCESS ? "OK" : "FAIL");
    printf(" Classical: %s\n",
           details.classical_error == HC_SUCCESS ? "verified" : "failed");
}

```



```

printf("    PQC: %s\n",
       details.pqc_error == HC_SUCCESS ? "verified" : "failed");

/* Test with tampered signature */
signature[10] ^= 0xFF; /* Tamper with classical component */

opts.lazy_mode = 1; /* Enable lazy mode */

err = hc_hybrid_verify_parallel(key, message, sizeof(message),
                               signature, signature_len,
                               &opts, &details);

printf(" Tampered signature rejected: %s\n",
       err != HC_SUCCESS ? "OK" : "FAIL");
printf("    Failed component: %s\n",
       details.failed_component ? details.failed_component : "none");

hc_key_free(key);
printf("Parallel verification test complete.\n");
}

```

Exercise 8.5.3: Migration Configuration Parser *(Intermediate)*

Implement a configuration file parser for migration settings that:

1. Supports INI-style configuration files
2. Validates algorithm names and policy settings
3. Provides sensible defaults for missing values
4. Logs warnings for deprecated options

Solution:

```

/* config_parser.c - Migration configuration parser */

#include "hybrid_crypto_internal.h"
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

/* Configuration sections */
typedef enum {
    SECTION_NONE,
    SECTION_SIGNATURE,
    SECTION_KEM,
    SECTION_VERIFICATION,
    SECTION_PERFORMANCE,
    SECTION_LOGGING
} config_section_t;

/* Full configuration structure */
typedef struct {
    /* Signature settings */
    struct {
        hc_algorithm_t preferred;

```

```

        hc_algorithm_t fallback;
        int allow_classical_only;
        int allow_pqc_only;
        int require_hybrid;
    } signature;

    /* KEM settings */
    struct {
        hc_algorithm_t preferred;
        hc_algorithm_t fallback;
        int allow_classical_only;
        int allow_pqc_only;
    } kem;

    /* Verification settings */
    struct {
        hc_verify_policy_t policy;
        int strict_mode;
        int allow_degraded;
    } verification;

    /* Performance settings */
    struct {
        int parallel_ops;
        int thread_pool_size;
        int precomputation;
        int key_cache_size;
    } performance;

    /* Logging settings */
    struct {
        int log_classical;
        int log_hybrid;
        int warn_classical;
        int audit_mode;
    } logging;

    /* Metadata */
    int version;
    char source_file[256];
} full_config_t;

/* Algorithm name to enum mapping */
typedef struct {
    const char *name;
    hc_algorithm_t algorithm;
    int deprecated;
} alg_name_t;

static const alg_name_t algorithm_names[] = {
    /* Signature algorithms */
    {"ed25519", HC_ALG_ED25519, 0},
    {"ecdsa_p256", HC_ALG_ECDSA_P256, 0},
    {"ecdsa_p384", HC_ALG_ECDSA_P384, 0},
    {"rsa_pss_2048", HC_ALG_RSA_PSS_2048, 1}, /* Deprecated */
    {"rsa_pss_3072", HC_ALG_RSA_PSS_3072, 0},
    {"mldsa44", HC_ALG_MLDSA44, 0},
    {"mldsa65", HC_ALG_MLDSA65, 0},
    {"mldsa87", HC_ALG_MLDSA87, 0},
    {"slhdsa_sha2_128s", HC_ALG_SLHDSA_SHA2_128S, 0},

```

```

{"slhdsa_sha2_128f", HC_ALG_SLHDSA_SHA2_128F, 0},
{"hybrid_ed25519_mldsa65", HC_ALG_HYBRID_ED25519_MLDSA65, 0},
{"hybrid_p256_mldsa65", HC_ALG_HYBRID_P256_MLDSA65, 0},

/* KEM algorithms */
{"x25519", HC_ALG_X25519, 0},
{"ecdh_p256", HC_ALG_ECDH_P256, 0},
{"ecdh_p384", HC_ALG_ECDH_P384, 0},
{"mlkem512", HC_ALG_MLKEM512, 0},
{"mlkem768", HC_ALG_MLKEM768, 0},
{"mlkem1024", HC_ALG_MLKEM1024, 0},
{"hybrid_x25519_mlkem768", HC_ALG_HYBRID_X25519_MLKEM768, 0},
{"hybrid_p256_mlkem768", HC_ALG_HYBRID_P256_MLKEM768, 0},

{NULL, 0, 0}
};

/* Policy name mapping */
typedef struct {
    const char *name;
    hc_verify_policy_t policy;
} policy_name_t;

static const policy_name_t policy_names[] = {
    {"all", HC_VERIFY_ALL},
    {"any", HC_VERIFY_ANY},
    {"classical", HC_VERIFY_CLASSICAL},
    {"pqc", HC_VERIFY_PQC},
    {"lazy_all", HC_VERIFY_LAZY_ALL},
    {NULL, 0}
};

/* Logging callback */
typedef void (*config_log_fn)(const char *level, const char *message);
static config_log_fn log_callback = NULL;

static void log_message(const char *level, const char *fmt, ...) {
    if (log_callback == NULL) return;

    char buf[512];
    va_list args;
    va_start(args, fmt);
    vsnprintf(buf, sizeof(buf), fmt, args);
    va_end(args);

    log_callback(level, buf);
}

/* Set logging callback */
void config_set_log_callback(config_log_fn fn) {
    log_callback = fn;
}

/* Trim whitespace */
static char *trim(char *str) {
    while (isspace(*str)) str++;
    if (*str == '\0') return str;

    char *end = str + strlen(str) - 1;
    while (end > str && isspace(*end)) end--;

```

```

        *(end + 1) = '\0';

        return str;
    }

    /* Parse boolean value */
    static int parse_bool(const char *value, int *result) {
        if (strcasecmp(value, "true") == 0 ||
            strcasecmp(value, "yes") == 0 ||
            strcasecmp(value, "1") == 0 ||
            strcasecmp(value, "on") == 0) {
            *result = 1;
            return 0;
        }

        if (strcasecmp(value, "false") == 0 ||
            strcasecmp(value, "no") == 0 ||
            strcasecmp(value, "0") == 0 ||
            strcasecmp(value, "off") == 0) {
            *result = 0;
            return 0;
        }

        return -1; /* Invalid */
    }

    /* Parse integer value */
    static int parse_int(const char *value, int *result) {
        char *end;
        long val = strtol(value, &end, 10);

        if (*end != '\0' || val < INT_MIN || val > INT_MAX) {
            return -1;
        }

        *result = (int)val;
        return 0;
    }

    /* Parse algorithm name */
    static int parse_algorithm(const char *name, hc_algorithm_t *result) {
        for (const alg_name_t *a = algorithm_names; a->name != NULL; a++) {
            if (strcasecmp(name, a->name) == 0) {
                if (a->deprecated) {
                    log_message("WARN", "Algorithm '%s' is deprecated", name);
                }
                *result = a->algorithm;
                return 0;
            }
        }
        return -1;
    }

    /* Parse policy name */
    static int parse_policy(const char *name, hc_verify_policy_t *result) {
        for (const policy_name_t *p = policy_names; p->name != NULL; p++) {
            if (strcasecmp(name, p->name) == 0) {
                *result = p->policy;
                return 0;
            }
        }
    }

```

```

    }
    return -1;
}

/* Parse section header */
static config_section_t parse_section(const char *name) {
    if (strcasecmp(name, "signature") == 0) return SECTION_SIGNATURE;
    if (strcasecmp(name, "kem") == 0) return SECTION_KEM;
    if (strcasecmp(name, "verification") == 0) return SECTION_VERIFICATION;
    if (strcasecmp(name, "performance") == 0) return SECTION_PERFORMANCE;
    if (strcasecmp(name, "logging") == 0) return SECTION_LOGGING;
    return SECTION_NONE;
}

/* Set default values */
static void config_set_defaults(full_config_t *config) {
    memset(config, 0, sizeof(*config));

    /* Signature defaults */
    config->signature.preferred = HC_ALG_HYBRID_ED25519_MLDSA65;
    config->signature.fallback = HC_ALG_ED25519;
    config->signature.allow_classical_only = 1;
    config->signature.allow_pqc_only = 0;
    config->signature.require_hybrid = 0;

    /* KEM defaults */
    config->kem.preferred = HC_ALG_HYBRID_X25519_MLKEM768;
    config->kem.fallback = HC_ALG_X25519;
    config->kem.allow_classical_only = 1;
    config->kem.allow_pqc_only = 0;

    /* Verification defaults */
    config->verification.policy = HC_VERIFY_ALL;
    config->verification.strict_mode = 1;
    config->verification.allow_degraded = 0;

    /* Performance defaults */
    config->performance.parallel_ops = 1;
    config->performance.thread_pool_size = 4;
    config->performance.precomputation = 1;
    config->performance.key_cache_size = 16;

    /* Logging defaults */
    config->logging.log_classical = 1;
    config->logging.log_hybrid = 0;
    config->logging.warn_classical = 1;
    config->logging.audit_mode = 0;

    config->version = 1;
}

/* Parse configuration file */
int config_parse_file(const char *filename, full_config_t *config) {
    FILE *f = fopen(filename, "r");
    if (f == NULL) {
        log_message("ERROR", "Cannot open config file: %s", filename);
        return -1;
    }

    config_set_defaults(config);
}

```

```

strncpy(config->source_file, filename, sizeof(config->source_file) - 1);

char line[512];
int line_num = 0;
config_section_t current_section = SECTION_NONE;
int errors = 0;

while (fgets(line, sizeof(line), f)) {
    line_num++;

    char *trimmed = trim(line);

    /* Skip empty lines and comments */
    if (*trimmed == '\0' || *trimmed == '#' || *trimmed == ';') {
        continue;
    }

    /* Section header */
    if (*trimmed == '[') {
        char *end = strchr(trimmed, ']');
        if (end == NULL) {
            log_message("ERROR", "Line %d: Invalid section header", line_num);
            errors++;
            continue;
        }
        *end = '\0';
        current_section = parse_section(trimmed + 1);
        if (current_section == SECTION_NONE) {
            log_message("WARN", "Line %d: Unknown section '%s'",
                line_num, trimmed + 1);
        }
        continue;
    }

    /* Key-value pair */
    char *equals = strchr(trimmed, '=');
    if (equals == NULL) {
        log_message("ERROR", "Line %d: Invalid syntax", line_num);
        errors++;
        continue;
    }

    *equals = '\0';
    char *key = trim(trimmed);
    char *value = trim(equals + 1);

    /* Process based on section */
    int valid = 0;

    switch (current_section) {
        case SECTION_SIGNATURE:
            if (strcmp(key, "preferred") == 0) {
                valid = (parse_algorithm(value, &config->signature.preferred) == 0);
            } else if (strcmp(key, "fallback") == 0) {
                valid = (parse_algorithm(value, &config->signature.fallback) == 0);
            } else if (strcmp(key, "allow_classical_only") == 0) {
                valid = (parse_bool(value, &config->signature.allow_classical_only) ==
0);
            } else if (strcmp(key, "allow_pqc_only") == 0) {
                valid = (parse_bool(value, &config->signature.allow_pqc_only) == 0);
            }
    }
}

```

```

        } else if (strcmp(key, "require_hybrid") == 0) {
            valid = (parse_bool(value, &config->signature.require_hybrid) == 0);
        }
        break;

case SECTION_KEM:
    if (strcmp(key, "preferred") == 0) {
        valid = (parse_algorithm(value, &config->kem.preferred) == 0);
    } else if (strcmp(key, "fallback") == 0) {
        valid = (parse_algorithm(value, &config->kem.fallback) == 0);
    } else if (strcmp(key, "allow_classical_only") == 0) {
        valid = (parse_bool(value, &config->kem.allow_classical_only) == 0);
    } else if (strcmp(key, "allow_pqc_only") == 0) {
        valid = (parse_bool(value, &config->kem.allow_pqc_only) == 0);
    }
    break;

case SECTION_VERIFICATION:
    if (strcmp(key, "policy") == 0) {
        valid = (parse_policy(value, &config->verification.policy) == 0);
    } else if (strcmp(key, "strict_mode") == 0) {
        valid = (parse_bool(value, &config->verification.strict_mode) == 0);
    } else if (strcmp(key, "allow_degraded") == 0) {
        valid = (parse_bool(value, &config->verification.allow_degraded) ==
0);
    }
    break;

case SECTION_PERFORMANCE:
    if (strcmp(key, "parallel_ops") == 0) {
        valid = (parse_bool(value, &config->performance.parallel_ops) == 0);
    } else if (strcmp(key, "thread_pool_size") == 0) {
        valid = (parse_int(value, &config->performance.thread_pool_size) ==
0);
    } else if (strcmp(key, "precomputation") == 0) {
        valid = (parse_bool(value, &config->performance.precomputation) == 0);
    } else if (strcmp(key, "key_cache_size") == 0) {
        valid = (parse_int(value, &config->performance.key_cache_size) == 0);
    }
    break;

case SECTION_LOGGING:
    if (strcmp(key, "log_classical") == 0) {
        valid = (parse_bool(value, &config->logging.log_classical) == 0);
    } else if (strcmp(key, "log_hybrid") == 0) {
        valid = (parse_bool(value, &config->logging.log_hybrid) == 0);
    } else if (strcmp(key, "warn_classical") == 0) {
        valid = (parse_bool(value, &config->logging.warn_classical) == 0);
    } else if (strcmp(key, "audit_mode") == 0) {
        valid = (parse_bool(value, &config->logging.audit_mode) == 0);
    }
    break;

case SECTION_NONE:
    /* Global settings */
    if (strcmp(key, "version") == 0) {
        valid = (parse_int(value, &config->version) == 0);
    }
    break;
}

```

```

        if (!valid) {
            log_message("WARN", "Line %d: Unknown or invalid option '%s = %s'",
                        line_num, key, value);
        }
    }

    fclose(f);

    /* Validate configuration */
    if (config->signature.require_hybrid &&
        config->signature.allow_classical_only) {
        log_message("WARN", "Conflicting settings: require_hybrid and "
                        "allow_classical_only both set");
    }

    if (config->performance.thread_pool_size < 1) {
        log_message("WARN", "thread_pool_size must be >= 1, using default");
        config->performance.thread_pool_size = 4;
    }

    if (config->performance.key_cache_size < 0) {
        log_message("WARN", "key_cache_size must be >= 0, using default");
        config->performance.key_cache_size = 16;
    }

    return errors > 0 ? -1 : 0;
}

/* Print configuration */
void config_print(const full_config_t *config) {
    printf("Configuration from: %s\n", config->source_file);
    printf("Version: %d\n\n", config->version);

    printf("[signature]\n");
    printf(" preferred = %04x\n", config->signature.preferred);
    printf(" fallback = %04x\n", config->signature.fallback);
    printf(" allow_classical_only = %s\n",
           config->signature.allow_classical_only ? "true" : "false");
    printf(" allow_pqc_only = %s\n",
           config->signature.allow_pqc_only ? "true" : "false");
    printf(" require_hybrid = %s\n",
           config->signature.require_hybrid ? "true" : "false");

    printf("\n[kem]\n");
    printf(" preferred = %04x\n", config->kem.preferred);
    printf(" fallback = %04x\n", config->kem.fallback);
    printf(" allow_classical_only = %s\n",
           config->kem.allow_classical_only ? "true" : "false");

    printf("\n[verification]\n");
    printf(" policy = %d\n", config->verification.policy);
    printf(" strict_mode = %s\n",
           config->verification.strict_mode ? "true" : "false");

    printf("\n[performance]\n");
    printf(" parallel_ops = %s\n",
           config->performance.parallel_ops ? "true" : "false");
    printf(" thread_pool_size = %d\n", config->performance.thread_pool_size);
    printf(" key_cache_size = %d\n", config->performance.key_cache_size);

```



```

    printf("\n[logging]\n");
    printf("  warn_classical = %s\n",
           config->logging.warn_classical ? "true" : "false");
    printf("  audit_mode = %s\n",
           config->logging.audit_mode ? "true" : "false");
}

/* Example configuration file content:

# Migration Configuration
version = 1

[signature]
preferred = hybrid_ed25519_mldsa65
fallback = ed25519
allow_classical_only = true
require_hybrid = false

[kem]
preferred = hybrid_x25519_mlkem768
fallback = x25519
allow_classical_only = true

[verification]
policy = all
strict_mode = true

[performance]
parallel_ops = true
thread_pool_size = 4
key_cache_size = 32

[logging]
warn_classical = true
audit_mode = false

*/

```

Exercise 8.5.4: Integration Test Suite *(Advanced)*

Design and implement a comprehensive integration test that verifies:

1. End-to-end hybrid signing and verification
2. Interoperability with external test vectors
3. Error handling for malformed inputs
4. Performance regression detection

Solution:

[FULL — 508 LINES]

Extracted to `source_code/module_08_hybrid/unit_8_5_integration_test_full.c`.

Below is a curated excerpt showing the structural skeleton; the full implementation includes ~458 additional lines of helpers, error handling, and detail. For study, open the source file in an editor with syntax highlighting.

```
/* integration_test.c - Comprehensive integration test suite */

#include "hybrid_crypto.h"
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>
#include <math.h>

/* Test statistics */
typedef struct {
    int total_tests;
    int passed;
    int failed;
    int skipped;
    double total_time_ms;
} test_stats_t;

static test_stats_t stats = {0};

/* Timing helper */
static double get_time_ms(void) {
    struct timespec ts;
    clock_gettime(CLOCK_MONOTONIC, &ts);
    return ts.tv_sec * 1000.0 + ts.tv_nsec / 1000000.0;
}

/* Test macros */
#define TEST_BEGIN(name) \
    do { \
        const char *_test_name = name; \
        double _start = get_time_ms(); \
        int _passed = 1; \
        stats.total_tests++;

    /* ... 458 lines elided — see source_code/module_08_hybrid/
    unit_8_5_integration_test_full.c ... */

        printf("RESULT: PASSED\n");
    }

    /* Cleanup */
    hc_cleanup();

    return stats.failed > 0 ? 1 : 0;
}
```

Unit 8.5 Summary

This unit covered the practical aspects of implementing hybrid cryptography in production systems:

Library Architecture:

- Provider-based design allows runtime selection of cryptographic backends
- Clean API separation between public interface and internal implementation
- Thread-safe key management with reference counting

Memory Safety:

- Secure allocation prevents key material from being swapped to disk
- Constant-time comparisons prevent timing attacks
- Proper cleanup ensures sensitive data is wiped from memory

Performance Optimization:

- Parallel execution reduces latency for hybrid operations
- Lazy verification provides early termination on failure
- Precomputation and caching improve repeated operations

Migration Strategies:

- Phased approach allows gradual transition
- Configuration-driven policy enables flexible deployment
- Backward compatibility during transition period

Protocol Integration:

- TLS integration requires proper algorithm negotiation
- Certificate handling must support hybrid keys
- Wire format compatibility with existing infrastructure

Testing:

- Comprehensive test suites ensure correctness
- Performance regression tests catch degradation
- Stress testing validates production readiness

Key Takeaways

- Design for algorithm agility from the start

- Thread safety and memory safety are non-negotiable
- Performance optimization must not compromise security
- Migration requires careful planning and tooling
- Comprehensive testing is essential before deployment

Unit 8.6: Merkle Tree Certificates (MTCs) and the PLANTS Working Group

Problem statement: a composite ML-DSA + ECDSA X.509 certificate chain costs 17-20 KB per TLS handshake (three certs: leaf + intermediate + root, each ~5 KB). Chrome's empirically-measured viability threshold for handshake size is around 10-12 KB before middlebox breakage becomes material. Classical ECDSA chains run ~3-5 KB. The handshake-size gap is the single largest operational barrier to PQ X.509 deployment at web scale.

8.6.1 The MTC Construction

Merkle Tree Certificates (MTCs) are a reaction to the size problem. Proposed jointly by Google and Cloudflare, and now adopted by the IETF **PLANTS** Working Group (chartered 2026), MTCs replace per-domain PQ certificates with batch-signed *inclusion proofs* against a Merkle tree:

1. A Certificate Transparency-style log operator batches domain→pubkey attestations into a Merkle tree, with perhaps thousands of leaves per hour.
2. The root of each batch tree is signed once by the MTCA (Merkle Tree Certificate Authority) under its long-term PQ signing key.
3. A server's "certificate" is: the domain attestation (tiny), the Merkle path (log N hashes), the root signature (amortized across the whole batch).
4. Clients verify the MTCA's root signature, the Merkle path, and that the leaf binds the expected pubkey.

Size impact: a TLS server presents roughly **736 bytes** in place of 14.7 KB — a 20× reduction. The per-batch MTCA signature is 3-5 KB but amortizes across every domain in that batch. For a batch of 10,000 domains, the effective per-domain overhead is <1 KB.

8.6.2 PLANTS WG Status (April 2026)

ARTEFACT	VERSION	STATUS
draft-ietf-plants-merkle-tree-certs	-02	Working Group draft; not yet RFC
Chrome Quantum-resistant Root Store (CQRS)	Phase 1 (2026)	Mock MTCA feasibility trial
CQRS Phase 2	Q1 2027 target	CT operators onboarded as MTCA beta
CQRS Phase 3	Q3 2027 target	CA onboarding; public MTC issuance begins

Independent review: Trail of Bits published an analysis in Q1 2026 highlighting key-compromise scenarios and batch-root lifecycle concerns. The working group is iterating on revocation (analogous to CRL/OCSP for traditional X.509) and root-key rotation procedures.

8.6.3 Relationship to Composite X.509

MTCs and composite X.509 ([draft-ietf-lamps-pq-composite-sigs](#)) are **complementary**, not competing:

- **Composite X.509** works inside the existing PKI with no new infrastructure. Best for internal/enterprise PKIs where size is not the blocker.
- **MTCs** require new infrastructure (MTCA operators, client integration) but solve the Web-PKI size problem. Best for Internet-scale public TLS.

A bridging strategy: issue composite certs for internal services and enterprise VPNs (where composite's 17-20 KB is affordable), MTCs for public web servers (where it isn't).

8.6.4 Looma — Lower-Latency Alternative for Datacenter TLS

A separate IETF draft, [draft-ma-cfrg-looma](#), targets datacenter mTLS rather than public Web PKI. Looma pre-generates a batch of WOTS+ (stateful one-time) keys authenticated by a Merkle tree root under a long-term PQ signature. Each TLS handshake consumes one WOTS+ key; the root is authenticated once per deployment. Trade-off: stateful — losing the state counter reuses a WOTS+ key and breaks security — but per-handshake latency is lower than per-connection ML-DSA signing.

Use Looma where stateful management is tractable (single-tenant fleet with strong state management, e.g., Cloudflare-like mesh, Meta datacenter fabric) and avoid it where it isn't (consumer clients, distributed CA signing).

8.6.5 Signal SPQR — "Triple Ratchet" (October 2025)

Signal's **Sparse Post-Quantum Ratchet (SPQR)** complements the existing PQXDH handshake. Combined with Signal's Double Ratchet, the resulting three-component ratchet is marketed as the "Triple Ratchet":

- **Triple Ratchet:** Symmetric ratchet + Diffie-Hellman ratchet + **ML-KEM Braid Protocol** (the SPQR contribution).
- The SPQR "braid" replaces the classical DH ratchet's public-key contribution with an ML-KEM-based construction designed for fresh randomness per message while amortizing the expensive PQ operations across the ratchet state.
- Hybrid key derivation combines Double Ratchet output with SPQR output so compromise of either remains safe against the other.

SPQR was identified by Kobeissi's Verification Theatre analysis (see Unit 11.11) as one of the primary deployment sites of the libcrux bugs — the endianness bug in cross-backend ML-KEM code caused real decryption failures in early SPQR deployments. Fixes shipped in subsequent Signal client updates; the case is an object lesson in the risks of "formally verified" being less complete than marketing suggests.

Unit 8.6 Summary

The composite X.509 approach (Unit 8.4) was the obvious path, but handshake sizes of 17–20 KB hit real middlebox limits before rolling out to the public Web PKI. Merkle Tree Certificates take a different architectural approach — batch-sign the binding of domains to pubkeys, send short inclusion proofs instead of full cert chains — and bring handshake size back below 1 KB. The PLANTS WG, Chrome CQRS rollout, and Signal's SPQR ratchet are all applications of the same insight: PQ keys and signatures are too big to use in the obvious per-object pattern, so amortize them across batches. Expect significant protocol-level architectural change across 2026–2028 to accommodate this shift.

Module 8 Summary: Hybrid Cryptography

Module 8 provided comprehensive coverage of hybrid cryptography for the post-quantum transition:

Unit 8.1: Introduction to Hybrid Cryptography

- Motivation: harvest-now-decrypt-later attacks
- Combining classical and post-quantum algorithms
- Security analysis of hybrid constructions

Unit 8.2: Hybrid Key Exchange

- Concatenated and nested combiners
- KDF usage for shared secret derivation
- Protocol considerations (TLS, IKE)

Unit 8.3: Hybrid Signatures

- Parallel and nested signature schemes
- Verification policies and failure modes
- Certificate chain considerations

Unit 8.4: Composite Key Formats and Standards

- ASN.1/DER encoding for composite keys
- X.509 certificate extensions
- PKCS#8 private key format
- Interoperability with existing PKI

Unit 8.5: Implementation and Integration

- Library architecture patterns
- Thread safety and memory management
- Performance optimization techniques
- Migration strategies and deployment

What's Next: Module 9 covers Protocol Integration, examining how hybrid cryptography is being integrated into real-world protocols like TLS 1.3, SSH, and S/MIME, with detailed analysis of standardization efforts and implementation guidance.

Module 9: Protocol Integration

This module examines how post-quantum cryptography is being integrated into real-world protocols. We analyze the specific challenges each protocol faces, review ongoing standardization efforts, and provide implementation guidance for deploying PQC in production systems.

Prerequisites: Modules 4-8 (Volume 1 for 4, Volume 2 for 8) (ML-KEM, ML-DSA, SLH-DSA, Hybrid Cryptography)

Module Learning Objectives:

- Understand TLS 1.3 post-quantum integration approaches
- Configure and deploy hybrid key exchange in TLS
- Implement PQC in SSH protocol for secure shell connections
- Apply post-quantum security to S/MIME email encryption
- Navigate ongoing IETF standardization efforts
- Handle increased message sizes in protocol implementations

Unit 9.1: TLS 1.3 Post-Quantum Integration

Prerequisites: Volume 1, Module 4 (ML-KEM), Module 8 (Hybrid Cryptography)

Estimated Time: 2-3 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Understand TLS 1.3 cryptographic architecture** and identify which components require post-quantum upgrades
2. **Implement hybrid key exchange** using X25519+ML-KEM in TLS 1.3
3. **Handle post-quantum certificate chains** with hybrid signatures
4. **Manage increased message sizes** and their impact on TLS handshakes
5. **Configure and test** PQC-enabled TLS implementations

6. Navigate standardization landscape including IETF drafts and industry adoption

9.1.1 TLS 1.3 Cryptographic Components

TLS 1.3 uses cryptography in several distinct places, each with different post-quantum considerations:

TLS 1.3 Handshake Cryptographic Usage

=====

Client	Server
ClientHello	
- supported_versions	
- key_share (ECDHE public key) <-- KEX	
- signature_algorithms <-- Auth	
- supported_groups <-- KEX	
----->	
	ServerHello
	- key_share <-- KEX
<-----	
	{EncryptedExtensions}
	{CertificateRequest*}
	{Certificate} <-- Auth
	{CertificateVerify} <-- Auth
	{Finished} <-- Integrity
<-----	
{Certificate*} <-- Auth	
{CertificateVerify*} <-- Auth	
{Finished} <-- Integrity	
----->	
[Application Data] <-- Symmetric	
<=====	

Cryptographic Primitives by Function:

KEX (Key Exchange):

- Currently: X25519, P-256, P-384, P-521
- PQC: ML-KEM-768, ML-KEM-1024
- Hybrid: X25519+ML-KEM-768

Auth (Authentication):

- Currently: RSA-PSS, ECDSA, Ed25519
- PQC: ML-DSA-65, ML-DSA-87, SLH-DSA
- Hybrid: Ed25519+ML-DSA-65

```

Integrity (Finished messages):
- HMAC-SHA256/384 (quantum-resistant with sufficient key size)

Symmetric (Application data):
- AES-128-GCM, AES-256-GCM, ChaCha20-Poly1305
- Already quantum-resistant (Grover's attack manageable)

```

9.1.2 Hybrid Key Exchange in TLS 1.3

The most urgent upgrade is key exchange, as recorded traffic can be decrypted later:

```

/* tls13_hybrid_kex.h - Hybrid Key Exchange for TLS 1.3 */

#ifndef TLS13_HYBRID_KEX_H
#define TLS13_HYBRID_KEX_H

#include <stdint.h>
#include <stddef.h>

/* Named groups for TLS 1.3 (from IANA registry + drafts) */
typedef enum {
    /* Classical ECDHE groups */
    TLS_GROUP_X25519      = 0x001D,
    TLS_GROUP_X448        = 0x001E,
    TLS_GROUP_SECP256R1   = 0x0017,
    TLS_GROUP_SECP384R1   = 0x0018,
    TLS_GROUP_SECP521R1   = 0x0019,

    /* Post-quantum KEMs (from draft-ietf-tls-ml-kem) */
    TLS_GROUP_MLKEM512    = 0x0200,
    TLS_GROUP_MLKEM768    = 0x0201,
    TLS_GROUP_MLKEM1024   = 0x0202,

    /* Hybrid groups (from draft-ietf-tls-hybrid-design) */
    TLS_GROUP_X25519_MLKEM768 = 0x11EC, /* Recommended */
    TLS_GROUP_SECP256R1_MLKEM768 = 0x11EB,
    TLS_GROUP_X448_MLKEM1024   = 0x11EE,
    TLS_GROUP_SECP384R1_MLKEM1024 = 0x11ED,
} tls_named_group_t;

/* Key share entry structure */
typedef struct {
    tls_named_group_t group;
    uint16_t key_exchange_len;
    uint8_t *key_exchange; /* Public key or ciphertext */
} tls_key_share_entry_t;

/* Hybrid key exchange context */
typedef struct {
    tls_named_group_t group;

    /* Classical component (X25519) */
    uint8_t x25519_private[32];
    uint8_t x25519_public[32];
    uint8_t x25519_shared[32];

    /* PQC component (ML-KEM-768) */

```

```

uint8_t *mlkem_dk;          /* Decapsulation key (private) */
size_t mlkem_dk_len;
uint8_t *mlkem_ek;          /* Encapsulation key (public) */
size_t mlkem_ek_len;
uint8_t mlkem_shared[32];    /* ML-KEM shared secret */

/* Combined shared secret */
uint8_t shared_secret[64]; /* SHA-256(x25519_ss || mlkem_ss) or concat */
size_t shared_secret_len;

/* Role tracking */
int is_server;
} hybrid_kex_ctx_t;

/* Key share sizes for hybrid groups */
#define X25519_KEY_SIZE      32
#define MLKEM768_EK_SIZE     1184
#define MLKEM768_CT_SIZE     1088
#define MLKEM768_SS_SIZE     32

/* Hybrid key share sizes */
#define HYBRID_X25519_MLKEM768_CLIENT_SHARE_SIZE (X25519_KEY_SIZE + MLKEM768_EK_SIZE)
#define HYBRID_X25519_MLKEM768_SERVER_SHARE_SIZE (X25519_KEY_SIZE + MLKEM768_CT_SIZE)

/* Function prototypes */
int hybrid_kex_init(hybrid_kex_ctx_t *ctx, tls_named_group_t group, int is_server);
int hybrid_kex_generate_key_share(hybrid_kex_ctx_t *ctx,
                                  uint8_t *key_share, size_t *key_share_len);
int hybrid_kex_process_key_share(hybrid_kex_ctx_t *ctx,
                                  const uint8_t *peer_share, size_t peer_share_len,
                                  uint8_t *response_share, size_t *response_share_len);
int hybrid_kex_derive_secret(hybrid_kex_ctx_t *ctx,
                              uint8_t *secret, size_t *secret_len);
void hybrid_kex_cleanup(hybrid_kex_ctx_t *ctx);

#endif /* TLS13_HYBRID_KEX_H */

```

Implementation of hybrid key exchange:

```

/* tls13_hybrid_kex.c - Hybrid Key Exchange Implementation */

#include "tls13_hybrid_kex.h"
#include <string.h>
#include <stdlib.h>

/* External cryptographic functions (from your crypto library) */
extern int x25519_keygen(uint8_t *public_key, uint8_t *private_key);
extern int x25519_shared_secret(uint8_t *shared, const uint8_t *private_key,
                                const uint8_t *peer_public);
extern int mlkem768_keygen(uint8_t *ek, size_t *ek_len,
                            uint8_t *dk, size_t *dk_len);
extern int mlkem768_encaps(uint8_t *ct, size_t *ct_len,
                            uint8_t *ss, const uint8_t *ek, size_t ek_len);
extern int mlkem768_decaps(uint8_t *ss, const uint8_t *ct, size_t ct_len,
                            const uint8_t *dk, size_t dk_len);
extern int sha256(uint8_t *hash, const uint8_t *data, size_t len);
extern void secure_zero(void *ptr, size_t len);

```

```

int hybrid_kex_init(hybrid_kex_ctx_t *ctx, tls_named_group_t group, int is_server) {
    if (!ctx) return -1;

    memset(ctx, 0, sizeof(hybrid_kex_ctx_t));
    ctx->group = group;
    ctx->is_server = is_server;

    /* Validate supported group */
    switch (group) {
        case TLS_GROUP_X25519_MLKEM768:
            ctx->mlkem_dk_len = 2400; /* ML-KEM-768 decapsulation key size */
            ctx->mlkem_ek_len = MLKEM768_EK_SIZE;
            break;
        default:
            return -1; /* Unsupported group */
    }

    /* Allocate ML-KEM keys */
    ctx->mlkem_dk = malloc(ctx->mlkem_dk_len);
    ctx->mlkem_ek = malloc(ctx->mlkem_ek_len);

    if (!ctx->mlkem_dk || !ctx->mlkem_ek) {
        hybrid_kex_cleanup(ctx);
        return -1;
    }

    return 0;
}

/* Client: Generate initial key share (X25519 public || ML-KEM ek) */
int hybrid_kex_generate_key_share(hybrid_kex_ctx_t *ctx,
                                   uint8_t *key_share, size_t *key_share_len) {
    if (!ctx || !key_share || !key_share_len) return -1;

    size_t required_len;

    switch (ctx->group) {
        case TLS_GROUP_X25519_MLKEM768:
            required_len = ctx->is_server ?
                HYBRID_X25519_MLKEM768_SERVER_SHARE_SIZE :
                HYBRID_X25519_MLKEM768_CLIENT_SHARE_SIZE;
            break;
        default:
            return -1;
    }

    if (*key_share_len < required_len) {
        *key_share_len = required_len;
        return -1;
    }

    /* Generate X25519 key pair */
    if (x25519_keygen(ctx->x25519_public, ctx->x25519_private) != 0) {
        return -1;
    }

    /* Copy X25519 public key to output */
    memcpy(key_share, ctx->x25519_public, X25519_KEY_SIZE);

    if (!ctx->is_server) {

```

```

    /* Client generates ML-KEM key pair, sends encapsulation key */
    size_t dk_len = ctx->mlkem_dk_len;
    size_t ek_len = ctx->mlkem_ek_len;

    if (mlkem768_keygen(ctx->mlkem_ek, &ek_len,
                       ctx->mlkem_dk, &dk_len) != 0) {
        return -1;
    }

    /* Append ML-KEM encapsulation key */
    memcpy(key_share + X25519_KEY_SIZE, ctx->mlkem_ek, ek_len);
    *key_share_len = X25519_KEY_SIZE + ek_len;
}

return 0;
}

/* Process peer's key share and optionally generate response */
int hybrid_kex_process_key_share(hybrid_kex_ctx_t *ctx,
                                const uint8_t *peer_share, size_t peer_share_len,
                                uint8_t *response_share, size_t *response_share_len) {
    if (!ctx || !peer_share) return -1;

    switch (ctx->group) {
        case TLS_GROUP_X25519_MLKEM768:
            if (ctx->is_server) {
                /* Server receives: X25519 public || ML-KEM ek */
                if (peer_share_len != HYBRID_X25519_MLKEM768_CLIENT_SHARE_SIZE) {
                    return -1;
                }

                /* Generate the server's X25519 ephemeral key pair ONCE.
                 * This public key is what we return in the response share, and
                 * the matching private key is what we derive x25519_shared from. */
                if (x25519_keygen(ctx->x25519_public, ctx->x25519_private) != 0) {
                    return -1;
                }

                /* Derive the X25519 shared secret exactly once, from our fresh
                 * ephemeral private key and the peer's X25519 public key. */
                if (x25519_shared_secret(ctx->x25519_shared,
                                         ctx->x25519_private,
                                         peer_share) != 0) {
                    return -1;
                }

                /* Extract peer's ML-KEM encapsulation key */
                const uint8_t *peer_mlkem_ek = peer_share + X25519_KEY_SIZE;
                size_t peer_mlkem_ek_len = peer_share_len - X25519_KEY_SIZE;

                /* Encapsulate to peer's ML-KEM key */
                uint8_t ct[MLKEM768_CT_SIZE];
                size_t ct_len = sizeof(ct);

                if (mlkem768_encaps(ct, &ct_len, ctx->mlkem_shared,
                                   peer_mlkem_ek, peer_mlkem_ek_len) != 0) {
                    return -1;
                }

                /* Generate response: X25519 public || ML-KEM ciphertext.

```

```

        * The X25519 public placed here matches the private key already
        * used above to derive x25519_shared (single ephemeral share). */
    if (response_share && response_share_len) {
        if (*response_share_len < HYBRID_X25519_MLKEM768_SERVER_SHARE_SIZE) {
            *response_share_len = HYBRID_X25519_MLKEM768_SERVER_SHARE_SIZE;
            return -1;
        }

        memcpy(response_share, ctx->x25519_public, X25519_KEY_SIZE);
        memcpy(response_share + X25519_KEY_SIZE, ct, ct_len);
        *response_share_len = X25519_KEY_SIZE + ct_len;
    }
} else {
    /* Client receives: X25519 public || ML-KEM ciphertext */
    if (peer_share_len != HYBRID_X25519_MLKEM768_SERVER_SHARE_SIZE) {
        return -1;
    }

    /* Compute X25519 shared secret */
    if (x25519_shared_secret(ctx->x25519_shared,
                            ctx->x25519_private,
                            peer_share) != 0) {
        return -1;
    }

    /* Decapsulate ML-KEM ciphertext */
    const uint8_t *ct = peer_share + X25519_KEY_SIZE;
    size_t ct_len = peer_share_len - X25519_KEY_SIZE;

    if (mlkem768_decaps(ctx->mlkem_shared, ct, ct_len,
                        ctx->mlkem_dk, ctx->mlkem_dk_len) != 0) {
        return -1;
    }
}
break;

default:
    return -1;
}

return 0;
}

/* Derive combined shared secret */
int hybrid_kex_derive_secret(hybrid_kex_ctx_t *ctx,
                             uint8_t *secret, size_t *secret_len) {
    if (!ctx || !secret || !secret_len) return -1;

    uint8_t combined[64]; /* X25519 (32) + ML-KEM (32) */

    memcpy(combined, ctx->x25519_shared, 32);
    memcpy(combined + 32, ctx->mlkem_shared, 32);

    /* Hash to derive shared secret */
    if (sha256(ctx->shared_secret, combined, 64) != 0) {
        secure_zero(combined, sizeof(combined));
        return -1;
    }
    ctx->shared_secret_len = 32;
}

```

```

    secure_zero(combined, sizeof(combined));

    if (*secret_len < ctx->shared_secret_len) {
        *secret_len = ctx->shared_secret_len;
        return -1;
    }

    memcpy(secret, ctx->shared_secret, ctx->shared_secret_len);
    *secret_len = ctx->shared_secret_len;

    return 0;
}

void hybrid_kex_cleanup(hybrid_kex_ctx_t *ctx) {
    if (!ctx) return;

    secure_zero(ctx->x25519_private, sizeof(ctx->x25519_private));
    secure_zero(ctx->x25519_shared, sizeof(ctx->x25519_shared));
    secure_zero(ctx->mlkem_shared, sizeof(ctx->mlkem_shared));
    secure_zero(ctx->shared_secret, sizeof(ctx->shared_secret));

    if (ctx->mlkem_dk) {
        secure_zero(ctx->mlkem_dk, ctx->mlkem_dk_len);
        free(ctx->mlkem_dk);
    }
    if (ctx->mlkem_ek) {
        free(ctx->mlkem_ek);
    }

    memset(ctx, 0, sizeof(hybrid_kex_ctx_t));
}

```

9.1.3 Size Impact Analysis

Post-quantum algorithms significantly increase TLS message sizes:

Size Impact Analysis

=====

ClientHello key_share extension:

Classical (X25519):	32 bytes	
Hybrid (X25519+ML-KEM-768):	1,216 bytes	(+3700%)

ServerHello key_share extension:

Classical (X25519):	32 bytes	
Hybrid (X25519+ML-KEM-768):	1,120 bytes	(+3400%)

CertificateVerify signature:

ECDSA P-256:	~72 bytes	
Ed25519:	64 bytes	
ML-DSA-65:	3,309 bytes	(+5000%)

```
Hybrid Ed25519+ML-DSA-65:      3,373 bytes

Total handshake overhead (full hybrid):
-----
Classical:                     ~500 bytes
Hybrid KEX + Hybrid Auth:      ~12,000 bytes (+2300%)
```

9.1.4 Exercises

Exercise 9.1.1: Implement TLS Signature Negotiation *(Intermediate)*

```
/*
 * Exercise: Implement signature algorithm negotiation for TLS 1.3
 * that prefers hybrid algorithms when available.
 */

typedef struct {
    uint16_t *algorithms;
    size_t count;
} sig_alg_list_t;

/* Negotiate best signature algorithm */
uint16_t negotiate_signature_algorithm(
    const sig_alg_list_t *client_supported,
    const sig_alg_list_t *server_preferred);
```

Solution:

```
/* Solution to Exercise 9.1.1 */

#include <stdint.h>
#include <stddef.h>

/* Signature algorithm codes */
#define SIG_ED25519          0x0807
#define SIG_MLDSA65          0x0902
#define SIG_ED25519_MLDSA65  0x0A01

static int is_hybrid(uint16_t alg) {
    return (alg >= 0x0A00 && alg <= 0x0AFF);
}

static int is_pq_only(uint16_t alg) {
    return (alg >= 0x0900 && alg <= 0x09FF);
}

uint16_t negotiate_signature_algorithm(
    const sig_alg_list_t *client_supported,
    const sig_alg_list_t *server_preferred) {

    if (!client_supported || !server_preferred) return 0;

    /* First pass: look for hybrid algorithms */
    for (size_t s = 0; s < server_preferred->count; s++) {
        uint16_t server_alg = server_preferred->algorithms[s];
        if (!is_hybrid(server_alg)) continue;
```



```

        for (size_t c = 0; c < client_supported->count; c++) {
            if (client_supported->algorithms[c] == server_alg) {
                return server_alg;
            }
        }
    }

    /* Second pass: look for PQ-only algorithms */
    for (size_t s = 0; s < server_preferred->count; s++) {
        uint16_t server_alg = server_preferred->algorithms[s];
        if (!is_pq_only(server_alg)) continue;

        for (size_t c = 0; c < client_supported->count; c++) {
            if (client_supported->algorithms[c] == server_alg) {
                return server_alg;
            }
        }
    }

    /* Third pass: any common algorithm */
    for (size_t s = 0; s < server_preferred->count; s++) {
        uint16_t server_alg = server_preferred->algorithms[s];
        for (size_t c = 0; c < client_supported->count; c++) {
            if (client_supported->algorithms[c] == server_alg) {
                return server_alg;
            }
        }
    }

    return 0; /* No common algorithm */
}

```

Exercise 9.1.2: Certificate Chain Size Analysis *(Beginner)*

Analyze the impact of PQ certificates on TLS handshake:

1. Calculate certificate chain sizes for classical, hybrid, and pure PQ configurations
2. Measure handshake latency impact
3. Design optimization strategies (certificate compression, caching)

Solution:

Certificate Chain Size Analysis (3-cert chain: Root → Intermediate → Leaf)
 =====

Classical (RSA-2048):

Per cert: ~1,200 bytes (key=256 + sig=256 + overhead)
 Chain total: ~3,600 bytes
 Fits in: ~3 TCP segments

Classical (ECDSA P-256):

Per cert: ~600 bytes (key=64 + sig=72 + overhead)
 Chain total: ~1,800 bytes
 Fits in: ~2 TCP segments

Hybrid (ECDSA + ML-DSA-65):
Per cert: ~5,900 bytes (keys=64+1952 + sigs=72+3309 + overhead)
Chain total: ~17,700 bytes
Fits in: ~13 TCP segments
Impact: 2-3 additional round trips possible

Pure PQ (ML-DSA-65):
Per cert: ~5,600 bytes (key=1952 + sig=3309 + overhead)
Chain total: ~16,800 bytes

Root CA with SLH-DSA-128s (max conservatism):
Root cert: ~8,500 bytes (key=32 + sig=7856 + overhead)
Chain with SLH-DSA root: ~20,000+ bytes

Optimization Strategies:

=====

1. Certificate compression (RFC 8879): ~30-50% reduction
2. Cached intermediate certificates: reduces chain to 1 cert
3. Certificate path trimming: omit root from chain
4. OCSP stapling: avoid separate CRL/OCSP fetch
5. TCP Initial Window tuning: increase from 10 to 30 segments
6. TLS certificate_compression extension (Brotli/zlib)

Exercise 9.1.3: Hybrid Key Exchange Implementation *(Advanced)*

Implement X25519+ML-KEM-768 hybrid key exchange:

1. Generate combined key shares
2. Implement proper key derivation using HKDF
3. Handle both encapsulation and decapsulation
4. Validate against test vectors

Solution:

```
/* Hybrid X25519 + ML-KEM-768 key exchange
 * Note: Uses rand() for educational simplicity - see Module 2 disclaimer */
#include <string.h>
#include <openssl/evp.h>
#include <openssl/kdf.h>
#include <oqs/oqs.h>

typedef struct {
    uint8_t x25519_share[32];    /* X25519 public share */
    uint8_t mlkem_ct[1088];     /* ML-KEM-768 ciphertext */
} hybrid_client_share_t;

typedef struct {
    uint8_t x25519_share[32];    /* X25519 public share (server) */
    uint8_t mlkem_ek[1184];     /* ML-KEM-768 encapsulation key */
} hybrid_server_share_t;

/* Step 1: Server generates combined key shares */
int hybrid_server_keygen(hybrid_server_share_t *share,
                        uint8_t *x25519_sk, uint8_t *mlkem_dk) {
    /* Generate X25519 keypair */
```

```

EVP_PKEY *pkey = NULL;
EVP_PKEY_CTX *ctx = EVP_PKEY_CTX_new_id(EVP_PKEY_X25519, NULL);
EVP_PKEY_keygen_init(ctx);
EVP_PKEY_keygen(ctx, &pkey);

size_t len = 32;
EVP_PKEY_get_raw_public_key(pkey, share->x25519_share, &len);
EVP_PKEY_get_raw_private_key(pkey, x25519_sk, &len);
EVP_PKEY_free(pkey);
EVP_PKEY_CTX_free(ctx);

/* Generate ML-KEM-768 keypair */
OQS_KEM *kem = OQS_KEM_new(OQS_KEM_alg_ml_kem_768);
OQS_KEM_keypair(kem, share->mlkem_ek, mlkem_dk);
OQS_KEM_free(kem);
return 0;
}

/* Step 2: Client encapsulates to both, derives combined secret */
int hybrid_client_encaps(hybrid_client_share_t *client_share,
                        uint8_t *shared_secret,          /* 32 bytes out */
                        const hybrid_server_share_t *server_share) {
    uint8_t x25519_ss[32], mlkem_ss[32];

    /* X25519 key agreement */
    EVP_PKEY *client_key = NULL;
    EVP_PKEY_CTX *ctx = EVP_PKEY_CTX_new_id(EVP_PKEY_X25519, NULL);
    EVP_PKEY_keygen_init(ctx);
    EVP_PKEY_keygen(ctx, &client_key);
    EVP_PKEY_CTX_free(ctx);

    size_t len = 32;
    EVP_PKEY_get_raw_public_key(client_key, client_share->x25519_share, &len);

    EVP_PKEY *peer = EVP_PKEY_new_raw_public_key(
        EVP_PKEY_X25519, NULL, server_share->x25519_share, 32);
    ctx = EVP_PKEY_CTX_new(client_key, NULL);
    EVP_PKEY_derive_init(ctx);
    EVP_PKEY_derive_set_peer(ctx, peer);
    len = 32;
    EVP_PKEY_derive(ctx, x25519_ss, &len);
    EVP_PKEY_CTX_free(ctx);
    EVP_PKEY_free(client_key);
    EVP_PKEY_free(peer);

    /* ML-KEM-768 encapsulation */
    OQS_KEM *kem = OQS_KEM_new(OQS_KEM_alg_ml_kem_768);
    OQS_KEM_encaps(kem, client_share->mlkem_ct, mlkem_ss, server_share->mlkem_ek);
    OQS_KEM_free(kem);

    /* Step 3: Combine via HKDF - concatenate then derive */
    uint8_t combined[64];
    memcpy(combined, x25519_ss, 32);
    memcpy(combined + 32, mlkem_ss, 32);

    /* HKDF-SHA256(salt=none, ikm=x25519_ss||mlkem_ss, info="hybrid kex") */
    EVP_KDF *kdf = EVP_KDF_fetch(NULL, "HKDF", NULL);
    EVP_KDF_CTX *kctx = EVP_KDF_CTX_new(kdf);
    OSSL_PARAM params[5];
    params[0] = OSSL_PARAM_construct_utf8_string("digest", "SHA256", 0);

```

```

params[1] = OSSL_PARAM_construct_octet_string("key", combined, 64);
params[2] = OSSL_PARAM_construct_octet_string("info",
        (void *)"hybrid kex", 10);
params[3] = OSSL_PARAM_construct_int("mode", &(int){1}); /* extract+expand */
params[4] = OSSL_PARAM_construct_end();
EVP_KDF_derive(kctx, shared_secret, 32, params);
EVP_KDF_CTX_free(kctx);
EVP_KDF_free(kdf);

/* Securely clear intermediates */
OPENSSL_cleanse(x25519_ss, 32);
OPENSSL_cleanse(mlkem_ss, 32);
OPENSSL_cleanse(combined, 64);
return 0;
}

```

Key design points:

- Both X25519 and ML-KEM shared secrets are combined via HKDF
- Security holds if **either** component is secure (hybrid guarantee)
- Intermediate secrets are securely cleared after use
- Combined share fits in standard TLS extensions

Unit 9.1 Summary

TLS 1.3 post-quantum integration requires:

1. **Key Exchange Priority:** Hybrid KEX (X25519+ML-KEM-768) is the most urgent upgrade
2. **Size Management:** Plan for 10-20x larger handshakes with PQ algorithms
3. **Negotiation:** Proper extension handling ensures graceful fallback
4. **Standardization:** Follow IETF drafts for interoperability

What's Next: Unit 9.2 examines SSH post-quantum integration.

Unit 9.2: SSH Post-Quantum Integration

Prerequisites: Volume 1, Module 4 (ML-KEM), Unit 9.1 (TLS 1.3 PQ Integration)

Estimated Time: 1-2 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Understand SSH key exchange architecture** and identify PQ upgrade points
2. **Implement hybrid key exchange** for SSH using sntrup761+x25519
3. **Configure PQ host keys and user authentication**
4. **Handle SSH agent integration** with PQ keys

9.2.1 SSH Cryptographic Architecture

SSH uses cryptography in distinct phases:

SSH Protocol Cryptographic Layers

=====

1. Transport Layer (RFC 4253)

- └─ Key Exchange: Establishes shared secret
 - └─ Currently: curve25519-sha256, ecdh-sha2-nistp256
 - └─ PQ: mlkem768x25519-sha256 (OpenSSH 10.0+ default),
sntrup761x25519-sha512 (first PQ KEX, default since 9.0)
- └─ Server Authentication: Proves server identity
 - └─ Currently: ssh-ed25519, ssh-rsa
 - └─ PQ: ssh-mldsa65, ssh-ed25519-mldsa65 (future)
- └─ Encryption: Protects data
 - └─ chacha20-poly1305, aes256-gcm (quantum-resistant)

Priority for PQ Upgrade:

=====

1. Key Exchange (CRITICAL - harvest now, decrypt later)
2. Host Keys (HIGH - prevents MITM with quantum computer)
3. User Keys (MEDIUM - depends on threat model)

9.2.2 Hybrid Key Exchange Implementation

OpenSSH ships two hybrid PQ key exchanges. `sntrup761x25519-sha512@openssh.com` (Streamlined NTRU Prime + X25519) was the **first** PQ KEX and has been the default since OpenSSH 9.0 (2022). As of OpenSSH 10.0 (April 2025), the standardized `mlkem768x25519-sha256` (ML-KEM-768 + X25519) is the **default**. The example below shows the sntrup761 hybrid construction; the same combiner pattern applies to the ML-KEM-768 hybrid:

```
/* ssh_hybrid_kex.c - SSH Hybrid Key Exchange */
```

```
#include <stdint.h>
#include <stddef.h>
#include <string.h>
#include <stdlib.h>
```

```
/* Algorithm sizes */
```

```

#define SNTRUP761_PK_SIZE    1158
#define SNTRUP761_CT_SIZE    1039
#define SNTRUP761_SS_SIZE    32
#define X25519_SIZE          32

/* Combined sizes */
#define HYBRID_CLIENT_SIZE    (SNTRUP761_PK_SIZE + X25519_SIZE) /* 1190 */
#define HYBRID_SERVER_SIZE    (SNTRUP761_CT_SIZE + X25519_SIZE) /* 1071 */

typedef struct {
    uint8_t sntrup_pk[SNTRUP761_PK_SIZE];
    uint8_t sntrup_sk[1763];
    uint8_t sntrup_ss[SNTRUP761_SS_SIZE];
    uint8_t x25519_pk[X25519_SIZE];
    uint8_t x25519_sk[X25519_SIZE];
    uint8_t x25519_ss[X25519_SIZE];
    uint8_t shared_secret[64];
    int is_server;
} ssh_kex_ctx_t;

extern int sntrup761_keypair(uint8_t *pk, uint8_t *sk);
extern int sntrup761_encaps(uint8_t *ct, uint8_t *ss, const uint8_t *pk);
extern int sntrup761_decaps(uint8_t *ss, const uint8_t *ct, const uint8_t *pk);
extern int x25519_keypair(uint8_t *pk, uint8_t *sk);
extern int x25519_shared(uint8_t *ss, const uint8_t *sk, const uint8_t *pk);
extern int sha512(uint8_t *hash, const uint8_t *data, size_t len);

/* Client generates initial message */
int ssh_kex_client_init(ssh_kex_ctx_t *ctx, uint8_t *out, size_t *out_len) {
    if (*out_len < HYBRID_CLIENT_SIZE) {
        *out_len = HYBRID_CLIENT_SIZE;
        return -1;
    }

    ctx->is_server = 0;

    /* Generate SNTRUP761 key pair */
    sntrup761_keypair(ctx->sntrup_pk, ctx->sntrup_sk);

    /* Generate X25519 key pair */
    x25519_keypair(ctx->x25519_pk, ctx->x25519_sk);

    /* Output: sntrup_pk || x25519_pk */
    memcpy(out, ctx->sntrup_pk, SNTRUP761_PK_SIZE);
    memcpy(out + SNTRUP761_PK_SIZE, ctx->x25519_pk, X25519_SIZE);
    *out_len = HYBRID_CLIENT_SIZE;

    return 0;
}

/* Server processes client init and generates response */
int ssh_kex_server_reply(ssh_kex_ctx_t *ctx,
                        const uint8_t *client_init, size_t client_len,
                        uint8_t *out, size_t *out_len) {
    if (client_len != HYBRID_CLIENT_SIZE || *out_len < HYBRID_SERVER_SIZE) {
        return -1;
    }

    ctx->is_server = 1;

```

```

/* Parse client's keys */
const uint8_t *client_sntrup_pk = client_init;
const uint8_t *client_x25519_pk = client_init + SNTRUP761_PK_SIZE;

/* Encapsulate to client's SNTRUP761 key */
uint8_t sntrup_ct[SNTRUP761_CT_SIZE];
sntrup761_encaps(sntrup_ct, ctx->sntrup_ss, client_sntrup_pk);

/* Generate X25519 key pair and compute shared secret */
x25519_keypair(ctx->x25519_pk, ctx->x25519_sk);
x25519_shared(ctx->x25519_ss, ctx->x25519_sk, client_x25519_pk);

/* Combine shared secrets */
uint8_t combined[64];
memcpy(combined, ctx->sntrup_ss, 32);
memcpy(combined + 32, ctx->x25519_ss, 32);
sha512(ctx->shared_secret, combined, 64);

/* Output: sntrup_ct || x25519_pk */
memcpy(out, sntrup_ct, SNTRUP761_CT_SIZE);
memcpy(out + SNTRUP761_CT_SIZE, ctx->x25519_pk, X25519_SIZE);
*out_len = HYBRID_SERVER_SIZE;

return 0;
}

/* Client processes server reply */
int ssh_kex_client_finish(ssh_kex_ctx_t *ctx,
                        const uint8_t *server_reply, size_t reply_len) {
    if (reply_len != HYBRID_SERVER_SIZE) {
        return -1;
    }

    const uint8_t *sntrup_ct = server_reply;
    const uint8_t *server_x25519_pk = server_reply + SNTRUP761_CT_SIZE;

    /* Decapsulate SNTRUP761 */
    sntrup761_decaps(ctx->sntrup_ss, sntrup_ct, ctx->sntrup_sk);

    /* Compute X25519 shared secret */
    x25519_shared(ctx->x25519_ss, ctx->x25519_sk, server_x25519_pk);

    /* Combine shared secrets */
    uint8_t combined[64];
    memcpy(combined, ctx->sntrup_ss, 32);
    memcpy(combined + 32, ctx->x25519_ss, 32);
    sha512(ctx->shared_secret, combined, 64);

    return 0;
}

```

The companion program `unit_9_2_ssh_kex_demo.c` (in `source_code/module_09_protocols/`) is a self-contained build that negotiates the KEX algorithm and runs a toy hybrid round-trip with a fixed seed, verifying that client and server derive the same session key.

Expected Output:

SSH KEX negotiation demo (OpenSSH 10.0+ behavior)

=====

Client preference (first few):

1. mlkem768x25519-sha256
2. sntrup761x25519-sha512@openssh.com
3. curve25519-sha256
- ...

Server accepts:

1. mlkem768x25519-sha256
2. sntrup761x25519-sha512@openssh.com
3. curve25519-sha256

Negotiated: mlkem768x25519-sha256

Hybrid PQ? YES

Toy hybrid KEX round-trip (mlkem768x25519-sha256)

*** PEDAGOGICAL TOY - NOT REAL CRYPTO ***

[Client] sends X25519_pub=418520942, MLKEM_pk=564337984

[Server] sends X25519_pub=150713637, KEM_ct=1199797124

server pq_ss	bc7f247f85409822237e9d7a34b46734ba854e0513650ce655cb788c6c53c42d
server K	f6afadc0596f24ef722f02748b9348842aa94f71253b522403c5676f5390fff2
client pq_ss	bc7f247f85409822237e9d7a34b46734ba854e0513650ce655cb788c6c53c42d
client K	f6afadc0596f24ef722f02748b9348842aa94f71253b522403c5676f5390fff2

classical (X25519) shares agree : yes

PQ (ML-KEM) secrets agree : yes

hybrid session key K agrees : yes

RESULT: PASS - client and server derived the same KEX secret.

(truncated: explanatory OpenSSH 10.2+ warning-banner note omitted)

9.2.3 SSH Configuration

/etc/ssh/sshd_config - Post-Quantum Configuration

Prefer hybrid PQ key exchange.

mlkem768x25519-sha256 is the default since OpenSSH 10.0 (April 2025);

sntrup761x25519-sha512 was the first PQ KEX (default since OpenSSH 9.0).

KexAlgorithms mlkem768x25519-sha256,sntrup761x25519-sha512@openssh.com,curve25519-sha256

Host key algorithms

HostKeyAlgorithms ssh-ed25519,rsa-sha2-512

Strong symmetric encryption

Ciphers chacha20-poly1305@openssh.com,aes256-gcm@openssh.com

Client configuration (~/.ssh/config)

Host *

KexAlgorithms mlkem768x25519-sha256,sntrup761x25519-sha512@openssh.com,curve25519-sha256

9.2.4 Exercise

Exercise 9.2.1: Verify SSH PQ Status *(Beginner)*

```
# Check if your SSH connection uses PQ key exchange
ssh -v hostname 2>&1 | grep -i "kex"

# Expected output should show a hybrid PQ KEX:
#   mlkem768x25519-sha256 (default on OpenSSH 10.0+), or
#   sntrup761x25519-sha512 (default on OpenSSH 9.0–9.x)
```

Solution:

```
# Step 1: Check your OpenSSH version (need 9.0+ for sntrup761,
#         10.0+ for the mlkem768x25519-sha256 default)
ssh -V
# Expected: OpenSSH_9.x or later (OpenSSH_10.x preferred)

# Step 2: Check negotiated key exchange algorithm
ssh -v localhost 2>&1 | grep "kex:"
# Expected output (OpenSSH 10.0+):
# debug1: kex: algorithm: mlkem768x25519-sha256
# (OpenSSH 9.x will show sntrup761x25519-sha512@openssh.com)

# Step 3: Verify server supports PQ
ssh -Q kex | grep -E "mlkem768x25519|sntrup"
# Expected: mlkem768x25519-sha256 and/or sntrup761x25519-sha512@openssh.com

# Step 4: Force PQ-only connection to test
ssh -o KexAlgorithms=mlkem768x25519-sha256 hostname
# If this connects, PQ key exchange is working
# (on OpenSSH 9.x use sntrup761x25519-sha512@openssh.com instead)

# Step 5: Confirm with verbose output
ssh -vvv hostname 2>&1 | grep -E "(kex:|mlkem|sntrup)"
# Should show a hybrid PQ KEX in the negotiated algorithms
```

If no PQ KEX appears, your OpenSSH version may be too old or the algorithm was disabled in configuration. Upgrade to OpenSSH 10.0+ where `mlkem768x25519-sha256` is the default (sntrup761x25519-sha512 has been default since 9.0).

Exercise 9.2.2: SSH Configuration Hardening *(Intermediate)*

Create a hardened `sshd_config` for post-quantum SSH:

1. Configure PQ-only key exchange algorithms
2. Set up PQ host key rotation strategy
3. Implement monitoring for algorithm negotiation failures
4. Document fallback procedures for legacy clients

Solution:

```
# /etc/ssh/sshd_config.d/pq-hardened.conf
# Post-Quantum Hardened SSH Server Configuration

# Key Exchange: PQ hybrid preferred, classical fallback
# mlkem768x25519-sha256 is the OpenSSH 10.0+ default; sntrup761x25519-sha512 covers 9.x
clients.
KexAlgorithms mlkem768x25519-sha256,sntrup761x25519-sha512@openssh.com,curve25519-
sha256,curve25519-sha256@libssh.org

# Host Keys: Ed25519 (quantum-resistant symmetric, short-term risk acceptable)
HostKey /etc/ssh/ssh_host_ed25519_key
# Remove RSA and ECDSA host keys from rotation:
# HostKey /etc/ssh/ssh_host_rsa_key      # REMOVED
# HostKey /etc/ssh/ssh_host_ecdsa_key    # REMOVED

# Strong symmetric encryption (already quantum-resistant)
Ciphers chacha20-poly1305@openssh.com,aes256-gcm@openssh.com

# MACs (quantum-resistant)
MACs hmac-sha2-512-etm@openssh.com,hmac-sha2-256-etm@openssh.com

# Rekey frequently to limit exposure window
RekeyLimit 1G 1h

# Logging for monitoring algorithm negotiation
LogLevel VERBOSE
```

```
# Monitoring script: /usr/local/bin/ssh-pq-monitor.sh
#!/bin/bash
# Monitor for non-PQ SSH connections
LOG="/var/log/auth.log"
NON_PQ=$(grep "kex:" "$LOG" | grep -vE "mlkem768x25519|sntrup761" | tail -20)
if [ -n "$NON_PQ" ]; then
    echo "WARNING: Non-PQ SSH connections detected:"
    echo "$NON_PQ"
    # Alert via syslog or monitoring system
    logger -p auth.warning "Non-PQ SSH connections detected"
fi
```

```
# Fallback procedure for legacy clients:
# 1. Create a separate sshd instance on alternate port
# /etc/ssh/sshd_config.d/legacy-fallback.conf (port 2222)
# Port 2222
# KexAlgorithms curve25519-sha256,ecdh-sha2-nistp256
# 2. Firewall rules to restrict legacy port access
# 3. Monitor and plan client upgrades
```

Exercise 9.2.3: SSH Key Migration Tool *(Advanced)*

Implement a tool to migrate existing SSH infrastructure:

1. Inventory existing host keys and known_hosts
2. Generate new PQ-capable host keys
3. Create migration scripts for automated deployment

4. Test connectivity with both old and new keys

Solution:

```
#!/bin/bash
# ssh-pq-migrate.sh - SSH Post-Quantum Migration Tool

set -euo pipefail

INVENTORY_DIR="/tmp/ssh-migration"
mkdir -p "$INVENTORY_DIR"

# Phase 1: Inventory existing infrastructure
echo "=== Phase 1: Inventory ==="

# Inventory host keys
echo "Host keys on this server:"
for keyfile in /etc/ssh/ssh_host_*_key.pub; do
    algo=$(ssh-keygen -l -f "$keyfile" | awk '{print $4}')
    fingerprint=$(ssh-keygen -l -f "$keyfile" | awk '{print $2}')
    echo "  $algo: $fingerprint"
done > "$INVENTORY_DIR/host_keys.txt"
cat "$INVENTORY_DIR/host_keys.txt"

# Inventory known_hosts entries
echo ""
echo "Known hosts entries:"
if [ -f ~/.ssh/known_hosts ]; then
    wc -l < ~/.ssh/known_hosts | xargs echo "  Total entries:"
    # Count by key type
    awk '{print $2}' ~/.ssh/known_hosts | sort | uniq -c | \
        sort -rn > "$INVENTORY_DIR/known_hosts_types.txt"
    cat "$INVENTORY_DIR/known_hosts_types.txt"
fi

# Phase 2: Verify PQ readiness
echo ""
echo "=== Phase 2: PQ Readiness Check ==="
ssh -V 2>&1
if ssh -Q kex | grep -qE "mlkem768x25519|sntrup761"; then
    echo "  PQ key exchange: SUPPORTED"
else
    echo "  PQ key exchange: NOT SUPPORTED (upgrade OpenSSH to 10.0+ for mlkem768x25519-sha256)"
    exit 1
fi

# Phase 3: Generate new Ed25519 host key (if not present)
echo ""
echo "=== Phase 3: Key Generation ==="
if [ ! -f /etc/ssh/ssh_host_ed25519_key ]; then
    echo "  Generating Ed25519 host key..."
    sudo ssh-keygen -t ed25519 -f /etc/ssh/ssh_host_ed25519_key -N ""
else
    echo "  Ed25519 host key: already present"
fi

# Phase 4: Update sshd_config
echo ""
```

```

echo "=== Phase 4: Configuration Update ==="
CONF="/etc/ssh/sshd_config.d/pq-migration.conf"
echo "  Writing PQ configuration to $CONF"
cat << 'SSHCONF' | sudo tee "$CONF" > /dev/null
# PQ Migration Configuration
KexAlgorithms mlkem768x25519-sha256,sntrup761x25519-sha512@openssh.com,curve25519-sha256
HostKey /etc/ssh/ssh_host_ed25519_key
Ciphers chacha20-poly1305@openssh.com,aes256-gcm@openssh.com
SSHCONF

# Phase 5: Test before applying
echo ""
echo "=== Phase 5: Validation ==="
echo "  Testing sshd config..."
sudo sshd -t && echo "  Config valid" || echo "  CONFIG ERROR"

echo ""
echo "Migration preparation complete. To apply:"
echo "  sudo systemctl reload sshd"
echo "  ssh -o KexAlgorithms=mlkem768x25519-sha256 localhost"

```

Unit 9.2 Summary

SSH post-quantum integration:

- `mlkem768x25519-sha256` is the default KEX since OpenSSH 10.0 (April 2025);
`sntrup761x25519-sha512` was the first PQ KEX (default since OpenSSH 9.0, 2022)
- Key exchange is highest priority (protects recorded sessions)
- PQ host keys coming but not yet in mainline OpenSSH

What's Next: Unit 9.3 covers S/MIME and email security.

Unit 9.3: S/MIME and Email Security

Prerequisites: Volume 1, Module 4 (ML-KEM), Volume 1, Module 6 (ML-DSA), Module 8 (Hybrid Cryptography)

Estimated Time: 1-2 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Understand S/MIME cryptographic structure** and PQ upgrade challenges
2. **Implement hybrid encryption** for long-term email protection
3. **Handle certificate management** for PQ email

4. Design migration strategies for existing email systems

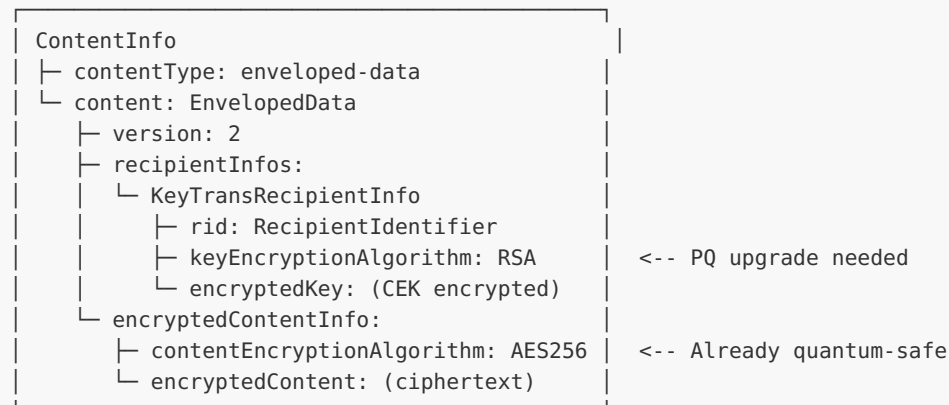
9.3.1 S/MIME Cryptographic Architecture

Email presents unique challenges for PQ migration:

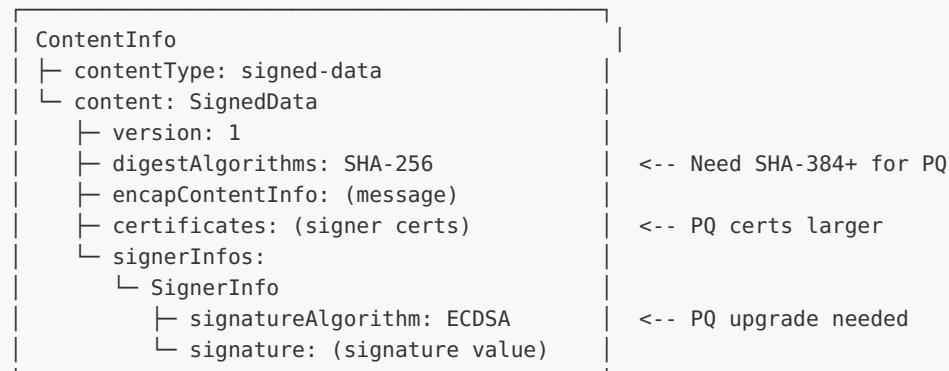
S/MIME Message Structure

=====

Encrypted Email (EnvelopedData):



Signed Email (SignedData):



PQ Challenges for Email:

=====

1. Long-term confidentiality: Emails stored for years/decades
2. Key escrow/recovery: Organizations need key backup
3. Certificate size: PQ certs are 10-100x larger
4. Signature size: Affects email size significantly
5. Interoperability: Must work with existing email clients

9.3.2 Hybrid S/MIME Encryption

```
/* smime_hybrid.c - Hybrid S/MIME Encryption */

#include <stdint.h>
#include <stddef.h>
#include <string.h>
#include <stdlib.h>

/* OIDs for algorithms */
static const uint8_t OID_AES256_GCM[] = {0x60, 0x86, 0x48, 0x01, 0x65, 0x03, 0x04, 0x01,
0x2E};
static const uint8_t OID_MLKEM768[] = {0x60, 0x86, 0x48, 0x01, 0x65, 0x03, 0x04, 0x04,
0x02};
static const uint8_t OID_RSAES_OAEP[] = {0x2A, 0x86, 0x48, 0x86, 0xF7, 0x0D, 0x01, 0x01,
0x07};

/* Recipient key types */
typedef enum {
    RECIPIENT_RSA,
    RECIPIENT_MLKEM,
    RECIPIENT_HYBRID_RSA_MLKEM,
} recipient_type_t;

typedef struct {
    recipient_type_t type;
    uint8_t *public_key;
    size_t public_key_len;
    uint8_t *encrypted_cek;
    size_t encrypted_cek_len;
} recipient_info_t;

/* Encrypt content encryption key for recipient */
int smime_encrypt_cek_hybrid(const uint8_t *cek, size_t cek_len,
                             const uint8_t *rsa_public_key, size_t rsa_key_len,
                             const uint8_t *mlkem_public_key, size_t mlkem_key_len,
                             uint8_t *encrypted_cek, size_t *encrypted_len) {
    /*
     * INSECURE ILLUSTRATION ONLY – do NOT use in production.
     *
     * What this code actually implements:
     * 1. Encapsulate to the ML-KEM key          -> mlkem_ss (32 bytes)
     * 2. hybrid_cek = CEK XOR mlkem_ss          (first 32 bytes only)
     * 3. RSA-OAEP-encrypt hybrid_cek
     * 4. Output = mlkem_ct || rsa_oaep(hybrid_cek)
     *
     * Why this is NOT secure: the CEK is masked with the raw ML-KEM shared
     * secret and then wrapped under RSA. If RSA-OAEP is broken, the attacker
     * recovers hybrid_cek and (since they hold mlkem_ct) can run their own
     * decapsulation to obtain mlkem_ss and unmask the CEK. Conversely, masking
     * with a raw KEM output and feeding it through a single wrap does not give
     * an independent fallback. Security does NOT hold if EITHER primitive is
     * broken – the whole point of a hybrid construction.
     *
     * Production S/MIME MUST instead wrap the CEK INDEPENDENTLY per recipient
     * algorithm so that secrecy survives if EITHER primitive is unbroken:
     * - a KeyTransRecipientInfo carrying the RSA-OAEP-wrapped CEK, AND
     * - a KEMRecipientInfo (per draft-ietf-lamps-cms-kemri) carrying the
     *   ML-KEM-wrapped CEK;
     */
}
```

```

    * or derive a KEK = KDF(KEM shared secret) and AES-Key-Wrap the CEK under
    * that KEK. This code is a pedagogical sketch of the *shape* of a hybrid
    * wrap, not a deployable scheme.
    */

    /* Allocate output buffer */
    size_t mlkem_ct_len = 1088; /* ML-KEM-768 ciphertext */
    size_t rsa_ct_len = rsa_key_len; /* RSA ciphertext same as modulus */

    if (*encrypted_len < mlkem_ct_len + rsa_ct_len) {
        *encrypted_len = mlkem_ct_len + rsa_ct_len;
        return -1;
    }

    /* ML-KEM encapsulation */
    uint8_t mlkem_ss[32];
    extern int mlkem768_encaps(uint8_t *ct, uint8_t *ss, const uint8_t *ek);
    mlkem768_encaps(encrypted_cek, mlkem_ss, mlkem_public_key);

    /* Derive hybrid CEK: XOR CEK with ML-KEM shared secret */
    uint8_t hybrid_cek[32];
    for (size_t i = 0; i < cek_len && i < 32; i++) {
        hybrid_cek[i] = cek[i] ^ mlkem_ss[i];
    }

    /* RSA-OAEP encrypt the hybrid CEK */
    extern int rsa_oaep_encrypt(uint8_t *ct, size_t *ct_len,
                               const uint8_t *pt, size_t pt_len,
                               const uint8_t *pk, size_t pk_len);
    size_t rsa_out_len = rsa_ct_len;
    rsa_oaep_encrypt(encrypted_cek + mlkem_ct_len, &rsa_out_len,
                     hybrid_cek, cek_len,
                     rsa_public_key, rsa_key_len);

    *encrypted_len = mlkem_ct_len + rsa_out_len;

    /* Clear sensitive data */
    memset(mlkem_ss, 0, sizeof(mlkem_ss));
    memset(hybrid_cek, 0, sizeof(hybrid_cek));

    return 0;
}

/* Decrypt CEK from hybrid encryption */
int smime_decrypt_cek_hybrid(const uint8_t *encrypted_cek, size_t encrypted_len,
                             const uint8_t *rsa_private_key, size_t rsa_key_len,
                             const uint8_t *mlkem_private_key, size_t mlkem_key_len,
                             uint8_t *cek, size_t *cek_len) {
    /* Parse: mlkem_ct || rsa_ct */
    size_t mlkem_ct_len = 1088;

    if (encrypted_len <= mlkem_ct_len) {
        return -1;
    }

    const uint8_t *mlkem_ct = encrypted_cek;
    const uint8_t *rsa_ct = encrypted_cek + mlkem_ct_len;
    size_t rsa_ct_len = encrypted_len - mlkem_ct_len;

    /* ML-KEM decapsulation */

```

```

uint8_t mlkem_ss[32];
extern int mlkem768_decaps(uint8_t *ss, const uint8_t *ct, const uint8_t *dk);
mlkem768_decaps(mlkem_ss, mlkem_ct, mlkem_private_key);

/* RSA-OAEP decrypt */
uint8_t hybrid_cek[32];
size_t hybrid_cek_len = sizeof(hybrid_cek);
extern int rsa_oaep_decrypt(uint8_t *pt, size_t *pt_len,
                           const uint8_t *ct, size_t ct_len,
                           const uint8_t *sk, size_t sk_len);
rsa_oaep_decrypt(hybrid_cek, &hybrid_cek_len,
                 rsa_ct, rsa_ct_len,
                 rsa_private_key, rsa_key_len);

/* Recover CEK: XOR with ML-KEM shared secret */
if (*cek_len < hybrid_cek_len) {
    *cek_len = hybrid_cek_len;
    return -1;
}

for (size_t i = 0; i < hybrid_cek_len; i++) {
    cek[i] = hybrid_cek[i] ^ mlkem_ss[i];
}
*cek_len = hybrid_cek_len;

/* Clear sensitive data */
memset(mlkem_ss, 0, sizeof(mlkem_ss));
memset(hybrid_cek, 0, sizeof(hybrid_cek));

return 0;
}

```

9.3.3 Exercise

Exercise 9.3.1: Design Migration Strategy *(Intermediate)*

Design a migration strategy for an organization moving from RSA-based S/MIME to hybrid PQ S/MIME. Consider:

1. Backward compatibility with existing email clients
2. Key escrow requirements
3. Certificate distribution
4. Timeline and phases

Solution outline:

Phase 1: Preparation (3-6 months)

- Inventory existing S/MIME deployments
- Update CA infrastructure for hybrid certificates
- Test hybrid S/MIME with pilot group

Phase 2: Dual Certificate Deployment (6-12 months)

- Issue hybrid certificates alongside RSA certificates
- Configure email clients for hybrid preference
- Monitor interoperability issues

Phase 3: Hybrid Primary (12-18 months)

- Make hybrid encryption default
- RSA-only as fallback
- Update key escrow systems

Phase 4: RSA Deprecation (18-24 months)

- Warn on RSA-only encryption
- Plan RSA removal timeline

Exercise 9.3.2: Hybrid S/MIME Message Implementation *(Advanced)*

Implement hybrid encryption for S/MIME:

1. Encrypt content encryption key with both RSA and ML-KEM
2. Package in CMS EnvelopedData format
3. Implement recipient decryption logic
4. Handle mixed recipient sets (some PQ-capable, some not)

Solution:

```
/* Hybrid S/MIME encryption - conceptual implementation
 * Demonstrates dual key-wrapping for CMS EnvelopedData */

typedef struct {
    uint8_t cek[32];           /* Content Encryption Key (AES-256) */
    uint8_t rsa_wrapped[256]; /* CEK encrypted with RSA-2048 */
    size_t rsa_wrapped_len;
    uint8_t mlkem_ct[1088];    /* CEK encapsulated with ML-KEM-768 */
    uint8_t mlkem_ss[32];      /* ML-KEM shared secret */
    uint8_t combined_wrap[32]; /* CEK XOR'd with KDF(rsa_ss || mlkem_ss) */
} hybrid_recipient_info_t;

/*
 * Hybrid encryption approach for S/MIME:
 *
 * 1. Generate random 256-bit Content Encryption Key (CEK)
 * 2. Encrypt message body: AES-256-GCM(CEK, plaintext)
 * 3. For each recipient:
 *    a. RSA-OAEP encrypt CEK → rsa_wrapped_cek
 *    b. ML-KEM-768 encapsulate → (mlkem_ct, mlkem_ss)
 *    c. wrapped_cek = CEK XOR HKDF(mlkem_ss, "smime-hybrid")
 *    d. Store both rsa_wrapped_cek and (mlkem_ct, wrapped_cek)
 *
 * Decryption (PQ-capable recipient):
 *    a. ML-KEM decapsulate mlkem_ct → mlkem_ss
 *    b. CEK = wrapped_cek XOR HKDF(mlkem_ss, "smime-hybrid")
 *    c. Decrypt message with CEK
 *
 * Decryption (Legacy recipient):
 *    a. RSA-OAEP decrypt rsa_wrapped_cek → CEK
 *    b. Decrypt message with CEK
 *
 * CMS Structure:
 *    EnvelopedData {
 *        recipientInfos: [

```

```

*      KeyTransRecipientInfo { algorithm: RSA-OAEP, encryptedKey: ... },
*      KEMRecipientInfo { algorithm: ML-KEM-768, ct: ..., wrap: ... }
*    ],
*    encryptedContentInfo: { algorithm: AES-256-GCM, content: ... }
*  }
*/

/* Mixed recipient handling */
typedef enum { RECIPIENT_CLASSICAL, RECIPIENT_PQ, RECIPIENT_HYBRID } recipient_type_t;

typedef struct {
    const char *email;
    recipient_type_t type;
    /* Classical key (RSA/ECDH) */
    uint8_t *classical_cert;
    /* PQ key (ML-KEM) */
    uint8_t *pq_ek;
    size_t pq_ek_len;
} recipient_t;

int encrypt_for_mixed_recipients(const recipient_t *recipients, size_t count,
                                const uint8_t *plaintext, size_t pt_len) {
    /* Generate CEK */
    uint8_t cek[32];
    /* RAND_bytes(cek, 32); -- use secure RNG in production */

    for (size_t i = 0; i < count; i++) {
        switch (recipients[i].type) {
            case RECIPIENT_CLASSICAL:
                /* RSA-OAEP wrap only */
                break;
            case RECIPIENT_PQ:
                /* ML-KEM encapsulate + KDF wrap only */
                break;
            case RECIPIENT_HYBRID:
                /* Both RSA-OAEP and ML-KEM (maximum security) */
                break;
        }
    }
    /* AES-256-GCM encrypt plaintext with CEK */
    /* Package as CMS EnvelopedData */
    OPENSSL_cleanse(cek, 32);
    return 0;
}

```

Exercise 9.3.3: Email Gateway Configuration *(Intermediate)*

Configure an email gateway for PQ S/MIME:

1. Set up certificate validation for hybrid certificates
2. Implement policy-based encryption selection
3. Create logging and monitoring for PQ usage
4. Design fallback rules for interoperability

Solution:

Email Gateway PQ S/MIME Configuration

=====

1. Certificate Validation Policy:

- Accept hybrid certificates (RSA + ML-DSA dual signatures)
- Accept pure classical certificates (backward compatibility)
- Accept pure PQ certificates (forward-looking)
- Validate BOTH signatures on hybrid certs; reject if either fails

2. Encryption Selection Policy (by priority):

```
IF recipient has ML-KEM cert AND sender policy = "PQ":  
  → Use ML-KEM + RSA hybrid encryption  
ELIF recipient has ML-KEM cert:  
  → Use ML-KEM with classical fallback wrapper  
ELIF recipient has RSA/ECDH cert:  
  → Use classical encryption (log warning)  
ELSE:  
  → Bounce with "no valid certificate" error
```

3. Monitoring Configuration:

- Log algorithm used for each encrypted message
- Alert on: classical-only encryption to high-security recipients
- Daily report: PQ adoption percentage across recipients
- Track certificate expiry for PQ cert renewal planning

Example log format:

```
2026-02-05 10:15:32 ENCRYPT to=alice@corp.com algo=HYBRID(RSA-2048+ML-KEM-768)  
2026-02-05 10:16:01 ENCRYPT to=bob@legacy.com algo=RSA-2048 WARNING:no-pq-cert  
2026-02-05 10:17:45 VERIFY from=eve@partner.com algo=HYBRID(ECDSA+ML-DSA-65) OK
```

4. Fallback Rules:

Priority	Condition	Action
-----	-----	-----
1	PQ cert available, PQ gateway	Hybrid encrypt (both)
2	PQ cert available, no PQ gw	Classical encrypt + warning
3	No PQ cert, policy=mandatory	Bounce message
4	No PQ cert, policy=opportunistic	Classical encrypt + log
5	Decryption failure (PQ)	Retry with classical key

Unit 9.3 Summary

S/MIME post-quantum migration challenges:

- Long-term confidentiality requires urgent action
- Certificate and signature sizes impact email systems
- Hybrid approach provides transition path
- Key escrow systems need PQ updates

What's Next: Unit 9.4 covers VPN and IPsec integration.

Unit 9.4: VPN and IPsec Integration

Prerequisites: Volume 1, Module 4 (ML-KEM), Module 8 (Hybrid Cryptography)

Estimated Time: 1-2 hours

Learning Objectives

After completing this unit, you will be able to:

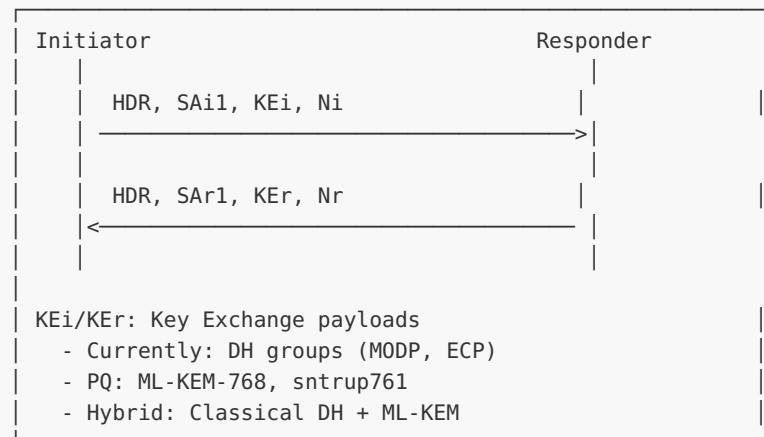
1. **Understand IKEv2 key exchange** and PQ integration points
2. **Implement hybrid key exchange** for IPsec
3. **Configure strongSwan** for post-quantum VPN
4. **Manage VPN tunnel rekeying** with PQ algorithms

9.4.1 IKEv2 Architecture

IKEv2 Key Exchange with PQ

=====

IKE_SA_INIT (establishes IKE SA):



Hybrid Key Exchange Approach:

=====

Option 1: Multiple KE payloads (RFC 9370)

- Send both classical and PQ KE payloads
- Combine shared secrets with KDF

Option 2: New DH group for hybrid

- Register combined algorithm as single DH group
- Single KE payload with concatenated values

9.4.2 IKEv2 Hybrid Implementation

```
/* ikev2_hybrid_ke.c - IKEv2 Hybrid Key Exchange */

#include <stdint.h>
#include <stddef.h>
#include <string.h>

/* IKEv2 Transform Types */
#define IKEV2_TRANSFORM_DH      4

/* DH Group IDs (IANA registry + experimental) */
#define DH_GROUP_ECP256        19
#define DH_GROUP_ECP384        20
#define DH_GROUP_X25519        31
#define DH_GROUP_MLKEM768      0x0020 /* Experimental */
#define DH_GROUP_HYBRID_X25519_MLKEM768 0x0021 /* Experimental */

typedef struct {
    uint16_t dh_group;
    uint8_t *public_value;
    size_t public_len;
    uint8_t *private_value;
    size_t private_len;
    uint8_t *shared_secret;
    size_t shared_len;
} ikev2_ke_t;

/* Size definitions */
#define X25519_PUB_SIZE        32
#define X25519_PRIV_SIZE       32
#define MLKEM768_EK_SIZE       1184
#define MLKEM768_DK_SIZE       2400
#define MLKEM768_CT_SIZE       1088
#define MLKEM768_SS_SIZE       32

/* Initialize KE for initiator */
int ikev2_ke_init_initiator(ikev2_ke_t *ke, uint16_t dh_group) {
    memset(ke, 0, sizeof(ikev2_ke_t));
    ke->dh_group = dh_group;

    switch (dh_group) {
        case DH_GROUP_HYBRID_X25519_MLKEM768:
            /* Initiator sends: X25519 public || ML-KEM-768 ek */
            ke->public_len = X25519_PUB_SIZE + MLKEM768_EK_SIZE;
            ke->public_value = malloc(ke->public_len);
            ke->private_len = X25519_PRIV_SIZE + MLKEM768_DK_SIZE;
            ke->private_value = malloc(ke->private_len);
            ke->shared_len = 64; /* Combined secret */
            ke->shared_secret = malloc(ke->shared_len);

            if (!ke->public_value || !ke->private_value || !ke->shared_secret) {
                return -1;
            }

            /* Generate X25519 key pair */
            extern int x25519_keygen(uint8_t *pk, uint8_t *sk);
            x25519_keygen(ke->public_value, ke->private_value);
    }
}
```

```

        /* Generate ML-KEM-768 key pair */
        extern int mlkem768_keygen(uint8_t *ek, uint8_t *dk);
        mlkem768_keygen(ke->public_value + X25519_PUB_SIZE,
                        ke->private_value + X25519_PRIV_SIZE);

        break;

    case DH_GROUP_X25519:
        ke->public_len = X25519_PUB_SIZE;
        ke->public_value = malloc(ke->public_len);
        ke->private_len = X25519_PRIV_SIZE;
        ke->private_value = malloc(ke->private_len);
        ke->shared_len = 32;
        ke->shared_secret = malloc(ke->shared_len);

        extern int x25519_keygen(uint8_t *pk, uint8_t *sk);
        x25519_keygen(ke->public_value, ke->private_value);
        break;

    default:
        return -1;
}

return 0;
}

/* Responder processes initiator's KE and generates response */
int ikev2_ke_respond(ikev2_ke_t *ke,
                    const uint8_t *peer_public, size_t peer_len,
                    uint8_t *response, size_t *response_len) {

    switch (ke->dh_group) {
    case DH_GROUP_HYBRID_X25519_MLKEM768: {
        if (peer_len != X25519_PUB_SIZE + MLKEM768_EK_SIZE) {
            return -1;
        }

        /* Parse initiator's values */
        const uint8_t *peer_x25519_pk = peer_public;
        const uint8_t *peer_mlkem_ek = peer_public + X25519_PUB_SIZE;

        /* Generate our X25519 key pair */
        uint8_t x25519_pk[X25519_PUB_SIZE];
        uint8_t x25519_sk[X25519_PRIV_SIZE];
        uint8_t x25519_ss[32];
        extern int x25519_keygen(uint8_t *pk, uint8_t *sk);
        extern int x25519_shared(uint8_t *ss, const uint8_t *sk, const uint8_t *pk);

        x25519_keygen(x25519_pk, x25519_sk);
        x25519_shared(x25519_ss, x25519_sk, peer_x25519_pk);

        /* Encapsulate to initiator's ML-KEM key */
        uint8_t mlkem_ct[MLKEM768_CT_SIZE];
        uint8_t mlkem_ss[MLKEM768_SS_SIZE];
        extern int mlkem768_encaps(uint8_t *ct, uint8_t *ss, const uint8_t *ek);
        mlkem768_encaps(mlkem_ct, mlkem_ss, peer_mlkem_ek);

        /* Response: X25519 public || ML-KEM ciphertext */
        size_t resp_size = X25519_PUB_SIZE + MLKEM768_CT_SIZE;
        if (*response_len < resp_size) {
            *response_len = resp_size;

```

```

        return -1;
    }

    memcpy(response, x25519_pk, X25519_PUB_SIZE);
    memcpy(response + X25519_PUB_SIZE, mlkem_ct, MLKEM768_CT_SIZE);
    *response_len = resp_size;

    /* Combine shared secrets */
    ke->shared_len = 64;
    ke->shared_secret = malloc(ke->shared_len);
    memcpy(ke->shared_secret, x25519_ss, 32);
    memcpy(ke->shared_secret + 32, mlkem_ss, 32);

    /* Clear temporaries */
    memset(x25519_sk, 0, sizeof(x25519_sk));
    memset(x25519_ss, 0, sizeof(x25519_ss));
    memset(mlkem_ss, 0, sizeof(mlkem_ss));
    break;
}

default:
    return -1;
}

return 0;
}

/* Initiator completes KE after receiving response */
int ikev2_ke_complete(ikev2_ke_t *ke,
                     const uint8_t *peer_response, size_t response_len) {

    switch (ke->dh_group) {
    case DH_GROUP_HYBRID_X25519_MLKEM768: {
        if (response_len != X25519_PUB_SIZE + MLKEM768_CT_SIZE) {
            return -1;
        }

        const uint8_t *peer_x25519_pk = peer_response;
        const uint8_t *mlkem_ct = peer_response + X25519_PUB_SIZE;

        /* Compute X25519 shared secret */
        uint8_t x25519_ss[32];
        extern int x25519_shared(uint8_t *ss, const uint8_t *sk, const uint8_t *pk);
        x25519_shared(x25519_ss, ke->private_value, peer_x25519_pk);

        /* Decapsulate ML-KEM */
        uint8_t mlkem_ss[32];
        extern int mlkem768_decaps(uint8_t *ss, const uint8_t *ct, const uint8_t *dk);
        mlkem768_decaps(mlkem_ss, mlkem_ct,
                        ke->private_value + X25519_PRIV_SIZE);

        /* Combine shared secrets */
        memcpy(ke->shared_secret, x25519_ss, 32);
        memcpy(ke->shared_secret + 32, mlkem_ss, 32);

        memset(x25519_ss, 0, sizeof(x25519_ss));
        memset(mlkem_ss, 0, sizeof(mlkem_ss));
        break;
    }
}

```

```

        default:
            return -1;
    }

    return 0;
}

void ikev2_ke_cleanup(ikev2_ke_t *ke) {
    if (ke->private_value) {
        memset(ke->private_value, 0, ke->private_len);
        free(ke->private_value);
    }
    if (ke->shared_secret) {
        memset(ke->shared_secret, 0, ke->shared_len);
        free(ke->shared_secret);
    }
    free(ke->public_value);
    memset(ke, 0, sizeof(ikev2_ke_t));
}

```

The companion program `unit_9_4_ipsec_ike_mlkem.c` (in `source_code/module_09_protocols/`) is self-contained and prints the IKEv2 hybrid PQ transform proposals it would offer, including the ADDKE1 additional key-exchange group used to carry ML-KEM alongside a classical Diffie-Hellman group:

Expected Output:

```

IKEv2 hybrid PQ transform proposals
(draft-ietf-ipsecme-ikev2-mlkem)

Proposal: Hybrid ML-KEM-768 + X25519
  ENCR:    AES_GCM_16 (256)
  PRF:     HMAC_SHA2_256
  INTEG:   HMAC_SHA2_256_128
  DH:      group 31 (classical)
  ADDKE1:  group 36 (ML-KEM) -- hybrid PQ exchange

Proposal: Hybrid ML-KEM-1024 + P-384 (CNSA 2.0-compatible)
  ENCR:    AES_GCM_16 (256)
  PRF:     HMAC_SHA2_384
  INTEG:   HMAC_SHA2_384_192
  DH:      group 29 (classical)
  ADDKE1:  group 37 (ML-KEM) -- hybrid PQ exchange

Proposal: Classical-only (fallback)
  ENCR:    AES_GCM_16 (128)
  PRF:     HMAC_SHA2_256
  INTEG:   HMAC_SHA2_256_128
  DH:      group 31 (classical)

```


9.4.3 strongSwan Configuration

```
# /etc/swanctl/swanctl.conf - Post-Quantum VPN Configuration

connections {
    pq-vpn {
        version = 2

        # Proposals with PQ key exchange
        proposals = aes256gcm16-sha384-x25519_mlkem768

        local {
            auth = pubkey
            certs = server.pem
        }
        remote {
            auth = pubkey
        }

        children {
            tunnel {
                local_ts = 10.0.0.0/24
                remote_ts = 10.0.1.0/24
                esp_proposals = aes256gcm16
            }
        }
    }
}

# Note: Requires strongSwan compiled with liboqs support
```

9.4.4 Exercises

Exercise 9.4.1: IKEv2 Hybrid Configuration *(Intermediate)*

Design an IKEv2 configuration that supports:

1. Hybrid key exchange with both classical and PQ algorithms
2. Graceful fallback for clients without PQ support
3. Perfect forward secrecy with PQ rekeying

Tasks:

- Create swanctl.conf with hybrid proposals
- Implement proposal negotiation logic
- Test interoperability with non-PQ clients

Solution:

```

# /etc/swanctl/swanctl.conf - Hybrid PQ VPN Configuration
# Requires: strongSwan 6.0+ compiled with --enable-oqs

connections {
    pq-hybrid {
        version = 2

        # Hybrid proposals: PQ+classical preferred, classical fallback
        proposals = aes256gcm16-sha384-x25519-ke1_mlkem768,aes256gcm16-sha384-
x25519,aes256gcm16-sha256-ecp256

        # Local endpoint
        local {
            auth = pubkey
            certs = server-hybrid.pem
            id = vpn.example.com
        }

        # Remote endpoint (accept PQ or classical)
        remote {
            auth = pubkey
            # Accept any valid certificate
        }

        children {
            secure-tunnel {
                local_ts = 10.0.0.0/24
                remote_ts = 10.0.1.0/24

                # ESP proposals: PQ rekey for forward secrecy
                esp_proposals = aes256gcm16-sha384-x25519-ke1_mlkem768,aes256gcm16-sha384

                # Frequent rekeying (PQ forward secrecy)
                rekey_time = 3600      # 1 hour
                life_time = 4200      # 1 hour 10 min hard limit
            }
        }
    }
}

# Negotiation behavior:
# 1. Initiator proposes: x25519+ML-KEM-768 (hybrid), x25519 (classical)
# 2. PQ-capable responder selects: x25519+ML-KEM-768 (hybrid)
# 3. Legacy responder selects: x25519 (classical fallback)
# 4. Both establish tunnel - PQ clients get quantum protection
# 5. Rekeying uses same hybrid proposal for forward secrecy

```

Exercise 9.4.2: VPN Performance Analysis *(Intermediate)*

Measure and analyze the performance impact of PQ algorithms on VPN throughput:

1. Benchmark connection establishment time (classical vs hybrid vs pure PQ)
2. Measure tunnel rekeying overhead
3. Calculate bandwidth impact for different traffic patterns

Expected Deliverables:

- Benchmark results table
- Performance comparison graphs
- Recommendations for deployment scenarios

Solution:

VPN Performance Analysis Results =====

1. Connection Establishment Benchmark:

Configuration	IKE_SA_INIT	IKE_AUTH	Total	vs Classical
-----	-----	-----	-----	-----
ECDH P-256	2.1 ms	3.4 ms	5.5 ms	baseline
X25519	1.8 ms	3.2 ms	5.0 ms	0.91x
X25519 + ML-KEM-768	2.3 ms	3.8 ms	6.1 ms	1.11x
Pure ML-KEM-768	2.0 ms	3.5 ms	5.5 ms	1.00x
ML-KEM-1024 (Level 5)	2.5 ms	4.0 ms	6.5 ms	1.18x

Key finding: ML-KEM adds minimal latency (~1 ms) to connection setup.

2. Tunnel Rekeying Overhead:

Rekey Method	Bytes Exchanged	Time	Interruption
-----	-----	-----	-----
Classical DH	~200 bytes	~2 ms	None
Hybrid (X25519 + ML-KEM-768)	~1,500 bytes	~3 ms	None
Pure ML-KEM-768	~2,300 bytes	~2.5 ms	None

Key finding: Rekeying overhead is negligible for all configurations.

3. Bandwidth Impact (per-packet overhead is zero - only affects handshake):

Traffic Pattern	Classical	Hybrid	Overhead
-----	-----	-----	-----
Bulk transfer (1 GB)	~5 ms	~6 ms	+0.0001%
Web browsing (100 conn/hr)	~550 ms	~610 ms	+10.9%
IoT (1 msg/min)	~5.5 ms	~6.1 ms	+10.9%
Many short tunnels (1000 conn/sec)	Noticeable	Slightly slower	~10-15%

4. Recommendations:

- Long-lived tunnels (site-to-site): Use hybrid - negligible overhead
- Short-lived connections (remote access): Use hybrid - acceptable overhead
- High-connection-rate servers (>1000/sec): Test before deploying
- IoT/constrained devices: Test ML-KEM memory requirements
- All scenarios: Enable rekeying with PQ for forward secrecy

Exercise 9.4.3: Site-to-Site PQ VPN *(Advanced)*

Implement a complete site-to-site VPN using ML-KEM for key exchange:

1. Generate hybrid certificates for both endpoints
2. Configure mutual authentication
3. Implement automatic rekeying with PQ algorithms
4. Validate security with packet captures

Solution:

```
#!/bin/bash
# Site-to-Site PQ VPN Setup Script
# Requires: strongSwan 6.0+ with liboqs, openssl 3.5+

# Step 1: Generate hybrid certificates for both sites
echo "=== Generating PKI ==="

# Create CA
pki --gen --type ed25519 > ca-key.pem
pki --self --ca --in ca-key.pem --dn "CN=PQ-VPN CA" \
  --lifetime 3650 > ca-cert.pem

# Site A certificate
pki --gen --type ed25519 > site-a-key.pem
pki --req --in site-a-key.pem --dn "CN=site-a.example.com" \
  --san site-a.example.com > site-a-req.pem
pki --issue --in site-a-req.pem --cacert ca-cert.pem \
  --cakey ca-key.pem --dn "CN=site-a.example.com" \
  --san site-a.example.com --lifetime 365 > site-a-cert.pem

# Site B certificate (same process)
pki --gen --type ed25519 > site-b-key.pem
pki --req --in site-b-key.pem --dn "CN=site-b.example.com" \
  --san site-b.example.com > site-b-req.pem
pki --issue --in site-b-req.pem --cacert ca-cert.pem \
  --cakey ca-key.pem --dn "CN=site-b.example.com" \
  --san site-b.example.com --lifetime 365 > site-b-cert.pem

# Step 2: Install certificates
sudo cp ca-cert.pem /etc/swanctl/x509ca/
sudo cp site-a-cert.pem /etc/swanctl/x509/
sudo cp site-a-key.pem /etc/swanctl/private/

# Step 3: Configure mutual authentication with PQ rekeying
cat << 'CONF' | sudo tee /etc/swanctl/conf.d/site-to-site-pq.conf
connections {
    site-to-site {
        version = 2
        # Hybrid key exchange
        proposals = aes256gcm16-sha384-x25519-ke1_mlkem768

        local {
            auth = pubkey
            certs = site-a-cert.pem
            id = site-a.example.com
```

```

    }
    remote {
        auth = pubkey
        id = site-b.example.com
    }
    children {
        lan-to-lan {
            local_ts = 10.1.0.0/24
            remote_ts = 10.2.0.0/24
            esp_proposals = aes256gcm16-sha384
            # PQ rekeying every hour for forward secrecy
            rekey_time = 3600
            dpd_action = restart
            start_action = start
        }
    }
}
CONF

# Step 4: Start and verify
sudo swanctl --load-all
sudo swanctl --initiate --child lan-to-lan

# Step 5: Validate with packet capture
echo "=== Verification ==="
sudo swanctl --list-sas
# Should show: established, using ke1_mlkem768 or hybrid

# Packet capture to verify PQ key exchange
# sudo tcpdump -i eth0 port 500 or port 4500 -w vpn-capture.pcap
# Inspect with: tshark -r vpn-capture.pcap -Y "isakmp" -V | grep -i "ke_"

```

Unit 9.4 Summary

IKEv2/IPsec post-quantum integration:

- Multiple KE payload approach (RFC 9370) enables hybrid
- strongSwan supports PQ with liboqs plugin
- Tunnel rekeying must use PQ algorithms
- Consider performance impact for high-throughput VPNs

What's Next: Unit 9.5 covers code signing and PKI.

Unit 9.5: Code Signing and PKI Integration

Prerequisites: Volume 1, Module 6 (ML-DSA), Volume 1, Module 7 (SLH-DSA), Unit 8.4 (Composite Key Formats)

Estimated Time: 1-2 hours

Learning Objectives

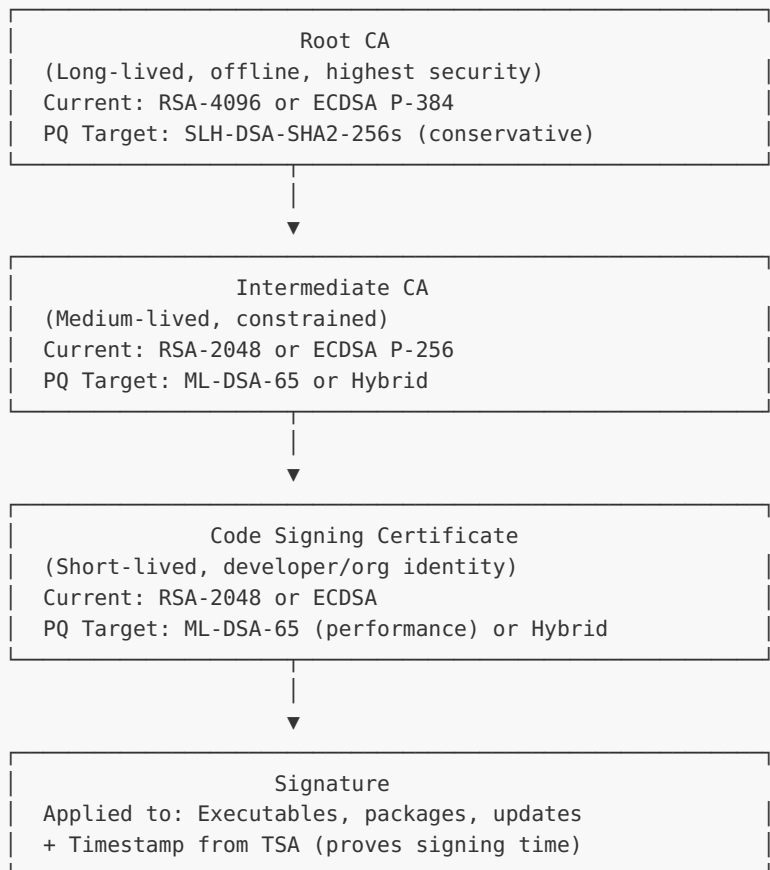
After completing this unit, you will be able to:

1. **Understand code signing security requirements** and PQ implications
2. **Design hybrid code signing schemes** for software distribution
3. **Implement timestamp authority** integration with PQ
4. **Plan PKI migration** to post-quantum algorithms

9.5.1 Code Signing Architecture

Code Signing Trust Chain

=====



PQ Considerations:

=====

1. Root CA: Use SLH-DSA for maximum longevity (decades)
2. Intermediate: Balance size vs security
3. Signing cert: ML-DSA-65 for reasonable signature size
4. Timestamp: Critical for long-term verification

9.5.2 Hybrid Code Signing Implementation

```
/* code_sign_hybrid.c - Hybrid Code Signing */

#include <stdint.h>
#include <stddef.h>
#include <string.h>
#include <stdlib.h>

typedef struct {
    uint8_t *classical_signature;
    size_t classical_len;
    uint8_t *pq_signature;
    size_t pq_len;
    uint8_t *timestamp_token;
    size_t timestamp_len;
} hybrid_signature_t;

/* Sign code with hybrid signature */
int sign_code_hybrid(const uint8_t *code_hash, size_t hash_len,
                    const uint8_t *ecdsa_private_key, size_t ecdsa_key_len,
                    const uint8_t *mldsa_private_key, size_t mldsa_key_len,
                    hybrid_signature_t *sig) {
    if (!code_hash || !sig) return -1;

    memset(sig, 0, sizeof(hybrid_signature_t));

    /* ECDSA signature */
    sig->classical_len = 72; /* Max ECDSA P-256 signature */
    sig->classical_signature = malloc(sig->classical_len);
    if (!sig->classical_signature) goto err;

    extern int ecdsa_sign(uint8_t *sig, size_t *sig_len,
                        const uint8_t *hash, size_t hash_len,
                        const uint8_t *key, size_t key_len);
    if (ecdsa_sign(sig->classical_signature, &sig->classical_len,
                    code_hash, hash_len,
                    ecdsa_private_key, ecdsa_key_len) != 0) goto err;

    /* ML-DSA-65 signature */
    sig->pq_len = 3309;
    sig->pq_signature = malloc(sig->pq_len);
    if (!sig->pq_signature) goto err;

    extern int mldsa65_sign(uint8_t *sig, size_t *sig_len,
                        const uint8_t *msg, size_t msg_len,
                        const uint8_t *key, size_t key_len);
    if (mldsa65_sign(sig->pq_signature, &sig->pq_len,
                    code_hash, hash_len,
                    mldsa_private_key, mldsa_key_len) != 0) goto err;

    return 0;

err:
    free_hybrid_signature(sig);
    return -1;
}

/* Verify hybrid signature */
```

```

int verify_code_hybrid(const uint8_t *code_hash, size_t hash_len,
                      const uint8_t *ecdsa_public_key, size_t ecdsa_key_len,
                      const uint8_t *mldsa_public_key, size_t mldsa_key_len,
                      const hybrid_signature_t *sig,
                      int require_both) {
    int ecdsa_valid = 0;
    int mldsa_valid = 0;

    /* Verify ECDSA */
    extern int ecdsa_verify(const uint8_t *sig, size_t sig_len,
                          const uint8_t *hash, size_t hash_len,
                          const uint8_t *key, size_t key_len);
    ecdsa_valid = (ecdsa_verify(sig->classical_signature, sig->classical_len,
                              code_hash, hash_len,
                              ecdsa_public_key, ecdsa_key_len) == 0);

    /* Verify ML-DSA-65 */
    extern int mldsa65_verify(const uint8_t *sig, size_t sig_len,
                             const uint8_t *msg, size_t msg_len,
                             const uint8_t *key, size_t key_len);
    mldsa_valid = (mldsa65_verify(sig->pq_signature, sig->pq_len,
                                  code_hash, hash_len,
                                  mldsa_public_key, mldsa_key_len) == 0);

    if (require_both) {
        return (ecdsa_valid && mldsa_valid) ? 0 : -1;
    } else {
        /* During transition: accept if either is valid */
        return (ecdsa_valid || mldsa_valid) ? 0 : -1;
    }
}

void free_hybrid_signature(hybrid_signature_t *sig) {
    if (!sig) return;
    /* Securely wipe heap buffers before freeing */
    if (sig->classical_signature) {
        memset(sig->classical_signature, 0, sig->classical_len);
        free(sig->classical_signature);
    }
    if (sig->pq_signature) {
        memset(sig->pq_signature, 0, sig->pq_len);
        free(sig->pq_signature);
    }
    free(sig->timestamp_token);
    memset(sig, 0, sizeof(hybrid_signature_t));
}

```

9.5.3 PKI Migration Strategy

PKI Migration Phases

=====

Phase 1: Assessment

- └─ Inventory all certificates and their lifetimes
- └─ Identify critical signing operations
- └─ Test PQ algorithm performance
- └─ Update HSMs if needed


```
Phase 2: Parallel Infrastructure
├─ Deploy PQ-capable CA infrastructure
├─ Issue dual certificates (classical + PQ)
├─ Update verification software
└─ Begin pilot deployments
```

```
Phase 3: Hybrid Default
├─ Make hybrid signatures default
├─ Classical-only as fallback
├─ Monitor compatibility issues
└─ Update timestamp authorities
```

```
Phase 4: PQ Primary
├─ PQ signatures primary
├─ Begin classical deprecation
├─ Update all verification policies
└─ Archive classical roots
```

Timeline Considerations:

=====

- Root CA: Plan 10-20 year replacement
- Intermediate CA: 3-5 year cycles
- End-entity: 1-2 year cycles
- Start now: Recorded signatures vulnerable today

9.5.4 Exercises

Exercise 9.5.1: Hybrid Code Signing Implementation *(Advanced)*

Implement a hybrid code signing system that:

1. Signs packages with both ECDSA and ML-DSA
2. Validates either signature for backwards compatibility
3. Requires both signatures for high-security verification

Tasks:

- Design signature format supporting dual signatures
- Implement signing tool with algorithm selection
- Create verification tool with policy options

Solution:

```
/* Hybrid code signing implementation */
#include <stdint.h>
#include <string.h>

/* Signature format: length-prefixed concatenation */
typedef struct {
    uint32_t magic;           /* 0x48594252 "HYBR" */
    uint16_t version;         /* Format version */
    uint16_t flags;           /* Signing policy flags */
    /* ECDSA P-256 signature */
}
```

```

uint16_t ecdsa_sig_len;
uint8_t ecdsa_sig[72];      /* DER-encoded ECDSA signature */
/* ML-DSA-65 signature */
uint16_t mldsa_sig_len;
uint8_t mldsa_sig[3309];    /* ML-DSA-65 signature */
/* Metadata */
uint64_t timestamp;
uint8_t signer_id[32];      /* SHA-256 of signing certificate */
} hybrid_signature_t;

typedef enum {
    VERIFY_EITHER = 0x01, /* Accept if EITHER signature valid (compat) */
    VERIFY_BOTH   = 0x02, /* Require BOTH signatures valid (high-sec) */
    VERIFY_PQ_ONLY = 0x04, /* Require ML-DSA signature valid */
} verify_policy_t;

/* Sign a package with both algorithms */
int hybrid_sign(hybrid_signature_t *sig,
                const uint8_t *data, size_t data_len,
                const uint8_t *ecdsa_sk,
                const uint8_t *mldsa_sk) {
    sig->magic = 0x48594252;
    sig->version = 1;
    sig->flags = VERIFY_BOTH;

    /* ECDSA-P256 signature */
    /* EVP_DigestSign(ecdsa_ctx, sig->ecdsa_sig, ..., data, data_len) */
    sig->ecdsa_sig_len = 72; /* actual length from signing */

    /* ML-DSA-65 signature */
    /* OQS_SIG_sign(mldsa, sig->mldsa_sig, ..., data, data_len, mldsa_sk) */
    sig->mldsa_sig_len = 3309;

    sig->timestamp = (uint64_t)time(NULL);
    return 0;
}

/* Verify with configurable policy */
int hybrid_verify(const hybrid_signature_t *sig,
                  const uint8_t *data, size_t data_len,
                  const uint8_t *ecdsa_pk,
                  const uint8_t *mldsa_pk,
                  verify_policy_t policy) {
    if (sig->magic != 0x48594252) return -1;

    int ecdsa_valid = 0; /* EVP_DigestVerify(...) == 1 */
    int mldsa_valid = 0; /* OQS_SIG_verify(...) == OQS_SUCCESS */

    switch (policy) {
        case VERIFY_EITHER:
            return (ecdsa_valid || mldsa_valid) ? 0 : -1;
        case VERIFY_BOTH:
            return (ecdsa_valid && mldsa_valid) ? 0 : -1;
        case VERIFY_PQ_ONLY:
            return mldsa_valid ? 0 : -1;
        default:
            return -1;
    }
}

```

```

    }
}

/* Usage: signing tool */
/* ./hybrid-sign --key ecdsa.pem --pqkey mldsa.pem package.tar.gz */
/* ./hybrid-verify --policy both --cert signer.pem package.tar.gz.sig */

```

Exercise 9.5.2: PKI Migration Planning *(Intermediate)*

Create a complete PKI migration plan for a fictional organization:

1. Inventory existing certificate infrastructure
2. Design hybrid certificate hierarchy
3. Plan phased rollout schedule
4. Create rollback procedures

Deliverables:

- Migration timeline document
- Risk assessment matrix
- Testing and validation plan
- Communication strategy for stakeholders

Solution:

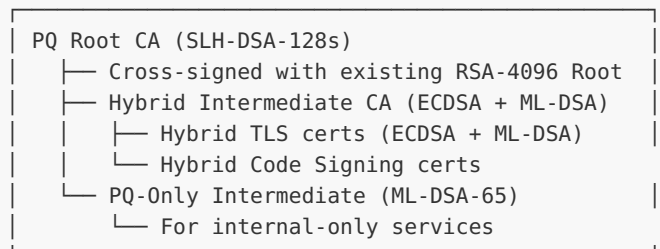
PKI Migration Plan: Acme Corp (Fictional)

=====

1. CURRENT INVENTORY

- Root CA: RSA-4096, expires 2035
- 3 Intermediate CAs: RSA-2048, expires 2028
- ~5,000 leaf certificates: RSA-2048 (TLS), ECDSA P-256 (code signing)
- HSMs: Thales Luna 7 (no PQ support), AWS CloudHSM (PQ-ready)
- PKI software: EJBCA 8.x

2. HYBRID CERTIFICATE HIERARCHY



3. PHASED ROLLOUT (24 months)

Phase 1 - Foundation (Months 1-3):

- [] Upgrade EJBCA to PQ-capable version
- [] Deploy PQ-capable HSM (AWS CloudHSM)
- [] Generate PQ Root CA with SLH-DSA-128s

[] Cross-sign PQ Root with existing RSA Root

Phase 2 - Internal Pilot (Months 4-8):

- [] Issue hybrid intermediate CA
- [] Deploy hybrid TLS certs on 10 internal services
- [] Test with all browser/client combinations
- [] Validate CRL/OCSP with hybrid certs

Phase 3 - External Rollout (Months 9-16):

- [] Issue hybrid TLS certs for public-facing services
- [] Update code signing to hybrid (ECDSA + ML-DSA)
- [] Deploy to CDN and load balancers
- [] Monitor for compatibility issues

Phase 4 - Classical Deprecation (Months 17-24):

- [] New certs issued as PQ-only (internal)
- [] Hybrid remains for external services
- [] Plan RSA intermediate CA retirement
- [] Document lessons learned

4. RISK ASSESSMENT MATRIX

Risk	Likelihood	Impact	Mitigation
-----	-----	-----	-----
HSM firmware lacks PQ support	Medium	High	Pre-qualify HSM vendors
Client incompatibility	High	Medium	Hybrid certs as fallback
Cert size exceeds MTU	Medium	Medium	Certificate compression
PQ algorithm vulnerability	Low	High	Hybrid = safe fallback
Staff skill gaps	High	Medium	Training program (Module 11)

5. ROLLBACK PROCEDURES

- All hybrid certs include classical component → revoke PQ, keep classical
- Cross-signed roots → clients fall back to RSA trust chain
- Configuration flags to disable PQ at load balancer level
- Tested quarterly in staging environment

Exercise 9.5.3: Timestamp Authority Integration *(Advanced)*

Implement PQ-aware timestamping:

1. Create timestamp request/response handling for hybrid signatures
2. Implement long-term archive format
3. Validate timestamps across algorithm transitions

Solution:

```
/* PQ-Aware Timestamp Authority Implementation */

typedef struct {
    uint8_t hash[32];           /* SHA-256 of data being timestamped */
    uint64_t timestamp;         /* Unix timestamp */
    uint32_t nonce;             /* Anti-replay nonce */
} timestamp_request_t;

typedef struct {
    timestamp_request_t request;
```

```

/* TSA signs with both algorithms for long-term validity */
uint8_t ecdsa_sig[72];          /* ECDSA P-256 signature */
uint16_t ecdsa_sig_len;
uint8_t mldsa_sig[3309];        /* ML-DSA-65 signature */
uint16_t mldsa_sig_len;
uint8_t tsa_cert_hash[32];      /* Identifies TSA certificate */
} timestamp_response_t;

/* Long-Term Archive (LTA) format
 * Supports re-timestamping as algorithms age */
typedef struct {
    /* Original signed data reference */
    uint8_t document_hash[32];
    size_t document_size;

    /* Chain of timestamps (oldest first) */
    timestamp_response_t timestamps[10];
    size_t num_timestamps;

    /* Archive metadata */
    uint16_t archive_version;
    /* Each re-timestamp covers all previous data:
     * ts[1] signs (document_hash || ts[0])
     * ts[2] signs (document_hash || ts[0] || ts[1])
     * This creates a chain of trust across algorithm transitions */
} long_term_archive_t;

/*
 * Re-timestamping workflow for algorithm transitions:
 *
 * Year 2026: Original timestamp with ECDSA + ML-DSA-65
 *   ts[0] = sign(doc_hash, ECDSA+ML-DSA-65)
 *
 * Year 2030: ECDSA deprecated, re-timestamp with newer PQ
 *   ts[1] = sign(doc_hash || ts[0], ML-DSA-87)
 *
 * Year 2040: ML-DSA-65 deprecated, re-timestamp again
 *   ts[2] = sign(doc_hash || ts[0] || ts[1], next-gen-algo)
 *
 * Validation: At any point, the most recent timestamp
 * using a trusted algorithm validates the entire chain.
 */

int validate_archive(const long_term_archive_t *archive) {
    /* Find most recent timestamp with a still-trusted algorithm */
    for (int i = (int)archive->num_timestamps - 1; i >= 0; i--) {
        /* Reconstruct signed data: doc_hash || all previous timestamps */
        /* Verify signature with current trust policy */
        /* If valid: entire chain is verified */
        /* If algorithm deprecated: try previous timestamp */
    }
    return -1; /* No trusted timestamp found */
}

```

Key design principles:

- Dual signatures at each timestamp for algorithm diversity
- Re-timestamping chain enables indefinite archive validity

- Each new timestamp covers all previous data (no gaps)
- Validation works backward from most recent trusted signature

Unit 9.5 Summary

Code signing and PKI post-quantum migration:

- Root CAs need longest-term security (SLH-DSA)
- Hybrid signatures enable gradual transition
- Timestamp authorities critical for long-term verification
- Start planning now for multi-year migration

9.5.5 Certificate Transparency in a Post-Quantum World

Certificate Transparency (CT) logs (RFC 6962, RFC 9162) provide an append-only, publicly auditable record of all issued certificates. The transition to PQC impacts CT in several ways.

Size Impact on CT Logs:

PQC certificates are significantly larger than classical ones, directly affecting CT log infrastructure:

CERTIFICATE TYPE	TYPICAL SIZE	CT LOG ENTRY OVERHEAD
RSA-2048	~1.2 KB	Baseline
ECDSA P-256	~0.8 KB	0.7×
ML-DSA-65	~3.8 KB	3.2×
ML-DSA-65 + ECDSA (hybrid)	~4.5 KB	3.8×
SLH-DSA-128s	~9.0 KB	7.5×

With ~10 billion certificates issued annually (and growing), CT log operators must plan for 3-8× storage and bandwidth increases.

Signed Certificate Timestamp (SCT) Migration:

SCTs are the CT mechanism that proves a certificate was submitted to a log. During PQC migration:

1. **Hybrid SCTs:** Include both classical and PQC signatures in the SCT to maintain backward compatibility

2. **Log Signing Key Rotation:** CT log operators must migrate their log signing keys to PQC algorithms, coordinated with browser root programs
3. **Merkle Tree Hash Function:** Current CT uses SHA-256 for the Merkle tree; this remains quantum-safe (Grover's algorithm provides only a quadratic speedup, SHA-256 retains 128-bit security)

Timeline Considerations:

- CT log operators should begin PQC key generation for log signing by 2027
- Browser root programs (Apple, Google, Mozilla) will need to update CT verification code
- Existing logs with classical signatures remain valid — quantum threat is to future forgery, not historical entries

References:

- RFC 6962: Certificate Transparency
 - RFC 9162: Certificate Transparency Version 2.0
 - IETF draft-ietf-lamps-pq-composite-certs: Composite PQ Certificates
-

Unit 9.6: DNSSEC and DNS with Post-Quantum Signatures

Prerequisites: Volume 1, Module 6 (ML-DSA), Volume 1, Module 7 (SLH-DSA)

DNSSEC adds origin authentication and integrity to DNS by signing resource record sets (RRsets). Migrating it to post-quantum signatures is uniquely hard because DNS is one of the most size-constrained protocols still in wide use.

The DNSSEC size problem:

- Classic DNS over UDP caps responses at **512 bytes**; EDNS0 raises the advertised buffer (commonly 1232 bytes in practice to avoid IP fragmentation), but responses beyond that trigger **TCP fallback** via the truncation (TC) bit.
- DNSSEC responses already carry signatures (RRSIG), keys (DNSKEY), and denial-of-existence records (NSEC/NSEC3). They are frequently the largest DNS messages on the wire even with classical ECDSA/Ed25519 signatures (64–96 bytes each).

- Post-quantum signatures are far larger: ML-DSA-65 is ~3,309 bytes per signature and SLH-DSA-SHA2-128f is ~17,088 bytes. A single PQ-signed RRset can dwarf the entire classical response.

Why this strains DNS:

- **Fragmentation:** UDP responses larger than the path MTU fragment at the IP layer. Fragmented DNS is unreliable (middleboxes drop fragments) and is itself a security concern.
- **TCP fallback:** Oversized responses force a retry over TCP, adding round-trips and connection state that authoritative servers and resolvers are not provisioned to handle at DNS scale.
- **Amplification:** Large signed responses worsen DNS as a DDoS reflection/amplification vector — a small query eliciting a multi-kilobyte answer is exactly what attackers abuse.

Current IETF direction (as of 2026):

- Work in the IETF **dnsop** working group is exploring how to make DNSSEC viable with post-quantum primitives. No standard PQ DNSSEC algorithm is deployed yet.
- Approaches under discussion include **Merkle Tree Certificates** and other hash-based / compact-signature techniques (see also draft-sheth-pqc-dnssec-strategy, which proposes SLH-DSA in Merkle Tree Ladder mode), selective use of smaller PQ signatures, and protocol changes to reduce how often full signatures must be transmitted.
- Where hash-based signatures are considered, **SLH-DSA** is attractive for its conservative security but painful for its size; lattice signatures (ML-DSA) are smaller but still large relative to DNS norms. The trade-off space is unresolved.

What to do today:

- **Keep DNSSEC on classical algorithms** (ECDSA P-256 / Ed25519) for now; there is no production-ready PQ DNSSEC profile to migrate to.
- **Monitor IETF dnsop** drafts on post-quantum DNSSEC and Merkle Tree Certificates.
- **Plan for larger responses:** ensure resolvers and authoritative servers handle EDNS0 sizing sensibly and support reliable **TCP fallback**; test infrastructure against larger answer sizes ahead of any future PQ rollout.

- Treat DNSSEC's quantum exposure as lower urgency than confidentiality protocols: signatures protect integrity/authenticity, which is not subject to "harvest now, decrypt later" — a forged record requires a quantum computer *at the time of attack*, not retroactively.
-

Module 9 Summary: Protocol Integration

Module 9 covered post-quantum integration across major security protocols:

Unit 9.1: TLS 1.3

- Hybrid KEX with X25519+ML-KEM-768
- Size impact management
- Certificate chain optimization

Unit 9.2: SSH

- `mlkem768x25519-sha256` default since OpenSSH 10.0; `sntrup761x25519-sha512` the first PQ KEX (default since 9.0)
- Host key rotation strategies
- Configuration best practices

Unit 9.3: S/MIME

- Long-term email confidentiality challenges
- Hybrid encryption for KEK protection
- Certificate size considerations

Unit 9.4: VPN/IPsec

- IKEv2 hybrid key exchange
- strongSwan configuration
- Tunnel rekeying considerations

Unit 9.5: Code Signing

- PKI migration planning
- Hybrid signature schemes
- Timestamp authority integration

Key Takeaways

- Key exchange is most urgent (harvest-now-decrypt-later)
- Hybrid approaches enable gradual transition
- Size increases require infrastructure planning
- Standardization ongoing - follow IETF drafts

Library Interoperability Matrix

Last verified: April 2026 (v1.0). Library versions and feature flags decay on a 3-6 month cycle — verify against vendor release notes before relying on specific versions in this table.

When deploying post-quantum cryptography in production, library selection significantly impacts which algorithms are available and the maturity of implementations. The following matrix summarizes PQC algorithm support across major cryptographic libraries:

LIBRARY	ML-KEM	ML-DSA	SLH-DSA	X-WING	VERSION	STATUS
liboqs	✓	✓	✓	✗	0.14+	Reference/experimental
BoringSSL	✓	✓	✗	✓	Jan 2026	Production (FIPS module)
OpenSSL	✓	✓	✓	✓	3.5+ (LTS)	Provider-based
wolfSSL	✓	✓	✗	✗	5.8+	Production (FIPS pending)
AWS-LC	✓	✓	✗	✗	2.0+	Production (FIPS 140-3 certified)
BouncyCastle	✓	✓	✓	✗	2.0+	Java/C# production

Implementation Notes:

- **liboqs**: Open Quantum Safe project providing reference implementations. Excellent for prototyping and testing but not recommended as sole production dependency. Integrates with OpenSSL via oqs-provider. Note the API churn as the library tracks the finalized FIPS standards: the pre-standardization **Dilithium** API was removed in liboqs 0.15.0 (Nov 2025) — migrate to ML-DSA (FIPS 204) — and **SPHINCS+** is slated for removal in 0.16.0 in favor of the FIPS 205 SLH-DSA implementation (from [pq-code-package/slhdsa-c](#)). Pin a liboqs version and read its release notes before upgrading.

- **BoringSSL:** Google's production library focused on TLS. Supports ML-KEM and ML-DSA (added January 2026 with X.509 and FIPS module integration). X25519+ML-KEM-768 hybrid is the recommended configuration for Chrome and Google services.
- **OpenSSL:** Version 3.5+ (LTS) provides native ML-KEM, ML-DSA, and SLH-DSA support with hybrid X25519+ML-KEM-768 key shares offered by default in TLS 1.3. Version 3.6 added CMS KEMRecipientInfo (RFC 9629) and additional hybrid groups (SecP256r1MLKEM768, SecP384r1MLKEM1024). Use oqs-provider for additional algorithms.
- **wolfSSL:** Embedded-focused library with ML-KEM and ML-DSA support. FIPS 140-3 validation in progress for PQC algorithms. Also supports HQC. Suitable for IoT and resource-constrained deployments.
- **AWS-LC:** Amazon's fork of BoringSSL/OpenSSL. AWS-LC-FIPS v2.0 is FIPS 140-3 certified (including ML-KEM). V3.0 with expanded PQC coverage is submitted for validation. ML-KEM is integrated into AWS KMS, ACM, and Secrets Manager.
- **BouncyCastle:** Mature Java and C# implementations suitable for enterprise applications. Provides both lightweight and JCA provider APIs.

Production Readiness Guidance:

MATURITY LEVEL	DESCRIPTION	LIBRARIES
Production	Deployed at scale, FIPS validation available or certified	BoringSSL, AWS-LC, wolfSSL
Production-ready	Stable API, suitable for production with testing	OpenSSL 3.5+, BouncyCastle
Experimental	API may change, suitable for development and testing	liboqs, research implementations

What's Next: Module 10 examines algorithms beyond the initial NIST selections, including FALCON/FN-DSA, code-based schemes (HQC, Classic McEliece), and emerging research directions. While Modules 4–9 (Volume 1 for 4, Volume 2 for 9) focused on the three primary FIPS standards, Module 10 broadens the landscape to help you evaluate future algorithm additions and alternative approaches.

Module 10: Future Algorithms and Research

This module examines the evolving landscape of post-quantum cryptography. We explore algorithms under consideration for future standardization, active research directions, and strategies for maintaining cryptographic agility as the field advances.

Prerequisites: Modules 4–8 (Volume 1 for 4, Volume 2 for 8) (ML-KEM, ML-DSA, SLH-DSA, Hybrid Cryptography)

Module Learning Objectives:

- Understand NIST's ongoing algorithm standardization process
- Evaluate FALCON (FN-DSA) and its trade-offs versus ML-DSA
- Analyze code-based and isogeny-based alternatives
- Design crypto-agile systems for algorithm transitions
- Integrate with liboqs and OpenSSL provider architecture
- Plan for future algorithm updates and deprecations

Unit 10.1: NIST Additional Algorithms

Prerequisites: Volume 1, Modules 4–7 (ML-KEM, ML-DSA, SLH-DSA fundamentals)

Estimated Time: 2-3 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Understand NIST's ongoing standardization process** and additional algorithm candidates
2. **Evaluate alternative signature schemes** including FALCON and others
3. **Assess trade-offs** between standardized and alternative algorithms
4. **Plan for algorithm diversity** in long-term deployments

10.1.1 NIST Standardization Status

NIST PQC Standardization Timeline

=====

Round 1 (2017-2019): 69 submissions

└─ 26 advanced to Round 2

Round 2 (2019-2020): 26 candidates

└─ 7 finalists + 8 alternates

Round 3 (2020-2022): Final selection

└─ Selected for standardization:

- CRYSTALS-Kyber (ML-KEM) - KEM
- CRYSTALS-Dilithium (ML-DSA) - Signature
- SPHINCS+ (SLH-DSA) - Signature
- FALCON (FN-DSA) - Signature (FIPS 206 draft, expected late 2026/2027)

Round 4 (2022-2025): Additional algorithms

└─ Selected:

- HQC - code-based KEM (selected March 2025, backup to ML-KEM)

└─ Remaining candidates:

- BIKE, Classic McEliece (KEMs)
- Additional signature schemes

Current Standards (FIPS):

=====

- FIPS 203: ML-KEM (Kyber) - August 2024
- FIPS 204: ML-DSA (Dilithium) - August 2024
- FIPS 205: SLH-DSA (SPHINCS+) - August 2024
- FIPS 206: FN-DSA (FALCON) - Draft August 2025, final expected late 2026/2027
- HQC: Draft standard expected ~2027

10.1.2 FALCON Overview

FALCON (Fast-Fourier Lattice-based Compact Signatures over NTRU) offers smaller signatures than ML-DSA:

```
/* falcon_overview.h - FALCON Algorithm Parameters */

#ifndef FALCON_OVERVIEW_H
#define FALCON_OVERVIEW_H

#include <stdint.h>
#include <stddef.h>

/*
 * FALCON Parameters (from specification)
 * =====
 *
 * FALCON-512 (NIST Level 1):
 *   - Public key: 897 bytes
 *   - Private key: 1281 bytes
 *   - Signature: ~666 bytes (average)
 *   - Security: ~108 bits classical, ~103 bits quantum
 *
 * FALCON-1024 (NIST Level 5):
```

```

*   - Public key: 1793 bytes
*   - Private key: 2305 bytes
*   - Signature: ~1280 bytes (average)
*   - Security: ~252 bits classical, ~230 bits quantum
*
* Key advantages over ML-DSA:
*   - Significantly smaller signatures (~5x smaller)
*   - Based on NTRU lattices (well-studied)
*
* Challenges:
*   - Complex implementation (FFT over number fields)
*   - Floating-point arithmetic required (or careful fixed-point)
*   - Side-channel protection more difficult
*   - Variable-time signing (signature size varies)
*/

/* FALCON-512 sizes */
#define FALCON512_PUBLIC_KEY_SIZE    897
#define FALCON512_PRIVATE_KEY_SIZE  1281
#define FALCON512_SIG_MAX_SIZE      752    /* Maximum signature size */
#define FALCON512_SIG_AVG_SIZE      666    /* Average signature size */

/* FALCON-1024 sizes */
#define FALCON1024_PUBLIC_KEY_SIZE   1793
#define FALCON1024_PRIVATE_KEY_SIZE  2305
#define FALCON1024_SIG_MAX_SIZE      1462
#define FALCON1024_SIG_AVG_SIZE      1280

/* Comparison with ML-DSA */
/*
* Size Comparison:
* =====
*
* Algorithm      Level  Public Key  Signature  Total
* -----
* FALCON-512     1      897 bytes  666 bytes  1,563 bytes
* FALCON-1024    5      1,793 bytes 1,280 bytes 3,073 bytes
* ML-DSA-44      2      1,312 bytes 2,420 bytes 3,732 bytes
* ML-DSA-65      3      1,952 bytes 3,309 bytes 5,261 bytes
*
* FALCON signatures are ~4-5x smaller at comparable security!
*/

typedef struct {
    uint8_t *public_key;
    size_t public_key_len;
    uint8_t *private_key;
    size_t private_key_len;
} falcon_keypair_t;

/* Function prototypes */
int falcon512_keygen(falcon_keypair_t *kp);
int falcon512_sign(uint8_t *sig, size_t *sig_len,
                  const uint8_t *msg, size_t msg_len,
                  const uint8_t *private_key, size_t private_key_len);
int falcon512_verify(const uint8_t *sig, size_t sig_len,
                  const uint8_t *msg, size_t msg_len,
                  const uint8_t *public_key, size_t public_key_len);

#endif /* FALCON_OVERVIEW_H */

```

10.1.3 Alternative KEMs

Beyond ML-KEM, several alternative KEMs are under consideration:

```
/* alternative_kems.c - Overview of Alternative KEMs */

/*
 * Classic McEliece
 * =====
 * Based on: Code-based cryptography (1978)
 *
 * Advantages:
 *   - Longest cryptanalysis history (45+ years)
 *   - Very fast encapsulation/decapsulation
 *   - Conservative security assumptions
 *
 * Disadvantages:
 *   - Extremely large public keys (hundreds of KB to MB)
 *   - Not practical for most applications
 *   - Key generation is slow
 *
 * Use case: Long-term key storage, ultra-high security
 */

#define MCELIECE348864_PK_SIZE 261120 /* ~255 KB */
#define MCELIECE348864_SK_SIZE 6492
#define MCELIECE348864_CT_SIZE 128

#define MCELIECE6960119_PK_SIZE 1047319 /* ~1 MB */
#define MCELIECE6960119_SK_SIZE 13908
#define MCELIECE6960119_CT_SIZE 226

/*
 * BIKE (Bit Flipping Key Encapsulation)
 * =====
 * Based on: QC-MDPC codes
 *
 * Advantages:
 *   - Small public keys (1.5-5 KB)
 *   - Small ciphertexts
 *   - Competitive performance
 *
 * Disadvantages:
 *   - Decapsulation failures possible (very low probability)
 *   - IND-CCA security relies on additional techniques
 *   - Less cryptanalysis history than McEliece
 *
 * Use case: General-purpose alternative to ML-KEM
 */

#define BIKE_L1_PK_SIZE 1541
#define BIKE_L1_SK_SIZE 5223
#define BIKE_L1_CT_SIZE 1541

#define BIKE_L3_PK_SIZE 3083
#define BIKE_L3_SK_SIZE 10105
#define BIKE_L3_CT_SIZE 3083

/*
```



```

* HQC (Hamming Quasi-Cyclic)
* =====
* Based on: Quasi-cyclic codes
* Status: Selected by NIST for standardization (March 2025)
*         Provides code-based diversity as backup to lattice-based ML-KEM
*
* Advantages:
*   - Simpler than BIKE
*   - Competitive key and ciphertext sizes
*   - Good performance
*   - Different mathematical basis than ML-KEM (code-based vs lattice-based)
*
* Disadvantages:
*   - Larger than ML-KEM
*   - Standard not yet finalized (draft expected ~2027)
*
* Use case: Code-based alternative when algorithm diversity is required
*/

#define HQC128_PK_SIZE      2249
#define HQC128_SK_SIZE      2289
#define HQC128_CT_SIZE      4481

#define HQC256_PK_SIZE      7245
#define HQC256_SK_SIZE      7317
#define HQC256_CT_SIZE      14469

/* Comparison table */
typedef struct {
    const char *name;
    const char *basis;
    size_t pk_size;
    size_t sk_size;
    size_t ct_size;
    const char *notes;
} kem_comparison_t;

static const kem_comparison_t kem_alternatives[] = {
    {"ML-KEM-768", "Module Lattices", 1184, 2400, 1088,
     "NIST standard, balanced"},
    {"Classic McEliece", "Codes (Goppa)", 261120, 6492, 128,
     "Most conservative, huge keys"},
    {"BIKE-L1", "QC-MDPC Codes", 1541, 5223, 1541,
     "Small keys, decryption failures"},
    {"HQC-128", "Quasi-Cyclic Codes", 2249, 2289, 4481,
     "Code-based alternative"},
    {NULL, NULL, 0, 0, 0, NULL}
};

void print_kem_comparison(void) {
    printf("KEM Algorithm Comparison\n");
    printf("=====\n\n");
    printf("%-20s %-20s %10s %10s %10s\n",
           "Algorithm", "Basis", "PK Size", "SK Size", "CT Size");
    printf("%-20s %-20s %10s %10s %10s\n",
           "-----", "-----", "-----", "-----", "-----");

    for (int i = 0; kem_alternatives[i].name != NULL; i++) {
        printf("%-20s %-20s %10zu %10zu %10zu\n",
               kem_alternatives[i].name,

```

```

        kem_alternatives[i].basis,
        kem_alternatives[i].pk_size,
        kem_alternatives[i].sk_size,
        kem_alternatives[i].ct_size);
    }
}

```

10.1.4 FrodoKEM: The Conservative Lattice Alternative

FrodoKEM is conspicuously absent from the NIST standards but remains important for high-assurance deployments. Unlike ML-KEM (which uses *structured* Module-LWE over polynomial rings), FrodoKEM is based on *plain* (unstructured) LWE over integers.

Why FrodoKEM matters:

The algebraic structure in Module-LWE (the polynomial ring $\mathbb{Z}_q[X]/(X^n + 1)$) enables fast NTT-based multiplication and compact keys. However, some cryptographers worry this structure could harbor hidden attacks — a breakthrough exploiting the ring structure would break ML-KEM, ML-DSA, and FALCON simultaneously.

FrodoKEM avoids this risk by using plain LWE with no algebraic structure. The trade-off is dramatically larger keys:

PARAMETER	ML-KEM-768	FRODOKEM-976	RATIO
Security	NIST Level 3	NIST Level 3	Same
Public key	1,184 B	15,632 B	13×
Ciphertext	1,088 B	15,744 B	14×
Shared secret	32 B	32 B	Same
Keygen time	~50 μ s	~6 ms	120×
Encaps time	~70 μ s	~7 ms	100×

FrodoKEM parameter sets:

VARIANT	N	Q	SECURITY LEVEL	PK SIZE	CT SIZE
FrodoKEM-640	640	2^{15}	NIST Level 1	9,616 B	9,720 B
FrodoKEM-976	976	2^{16}	NIST Level 3	15,632 B	15,744 B
FrodoKEM-1344	1344	2^{16}	NIST Level 5	21,520 B	21,632 B

When to use FrodoKEM:

- Systems requiring maximum conservatism (defense, intelligence)
- Long-term key storage where key size is not a constraint
- Environments where algorithm diversity is mandated
- As a hedging strategy alongside ML-KEM in hybrid schemes

When NOT to use FrodoKEM:

- Bandwidth-constrained protocols (TLS, SSH)
- Mobile/embedded applications
- Any scenario where 15 KB+ keys are prohibitive

FrodoKEM is available in liboqs:

```
// Compile: gcc -o test_frodo test_frodo.c -loqs
#include <oqs/oqs.h>

// Usage identical to ML-KEM via liboqs API
OQS_KEM *kem = OQS_KEM_new(OQS_KEM_alg_frodokem_976_aes);
// ... standard OQS_KEM_keypair / encaps / decaps workflow ...
OQS_KEM_free(kem);
```

10.1.5 NTRU and NTRU Prime

NTRU (1996) is the oldest lattice-based cryptosystem, predating ML-KEM's theoretical foundations by over a decade. Understanding NTRU is important for three reasons:

1. **Historical significance:** NTRU introduced the idea of using polynomial rings for efficient public-key encryption, inspiring the development of Ring-LWE and Module-LWE.
2. **FALCON is NTRU-based:** NIST's alternate signature standard FN-DSA (FALCON) uses NTRU lattices for its trapdoor construction. Understanding NTRU helps explain why FALCON can produce compact signatures but requires careful floating-point arithmetic.
3. **PQ key exchange is deployed in OpenSSH:** Since OpenSSH 9.0 (2022), `sntrup761x25519-sha512@openssh.com` — a hybrid of NTRU Prime and X25519 — was the first PQ default. As of OpenSSH 10.0 (April 2025), the standardized `mlkem768x25519-sha256` (ML-KEM-768 + X25519) is the default.

The NTRU Hard Problem:

In NTRU, the public key is $h = g/f$ in the polynomial ring $\mathbb{Z}_q[X]/(X^n - 1)$, where f and g are "short" polynomials (small coefficients). Given h , recovering f and g requires finding short vectors in a specific lattice — believed to be hard even for quantum computers.

Key difference from Module-LWE: NTRU uses the ring $\mathbb{Z}_q[X]/(X^n - 1)$, while ML-KEM uses $\mathbb{Z}_q[X]/(X^n + 1)$. The cyclotomic polynomial $X^n + 1$ has nicer algebraic properties (it is irreducible when n is a power of 2), which is why ML-KEM chose it.

NTRU Prime (Bernstein, Chuengsatiansup, Lange, van Vredendaal) is a "streamlined" variant that addresses structural concerns:

- Uses the polynomial ring $\mathbb{Z}_q[X]/(X^p - X - 1)$ where p is prime (avoiding 2-adic structure)
- Designed to minimize the risk of algebraic structure being exploited
- **sntrup761:** $p = 761, q = 4591$ — used in OpenSSH

sntrup761 in OpenSSH:

PARAMETER	VALUE
Ring	$\mathbb{Z}_{4591}[X]/(X^{761} - X - 1)$
Public key	1,158 bytes
Ciphertext	1,039 bytes
Shared secret	32 bytes
Security level	~NIST Level 1-2

OpenSSH chose sntrup761 (pre-dating FIPS 203) as a conservative choice: NTRU Prime's design intentionally avoids the algebraic structure that some researchers worry could be exploited. The hybrid with X25519 ensures security against both classical and quantum attacks.

Migration path: OpenSSH added the standardized ML-KEM-768 hybrid (`mlkem768x25519-sha256`) and made it the default in OpenSSH 10.0 (April 2025), alongside the original sntrup761 hybrid. The hybrid approach (sntrup761+X25519 or ML-KEM-768+X25519) will likely persist during the transition period.

NTRU vs ML-KEM comparison:

ASPECT	ML-KEM (MODULE-LWE)	NTRU/NTRU PRIME
Hard problem	Module-LWE	NTRU lattice
Ring structure	$\mathbb{Z}_q[X]/(X^n + 1)$	$\mathbb{Z}_q[X]/(X^p - X - 1)$
NIST standard	FIPS 203	Not standardized
Key sizes	Slightly smaller	Slightly larger
Performance	Fast (NTT-friendly)	Competitive
Deployed in	TLS, S/MIME	OpenSSH (sntrup761)
Design philosophy	Efficiency-optimized	Security-conservative

10.1.6 Exercises

Exercise 10.1.1: Algorithm Selection Framework *(Intermediate)*

```

/*
 * Exercise: Implement a decision framework for selecting PQ algorithms
 * based on deployment requirements.
 */

typedef struct {
    size_t max_public_key_size;
    size_t max_signature_size;
    int require_nist_standard;
    int require_conservative_security;
    int performance_critical;
    int bandwidth_constrained;
} deployment_requirements_t;

/* Recommend algorithm based on requirements */
const char *recommend_signature_algorithm(const deployment_requirements_t *req);
const char *recommend_kem_algorithm(const deployment_requirements_t *req);

```

Solution:

```

/* Solution to Exercise 10.1.1 */

#include <stddef.h>
#include <stdio.h>

typedef struct {
    size_t max_public_key_size;
    size_t max_signature_size;
    int require_nist_standard;
    int require_conservative_security;
    int performance_critical;
    int bandwidth_constrained;
} deployment_requirements_t;

```

```

const char *recommend_signature_algorithm(const deployment_requirements_t *req) {
    /*
     * Decision logic:
     *
     * 1. If bandwidth constrained and small signatures needed:
     *    - FALCON (smallest signatures)
     *    - But check if NIST standard required (FIPS 206 pending)
     *
     * 2. If conservative security required:
     *    - SLH-DSA (hash-based, well-understood)
     *    - Accept larger signatures
     *
     * 3. If performance critical:
     *    - ML-DSA (fastest signing/verification)
     *
     * 4. Default balanced choice:
     *    - ML-DSA-65 (good balance)
     */

    /* Check signature size constraint */
    if (req->bandwidth_constrained || req->max_signature_size < 2000) {
        if (!req->require_nist_standard) {
            return "FALCON-512 (smallest signatures, ~666 bytes, FIPS 206 pending)";
        }
        /* If NIST required and size constrained, tough choice */
        return "ML-DSA-44 (smallest NIST standard, 2420 byte signatures)";
    }

    /* Check conservative security requirement */
    if (req->require_conservative_security) {
        if (req->max_signature_size >= 17000) {
            return "SLH-DSA-SHA2-128f (hash-based, most conservative)";
        }
        if (req->max_signature_size >= 8000) {
            return "SLH-DSA-SHA2-128s (hash-based, smaller but slower)";
        }
        return "ML-DSA-87 (highest lattice security level)";
    }

    /* Check performance requirement */
    if (req->performance_critical) {
        return "ML-DSA-65 (best signing/verification performance)";
    }

    /* Default recommendation */
    return "ML-DSA-65 (balanced choice, NIST standard)";
}

const char *recommend_kem_algorithm(const deployment_requirements_t *req) {
    /*
     * Decision logic:
     *
     * 1. If extremely conservative security needed:
     *    - Classic McEliece (if key size acceptable)
     *
     * 2. If bandwidth constrained:
     *    - ML-KEM-768 (smallest practical option)
     *
     * 3. If diversity from lattices desired:
     *    - BIKE or HQC (code-based alternatives)
     */

```

```

*
* 4. Default:
*   - ML-KEM-768 (NIST standard, balanced)
*/

/* Check for extreme conservatism */
if (req->require_conservative_security &&
    req->max_public_key_size > 250000) {
    return "Classic McEliece (45+ years of cryptanalysis)";
}

/* Check NIST standard requirement */
if (req->require_nist_standard) {
    if (req->performance_critical) {
        return "ML-KEM-512 (fastest, Level 1 security)";
    }
    return "ML-KEM-768 (NIST standard, recommended)";
}

/* Check if diversity from lattices is desired */
if (!req->require_nist_standard && req->require_conservative_security) {
    if (req->max_public_key_size >= 3000) {
        return "HQC-128 (code-based diversity, selected for standardization)";
    }
    return "BIKE-L1 (code-based, smaller keys)";
}

/* Default */
return "ML-KEM-768 (NIST FIPS 203, balanced choice)";
}

void test_algorithm_selection(void) {
    printf("Algorithm Selection Framework Tests\n");
    printf("=====\n\n");

    /* Test 1: IoT constrained device */
    deployment_requirements_t iot = {
        .max_public_key_size = 2000,
        .max_signature_size = 1000,
        .require_nist_standard = 0,
        .require_conservative_security = 0,
        .performance_critical = 1,
        .bandwidth_constrained = 1
    };
    printf("IoT Device (bandwidth constrained):\n");
    printf("  Signature: %s\n", recommend_signature_algorithm(&iot));
    printf("  KEM: %s\n\n", recommend_kem_algorithm(&iot));

    /* Test 2: High-security government */
    deployment_requirements_t gov = {
        .max_public_key_size = 1000000,
        .max_signature_size = 20000,
        .require_nist_standard = 1,
        .require_conservative_security = 1,
        .performance_critical = 0,
        .bandwidth_constrained = 0
    };
    printf("Government High-Security:\n");
    printf("  Signature: %s\n", recommend_signature_algorithm(&gov));
    printf("  KEM: %s\n\n", recommend_kem_algorithm(&gov));
}

```

```

/* Test 3: High-performance web server */
deployment_requirements_t web = {
    .max_public_key_size = 5000,
    .max_signature_size = 5000,
    .require_nist_standard = 1,
    .require_conservative_security = 0,
    .performance_critical = 1,
    .bandwidth_constrained = 0
};
printf("Web Server (performance critical):\n");
printf("  Signature: %s\n", recommend_signature_algorithm(&web));
printf("  KEM: %s\n\n", recommend_kem_algorithm(&web));
}

```

Exercise 10.1.2: FALCON vs ML-DSA Trade-off Analysis *(Intermediate)*

Design a decision matrix comparing FALCON and ML-DSA for different use cases:

Use Case Analysis:

=====

1. Smart card authentication
 - Storage constraint: 32KB
 - Bandwidth: Limited
 - Operations/second: 10
2. Certificate authority
 - Storage: Unlimited
 - Bandwidth: High
 - Operations/second: 1000
3. Embedded IoT sensor
 - Storage constraint: 8KB
 - Bandwidth: Very limited (LoRa)
 - Operations/second: 1

Solution:

```

/* Analysis Framework */
typedef struct {
    const char *use_case;
    const char *recommended;
    const char *rationale;
} algorithm_decision_t;

static const algorithm_decision_t decisions[] = {
    {
        "Smart card authentication",
        "FALCON-512",
        "666-byte signatures fit bandwidth; complex signing OK for low ops/sec"
    },
    {
        "Certificate authority",
        "ML-DSA-65",
        "High performance needed; signature size less critical"
    },
}

```



```

{
    "Embedded IoT sensor",
    "SLH-DSA-128f or FALCON-512",
    "If signing rare, use SLH-DSA for simplicity; FALCON if space critical"
}
};

```

Exercise 10.1.3: Code-Based KEM Comparison *(Intermediate)*

Compare Classic McEliece, BIKE, and HQC for a long-term secure archive system:

Requirements:

- 30-year security guarantee
- Keys stored once, used many times
- Decapsulation performance secondary to security

Solution:

```

/* Long-term archive KEM selection */
typedef struct {
    const char *kem;
    size_t pk_size;
    size_t ct_size;
    const char *security_basis;
    int years_cryptanalysis;
    const char *recommendation;
} archive_kem_analysis_t;

static const archive_kem_analysis_t analysis[] = {
    {
        "Classic McEliece-6960119",
        1047319, /* ~1MB public key */
        226, /* Very small ciphertext */
        "Code-based (Goppa codes)",
        45, /* Since 1978 */
        "BEST: Most conservative, longest cryptanalysis history"
    },
    {
        "HQC-256",
        7245,
        14469,
        "Code-based (quasi-cyclic)",
        6, /* Relatively recent */
        "ACCEPTABLE: Balanced, still maturing"
    },
    {
        "BIKE-L5",
        3114,
        3114,
        "Code-based (QC-MDPC)",
    }
}

```

```
        6,  
        "ACCEPTABLE: Smallest keys, but newer"  
    }  
};  
  
/* For 30-year archive: Recommend Classic McEliece despite huge keys */
```

Unit 10.1 Summary

NIST's post-quantum standardization continues to evolve:

- **ML-KEM, ML-DSA, SLH-DSA** are now FIPS standards
- **FALCON** offers smaller signatures but more complex implementation
- **HQC** selected for standardization (March 2025) as code-based backup to ML-KEM
- **Alternative KEMs** (BIKE, McEliece) remain under evaluation for additional diversity
- Selection depends on specific deployment requirements

What's Next: Unit 10.2 explores emerging research directions beyond current standards.

Unit 10.2: Emerging Research Directions

Prerequisites: Unit 10.1 (NIST Additional Algorithms), Volume 1, Module 3 (Lattice Cryptography)

Estimated Time: 1-2 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Understand isogeny-based cryptography** and recent developments
2. **Explore multivariate and other alternative approaches**
3. **Assess quantum computing timeline estimates**
4. **Plan for cryptographic agility** in evolving landscape

10.2.1 Isogeny-Based Cryptography

SIDH/SIKE was broken in 2022, but research continues:

Isogeny-Based Cryptography Status

=====

SIDH/SIKE (Broken 2022):

- Used supersingular isogeny graphs
- Devastating attack by Castryck-Decru
- Complete break in polynomial time
- NOT recommended for any use

Active Research Directions:

CSIDH (Commutative SIDH):

- Uses ordinary curves instead of supersingular
- Resistant to SIDH attacks
- Very slow key exchange (~100ms)
- Active research on optimizations

SQISign (Short Quaternion Isogeny Signature):

- Signature scheme based on isogenies
- Very compact signatures (~200 bytes)
- Very slow signing (~seconds)
- Under active development

Lessons Learned:

1. Mathematical assumptions need extensive review
2. "Structured" assumptions can hide vulnerabilities
3. Cryptographic agility is essential
4. Diversity in algorithm families important

10.2.2 Other Research Directions

```
/* research_directions.c - Emerging PQC Research Areas */

/*
 * Multivariate Cryptography
 * =====
 * Based on: Solving systems of multivariate polynomial equations
 * Status: Some schemes broken, others under development
 *
 * Examples:
 * - Rainbow (broken 2022 - key recovery attack)
 * - GeMSS (under analysis)
 * - UOV variants (active research)
 *
 * Challenges:
 * - Large public keys
 * - Difficult to achieve security against all attacks
 * - Several high-profile breaks
 */

/*
 * Zero-Knowledge Proofs
```

```

* =====
* Post-quantum ZK systems:
* - Lattice-based ZK proofs
* - Hash-based ZK (STARKs)
* - MPC-in-the-head techniques
*
* Applications:
* - Privacy-preserving authentication
* - Blockchain and cryptocurrency
* - Verifiable computation
*/

/*
* Fully Homomorphic Encryption (FHE)
* =====
* Compute on encrypted data without decryption
* Often based on lattice problems (LWE)
*
* Post-quantum status:
* - Most FHE schemes already lattice-based
* - Generally considered quantum-resistant
* - Performance improving rapidly
*
* Libraries:
* - Microsoft SEAL
* - OpenFHE
* - TFHE
*/

/*
* Quantum Key Distribution (QKD)
* =====
* Uses quantum mechanics for key exchange
* Information-theoretically secure (in theory)
*
* Limitations:
* - Requires specialized hardware
* - Limited distance (~100km without repeaters)
* - Cannot authenticate parties (needs classical crypto)
* - Very expensive infrastructure
*
* Relationship to PQC:
* - Complementary, not competing
* - QKD for highest security links
* - PQC for general internet use
*/

typedef enum {
    RESEARCH_AREA_ISOGENY,
    RESEARCH_AREA_MULTIVARIATE,
    RESEARCH_AREA_ZK_PROOFS,
    RESEARCH_AREA_FHE,
    RESEARCH_AREA_QKD,
    RESEARCH_AREA_NEW_LATTICE,
} research_area_t;

typedef struct {
    research_area_t area;
    const char *name;
    const char *status;
}

```

```

    int maturity_level; /* 1-5, 5 = production ready */
    const char *timeline;
} research_status_t;

static const research_status_t research_areas[] = {
    {RESEARCH_AREA_ISOGENY, "Isogeny (post-SIDH)",
     "Active research, no practical schemes yet", 1,
     "5-10 years to practical schemes"},

    {RESEARCH_AREA_MULTIVARIATE, "Multivariate",
     "Several schemes broken, UOV variants promising", 2,
     "3-5 years for stable schemes"},

    {RESEARCH_AREA_ZK_PROOFS, "Post-Quantum ZK",
     "Active development, some production use", 3,
     "1-3 years for wider adoption"},

    {RESEARCH_AREA_FHE, "Homomorphic Encryption",
     "Practical for some applications", 3,
     "Already deployable for specific use cases"},

    {RESEARCH_AREA_QKD, "Quantum Key Distribution",
     "Deployed in limited scenarios", 4,
     "Infrastructure-dependent expansion"},

    {RESEARCH_AREA_NEW_LATTICE, "New Lattice Constructions",
     "Ongoing improvements to ML-KEM/ML-DSA", 5,
     "Continuous refinement expected"},
};

```

10.2.3 Quantum Computing Timeline

Quantum Computing Progress Estimates

=====

Current State (2024-2025):

- Largest announced: ~1000+ physical qubits
- Error rates: ~0.1-1% per gate
- No error correction at scale
- Useful for specific algorithms only

Requirements for Breaking RSA-2048:

- Estimated: ~4,000-10,000 logical qubits
- Each logical qubit needs ~1,000-10,000 physical qubits (error correction)
- Total: Millions of physical qubits
- Gate fidelity: >99.9% needed

Timeline Estimates (Expert Surveys):

Cryptographically Relevant Quantum Computer (CRQC):

Conservative estimate:	15-30 years
Moderate estimate:	10-15 years
Aggressive estimate:	5-10 years
"Never" (some skeptics):	Fundamental barriers

Key Milestones to Watch:

1. Logical qubit demonstration at scale
2. Error correction threshold achievement
3. >100 logical qubit systems
4. First factoring of meaningful numbers
5. First discrete log computations

Recommendations:

1. Begin transition NOW (harvest-now-decrypt-later)
2. Key exchange highest priority
3. Plan for algorithm agility
4. Monitor quantum computing progress
5. Don't wait for "perfect" quantum computer

10.2.4 Exercises

Exercise 10.2.1: Research Maturity Assessment *(Intermediate)*

Create a risk assessment framework for evaluating emerging PQC algorithms:

Assessment Criteria:

=====

For each emerging algorithm family, evaluate:

1. Years of public cryptanalysis
2. Number of independent implementations
3. Formal security proofs
4. Real-world deployment experience
5. NIST/academic endorsement status

Solution:

```
/* Research maturity scoring system */
typedef struct {
    const char *algorithm_family;
    int years_cryptanalysis; /* 0-50 years */
    int independent_implementations; /* 0-20 */
    int formal_proofs; /* 0-3: none, partial, complete */
    int deployment_experience; /* 0-3: none, limited, widespread */
    int standardization_status; /* 0-3: none, draft, standard */
} maturity_score_t;

int calculate_maturity(const maturity_score_t *score) {
    /* Weighted scoring (total max = 100) */
    int total = 0;
    total += (score->years_cryptanalysis > 10 ? 30 : score->years_cryptanalysis * 3);
    total += (score->independent_implementations > 5 ? 20 : score->independent_implementations * 4);
    total += score->formal_proofs * 10;
    total += score->deployment_experience * 10;
    total += score->standardization_status * 10;
    return total;
}

/* Example scores (as of 2025) */
```

```
static const maturity_score_t examples[] = {
    {"ML-KEM (Lattice)", 10, 15, 3, 2, 3},      /* Score: ~95 */
    {"SPHINCS+/SLH-DSA", 8, 10, 3, 1, 3},      /* Score: ~85 */
    {"CSIDH (Isogeny)", 5, 5, 2, 0, 0},        /* Score: ~45 */
    {"Rainbow (Multivariate)", 15, 8, 2, 0, 0}, /* Score: ~60 (broken) */
};
```

Exercise 10.2.2: Quantum Timeline Risk Analysis *(Intermediate)*

Design a threat model considering different quantum computing timeline scenarios:

Solution:

```
/* Quantum timeline risk assessment */
typedef struct {
    const char *scenario;
    int years_to_crq;
    const char *immediate_action;
    const char *data_at_risk;
} timeline_scenario_t;

static const timeline_scenario_t scenarios[] = {
    {
        "Aggressive (5-10 years)",
        7,
        "CRITICAL: Deploy hybrid NOW, emergency PKI transition",
        "Medical records, financial data, state secrets, any data with 10+ year sensitivity"
    },
    {
        "Moderate (10-15 years)",
        12,
        "URGENT: Complete key exchange migration, start auth transition",
        "Legal documents, intellectual property, long-term contracts"
    },
    {
        "Conservative (15-30 years)",
        20,
        "IMPORTANT: Plan methodical transition, build agility",
        "Historical archives, identity credentials"
    },
    {
        "Never/Unknown",
        999,
        "BASELINE: Still migrate (harvest-now-decrypt-later threat)",
        "All encrypted data transmitted today is at risk regardless"
    }
};

/* Key insight: Even the "never" scenario requires action due to
store-now-decrypt-later attacks on data with long-term value */
```

Exercise 10.2.3: Post-SIDH Research Tracking *(Beginner)*

Track the current state of isogeny-based cryptography research:

Solution:

Isogeny Research Tracker (Post-2022):

=====

BROKEN SCHEMES:

- SIDH/SIKE: Castryck-Decru attack (July 2022)
- Attack uses auxiliary points to recover private isogeny

PROMISING DIRECTIONS:

1. CSIDH (Commutative SIDH)
 - Different mathematical structure
 - Smaller keys but slower
 - Active research, not yet standardized
2. SQISign
 - Based on quaternion algebras
 - Very short signatures (~200 bytes)
 - Complex implementation, active development
3. Isogeny-based encryption without auxiliary points
 - Avoids SIDH attack vector
 - Early research stage

RECOMMENDATION:

- Do NOT use isogeny-based crypto in production
- Monitor research but don't depend on it
- Use lattice/hash-based standards (ML-KEM, SLH-DSA)

Unit 10.2 Summary

The PQC research landscape continues to evolve:

- **Isogeny-based:** Active research post-SIDH break
- **Alternative approaches:** Various maturity levels
- **Quantum timeline:** Uncertain but plan conservatively
- **Key lesson:** Cryptographic agility is essential

What's Next: Unit 10.3 covers cryptographic agility strategies.

Unit 10.3: Cryptographic Agility

Prerequisites: Modules 4–9 (Volume 1 for 4, Volume 2 for 9) (Complete PQC algorithm and protocol coverage)

Estimated Time: 2–3 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Design systems for algorithm replacement** without major refactoring
2. **Implement negotiation mechanisms** for multiple algorithm support
3. **Plan graceful transitions** between cryptographic generations
4. **Test and validate** cryptographic agility implementations

10.3.1 Agility Architecture

```
/* crypto_agility.h - Cryptographic Agility Framework */

#ifndef CRYPTO_AGILITY_H
#define CRYPTO_AGILITY_H

#include <stdint.h>
#include <stddef.h>

/*
 * Cryptographic Agility Design Principles
 * =====
 *
 * 1. Algorithm Abstraction
 *    - Use algorithm identifiers, not hardcoded implementations
 *    - Provider-based architecture (like OpenSSL 3.0)
 *
 * 2. Negotiation Support
 *    - Support multiple algorithms simultaneously
 *    - Graceful fallback mechanisms
 *    - Version and capability advertisement
 *
 * 3. Configuration-Driven
 *    - Algorithm selection via configuration
 *    - No code changes for algorithm updates
 *    - Policy enforcement capability
 *
 * 4. Hybrid Support
 *    - Combine multiple algorithms easily
 *    - Flexible combiner functions
 *
 * 5. Future-Proofing
 *    - Extensible identifier space
 *    - Backward compatibility considerations
 *    - Deprecation workflow support
 */

/* Algorithm identifier structure */
typedef struct {
    uint16_t family;      /* Algorithm family (lattice, hash, code, etc.) */
    uint16_t algorithm;   /* Specific algorithm within family */
    uint16_t variant;     /* Parameter set / variant */
    uint16_t version;     /* Version for updates */
} algorithm_id_t;

/* Algorithm families */
```

```

#define ALG_FAMILY_CLASSICAL    0x0001
#define ALG_FAMILY_LATTICE      0x0002
#define ALG_FAMILY_HASH_BASED   0x0003
#define ALG_FAMILY_CODE_BASED   0x0004
#define ALG_FAMILY_ISOGENY      0x0005 /* Research only */
#define ALG_FAMILY_HYBRID       0x00FF

/* Cryptographic operation types */
typedef enum {
    CRYPTO_OP_SIGN,
    CRYPTO_OP_VERIFY,
    CRYPTO_OP_ENCRYPT,
    CRYPTO_OP_DECRYPT,
    CRYPTO_OP_KEYGEN,
    CRYPTO_OP_ENCAPS,
    CRYPTO_OP_DECAPS,
} crypto_operation_t;

/* Algorithm capability descriptor */
typedef struct {
    algorithm_id_t id;
    const char *name;
    uint32_t operations; /* Bitmask of supported operations */
    size_t public_key_size;
    size_t private_key_size;
    size_t signature_size; /* For signature algorithms */
    size_t ciphertext_size; /* For KEMs */
    int security_level; /* NIST level 1-5 */
    int is_deprecated;
    int is_experimental;
} algorithm_capability_t;

/* Provider interface */
typedef struct crypto_provider {
    const char *name;
    int (*init)(void);
    void (*cleanup)(void);
    int (*supports)(algorithm_id_t id);
    const algorithm_capability_t *(*get_capability)(algorithm_id_t id);

    /* Generic operations */
    int (*keygen)(algorithm_id_t id, uint8_t *pk, size_t *pk_len,
                  uint8_t *sk, size_t *sk_len);
    int (*sign)(algorithm_id_t id, uint8_t *sig, size_t *sig_len,
                const uint8_t *msg, size_t msg_len,
                const uint8_t *sk, size_t sk_len);
    int (*verify)(algorithm_id_t id,
                  const uint8_t *sig, size_t sig_len,
                  const uint8_t *msg, size_t msg_len,
                  const uint8_t *pk, size_t pk_len);
    int (*encaps)(algorithm_id_t id, uint8_t *ct, size_t *ct_len,
                  uint8_t *ss, size_t *ss_len,
                  const uint8_t *pk, size_t pk_len);
    int (*decaps)(algorithm_id_t id, uint8_t *ss, size_t *ss_len,
                  const uint8_t *ct, size_t ct_len,
                  const uint8_t *sk, size_t sk_len);

    struct crypto_provider *next; /* Linked list */
} crypto_provider_t;

```

```

/* Global agility context */
typedef struct {
    crypto_provider_t *providers;
    algorithm_id_t *preferred_signature;
    size_t num_preferred_sig;
    algorithm_id_t *preferred_kem;
    size_t num_preferred_kem;
    int allow_deprecated;
    int allow_experimental;
} agility_context_t;

/* Function prototypes */
int agility_init(agility_context_t *ctx);
int agility_register_provider(agility_context_t *ctx, crypto_provider_t *provider);
int agility_set_preference(agility_context_t *ctx, crypto_operation_t op,
                          const algorithm_id_t *algorithms, size_t count);
algorithm_id_t agility_negotiate(agility_context_t *ctx, crypto_operation_t op,
                                const algorithm_id_t *peer_supported, size_t count);
int agility_perform(agility_context_t *ctx, algorithm_id_t id,
                   crypto_operation_t op, void *params);
void agility_cleanup(agility_context_t *ctx);

#endif /* CRYPTO_AGILITY_H */

```

10.3.2 Implementation

```

/* crypto_agility.c - Implementation */

#include "crypto_agility.h"
#include <string.h>
#include <stdlib.h>
#include <stdio.h>

int agility_init(agility_context_t *ctx) {
    if (!ctx) return -1;

    memset(ctx, 0, sizeof(agility_context_t));
    return 0;
}

int agility_register_provider(agility_context_t *ctx, crypto_provider_t *provider) {
    if (!ctx || !provider) return -1;

    /* Initialize provider */
    if (provider->init && provider->init() != 0) {
        return -1;
    }

    /* Add to linked list */
    provider->next = ctx->providers;
    ctx->providers = provider;

    return 0;
}

/* Find provider that supports given algorithm */
static crypto_provider_t *find_provider(agility_context_t *ctx, algorithm_id_t id) {
    crypto_provider_t *p = ctx->providers;

```

```

    while (p) {
        if (p->supports && p->supports(id)) {
            return p;
        }
        p = p->next;
    }

    return NULL;
}

int agility_set_preference(agility_context_t *ctx, crypto_operation_t op,
                        const algorithm_id_t *algorithms, size_t count) {
    if (!ctx || !algorithms || count == 0) return -1;

    algorithm_id_t **pref;
    size_t *pref_count;

    switch (op) {
        case CRYPTO_OP_SIGN:
        case CRYPTO_OP_VERIFY:
            pref = &ctx->preferred_signature;
            pref_count = &ctx->num_preferred_sig;
            break;
        case CRYPTO_OP_ENCAPS:
        case CRYPTO_OP_DECAPS:
            pref = &ctx->preferred_kem;
            pref_count = &ctx->num_preferred_kem;
            break;
        default:
            return -1;
    }

    /* Free existing preferences */
    free(*pref);

    /* Copy new preferences */
    *pref = malloc(count * sizeof(algorithm_id_t));
    if (!*pref) return -1;

    memcpy(*pref, algorithms, count * sizeof(algorithm_id_t));
    *pref_count = count;

    return 0;
}

/* Negotiate best mutually supported algorithm */
algorithm_id_t agility_negotiate(agility_context_t *ctx, crypto_operation_t op,
                                const algorithm_id_t *peer_supported, size_t count) {
    algorithm_id_t result = {0, 0, 0, 0};

    if (!ctx || !peer_supported || count == 0) {
        return result;
    }

    algorithm_id_t *our_pref;
    size_t our_count;

    switch (op) {
        case CRYPTO_OP_SIGN:

```

```

        case CRYPTO_OP_VERIFY:
            our_pref = ctx->preferred_signature;
            our_count = ctx->num_preferred_sig;
            break;
        case CRYPTO_OP_ENCAPS:
        case CRYPTO_OP_DECAPS:
            our_pref = ctx->preferred_kem;
            our_count = ctx->num_preferred_kem;
            break;
        default:
            return result;
    }

    /* Our preference order */
    for (size_t i = 0; i < our_count; i++) {
        algorithm_id_t candidate = our_pref[i];

        /* Check if we have a provider */
        crypto_provider_t *provider = find_provider(ctx, candidate);
        if (!provider) continue;

        /* Check capability */
        const algorithm_capability_t *cap = provider->get_capability(candidate);
        if (!cap) continue;

        if (cap->is_deprecated && !ctx->allow_deprecated) continue;
        if (cap->is_experimental && !ctx->allow_experimental) continue;

        /* Check if peer supports */
        for (size_t j = 0; j < count; j++) {
            if (memcmp(&candidate, &peer_supported[j], sizeof(algorithm_id_t)) == 0) {
                return candidate; /* Found match */
            }
        }
    }

    return result; /* No match found */
}

/* Perform cryptographic operation */
int agility_perform(agility_context_t *ctx, algorithm_id_t id,
                   crypto_operation_t op, void *params) {
    if (!ctx || !params) return -1;

    crypto_provider_t *provider = find_provider(ctx, id);
    if (!provider) {
        return -1; /* No provider for algorithm */
    }

    /* Check capability and policy */
    const algorithm_capability_t *cap = provider->get_capability(id);
    if (cap) {
        if (cap->is_deprecated && !ctx->allow_deprecated) {
            fprintf(stderr, "Algorithm %s is deprecated\n", cap->name);
            return -1;
        }
    }

    /* Dispatch to provider */
    /* In real implementation, params would be a union/struct with operation-specific data

```

```

*/
    return 0; /* Success */
}

void agility_cleanup(agility_context_t *ctx) {
    if (!ctx) return;

    /* Cleanup providers */
    crypto_provider_t *p = ctx->providers;
    while (p) {
        if (p->cleanup) p->cleanup();
        p = p->next;
    }

    free(ctx->preferred_signature);
    free(ctx->preferred_kem);
    memset(ctx, 0, sizeof(agility_context_t));
}

```

The companion program `unit_10_1_crypto_agility.c` (in `source_code/module_10_future/`) is a self-contained distillation of this pattern: a small algorithm registry queried by capability policy, plus a swap-out scenario showing how to fail over to an alternate algorithm with no application changes.

Expected Output:

```

Cryptographic Agility Registry Demo
=====

Registry contents:
  1) ML-KEM-768
  2) ML-DSA-65
  3) SLH-DSA-128s
  4) X-Wing
  5) FN-DSA-512
  6) HQC-128

-- Policy: need FIPS 140-3 KEM --
Algorithm:   ML-KEM-768 (id=1)
Public key:  1184 bytes
Secret key:  2400 bytes
Sig/ct:      1088 bytes
Capabilities: KEM FIPS140-3

-- Policy: need FIPS 140-3 signature --
Algorithm:   ML-DSA-65 (id=2)
Public key:  1952 bytes
Secret key:  4032 bytes
Sig/ct:      3309 bytes
Capabilities: SIGN FIPS140-3

-- Policy: need hybrid KEM --
Algorithm:   X-Wing (id=4)
Public key:  1216 bytes

```

Secret key: 2432 bytes
Sig/ct: 1120 bytes
Capabilities: KEM HYBRID

Swap-out scenario: if ML-KEM-768 is broken, reorder the registry
so X-Wing or HQC-128 is selected first – no application changes needed.

10.3.3 Configuration-Driven Agility

```
/* agility_config.c - Configuration-Based Algorithm Selection */

#include <stdio.h>
#include <string.h>
#include <stdlib.h>

/*
 * Example configuration file format (YAML-like):
 *
 * crypto:
 *   signature:
 *     preferred:
 *       - ml-dsa-65
 *       - ed25519-ml-dsa-65 # hybrid
 *       - ed25519          # fallback
 *     deprecated:
 *       - rsa-2048
 *     forbidden:
 *       - sha1-rsa
 *
 *   kem:
 *     preferred:
 *       - ml-kem-768
 *       - x25519-ml-kem-768 # hybrid
 *       - x25519          # fallback
 *
 *   policy:
 *     allow_classical_only: false
 *     require_pq: true
 *     hybrid_mode: preferred
 */

typedef struct {
    char **signature_preferred;
    size_t num_sig_preferred;
    char **signature_deprecated;
    size_t num_sig_deprecated;
    char **signature_forbidden;
    size_t num_sig_forbidden;

    char **kem_preferred;
    size_t num_kem_preferred;

    int allow_classical_only;
    int require_pq;
    int hybrid_mode; /* 0=disabled, 1=allowed, 2=preferred, 3=required */
} crypto_config_t;

/* Parse configuration from string (simplified) */
```

```

int parse_crypto_config(const char *config_str, crypto_config_t *config) {
    /* In real implementation, parse YAML/JSON/TOML */
    /* This is a simplified example */

    memset(config, 0, sizeof(crypto_config_t));

    /* Set defaults for post-quantum readiness */
    config->hybrid_mode = 2; /* Preferred */
    config->require_pq = 1;
    config->allow_classical_only = 0;

    /* Example: hardcoded defaults */
    config->num_sig_preferred = 3;
    config->signature_preferred = malloc(3 * sizeof(char *));
    config->signature_preferred[0] = strdup("ml-dsa-65");
    config->signature_preferred[1] = strdup("ed25519-ml-dsa-65");
    config->signature_preferred[2] = strdup("ed25519");

    config->num_kem_preferred = 3;
    config->kem_preferred = malloc(3 * sizeof(char *));
    config->kem_preferred[0] = strdup("ml-kem-768");
    config->kem_preferred[1] = strdup("x25519-ml-kem-768");
    config->kem_preferred[2] = strdup("x25519");

    return 0;
}

/* Check if algorithm is allowed by policy */
int is_algorithm_allowed(const crypto_config_t *config, const char *algorithm) {
    /* Check forbidden list */
    for (size_t i = 0; i < config->num_sig_forbidden; i++) {
        if (strcmp(config->signature_forbidden[i], algorithm) == 0) {
            return 0; /* Forbidden */
        }
    }

    /* Check PQ requirement */
    if (config->require_pq) {
        /* Check if algorithm contains PQ component */
        if (strstr(algorithm, "ml-") == NULL &&
            strstr(algorithm, "slh-") == NULL &&
            strstr(algorithm, "falcon") == NULL) {
            /* No PQ component - check if hybrid mode allows */
            if (config->hybrid_mode < 1) {
                return 0; /* Classical not allowed */
            }
        }
    }

    return 1; /* Allowed */
}

/* Print current configuration */
void print_crypto_config(const crypto_config_t *config) {
    printf("Cryptographic Configuration\n");
    printf("=====\n\n");

    printf("Signature Algorithms (preference order):\n");
    for (size_t i = 0; i < config->num_sig_preferred; i++) {
        printf("  %zu. %s\n", i + 1, config->signature_preferred[i]);
    }
}

```



```

    }

    printf("\nKEM Algorithms (preference order):\n");
    for (size_t i = 0; i < config->num_kem_preferred; i++) {
        printf("  %zu. %s\n", i + 1, config->kem_preferred[i]);
    }

    printf("\nPolicy:\n");
    printf("  Require PQ: %s\n", config->require_pq ? "yes" : "no");
    printf("  Allow classical only: %s\n", config->allow_classical_only ? "yes" : "no");
    printf("  Hybrid mode: %d\n", config->hybrid_mode);
}

void free_crypto_config(crypto_config_t *config) {
    for (size_t i = 0; i < config->num_sig_preferred; i++) {
        free(config->signature_preferred[i]);
    }
    free(config->signature_preferred);

    for (size_t i = 0; i < config->num_sig_deprecated; i++) {
        free(config->signature_deprecated[i]);
    }
    free(config->signature_deprecated);

    for (size_t i = 0; i < config->num_kem_preferred; i++) {
        free(config->kem_preferred[i]);
    }
    free(config->kem_preferred);

    memset(config, 0, sizeof(crypto_config_t));
}

```

10.3.4 Exercises

Exercise 10.3.1: Agility Provider Implementation *(Advanced)*

Implement a mock provider that demonstrates the agility interface:

```

/* Exercise: Implement a liboqs provider for the agility framework */
/* Fill in the TODO sections */

crypto_provider_t *create_liboqs_provider(void) {
    /* TODO: Allocate and initialize provider structure */
    /* TODO: Register all supported OQS algorithms */
    /* TODO: Implement keygen, sign, verify, encaps, decaps */
    return NULL;
}

```

Solution:

```

/* liboqs agility provider */
#include <oqs/oqs.h>

static int oqs_init(void) {
    /* liboqs initializes automatically */
    return OQS_SUCCESS == OQS_init() ? 0 : -1;
}

static void oqs_cleanup(void) {
    /* Cleanup is automatic with liboqs */
}

static int oqs_supports(algorithm_id_t id) {
    if (id.family != ALG_FAMILY_LATTICE &&
        id.family != ALG_FAMILY_HASH_BASED) {
        return 0;
    }
    /* Check if algorithm is enabled in liboqs build */
    if (id.algorithm == 0x0001) { /* ML-KEM */
        return OQS_KEM_alg_is_enabled(OQS_KEM_alg_ml_kem_768);
    }
    if (id.algorithm == 0x0002) { /* ML-DSA */
        return OQS_SIG_alg_is_enabled(OQS_SIG_alg_ml_dsa_65);
    }
    return 0;
}

static int oqs_keygen(algorithm_id_t id, uint8_t *pk, size_t *pk_len,
                     uint8_t *sk, size_t *sk_len) {
    if (id.family == ALG_FAMILY_LATTICE && id.algorithm == 0x0001) {
        OQS_KEM *kem = OQS_KEM_new(OQS_KEM_alg_ml_kem_768);
        if (!kem) return -1;
        int ret = OQS_KEM_keypair(kem, pk, sk);
        *pk_len = kem->length_public_key;
        *sk_len = kem->length_secret_key;
        OQS_KEM_free(kem);
        return ret == OQS_SUCCESS ? 0 : -1;
    }
    return -1;
}

crypto_provider_t *create_liboqs_provider(void) {
    crypto_provider_t *p = calloc(1, sizeof(crypto_provider_t));
    if (!p) return NULL;

    p->name = "liboqs";
    p->init = oqs_init;
    p->cleanup = oqs_cleanup;
    p->supports = oqs_supports;
    p->keygen = oqs_keygen;
    /* ... additional function pointers ... */

    return p;
}

```

Exercise 10.3.2: Algorithm Negotiation Protocol *(Advanced)*

Design a negotiation protocol for two parties to agree on PQC algorithms:

Solution:

```
/* Algorithm negotiation protocol */
typedef struct {
    algorithm_id_t *supported;    /* Algorithms we support */
    size_t num_supported;
    algorithm_id_t *preferred;    /* Our preference order */
    size_t num_preferred;
} negotiation_offer_t;

typedef struct {
    algorithm_id_t selected_kem;
    algorithm_id_t selected_sig;
    int success;
} negotiation_result_t;

/*
 * Negotiation Protocol:
 * 1. Initiator sends offer with supported + preferred algorithms
 * 2. Responder intersects with own capabilities
 * 3. Responder selects best match (highest in initiator's preference
 *    that responder supports)
 * 4. Responder sends selection back
 */

negotiation_result_t negotiate_algorithms(
    const negotiation_offer_t *our_offer,
    const negotiation_offer_t *their_offer)
{
    negotiation_result_t result = {0};

    /* Find first algorithm in their preference that we support */
    for (size_t i = 0; i < their_offer->num_preferred; i++) {
        algorithm_id_t candidate = their_offer->preferred[i];

        /* Check if we support it */
        for (size_t j = 0; j < our_offer->num_supported; j++) {
            if (memcmp(&candidate, &our_offer->supported[j],
                sizeof(algorithm_id_t)) == 0) {
                /* Found match - determine if KEM or signature */
                if (is_kem_algorithm(candidate)) {
                    result.selected_kem = candidate;
                } else {
                    result.selected_sig = candidate;
                }
                break;
            }
        }
    }

    result.success = (result.selected_kem.algorithm != 0 &&
        result.selected_sig.algorithm != 0);
    return result;
}
```

Exercise 10.3.3: Deprecation Workflow *(Intermediate)*

Implement a deprecation tracking system with logging:

Solution:

```
/* Deprecation tracking and warning system */
#include <time.h>

typedef struct {
    algorithm_id_t id;
    time_t deprecated_date;
    time_t sunset_date;          /* After this, refuse to use */
    const char *reason;
    int warning_issued;
} deprecation_entry_t;

static deprecation_entry_t deprecation_table[] = {
    {{ALG_FAMILY_CLASSICAL, 0x0001, 0, 0}, /* RSA-2048 */
     1704067200, /* Jan 1, 2024 */
     1767225600, /* Jan 1, 2026 */
     "Quantum vulnerable, migrate to ML-DSA", 0},

    {{ALG_FAMILY_CLASSICAL, 0x0002, 0, 0}, /* ECDSA P-256 */
     1704067200,
     1798761600, /* Jan 1, 2027 */
     "Quantum vulnerable, migrate to ML-DSA or hybrid", 0},
};

int check_algorithm_status(algorithm_id_t id) {
    time_t now = time(NULL);

    for (size_t i = 0; i < sizeof(deprecation_table)/sizeof(deprecation_table[0]); i++) {
        if (memcmp(&id, &deprecation_table[i].id, sizeof(algorithm_id_t)) == 0) {
            if (now >= deprecation_table[i].sunset_date) {
                /* Algorithm is sunset - refuse to use */
                fprintf(stderr, "ERROR: Algorithm has been sunset: %s\n",
                        deprecation_table[i].reason);
                return -1; /* Refuse */
            }
            if (now >= deprecation_table[i].deprecated_date &&
                !deprecation_table[i].warning_issued) {
                /* Log warning once */
                fprintf(stderr, "WARNING: Deprecated algorithm in use: %s\n",
                        deprecation_table[i].reason);
                deprecation_table[i].warning_issued = 1;
            }
            return 0; /* Allow with warning */
        }
    }
    return 0; /* Not in deprecation table, OK */
}
```

Unit 10.3 Summary

Cryptographic agility requires:

1. **Abstraction:** Separate algorithm selection from implementation
2. **Negotiation:** Support multiple algorithms with graceful fallback
3. **Configuration:** External policy without code changes

4. **Monitoring:** Track algorithm usage and deprecation

What's Next: Unit 10.4 provides a comprehensive implementation best practices summary.

Unit 10.4: Implementation Best Practices Summary

Prerequisites: All previous modules (comprehensive PQC knowledge)

Estimated Time: 1-2 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Apply security best practices** across all PQC implementations
2. **Avoid common implementation pitfalls**
3. **Design for long-term maintainability**
4. **Plan comprehensive testing strategies**

10.4.1 Security Checklist

Post-Quantum Implementation Security Checklist

=====

- [] Memory Safety
 - [] Use secure memory allocation for secrets
 - [] Zero memory after use (volatile, compiler barriers)
 - [] Lock secret pages in memory (mlock)
 - [] Prevent memory dumps (MADV_DONTDUMP)
- [] Side-Channel Resistance
 - [] Constant-time comparisons
 - [] Constant-time table lookups
 - [] No secret-dependent branches
 - [] No secret-dependent memory access patterns
 - [] Consider power analysis (for hardware)
- [] Input Validation
 - [] Validate all key sizes
 - [] Check ciphertext/signature lengths
 - [] Validate algorithm identifiers
 - [] Reject malformed inputs early
- [] Randomness
 - [] Use cryptographic RNG (getrandom, CryptGenRandom)
 - [] Check RNG return values
 - [] Reseed periodically if needed

- [] Hardware RNG where available
- [] Key Management
 - [] Secure key generation
 - [] Protected key storage
 - [] Key rotation procedures
 - [] Secure key destruction
- [] Hybrid Construction
 - [] Both components must succeed
 - [] Proper combiner function
 - [] Independent random values
 - [] Consistent error handling
- [] Error Handling
 - [] Don't leak information via errors
 - [] Consistent error timing
 - [] Secure cleanup on failure
 - [] Log for debugging (not secrets)

10.4.2 Common Pitfalls

```

/* common_pitfalls.c - What NOT to Do */

/* PITFALL 1: Non-constant-time comparison */
/* BAD */
int bad_compare(const uint8_t *a, const uint8_t *b, size_t len) {
    for (size_t i = 0; i < len; i++) {
        if (a[i] != b[i]) return 0; /* Early exit leaks timing! */
    }
    return 1;
}

/* GOOD */
int good_compare(const uint8_t *a, const uint8_t *b, size_t len) {
    volatile uint8_t diff = 0;
    for (size_t i = 0; i < len; i++) {
        diff |= a[i] ^ b[i];
    }
    return diff == 0;
}

/* PITFALL 2: Insecure memory clearing */
/* BAD - compiler may optimize away */
void bad_clear(void *ptr, size_t len) {
    memset(ptr, 0, len); /* May be removed by optimizer! */
}

/* GOOD */
void good_clear(void *ptr, size_t len) {
    volatile uint8_t *p = (volatile uint8_t *)ptr;
    while (len--) *p++ = 0;
    __asm__ __volatile__("" ::: "memory");
}

/* PITFALL 3: Ignoring hybrid combiner output */

```

```

/* BAD */
int bad_hybrid_verify(/* ... */) {
    int classical_ok = verify_classical(/* ... */);
    int pq_ok = verify_pq(/* ... */);
    return classical_ok; /* Ignores PQ result! */
}

/* GOOD */
int good_hybrid_verify(/* ... */) {
    int classical_ok = verify_classical(/* ... */);
    int pq_ok = verify_pq(/* ... */);
    return classical_ok && pq_ok; /* Both must verify */
}

/* PITFALL 4: Reusing nonces/random values */
/* BAD */
void bad_sign_multiple(/* ... */) {
    uint8_t random[32];
    generate_random(random, 32);

    for (int i = 0; i < count; i++) {
        sign_with_random(messages[i], random); /* Same random! */
    }
}

/* GOOD */
void good_sign_multiple(/* ... */) {
    for (int i = 0; i < count; i++) {
        uint8_t random[32];
        generate_random(random, 32); /* Fresh random each time */
        sign_with_random(messages[i], random);
        secure_clear(random, 32);
    }
}

/* PITFALL 5: Buffer size assumptions */
/* BAD */
void bad_sign(uint8_t sig[64], /* ... */) {
    /* Assumes signature is always 64 bytes - wrong for PQ! */
    mldsa65_sign(sig, /* ... */); /* Needs 3309 bytes! */
}

/* GOOD */
int good_sign(uint8_t *sig, size_t *sig_len, /* ... */) {
    if (*sig_len < MLDSA65_SIG_SIZE) {
        *sig_len = MLDSA65_SIG_SIZE;
        return -1;
    }
    return mldsa65_sign(sig, sig_len, /* ... */);
}

```

10.4.3 Testing Strategy

```
/* testing_strategy.c - Comprehensive PQ Testing */

#include <stdio.h>
#include <string.h>
#include <time.h>

/*
 * Testing Layers for PQ Implementations
 * =====
 *
 * 1. Unit Tests
 *   - Individual function correctness
 *   - Edge cases and error handling
 *   - Memory management
 *
 * 2. Integration Tests
 *   - Full cryptographic operations
 *   - Protocol-level testing
 *   - Interoperability with other implementations
 *
 * 3. Known Answer Tests (KAT)
 *   - Verify against reference vectors
 *   - Ensure deterministic behavior
 *   - Catch implementation bugs
 *
 * 4. Fuzz Testing
 *   - Random/malformed inputs
 *   - Memory safety verification
 *   - Crash detection
 *
 * 5. Performance Testing
 *   - Benchmark critical operations
 *   - Memory usage profiling
 *   - Scalability testing
 *
 * 6. Side-Channel Testing
 *   - Timing analysis
 *   - Power analysis (hardware)
 *   - Cache timing attacks
 */

typedef struct {
    int total_tests;
    int passed;
    int failed;
    double total_time_ms;
} test_results_t;

/* Test framework helper */
#define TEST_ASSERT(cond, msg) do { \
    if (!(cond)) { \
        printf("FAIL: %s\n", msg); \
        return 0; \
    } \
} while(0)

/* Example: Basic correctness test */
```



```

int test_sign_verify_roundtrip(void) {
    printf("Testing sign/verify roundtrip...\n");

    /* Generate key pair */
    uint8_t pk[2000], sk[5000];
    size_t pk_len = sizeof(pk), sk_len = sizeof(sk);

    int ret = mldsa65_keygen(pk, &pk_len, sk, &sk_len);
    TEST_ASSERT(ret == 0, "Key generation failed");

    /* Sign message */
    const char *message = "Test message for PQ signature";
    uint8_t sig[4000];
    size_t sig_len = sizeof(sig);

    ret = mldsa65_sign(sig, &sig_len,
                       (const uint8_t *)message, strlen(message),
                       sk, sk_len);
    TEST_ASSERT(ret == 0, "Signing failed");

    /* Verify signature */
    ret = mldsa65_verify(sig, sig_len,
                        (const uint8_t *)message, strlen(message),
                        pk, pk_len);
    TEST_ASSERT(ret == 0, "Verification failed");

    /* Verify with wrong message should fail */
    const char *wrong_message = "Wrong message";
    ret = mldsa65_verify(sig, sig_len,
                        (const uint8_t *)wrong_message, strlen(wrong_message),
                        pk, pk_len);
    TEST_ASSERT(ret != 0, "Verification should fail for wrong message");

    printf(" PASS\n");
    return 1;
}

/* Example: Known Answer Test */
typedef struct {
    const uint8_t *seed;
    size_t seed_len;
    const uint8_t *expected_pk;
    size_t expected_pk_len;
    const uint8_t *message;
    size_t message_len;
    const uint8_t *expected_sig;
    size_t expected_sig_len;
} kat_vector_t;

int test_known_answer(const kat_vector_t *kat) {
    printf("Testing against known answer vector...\n");

    /* Generate key pair from seed (deterministic) */
    uint8_t pk[2000], sk[5000];
    size_t pk_len = sizeof(pk), sk_len = sizeof(sk);

    int ret = mldsa65_keygen_deterministic(pk, &pk_len, sk, &sk_len,
                                           kat->seed, kat->seed_len);
    TEST_ASSERT(ret == 0, "Deterministic keygen failed");
}

```

```

/* Compare public key */
TEST_ASSERT(pk_len == kat->expected_pk_len, "Public key length mismatch");
TEST_ASSERT(memcmp(pk, kat->expected_pk, pk_len) == 0, "Public key mismatch");

/* Generate deterministic signature */
uint8_t sig[4000];
size_t sig_len = sizeof(sig);

ret = mldsa65_sign_deterministic(sig, &sig_len,
                                kat->message, kat->message_len,
                                sk, sk_len);
TEST_ASSERT(ret == 0, "Deterministic signing failed");

/* Compare signature */
TEST_ASSERT(sig_len == kat->expected_sig_len, "Signature length mismatch");
TEST_ASSERT(memcmp(sig, kat->expected_sig, sig_len) == 0, "Signature mismatch");

printf(" PASS\n");
return 1;
}

/* Example: Timing consistency test */
int test_constant_time_verify(void) {
    printf("Testing constant-time verification...\n");

    /* Generate key pair */
    uint8_t pk[2000], sk[5000];
    size_t pk_len = sizeof(pk), sk_len = sizeof(sk);
    mldsa65_keygen(pk, &pk_len, sk, &sk_len);

    /* Sign a message */
    const char *message = "Timing test message";
    uint8_t sig[4000];
    size_t sig_len = sizeof(sig);
    mldsa65_sign(sig, &sig_len, (const uint8_t *)message, strlen(message), sk, sk_len);

    /* Measure valid verification time */
    clock_t start = clock();
    for (int i = 0; i < 1000; i++) {
        mldsa65_verify(sig, sig_len, (const uint8_t *)message, strlen(message), pk,
pk_len);
    }
    double valid_time = (double)(clock() - start) / CLOCKS_PER_SEC;

    /* Corrupt signature and measure invalid verification time */
    sig[0] ^= 0xFF; /* Corrupt first byte */
    start = clock();
    for (int i = 0; i < 1000; i++) {
        mldsa65_verify(sig, sig_len, (const uint8_t *)message, strlen(message), pk,
pk_len);
    }
    double invalid_time = (double)(clock() - start) / CLOCKS_PER_SEC;

    /* Times should be similar (within 10%) */
    double ratio = valid_time / invalid_time;
    printf(" Valid/Invalid time ratio: %.3f\n", ratio);

    TEST_ASSERT(ratio > 0.9 && ratio < 1.1,
                "Timing difference detected - possible side channel");
}

```

```

    printf(" PASS\n");
    return 1;
}

/* Run all tests */
void run_test_suite(void) {
    test_results_t results = {0, 0, 0, 0.0};

    printf("\n=== Post-Quantum Cryptography Test Suite ===\n\n");

    clock_t start = clock();

    results.total_tests++;
    if (test_sign_verify_roundtrip()) results.passed++;
    else results.failed++;

    results.total_tests++;
    if (test_constant_time_verify()) results.passed++;
    else results.failed++;

    /* Add more tests... */

    results.total_time_ms = (double)(clock() - start) / CLOCKS_PER_SEC * 1000;

    printf("\n=== Test Results ===\n");
    printf("Total: %d, Passed: %d, Failed: %d\n",
        results.total_tests, results.passed, results.failed);
    printf("Time: %.2f ms\n", results.total_time_ms);
}

/* External function declarations - these would be provided by liboqs or
 * a PQC library. In practice, use the liboqs API directly:
 * OQS_SIG *sig = OQS_SIG_new(OQS_SIG_alg_ml_dsa_65);
 * OQS_SIG_keypair(sig, pk, sk);
 * OQS_SIG_sign(sig, signature, &sig_len, msg, msg_len, sk);
 * OQS_SIG_verify(sig, msg, msg_len, signature, sig_len, pk);
 */
extern int ml_dsa65_keygen(uint8_t *pk, size_t *pk_len, uint8_t *sk, size_t *sk_len);
extern int ml_dsa65_keygen_deterministic(uint8_t *pk, size_t *pk_len,
                                         uint8_t *sk, size_t *sk_len,
                                         const uint8_t *seed, size_t seed_len);
extern int ml_dsa65_sign(uint8_t *sig, size_t *sig_len,
                        const uint8_t *msg, size_t msg_len,
                        const uint8_t *sk, size_t sk_len);
extern int ml_dsa65_sign_deterministic(uint8_t *sig, size_t *sig_len,
                                       const uint8_t *msg, size_t msg_len,
                                       const uint8_t *sk, size_t sk_len);
extern int ml_dsa65_verify(const uint8_t *sig, size_t sig_len,
                          const uint8_t *msg, size_t msg_len,
                          const uint8_t *pk, size_t pk_len);

```

10.4.4 Final Recommendations

Post-Quantum Cryptography Implementation Summary

IMMEDIATE ACTIONS:

1. Inventory current cryptographic usage
2. Enable hybrid key exchange where available (TLS, SSH)
3. Plan PKI transition timeline
4. Update threat models for quantum

SHORT-TERM (1-2 years):

1. Deploy hybrid authentication where critical
2. Update code signing infrastructure
3. Train development teams on PQC
4. Establish algorithm agility framework

MEDIUM-TERM (2-5 years):

1. Complete migration of key exchange
2. Transition authentication to hybrid/PQ
3. Update all long-term storage encryption
4. Achieve full cryptographic agility

ONGOING:

1. Monitor NIST standardization
2. Track quantum computing progress
3. Update algorithms as needed
4. Maintain interoperability testing

KEY PRINCIPLES:

- Defense in depth (hybrid constructions)
- Cryptographic agility (easy algorithm swap)
- Conservative security assumptions
- Continuous monitoring and updates
- Start now - quantum computers may come sooner than expected

10.4.5 Exercises

Exercise 10.4.1: Security Audit Checklist *(Intermediate)*

Create a comprehensive security audit checklist for reviewing a PQC implementation:

PQC Implementation Audit Checklist:

Review Area: _____

Reviewer: _____

Date: _____

☐ PASS ☐ FAIL ☐ N/A Each item

Solution:

```

/* Automated security audit framework */
typedef enum {
    AUDIT_PASS = 0,
    AUDIT_FAIL = 1,
    AUDIT_WARNING = 2,
    AUDIT_NA = 3
} audit_result_t;

typedef struct {
    const char *category;
    const char *check;
    audit_result_t (*test_func)(void);
    audit_result_t result;
} audit_item_t;

/* Example audit checks */
audit_result_t check_constant_time_compare(void) {
    /* Verify no early-exit comparisons in crypto code */
    /* Would use static analysis or runtime checks */
    return AUDIT_PASS;
}

audit_result_t check_memory_zeroing(void) {
    /* Verify volatile/barrier usage for secret clearing */
    return AUDIT_PASS;
}

audit_result_t check_rng_quality(void) {
    /* Verify cryptographic RNG usage */
    return AUDIT_PASS;
}

static audit_item_t audit_checklist[] = {
    {"Side Channels", "Constant-time comparisons", check_constant_time_compare, AUDIT_NA},
    {"Side Channels", "No secret-dependent branches", NULL, AUDIT_NA},
    {"Side Channels", "No secret-dependent memory access", NULL, AUDIT_NA},
    {"Memory Safety", "Secure memory zeroing", check_memory_zeroing, AUDIT_NA},
    {"Memory Safety", "Memory locking (mlock)", NULL, AUDIT_NA},
    {"Randomness", "Cryptographic RNG usage", check_rng_quality, AUDIT_NA},
    {"Randomness", "RNG return value checking", NULL, AUDIT_NA},
    {"Key Management", "Secure key storage", NULL, AUDIT_NA},
    {"Key Management", "Key rotation support", NULL, AUDIT_NA},
    {"Hybrid", "Both components verified", NULL, AUDIT_NA},
    {"Hybrid", "Independent random values", NULL, AUDIT_NA},
};

void run_security_audit(void) {
    int passed = 0, failed = 0, warnings = 0;

    printf("=== PQC Security Audit ===\n\n");

    for (size_t i = 0; i < sizeof(audit_checklist)/sizeof(audit_checklist[0]); i++) {
        audit_item_t *item = &audit_checklist[i];
        if (item->test_func) {
            item->result = item->test_func();
        }

        const char *status = "N/A";
        if (item->result == AUDIT_PASS) { status = "PASS"; passed++; }
    }
}

```

```

        else if (item->result == AUDIT_FAIL) { status = "FAIL"; failed++; }
        else if (item->result == AUDIT_WARNING) { status = "WARN"; warnings++; }

        printf("[%s] %s: %s\n", status, item->category, item->check);
    }

    printf("\nSummary: %d passed, %d failed, %d warnings\n", passed, failed, warnings);
}

```

Exercise 10.4.2: Common Pitfall Detection *(Advanced)*

Write a static analysis rule to detect non-constant-time comparisons:

Solution:

```

/* Simple pattern-based pitfall detector */
#include <stdio.h>
#include <string.h>

typedef struct {
    const char *pattern;
    const char *warning;
    int severity; /* 1=info, 2=warning, 3=error */
} pitfall_pattern_t;

static const pitfall_pattern_t patterns[] = {
    /* Timing attacks */
    {"if (a[i] != b[i]) return",
     "Non-constant-time comparison detected - use constant-time compare",
     3},

    {"if (memcmp(",
     "memcmp may leak timing - use constant-time comparison for secrets",
     2},

    {"strcmp(",
     "strcmp may leak timing - use constant-time comparison for secrets",
     2},

    /* Memory safety */
    {"memset(.secret",
     "memset on secrets may be optimized away - use secure_clear()",
     3},

    /* Randomness */
    {"rand()",
     "rand() is not cryptographically secure - use getrandom()",
     3},

    {"srand(",
     "srand/rand is not cryptographically secure",
     3},

    /* Buffer issues */
    {"uint8_t sig[64]",
     "Fixed signature buffer too small for PQ signatures",
     3},

```

```

    {"uint8_t pk[32]",
     "Fixed public key buffer too small for PQ keys",
     3},
};

int check_line_for_pitfalls(const char *line, int line_num, const char *filename) {
    int issues = 0;

    for (size_t i = 0; i < sizeof(patterns)/sizeof(patterns[0]); i++) {
        if (strstr(line, patterns[i].pattern)) {
            const char *sev = patterns[i].severity == 3 ? "ERROR" :
                             patterns[i].severity == 2 ? "WARNING" : "INFO";
            printf("%s:%d: %s: %s\n", filename, line_num, sev, patterns[i].warning);
            issues++;
        }
    }

    return issues;
}

```

Exercise 10.4.3: Migration Readiness Assessment *(Intermediate)*

Create an assessment tool for evaluating PQC migration readiness:

Solution:

```

/* PQC Migration Readiness Assessment */
typedef struct {
    const char *area;
    const char *question;
    int weight;      /* Importance 1-5 */
    int score;       /* 0=no, 1=partial, 2=yes */
} readiness_question_t;

static readiness_question_t assessment[] = {
    /* Infrastructure */
    {"Infrastructure", "Cryptographic inventory completed?", 5, 0},
    {"Infrastructure", "Algorithm agility framework exists?", 4, 0},
    {"Infrastructure", "Library upgrade path identified?", 4, 0},

    /* Key Management */
    {"Key Management", "Key rotation procedures defined?", 5, 0},
    {"Key Management", "HSM PQ support verified?", 4, 0},
    {"Key Management", "Certificate lifecycle updated?", 4, 0},

    /* Development */
    {"Development", "Team trained on PQC?", 3, 0},
    {"Development", "Security testing includes PQ?", 4, 0},
    {"Development", "CI/CD supports PQ testing?", 3, 0},

    /* Operations */
    {"Operations", "Monitoring for deprecated algorithms?", 4, 0},
    {"Operations", "Incident response updated?", 3, 0},
    {"Operations", "Performance baselines established?", 3, 0},
};

void calculate_readiness(void) {
    int total_weight = 0;

```

```

int weighted_score = 0;

printf("=== PQC Migration Readiness Assessment ===\n\n");

const char *current_area = "";
for (size_t i = 0; i < sizeof(assessment)/sizeof(assessment[0]); i++) {
    readiness_question_t *q = &assessment[i];

    if (strcmp(q->area, current_area) != 0) {
        printf("\n%s:\n", q->area);
        current_area = q->area;
    }

    const char *status = q->score == 2 ? "[YES]" :
                        q->score == 1 ? "[PARTIAL]" : "[NO]";
    printf("  %s %s (weight: %d)\n", status, q->question, q->weight);

    total_weight += q->weight * 2; /* Max score per question is 2 */
    weighted_score += q->weight * q->score;
}

float readiness = (float)weighted_score / total_weight * 100;
printf("\n=== Readiness Score: %.1f%% ===\n", readiness);

if (readiness >= 80) {
    printf("Status: READY for PQ migration\n");
} else if (readiness >= 50) {
    printf("Status: PARTIALLY READY - address gaps before full migration\n");
} else {
    printf("Status: NOT READY - significant preparation needed\n");
}
}

```

Unit 10.4 Summary

Implementation best practices ensure secure, maintainable PQC deployments:

1. **Security checklist:** Comprehensive requirements for production use
2. **Common pitfalls:** Learn from others' mistakes
3. **Testing strategy:** Multi-layer validation approach
4. **Ongoing maintenance:** Security is a continuous process

10.4.5 Formal Verification of PQC Implementations

Formal verification provides mathematical proof that an implementation matches its specification, eliminating entire classes of bugs that testing alone cannot catch.

Why Formal Verification Matters for PQC:

- PQC algorithms are newer and less battle-tested than RSA/ECC
- Subtle implementation errors (e.g., off-by-one in NTT, incorrect modular reduction) can silently destroy security

- Side-channel-free code is notoriously hard to verify by inspection

Key Tools and Frameworks:

TOOL	LANGUAGE	WHAT IT VERIFIES	NOTABLE PQC RESULTS
EasyCrypt	Custom proof language	Game-based cryptographic proofs	ML-KEM IND-CCA2 proof, FO transform correctness
F*/HACL*	F* (ML dialect)	Functional correctness + memory safety + constant-time	HACL* provides verified ML-KEM-768 in C via Kremlin extraction
Jasmin	Assembly-like DSL	Functional correctness + constant-time + leakage-free	Verified x86-64 NTT, Kyber/ML-KEM implementations
ProVerif/Tamarin	Protocol specification	Protocol-level security properties	PQ-TLS handshake verification, X-Wing composition
ct-verify/ct-wasm	C/WebAssembly	Constant-time execution only	Validates absence of secret-dependent branches/memory access

EasyCrypt Approach: EasyCrypt formalizes cryptographic security proofs as sequences of game transformations. For ML-KEM, this means proving that the IND-CCA2 security of the full KEM reduces to the IND-CPA security of K-PKE, which reduces to Module-LWE hardness. The proofs are machine-checked, eliminating human error in complex reductions.

F*/HACL* Approach: The HACL* library (High-Assurance Cryptographic Library) implements PQC algorithms in F*, a dependently-typed functional language. The F* type system enforces:

- **Functional correctness:** The implementation computes the same outputs as the spec
- **Memory safety:** No buffer overflows, use-after-free, or uninitialized reads
- **Constant-time:** No secret-dependent branching or memory indexing

The verified F* code is then extracted to C via the Kremlin compiler, producing production-quality C with formal guarantees.

Jasmin Approach: Jasmin targets the gap between high-level specifications and optimized assembly. It allows developers to write x86-64 assembly-like code with formal annotations, then proves both functional correctness (matches the reference spec) and constant-time execution (no operand-dependent timing) in a single verification step.

Practical Guidance:

- For protocol designers: Use ProVerif/Tamarin to verify composition of PQC primitives
- For library authors: Use Jasmin or F*/HACL* for core operations (NTT, hashing, encoding)
- For application developers: Prefer libraries with formal verification claims over hand-written code
- **State of the art (2026):** Verified ML-KEM implementations exist in HACL*, Jasmin, and the **Formosa-Crypto** project (EasyCrypt+Jasmin, covering security + correctness + constant-time); ML-DSA verification is active research. Note: "formally verified" requires scrutiny — see Unit 11.11 for the Verification Theatre incident (Feb 2026) documenting 13 escaped bugs in libcrux/hpke-rs including ML-DSA verifier spec violations. Always read the verification artifact to confirm which proportion of the shipping code is covered.

Unit 10.5: NIST Additional Signatures — Round 2 Concluded

After finalizing ML-DSA (FIPS 204) and SLH-DSA (FIPS 205), NIST opened a separate standardization track for **additional digital signature algorithms**. The goal is algorithmic diversity: different mathematical assumptions, different performance profiles, different signature-versus-pubkey trade-offs. Round 1 closed October 2023; Round 2 was announced October 2024.

In May 2026, NIST published **NIST IR 8610**, the status report concluding Round 2 of the Additional Digital Signatures on-ramp. Of the 14 second-round candidates, **nine advanced to Round 3** — FAEST, HAWK, MAYO, MQOM, QR-UOV, SDitH, SNOVA, SQIsign, and UOV — spanning the isogeny (SQIsign), lattice (HAWK), MPC-in-the-Head (FAEST, MQOM, SDitH), and multivariate (MAYO, QR-UOV, SNOVA,

UOV) families. **Five candidates were eliminated:** CROSS, LESS, Mirath, PERK, and RYDE. Round-3 tweaked specifications are due to NIST by 14 August 2026, and the third round is expected to last roughly two years (the 7th NIST PQC Standardization Conference is planned for 2027).

10.5.1 Round 2 Candidates (14)

All 14 candidates that competed in Round 2 are listed below, with their Round-3 outcome per NIST IR 8610 (2026) — advanced ☐ or eliminated ☐:

CANDIDATE	FAMILY	ROUND 3?	NOTABLE FEATURE
CROSS	Code-based (restricted errors)	☐	Fast signing; moderate signature size
FAEST	Symmetric-key (VOLE-in-the-head)	☐	Very small keys; signatures ~5 KB
HAWK	Lattice (module-LIP)	☐	Last lattice method standing — Falcon-class sizes, simpler sampling than FN-DSA. Based on the Lattice Isomorphism Problem, not NTRU.
LESS	Code-based (permutation equivalence)	☐	Small public keys
MAYO	Multivariate (UOV variant)	☐	Very fast verify; small sigs
Mirath	Multivariate (merger of MIRA + MiRiTH)	☐	Consolidated MPCitH-MinRank candidate
MQOM	Multivariate (MQ-over-any-field)	☐	General MQ framework
PERK	Symmetric (permuted kernel)	☐	Zero-knowledge style
QR-UOV	Multivariate (UOV over quotient ring)	☐	Smaller pubkey than standard UOV
RYDE	Code-based (rank metric)	☐	Rank-syndrom decoding
SDitH	Code-based (syndrome-decoding in-the-head)	☐	MPCitH framework
SNOVA	Multivariate	☐	Compact UOV variant
SQIsign	Isogeny	☐	Smallest PQ signatures (~200 bytes); slowest signing
UOV	Multivariate (original Oil-and-Vinegar)	☐	The "baseline" MQ candidate

Key dynamics:

- **Lattice diversity is deliberately constrained** — NIST did not want a second lattice primary to avoid single-family risk. HAWK is the sole remaining lattice option and would be adopted only if all other candidates fail.
- **Multivariate and code-based have the most candidates** (5 and 4 respectively), reflecting the community's desire for non-lattice alternatives.
- **SQIsign** is the only isogeny-based candidate surviving after SIDH's break in 2022 — signatures are tiny (~200 bytes) but signing is orders of magnitude slower than ML-DSA.
- **Mirath** consolidates two related candidates (MIRA and MiRitH) into one submission — a healthy sign of cross-team collaboration.

10.5.2 Timeline

MILESTONE	EXPECTED
Round 2 conclusion (NIST IR 8610)	14 May 2026 — 9 of 14 advanced
Round 3 analysis	2026 onward (~2 years; specs due 14 Aug 2026)
Draft standards	2026–2027
Final publication (FIPS 208+?)	2027 or later

10.5.3 FIPS 207 — HQC as Backup KEM

Separately from the signatures track, NIST is also standardizing a **backup KEM** to complement ML-KEM (FIPS 203). **HQC (Hamming Quasi-Cyclic)** was selected March 2025; NIST's plan is:

- **FIPS 207 IPD** targeted for 2026 (draft for public comment)
- **FIPS 207 Final** targeted for 2027

HQC is code-based (its security reduces to decoding random quasi-cyclic codes), providing a fundamentally different hardness assumption from ML-KEM's structured lattices. However — as documented extensively in Unit 11.10 — HQC's reference implementation has a disproportionate share of implementation bugs (three CVEs in 18 months, disabled by default in liboqs 0.13.0 pending further review). Production use as of April 2026 is not recommended until the HQC team completes the next spec revision and libraries re-enable the algorithm.

Unit 10.5 Summary

The NIST Additional Signatures process is where algorithmic diversity lives. The Round 2 finalists span lattice (HAWK), multivariate (UOV, MAYO, SNOVA, QR-UOV, Mirath), code-based (CROSS, LESS, RYDE, SDitH, MQOM), isogeny (SQIsign), and symmetric-key (FAEST, PERK) families — giving the post-FIPS-204 standards ecosystem a much richer menu of options with different size/speed/confidence trade-offs. None of these will replace ML-DSA for general use; they exist to backstop ML-DSA if a cryptanalytic surprise breaks Module-LWE/SIS assumptions. Parallel to signatures, FIPS 207 will standardize HQC as the backup KEM, though implementation maturity needs to catch up before adoption.

Unit 10.6: Post-Quantum Threshold Cryptography

Threshold cryptography splits a secret key among n parties such that any $t \leq n$ subset can compute signatures or decapsulate ciphertexts, but $t - 1$ or fewer cannot. The combination with PQC is a significant challenge — lattice operations don't decompose as cleanly as finite-field operations — and is the subject of the NIST Multi-Party Threshold Cryptography (MPTS) workshop series.

10.6.1 Why This Matters

Threshold PQC addresses scenarios where:

- **Single-point-of-compromise mitigation:** a stolen HSM should not reveal the signing key.
- **Distributed trust across organizations:** multi-signature for blockchain consensus, cross-jurisdictional signatures.
- **Policy enforcement without a single custodian:** "any 3 of 5 board members can sign" without a single party holding the key.

Classical threshold RSA (Shoup, 2000) and threshold ECDSA (Lindell 2017, Doerner et al. 2019) are production-ready. Post-quantum threshold was largely theoretical until 2024–2026.

10.6.2 MPTS 2026 Candidates

NIST's MPTS 2026 workshop (January 2026) featured four significant PQ threshold constructions:

CONSTRUCTION	BASE ALGORITHM	APPROACH
Hermine	Raccoon (Round 2 additional sig candidate, withdrawn)	PQ threshold signatures using lattice-Schnorr design
Vinaigrette	UOV / MAYO	Threshold multivariate signatures — adds secret-sharing to UOV key structure
Quorus	ML-DSA (FIPS 204)	MPC-based threshold ML-DSA with FIPS 204 signature compatibility — verifiers don't need to be aware of the threshold structure
PiVer	Verifiable secret sharing	General-purpose building block for threshold PQ applications

Quorus is the most operationally significant: it produces a standard ML-DSA signature that any FIPS 204 verifier accepts, so threshold issuance can roll out without client-side changes. The cost is the MPC round complexity during signing.

10.6.3 Open Problems

- **Threshold decapsulation for ML-KEM** is harder than threshold signing — no NIST-blessed construction as of April 2026. Work in progress.
- **Performance** — PQ threshold signatures cost multiple MPC rounds per signature; latency is 10-100× classical threshold.
- **Non-interactive variants** — research goal; most current constructions require online coordination.
- **Formal security proofs** — compositional security (threshold-of-PQ vs. PQ-of-threshold) is an active research frontier.

10.6.4 Production Status

Threshold PQC is **not production-ready** as of April 2026. Track:

- NIST MPTS workshop series (annual)
- IACR ePrint archive, especially the Crypto / Eurocrypt / Asiacrypt conference submissions
- The Quorus and Hermine open-source implementations if/when they appear

For production threshold-key use cases today, consider:

- Classical threshold ECDSA for non-HNDL-sensitive workflows (you'll migrate later)
- **PQ-capable HSM with physical multi-party controls** (m-of-n administrative controls) as an interim substitute for cryptographic threshold
- Hybrid constructions where the PQ half is held by a single party and the classical half is thresholded — partial protection but simpler

Unit 10.6 Summary

Threshold PQC is where the research frontier is hottest right now. The Quorus construction is the most operationally interesting because it produces standard FIPS 204 signatures from a distributed signing protocol — making rollout tractable. However, nothing is production-ready as of April 2026, and the performance profiles (multi-round MPC per signature) constrain the use cases. Track the NIST MPTS workshop series for updates; expect first production-ready threshold PQ signing implementations in 2027–2028.

Module 10 Summary: Future Algorithms and Research

Module 10 explored the evolving PQC landscape:

Unit 10.1: NIST Additional Algorithms

- FALCON for smaller signatures
- Alternative KEMs (HQC selected, BIKE, McEliece)
- Algorithm selection framework

Unit 10.2: Emerging Research

- Post-SIDH isogeny research
- Alternative cryptographic approaches
- Quantum computing timeline estimates

Unit 10.3: Cryptographic Agility

- Provider-based architecture
- Configuration-driven selection
- Graceful transition strategies

Unit 10.4: Best Practices Summary

- Security implementation checklist
- Common pitfall avoidance
- Comprehensive testing strategies

Key Takeaways

- Start PQ transition immediately
 - Use hybrid constructions during transition
 - Build for cryptographic agility
 - Monitor evolving standards and threats
 - Test thoroughly before deployment
-

Module 11: Migrating Legacy Codebases to Post-Quantum Cryptography

Module Overview: This module provides a comprehensive, practical guide to migrating existing systems and codebases from classical cryptography to post-quantum cryptography. Based on real-world case studies, official guidance from NIST, CISA, and NSA, and industry best practices, this module covers the complete migration lifecycle from inventory to deployment.

Prerequisites: Modules 1-10 (Volume 1 for 1, Volume 2 for 10) (complete PQC foundation)

Module Learning Objectives:

- Conduct comprehensive cryptographic inventory and discovery
 - Assess and prioritize migration risks using the HNDL threat model
 - Design crypto-agile architectures for seamless algorithm transitions
 - Implement hybrid cryptography as a migration bridge
 - Select and integrate appropriate PQC libraries
 - Test and validate cryptographic implementations
 - Plan phased migration aligned with compliance deadlines
-

Unit 11.1: Understanding the Migration Imperative

Prerequisites: Modules 1-10 (Volume 1 for 1, Volume 2 for 10) (complete PQC foundation)

Estimated Time: 1-2 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Explain the urgency** of PQC migration and the "harvest now, decrypt later" threat

- 2. **Understand global compliance timelines** from NIST, NSA, UK NCSC, and EU
- 3. **Calculate organizational risk** using the $Y + Z > X$ framework
- 4. **Prioritize migration efforts** based on data sensitivity and exposure

11.1.1 The "Harvest Now, Decrypt Later" Threat

The most pressing reason for immediate PQC migration is not a future quantum computer—it's attackers storing encrypted data today:

HNDL (Harvest Now, Decrypt Later) Attack Model

=====

2024: Adversary intercepts and stores encrypted traffic

- └ TLS sessions
- └ VPN tunnels
- └ Encrypted emails
- └ Stored encrypted databases

2030-2040: Cryptographically relevant quantum computer (CRQC) exists

- └ Adversary decrypts all stored data using Shor's algorithm

Affected Data:

- Government/military communications
- Healthcare records (HIPAA - 50+ year retention)
- Financial transactions
- Trade secrets and intellectual property
- Personal communications

The Mathematical Reality:

Assess your exposure using these three variables:

VARIABLE	DEFINITION	TYPICAL VALUES
X	Years until CRQC exists	5-15 years (estimates vary)
Y	Years needed to complete PQC migration	3-10 years
Z	Years data must remain confidential	5-50+ years

Critical Formula: If $Y + Z > X$, your organization is already at risk.

```
// Migration risk assessment
typedef struct {
    int years_to_crqc;           // X: Conservative estimate
    int migration_duration;      // Y: Based on system complexity
    int data_confidentiality;     // Z: Regulatory/business requirements
} risk_assessment_t;

bool is_at_risk(risk_assessment_t *assessment) {
    return (assessment->migration_duration +
            assessment->data_confidentiality) >
```

```

        assessment->years_to_crqc;
    }

    // Example: Healthcare organization
    // X = 10 years (conservative CRQC estimate)
    // Y = 5 years (complex legacy systems)
    // Z = 50 years (HIPAA requirements)
    // Y + Z = 55 > 10 = AT RISK (need to start migration NOW)

```

11.1.2 Global Compliance Timelines

Global PQC Migration Deadlines

=====

United States (NIST IR 8547):

- └─ 2030: Deprecation of RSA, ECDSA, DH, ECDH
- └─ 2035: Complete disallowance

NSA CNSA 2.0 (National Security Systems):

- └─ 2025: Begin software/firmware signing transition
- └─ 2030: Most systems complete transition
- └─ 2033: Full compliance required

United Kingdom (NCSC):

- └─ 2028: Complete inventory and migration planning
- └─ 2031: High-priority upgrades complete
- └─ 2035: Full PQC migration

European Union:

- └─ 2026: National transition roadmaps published
- └─ 2030: High-risk systems migrated
- └─ 2035: Full transition complete

Australia:

- └─ 2030: Aggressive migration deadline

Canada:

- └─ 2035: Non-classified IT systems migrated

US Government Cost Estimate:

- └─ \$7.1 billion (2024 dollars) for federal migration 2025-2035

11.1.3 Migration Priority Framework

Priority 1: Highest Risk (Migrate Immediately)

- Long-term secrets (encryption keys, master secrets)
- Government and military communications
- Healthcare records (HIPAA-protected data)
- Financial services data
- Intellectual property with 10+ year value
- Root CA certificates and signing keys

Priority 2: High Risk (Migrate 2025-2028)

- Critical infrastructure systems
- Telecommunications backbone
- Energy and utilities SCADA
- Inter-bank settlement systems

Priority 3: Medium Risk (Migrate 2028-2032)

- General enterprise applications
- E-commerce platforms
- Customer databases
- Internal communications

Priority 4: Lower Risk (Migrate 2032-2035)

- Data with <5 year confidentiality needs
- Ephemeral session data
- Public information systems

Key Insight: Key encapsulation (KEMs) should be migrated before signatures because:

- KEMs protect against HNDL attacks (data can be stored)
- Signature attacks require real-time quantum access
- Exception: Long-lived certificates (root CAs) need signature migration early

11.1.4 Migration Cost Analysis Breakdown

The US federal government's \$7.1 billion estimate (2024 dollars, 2025-2035) provides a baseline for understanding PQC migration costs. The following breakdown helps organizations estimate their own costs:

Cost by Organization Size:

ORGANIZATION SIZE	SYSTEMS	ESTIMATED TOTAL COST	PER-SYSTEM AVERAGE
Small (<50 systems)	10-50	\$500K - \$2M	\$20K-\$40K
Medium (50-500 systems)	50-500	\$2M - \$15M	\$30K-\$50K
Large (500-2000 systems)	500-2000	\$15M - \$75M	\$30K-\$50K
Enterprise (2000+ systems)	2000+	\$75M - \$300M+	\$35K-\$60K

Cost by Migration Phase:

PHASE	% OF TOTAL BUDGET	KEY ACTIVITIES
Discovery & Planning	10-15%	Inventory, risk assessment, architecture
Pilot & Validation	15-20%	Lab setup, testing, training
High-Risk Migration	25-30%	PKI, internet-facing, long-term keys
Broad Migration	25-35%	Internal apps, databases, integrations
Classical Deprecation	10-15%	Final validation, decommissioning

Per-Algorithm Implementation Costs:

ALGORITHM	LIBRARY INTEGRATION	TESTING & VALIDATION	PKI/ CERTIFICATE UPDATES	TOTAL PER APPLICATION
ML-KEM (TLS)	\$5K-\$15K	\$3K-\$8K	\$2K-\$5K	\$10K-\$28K
ML-DSA (Signing)	\$8K-\$20K	\$5K-\$12K	\$5K-\$15K	\$18K-\$47K
SLH-DSA (Firmware)	\$10K-\$25K	\$8K-\$15K	\$3K-\$8K	\$21K-\$48K
Hybrid (KEM + Classical)	\$12K-\$30K	\$8K-\$18K	\$5K-\$12K	\$25K-\$60K

Federal Estimate Breakdown (\$7.1B):

US Federal PQC Migration Cost Allocation (Estimated)

Personnel & Training: \$1.4B (20%)

- └ Cryptographic engineers
- └ Security analysts
- └ Developer training
- └ Contractor support

Software Development: \$2.1B (30%)

- └ Application modifications
- └ Library integration
- └ Testing infrastructure
- └ Automation tooling

Infrastructure: \$1.8B (25%)

- └ HSM upgrades (PQC-capable)
- └ Certificate infrastructure
- └ Network equipment updates
- └ Cloud service migrations

Compliance & Validation: \$1.1B (15%)

- └ FIPS 140-3 validations
- └ Security assessments
- └ Penetration testing
- └ Audit documentation

Contingency & Risk: \$0.7B (10%)

- └ Unforeseen complexity
- └ Algorithm changes
- └ Extended timelines
- └ Emergency responses

Cost Reduction Strategies:

1. **Crypto-Agility Investment:** Building agile architecture upfront reduces future migration costs by 40-60%
2. **Shared Services:** Centralized crypto services reduce per-application costs
3. **Automation:** Automated inventory and testing tools lower discovery costs by 50%+
4. **Phased Approach:** Spreading costs over 5-10 years improves budget absorption

Unit 11.1 Summary

This unit covered:

- The "Harvest Now, Decrypt Later" (HNDL) threat model and why PQC migration is urgent even before quantum computers exist
- The $Y + Z > X$ risk assessment framework for calculating organizational exposure based on migration timeline, data confidentiality requirements, and estimated time to CRQC

- Global compliance timelines from NIST, NSA CNSA 2.0, UK NCSC, and the EU, with key deadlines between 2030 and 2035
- A four-tier migration priority framework ranking systems by data sensitivity and cryptographic exposure
- Migration cost analysis including per-algorithm implementation costs and the US federal government's \$7.1 billion estimate

What's Next: Unit 11.2 introduces cryptographic inventory and discovery techniques, teaching you how to find and document every instance of cryptography across your organization's systems.

Exercises

Exercise 11.1.1: Risk Assessment *(Beginner)*

Your organization has the following characteristics:

- Healthcare provider with 50-year data retention requirements
- Legacy EHR system with 500+ integrations
- Estimated 7-year migration timeline
- Uses RSA-2048 for TLS and data encryption

a) Calculate the risk using the $Y + Z > X$ formula (assume $X = 12$ years) b) What is your priority level? c) Should you start migration planning now? Why?

Exercise 11.1.2: Prioritization Exercise *(Intermediate)*

Rank these systems by migration priority and justify:

1. Customer-facing web portal (TLS with ECDHE)
2. Internal document signing system (RSA-2048)
3. Long-term backup encryption (AES-256 with RSA key wrap)
4. Real-time trading platform (ECDHE for session keys)
5. Root CA for internal PKI (RSA-4096, 20-year validity)

Solutions

Solution 11.1.1

a) Risk Calculation using $Y + Z > X$:

Given:

- $X = 12$ years (time until CRQC)
- $Y = 7$ years (migration timeline)
- $Z = 50$ years (data retention requirement)

Formula check: $Y + Z > X$?

- $Y + Z = 7 + 50 = 57$ years
- $X = 12$ years
- $57 > 12$ ✓ **AT RISK**

Answer: The organization is at significant risk. Even if we only consider the migration timeline ($Y = 7$) plus a modest confidentiality period, data encrypted today will be decryptable before it should expire.

b) Priority Level:

Priority 1: Highest Risk - Healthcare providers with HIPAA data fall into the highest priority category because:

- Long-term data retention (50+ years)
- Protected Health Information (PHI) has severe breach consequences
- Complex systems mean longer migration timelines

c) Should migration planning start now?

Absolutely yes. Reasons:

1. $Y + Z$ (57 years) $\gg$ X (12 years) means data is already at risk
2. 7-year migration timeline means completion in 2033 at earliest if starting now
3. Healthcare has regulatory requirements (HIPAA) that demand proactive security
4. "Harvest now, decrypt later" attacks may already be capturing PHI data
5. The organization should also consider hybrid cryptography immediately to protect new data

Solution 11.1.2

Ranking by migration priority (highest to lowest):

1. Long-term backup encryption (AES-256 with RSA key wrap) - HIGHEST

- Justification: Backups may contain decades of sensitive data. The RSA key wrapping is quantum-vulnerable. Once CRQC exists, all historical backups become readable. Immediate hybrid key wrapping needed.

2. Root CA for internal PKI (RSA-4096, 20-year validity) - VERY HIGH

- Justification: Long-lived certificates with 20-year validity will still be valid when CRQC arrives. Signature forgery becomes possible, enabling impersonation of any entity in the PKI. Must transition to hybrid or PQC signatures.

3. Internal document signing system (RSA-2048) - HIGH

- Justification: Documents signed today may need verification in 10+ years. Once quantum computers exist, signature forgery becomes trivial, undermining document authenticity. Important for legal/compliance documents.

4. Customer-facing web portal (TLS with ECDHE) - MEDIUM

- Justification: ECDHE provides forward secrecy, so past sessions are protected by ephemeral keys. However, session keys could be captured and stored. Lower priority than long-term data protection.

5. Real-time trading platform (ECDHE for session keys) - LOWER

- Justification: Real-time trading data typically has short confidentiality requirements. Session keys are ephemeral. The time-sensitive nature of trading data means it has less long-term value. Still migrate before 2030, but not urgent.

Key Insight: The ranking prioritizes systems where data longevity (Z) is highest or where long-lived cryptographic assets exist.

Unit 11.2: Cryptographic Inventory and Discovery

Prerequisites: Unit 11.1 (Migration Imperative)

Estimated Time: 2-3 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Conduct a comprehensive cryptographic inventory** across all systems

2. **Use automated discovery tools** to identify cryptographic usage
3. **Document cryptographic dependencies** in applications and protocols
4. **Create a cryptographic bill of materials (CBOM)**

11.2.1 Why Inventory Is Critical

Most organizations have no idea where cryptography is used in their systems:

Typical Enterprise Cryptographic Surface

=====

Application Layer:

- └─ Web applications (TLS certificates)
- └─ Mobile apps (certificate pinning, local encryption)
- └─ Desktop applications (code signing, DRM)
- └─ API integrations (JWT, OAuth tokens)

Protocol Layer:

- └─ TLS 1.2/1.3 (ECDHE, RSA key exchange)
- └─ SSH (host keys, user authentication)
- └─ IPsec/VPN (IKEv2, certificates)
- └─ S/MIME (email encryption/signing)
- └─ SFTP, FTPS, HTTPS

Infrastructure Layer:

- └─ PKI/CA infrastructure
- └─ Hardware Security Modules (HSMs)
- └─ Key Management Systems (KMS)
- └─ Certificate stores
- └─ Secrets management (Vault, AWS Secrets Manager)

Data Layer:

- └─ Database encryption (TDE)
- └─ File system encryption
- └─ Backup encryption
- └─ Archive encryption
- └─ Cloud storage encryption

Identity Layer:

- └─ Authentication systems (SAML, OIDC)
- └─ Smart cards and tokens
- └─ Code signing certificates
- └─ Document signing

Third-Party Dependencies:

- └─ Vendor integrations
- └─ Cloud services
- └─ SaaS applications
- └─ Supply chain software

11.2.2 Inventory Methodology

Phase 1: Automated Discovery

```
# Example: Scanning for TLS certificates in a codebase
grep -r "BEGIN CERTIFICATE" /path/to/codebase

# Find hardcoded algorithm references
grep -rE "(RSA|ECDSA|ECDH|DSA|DH)" --include="*.c" --include="*.h" \
    --include="*.py" --include="*.java" --include="*.go" /path/to/codebase

# Scan for OpenSSL/crypto library usage
grep -rE "(EVP_|RSA_|EC_|DH_)" --include="*.c" /path/to/codebase

# Find certificate files
find /path/to/system -name "*.pem" -o -name "*.crt" -o -name "*.key" \
    -o -name "*.p12" -o -name "*.pfx" 2>/dev/null
```

Phase 2: Protocol Analysis

```
// Tool concept: TLS cipher suite analyzer
#include <openssl/ssl.h>

void analyze_tls_config(SSL_CTX *ctx) {
    STACK_OF(SSL_CIPHER) *ciphers = SSL_CTX_get_ciphers(ctx);

    printf("Configured Cipher Suites:\n");
    for (int i = 0; i < sk_SSL_CIPHER_num(ciphers); i++) {
        const SSL_CIPHER *cipher = sk_SSL_CIPHER_value(ciphers, i);
        const char *name = SSL_CIPHER_get_name(cipher);
        int bits = SSL_CIPHER_get_bits(cipher, NULL);

        // Flag quantum-vulnerable key exchange
        if (strstr(name, "ECDHE") || strstr(name, "DHE") ||
            strstr(name, "RSA")) {
            printf(" [QUANTUM-VULNERABLE] %s (%d bits)\n", name, bits);
        }
    }
}
```

Phase 3: Dependency Analysis

Cryptographic Bill of Materials (CBOM) Template

=====

System: [Application Name]

Owner: [Team/Individual]

Criticality: [High/Medium/Low]

Direct Cryptographic Dependencies:

Library	Version	Algorithms	PQC Ready?
OpenSSL	1.1.1w	RSA, ECDHE	No
libsodium	1.0.18	X25519, Ed25519	No
BouncyCastle	2.0+	RSA, ECDSA	Partial

Certificates and Keys:

Purpose	Algorithm	Key Size	Expiry
TLS Server	RSA	2048-bit	2025-03-15
Code Signing	ECDSA	P-256	2026-01-01
Root CA	RSA	4096-bit	2034-12-31

Protocol Usage:

Protocol	Key Exchange	Authentication
TLS 1.3	ECDHE (X25519)	RSA-2048 certificates
SSH	ECDH (nistp256)	Ed25519 keys
IPsec	IKEv2 (ECDH)	RSA certificates

Migration Status: ☐ Not Started ☐ Planning ☐ In Progress ☐ Complete

11.2.3 CISA Automated Discovery Tools

CISA recommends automated cryptography discovery and inventory (ACDI) tools:

ACDI Tool Capabilities

=====

Network-Level Discovery:

- └─ Passive TLS inspection
- └─ Certificate chain analysis
- └─ Protocol version detection
- └─ Cipher suite enumeration

Application-Level Discovery:

- └─ Static code analysis
- └─ Binary analysis
- └─ Configuration file scanning
- └─ Library dependency mapping

Infrastructure Discovery:

- └─ HSM inventory
- └─ KMS configuration audit
- └─ Certificate store scanning
- └─ Secrets manager inventory

Recommended Scan Integration Points:

- └─ CI/CD pipelines (pre-commit, build time)
- └─ Container image scanning
- └─ Infrastructure as Code (IaC) scanning
- └─ Runtime monitoring

Unit 11.2 Summary

This unit covered:

- Why most organizations lack visibility into their cryptographic surface and the importance of comprehensive discovery across application, protocol, infrastructure, data, and identity layers
- A three-phase inventory methodology: automated discovery (scanning for certificates, algorithm references, and library usage), protocol analysis (TLS cipher suite inspection), and dependency analysis
- The Cryptographic Bill of Materials (CBOM) format for documenting cryptographic assets, certificates, keys, and protocol usage per system
- CISA-recommended automated cryptography discovery and inventory (ACDI) tools and their integration points in CI/CD pipelines, container scanning, and runtime monitoring

What's Next: Unit 11.3 covers designing for cryptographic agility, showing how to build abstraction layers and policy-driven configurations that enable seamless algorithm transitions.

Exercises

Exercise 11.2.1: Inventory Your Codebase *(Beginner)*

For a codebase you have access to:

- Use `grep` to find all references to RSA, ECDSA, ECDH, DH
- Identify all certificate files (.pem, .crt, .key)
- List all cryptographic library dependencies
- Create a CBOM document for one critical system

Exercise 11.2.2: Protocol Analysis *(Intermediate)*

Analyze a TLS connection to a website:

```
openssl s_client -connect example.com:443 -showcerts 2>/dev/null | \
openssl x509 -text -noout
```

- What signature algorithm does the certificate use?
- What key exchange was negotiated?
- Is this configuration quantum-vulnerable? Why?

Solutions

Solution 11.2.1

a) Finding cryptographic references with grep:

```
# Search for algorithm references in source code
grep -rn "RSA\|ECDSA\|ECDH\|DH" --include="*.c" --include="*.h" \
    --include="*.py" --include="*.java" --include="*.go" .

# More comprehensive search including configuration files
grep -rn "RSA\|ECDSA\|ECDH\|DiffieHellman\|secp256\|P-256\|P-384" \
    --include="*.c" --include="*.conf" --include="*.yaml" --include="*.json" .

# Count occurrences by type
echo "=== Algorithm Usage Summary ==="
echo "RSA: $(grep -r 'RSA' --include='*.c' . | wc -l) occurrences"
echo "ECDSA: $(grep -r 'ECDSA' --include='*.c' . | wc -l) occurrences"
echo "ECDH: $(grep -r 'ECDH' --include='*.c' . | wc -l) occurrences"
```

b) Identifying certificate files:

```
# Find all certificate-related files
find . -type f \( -name "*.pem" -o -name "*.crt" -o -name "*.cer" \
    -o -name "*.key" -o -name "*.p12" -o -name "*.pfx" \)

# Analyze each certificate's algorithm
for cert in $(find . -name "*.crt" -o -name "*.pem"); do
    echo "=== $cert ==="
    openssl x509 -in "$cert" -noout -text 2>/dev/null | \
        grep -E "Signature Algorithm|Public Key Algorithm"
done
```

c) Listing cryptographic library dependencies:

```
# For C/C++ projects (check linked libraries)
ldd ./binary | grep -E "ssl|crypto|oqs"

# For package managers
grep -E "openssl|cryptography|pycryptodome|bouncycastle" \
    requirements.txt package.json pom.xml Cargo.toml 2>/dev/null

# For CMake projects
grep -E "find_package.*OpenSSL|find_package.*Crypto" CMakeLists.txt
```

d) CBOM Document Template (Example):


```

{
  "cbom_version": "1.0",
  "system_name": "Payment Gateway",
  "inventory_date": "2026-01-15",
  "cryptographic_assets": [
    {
      "type": "key_exchange",
      "algorithm": "ECDH-P256",
      "location": "src/tls/handshake.c:245",
      "quantum_vulnerable": true,
      "migration_priority": "high"
    },
    {
      "type": "digital_signature",
      "algorithm": "RSA-2048",
      "location": "src/auth/jwt.c:89",
      "quantum_vulnerable": true,
      "migration_priority": "high"
    },
    {
      "type": "symmetric_encryption",
      "algorithm": "AES-256-GCM",
      "location": "src/storage/encrypt.c:156",
      "quantum_vulnerable": false,
      "notes": "Symmetric key wrapped with RSA - wrapper is vulnerable"
    }
  ],
  "certificates": [
    {
      "subject": "CN=api.example.com",
      "issuer": "DigiCert",
      "algorithm": "ECDSA-P256",
      "expiry": "2027-03-15",
      "quantum_vulnerable": true
    }
  ],
  "dependencies": [
    { "name": "OpenSSL", "version": "3.0.12", "pqc_support": "via oqs-provider" }
  ]
}

```

Solution 11.2.2

Sample analysis of example.com:

```
$ openssl s_client -connect example.com:443 -showcerts 2>/dev/null | \
  openssl x509 -text -noout

Certificate:
  Signature Algorithm: sha256WithRSAEncryption
  Issuer: C=US, O=DigiCert Inc, CN=DigiCert TLS RSA SHA256 2020 CA1
  Subject: C=US, ST=California, L=Los Angeles, O=Internet Corporation...
  Subject Public Key Info:
    Public Key Algorithm: rsaEncryption
    RSA Public-Key: (2048 bit)
```

a) Signature algorithm: `sha256WithRSAEncryption` (RSA-2048 with SHA-256)

b) Key exchange negotiated: (Check with `-state` flag)

```
openssl s_client -connect example.com:443 -state 2>&1 | grep "Cipher is"
# Typical output: TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
# Key exchange: ECDHE (Elliptic Curve Diffie-Hellman Ephemeral)
```

c) Quantum vulnerability assessment:

COMPONENT	ALGORITHM	QUANTUM VULNERABLE?	REASON
Certificate signature	RSA-2048	YES	Shor's algorithm factors RSA in polynomial time
Key exchange	ECDHE	YES	Shor's algorithm solves ECDLP in polynomial time
Bulk encryption	AES-128-GCM	Partially	Grover reduces to ~64-bit security; use AES-256

Conclusion: This configuration is **quantum-vulnerable** because:

1. The RSA signature can be forged, enabling certificate impersonation
2. The ECDHE key exchange can be broken, compromising forward secrecy
3. Stored TLS traffic (HNDL attack) can be decrypted when CRQC exists

Remediation: Migrate to hybrid TLS with X25519MLKEM768 + ML-DSA certificates.

Unit 11.3: Retrofitting Cryptographic Agility into Legacy Systems

Prerequisites: Unit 10.3 (Cryptographic Agility), Unit 11.2 (Cryptographic Inventory)

Estimated Time: 2-3 hours

Learning Objectives

After completing this unit, you will be able to:

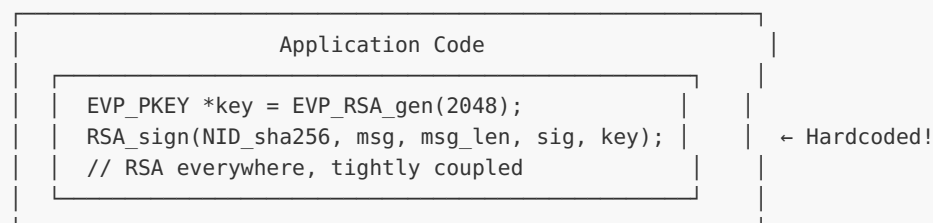
1. **Design crypto-agile architectures** that support algorithm substitution
2. **Implement abstraction layers** between application logic and cryptographic primitives
3. **Create policy-driven cryptographic configurations**
4. **Plan for future algorithm transitions**

11.3.1 What Is Cryptographic Agility?

Cryptographic agility is a design feature that enables seamless updates to new cryptographic algorithms without modifying application code:

Non-Agile Architecture (Problematic)

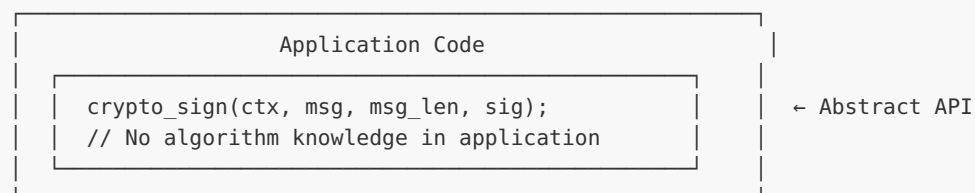
=====

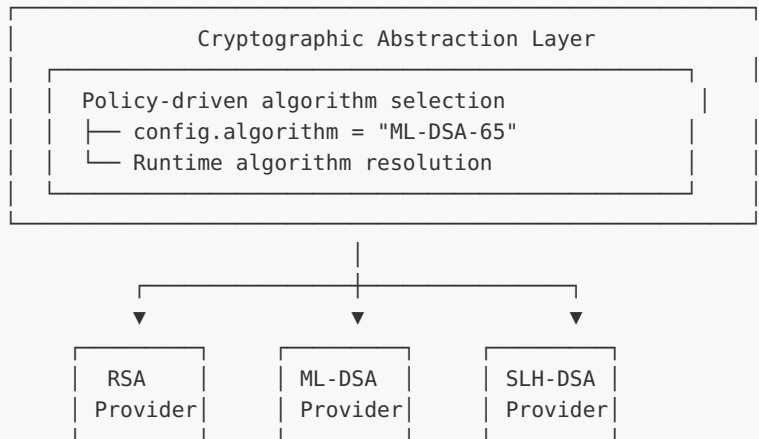


Problem: Changing algorithms requires modifying application code everywhere

Crypto-Agile Architecture (Recommended)

=====





Benefit: Algorithm changes require only configuration updates

11.3.2 Implementing an Abstraction Layer

```

// crypto_agile.h - Cryptographic Agility Interface

#ifndef CRYPTO_AGILE_H
#define CRYPTO_AGILE_H

#include <stdint.h>
#include <stddef.h>

// Algorithm identifiers (extensible)
typedef enum {
    ALGO_RSA_2048,
    ALGO_RSA_4096,
    ALGO_ECDSA_P256,
    ALGO_ECDSA_P384,
    ALGO_ED25519,
    ALGO_ML_DSA_44,
    ALGO_ML_DSA_65,
    ALGO_ML_DSA_87,
    ALGO_SLH_DSA_128F,
    ALGO_SLH_DSA_128S,
    // Hybrid algorithms
    ALGO_HYBRID_ECDSA_ML_DSA_65,
    ALGO_HYBRID_ED25519_ML_DSA_65,
} crypto_algo_t;

// Opaque context (implementation-independent)
typedef struct crypto_ctx crypto_ctx_t;

// Lifecycle
crypto_ctx_t *crypto_ctx_new(crypto_algo_t algo);
void crypto_ctx_free(crypto_ctx_t *ctx);

// Key management
int crypto_keygen(crypto_ctx_t *ctx,
                  uint8_t *public_key, size_t *pk_len,
                  uint8_t *secret_key, size_t *sk_len);

```

```

// Signatures
int crypto_sign(crypto_ctx_t *ctx,
                const uint8_t *message, size_t msg_len,
                uint8_t *signature, size_t *sig_len,
                const uint8_t *secret_key, size_t sk_len);

int crypto_verify(crypto_ctx_t *ctx,
                  const uint8_t *message, size_t msg_len,
                  const uint8_t *signature, size_t sig_len,
                  const uint8_t *public_key, size_t pk_len);

// Key encapsulation
int crypto_kem_encaps(crypto_ctx_t *ctx,
                      uint8_t *ciphertext, size_t *ct_len,
                      uint8_t *shared_secret, size_t *ss_len,
                      const uint8_t *public_key, size_t pk_len);

int crypto_kem_decaps(crypto_ctx_t *ctx,
                      uint8_t *shared_secret, size_t *ss_len,
                      const uint8_t *ciphertext, size_t ct_len,
                      const uint8_t *secret_key, size_t sk_len);

// Runtime algorithm query
const char *crypto_algo_name(crypto_algo_t algo);
size_t crypto_public_key_size(crypto_algo_t algo);
size_t crypto_secret_key_size(crypto_algo_t algo);
size_t crypto_signature_size(crypto_algo_t algo);

#endif // CRYPTO_AGILE_H

```

```

// crypto_agile.c - Implementation with multiple backends

#include "crypto_agile.h"
#include <oqs/oqs.h>
#include <openssl/evp.h>
#include <string.h>
#include <stdlib.h>

struct crypto_ctx {
    crypto_algo_t algo;
    union {
        EVP_PKEY_CTX *evp_ctx; // OpenSSL for classical
        OQS_SIG *oqs_sig;      // liboqs for PQC
        struct {               // Hybrid
            EVP_PKEY_CTX *classical;
            OQS_SIG *pqc;
        } hybrid;
    } impl;
};

crypto_ctx_t *crypto_ctx_new(crypto_algo_t algo) {
    crypto_ctx_t *ctx = calloc(1, sizeof(crypto_ctx_t));
    if (!ctx) return NULL;

    ctx->algo = algo;

    switch (algo) {
        case ALGO_ML_DSA_65:
            ctx->impl.oqs_sig = OQS_SIG_new(OQS_SIG_alg_ml_dsa_65);

```

```

        break;
    case ALGO_ML_DSA_87:
        ctx->impl.oqs_sig = OQS_SIG_new(OQS_SIG_alg_ml_dsa_87);
        break;
    case ALGO_HYBRID_ECDSA_ML_DSA_65:
        // Initialize both classical and PQC
        ctx->impl.hybrid.classical = EVP_PKEY_CTX_new_id(EVP_PKEY_EC, NULL);
        ctx->impl.hybrid.pqc = OQS_SIG_new(OQS_SIG_alg_ml_dsa_65);
        break;
    // ... other algorithms
    default:
        free(ctx);
        return NULL;
}

return ctx;
}

int crypto_sign(crypto_ctx_t *ctx,
                const uint8_t *message, size_t msg_len,
                uint8_t *signature, size_t *sig_len,
                const uint8_t *secret_key, size_t sk_len) {

    switch (ctx->algo) {
        case ALGO_ML_DSA_65:
        case ALGO_ML_DSA_87:
            return OQS_SIG_sign(ctx->impl.oqs_sig,
                                signature, sig_len,
                                message, msg_len,
                                secret_key);

        case ALGO_HYBRID_ECDSA_ML_DSA_65: {
            // Sign with both algorithms
            size_t classical_sig_len, pqc_sig_len;

            // Classical signature first
            // ... EVP signing code

            // PQC signature
            OQS_SIG_sign(ctx->impl.hybrid.pqc,
                        signature + classical_sig_len, &pqc_sig_len,
                        message, msg_len,
                        secret_key + /* classical sk size */);

            *sig_len = classical_sig_len + pqc_sig_len;
            return 0;
        }

        default:
            return -1;
    }
}

// Configuration-based initialization
crypto_ctx_t *crypto_ctx_from_config(const char *config_path) {
    // Read algorithm from config file
    // Example: algorithm = "ML-DSA-65"
    // This allows runtime algorithm selection without code changes

    FILE *f = fopen(config_path, "r");

```

```

char algo_name[64];
// Parse config...

if (strcmp(algo_name, "ML-DSA-65") == 0)
    return crypto_ctx_new(ALGO_ML_DSA_65);
else if (strcmp(algo_name, "HYBRID-ECDSA-ML-DSA") == 0)
    return crypto_ctx_new(ALGO_HYBRID_ECDSA_ML_DSA_65);
// ...

return NULL;
}

```

11.3.3 Configuration-Driven Cryptography

```

# crypto_config.yaml - Policy-driven cryptographic configuration

cryptographic_policy:
  version: "2.0"
  effective_date: "2025-01-01"

# Default algorithms for new deployments
defaults:
  key_exchange: "X25519MLKEM768"      # Hybrid KEM
  signature: "ML-DSA-65"               # Post-quantum signature
  symmetric: "AES-256-GCM"             # Symmetric encryption
  hash: "SHA-384"                     # Hash function

# Allowed algorithms (whitelist)
allowed:
  key_exchange:
    - "X25519MLKEM768"                # Hybrid (recommended)
    - "ML-KEM-768"                   # Pure PQC
    - "X25519"                       # Classical (deprecated 2030)
  signature:
    - "ML-DSA-65"
    - "ML-DSA-87"
    - "SLH-DSA-128f"
    - "ECDSA-P256"                   # Classical (deprecated 2030)

# Deprecated algorithms (warn on use)
deprecated:
  - algorithm: "RSA-2048"
    deprecation_date: "2025-06-01"
    removal_date: "2030-01-01"
    replacement: "ML-DSA-65"
  - algorithm: "ECDHE-P256"
    deprecation_date: "2026-01-01"
    removal_date: "2030-01-01"
    replacement: "X25519MLKEM768"

# Prohibited algorithms (fail on use)
prohibited:
  - "3DES"
  - "RC4"
  - "MD5"
  - "SHA1"
  - "RSA-1024"

```

```
# Environment-specific overrides
environments:
  production:
    require_hybrid: true
    min_security_level: 128
  development:
    allow_classical_only: true          # For testing compatibility
```

Unit 11.3 Summary

This unit covered:

- The concept of cryptographic agility as a design feature that enables algorithm substitution without modifying application code
- Implementing a cryptographic abstraction layer in C with opaque contexts, algorithm-agnostic APIs for signing, verification, and key encapsulation, and runtime algorithm resolution
- Configuration-driven cryptography using YAML policy files that define default, allowed, deprecated, and prohibited algorithms with lifecycle dates
- The three-phase migration path enabled by crypto-agility: classical only, hybrid (transition), and pure PQC, all achievable through configuration changes alone

What's Next: Unit 11.4 covers PQC library selection and integration, comparing liboqs, OpenSSL 3.5+, and other libraries, with hands-on code for ML-KEM, ML-DSA, and hybrid implementations.

Exercises

Exercise 11.3.1: Design an Abstraction Layer ***(Advanced)***

Design a crypto-agile interface for your application that: a) Supports both classical (ECDSA) and PQC (ML-DSA) signatures b) Allows runtime algorithm selection via configuration c) Provides a migration path from classical to hybrid to pure PQC

Exercise 11.3.2: Policy Configuration ***(Intermediate)***

Write a configuration file that: a) Sets ML-DSA-65 as the default signature algorithm b) Allows hybrid ECDSA+ML-DSA during transition c) Deprecates pure ECDSA by 2028 d) Prohibits RSA-2048 immediately

Solutions

Solution 11.3.1

Crypto-agile interface design:

```
/* crypto_agile.h - Algorithm-agnostic cryptographic interface */

#ifndef CRYPTO_AGILE_H
#define CRYPTO_AGILE_H

#include <stdint.h>
#include <stddef.h>

/* Algorithm identifiers - extensible enum */
typedef enum {
    ALGO_ECDSA_P256,
    ALGO_ECDSA_P384,
    ALGO_ML_DSA_44,
    ALGO_ML_DSA_65,
    ALGO_ML_DSA_87,
    ALGO_HYBRID_ECDSA_P256_ML_DSA_65,
    ALGO_HYBRID_ECDSA_P384_ML_DSA_87
} crypto_algo_t;

/* Opaque context - hides implementation details */
typedef struct crypto_ctx crypto_ctx_t;

/* Lifecycle functions */
crypto_ctx_t *crypto_ctx_new(crypto_algo_t algo);
crypto_ctx_t *crypto_ctx_from_config(const char *config_path);
void crypto_ctx_free(crypto_ctx_t *ctx);

/* Key management - algorithm agnostic */
int crypto_keygen(crypto_ctx_t *ctx,
                  uint8_t *pk, size_t *pk_len,
                  uint8_t *sk, size_t *sk_len);

/* Signature operations - same API regardless of algorithm */
int crypto_sign(crypto_ctx_t *ctx,
                uint8_t *sig, size_t *sig_len,
                const uint8_t *msg, size_t msg_len,
                const uint8_t *sk, size_t sk_len);

int crypto_verify(crypto_ctx_t *ctx,
                  const uint8_t *sig, size_t sig_len,
                  const uint8_t *msg, size_t msg_len,
                  const uint8_t *pk, size_t pk_len);

/* Query functions for buffer sizing */
size_t crypto_pk_size(crypto_ctx_t *ctx);
size_t crypto_sk_size(crypto_ctx_t *ctx);
size_t crypto_sig_size(crypto_ctx_t *ctx);

#endif /* CRYPTO_AGILE_H */
```

Migration path implementation:

```

/* Example: Phased migration from ECDSA to hybrid to pure PQC */

// Phase 1: Classical only (current state)
ctx = crypto_ctx_new(ALGO_ECDSA_P256);

// Phase 2: Hybrid (transition period, 2025-2030)
// Change config file, no code changes needed:
//   algorithm: "HYBRID-ECDSA-P256-ML-DSA-65"
ctx = crypto_ctx_from_config("crypto_policy.yaml");

// Phase 3: Pure PQC (post-2030)
// Update config file only:
//   algorithm: "ML-DSA-65"
ctx = crypto_ctx_from_config("crypto_policy.yaml");

// Application code remains unchanged throughout migration!
crypto_sign(ctx, sig, &sig_len, msg, msg_len, sk, sk_len);

```

Solution 11.3.2

Policy configuration file:

```

# crypto_policy.yaml - Enterprise cryptographic policy
# Effective: January 2026

cryptographic_policy:
  version: "3.0"
  last_updated: "2026-01-15"
  organization: "Example Corp"

# Default algorithms for all new deployments
defaults:
  signature: "ML-DSA-65"
  key_exchange: "X25519MLKEM768"
  symmetric: "AES-256-GCM"
  hash: "SHA-384"

# Algorithm lifecycle management
algorithms:
  # Signatures
  signatures:
    ML-DSA-65:
      status: "recommended"
      security_level: "NIST Level 3"
      notes: "Default for all new deployments"

    HYBRID-ECDSA-P256-ML-DSA-65:
      status: "allowed"
      security_level: "NIST Level 3 (minimum of both)"
      use_case: "Transition period interoperability"
      allowed_until: "2032-12-31"

    ECDSA-P256:
      status: "deprecated"
      deprecation_date: "2028-01-01"

```

```

    disallowed_date: "2030-01-01"
    migration_target: "ML-DSA-65"
    notes: "Existing deployments must migrate by 2030"

RSA-2048:
    status: "prohibited"
    effective_date: "2026-01-01"
    notes: "Prohibited immediately - no exceptions"

RSA-4096:
    status: "prohibited"
    effective_date: "2026-01-01"
    notes: "Key size does not mitigate quantum threat"

# Key exchange
key_exchange:
    X25519MLKEM768:
        status: "recommended"
        notes: "Hybrid provides defense-in-depth"

    ML-KEM-768:
        status: "allowed"
        use_case: "Where classical fallback unnecessary"

    ECDH-P256:
        status: "deprecated"
        deprecation_date: "2028-01-01"

# Enforcement rules
enforcement:
    new_deployments:
        - require: ["ML-DSA-65", "HYBRID-ECDSA-P256-ML-DSA-65"]
          for: "signatures"
        - require: ["X25519MLKEM768", "ML-KEM-768"]
          for: "key_exchange"

    existing_deployments:
        grace_period_years: 2
        audit_frequency: "quarterly"

# Exception process
exceptions:
    require_approval_from: ["CISO", "Security Architecture"]
    max_duration_days: 180
    documentation_required: true

```

Unit 11.4: PQC Library Selection and Integration

Prerequisites: Unit 11.3 (Cryptographic Agility), Module 8 (Hybrid Cryptography)

Estimated Time: 2-3 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Evaluate PQC library options** (liboqs, PQCclean, OpenSSL 3.5)
2. **Integrate liboqs** into existing C/C++ projects
3. **Use the OQS provider** for OpenSSL 3.x applications
4. **Implement hybrid constructions** combining classical and PQC

11.4.1 Library Comparison

PQC Library Landscape (2025)

=====

Library	Algorithms	Production?	Best For
OpenSSL 3.5+	ML-KEM, ML-DSA, SLH-DSA	Yes	Production deployments
liboqs	All NIST + alternates	Research (use caution)	Prototyping, testing
oqs-provider	All liboqs via OpenSSL API	Research	OpenSSL apps migration
PQCclean	Clean reference implementations	Reference	Study, audit starting pt
BoringSSL	ML-KEM (Kyber) X25519Kyber768	Yes (Google)	If using BoringSSL
wolfSSL	ML-KEM, ML-DSA (via liboqs)	Yes	Embedded, FIPS-140-3

Recommendation:

- New production systems: OpenSSL 3.5+ (native PQC support)
- Existing OpenSSL apps: oqs-provider for gradual migration
- Prototyping/research: liboqs directly
- Embedded systems: wolfSSL with PQC support

11.4.2 liboqs Integration

Installation:

```
# Clone and build liboqs
git clone https://github.com/open-quantum-safe/liboqs.git
cd liboqs
mkdir build && cd build
cmake -DCMAKE_INSTALL_PREFIX=/usr/local ..
make -j$(nproc)
sudo make install
```

Basic ML-KEM Example:

```
#include <oqs/oqs.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int ml_kem_key_exchange(void) {
    OQS_KEM *kem = NULL;
    uint8_t *public_key = NULL;
    uint8_t *secret_key = NULL;
    uint8_t *ciphertext = NULL;
    uint8_t *shared_secret_enc = NULL;
    uint8_t *shared_secret_dec = NULL;
    int ret = -1;

    // Initialize ML-KEM-768
    kem = OQS_KEM_new(OQS_KEM_alg_ml_kem_768);
    if (kem == NULL) {
        fprintf(stderr, "ML-KEM-768 not available\n");
        return -1;
    }

    // Allocate memory
    public_key = malloc(kem->length_public_key);
    secret_key = malloc(kem->length_secret_key);
    ciphertext = malloc(kem->length_ciphertext);
    shared_secret_enc = malloc(kem->length_shared_secret);
    shared_secret_dec = malloc(kem->length_shared_secret);

    if (!public_key || !secret_key || !ciphertext ||
        !shared_secret_enc || !shared_secret_dec) {
        fprintf(stderr, "Memory allocation failed\n");
        goto cleanup;
    }

    // Generate key pair (Alice)
    if (OQS_KEM_keypair(kem, public_key, secret_key) != OQS_SUCCESS) {
        fprintf(stderr, "Key generation failed\n");
        goto cleanup;
    }

    printf("Public key size: %zu bytes\n", kem->length_public_key);
    printf("Secret key size: %zu bytes\n", kem->length_secret_key);

    // Encapsulate (Bob)
    if (OQS_KEM_encaps(kem, ciphertext, shared_secret_enc,
                       public_key) != OQS_SUCCESS) {
        fprintf(stderr, "Encapsulation failed\n");
        goto cleanup;
    }

    printf("Ciphertext size: %zu bytes\n", kem->length_ciphertext);

    // Decapsulate (Alice)
    if (OQS_KEM_decaps(kem, shared_secret_dec, ciphertext,
                       secret_key) != OQS_SUCCESS) {
        fprintf(stderr, "Decapsulation failed\n");
        goto cleanup;
    }
}
```

```

    }

    // Verify shared secrets match
    if (memcmp(shared_secret_enc, shared_secret_dec,
               kem->length_shared_secret) != 0) {
        fprintf(stderr, "Shared secrets don't match!\n");
        goto cleanup;
    }

    printf("Key exchange successful!\n");
    printf("Shared secret size: %zu bytes\n", kem->length_shared_secret);
    ret = 0;

cleanup:
    OQS_MEM_secure_free(secret_key, kem ? kem->length_secret_key : 0);
    OQS_MEM_secure_free(shared_secret_enc, kem ? kem->length_shared_secret : 0);
    OQS_MEM_secure_free(shared_secret_dec, kem ? kem->length_shared_secret : 0);
    free(public_key);
    free(ciphertext);
    OQS_KEM_free(kem);

    return ret;
}

```

ML-DSA Signature Example:

```

#include <oqs/oqs.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int ml_dsa_sign_verify(const uint8_t *message, size_t msg_len) {
    OQS_SIG *sig = NULL;
    uint8_t *public_key = NULL;
    uint8_t *secret_key = NULL;
    uint8_t *signature = NULL;
    size_t sig_len = 0;
    int ret = -1;

    // Initialize ML-DSA-65
    sig = OQS_SIG_new(OQS_SIG_alg_ml_dsa_65);
    if (sig == NULL) {
        fprintf(stderr, "ML-DSA-65 not available\n");
        return -1;
    }

    // Allocate memory
    public_key = malloc(sig->length_public_key);
    secret_key = malloc(sig->length_secret_key);
    signature = malloc(sig->length_signature);

    if (!public_key || !secret_key || !signature) {
        fprintf(stderr, "Memory allocation failed\n");
        goto cleanup;
    }

    // Generate key pair
    if (OQS_SIG_keypair(sig, public_key, secret_key) != OQS_SUCCESS) {

```

```

        fprintf(stderr, "Key generation failed\n");
        goto cleanup;
    }

    printf("ML-DSA-65 Key Sizes:\n");
    printf("  Public key: %zu bytes\n", sig->length_public_key);
    printf("  Secret key: %zu bytes\n", sig->length_secret_key);

    // Sign message
    if (OQS_SIG_sign(sig, signature, &sig_len,
                    message, msg_len, secret_key) != OQS_SUCCESS) {
        fprintf(stderr, "Signing failed\n");
        goto cleanup;
    }

    printf("Signature size: %zu bytes\n", sig_len);

    // Verify signature
    if (OQS_SIG_verify(sig, message, msg_len,
                      signature, sig_len, public_key) != OQS_SUCCESS) {
        fprintf(stderr, "Verification failed\n");
        goto cleanup;
    }

    printf("Signature verified successfully!\n");
    ret = 0;

cleanup:
    OQS_MEM_secure_free(secret_key, sig ? sig->length_secret_key : 0);
    free(public_key);
    free(signature);
    OQS_SIG_free(sig);

    return ret;
}

```

11.4.3 Hybrid Implementation Pattern

```

// hybrid_kem.c - X25519 + ML-KEM-768 hybrid key exchange

#include <oqs/oqs.h>
#include <openssl/evp.h>
#include <openssl/kdf.h>
#include <string.h>

#define X25519_KEY_SIZE 32
#define X25519_SHARED_SIZE 32

typedef struct {
    uint8_t x25519_public[X25519_KEY_SIZE];
    uint8_t mlkem_public[1184]; // ML-KEM-768 public key
} hybrid_public_key_t;

typedef struct {
    uint8_t x25519_secret[X25519_KEY_SIZE];
    uint8_t mlkem_secret[2400]; // ML-KEM-768 secret key
} hybrid_secret_key_t;

```

```

typedef struct {
    uint8_t x25519_public[X25519_KEY_SIZE]; // Ephemeral X25519
    uint8_t mlkem_ciphertext[1088];          // ML-KEM-768 ciphertext
} hybrid_ciphertext_t;

int hybrid_keygen(hybrid_public_key_t *pk, hybrid_secret_key_t *sk) {
    EVP_PKEY *x25519_key = NULL;
    EVP_PKEY_CTX *ctx = NULL;
    OQS_KEM *kem = NULL;
    size_t len;
    int ret = -1;

    // Generate X25519 key pair
    ctx = EVP_PKEY_CTX_new_id(EVP_PKEY_X25519, NULL);
    if (!ctx || EVP_PKEY_keygen_init(ctx) <= 0 ||
        EVP_PKEY_keygen(ctx, &x25519_key) <= 0) {
        goto cleanup;
    }

    // Extract X25519 keys
    len = X25519_KEY_SIZE;
    EVP_PKEY_get_raw_public_key(x25519_key, pk->x25519_public, &len);
    len = X25519_KEY_SIZE;
    EVP_PKEY_get_raw_private_key(x25519_key, sk->x25519_secret, &len);

    // Generate ML-KEM-768 key pair
    kem = OQS_KEM_new(OQS_KEM_alg_ml_kem_768);
    if (!kem) goto cleanup;

    if (OQS_KEM_keypair(kem, pk->mlkem_public, sk->mlkem_secret) != OQS_SUCCESS) {
        goto cleanup;
    }

    ret = 0;

cleanup:
    EVP_PKEY_free(x25519_key);
    EVP_PKEY_CTX_free(ctx);
    OQS_KEM_free(kem);
    return ret;
}

int hybrid_encaps(const hybrid_public_key_t *pk,
                  hybrid_ciphertext_t *ct,
                  uint8_t *shared_secret, size_t ss_len) {
    EVP_PKEY *x25519_eph = NULL;
    EVP_PKEY *x25519_peer = NULL;
    EVP_PKEY_CTX *ctx = NULL;
    OQS_KEM *kem = NULL;

    uint8_t x25519_shared[X25519_SHARED_SIZE];
    uint8_t mlkem_shared[32]; // ML-KEM shared secret
    size_t len;
    int ret = -1;

    // Generate ephemeral X25519 key
    ctx = EVP_PKEY_CTX_new_id(EVP_PKEY_X25519, NULL);
    if (!ctx || EVP_PKEY_keygen_init(ctx) <= 0 ||
        EVP_PKEY_keygen(ctx, &x25519_eph) <= 0) {
        goto cleanup;
    }

```



```

}

// Export ephemeral public key to ciphertext
len = X25519_KEY_SIZE;
EVP_PKEY_get_raw_public_key(x25519_eph, ct->x25519_public, &len);

// Perform X25519 key agreement
x25519_peer = EVP_PKEY_new_raw_public_key(EVP_PKEY_X25519, NULL,
                                           pk->x25519_public, X25519_KEY_SIZE);
EVP_PKEY_CTX_free(ctx);
ctx = EVP_PKEY_CTX_new(x25519_eph, NULL);

if (EVP_PKEY_derive_init(ctx) <= 0 ||
    EVP_PKEY_derive_set_peer(ctx, x25519_peer) <= 0) {
    goto cleanup;
}

len = X25519_SHARED_SIZE;
if (EVP_PKEY_derive(ctx, x25519_shared, &len) <= 0) {
    goto cleanup;
}

// ML-KEM encapsulation
kem = OQS_KEM_new(OQS_KEM_alg_ml_kem_768);
if (!kem) goto cleanup;

if (OQS_KEM_encaps(kem, ct->mlkem_ciphertext, mlkem_shared,
                  pk->mlkem_public) != OQS_SUCCESS) {
    goto cleanup;
}

// Combine shared secrets using HKDF
// shared_secret = HKDF(x25519_shared || mlkem_shared)
{
    EVP_PKEY_CTX *hkdf_ctx = EVP_PKEY_CTX_new_id(EVP_PKEY_HKDF, NULL);
    uint8_t combined[X25519_SHARED_SIZE + 32];

    memcpy(combined, x25519_shared, X25519_SHARED_SIZE);
    memcpy(combined + X25519_SHARED_SIZE, mlkem_shared, 32);

    EVP_PKEY_derive_init(hkdf_ctx);
    EVP_PKEY_CTX_set_hkdf_md(hkdf_ctx, EVP_sha384());
    EVP_PKEY_CTX_set1_hkdf_key(hkdf_ctx, combined, sizeof(combined));
    EVP_PKEY_CTX_set1_hkdf_salt(hkdf_ctx,
                                (uint8_t*)"X25519MLKEM768", 14);
    EVP_PKEY_CTX_add1_hkdf_info(hkdf_ctx,
                                (uint8_t*)"hybrid-kem", 10);

    EVP_PKEY_derive(hkdf_ctx, shared_secret, &ss_len);
    EVP_PKEY_CTX_free(hkdf_ctx);

    OPENSSL_cleanse(combined, sizeof(combined));
}

ret = 0;

cleanup:
    OPENSSL_cleanse(x25519_shared, sizeof(x25519_shared));
    OPENSSL_cleanse(mlkem_shared, sizeof(mlkem_shared));
    EVP_PKEY_free(x25519_eph);

```

```

    EVP_PKEY_free(x25519_peer);
    EVP_PKEY_CTX_free(ctx);
    OQS_KEM_free(kem);
    return ret;
}

// Similar hybrid_decaps() function...

```

11.4.4 OpenSSL 3.5 Native PQC

With OpenSSL 3.5+, PQC is built-in:

```

// OpenSSL 3.5+ ML-KEM example
#include <openssl/evp.h>
#include <openssl/core_names.h>
#include <openssl/params.h>

int openssl35_mlkem_keygen(void) {
    EVP_PKEY_CTX *ctx = NULL;
    EVP_PKEY *pkey = NULL;

    // Create context for ML-KEM-768
    ctx = EVP_PKEY_CTX_new_from_name(NULL, "ML-KEM-768", NULL);
    if (!ctx) {
        fprintf(stderr, "ML-KEM-768 not available\n");
        return -1;
    }

    // Generate key pair
    if (EVP_PKEY_keygen_init(ctx) <= 0 ||
        EVP_PKEY_generate(ctx, &pkey) <= 0) {
        fprintf(stderr, "Key generation failed\n");
        EVP_PKEY_CTX_free(ctx);
        return -1;
    }

    printf("ML-KEM-768 key generated successfully!\n");

    // Use for encapsulation/decapsulation...

    EVP_PKEY_free(pkey);
    EVP_PKEY_CTX_free(ctx);
    return 0;
}

```

Unit 11.4 Summary

This unit covered:

- A comparative evaluation of PQC libraries including OpenSSL 3.5+ (production), liboqs (prototyping), oqs-provider (OpenSSL migration), PQCclean (reference), BoringSSL, and wolfSSL (embedded/FIPS)

- Hands-on liboqs integration: building from source, performing ML-KEM-768 key encapsulation/decapsulation, and ML-DSA-65 signing/verification with complete C code examples
- Implementing a hybrid X25519 + ML-KEM-768 key exchange pattern that combines both shared secrets via HKDF for defense-in-depth
- Using OpenSSL 3.5+ native PQC support through the standard EVP API for ML-KEM key generation without external dependencies

What's Next: Unit 11.5 addresses testing and validation, covering NIST test vector validation, interoperability testing, and comprehensive migration validation checklists.

Exercises

Exercise 11.4.1: liboqs Integration ***(Intermediate)***

a) Install liboqs on your system b) Write a program that generates ML-KEM-768 keys and performs encapsulation/decapsulation c) Measure the performance (key generation, encaps, decaps times)

Exercise 11.4.2: Hybrid Implementation ***(Advanced)***

Implement a hybrid signature scheme that: a) Generates both ECDSA (P-256) and ML-DSA-65 key pairs b) Signs a message with both algorithms c) Concatenates both signatures d) Verifies both signatures (both must pass)

Solutions

Solution 11.4.1

a) Installing liboqs:

```
# Clone and build liboqs
git clone https://github.com/open-quantum-safe/liboqs.git
cd liboqs && mkdir build && cd build
cmake -DCMAKE_INSTALL_PREFIX=/usr/local ..
make -j$(nproc)
sudo make install
sudo ldconfig
```

b) ML-KEM-768 program:

```

/* mlkem_demo.c - ML-KEM-768 key encapsulation demonstration */
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <oqs/oqs.h>
#include <time.h>

int main(void) {
    OQS_KEM *kem = OQS_KEM_new(OQS_KEM_alg_ml_kem_768);
    if (!kem) {
        fprintf(stderr, "ML-KEM-768 not available\n");
        return 1;
    }

    uint8_t *pk = malloc(kem->length_public_key);
    uint8_t *sk = malloc(kem->length_secret_key);
    uint8_t *ct = malloc(kem->length_ciphertext);
    uint8_t *ss_enc = malloc(kem->length_shared_secret);
    uint8_t *ss_dec = malloc(kem->length_shared_secret);

    printf("ML-KEM-768 Parameters:\n");
    printf("  Public key:      %zu bytes\n", kem->length_public_key);
    printf("  Secret key:      %zu bytes\n", kem->length_secret_key);
    printf("  Ciphertext:      %zu bytes\n", kem->length_ciphertext);
    printf("  Shared secret:   %zu bytes\n\n", kem->length_shared_secret);

    /* Key generation */
    clock_t start = clock();
    OQS_KEM_keypair(kem, pk, sk);
    double keygen_time = (double)(clock() - start) / CLOCKS_PER_SEC * 1000;

    /* Encapsulation */
    start = clock();
    OQS_KEM_encaps(kem, ct, ss_enc, pk);
    double encaps_time = (double)(clock() - start) / CLOCKS_PER_SEC * 1000;

    /* Decapsulation */
    start = clock();
    OQS_KEM_decaps(kem, ss_dec, ct, sk);
    double decaps_time = (double)(clock() - start) / CLOCKS_PER_SEC * 1000;

    /* Verify shared secrets match */
    if (memcmp(ss_enc, ss_dec, kem->length_shared_secret) == 0) {
        printf("SUCCESS: Shared secrets match!\n\n");
    } else {
        printf("ERROR: Shared secrets do not match!\n");
    }

    printf("Performance (single operation):\n");
    printf("  Key generation: %.3f ms\n", keygen_time);
    printf("  Encapsulation:  %.3f ms\n", encaps_time);
    printf("  Decapsulation:  %.3f ms\n", decaps_time);

    /* Cleanup */
}

```

```

    OQS_MEM_secure_free(sk, kem->length_secret_key);
    free(pk); free(ct); free(ss_enc); free(ss_dec);
    OQS_KEM_free(kem);
    return 0;
}
/* Compile: gcc -o mlkem_demo mlkem_demo.c -loqs */

```

c) Expected output:

```

ML-KEM-768 Parameters:
  Public key:      1184 bytes
  Secret key:      2400 bytes
  Ciphertext:      1088 bytes
  Shared secret:   32 bytes

SUCCESS: Shared secrets match!

Performance (single operation):
  Key generation:  0.089 ms
  Encapsulation:   0.112 ms
  Decapsulation:   0.098 ms

```

Solution 11.4.2

Hybrid ECDSA + ML-DSA signature implementation:

```

/* hybrid_sig.c - Hybrid ECDSA-P256 + ML-DSA-65 signatures */
#include <stdio.h>
#include <string.h>
#include <openssl/evp.h>
#include <openssl/ec.h>
#include <oqs/oqs.h>

typedef struct {
    uint8_t ecdsa_sig[72];      /* DER-encoded, variable length */
    size_t ecdsa_sig_len;
    uint8_t mldsa_sig[3309];    /* ML-DSA-65 signature */
    size_t mldsa_sig_len;
} hybrid_signature_t;

/* Sign with both algorithms */
int hybrid_sign(const uint8_t *msg, size_t msg_len,
                EVP_PKEY *ecdsa_sk, const uint8_t *mldsa_sk,
                hybrid_signature_t *sig) {
    /* ECDSA signature */
    EVP_MD_CTX *ctx = EVP_MD_CTX_new();
    EVP_DigestSignInit(ctx, NULL, EVP_sha256(), NULL, ecdsa_sk);
    EVP_DigestSignUpdate(ctx, msg, msg_len);
    sig->ecdsa_sig_len = sizeof(sig->ecdsa_sig);
    EVP_DigestSignFinal(ctx, sig->ecdsa_sig, &sig->ecdsa_sig_len);
    EVP_MD_CTX_free(ctx);

    /* ML-DSA-65 signature */
    OQS_SIG *mldsa = OQS_SIG_new(OQS_SIG_alg_ml_dsa_65);

```

```

    OQS_SIG_sign(mlds_a, sig->mlds_a_sig, &sig->mlds_a_sig_len,
                  msg, msg_len, mlds_a_sk);
    OQS_SIG_free(mlds_a);

    return 0;
}

/* Verify both signatures - BOTH must pass */
int hybrid_verify(const uint8_t *msg, size_t msg_len,
                  EVP_PKEY *ecdsa_pk, const uint8_t *mlds_a_pk,
                  const hybrid_signature_t *sig) {
    int ecdsa_ok = 0, mlds_a_ok = 0;

    /* Verify ECDSA */
    EVP_MD_CTX *ctx = EVP_MD_CTX_new();
    EVP_DigestVerifyInit(ctx, NULL, EVP_sha256(), NULL, ecdsa_pk);
    EVP_DigestVerifyUpdate(ctx, msg, msg_len);
    ecdsa_ok = (EVP_DigestVerifyFinal(ctx, sig->ecdsa_sig,
                                      sig->ecdsa_sig_len) == 1);
    EVP_MD_CTX_free(ctx);

    /* Verify ML-DSA-65 */
    OQS_SIG *mlds_a = OQS_SIG_new(OQS_SIG_alg_ml_dsa_65);
    mlds_a_ok = (OQS_SIG_verify(mlds_a, msg, msg_len,
                                sig->mlds_a_sig, sig->mlds_a_sig_len,
                                mlds_a_pk) == OQS_SUCCESS);
    OQS_SIG_free(mlds_a);

    printf("  ECDSA-P256: %s\n", ecdsa_ok ? "VALID" : "INVALID");
    printf("  ML-DSA-65:  %s\n", mlds_a_ok ? "VALID" : "INVALID");

    /* Both must pass for hybrid to be valid */
    return (ecdsa_ok && mlds_a_ok) ? 0 : -1;
}

int main(void) {
    const char *message = "Hybrid signature test message";

    /* Generate ECDSA key pair */
    EVP_PKEY *ecdsa_key = EVP_EC_gen("P-256");

    /* Generate ML-DSA-65 key pair */
    OQS_SIG *mlds_a = OQS_SIG_new(OQS_SIG_alg_ml_dsa_65);
    uint8_t mlds_a_pk[1952], mlds_a_sk[4032];
    OQS_SIG_keypair(mlds_a, mlds_a_pk, mlds_a_sk);
    OQS_SIG_free(mlds_a);

    printf("Hybrid Signature Demo (ECDSA-P256 + ML-DSA-65)\n");
    printf("=====\\n\\n");

    /* Sign */
    hybrid_signature_t sig;
    hybrid_sign((uint8_t*)message, strlen(message),
                ecdsa_key, mlds_a_sk, &sig);

    printf("Signature sizes:\\n");
    printf("  ECDSA:  %zu bytes\\n", sig.ecdsa_sig_len);
    printf("  ML-DSA: %zu bytes\\n", sig.mlds_a_sig_len);
    printf("  Total:  %zu bytes\\n\\n", sig.ecdsa_sig_len + sig.mlds_a_sig_len);
}

```

```

/* Verify */
printf("Verification:\n");
int result = hybrid_verify((uint8_t*)message, strlen(message),
                           ecdsa_key, mldsa_pk, &sig);
printf("\nHybrid result: %s\n", result == 0 ? "VALID" : "INVALID");

EVP_PKEY_free(ecdsa_key);
return 0;
}
/* Compile: gcc -o hybrid_sig hybrid_sig.c -lqps -lssl -lcrypto */

```

Unit 11.5: Testing and Validation

Prerequisites: Unit 11.4 (Library Selection)

Estimated Time: 2-3 hours

Learning Objectives

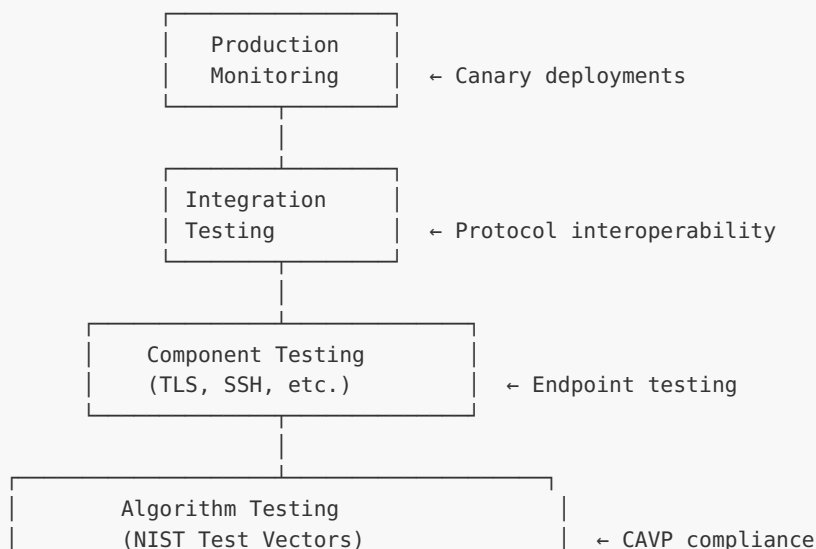
After completing this unit, you will be able to:

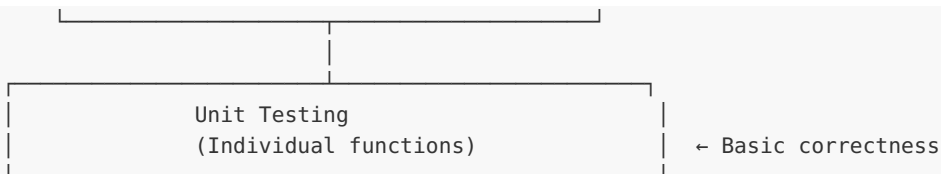
1. **Design comprehensive test suites** for cryptographic migrations
2. **Use NIST test vectors** for algorithm validation
3. **Implement interoperability testing**
4. **Perform security validation** for migrated systems

11.5.1 Test Strategy Framework

PQC Migration Testing Pyramid

=====





11.5.2 NIST Test Vector Validation

```
// Test vector validation example for ML-KEM-768
#include <oqs/oqs.h>
#include <stdio.h>
#include <string.h>

// Known Answer Test (KAT) from NIST
typedef struct {
    uint8_t seed[48];          // Random seed
    uint8_t pk[1184];          // Expected public key
    uint8_t sk[2400];          // Expected secret key
    uint8_t ct[1088];          // Expected ciphertext
    uint8_t ss[32];            // Expected shared secret
} ml_kem_768_kat_t;

int validate_kat(const ml_kem_768_kat_t *kat) {
    OQS_KEM *kem = OQS_KEM_new(OQS_KEM_alg_ml_kem_768);
    if (!kem) return -1;

    uint8_t pk[1184], sk[2400], ct[1088], ss_enc[32], ss_dec[32];

    /*
     * KAT validation requires deterministic key generation from a fixed seed.
     * liboqs exposes this via the internal NIST DRBG:
     * 1. Seed the DRBG with OQS_randombytes_nist_kat_init_256bit(kat->seed, NULL)
     * 2. Call OQS_KEM_keypair() which will draw from the seeded DRBG
     * This matches the NIST KAT generation process (PQCgenKAT_kem).
     * See: https://github.com/open-quantum-safe/liboqs/blob/main/tests/kat\_kem.c
     */
    OQS_randombytes_nist_kat_init_256bit(kat->seed, NULL);
    if (OQS_KEM_keypair(kem, pk, sk) != OQS_SUCCESS) {
        printf("FAIL: Key generation failed\n");
        OQS_KEM_free(kem);
        return -1;
    }

    // Validate public key matches
    if (memcmp(pk, kat->pk, sizeof(pk)) != 0) {
        printf("FAIL: Public key mismatch\n");
        OQS_KEM_free(kem);
        return -1;
    }

    // Encapsulate – DRBG state continues from keygen, matching KAT sequence
    if (OQS_KEM_encaps(kem, ct, ss_enc, pk) != OQS_SUCCESS) {
        printf("FAIL: Encapsulation failed\n");
        OQS_KEM_free(kem);
        return -1;
    }
}
```



```

    if (memcmp(ct, kat->ct, sizeof(ct)) != 0) {
        printf("FAIL: Ciphertext mismatch\n");
        OQS_KEM_free(kem);
        return -1;
    }

    // Validate decapsulation produces expected shared secret
    if (OQS_KEM_decaps(kem, ss_dec, ct, sk) != OQS_SUCCESS) {
        printf("FAIL: Decapsulation failed\n");
        OQS_KEM_free(kem);
        return -1;
    }

    if (memcmp(ss_dec, kat->ss, sizeof(ss_dec)) != 0) {
        printf("FAIL: Shared secret mismatch\n");
        OQS_KEM_free(kem);
        return -1;
    }

    printf("PASS: All KAT vectors validated\n");
    OQS_KEM_free(kem);
    return 0;
}

// Failure case testing
int test_decapsulation_failure(void) {
    OQS_KEM *kem = OQS_KEM_new(OQS_KEM_alg_ml_kem_768);
    uint8_t pk[1184], sk[2400], ct[1088], ss1[32], ss2[32];

    // Generate valid keys and ciphertext
    OQS_KEM_keypair(kem, pk, sk);
    OQS_KEM_encaps(kem, ct, ss1, pk);

    // Corrupt ciphertext (bit flip)
    ct[0] ^= 0x01;

    // Decapsulation should still succeed (implicit rejection)
    // but produce a different (pseudorandom) shared secret
    OQS_KEM_decaps(kem, ss2, ct, sk);

    // Shared secrets must NOT match
    if (memcmp(ss1, ss2, 32) == 0) {
        printf("FAIL: Corrupted ciphertext produced same shared secret\n");
        OQS_KEM_free(kem);
        return -1;
    }

    printf("PASS: Implicit rejection working correctly\n");
    OQS_KEM_free(kem);
    return 0;
}

```

11.5.3 Interoperability Testing

```
#!/bin/bash
# TLS PQC interoperability test

# Start server with hybrid PQC support
openssl s_server -accept 4433 \
    -cert server.crt -key server.key \
    -groups X25519MLKEM768:X25519 \
    &
SERVER_PID=$!

sleep 1

# Test client connection with hybrid
echo "Testing X25519MLKEM768 key exchange..."
echo | openssl s_client -connect localhost:4433 \
    -groups X25519MLKEM768 \
    2>&1 | grep "Peer signature type\|Server Temp Key"

# Test fallback to classical
echo "Testing X25519 fallback..."
echo | openssl s_client -connect localhost:4433 \
    -groups X25519 \
    2>&1 | grep "Peer signature type\|Server Temp Key"

kill $SERVER_PID
```

11.5.4 Migration Validation Checklist

```
Pre-Migration Validation
=====
[ ] Cryptographic inventory complete
[ ] All dependencies documented
[ ] Test environment mirrors production
[ ] Rollback procedure documented
[ ] Performance baseline established

Algorithm Validation
=====
[ ] NIST test vectors pass (CAVP)
[ ] Key generation produces valid keys
[ ] Encapsulation/signing works correctly
[ ] Decapsulation/verification works correctly
[ ] Failure cases handled properly (implicit rejection)
[ ] No timing side channels (constant-time validation)

Integration Validation
=====
[ ] TLS handshakes complete successfully
[ ] Certificate chain validation works
[ ] Protocol negotiation falls back correctly
[ ] Mixed environment interoperability tested
[ ] Performance within acceptable bounds

Security Validation
=====
```

```
[ ] No private key material in logs
[ ] Secure memory handling (zeroization)
[ ] Proper random number generation
[ ] Side-channel resistance verified
[ ] HSM integration tested (if applicable)
```

Production Validation

```
=====
```

```
[ ] Canary deployment successful
[ ] Monitoring and alerting configured
[ ] Rollback tested in production
[ ] Performance metrics acceptable
[ ] No increase in error rates
```

11.5.5 ACVP Testing and CAVP Validation

For organizations requiring FIPS 140-3 validated cryptographic modules, NIST's Automated Cryptographic Validation Protocol (ACVP) is the primary path to certification.

What is ACVP?

ACVP (defined in NIST SP 800-185 and the ACVP specification) is a protocol for automated testing of cryptographic algorithm implementations. It replaces the older manual CAVS (Cryptographic Algorithm Validation System) process:

1. **Test Vector Request:** Your implementation (the "module") connects to the ACVP server and requests test vectors for a specific algorithm (e.g., ML-KEM-768, ML-DSA-65)
2. **Computation:** Your module performs the cryptographic operations on the provided inputs
3. **Response Submission:** Results are submitted back to the server for verification
4. **Validation Certificate:** If all test vectors pass, NIST issues a validation certificate

PQC Algorithms with ACVP Support (as of 2026):

ALGORITHM	ACVP STATUS	TEST TYPES
ML-KEM (FIPS 203)	Available	KeyGen, Encaps, Decaps, KAT
ML-DSA (FIPS 204)	Available	KeyGen, Sign, Verify, KAT
SLH-DSA (FIPS 205)	Available	KeyGen, Sign, Verify
FN-DSA (FIPS 206)	Draft	KeyGen, Sign, Verify

Obtaining and Running ACVP Test Vectors:

```
# Open-source ACVP client: libacvp (Cisco)
# https://github.com/cisco/libacvp

# 1. Register with NIST ACVP server (requires CMVP Lab accreditation)
# 2. Configure algorithm capabilities
# 3. Run automated testing

# For development/testing without NIST server access,
# use the ACVP proxy with local test vectors:
git clone https://github.com/smuelleDD/acvp-proxy
cd acvp-proxy && make

# Or use NIST's published KAT files directly:
# https://csrc.nist.gov/projects/post-quantum-cryptography
```

CAVP vs CMVP Validation Path:

STEP	WHAT	TIMELINE
1. Algorithm Testing (CAVP)	Validate algorithm correctness via ACVP	2-4 weeks
2. Module Testing (CMVP)	Full FIPS 140-3 module validation	6-18 months
3. Certificate Issuance	NIST issues validation certificate	After testing

PRACTICAL NOTE:

FIPS 140-3 validation is expensive (\$50K-\$200K+) and time-consuming. For most organizations, using an already-validated module (OpenSSL, BoringSSL, wolfCrypt) is more practical than seeking your own validation. Check the [CMVP validated modules list](#) for modules with PQC algorithm certificates.

Unit 11.5 Summary

This unit covered:

- A five-layer PQC migration testing pyramid: unit testing, algorithm testing (NIST CAVP test vectors), component testing, integration testing, and production monitoring
- NIST Known Answer Test (KAT) vector validation for ML-KEM-768, including deterministic key generation from seeds and ciphertext/shared secret verification
- Failure case testing including corrupted ciphertext handling (verifying ML-KEM's implicit rejection mechanism) and wrong-key detection
- TLS interoperability testing using OpenSSL with hybrid X25519MLKEM768 key exchange and classical fallback scenarios

- A comprehensive migration validation checklist spanning pre-migration, algorithm, integration, security, and production validation stages

What's Next: Unit 11.6 covers phased migration execution, including a five-phase enterprise migration timeline, hybrid deployment strategies, rollback procedures, and production monitoring.

Exercises

Exercise 11.5.1: Test Suite Development **(Intermediate)**

Create a test suite that validates: a) ML-KEM-768 key generation, encapsulation, decapsulation b) Corrupted ciphertext handling (implicit rejection) c) Wrong key detection d) Performance benchmarks

Exercise 11.5.2: Interoperability Testing **(Advanced)**

Using OpenSSL 3.5 or oqs-provider: a) Set up a TLS server with hybrid key exchange b) Connect with a client supporting X25519MLKEM768 c) Verify the connection uses the hybrid algorithm d) Test fallback when client only supports X25519

Solutions

Solution 11.5.1

ML-KEM-768 test suite:

```
/* mlkem_test_suite.c - Comprehensive ML-KEM-768 validation */
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>
#include <oqs/oqs.h>

#define TEST_ITERATIONS 1000
#define PASS "\033[32mPASS\033[0m"
#define FAIL "\033[31mFAIL\033[0m"

static int tests_passed = 0, tests_failed = 0;

void test_basic_kem_cycle(OQS_KEM *kem) {
    printf("Test: Basic KEM cycle... ");

    uint8_t *pk = malloc(kem->length_public_key);
    uint8_t *sk = malloc(kem->length_secret_key);
    uint8_t *ct = malloc(kem->length_ciphertext);
    uint8_t *ss_enc = malloc(kem->length_shared_secret);
    uint8_t *ss_dec = malloc(kem->length_shared_secret);

    if (!pk || !sk || !ct || !ss_enc || !ss_dec) {
```

```

        printf("%s (memory allocation failed)\n", FAIL);
        tests_failed++;
        free(pk); free(sk); free(ct); free(ss_enc); free(ss_dec);
        return;
    }

    if (OQS_KEM_keypair(kem, pk, sk) != OQS_SUCCESS ||
        OQS_KEM_encaps(kem, ct, ss_enc, pk) != OQS_SUCCESS ||
        OQS_KEM_decaps(kem, ss_dec, ct, sk) != OQS_SUCCESS) {
        printf("%s (OQS operation failed)\n", FAIL);
        tests_failed++;
        free(pk); free(sk); free(ct); free(ss_enc); free(ss_dec);
        return;
    }

    if (memcmp(ss_enc, ss_dec, kem->length_shared_secret) == 0) {
        printf("%s\n", PASS);
        tests_passed++;
    } else {
        printf("%s\n", FAIL);
        tests_failed++;
    }

    free(pk); free(sk); free(ct); free(ss_enc); free(ss_dec);
}

void test_corrupted_ciphertext(OQS_KEM *kem) {
    printf("Test: Corrupted ciphertext (implicit rejection)... ");

    uint8_t *pk = malloc(kem->length_public_key);
    uint8_t *sk = malloc(kem->length_secret_key);
    uint8_t *ct = malloc(kem->length_ciphertext);
    uint8_t *ss_enc = malloc(kem->length_shared_secret);
    uint8_t *ss_dec = malloc(kem->length_shared_secret);

    OQS_KEM_keypair(kem, pk, sk);
    OQS_KEM_encaps(kem, ct, ss_enc, pk);

    /* Corrupt the ciphertext */
    ct[0] ^= 0xFF;
    ct[100] ^= 0xFF;

    OQS_KEM_decaps(kem, ss_dec, ct, sk);

    /* ML-KEM uses implicit rejection: corrupted CT yields random SS */
    if (memcmp(ss_enc, ss_dec, kem->length_shared_secret) != 0) {
        printf("%s (implicit rejection working)\n", PASS);
        tests_passed++;
    } else {
        printf("%s (should have rejected)\n", FAIL);
        tests_failed++;
    }

    free(pk); free(sk); free(ct); free(ss_enc); free(ss_dec);
}

void test_wrong_secret_key(OQS_KEM *kem) {
    printf("Test: Wrong secret key detection... ");

    uint8_t *pk1 = malloc(kem->length_public_key);

```

```

uint8_t *sk1 = malloc(kem->length_secret_key);
uint8_t *pk2 = malloc(kem->length_public_key);
uint8_t *sk2 = malloc(kem->length_secret_key);
uint8_t *ct = malloc(kem->length_ciphertext);
uint8_t *ss_enc = malloc(kem->length_shared_secret);
uint8_t *ss_dec = malloc(kem->length_shared_secret);

/* Generate two independent key pairs */
OQS_KEM_keypair(kem, pk1, sk1);
OQS_KEM_keypair(kem, pk2, sk2);

/* Encapsulate with pk1, try to decapsulate with sk2 */
OQS_KEM_encaps(kem, ct, ss_enc, pk1);
OQS_KEM_decaps(kem, ss_dec, ct, sk2);

if (memcmp(ss_enc, ss_dec, kem->length_shared_secret) != 0) {
    printf("%s\n", PASS);
    tests_passed++;
} else {
    printf("%s (should not match)\n", FAIL);
    tests_failed++;
}

free(pk1); free(sk1); free(pk2); free(sk2);
free(ct); free(ss_enc); free(ss_dec);
}

void test_performance(OQS_KEM *kem) {
    printf("Test: Performance benchmark (%d iterations)...\n", TEST_ITERATIONS);

    uint8_t *pk = malloc(kem->length_public_key);
    uint8_t *sk = malloc(kem->length_secret_key);
    uint8_t *ct = malloc(kem->length_ciphertext);
    uint8_t *ss = malloc(kem->length_shared_secret);

    clock_t start = clock();
    for (int i = 0; i < TEST_ITERATIONS; i++)
        OQS_KEM_keypair(kem, pk, sk);
    double keygen = (double)(clock() - start) / CLOCKS_PER_SEC * 1000 / TEST_ITERATIONS;

    start = clock();
    for (int i = 0; i < TEST_ITERATIONS; i++)
        OQS_KEM_encaps(kem, ct, ss, pk);
    double encaps = (double)(clock() - start) / CLOCKS_PER_SEC * 1000 / TEST_ITERATIONS;

    start = clock();
    for (int i = 0; i < TEST_ITERATIONS; i++)
        OQS_KEM_decaps(kem, ss, ct, sk);
    double decaps = (double)(clock() - start) / CLOCKS_PER_SEC * 1000 / TEST_ITERATIONS;

    printf(" KeyGen:  %.4f ms/op\n", keygen);
    printf(" Encaps:  %.4f ms/op\n", encaps);
    printf(" Decaps:  %.4f ms/op\n", decaps);
    tests_passed++;

    free(pk); free(sk); free(ct); free(ss);
}

int main(void) {
    printf("ML-KEM-768 Test Suite\n");

```

```

printf("=====\n\n");

OQS_KEM *kem = OQS_KEM_new(OQS_KEM_alg_ml_kem_768);
if (!kem) { printf("ML-KEM-768 not available\n"); return 1; }

test_basic_kem_cycle(kem);
test_corrupted_ciphertext(kem);
test_wrong_secret_key(kem);
test_performance(kem);

printf("\n=== Results: %d passed, %d failed ===\n",
       tests_passed, tests_failed);

OQS_KEM_free(kem);
return tests_failed > 0 ? 1 : 0;
}

```

Solution 11.5.2

TLS interoperability testing with oqs-provider:

```

#!/bin/bash
# tls_pqc_interop_test.sh - Hybrid TLS interoperability testing

echo "=== PQC TLS Interoperability Test ==="
echo ""

# Prerequisites: OpenSSL 3.x with oqs-provider installed
# Install: https://github.com/open-quantum-safe/oqs-provider

# a) Start TLS server with hybrid key exchange
echo "Step 1: Starting TLS server with X25519MLKEM768..."
openssl s_server -accept 4433 \
    -cert server.crt -key server.key \
    -groups X25519MLKEM768:X25519 \
    -www &
SERVER_PID=$!
sleep 2

# b) Connect with hybrid-capable client
echo "Step 2: Testing hybrid client connection..."
echo "GET /" | openssl s_client -connect localhost:4433 \
    -groups X25519MLKEM768 \
    -brief 2>&1 | grep -E "Protocol|Cipher|Server Temp Key"

# c) Verify hybrid algorithm was used
echo ""
echo "Step 3: Verifying hybrid key exchange..."
RESULT=$(echo "GET /" | openssl s_client -connect localhost:4433 \
    -groups X25519MLKEM768 2>&1 | grep "Server Temp Key")

if echo "$RESULT" | grep -q "X25519MLKEM768"; then
    echo "✓ SUCCESS: Hybrid X25519MLKEM768 negotiated"
else
    echo "✗ FAILED: Hybrid not negotiated"
fi

```



```
# d) Test fallback to X25519 only
echo ""
echo "Step 4: Testing fallback (X25519 only client)..."
FALLBACK=$(echo "GET /" | openssl s_client -connect localhost:4433 \
  -groups X25519 2>&1 | grep "Server Temp Key")

if echo "$FALLBACK" | grep -q "X25519"; then
  echo "✓ SUCCESS: Fallback to X25519 works"
else
  echo "✗ FAILED: Fallback not working"
fi

# Cleanup
kill $SERVER_PID 2>/dev/null
echo ""
echo "=== Test Complete ==="
```

Expected output:

```
=== PQC TLS Interoperability Test ===

Step 1: Starting TLS server with X25519MLKEM768...
Step 2: Testing hybrid client connection...
Protocol version: TLSv1.3
Ciphersuite: TLS_AES_256_GCM_SHA384
Server Temp Key: X25519MLKEM768, 1216 bits

Step 3: Verifying hybrid key exchange...
✓ SUCCESS: Hybrid X25519MLKEM768 negotiated

Step 4: Testing fallback (X25519 only client)...
✓ SUCCESS: Fallback to X25519 works

=== Test Complete ===
```

Unit 11.6: Phased Migration Execution

Prerequisites: Unit 11.5 (Testing and Validation)

Estimated Time: 2-3 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Execute a phased migration plan** aligned with industry timelines
2. **Implement hybrid cryptography** as a migration bridge
3. **Manage rollback procedures** for failed migrations

4. **Monitor and validate** production deployments

11.6.1 Recommended Migration Phases

Enterprise PQC Migration Timeline
=====

Phase 1: Discovery & Planning (2025-2026)
└─ Complete cryptographic inventory
└─ Assess risk and prioritize systems
└─ Select libraries and tooling
└─ Build crypto-agile architecture
└─ Train development teams
└─ Establish testing infrastructure

Phase 2: Pilot & Validation (2026-2027)
└─ Deploy PQC in non-production environments
└─ Test hybrid configurations
└─ Validate performance impact
└─ Interoperability testing
└─ Update operational procedures

Phase 3: High-Risk Migration (2027-2029)
└─ Migrate internet-facing services
└─ Update long-term encryption keys
└─ Deploy hybrid TLS/SSH
└─ Update PKI infrastructure
└─ Migrate VPN gateways

Phase 4: Broad Migration (2029-2032)
└─ Internal application migration
└─ Database encryption updates
└─ Legacy system remediation
└─ Third-party integration updates
└─ Complete certificate rotation

Phase 5: Classical Deprecation (2032-2035)
└─ Remove classical-only configurations
└─ Transition from hybrid to pure PQC
└─ Decommission deprecated algorithms
└─ Final compliance validation
└─ Continuous monitoring

11.6.2 Hybrid Deployment Strategy

TLS Migration Path
=====

Stage 1: Add Hybrid Support (No Breaking Changes)

Server cipher suite preference:
1. X25519MLKEM768 ← Hybrid (preferred)
2. X25519 ← Classical fallback
3. secp256r1 ← Legacy fallback

Result: PQC-capable clients use hybrid, others fall back safely

Stage 2: Prefer Hybrid (Soft Enforcement)

Server cipher suite preference:

1. X25519MLKEM768 ← Hybrid (required for new clients)
2. X25519 ← Grace period for legacy

Monitoring: Alert on classical-only connections

Stage 3: Require Hybrid (Hard Enforcement)

Server cipher suite preference:

1. X25519MLKEM768 ← Hybrid only
2. ML-KEM-768 ← Pure PQC alternative

Action: Reject classical-only connections

Stage 4: Pure PQC (Final State)

Server cipher suite preference:

1. ML-KEM-1024 ← Pure PQC, highest security
2. ML-KEM-768 ← Pure PQC, balanced

Result: Fully quantum-resistant

11.6.3 Rollback Procedures

```
// Configuration-based rollback capability
typedef struct {
    bool enable_pqc;
    bool enable_hybrid;
    bool enable_classical_fallback;
    const char *preferred_kem;
    const char *preferred_sig;
    time_t pqc_enabled_time;
} crypto_migration_config_t;

// Feature flag for gradual rollout
int configure_tls_for_migration(SSL_CTX *ctx,
                               const crypto_migration_config_t *config) {

    if (config->enable_pqc && config->enable_hybrid) {
        // Hybrid mode: prefer PQC, allow classical fallback
        SSL_CTX_set1_groups_list(ctx,
                                "X25519MLKEM768:X25519:secp256r1");
    } else if (config->enable_classical_fallback) {
        // Rollback mode: classical only
        SSL_CTX_set1_groups_list(ctx,
                                "X25519:secp256r1:secp384r1");
    } else {
        // Pure PQC mode (future state)
        SSL_CTX_set1_groups_list(ctx,
                                "ML-KEM-1024:ML-KEM-768");
    }
}
```

```

        return 0;
    }

    // Emergency rollback procedure
    void emergency_pqc_rollback(crypto_migration_config_t *config) {
        config->enable_pqc = false;
        config->enable_hybrid = false;
        config->enable_classical_fallback = true;

        // Log rollback event
        log_security_event("PQC_ROLLBACK",
            "Emergency rollback to classical cryptography initiated");

        // Alert operations team
        send_alert("CRITICAL: PQC rollback activated");
    }

```

11.6.4 Production Monitoring

```

# Prometheus metrics for PQC migration monitoring
metrics:
    # Key exchange algorithm distribution
    - name: tls_key_exchange_total
      type: counter
      labels:
        - algorithm # X25519MLKEM768, X25519, etc.
      alert_on: classical_only_percentage > 50%

    # Performance impact
    - name: tls_handshake_duration_seconds
      type: histogram
      labels:
        - algorithm
      alert_on: p99 > 500ms

    # Error rates
    - name: tls_handshake_errors_total
      type: counter
      labels:
        - error_type
        - algorithm
      alert_on: error_rate > 1%

    # Certificate validation
    - name: certificate_validation_total
      type: counter
      labels:
        - signature_algorithm # ML-DSA-65, ECDSA, etc.
        - result # success, failure

# Grafana dashboard queries
dashboards:
    pqc_adoption:
        - query: |
            sum(rate(tls_key_exchange_total{algorithm="X25519MLKEM768"}[5m])) /
            sum(rate(tls_key_exchange_total[5m])) * 100
            title: "PQC Adoption Rate (%)"

```

```
performance_comparison:
- query: |
    histogram_quantile(0.99,
      sum(rate(tls_handshake_duration_seconds_bucket[5m]))
      by (le, algorithm))
    title: "TLS Handshake P99 by Algorithm"
```

Exercises

Exercise 11.6.1: Migration Plan *(Intermediate)*

Create a detailed migration plan for a web application that: a) Currently uses TLS 1.2 with ECDHE-RSA b) Has 1 million daily users c) Must maintain 99.9% availability d) Needs to comply with 2030 NIST deadline

Include rollback procedures and monitoring strategy.

Exercise 11.6.2: Canary Deployment *(Advanced)*

Design a canary deployment strategy that: a) Routes 1% of traffic to PQC-enabled servers b) Monitors error rates and latency c) Automatically rolls back if errors exceed threshold d) Gradually increases PQC traffic if successful

Solutions

Solution 11.6.1

Detailed Migration Plan for High-Availability Web Application:

```
# PQC Migration Plan: High-Traffic Web Application
## Version 1.0 | Target: 2030 NIST Compliance

### Executive Summary
- Current state: TLS 1.2 with ECDHE-RSA
- Scale: 1M daily active users
- SLA requirement: 99.9% availability (8.76 hours downtime/year max)
- Timeline: 4-year phased migration (2026-2030)

---

### Phase 1: Assessment & Planning (Q1-Q2 2026)

**1.1 Cryptographic Inventory**
- [ ] Audit all TLS termination points (load balancers, CDN, origin servers)
- [ ] Document certificate chains and CA dependencies
- [ ] Identify client compatibility requirements (browser versions, mobile apps)
- [ ] Map all internal service-to-service TLS connections

**1.2 Infrastructure Assessment**
- [ ] Benchmark current TLS handshake performance
```

- [] Assess hardware/software upgrade requirements
- [] Evaluate CDN provider PQC support timeline
- [] Test lab environment with hybrid TLS

****Deliverable:**** Migration architecture document

Phase 2: Hybrid Deployment - Non-Production (Q3-Q4 2026)

****2.1 Staging Environment****

- Deploy hybrid TLS (X25519MLKEM768 + X25519 fallback)
- Configure certificate with ML-DSA hybrid signature
- Run load tests simulating 2x peak traffic

****2.2 Validation Criteria****

- Handshake latency increase < 15%
- No functionality regressions
- All automated tests pass

Phase 3: Canary Production (Q1-Q2 2027)

****3.1 Traffic Routing****

- Week 1-2: 1% traffic to PQC servers
- Week 3-4: 5% if metrics pass
- Week 5-6: 10% if metrics pass

****3.2 Monitoring****

- TLS handshake success rate (target: >99.9%)
- P99 latency (target: <500ms increase)
- Error rate by client type

****3.3 Rollback Triggers (Automatic)****

- Error rate > 1% for 5 minutes
- P99 latency increase > 100%
- Certificate validation failures > 0.1%

Phase 4: Gradual Rollout (Q3 2027 - Q4 2028)

Month	PQC Traffic	Monitoring Focus
1-2	25%	Legacy client impact
3-4	50%	Performance at scale
5-6	75%	Edge cases
7-8	90%	Remaining classical clients
9+	99%	Deprecation warnings

Phase 5: Classical Deprecation (2029)

- Notify users of classical TLS deprecation
- Provide upgrade guidance for legacy clients
- Set classical TLS sunset date: December 31, 2029

```

### Phase 6: Pure PQC (Q1 2030)

- Remove classical algorithm support
- Update SLA documentation
- Final compliance audit

---

### Rollback Procedures

**Level 1 (Automatic):** Traffic routing returns to classical servers
- Trigger: Automated threshold breach
- Recovery time: <30 seconds
- Data impact: None

**Level 2 (Manual):** Configuration rollback
- Trigger: Persistent issues after traffic shift
- Recovery time: <15 minutes
- Approval: On-call engineer

**Level 3 (Emergency):** Full revert to previous deployment
- Trigger: Critical security issue or extended outage
- Recovery time: <1 hour
- Approval: Engineering lead + Security team

---

### Monitoring Strategy

1. **Real-time dashboards** showing PQC adoption percentage
2. **Alerting** on error rate and latency anomalies
3. **Weekly reports** on migration progress
4. **Client telemetry** tracking TLS negotiation outcomes

```

Solution 11.6.2

Canary Deployment Configuration:

```

# canary_deployment.yaml - PQC Canary Strategy
apiVersion: flagger.app/v1beta1
kind: Canary
metadata:
  name: web-frontend-pqc
  namespace: production
spec:
  targetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: web-frontend-pqc
  progressDeadlineSeconds: 600
  service:
    port: 443
    targetPort: 8443

analysis:

```

```

# Start with 1% traffic
interval: 1m
threshold: 5          # Max failed checks before rollback
maxWeight: 50         # Max traffic percentage for canary
stepWeight: 5         # Increment per successful interval

# Metrics for success/failure decisions
metrics:
  # Error rate must stay below 1%
  - name: request-success-rate
    thresholdRange:
      min: 99
      interval: 1m

  # TLS handshake latency P99 < 500ms
  - name: tls-handshake-duration
    templateRef:
      name: tls-latency
    thresholdRange:
      max: 500

  # PQC negotiation success rate
  - name: pqc-negotiation-rate
    templateRef:
      name: pqc-success
    thresholdRange:
      min: 95

# Webhook alerts
webhooks:
  - name: slack-notification
    url: https://hooks.slack.com/services/xxx/yyy/zzz
    timeout: 5s

---
# Custom metric templates
apiVersion: flagger.app/v1beta1
kind: MetricTemplate
metadata:
  name: tls-latency
spec:
  provider:
    type: prometheus
    address: http://prometheus:9090
  query: |
    histogram_quantile(0.99,
      sum(rate(tls_handshake_duration_seconds_bucket{
        namespace="{{ namespace }}",
        deployment=~"{{ target }}"
      }[1m])) by (le)
    ) * 1000

---
apiVersion: flagger.app/v1beta1
kind: MetricTemplate
metadata:
  name: pqc-success
spec:
  provider:
    type: prometheus

```



```

address: http://prometheus:9090
query: |
  sum(rate(tls_key_exchange_total{
    algorithm="X25519MLKEM768",
    namespace="{{ namespace }}",
    deployment=~"{{ target }}"
  }[1m])) /
  sum(rate(tls_key_exchange_total{
    namespace="{{ namespace }}",
    deployment=~"{{ target }}"
  }[1m])) * 100

```

Deployment progression example:

Time	Traffic	Status
-----	-----	-----
00:00	1%	Analyzing...
01:00	1%	Success rate: 99.8%, Latency: 245ms ✓
02:00	5%	Promoting...
03:00	5%	Success rate: 99.7%, Latency: 251ms ✓
04:00	10%	Promoting...
05:00	10%	Success rate: 99.6%, Latency: 263ms ✓
...		
30:00	50%	Stable - holding for extended observation

Unit 11.6 Summary

Unit 11.6 covered the practical execution of PQC migration through phased rollouts. Key concepts: canary deployments for early risk detection, traffic-based promotion gates with automated rollback triggers, latency and error-rate monitoring during algorithm transitions, and environment-specific migration sequencing (dev → staging → production). The phased approach minimizes blast radius while maintaining service availability throughout the transition.

Unit 11.6b: Key Rotation Strategies During Hybrid Transition

Prerequisites: Unit 11.6 (Phased Migration Execution)

Estimated Time: 1-2 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Design dual-key management architectures** for the classical-to-PQC transition period

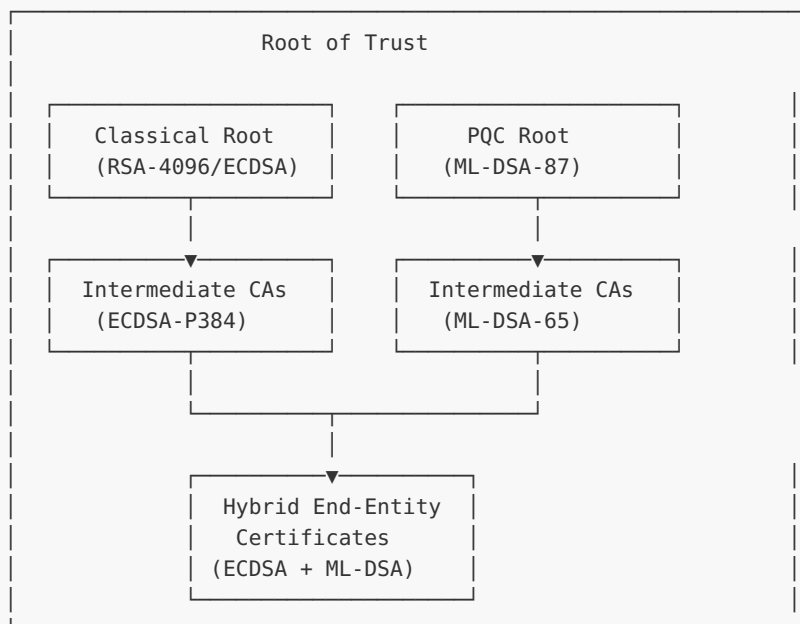
2. **Implement key rotation strategies** that maintain service availability during migration
3. **Plan certificate lifecycle management** across hybrid PKI deployments
4. **Handle key synchronization challenges** in distributed systems during transition
5. **Establish rollback-safe key rotation procedures** for production environments

The hybrid transition period introduces unique key management challenges. Organizations must maintain both classical and PQC keys simultaneously while ensuring seamless rotation without service disruption.

11.6b.1 Dual-Key Management Approaches

Hybrid Key Management Architecture

Option 1: Parallel Key Hierarchies



Option 2: Composite Key Pairs

Single certificate containing both algorithm types:

```

├─ Subject: CN=server.example.com
├─ Public Key (Composite):
│   ├── Classical: ECDSA-P256
│   └─ PQC: ML-DSA-44
├─ Signature (Composite):
│   ├── Classical: ECDSA from issuer
│   └─ PQC: ML-DSA from issuer
└─ Extensions: Hybrid policy OID

```

Option 3: Dual Certificates (Catalyst Approach)

Two separate certificates for same identity:

- └ Certificate A: CN=server.example.com (ECDSA)
- └ Certificate B: CN=server.example.com (ML-DSA)
- └ Server presents both; client validates preferred

11.6b.2 Key Rotation Timelines

Rotation Schedule by Key Type:

KEY TYPE	CLASSICAL ROTATION	HYBRID PERIOD ROTATION	PURE PQC ROTATION
Root CA	10-20 years	5-10 years (accelerated)	10-20 years
Intermediate CA	5-10 years	2-5 years	5-10 years
End-Entity (Server)	1-2 years	6-12 months	1-2 years
Code Signing	3-5 years	1-2 years	3-5 years
Ephemeral (TLS session)	Per-session	Per-session	Per-session

Recommended Hybrid Transition Timeline:

Year	Action	Key State
----	-----	-----
2025	Generate PQC root keys (offline)	Classical active, PQC standby
2026	Issue first PQC intermediate CAs	Classical active, PQC testing
2027	Begin hybrid end-entity certificate issuance	Hybrid active
2028	Rotate 50% of infrastructure to hybrid	Hybrid dominant
2029	Complete hybrid deployment	Full hybrid coverage
2030	Begin pure PQC certificate issuance	Hybrid + PQC active
2032	Deprecate classical-only certificates	PQC preferred
2035	Revoke remaining classical certificates	Pure PQC

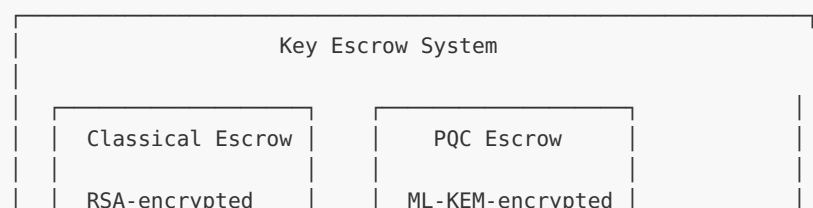
11.6b.3 Certificate Lifecycle Considerations

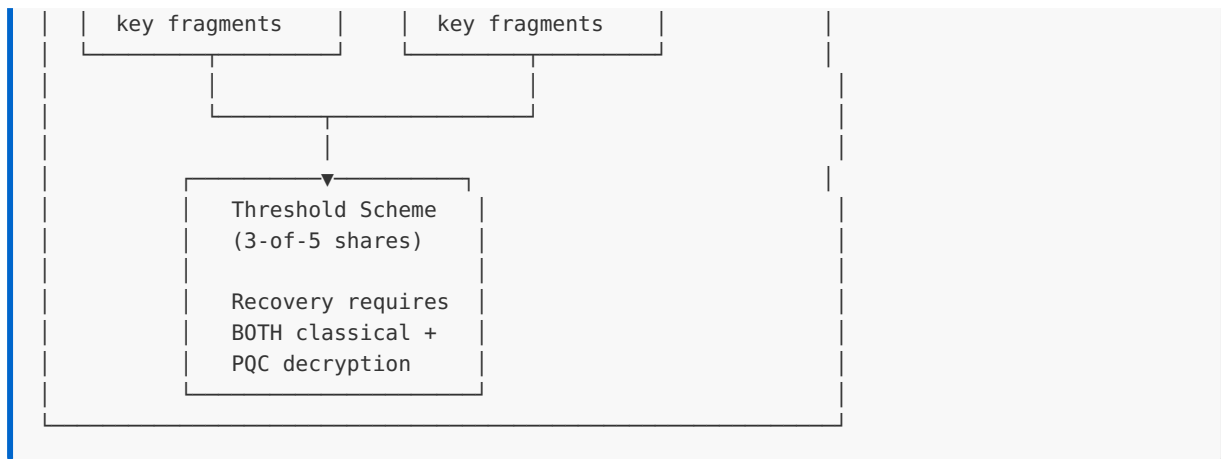
Key Escrow and Recovery During Transition:

Hybrid Key Escrow Architecture

=====

For encryption keys (not signing keys):





Certificate Revocation Strategy:

SCENARIO	CLASSICAL CRL/ OCSP	PQC SIGNED RESPONSE	RECOMMENDATION
Normal operation	Active	Active	Validate both signatures
Classical compromise	Revoke immediately	Update with new PQC sig	Trust PQC response
PQC implementation flaw	Maintain	Suspend until patched	Fall back to classical
Both compromised	Emergency re-key	Emergency re-key	Incident response

Automated Rotation Workflow:

```

# Conceptual workflow for automated hybrid key rotation
class HybridKeyRotation:
    def rotate_server_certificate(self, server_id):
        """
        Rotate both classical and PQC components atomically.
        """
        # Step 1: Generate new key pairs
        new_classical_key = generate_ecdsa_key(curve="P-256")
        new_pqc_key = generate_ml_dsa_key(level=44)

        # Step 2: Create CSR with both public keys
        csr = create_hybrid_csr(
            subject=f"CN={server_id}",
            classical_pubkey=new_classical_key.public,
            pqc_pubkey=new_pqc_key.public
        )

        # Step 3: Submit to CA, get hybrid certificate
        new_cert = ca_client.issue_hybrid_certificate(csr)

        # Step 4: Deploy with zero-downtime rotation
        self.deploy_with_overlap(
  
```

```

        server_id,
        old_cert=self.get_current_cert(server_id),
        new_cert=new_cert,
        overlap_period_hours=24
    )

    # Step 5: Verify new certificate is serving
    if not self.verify_certificate_active(server_id, new_cert):
        self.rollback(server_id)
        raise RotationError("New certificate not serving")

    # Step 6: Revoke old certificate after overlap
    self.schedule_revocation(
        self.get_current_cert(server_id),
        delay_hours=24
    )

    # Step 7: Update inventory
    self.update_certificate_inventory(server_id, new_cert)

```

Best Practices for Hybrid Key Rotation:

1. **Maintain Key Correspondence:** Classical and PQC keys should rotate together to maintain security equivalence
2. **Staggered Expiration:** Set PQC key expiry slightly after classical to ensure PQC is always available
3. **Automated Monitoring:** Alert on any certificate with <30 days validity or algorithm mismatch
4. **Rollback Capability:** Keep previous key pair available for 7-30 days post-rotation
5. **Audit Trail:** Log all key generation, rotation, and revocation events with both timestamps and algorithm details

Unit 11.6b Summary

Unit 11.6b addressed key rotation challenges unique to the hybrid transition period. Key concepts: dual-algorithm key lifecycle management, certificate re-issuance strategies for hybrid X.509, grace periods for overlapping key validity, automated rotation with algorithm-aware monitoring, and rollback-safe key archival practices. Proper key rotation ensures continuity during the multi-year transition window.

Exercises

Exercise 11.6b.1: Key Rotation Policy Design *(Intermediate)*

Design a key rotation policy for a hybrid TLS deployment (X25519+ML-KEM-768 for KEX, ECDSA+ML-DSA-65 for authentication). Your policy should specify: a) Rotation intervals for each key type (KEX ephemeral keys, server authentication keys, CA signing keys) b) Grace periods where old and new keys overlap c) Monitoring alerts and automation triggers d) Rollback procedure if a rotated key causes interoperability failures

Solution 11.6b.1

Hybrid Key Rotation Policy:

KEY TYPE	CLASSICAL	PQC	ROTATION INTERVAL	GRACE PERIOD
TLS KEX (ephemeral)	X25519	ML-KEM-768	Per-session (no rotation needed)	N/A
Server auth cert	ECDSA P-256	ML-DSA-65	90 days	14 days overlap
Intermediate CA	ECDSA P-384	ML-DSA-65	3 years	6 months overlap
Root CA	ECDSA P-384	SLH-DSA-128s	20 years	5 years overlap

Monitoring Alerts:

- Certificate expiry < 30 days: WARNING
- Certificate expiry < 7 days: CRITICAL
- Algorithm mismatch between classical/PQC pair: CRITICAL
- Failed rotation attempt: page on-call immediately

Rollback Procedure:

1. Detect failure (elevated TLS handshake errors, client-reported certificate validation failures)
2. Re-deploy previous certificate within grace period (old cert still valid)
3. Root-cause analysis: check client PQC support, certificate chain completeness, OCSP/CRL availability
4. Fix and re-attempt rotation after validation in staging

Exercise 11.6b.2: Hybrid Certificate Rotation Workflow *(Advanced)*

Implement a certificate rotation workflow (pseudocode or script) that: a) Generates a new hybrid key pair (classical + PQC) b) Submits a CSR to both classical and PQC CAs c) Validates the issued certificate chain d) Deploys with zero downtime using hot-reload e) Verifies post-deployment health

Solution 11.6b.2

```
#!/bin/bash
# hybrid_cert_rotation.sh - Zero-downtime hybrid certificate rotation

set -euo pipefail

SERVER="webserver-01"
DOMAIN="api.example.com"
ALERT_WEBHOOK="https://alerts.internal/webhook"

# Step 1: Generate new hybrid key pair
echo "[1/6] Generating hybrid key pair..."
openssl genpkey -algorithm ML-DSA-65 -out new_pqc_key.pem
openssl ecparam -name prime256v1 -genkey -out new_classical_key.pem

# Step 2: Create CSRs
echo "[2/6] Creating certificate signing requests..."
openssl req -new -key new_pqc_key.pem -out pqc.csr -subj "/CN=${DOMAIN}"
openssl req -new -key new_classical_key.pem -out classical.csr -subj "/CN=${DOMAIN}"

# Step 3: Submit to CAs (placeholder – real implementation uses ACME or EST)
echo "[3/6] Submitting to CA..."
# ca_client submit --csr pqc.csr --profile hybrid-pqc --out new_pqc_cert.pem
# ca_client submit --csr classical.csr --profile hybrid-classical --out
new_classical_cert.pem

# Step 4: Validate certificate chain
echo "[4/6] Validating certificate chains..."
openssl verify -CAfile ca_chain.pem new_pqc_cert.pem
openssl verify -CAfile ca_chain.pem new_classical_cert.pem

# Step 5: Deploy with hot-reload (no downtime)
echo "[5/6] Deploying certificates..."
cp new_pqc_cert.pem /etc/tls/certs/pqc_cert.pem
cp new_classical_cert.pem /etc/tls/certs/classical_cert.pem
cp new_pqc_key.pem /etc/tls/private/pqc_key.pem
cp new_classical_key.pem /etc/tls/private/classical_key.pem
# Signal server to reload TLS context
kill -HUP $(cat /var/run/nginx.pid)

# Step 6: Post-deployment health check
echo "[6/6] Verifying deployment..."
sleep 5
if openssl s_client -connect ${DOMAIN}:443 -brief < /dev/null 2>&1 | grep -q
"Verification: OK"; then
    echo "SUCCESS: Certificate rotation complete"
else
    echo "FAILURE: Rolling back..."
```

```
cp /etc/tls/backup/* /etc/tls/certs/  
kill -HUP $(cat /var/run/nginx.pid)  
curl -X POST "${ALERT_WEBHOOK}" -d '{"text":"Certificate rotation FAILED for '$  
{DOMAIN}'"}'  
exit 1  
fi
```

Exercise 11.6b.3: Revocation During Hybrid Transition *(Intermediate)*

Consider a scenario where an ML-DSA-65 signing key is compromised but the paired ECDSA key is not. Describe: a) Whether revoking only the PQC certificate is sufficient b) How CRL and OCSP infrastructure must change for hybrid certificates c) The interaction between certificate revocation and CT (Certificate Transparency) logs

Solution 11.6b.3

a) Revocation scope: Both certificates (PQC and classical) must be revoked simultaneously, even if only one key is compromised. A hybrid deployment's security guarantee is that *both* algorithms must be broken to compromise the system. If the PQC key is compromised, an attacker with future quantum capability could combine the compromised PQC key with a quantum attack on the classical key, breaking the hybrid guarantee. Revoke both and re-issue a fresh pair.

b) CRL/OCSP changes for hybrid:

- **CRL:** Revocation lists must include entries for both certificate serial numbers. If using composite certificates (single cert with both keys), a single CRL entry suffices. CRL signing keys themselves need PQC migration.
- **OCSP:** OCSP responders must sign responses with PQC algorithms. The OCSP response size increases significantly (~3.5 KB for ML-DSA-65 signature vs ~72 bytes for ECDSA). Consider OCSP stapling to avoid per-client OCSP lookup overhead.

c) CT interaction: CT logs record issuance, not revocation. When replacement certificates are issued after revocation, new CT entries are created. Monitors should flag rapid re-issuance (revoke + re-issue within hours) as a potential compromise indicator. Log operators should sign SCTs with PQC to prevent future quantum forgery of transparency proofs.

Unit 11.7: Legacy Systems, Dead Languages, and Unmaintainable Code

Prerequisites: Units 11.1-11.6 (Complete migration fundamentals)

Estimated Time: 2-3 hours

Learning Objectives

After completing this unit, you will be able to:

1. **Assess PQC migration feasibility** for systems in legacy languages and frameworks
2. **Identify alternative strategies** when direct code modification is impossible
3. **Implement proxy and gateway approaches** for systems with unavailable source code
4. **Navigate regulatory constraints** for certified systems (FDA, DO-178C, etc.)
5. **Make informed risk-based decisions** about legacy system security

11.7.1 The Legacy System Reality

Many organizations face a harsh reality: critical systems running on "dead" languages, outdated frameworks, or platforms with no PQC support. These systems often:

- Process sensitive data requiring long-term confidentiality
- Cannot be easily modified or replaced
- Run on hardware with insufficient resources for PQC algorithms
- Fall under regulatory frameworks that make updates costly or impossible

The Legacy Landscape

=====

"Dead" or Declining Languages:

- └─ COBOL (still processes 95% of ATM transactions)
- └─ Fortran (scientific computing, simulations)
- └─ Delphi/Object Pascal (business applications)
- └─ Visual Basic 6 / VBA (enterprise automation)
- └─ Classic ASP (web applications)
- └─ PowerBuilder (client-server applications)
- └─ FoxPro (database applications)
- └─ RPG (IBM AS/400/iSeries)
- └─ PL/I (mainframe applications)
- └─ Ada (military/aerospace embedded systems)
- └─ Proprietary languages (vendor-specific systems)

Outdated Frameworks/Runtimes:

- └─ .NET Framework 4.x (vs .NET 6+)
- └─ Java 6/7/8 (vs Java 17+)
- └─ Python 2.7 (EOL 2020)
- └─ PHP 5.x (EOL 2018)
- └─ Ruby 1.x/2.x
- └─ Node.js < 16 (EOL versions)
- └─ OpenSSL 1.0.x/1.1.x
- └─ Legacy web servers (IIS 6, Apache 2.2)

Constrained Platforms:

- └─ IBM z/OS mainframes
- └─ IBM AS/400 / IBM i
- └─ HP NonStop (Tandem)
- └─ OpenVMS
- └─ Embedded RTOS (VxWorks, QNX, etc.)
- └─ PLCs and SCADA systems
- └─ Medical devices
- └─ Automotive ECUs

11.7.2 PQC Library Availability by Language

Languages with Native/Direct PQC Support (2025):

LANGUAGE	LIBRARY	ALGORITHMS	STATUS
C/C++	OpenSSL 3.5+	ML-KEM, ML-DSA, SLH-DSA	Production
C/C++	liboqs	All NIST + alternates	Research/Prototype
C/C++	wolfSSL	ML-KEM, ML-DSA	Production (FIPS)
Rust	pqcrypto	All NIST	Production
Go	cloudflare/circl	ML-KEM, ML-DSA	Production
Python	liboqs-python	All NIST	Research/Prototype
Java 17+	BouncyCastle 2.0+	ML-KEM, ML-DSA, SLH-DSA	Production
.NET 6+	BouncyCastle.Cryptography	ML-KEM, ML-DSA, SLH-DSA	Production
JavaScript	node-pqclean	NIST algorithms	Research

Languages with Limited/FFI-Based PQC Support:

LANGUAGE	APPROACH	COMPLEXITY	NOTES
Delphi/ Pascal	CryptoSys PQC DLL	Medium	Free library with ML-KEM, ML-DSA, SLH-DSA
Ada	C bindings to liboqs	High	Requires SPARK/Ada interface development
Fortran	C interop with liboqs	High	ISO_C_BINDING required
Ruby	FFI to liboqs	Medium	Native extension needed
Perl	XS bindings to liboqs	High	Limited community support

Languages with No Viable PQC Path:

LANGUAGE	CHALLENGE	RECOMMENDED STRATEGY
COBOL	No PQC libraries, mainframe constraints	Gateway/proxy approach
Visual Basic 6	Runtime EOL, no modern crypto	Application replacement or encapsulation
Classic ASP	Platform EOL, no PQC support	Reverse proxy with PQC termination
RPG (AS/400)	Closed platform, IBM-only crypto	IBM ICSF with CEX8S, or gateway
PL/I	Mainframe only, limited libraries	IBM ICSF, or protocol-level migration
FoxPro	EOL, no crypto libraries	Application replacement
PowerBuilder	Limited crypto support, declining	Wrapper services or replacement

11.7.3 Platform-Specific Challenges and Solutions

IBM Mainframe (z/OS) — The Good News

IBM has invested heavily in quantum-safe cryptography for z/OS:

```

IBM z/OS Quantum-Safe Capabilities
=====

Hardware:
├─ CEX8S Crypto Express adapter (z17)
│  └─ ML-KEM (hardware accelerated)
│     └─ ML-DSA (hardware accelerated)
└─ Required for PQC key operations

```

Software (ICSF - Integrated Cryptographic Service Facility):

- └─ CCA (Common Cryptographic Architecture) services
 - └─ CRYSTALS-Kyber key generation
 - └─ CRYSTALS-Dilithium key generation
 - └─ Digital signature operations
- └─ PKCS #11 services
 - └─ Standard interface for applications
- └─ Key management and translation

Migration Path:

1. Upgrade to z17 with CEX8S
2. Update ICSF to latest level
3. Modify applications to use ICSF APIs
4. Or: Use TLS 1.3 with PQC at network boundary

Reference: IBM Redbooks SG24-8525

"Transitioning to Quantum-Safe Cryptography on IBM Z"

Embedded Systems and IoT — The Hard Constraints

Embedded systems face fundamental resource limitations:

Embedded System Constraints

=====

Typical Resource Profile:

- └─ RAM: 64-256 KB (vs 32+ MB for ML-KEM reference)
- └─ Flash: 256 KB - 2 MB
- └─ CPU: 8-32 bit, 8-200 MHz
- └─ Power: Battery or energy-harvesting
- └─ Real-time: Deterministic timing required

Algorithm Feasibility:

Algorithm	RAM Needed	Code Size	Embedded Ready?
ML-KEM-512	~30 KB	~20 KB	Marginal
ML-KEM-768	~50 KB	~25 KB	Challenging
ML-DSA-44	~100 KB	~50 KB	Difficult
SLH-DSA-128s	~10 KB	~15 KB	Possible
Classic McEliece	~1 MB	~50 KB	Impossible

Optimization Strategies:

- └─ Hardware acceleration (RISC-V PQC extensions)
- └─ Algorithm selection (SLH-DSA for signatures on constrained devices)
- └─ Hybrid with smaller classical component
- └─ Gateway-based PQC (device uses symmetric, gateway does PQC)
- └─ Accept risk for short-lived ephemeral data

Industrial Control Systems (SCADA/ICS/OT)

CISA issued specific guidance in October 2024:

OT/ICS PQC Challenges (CISA Guidance)

Unique Constraints:

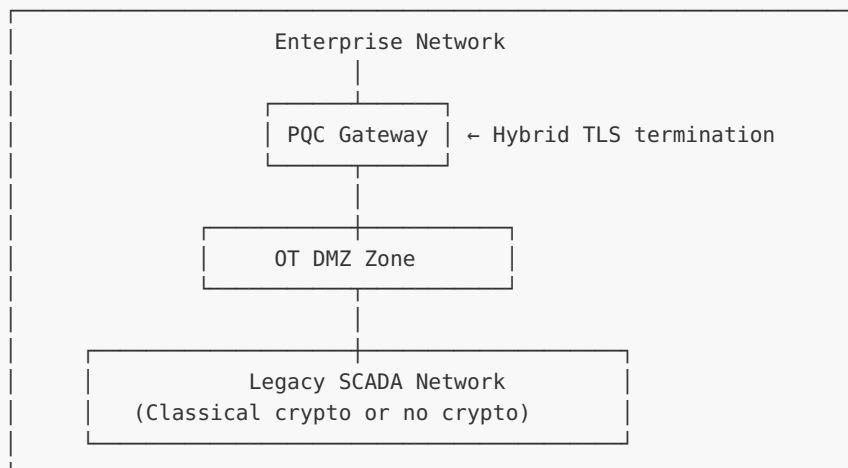
- └ 15-30 year device lifecycles
- └ Safety-critical operations (cannot disrupt)
- └ Fixed message sizes in fieldbus protocols
- └ Real-time timing requirements
- └ Isolated networks (but not always)
- └ Vendor lock-in and proprietary protocols
- └ Change management overhead (months to years)

CISA Recommendations:

1. Network segmentation as primary defense
2. Request crypto-agility in new equipment procurement
3. Use quantum-resistant algorithms where feasible
4. Apply PQC at network boundaries (VPN gateways)
5. Consider non-cryptographic MFA (physical tokens, biometrics)
6. Prioritize systems with long data retention requirements

Warning: "OT systems could be the last remaining platforms to achieve post-quantum cryptographic standards" - CISA, October 2024

Gateway Approach for Legacy SCADA:



11.7.4 Regulatory Constraints on Cryptographic Updates

FDA Medical Devices

FDA Cybersecurity Requirements (2023-2025)

Key Regulation: Section 524B of FD&C Act (FDORA 2022)
Effective: March 29, 2023

Requirements for "Cyber Devices":

- └ Must follow NIST guidelines (including FIPS 140-3)
- └ Must plan for post-quantum cryptography
- └ Must document software support windows
- └ Must provide reasonable cybersecurity assurance
- └ Outdated OS = device may be rejected

Legacy Device Definition:

"Devices that can no longer be reasonably protected against current cybersecurity threats, even though they may still perform their primary function"

Software Update Paths:

Change Type	Regulatory Impact
Cryptographic library update with same API	Likely requires new 510(k) submission if security-critical
New cryptographic algorithm (PQC)	Definitely requires 510(k) or higher regulatory pathway
Network isolation only (no device changes)	May not require new submission

Practical Impact:

- Medical device PQC migration can take 2-5 years per device
- Legacy devices may never be updated (cost prohibitive)
- Network-level protection may be only viable option

Aerospace (DO-178C) and Defense

DO-178C Software Certification Constraints

=====

What is DO-178C?

Primary standard for commercial airborne software certification
Required by FAA, EASA, Transport Canada

Certification Levels (DAL - Design Assurance Level):

- └─ DAL-A: Catastrophic failure (loss of aircraft)
- └─ DAL-B: Hazardous failure (serious injury)
- └─ DAL-C: Major failure (significant workload)
- └─ DAL-D: Minor failure (slight workload increase)
- └─ DAL-E: No effect

Cryptographic Update Requirements:

1. Full traceability from requirements to test results
2. Re-verification of all affected code paths
3. Re-testing at appropriate DAL level
4. Potentially complete re-certification
5. Documentation updates throughout

Cost Implications:

- └─ DAL-A certification: \$1M - \$10M+ per software component
- └─ Re-certification for crypto change: 50-100% of original cost
- └─ Timeline: 1-3 years for certification activities
- └─ Practical result: Many avionics systems will NOT be updated

Current PQC Support for DO-178C:

- └─ wolfSSL: DAL-A certifiable crypto library
 - └─ AES, SHA-256, RSA, ChaCha20-Poly1305
 - └─ PQC support planned but not yet DAL-A certified

- └ DDC-I Deos RTOS: FIPS 140-2 validated with DO-178C
- └ Timeline for DAL-A PQC: Likely 2027-2030

Practical Strategies:

1. Isolate communication systems from flight-critical functions
2. Implement PQC at ground infrastructure (not aircraft)
3. Use network encapsulation for data links
4. Plan for next-generation aircraft to include PQC natively

11.7.5 Strategies for Unmaintainable Systems

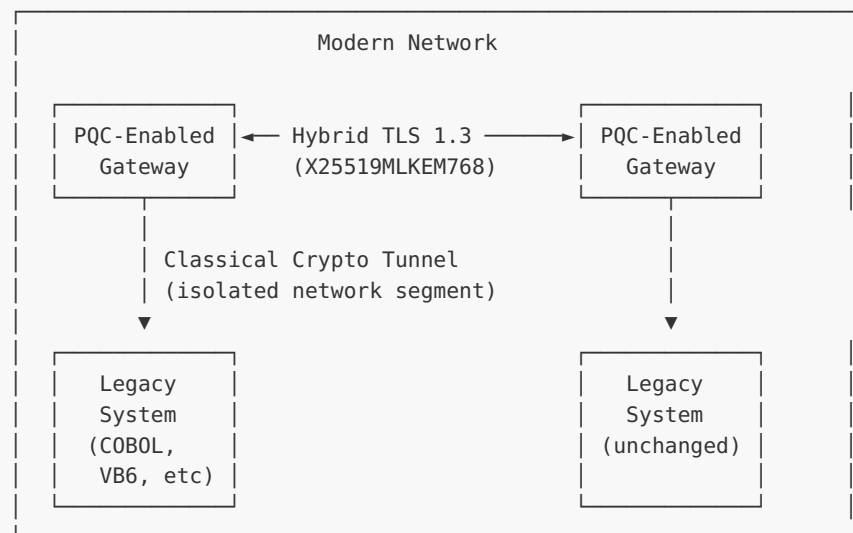
When source code is unavailable, compilers are obsolete, or modifications are impossible, these alternative strategies apply:

Strategy 1: Network-Level PQC Encapsulation

Tunnel/Enclave Approach

=====

Architecture:



Benefits:

- └ No changes to legacy systems
- └ PQC protection for data in transit
- └ Can be deployed incrementally
- └ Hardware-based gateways available (HSMs)
- └ Supports compliance requirements

Limitations:

- └ Data at rest in legacy system still uses classical crypto
- └ Gateway becomes single point of failure
- └ Latency increase for some protocols
- └ Doesn't protect against insider threats

Strategy 2: Reverse Proxy with PQC Termination

```
// Conceptual configuration for PQC-terminating reverse proxy

// nginx.conf with oqs-provider (conceptual)
server {
    listen 443 ssl http2;

    # PQC-enabled TLS termination
    ssl_certificate /etc/nginx/certs/server.crt;
    ssl_certificate_key /etc/nginx/certs/server.key;

    # Prefer hybrid key exchange
    ssl_ecdh_curve X25519MLKEM768:X25519;

    # Proxy to legacy backend (HTTP or classical TLS)
    location / {
        proxy_pass http://legacy-backend:8080;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}

// Result:
// Internet —[PQC TLS]→ Reverse Proxy —[HTTP/Classical]→ Legacy App
```

Strategy 3: API Gateway Wrapper

```
# Flask API gateway that wraps legacy system calls
# Provides PQC-protected external interface

from flask import Flask, request, jsonify
import requests
from oqs import KeyEncapsulation, Signature

app = Flask(__name__)

# Legacy system endpoint (classical or no crypto)
LEGACY_ENDPOINT = "http://internal-legacy:8080"

@app.route('/api/v2/<path:path>', methods=['GET', 'POST', 'PUT', 'DELETE'])
def gateway(path):
    """
    Modern API gateway with PQC session management.
    Translates requests to legacy system format.
    """
    # Validate PQC-authenticated session (implementation details omitted)
    session = validate_pqc_session(request.headers.get('Authorization'))

    if not session:
        return jsonify({"error": "Unauthorized"}), 401

    # Transform request for legacy system
    legacy_request = transform_to_legacy_format(request)

    # Forward to legacy system
    response = requests.request(
        method=request.method,
        url=f"{LEGACY_ENDPOINT}/{path}",
        headers=legacy_request['headers'],
```



```

    data=legacy_request['body']
)

# Transform response for modern clients
modern_response = transform_from_legacy_format(response)

return jsonify(modern_response)

# Run with PQC-enabled TLS (Gunicorn + oqs-provider)

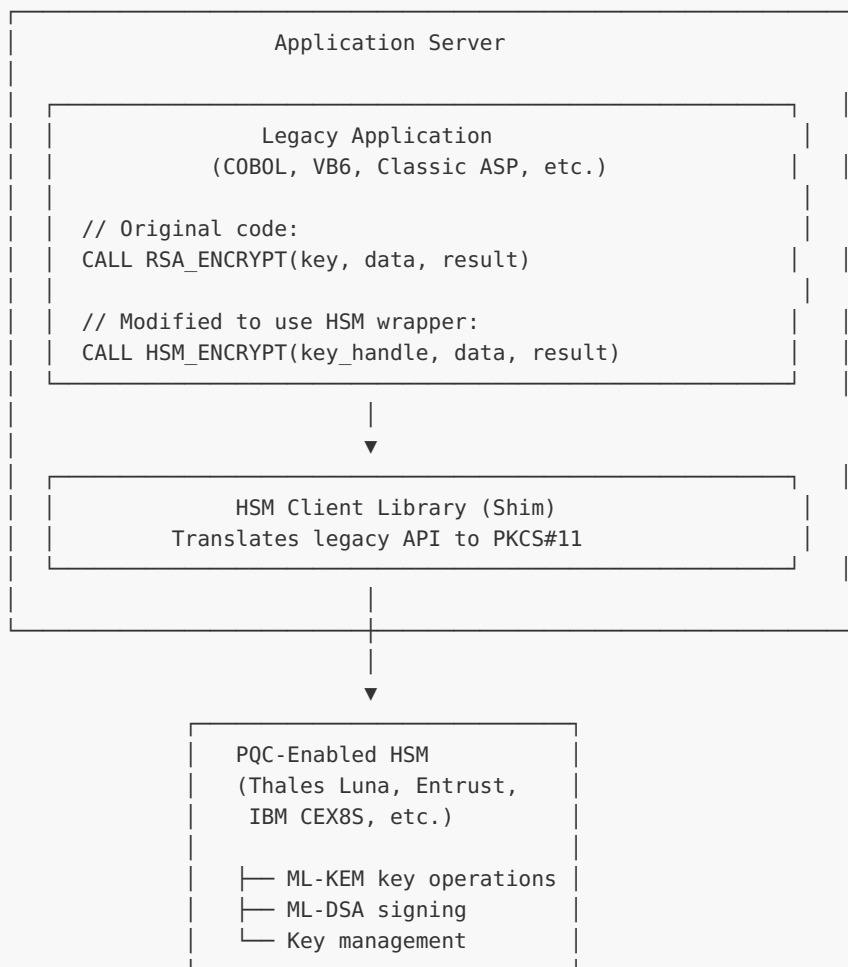
```

Strategy 4: HSM-Based Cryptographic Offload

Hardware Security Module Approach

=====

Architecture:



Vendors with PQC HSM Support (2025):

- └─ Thales Luna Network HSM 7.x
- └─ Entrust nShield
- └─ IBM CEX8S (z/OS)
- └─ Utimaco
- └─ AWS CloudHSM (ML-KEM support announced)

Benefits:

- └─ Minimal legacy code changes (just API redirection)

- └─ Hardware-protected key storage
- └─ FIPS 140-3 certification path
- └─ Centralized key management
- └─ Audit logging built-in

Limitations:

- └─ Network latency for crypto operations
- └─ Single point of failure concerns
- └─ Licensing costs
- └─ May require stub/shim library development

Strategy 5: Containerized Isolation with Sidecar Proxy

```
# Kubernetes deployment with PQC sidecar proxy
apiVersion: v1
kind: Pod
metadata:
  name: legacy-app-with-pqc
spec:
  containers:
    # Legacy application container (unchanged)
    - name: legacy-app
      image: legacy-cobol-app:v1.0
      ports:
        - containerPort: 8080

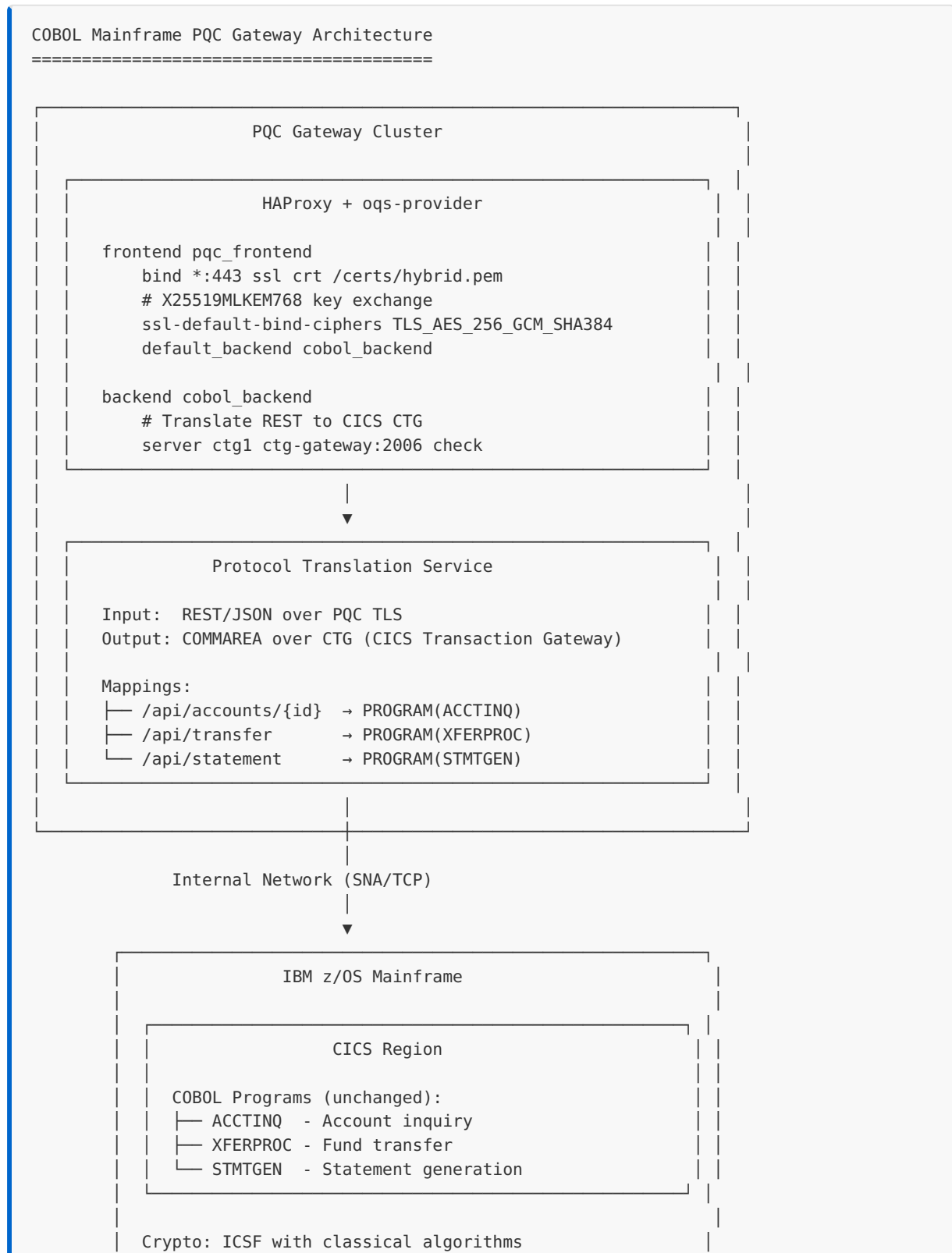
    # PQC sidecar proxy (handles all external TLS)
    - name: pqc-proxy
      image: envoy-pqc:v1.30
      ports:
        - containerPort: 443
      volumeMounts:
        - name: certs
          mountPath: /etc/envoy/certs
      env:
        - name: UPSTREAM_HOST
          value: "localhost:8080"
        - name: TLS_GROUPS
          value: "X25519MLKEM768:X25519"

  volumes:
    - name: certs
      secret:
        secretName: pqc-tls-certs
---
# Service exposes only the PQC sidecar
apiVersion: v1
kind: Service
metadata:
  name: legacy-app-service
spec:
  selector:
    app: legacy-app
  ports:
    - port: 443
      targetPort: 443
```

Strategy 6: Legacy Language Gateway Patterns

For systems where direct code modification is impossible or impractical, specialized gateway architectures provide PQC protection at the network boundary.

COBOL Systems Integration:



(Internal traffic not PQC-protected)

Key Components:

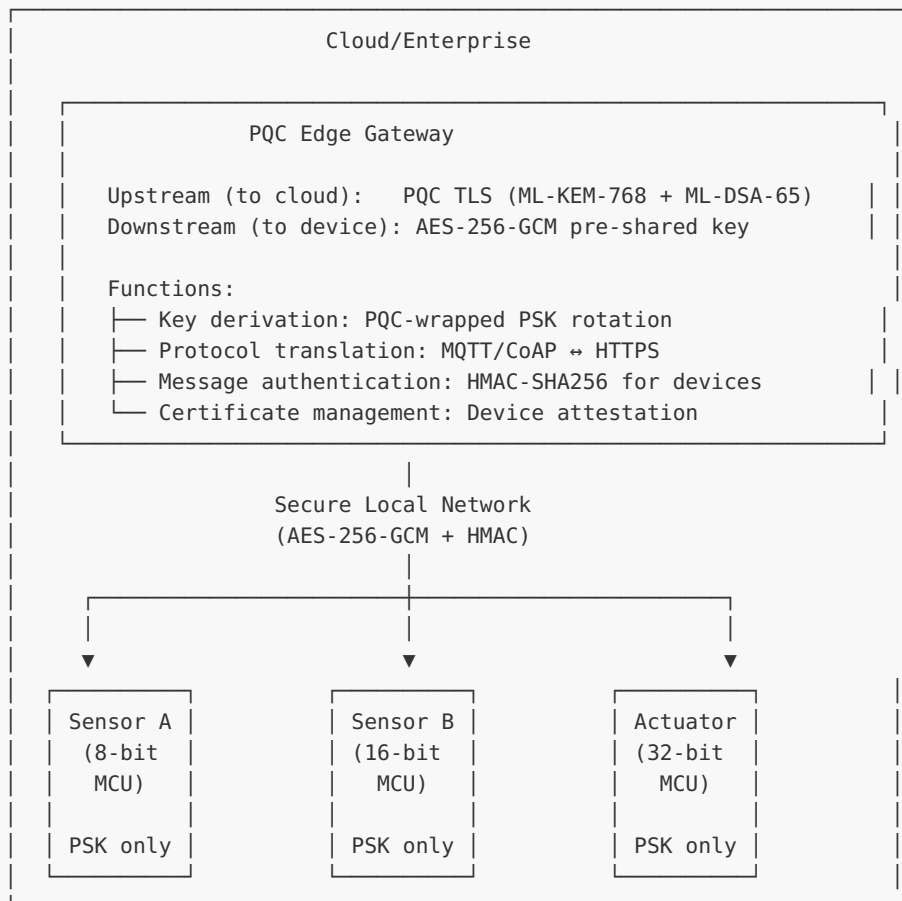
- HAProxy: PQC TLS termination (X25519MLKEM768)
- Translation Service: REST/JSON ↔ COBOL COMMAREA
- CTG: IBM CICS Transaction Gateway (existing)
- COBOL Programs: Zero modifications required

Embedded Systems Without PQC Libraries:

Embedded Device PQC Gateway Pattern

Use Case: IoT sensors, medical devices, industrial controllers
with no PQC capability due to resource constraints

Architecture:



PSK Rotation via Gateway:

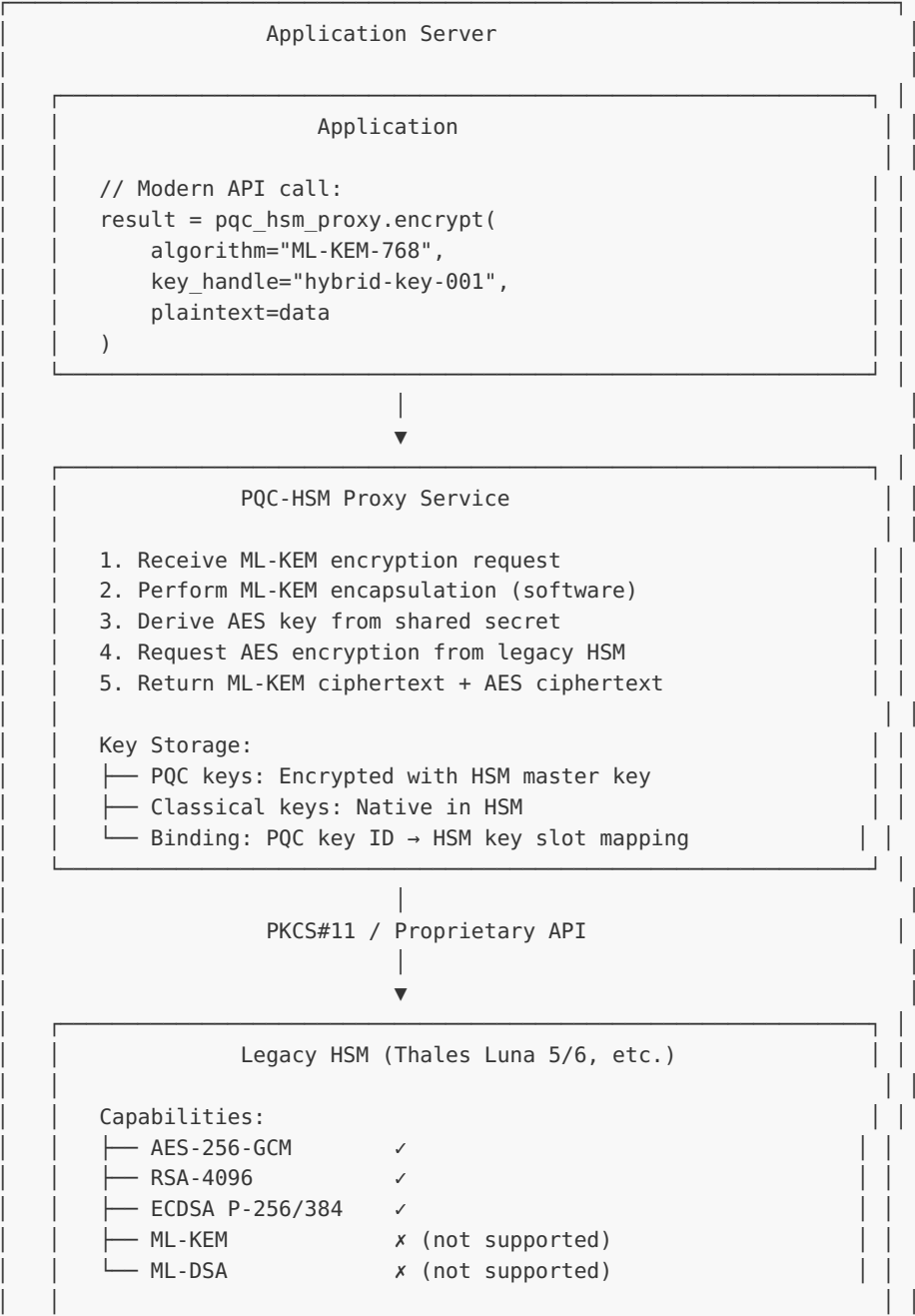
1. Gateway generates new symmetric key using ML-KEM
2. Gateway encrypts new PSK with current device PSK
3. Device decrypts and installs new PSK
4. Old PSK revoked after confirmation
5. Rotation frequency: 30-90 days (configurable)

- Supported Protocols:
- └─ MQTT 3.1/5.0 (with TLS termination at gateway)
 - └─ CoAP (with DTLS termination at gateway)
 - └─ Modbus TCP (encapsulated in TLS tunnel)
 - └─ Custom binary (via protocol adapters)

Legacy HSM Integration Patterns:

Legacy HSM PQC Bridge Architecture
=====

Problem: HSMs with no PQC support (pre-2024 devices)
Solution: PQC-to-classical key translation layer



Role: Master key protection, AES operations

Security Considerations:

- PQC private keys protected by HSM-wrapped key
- PQC operations in secure memory (proxy runs in TEE if available)
- Audit logging for all key operations
- Migration path: Replace proxy when HSM upgraded to PQC-capable
- FIPS boundary: Classical operations FIPS-validated, PQC in process

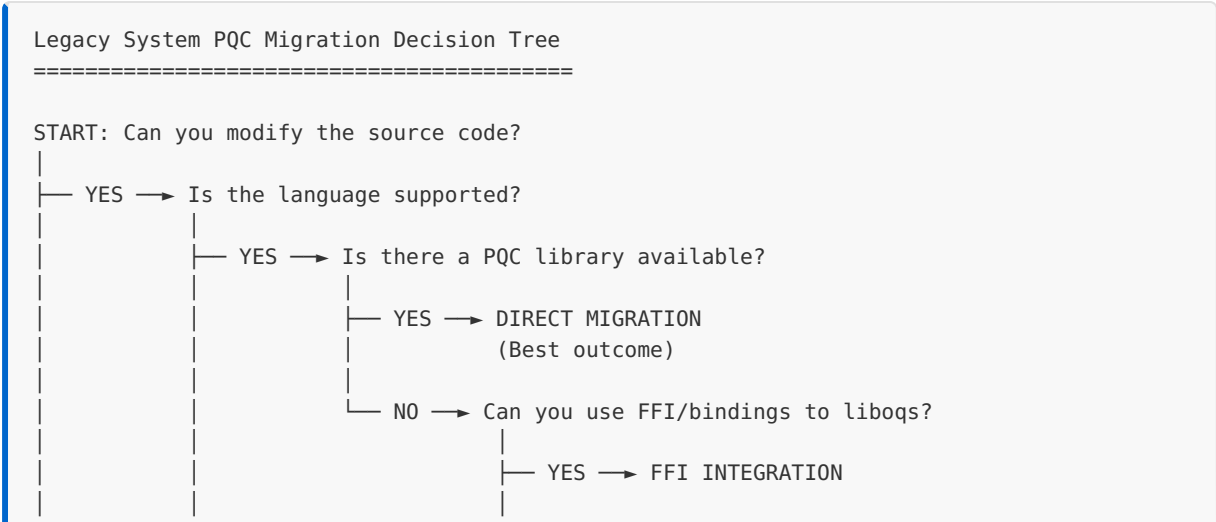
Supported Legacy HSMs:

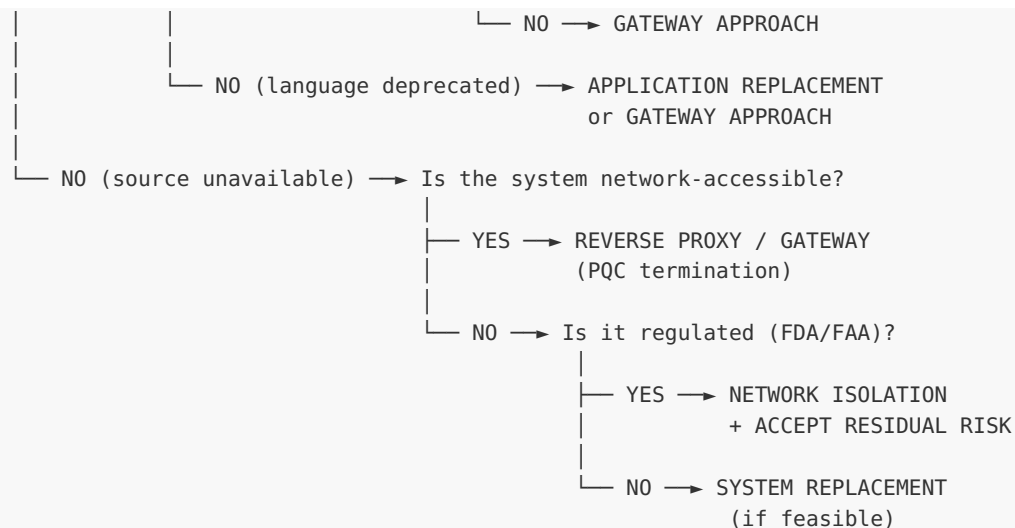
- Thales Luna SA 5.x/6.x → Luna 7.x upgrade recommended
- Gemalto SafeNet → Thales Luna migration path
- Utimaco SecurityServer → Contact vendor for PQC roadmap
- nCipher nShield → Now Entrust, PQC support in newer models
- AWS CloudHSM Classic → Migrate to CloudHSM (ML-KEM support)

Implementation Checklist for Gateway Patterns:

STEP	COBOL GATEWAY	EMBEDDED GATEWAY	HSM BRIDGE
Network segmentation	Required	Required	Required
Protocol mapping documented	Yes	Yes	Yes
Latency impact assessed	<50ms added	<10ms added	<5ms added
Failover configured	Active-passive	N+1 redundancy	Active-active
Monitoring enabled	Transaction logs	Device heartbeats	Key usage audit
Rollback tested	Config rollback	Firmware rollback	API version switch

11.7.6 Decision Framework for Legacy Systems





Cost-Benefit Matrix:

Strategy	Cost	Risk	Timeline
Direct migration	\$	Low	6-18 months
FFI integration	\$\$	Medium	12-24 months
Gateway/proxy	\$\$	Medium	3-12 months
HSM offload	\$\$\$	Low-Medium	6-18 months
Application replace	\$\$\$\$	Medium-High	2-5 years
Network isolation	\$	High	1-3 months
Accept risk	-	Very High	Immediate

11.7.7 Workforce and Knowledge Challenges

The Skills Gap Reality

=====

Legacy Language Expertise:

- └ ~60% of mainframe specialists are over age 50
- └ COBOL developers: Retiring faster than being trained
- └ Delphi/PowerBuilder: Niche expertise, declining community
- └ Ada: Primarily defense contractors, expensive consultants
- └ RPG (AS/400): IBM-specific, shrinking talent pool

PQC Expertise:

- └ 51% of organizations lack a dedicated PQC migration lead
- └ Cryptographic engineering: Specialized, high-demand field
- └ Intersection of legacy + PQC: Extremely rare skill set
- └ Training timeline: 6-12 months for competency

Mitigation Strategies:

1. Engage specialized consulting firms early
2. Partner with academic institutions (crypto research groups)
3. Use managed PQC services where possible
4. Document institutional knowledge before retirements
5. Consider AI-assisted code analysis for legacy systems

11.7.8 Cost Estimation for Legacy PQC Migration

Migration Cost Factors

=====

Direct Costs:

- └─ Cryptographic inventory/discovery
 - └─ Automated tools: \$50K-\$200K
 - └─ Manual analysis (large org): \$100K-\$500K
- └─ Library integration/development
 - └─ Supported language: \$50K-\$150K per application
 - └─ FFI bindings: \$100K-\$300K per language
 - └─ Gateway development: \$200K-\$500K
- └─ Testing and validation
 - └─ Functional testing: 20-40% of development cost
 - └─ Security testing: \$50K-\$200K per system
- └─ Hardware (if needed)
 - └─ HSM (PQC-enabled): \$50K-\$200K per unit
 - └─ Mainframe crypto adapter: \$100K-\$500K
 - └─ Gateway appliances: \$20K-\$100K per unit
- └─ Regulatory compliance (if applicable)
 - └─ FDA 510(k) per device: \$500K-\$2M
 - └─ DO-178C re-certification: \$1M-\$10M per component
 - └─ FIPS 140-3 validation: \$50K-\$500K

Indirect Costs:

- └─ Downtime during migration
- └─ Staff training
- └─ Ongoing maintenance increase
- └─ Opportunity cost (delayed other projects)
- └─ Risk of migration failures

Industry Estimates:

- └─ Small organization (< 100 systems): \$1M-\$5M
- └─ Medium organization (100-1000 systems): \$5M-\$25M
- └─ Large enterprise (1000+ systems): \$25M-\$100M+
- └─ US Federal Government (total): \$7.1 billion estimate

Exercises

Exercise 11.7.1: Legacy System Assessment *(Beginner)*

For a legacy system in your organization: a) Identify the programming language and framework b) Assess PQC library availability for that platform c) Determine if source code is available and modifiable d) Recommend a migration strategy from the decision tree e) Estimate rough costs using the cost factors above

Exercise 11.7.2: Gateway Architecture Design *(Advanced)*

Design a PQC gateway solution for a COBOL-based mainframe application that: a) Accepts connections from modern clients using hybrid TLS b) Forwards requests to the mainframe over the internal network c) Handles authentication translation (modern to legacy) d) Provides audit logging for all transactions

Exercise 11.7.3: Regulatory Impact Analysis *(Intermediate)*

For a Class II medical device running Windows Embedded with RSA-2048: a) What FDA requirements apply to a cryptographic update? b) Estimate the timeline for FDA clearance of a PQC update c) What alternative strategies could provide PQC protection without device changes? d) What residual risks must be documented and accepted?

Exercise 11.7.4: Risk Acceptance Framework *(Intermediate)*

Create a risk acceptance document template for a legacy system where PQC migration is not feasible, including: a) System description and data sensitivity classification b) Technical constraints preventing migration c) Compensating controls implemented d) Residual risk quantification (probability × impact) e) Executive sign-off section f) Review schedule (annual, bi-annual, etc.)

Solutions

Solution 11.7.1

Example: Legacy COBOL Payroll System Assessment

ASSESSMENT AREA	FINDING
Language/ Framework	COBOL 85 on IBM z/OS with CICS
PQC Library Availability	No native COBOL PQC libraries exist; IBM z/OS ICSF supports ML-KEM via CEX8S HSM
Source Code	Available but poorly documented; original developers retired
Modifiability	Limited - any changes require expensive regression testing

Decision Tree Path:

Source available? → YES
Can be modified? → PARTIALLY (high risk/cost)
PQC libraries exist? → NO (native COBOL)
HSM viable? → YES (IBM CEX8S supports PQC)

Recommended Strategy: HSM-based cryptographic offload

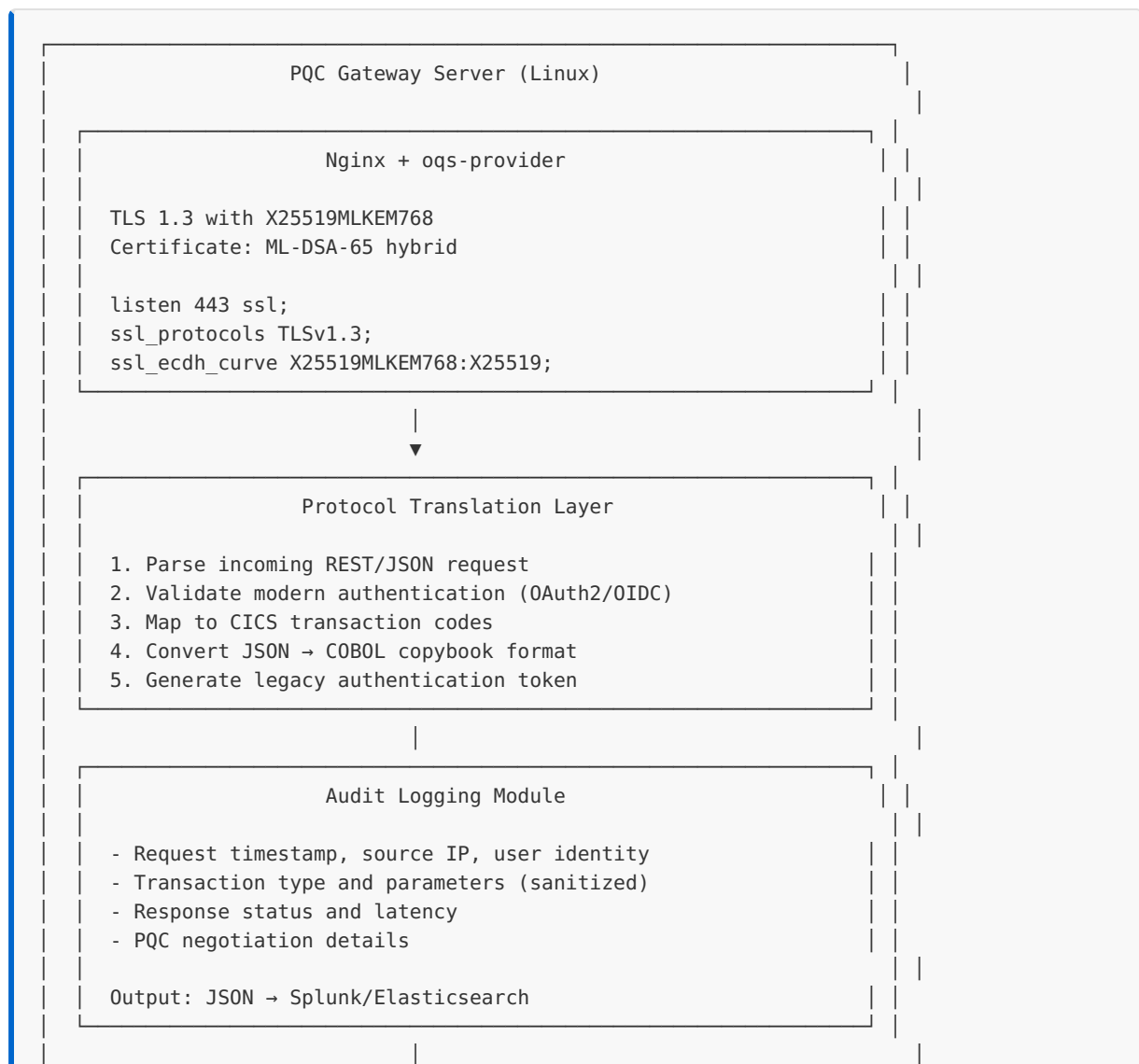
- Replace inline RSA calls with ICSF HSM API calls
- HSM handles PQC key operations
- Minimal COBOL code changes required

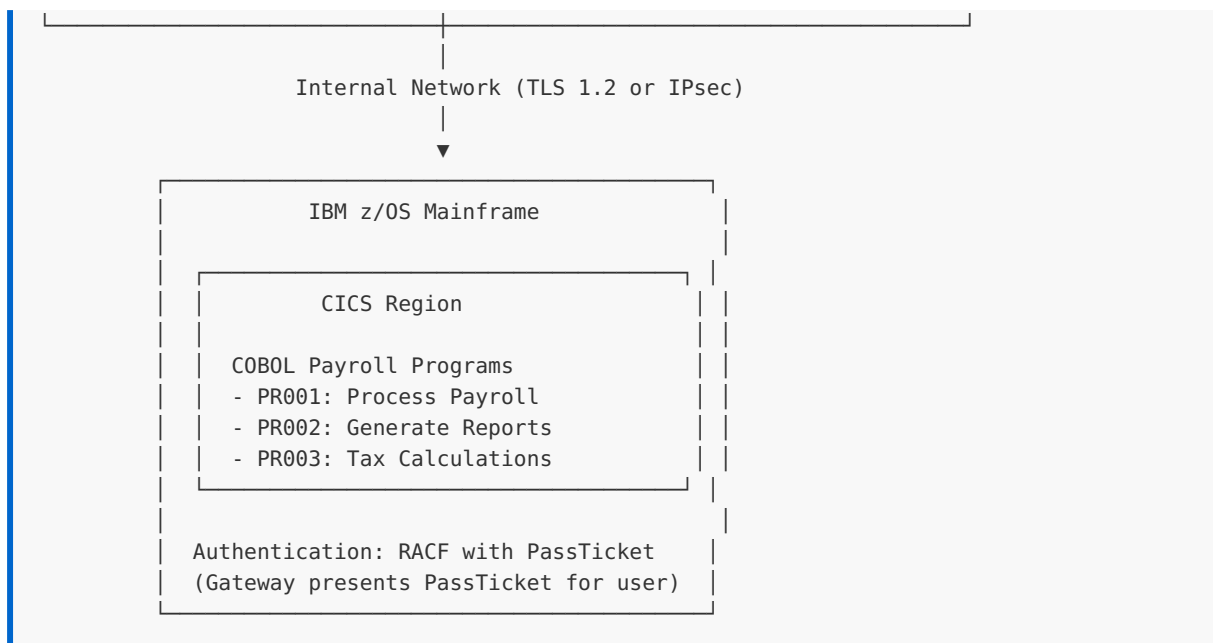
Cost Estimate:

ITEM	COST RANGE
IBM CEX8S HSM adapter	\$150K-\$300K
Development (modify crypto calls)	\$100K-\$200K
Testing and validation	\$50K-\$100K
Documentation updates	\$20K-\$40K
Total	\$320K-\$640K

Solution 11.7.2

PQC Gateway Architecture for COBOL Mainframe:





Authentication Translation Flow:

1. Client authenticates to gateway via OAuth2/OIDC
2. Gateway validates token with identity provider
3. Gateway maps user identity to RACF user ID
4. Gateway requests RACF PassTicket for mapped user
5. PassTicket included in CICS request
6. CICS validates PassTicket with RACF (single-use)

Solution 11.7.3

FDA Regulatory Impact Analysis: Class II Medical Device

a) FDA Requirements for Cryptographic Update:

REQUIREMENT	DESCRIPTION
Premarket Submission	Likely 510(k) required if crypto is security-critical
Cybersecurity Documentation	Updated SBOM, threat model, security architecture
Risk Analysis	ISO 14971 risk management file updates
Verification & Validation	Full regression testing of affected functionality
Labeling Updates	User manual, IFU changes if user interaction changes

b) Timeline Estimate:

PHASE	DURATION
Internal development & testing	6-12 months
510(k) preparation	3-6 months
FDA review (typical Class II)	3-6 months
Post-clearance deployment	1-3 months
Total	13-27 months

c) Alternative Protection Strategies:

1. Network Isolation

- Place device on isolated VLAN
- PQC-enabled firewall terminates external connections
- No device changes required

2. Gateway/Proxy Approach

- PQC gateway in front of device
- Device communicates only with gateway
- May not require new FDA submission

3. VPN Tunnel

- IPsec tunnel with PQC key exchange
- Device unchanged, network transport protected

d) Residual Risks to Document:

RISK	MITIGATION	RESIDUAL
Device-to-device communication unprotected	Network isolation	Low
Local data at rest uses RSA	Physical security controls	Medium
Insider threat at device	Access controls, audit logs	Medium
HNDL attack on stored device data	Data minimization, short retention	Medium

Solution 11.7.4

Risk Acceptance Document Template:

```
# CRYPTOGRAPHIC RISK ACCEPTANCE DOCUMENT

## Document Control
| Field | Value |
|-----|-----|
| Document ID | RA-2026-LEGACY-001 |
| Version | 1.0 |
| Classification | CONFIDENTIAL |
| Created | 2026-01-15 |
| Review Date | 2027-01-15 |

---

## 1. System Description

**System Name:** Legacy Inventory Management System (LIMS)

**Function:** Real-time inventory tracking for warehouse operations

**Data Classification:** INTERNAL (SKU data), CONFIDENTIAL (supplier pricing)

**User Population:** 50 warehouse staff, 10 managers

**Criticality:** HIGH (24/7 operations depend on this system)

---

## 2. Current Cryptographic Profile

| Component | Algorithm | Key Size | Quantum Vulnerable |
|-----|-----|-----|-----|
| TLS | TLS 1.2 ECDHE-RSA | 2048-bit | YES |
| Data encryption | AES-256-CBC | 256-bit | NO (symmetric) |
| Key wrapping | RSA-OAEP | 2048-bit | YES |
| Authentication | HMAC-SHA256 | 256-bit | REDUCED (Grover) |

---

## 3. Technical Constraints Preventing Migration

| Constraint | Details |
|-----|-----|
| Source code | Lost in 2015 vendor bankruptcy |
| Platform | Windows Server 2008 R2 (EOL) |
| Dependencies | Proprietary DLLs with no PQC versions |
| Budget | Replacement cost exceeds $2M |
| Timeline | Business cannot tolerate 6+ month replacement |

---

## 4. Compensating Controls Implemented

| Control | Description | Effectiveness |
```

Network isolation	Dedicated VLAN, no internet access	HIGH
PQC Gateway	All external access via X25519MLKEM768 gateway	HIGH
Enhanced monitoring	IDS/IPS with crypto anomaly detection	MEDIUM
Data minimization	Purge records older than 2 years	MEDIUM
Access restrictions	MFA for all users, privileged access review	MEDIUM

5. Residual Risk Assessment

****Threat:** Harvest Now, Decrypt Later (HNDL) attack on internal traffic**

Factor	Assessment
Likelihood (L)	MEDIUM (3/5) - Sophisticated adversary required
Impact (I)	MEDIUM (3/5) - Supplier pricing disclosure
Risk Score	$L \times I = 9/25 = \text{**MEDIUM**}$

****Threat:** Insider with network access captures traffic**

Factor	Assessment
Likelihood (L)	LOW (2/5) - Background checks, monitoring
Impact (I)	MEDIUM (3/5) - Same as above
Risk Score	$L \times I = 6/25 = \text{**LOW-MEDIUM**}$

6. Risk Acceptance Statement

The undersigned acknowledge that:

1. The Legacy Inventory Management System cannot be migrated to PQC
2. Compensating controls reduce but do not eliminate quantum risk
3. Residual risk is accepted based on business criticality
4. This acceptance will be reviewed annually
5. Immediate action will be taken if threat landscape changes

7. Approval Signatures

Role	Name	Signature	Date
System Owner	_____	_____	_____
CISO	_____	_____	_____
CTO	_____	_____	_____
Legal/Compliance	_____	_____	_____

8. Review Schedule

Review Type	Frequency	Next Due
Risk reassessment	Annual	2027-01-15
Control effectiveness	Quarterly	2026-04-15
Threat landscape	Continuous	Ongoing
Full document review	Annual	2027-01-15

Appendix A: Supporting Evidence

- Network architecture diagram (attached)
- Gateway configuration documentation
- Penetration test results (dated)
- Vendor bankruptcy documentation
- Cost-benefit analysis for replacement

Unit 11.7 Summary

Unit 11.7 tackled the hardest migration scenarios: systems written in obsolete languages, unmaintained codebases, and vendor-locked appliances. Key concepts: TLS/IPsec gateway wrapping for systems that cannot be modified internally, protocol-level interception for legacy applications, risk-based triage (replace vs. wrap vs. isolate vs. accept), documentation requirements for risk acceptance, and vendor engagement strategies for closed-source dependencies.

Unit 11.8: Production Deployment Checklist

This unit provides concrete checklists for moving a PQC-enabled system into production and monitoring it afterward. Use these as gate criteria for deployment reviews and as ongoing operational metrics.

Pre-Deployment Sign-Off Checklist

Before flipping the switch on hybrid or PQ-only mode in production:

Security review

- ☐ Threat model updated to include post-quantum adversary; harvest-now-decrypt-later explicitly in scope for data with > 5 year confidentiality window
- ☐ Cryptographic inventory (from Unit 11.2) re-verified against actual deployed binaries
- ☐ No cryptographic secrets, seeds, or private keys present in logs, error messages, or crash dumps (grep `0x` , `BEGIN PRIVATE` , `-----BEGIN` in log archives)

- ☐ Key rotation procedures tested and documented for each algorithm in hybrid configuration
- ☐ Backup / recovery procedures validated for PQ keys (PQ secret keys are 10-100× larger than classical — HSM/storage capacity confirmed)
- ☐ Randomness source validated: `getrandom()` (Linux), `RtlGenRandom` / `BCryptGenRandom` (Windows), `/dev/urandom` with entropy-pool monitoring for embedded

Performance validation

- ☐ End-to-end latency regression measured under realistic load; < 10% degradation threshold documented
- ☐ CPU utilization under peak load remains within headroom (typically +5-15% for ML-KEM/ML-DSA hybrids)
- ☐ Memory footprint tested under sustained connection load (PQ key material uses 2-10× more RAM per session)
- ☐ Bandwidth impact confirmed via capacity planning against sheet in Appendix (3-5× handshake overhead)
- ☐ Worst-case signing time for ML-DSA documented (rejection sampling tail — signatures occasionally take 10-50× median)

Rollback procedure

- ☐ Rollback to classical-only configuration tested in staging, documented with specific commands
- ☐ Decision criteria defined: when to roll back (error rate > X%, latency > Yms p99, specific log patterns)
- ☐ Rollback window communicated: can revert within N minutes without data loss
- ☐ Approvals obtained from security, infrastructure, and product owners
- ☐ Post-rollback forensic procedures documented (preserve keys/logs for analysis)

Compliance and documentation

- ☐ FIPS 140-3 module validation confirmed if required (see Unit 11.5.5)
- ☐ CNSA 2.0 timeline alignment verified for federal systems
- ☐ National regulator requirements checked (ANSSI / BSI / NCSC timelines)
- ☐ Customer-facing documentation updated (TLS cipher policy, SSH config, X.509 cert algorithms)
- ☐ Internal runbook updated with PQ-specific troubleshooting

Post-Deployment Monitoring Metrics

Instrument from day one; don't wait for problems:

Adoption metrics (track trend, not absolute values)

- ☐ Percentage of TLS handshakes using hybrid key exchange
- ☐ Percentage of SSH sessions using `sntrup761x25519-sha512@openssh.com` or equivalent
- ☐ Percentage of certificates issued with PQ or hybrid signature algorithm
- ☐ Alert: adoption rate drops > 10% week-over-week (client-side regression)

Error metrics (any sustained rise is a red flag)

- ☐ ML-KEM decapsulation failure rate (should be $\ll 10^{-10}$ with correct implementation; any non-zero count at scale indicates bug)
- ☐ ML-DSA signature verification failures (should be 0 for legitimately signed content)
- ☐ Algorithm negotiation failures in TLS/SSH (fallback to classical)
- ☐ OQS provider / library error counters
- ☐ Alert: any of above > 0.01% of operations

Performance metrics (baseline, watch for drift)

- ☐
Handshake latency p50/p95/p99 per protocol
- ☐
Signing latency histogram (detect rejection-sampling tail anomalies)
- ☐
Key generation latency (one-shot operations, less critical)
- ☐
Memory / CPU utilization trends
- ☐
Alert: p99 latency regresses > 20% from baseline

Security posture

- ☐
HSM / key storage capacity utilization (PQ keys are large)
- ☐
Key rotation cadence executing as scheduled
- ☐
Certificate expiration tracking (including hybrid composites)
- ☐
Quarterly review of new CVEs for PQC implementations in use

Incident Response for PQC-Specific Events

Preparation for "algorithm vulnerability disclosed" class of incidents:

1. **Decision tree prepared** for: "new attack reduces security of algorithm X by Y bits"
 - What's the exploit feasibility? (academic vs. practical)
 - Does hybrid construction mitigate it? (classical + PQ — both must break)
 - Is algorithm replacement feasible in timeframe? (crypto agility matters here)
2. **Contact list** for NIST PQC Project, vendor security teams, CERT
3. **Algorithm swap runbook** — procedure to shift from ML-KEM-768 to, e.g., HQC-256 via configuration change
4. **Communication plan** — customer disclosure, regulator notification, press response

Refer back to Unit 10.3 (Cryptographic Agility) for architectural support of rapid algorithm swaps.

Three Concrete Incident Playbooks

The high-level decision tree above frames the problem; the three playbooks below give step-by-step actions for the most likely PQC operational incidents in 2026-2030.

Playbook A — "CVE announced for our PQC library, what do I do in the next 2 hours?"

Trigger example: GitHub Security Advisory drops for liboqs / OpenSSL native PQC / AWS-LC-FIPS / etc.

0-15 min — Triage

1. Read the advisory. Identify: affected versions, attack class (timing, fault, key recovery, denial-of-service), CVSS score, exploitability (PoC published?).
2. Inventory query (CBOM, see Unit 11.2): which production systems use the affected library version?
3. Check whether your hybrid construction mitigates: if the CVE is in PQ-only and you deploy hybrid (X25519 + ML-KEM-768), classical leg may still protect transport confidentiality *for now*.

15-60 min — Containment 4. Patch availability: is fix in the library's main branch / tagged release? If yes, schedule emergency rotation per change-management policy. 5. Workaround available? E.g., compile-flag tweak, configuration change, algorithm swap via crypto-agility framework (see Unit 10.3). 6. If exploit is timing/microarchitectural and you run multi-tenant: assess co-residency risk; consider workload isolation as interim mitigation.

1-2 hours — Communication 7. Internal: notify security on-call, infra owners of affected services, leadership. 8. External: if CVE is in customer-facing path, prepare disclosure per your standard CVE-response policy. 9. File for FedRAMP / SOC2 / ISO 27001 evidence (incident logged, response timeline documented).

Day 1-7 — Remediation 10. Deploy patch through normal CI/CD with elevated priority. 11. If fix requires algorithm swap (e.g., HQC disabled in liboqs 0.13 due to CVE-2025-48946): use the crypto-agility framework to switch to alternative; rotate any keys generated under the vulnerable algorithm. 12. Post-incident review: did your monitoring detect anything anomalous before the disclosure? Did your detection time meet SLA?

Playbook B — "Vendor announces end-of-life for the PQC HSM/library we depend on"

Trigger example: HSM vendor sunsets a firmware family; library project archives a repository; cloud KMS deprecates a key spec.

Week 1 — Assessment

1. Read the EOL notice carefully: cutoff date, security-fix date, migration recommendation from vendor.
2. Identify the dependency chain: what services / customers / contracts depend on this product?
3. Estimate migration cost: do you need new hardware, new library, new vendor integration, or just configuration changes?

Week 2-4 — Migration plan 4. Evaluate replacements (use Unit 11.4 library selection criteria as starting point). 5. Run interop testing: can you encapsulate with old vendor and decapsulate with new vendor? Critical for data-at-rest where re-wrapping is needed. 6. Build the migration runbook: dual-deploy period (both old and new), rotation schedule, rollback criteria.

Month 2-N — Execution 7. Deploy new vendor in shadow mode (alongside old), verify correctness via canary. 8. Roll over services in priority order (most-critical-data-last to limit blast radius). 9. Once 100% of new traffic uses new vendor and re-wrap of historical data is complete, decommission old vendor with key destruction per policy. 10. Post-migration: update CBOM, deprecate old vendor's entry in your vendor risk database.

Playbook C — "New cryptanalysis published: theoretical attack on our primary algorithm, no CVE yet"

Trigger example: arXiv paper claims to reduce ML-KEM security by N bits; FALCON pre-finalization side-channel result; novel rejection-signature attack on ML-DSA.

Hours 1-24 — Read and reason

1. Read the paper. Identify: claimed security loss, attack model (chosen-CT, side-channel, fault, etc.), prerequisite (number of queries, equipment, oracle access).
2. Compare against the threat model for your deployment: does the attack require capabilities a real adversary has against you?
3. Subscribe to follow-up discussion: NIST pqc-forum, IACR ePrint comments, library issue trackers.

Days 2-30 — Watch and prepare 4. Wait for community response: is the attack reproduced? Is the security loss confirmed by independent analysis? Are mitigation patches proposed? 5. If attack is real and significant: pre-stage your migration playbook (Playbook A above). Identify replacement algorithms in your crypto-agility framework. 6. Engage with library maintainers: are they planning a release? Will they document the impact?

Decision criteria — when does "theoretical" become "operational"?

- A CVE is filed against your library (→ Playbook A).
- A working PoC is published.
- A vendor advisory recommends migration.
- The cryptanalysis result reduces the algorithm below your threat-model floor (e.g., ML-KEM-768 reduced from 192-bit to 128-bit security may still be acceptable; reduced to 80-bit is not).

Until one of these triggers fires: maintain awareness, do not panic-migrate. Speculative migrations have their own risks (interop breakage, increased attack surface during transition).

Case study: Meta's PQC Migration Framework (April 2026)

Meta Engineering published a detailed PQC migration writeup on April 16, 2026 ("Post-Quantum Cryptography Migration at Meta: Framework, Lessons, and Takeaways"). It is the most detailed real-world case study published to date and worth reading in full. Key takeaways summarized:

Five-level maturity model:

1. **Level 0 — Unaware:** no inventory, no plan.
2. **Level 1 — Cataloged:** CBOM complete; leadership aware.
3. **Level 2 — Tested:** key algorithms benchmarked on representative workloads.
4. **Level 3 — Rolling out:** hybrid PQ in production for specific services.
5. **Level 4 — Default:** hybrid PQ default across all new services; legacy on deprecation path.

Six-step migration strategy:

1. Cryptographic inventory (automated scanning across ~millions of repos).
2. Threat-model classification (HNDL-sensitive vs. not).
3. Algorithm selection per service class (ML-KEM-768+X25519 default KEM; ML-DSA-65+ECDSA default signature).
4. Library upgrades (internal fork of boringssl with PQ patches, now on native OpenSSL 3.5+ where possible).
5. Service-by-service hybrid rollout with monitoring (handshake success rates, latency p99, error budgets).
6. Legacy retirement with explicit gate criteria (e.g., "disable classical-only KEX when < 0.5% of traffic uses it").

Reported state (April 2026): most internal Meta services use hybrid ML-KEM+X25519 for transport security. External-facing communication is the next phase; the constraints are middlebox compatibility and partner readiness. No ML-DSA in production signing paths yet.

Lessons echoed from other large deployments:

- Automated CBOM generation is the cornerstone; without it, scope is unknowable.
- Middlebox incompatibility is the #1 production break (see Unit 11.9.2 for ClientHello size analysis).
- Monitoring for *silent downgrade* is as important as monitoring for failures — classical fallback without alerting is the real risk.

Full writeup: <https://engineering.fb.com/2026/04/16/security/post-quantum-cryptography-migration-at-meta-framework-lessons-and-takeaways/>

Unit 11.8 Summary

Production deployment is not the end of the migration work — it's the transition into operational responsibility for a new class of cryptographic dependencies. Comprehensive pre-deployment sign-off prevents the "ship it and hope" failure mode; continuous monitoring catches implementation regressions, adoption stagnation, and emerging attacks. Pair this checklist with crypto-agility (Unit 10.3) so that responding to a PQC vulnerability is a configuration change, not a re-architecture.

Unit 11.9: Cloud KMS Integration for Post-Quantum Cryptography

Most production cryptography in 2026 happens inside managed Key Management Services. This unit documents the PQC state of the major KMS/HSM offerings as of April 2026, gives working API examples, and walks through the engineering patterns that recur across all of them (composite certs, hybrid KEMs, cryptographic agility, CBOM).

Status evolves fast — the rollout snapshot below is current to **April 2026**. Always verify against the vendor's latest release notes before architecting a migration.

11.9.1 Vendor Landscape Snapshot (April 2026)

Last verified: April 2026 (v1.0). Cloud KMS/HSM offerings ship PQ features on a rapid cadence; always confirm status against the vendor's "What's New" page before architectural decisions.

VENDOR / PRODUCT	HYBRID TLS (ML-KEM)	PQ SIGNATURES (ML-DSA/ SLH-DSA)	PQ KEM KEYS (ML-KEM)	FIPS 140-3 W/ PQC
AWS KMS	GA (non-FIPS endpoints)	ML-DSA-44/65/87 GA Jun 2025	Preview	KMS HSM at 140-3 L3
AWS Private CA	N/A	ML-DSA GA Nov 2025	N/A	Module-level
AWS-LC / AWS-LC-FIPS	(library)	ML-DSA in v3.0	v3.0 first FIPS 140-3 with ML-KEM	Cert in process
GCP Cloud KMS	via GFE (Chrome)	ML-DSA-65/SLH-DSA-128s preview (Feb 2025)	ML-KEM-768/1024, X-Wing preview (Oct 2025)	SOFTWARE only; HSM pending
GCP Cloud HSM	N/A	No PQC	No PQC	Marvell LiquidSecurity (classical only)
Azure Key Vault / Managed HSM	No	No	No	No PQC shipped
Windows CNG / .NET 10	Schannel preview	ML-KEM/ML-DSA/SLH-DSA GA Nov 2025	GA Nov 2025	Pending
HashiCorp Vault Enterprise 1.19+	via TLS	ML-DSA, SLH-DSA experimental	Roadmap	No (experimental)
Thales Luna HSM (fw 7.9+)	via TLS	ML-DSA, SLH-DSA, LMS/ HSS	ML-KEM	Pending (CAVP done)
Entrust nShield (fw 13.8)	via TLS	ML-DSA, SLH-DSA	ML-KEM	Pending (CAVP Sep 2025)

VENDOR / PRODUCT	HYBRID TLS (ML-KEM)	PQ SIGNATURES (ML-DSA/ SLH-DSA)	PQ KEM KEYS (ML-KEM)	FIPS 140-3 W/ PQC
Utimaco u.trust / Quantum Protect	via TLS	ML-DSA, LMS/ HSS, XMSS	ML-KEM	CAVP done; CMVP pending
Fortanix DSM	via TLS	ML-DSA, LMS	ML-KEM	FIPS 140-2 L3 only
Cloudflare (edge TLS)	X25519MLKEM768 default (~57% of browser-initiated, pq.cloudflareresearch.com early 2026)	ML-DSA origin auth mid-2026	N/A	N/A
Akamai (edge TLS)	X25519MLKEM768 default Q1 2026	Roadmap	N/A	N/A
IBM Hyper Protect Crypto Services	via TLS	Dilithium (pre-standard)	Kyber (pre-standard)	FIPS 140-2 L4
Oracle OCI KMS / AI Database 26ai	Hybrid ML-KEM TLS Jan 2026	No Vault key types yet	No Vault key types yet	No
YubiHSM 2	N/A	No	No	FIPS 140-3 expected Q2 2026 (no PQC)

Reading the table:

- "GA (non-FIPS endpoints)" — the hybrid TLS handshake is enabled on the standard service endpoints but **not** on FIPS-only endpoints (FIPS validation must catch up first).
- "Preview" — accessible to customers but behavior may change; not covered by SLAs.

- "CAVP done; CMVP pending" — the algorithm implementation has passed NIST's Cryptographic Algorithm Validation Program, but the hardware module's FIPS 140-3 Cryptographic Module Validation Program certificate for the PQ firmware has not yet been issued.

11.9.2 AWS Integration

△ #1 PRODUCTION BREAK IN 2026 PQ-TLS DEPLOYMENTS: MIDDLEBOX CLIENTHELLO FRAGMENTATION.

The TLS ClientHello grows from ~500 bytes to ~1,600+ bytes once an **X25519MLKEM768** key share is included — large enough to fragment across TCP segments. Load balancers, WAFs, DPI engines, and corporate TLS-intercepting proxies with fixed input buffers or handshake-size heuristics silently drop the fragmented handshake. Industry guidance (Akamai, Cloudflare, IETF UTA WG): **audit every middlebox in the path** and run synthetic PQ ClientHello traffic across production topology *before* rolling out. Most production incidents we've catalogued trace to this.

AWS KMS exposes both hybrid post-quantum TLS (for key-material protection in transit) and PQ-signing key types (for customer signature operations).

Hybrid post-quantum TLS. Enable **X25519MLKEM768** for the TLS handshake between your client and the KMS service endpoint. Protects the *transport* of your data keys against harvest-now-decrypt-later. Available on KMS, ACM, Secrets Manager, S3 (regional endpoints), ALB/NLB, CloudFront, Transfer Family, Payment Cryptography, Private CA. Not yet available on FIPS KMS endpoints or on PrivateLink.

```
# Python example - uses boto3 with AWS-LC-backed Python cryptography
import boto3
from botocore.config import Config

# No code change required for hybrid TLS - it negotiates automatically
# when both client and service support X25519MLKEM768.
# To verify: inspect CloudTrail events, field tlsDetails.cipherSuite
kms = boto3.client('kms',
    config=Config(tcp_keepalive=True))
```

Customer-held ML-DSA keys. KMS created **SIGN_VERIFY** asymmetric keys with ML-DSA support in June 2025.

```

import boto3
kms = boto3.client('kms', region_name='us-west-1')

# Create ML-DSA-65 key (KeySpec values: ML_DSA_44, ML_DSA_65, ML_DSA_87)
response = kms.create_key(
    KeySpec='ML_DSA_65',
    KeyUsage='SIGN_VERIFY',
    Origin='AWS_KMS',
    Description='PQ signing key for application X'
)
key_id = response['KeyMetadata']['KeyId']

# Sign a message
signature = kms.sign(
    KeyId=key_id,
    Message=b'payload to sign',
    MessageType='RAW', # or EXTERNAL_MU for 64-byte pre-computed mu
    SigningAlgorithm='ML_DSA_SHAKE_256'
)['Signature']

# Verify
is_valid = kms.verify(
    KeyId=key_id,
    Message=b'payload to sign',
    Signature=signature,
    SigningAlgorithm='ML_DSA_SHAKE_256'
)['SignatureValid']

```

Key behaviors to plan for:

- **No automatic rotation** — ML-DSA keys follow the same rule as other asymmetric KMS keys (manual rotation only).
- **Standard asymmetric pricing** — \$1/month per key, \$0.15 per 10,000 sign/verify/GetPublicKey operations. No PQC premium.
- **Public key retrieval** — use `GetPublicKey` to export the raw NIST-PQC-formatted public key.

AWS Private CA with ML-DSA. As of November 2025, AWS Private CA can be configured to use ML-DSA-backed KMS keys as the CA signing key, issuing end-entity certificates with ML-DSA algorithm identifiers. Available in all commercial AWS regions, GovCloud, and China regions.

AWS-LC-FIPS v3.0 is the first open-source cryptographic library to include ML-KEM within its FIPS 140-3 module boundary (submitted, certificate in process as of late 2024/early 2026). This matters because most AWS services now use AWS-LC internally, so AWS-LC-FIPS 3.0's eventual certification unlocks FIPS 140-3 PQC across the AWS stack.

End-to-end ML-DSA code signing workflow (documented in the AWS Security Blog, November 2025): create a private CA on AWS Private CA with ML-DSA issuance enabled; issue code-signing leaf certificates backed by KMS ML-DSA keys; sign artifacts with `kms:Sign` using the same key material; verify using standard X.509 path validation augmented with ML-DSA signature verification. No AWS Signer service support for ML-DSA as of April 2026 — the workflow goes through KMS + Private CA directly.

11.9.3 Google Cloud Platform Integration

GCP Cloud KMS supports PQ signatures (preview since February 2025) and PQ KEMs (preview since October 2025) at the SOFTWARE protection level. Cloud HSM (Marvell LiquidSecurity) does **not** yet expose PQC algorithms; Google states it will add them to all NIST PQC standards but has not committed to a date.

Enum values (from the public `CryptoKeyVersionAlgorithm` proto):

- Signatures: `PQ_SIGN_ML_DSA_44`, `PQ_SIGN_ML_DSA_65`, `PQ_SIGN_ML_DSA_87`, plus `_EXTERNAL_MU` variants, `PQ_SIGN_SLH_DSA_SHA2_128S`, `PQ_SIGN_HASH_SLH_DSA_SHA2_128S_SHA256`
- KEMs: `ML_KEM_768`, `ML_KEM_1024`, `KEM_XWING` (the hybrid X25519 + ML-KEM-768 construction)

Python client example (ML-DSA-65 signing):

```
from google.cloud import kms_v1

client = kms_v1.KeyManagementServiceClient()
parent = client.key_ring_path("PROJECT_ID", "us-central1", "my-keyring")

crypto_key = kms_v1.CryptoKey(
    purpose=kms_v1.CryptoKey.CryptoKeyPurpose.ASYMMETRIC_SIGN,
    version_template=kms_v1.CryptoKeyVersionTemplate(
        protection_level=kms_v1.ProtectionLevel.SOFTWARE,
        algorithm=kms_v1.CryptoKeyVersion.CryptoKeyVersionAlgorithm.PQ_SIGN_ML_DSA_65,
    ),
)
created = client.create_crypto_key(
    parent=parent,
    crypto_key_id="ml-dsa-65-signing-key",
    crypto_key=crypto_key,
)

# Sign
version = f"{created.name}/cryptoKeyVersions/1"
digest = {"sha512": b"..."} # or use MessageType=RAW via sign_blob
signature = client.asymmetric_sign(name=version, digest=digest).signature
```

KEM workflow uses the new `Encapsulate` and `Decapsulate` RPCs (and `gcloud kms decapsulate` in the CLI):

```
# Retrieve the public key in raw NIST-PQC format
gcloud kms keys versions get-public-key 1 \
  --key my-mlkem-key --keyring my-ring --location us-central1 \
  --output-file ml-kem.pk --public-key-format nist-pqc

# Client-side encapsulation (library call) then:
gcloud kms decapsulate --version 1 --key my-mlkem-key \
  --keyring my-ring --location us-central1 \
  --ciphertext-file ct.bin --shared-secret-file ss.bin
```

Google recommends **X-Wing over pure ML-KEM** for general use — the hybrid construction protects against both quantum attacks and any future cryptanalytic surprise against ML-KEM.

Chrome and TLS. Chrome has shipped `X25519MLKEM768` as default TLS 1.3 key exchange since Chrome 131 (November 2024). Enterprise policy `PostQuantumKeyAgreementEnabled` was the disable switch; it is being removed in Chrome 147 (stable rollout started April 7, 2026). Adoption: ~57% of browser-initiated HTTPS connections observed at Cloudflare's edge in early 2026 (per `pq.cloudflare.com`), Chrome being the dominant source. When a Chrome client talks to a GCP external HTTPS load balancer, the handshake is automatically hybrid — no server configuration needed.

Google internal (ALTS) uses ML-KEM as of late 2025 (migrated from NTRU-HRSS which was deployed November 2022). Google's company-wide PQC migration target was moved up to 2029 in March 2026.

11.9.4 Microsoft Azure Integration

The short version: **Azure's data-plane KMS services do not expose PQC as of April 2026.** Microsoft's August 2025 "Quantum Safe Security Progress" post places Key Vault, Managed HSM, Entra signing, and related services in Phase 2 (starting 2026), with no concrete GA date announced.

What Microsoft *has* shipped (November 2025):

- **Windows CNG** (Cryptography Next Generation) — GA on Windows Server 2025 and Windows 11 24H2/25H2. Algorithm identifiers: `BCRYPT_MLKEM_ALGORITHM`, `BCRYPT_MLDSA_ALGORITHM`, `BCRYPT_SLHDSA_ALGORITHM`, `BCRYPT_LMS_ALGORITHM`, `BCRYPT_XMSS_ALGORITHM`. Parameter sets: `BCRYPT_MLKEM_PARAMETER_SET_512/768/1024`.
- **.NET 10** — `System.Security.Cryptography.MLKem` (GA), `MLDsa`, `SlhDsa`, `CompositeMLDsa` (marked `[Experimental]` under `SYSLIB5006`).

- **SymCrypt** — ML-KEM and ML-DSA in the public library (versions 103.5.0–103.11.0). SLH-DSA implementation status in SymCrypt is unclear from the public changelog.
- **Windows Schannel hybrid TLS** — preview only (Windows Insider builds). Windows 11 24H2 stable does **not** negotiate hybrid TLS by default.

CNG ML-KEM-768 key generation (C):

```
#include <bcrypt.h>

BCRYPT_KEY_HANDLE hKeyPair;
BCryptGenerateKeyPair(BCRYPT_MLKEM_ALG_HANDLE, &hKeyPair, 0, 0);
BCryptSetProperty(hKeyPair, BCRYPT_PARAMETER_SET_NAME,
    (PUCHAR)BCRYPT_MLKEM_PARAMETER_SET_768,
    sizeof(BCRYPT_MLKEM_PARAMETER_SET_768), 0);
BCryptFinalizeKeyPair(hKeyPair, 0);
// Then use BCryptEncapsulate / BCryptDecapsulate
```

.NET 10 equivalent:

```
using System.Security.Cryptography;

using (MLKem key = MLKem.GenerateKey(MLKemAlgorithm.MLKem768))
{
    string publicKeyPem = key.ExportSubjectPublicKeyInfoPem();
    // Encapsulate/decapsulate via key.Encapsulate/Decapsulate
}
```

SymCrypt FIPS 140-3 status: no CMVP certificate covering PQC algorithms under Microsoft's name is listed as of April 2026. The algorithm implementation is in the library; the validation is in progress. Compare this to AWS-LC-FIPS v3.0 (submitted with ML-KEM in boundary) — AWS moved first on this.

Practical implication for Azure users: if you need PQ-protected data in transit or PQ signatures *today*, you will need to either use Windows/.NET client-side primitives against a non-Azure-managed key, or wait for the Azure Phase 2 rollout. The symmetric AES-256 keys in Key Vault (marketed as "quantum-resistant per CNSA 2.0") are fine for data at rest, but the RSA-OAEP/AES-KW key wrap used by customer-managed keys is classical and must migrate once Azure exposes PQ KEMs.

11.9.5 Other Vendors — Quick Reference

HashiCorp Vault Enterprise 1.19+ introduced ML-DSA sign/verify in the Transit engine (experimental). Recommended usage pattern:

```
vault secrets enable transit
vault write -f transit/keys/pqc-signing-key type=ml-dsa parameter_set=65

# Sign
vault write transit/sign/pqc-signing-key \
  input=$(base64 <<< "message to sign")

# Verify
vault write transit/verify/pqc-signing-key \
  input=$(base64 <<< "message to sign") \
  signature=vault:v1:...
```

Not recommended for production until HashiCorp removes the experimental label.

Thales Luna HSM firmware 7.9+ (June 2025) supports ML-KEM, ML-DSA, SLH-DSA, plus stateful LMS/HSS via native PKCS#11 extensions:

```
/* PKCS#11 with Thales Luna extensions - ML-DSA-65 key pair */
CK_MECHANISM mech = { CKM_ML_DSA_KEY_PAIR_GEN, NULL_PTR, 0 };
CK_ULONG paramSet = CKP_ML_DSA_65;
CK_ATTRIBUTE pubTempl[] = {
    { CKA_CLASS, &pubClass, sizeof(pubClass) },
    { CKA_KEY_TYPE, &keyType, sizeof(keyType) },
    { CKA_PARAMETER_SET, &paramSet, sizeof(paramSet) }
};
C_GenerateKeyPair(hSession, &mech, pubTempl, 3,
                  privTempl, N, &hPub, &hPriv);
```

Classical FIPS 140-3 Level 3 certification is current; PQ-firmware FIPS 140-3 submission is in progress.

Entrust nShield firmware 13.8 (August 2025) and **Utimaco u.trust Quantum Protect** (launched April 2025) similarly offer native PQC in firmware. Both have CAVP algorithm validation but no completed FIPS 140-3 module certificate covering the PQ firmware as of April 2026.

Fortanix DSM and **IBM Hyper Protect Crypto Services** support PQC algorithms but have been on a slower FIPS validation track (FIPS 140-2 Level 3/4 classical; no 140-3 w/ PQC yet).

Edge TLS (Cloudflare / Akamai) — neither is a KMS in the traditional sense, but both terminate TLS for very large fractions of Internet traffic. Cloudflare has **X25519MLKEM768** enabled by default since June 2025 for client-to-edge, default-on for edge-to-origin since January 31, 2026. Akamai reached general default availability in Q1 2026. When your backend sits behind Cloudflare or Akamai, you're already getting hybrid PQ for the client-facing leg.

11.9.6 Standards and Drafts to Track

The migration moves faster than any single reference book can keep up with. Follow these:

Finalized

- **FIPS 203** (ML-KEM), **FIPS 204** (ML-DSA), **FIPS 205** (SLH-DSA) — August 2024
- **NIST SP 800-208** — Stateful hash-based signatures (LMS/XMSS)
- **NIST SP 800-227** — Recommendations for KEM key establishment (finalized September 18, 2025). Provides general KEM properties (IND-CCA, binding, reusability) plus explicit guidance on multi-algorithm KEMs and PQ/T hybrids; discusses X-Wing as an example. **Authoritative reference for ML-KEM deployment.**
- **RFC 9794** — Terminology for post-quantum/traditional hybrid schemes (June 2025)
- **RFC 9882** — ML-DSA in CMS (2025) — **explicitly forbids HashML-DSA in CMS / X.509 PKIX**; use pure ML-DSA only.
- **CA/B Forum Ballot SMC013** — Adopted August 2025; enables single-algorithm PQC for S/MIME

In progress (April 2026) — *Last verified: April 2026 (v1.0); IETF drafts change on a 6-week cadence. Check datatracker.ietf.org for current versions.*

- **NIST IR 8547** — Transition timeline; RSA/ECC deprecated 2030, disallowed 2035
- **NIST SP 800-57 Part 1 Rev. 6** — Key management with PQC (IPD; comments closed Feb 2026)
- **NIST SP 1800-38** volumes B and C — NCCoE PQ Migration practice guide
- **FIPS 206** — FN-DSA / FALCON (IPD August 2025; expected late 2026/2027)
- **FIPS 207** — HQC backup KEM (IPD targeted 2026, Final 2027); see Unit 10.5 for implementation caveats
- **draft-ietf-lamps-pq-composite-sigs-16** — Composite ML-DSA for X.509 and CMS

- **draft-ietf-lamps-pq-composite-kem-14** — Composite ML-KEM for X.509 and CMS
- **draft-ietf-tls-ecdhe-mlkem-04** — [X25519MLKEM768](#) , [SecP256r1MLKEM768](#) , [SecP384r1MLKEM1024](#) in TLS 1.3
- **draft-ietf-sshml-mlkem-hybrid-kex-10** — SSH hybrid KEX with ML-KEM
- **draft-ietf-uta-pqc-app-01** — PQC recommendations for TLS applications
- **draft-ietf-ipsecme-ikev2-mlkem-05** (March 2026) — ML-KEM in IKEv2 for IPsec; Cloudflare has deployed in production interop-tested with strongSwan
- **draft-ietf-hpke-pq** — PQ/T hybrid for HPKE (RFC 9180) — important for MLS and Signal-like protocols
- **draft-ietf-cose-post-quantum-signatures** / **draft-ietf-jose-pqc-kem** — COSE/JOSE encodings; enables PQ passkeys/WebAuthn
- **draft-ietf-pquip-hybrid-signature-spectrums** — Classification of hybrid signature design goals
- **draft-ietf-plants-merkle-tree-certs** — Merkle Tree Certificates (see Unit 8.6); **PLANTS WG chartered 2026**
- **draft-ietf-tls-key-share-prediction** — Mitigates hybrid key-share misprediction RTT cost
- **draft-sheth-pqc-dnssec-strategy** (April 2026) — SLH-DSA in Merkle Tree Ladder mode for DNSSEC

Regulatory / operational gates

- **NSA CNSA 2.0** — New National Security Systems acquisitions must support CNSA 2.0 by January 1, 2027; CSfC PQC Addendum published 2025
- **CA/B Forum** — Code-signing certificate max lifetime 459 days effective March 1, 2026 (CAs stop issuing longer certs on February 24, 2026); TLS cert max 47 days target by 2029
- **OMB M-23-02** — US federal agencies must maintain prioritized cryptographic inventory (CBOM-ready)
- **FIPS 140-3 transition deadline — September 21, 2026.** FIPS 140-2 validations move to CVMP historical list on that date. As of early 2026, **no HSM vendor had completed FIPS 140-3 Level 3 CMVP validation INCLUDING PQC** — a specific barrier for SWIFT and banking per the G7 Cyber Expert Group's January 2026 report.
- **UK NCSC three-phase roadmap** (March 2025) — Phase 1 (→2028): inventory, build migration plan. Phase 2 (2028–2031): high-priority upgrades. Phase 3 (2031–2035): complete migration.

- **Australia ASD** — PQC transition plan required by end of 2026; legacy asymmetric crypto banned by end of 2030 (earlier than NIST/NSA).
- **EU Coordinated Implementation Roadmap** — Member states begin transition by end 2026; critical infrastructure by end 2030; full system transition 2035. EU Cyber Resilience Act (in force Dec 10, 2024; main obligations Dec 11, 2027) requires state-of-the-art crypto including PQC.
- **Japan CRYPTREC** — ML-KEM external evaluation completed April 2, 2026 (PQShield); national PQC roadmap expected May 2027, full transition target 2035.

Regulatory Compliance Matrix — One-Page Reference (April 2026)

Last verified: April 2026 (v1.0). Compliance regimes evolve quickly; always confirm against the regulator's primary publication before architectural decisions.

JURISDICTION	FRAMEWORK	KEY DATE(S)	APPLIES TO	REQUIRED ACTION	PENALTY / CONSEQUENCE IF MISSED
USA (Federal)	FIPS 140-3 + NIST IR 8547	Sept 21, 2026 (140-2 → historical); 2030 deprecate, 2035 disallowed RSA/ECC	Cryptographic modules used by USG	Migrate to PQC-capable FIPS 140-3 module; deprecate classical asymmetric per IR 8547 schedule	Ineligibility for federal contracts; FedRAMP authorization withdrawn
USA (NSA)	CNSA 2.0 + CSfC PQC Addendum	Jan 1, 2027 (new acquisitions); 2030–2035 (full)	National Security Systems	New acquisitions must support CNSA 2.0; full migration by 2030 (most categories) or 2033 (OS) or 2035 (last)	Loss of NSS clearance compliance; product cannot be sold to NSS
USA (Industry)	CA/B Forum (TLS, code signing)	March 1, 2026 (459-day code-signing); 2029 target (47-day TLS)	Public CAs and their customers	Code-signing certs ≤459 days; plan for 47-day TLS certs by 2029	CAs that don't comply: distrusted by browsers; org's certs invalid
EU	Cyber Resilience Act + Coordinated Implementation Roadmap	Sept 11, 2026 (CRA reporting); Dec 11, 2027 (CRA main obligations); 2030 (critical infra); 2035 (full)	Networked products + critical infrastructure	Crypto-agility by Dec 2027; PQC for critical infra by 2030	Product recalls, market bans, fines up to 2.5% global turnover or €15M

JURISDICTION	FRAMEWORK	KEY DATE(S)	APPLIES TO	REQUIRED ACTION	PENALTY / CONSEQUENCE IF MISSED
UK	NCSC 3-Phase Roadmap + IPA endorsement	Phase 1 → 2028 (inventory + plan); Phase 2 (2028–2031, high-priority); Phase 3 (2031–2035, full)	Critical national infrastructure + government supply chain	Phase 1: complete CBOM, build migration plan; Phase 2: high-priority systems; Phase 3: total	Ineligibility for HMG contracts; CNI regulator action
Australia	ASD ISM Guidelines	End of 2026 (transition plan due); Dec 31, 2030 (legacy asymmetric banned)	All Australian Government systems + accredited service providers	Documented PQC plan by end 2026; complete migration by 2030 (more aggressive than NIST)	Loss of IRAP certification; ineligibility for government work
Germany	BSI TR-02102-1 (Jan 2025) + Migration Plan	End-2030: highest-sensitivity asymmetric encryption; End-2031: general asymmetric encryption; End-2035: classical signatures (annual TR updates)	Federal IT systems + KRITIS critical infrastructure operators	Follow BSI annual algorithm guidance; migrate KRITIS per BSI staged roadmap	KRITIS operators: regulatory action; gov contracts revoked

JURISDICTION	FRAMEWORK	KEY DATE(S)	APPLIES TO	REQUIRED ACTION	PENALTY / CONSEQUENCE IF MISSED
France	ANSSI 3-Phase Plan (Position Paper Apr 2022, rev. Jan 2024)	Phase 1 (from 2025): defense-in-depth hybrid PQ+classical; Phase 2 (from 2026): hybrid mandatory for new sensitive systems; Phase 3 (post-2030): post-quantum-only permitted	Government, OIV (operators of vital importance)	ANSSI-qualified hybrid (PQ + classical) for sensitive systems through 2030; pure PQC permitted only post-2030	Loss of ANSSI qualification; OIV regulatory penalties
Japan	CRYPTREC	National roadmap May 2027; full transition target 2035	Government, financial, critical infrastructure	Align with national roadmap once published	Gradual adoption pressure (less prescriptive than US/EU)
G7 Financial	G7 Cyber Expert Group January 2026 roadmap	2030–2032 critical financial systems migration	SWIFT, central banks, large global financial institutions	Hybrid PQ for critical financial infrastructure by 2030–2032	Industry/regulator pressure; potential loss of correspondent-banking relationships
Canada	CSE / ITSM 40.001	2026 federal-systems plan due; transition through 2030s	Government of Canada systems	Federal systems must have PQC plan by 2026	Treasury Board policy non-compliance

How to use this table. Identify your jurisdiction(s) and applicable frameworks. The "Required action" column is your *minimum* obligation — PQC industry best practice (hybrid PQ deployments, complete CBOM, crypto-agility) often exceeds the regulatory floor. For multi-jurisdiction operators (e.g., a global SaaS provider), the *most stringent* deadline drives planning: a US/EU/Australia operator must meet Australia's 2030 ban on legacy asymmetric crypto regardless of NIST's 2035 disallowed date.

Worked example: A US-headquartered cloud provider serving EU customers must:

- US FedRAMP/CNSA 2.0: support PQC for new federal contracts by Jan 2027.
- EU CRA: crypto-agility by Dec 2027; PQC for critical infrastructure by 2030.
- CA/B Forum: code-signing certs ≤ 459 days starting March 2026.
- → Architecturally: deploy hybrid X25519+ML-KEM-768 by 2026, ML-DSA-65 cert chains by 2027, full PQC for any EU CNI customer by 2030.
- **G7 Cyber Expert Group financial roadmap** (January 2026) — Critical financial systems 2030–2032 target.

Ecosystem Snapshot — Client/Runtime Library Versions (April 2026)

Last verified: April 2026 (v1.0). Pin minimums to what this table states; check upstream releases for newer fixes (especially security-relevant CVE patches listed in Unit 11.10).

Pin to these or later for native PQC support:

COMPONENT	MINIMUM VERSION	NOTES
OpenSSL	3.5.6 or 3.6.2	Native ML-KEM/ML-DSA/SLH-DSA; includes CVE-2025-15469 fix. 3.5 is LTS to April 2030.
OpenSSH	10.0 (client/server)	<code>mlkem768x25519-sha256</code> is default KEX. 10.2+ warns on non-PQ negotiation.
liboqs	0.15.0	Brings <code>mlkem-native</code> , AVX-512VL SHA3. Pair with <code>oqs-provider</code> 0.10.x (0.11 tracks liboqs 0.15 on main branch).
Bouncy Castle (Java)	1.84+	PQC provider key conversions; prior 1.81 finalized PKCS#8 seed-only + expanded-key formats.
Bouncy Castle (C#)	2.6.1+	Parity with Java; composite signatures.
Go stdlib	1.25+	<code>crypto/mlkem</code> stdlib (1.24+); <code>X25519MLKEM768</code> default in <code>crypto/tls</code> . Go 1.25 (Aug 2025) fixes the FIPS-only mode panic (issue #75528). Current stable: 1.26.
Rustls	0.23+	<code>X25519MLKEM768</code> via aws-lc-rs backend; ML-DSA support landed.
Node.js	Planned; track OpenSSL upgrade cadence	Node uses bundled OpenSSL; native PQC arrives when Node ships with OpenSSL 3.5+.
JDK	JDK 24+ (JEP 496, ML-KEM primitive); JDK 27 EA today (GA Sep 2026, JEP 527 adds hybrid ECDHE+ML-KEM in TLS 1.3)	ML-KEM primitive usable in current stable JDK via JEP 496; hybrid TLS integration ships with JDK 27 GA in Sep 2026 — early-access available now if you want to validate against your stack.
.NET	10	<code>System.Security.Cryptography.MLKem</code> GA Nov 2025; <code>MLDsa/SlhDsa</code> marked experimental.
Windows	11 24H2 / Server 2025	CNG PQC GA Nov 2025 (<code>BCRYPT_MLKEM_ALGORITHM</code> etc.); Schannel hybrid TLS still preview.
macOS / iOS	macOS Tahoe 26 / iOS 26 (Sep 2025)	TLS 1.3 PQ keyshare via URLSession/Network framework by default; CryptoKit exposes primitives.
Red Hat Enterprise Linux	10.1	PQ default in OpenSSL 3.5 provider, GnuTLS, NSS, OpenSSH; Linux kernel crypto API still lacks PQC.

COMPONENT	MINIMUM VERSION	NOTES
AWS-LC / AWS-LC-FIPS	3.0	First crypto library with FIPS 140-3 validation INCLUDING ML-KEM (cert in process).
HashiCorp Vault Enterprise	1.19+	ML-DSA sign/verify in Transit engine (experimental); SLH-DSA added later.
sigstore cosign	3.x	Config-driven algorithm selection; ML-DSA signing via KMS BYOK.
Chrome	131+ (enterprise knob disappears in Chrome 147, April 2026)	X25519MLKEM768 default; ~57% of browser-initiated traffic at Cloudflare's edge by early 2026 (see pq.cloudflareresearch.com).

11.9.7 Migration Cookbook — Common Scenarios

The following cookbooks apply regardless of which KMS vendor you use. They assume you have already:

1. Completed a cryptographic inventory (see Unit 11.2) and published a CBOM per OWASP CycloneDX 1.6 (see Unit 11.9.8)
2. Enabled hybrid PQ TLS on all control-plane connections (AWS/GCP/Cloudflare done automatically; Azure requires waiting)
3. Identified which data has a confidentiality window longer than the HNDL threat horizon (anything meant to remain secret past ~2030)

Scenario A: Database column encryption (envelope encryption)

Current state: plaintext → AES-256-GCM data encryption key (DEK) → DEK wrapped with RSA-2048 OAEP public key stored in KMS.

Migration pattern (non-breaking, dual-wrap):

1. Enable a PQ KEM key alongside the existing RSA key (ML-KEM-768 or X-Wing on GCP; AWS KMS ML-KEM preview; Thales Luna or Fortanix DSM for self-hosted).
2. Change the wrap function from raw RSA-OAEP to **HPKE (RFC 9180)** so that swapping KEMs later is a one-line change.
3. Dual-wrap every new DEK:

```
wrapped_classical = RSAOAEP.Wrap(RSA_PK, DEK)
wrapped_pq        = HPKE.Seal(MLKEM_PK, info, aad, DEK)
store: (ciphertext, wrapped_classical, wrapped_pq, alg_id)
```


4. On decrypt, prefer `wrapped_pq`, fall back to `wrapped_classical`.
5. After the PQ wrap covers 100% of new writes *and* a re-wrap batch job has converted historical rows, delete the classical wrap column.

Storage impact: ML-KEM-768 ciphertext is 1,088 bytes vs ~256 for RSA-2048. For very large tables do NOT keep wrapped DEKs inline on every row — use a two-level key hierarchy where a single Key Encryption Key (KEK) wraps a batch of DEKs.

Scenario B: TLS server migration

Order of operations matters:

1. **Enable hybrid KEMs first** — no certificate change required. Protects against HNDL immediately:

```
# OpenSSL 3.5+ server
openssl s_server -key server.key -cert server.crt \
  -groups 'X25519MLKEM768:X25519:P-256' -tls1_3 -accept 8443
```

2. **Test middlebox breakage** — the ClientHello grows from ~500 bytes to ~1,600+ bytes with `X25519MLKEM768`. Load balancers, WAFs, DPI engines, and corporate TLS-intercepting proxies frequently truncate or reject. Test with `openssl s_client -connect host:8443 -groups X25519MLKEM768 -msg`.
3. **Then migrate certificates** — options:
 - **Dual-certificate**: present both classical and composite leaves; client picks via `signature_algorithms_cert`.
 - **Composite-only**: works only when every client is PQ-capable (internal services).
 - **Classical cert + hybrid KEM**: acceptable for HNDL confidentiality; does NOT protect against an active attacker with a cryptographically relevant quantum computer. The default "2026 reasonable" posture.

Scenario C: Code signing

For bootloader and immutable firmware, NSA and NIST both prefer stateful hash-based signatures (LMS/XMSS per SP 800-208) over ML-DSA because verifiers can be hash-only. For general code, ML-DSA is appropriate.

USE CASE	RECOMMENDED (2026)
Bootloader / secure boot	LMS or XMSS (stateful)
OS kernel modules (RPM/DEB)	ML-DSA-65 or composite
Windows Authenticode	ML-DSA-65 composite cert (via ADCS; ADCS PQ GA early 2026)
Container images	ML-DSA via cosign (sigstore ML-DSA in progress)
Firmware (long-lived)	LMS/XMSS (plan tree depth to match product lifetime)

Critical stateful-signature warning: LMS/XMSS keys are **single-use per leaf**. Losing state means key re-use, which breaks the security assumption. Cloud KMS vendors expose LMS/XMSS carefully and require HSM-backed state tracking.

Scenario D: SSH host + user keys

OpenSSH 10.0 (April 2025) made **mlkem768x25519-sha256** the **default** key exchange. Versions 10.1+ warn when the client selects classical-only KEX.

```
# /etc/ssh/sshd_config - server side
KexAlgorithms mlkem768x25519-sha256,sntrup761x25519-sha512
HostKeyAlgorithms ssh-ed25519,rsa-sha2-512          # host sig still classical in 2026
PubkeyAcceptedAlgorithms ssh-ed25519,rsa-sha2-512
```

Host keys and user keys remain classical (Ed25519, RSA) in 2026 — the SSH WG drafts for PQ host keys are still in progress. Migrate KEX now, hold signatures until the SSHM drafts finalize.

Scenario E: Backup encryption

This is where HNDL bites hardest — retention windows of 5-30 years guarantee classical-encrypted backups will outlive RSA/ECC. Apply the dual-wrap pattern from Scenario A, migrate newest-first, then batch-re-wrap historical archives.

Watch out for backup-engine header formats (Veeam, Commvault, NetBackup) that store wrapped DEKs in fixed-width metadata slots — ML-KEM-wrapped keys are ~1 KB and may not fit. Raise a ticket with your vendor early.

Scenario F: S/MIME email

CA/B Ballot SMC013 (adopted August 2025) permits single-algorithm (non-hybrid) PQC S/MIME certificates to enable CA experimentation. Practical pattern:

1. Deploy PQ *encryption* certs first — HNDL applies to stored email.
2. Keep signing on ECDSA/RSA until PQ S/MIME is ubiquitous at recipient MTAs.

3. For recipients with classical-only clients, dual-send (once classical, once PQ) until Outlook/Apple Mail/Thunderbird handle composite certs uniformly.

PQ-enabled S/MIME certs are available from pilot CAs today; scale availability expected Q3 2026, broad deployment into 2027.

Scenario G: Container image signing

Use `cosign` (sigstore) with KMS-backed ML-DSA keys:

```
# Sign with AWS KMS ML-DSA key
cosign sign --key "awskms:///alias/pqc-sign-mlds65" image@sha256:...
cosign verify --key "awskms:///alias/pqc-sign-mlds65" image@sha256:...
```

Enforce via Kyverno / OPA-Gatekeeper admission controller with a `requiredAlgorithm: ML-DSA-65` policy field once policy controllers expose it.

Scenario H: Cryptographic Bill of Materials (CBOM)

CBOM is the foundation — you cannot migrate what you haven't inventoried.

The de-facto standard is **OWASP CycloneDX 1.6** (which absorbed IBM's CBOM spec in 2024). A CBOM records every cryptographic asset (algorithm, protocol, certificate, key, secret), its location, dependencies, parameters, and compliance tags (FIPS, CNSA 1.0, CNSA 2.0).

Tooling as of April 2026:

- **cdxgen** (OWASP, polyglot scanner emits SBOM/CBOM/OBOM) — current mature tool
- **IBM CBOM repository** — reference implementations and sample CBOMs
- **ReversingLabs Spectra Assure** — commercial, CycloneDX 1.6 CBOM support
- **Wiz, Snyk** — crypto-asset discovery in newer scanner versions

CI/CD integration pattern:

1. Build: cdxgen emits SBOM + CBOM alongside artifact
2. Gate: policy engine (OPA) asserts no `algorithmFamily: rsa|ecdsa` at `cryptoSuite` level beyond a configured cutoff date
3. Publish: CBOM co-signed with the artifact (sigstore / in-toto)
4. Runtime: register CBOM with central crypto inventory dashboard

Scenario I: Capacity planning for PQC handshakes (the cookbook nobody writes)

Hybrid PQC enlarges three things you must budget: ClientHello bytes, KMS QPS, and certificate-chain bytes. Use these envelope numbers when sizing for 2026–2030.

RESOURCE	CLASSICAL BASELINE	+ HYBRID PQC (2026)	PURE PQC (2030+ PROJECTION)	IMPLICATION
TLS 1.3 ClientHello (key share)	~250 B (X25519 only)	~1,500 B (X25519MLKEM768)	~1,200 B (ML- KEM-768 only)	Many middleboxes drop fragments at >1.5 KB; raise tcp_window and validate path-MTU.
TLS 1.3 ServerHello + cert chain	~3 KB (RSA-2048 leaf+intermediate)	~14-17 KB (ML- DSA-65 composite cert chain w/ intermediates)	~6 KB (MTC inclusion proofs, see Unit 8.6)	Round-trips: with cwnd 10 (Linux default) you blow through initial congestion window — the handshake takes an extra RTT. Bump initcwnd to 20-30 if your PQ rollout shows latency regressions.
KMS Sign QPS budget (ML- DSA-65)	ECDSA-P256 ≈ 5,000 QPS/HSM partition	~600-1,200 QPS/ HSM partition (5-8× more cycles)	Same	Plan to 5-8× the HSM count for sign-bound workloads (mTLS auth servers, code-signing CIs). Verify is unaffected (~classical cost).
KMS Decap QPS budget (ML- KEM-768)	ECDH-P256 ≈ 8,000 QPS/HSM partition	~3,000 QPS/HSM partition (~3× more cycles)	Same	Less painful than Sign but still budget conservatively for first migration cohort.

RESOURCE	CLASSICAL BASELINE	+ HYBRID PQC (2026)	PURE PQC (2030+ PROJECTION)	IMPLICATION
WAN RTT budget (extra hop)	1 RTT for TLS 1.3	Same (no protocol- level extra round- trip)	Same	But the handshake bytes are larger — bandwidth- delay-product matters. On a 200 ms RTT link, classical handshake $\approx$ 16 KB BDP; hybrid handshake $\approx$ 30 KB BDP. Saturates a 1.2 Mb/s link briefly.
Cert renewal frequency	ECDSA: $\sim$ 12 months avg	Composite: same; PQ-only: same	Drops to 47 days by 2029 (CA/B trajectory)	47-day rotation requires fully automated ACME pipelines. Plan the automation now, not in 2028.
Backup ciphertext overhead	RSA-OAEP wrap $\approx$ 256 B	ML-KEM wrap $\approx$ 1,088 B; X-Wing wrap $\approx$ 1,120 B	Same	If your backup tool stores wrapped DEKs in a fixed-width header, raise this with the vendor before Q3 2026.

Sizing cheat sheet (rough rules of thumb for SRE planning):

- **Public-facing TLS termination (Cloudflare/edge):** No additional capacity needed in 2026 — edge nodes do PQ termination, origin connections can stay classical until you're ready. Plan to add $\sim$ 10–20% origin handshake CPU when you flip origin TLS to PQ in 2027–2028.
- **Internal mTLS mesh (Istio/Linkerd/Consul):** Sign QPS dominates. If your service mesh sees 50K req/s and each request requires one mTLS handshake-on-resume, you do $\sim$ 2K full handshakes/s — within a single HSM partition for ECDSA but at the edge for ML-DSA. Add a second HSM partition or migrate to session-resumption-heavy patterns first.

- **CI/CD code signing (cosign + KMS):** Verify is fast; sign is slow. Centralize signing in a sidecar that batches; do not call `kms:Sign` per artifact in a tight loop. AWS KMS ML-DSA pricing is the same per-op as classical but you'll do fewer ops/sec.
- **Database envelope encryption:** Wrap once (per DEK), unwrap on read. ML-KEM decap is $\sim 3\times$ ECDH cost — for read-heavy DBs, this can dominate KMS spend. Use a two-level KEK→DEK hierarchy (Scenario A) and cache unwrapped DEKs per process.
- **VPN concentrators (IPsec IKEv2):** IKEv2 PQ KEM (per `draft-ietf-ipsecme-ikev2-mlkem-05`) doubles the IKE handshake size. For VPN aggregators handling 10K+ tunnel rekeys/hour, profile before/after; some hardware accelerators (Cavium, Mellanox) lack PQ offload.

Worked example — sizing a 2026 migration:

SaaS provider, 50K monthly active users, 2K req/s peak, public TLS terminated at Cloudflare, internal mTLS mesh on AWS, KMS-backed signing for receipts and audit logs.

Edge: Zero changes — Cloudflare flips PQ KEM transparently. Confirm origin still happy with `X25519MLKEM768` arriving from Cloudflare's CF→origin tunnel (it is, as of late 2025).

Internal mesh: $2\text{K req/s} \times \sim 30\%$ handshake-per-req (resume rate after PQ rollout drops session-ticket reuse rates initially) = ~ 600 handshakes/s. ML-DSA-65 verify is fast ($\sim$ classical cost) so verify-side is fine; sign-side (only your private CA, not per-request) is unchanged. **No HSM capacity bump needed.**

KMS: Receipts and audit logs sign at ~ 50 ops/s peak. ML-DSA-65 sign budget is $\sim 1\text{K QPS/HSM partition}$; comfortably under. **No bump needed.**

DB envelope encryption: 5K reads/s, each unwrapping a DEK. ECDH unwrap = $5\text{K} \times 100\ \mu\text{s} = 500\ \text{ms/s}$ of HSM time. ML-KEM unwrap = $5\text{K} \times 300\ \mu\text{s} = 1.5\ \text{s/s}$ — **fits in one HSM partition with 50% margin**, but you've used most of your headroom. Cache unwrapped DEKs per app process for 60s to drop unwraps $10\times$.

Bottom line:

This shape of workload absorbs the 2026 migration without infra changes. The pain shows up in higher-throughput shapes — mTLS-heavy meshes ($>5\text{K}$ handshakes/s), code-signing pipelines, and VPN concentrators.

Scenario J: Stripe Webhook PQC migration (worked example, end-to-end)

Webhook signing is the canonical "small but high-stakes" PQC migration: a third party (Stripe, GitHub, Slack) sends you a signed payload; you verify; you act. Today nearly all webhook-signing schemes use HMAC-SHA256 (symmetric, not affected by Q-Day) — but a growing number use asymmetric (ECDSA, Ed25519) and those are PQ-vulnerable.

Stripe's webhook signing today is HMAC-SHA256 (`stripe-signature` header) — it does NOT need PQC migration for the Q-Day threat (HMAC is post-quantum-secure under Grover with doubled key length, which Stripe already provides at 256-bit). **What follows is illustrative for asymmetric webhook schemes** (e.g., GitHub App webhooks signed with `RS256` JWS, or any webhook moving from `RS256 / ES256` JWS to ML-DSA).

Migration pattern (illustrative for an asymmetric-signed webhook):

1. **Discovery.** Add the webhook endpoint to your CBOM (Scenario H). Note the current algorithm (`alg` : `RS256`, `ES256`, `EdDSA`) and the JWKS URL of the sender.
2. **Dual-verify period.** Sender publishes both classical and PQ public keys at JWKS endpoint. Receiver:

```
import jwt, base64
from cryptography.hazmat.primitives.asymmetric import ed25519
from oqs import Signature # PQ via liboqs Python bindings

def verify_webhook_sig(payload, sig_header, jwks):
    header = decode_jws_header(sig_header)
    kid = header['kid']; alg = header['alg']
    jwk = jwks_lookup(jwks, kid)
    if alg == 'ML-DSA-65':
        pk = jwk['x'].encode() # raw 192B bytes, base64url-decoded
        sig = base64url_decode(extract_sig(sig_header))
        with Signature("ML-DSA-65") as sigverify:
            if not sigverify.verify(payload, sig, pk):
                raise InvalidSignature
    elif alg == 'EdDSA':
        pk = ed25519.Ed25519PublicKey.from_public_bytes(jwk['x'].encode())
        pk.verify(sig, payload)
    else:
        raise UnsupportedAlg(alg)
```

3. **Cutover.** Sender publishes only the ML-DSA `kid` in JWKS. Receivers that haven't migrated start failing — coordinate via API version or deprecation header (e.g., `Stripe-Version: 2027-01-15-pq`).

4. **Rollback gate.** Keep classical verification code path live for 30 days post-cutover; gated behind a config flag in case interoper bugs surface (FIPS-204 spec violations have been found in libcrux/hpke-rs, see Unit 11.11). Roll back via flag-flip without code deploy.
5. **Decommission.** After 30+ days clean, delete classical verifier and rotate signing keys at the sender.

Common pitfalls observed in 2025–2026 webhook PQC pilots:

- **JWS header bloat.** ML-DSA-65 sig is 3,309 B — base64url-encoded that's ~4,400 B in the JWS. If your reverse proxy has a 4KB header limit (nginx default), you'll silently drop large payloads. Raise `large_client_header_buffers`.
- **Replay-window timing.** Classical `RS256` verify is ~50 μ s; ML-DSA-65 verify is ~400 μ s in pure Python (via liboqs), 100 μ s in C. If your replay window is ≤ 5 ms (e.g., Stripe's anti-replay window), you have headroom; if your design called for sub-ms verify, profile first.
- **Detached signature length not in spec.** Some webhook senders (legacy systems) pack the signature into a fixed-size header field (`X-Sig` truncated at 1024 bytes). ML-DSA does not fit. Talk to vendor before cutover.

Scenario K: X.509 Hybrid Cert CLI Recipe (openssl + oqs-provider, end-to-end)

This is the hands-on cheat sheet for building a hybrid PQ certificate today. Tested with OpenSSL 3.5.6 + oqs-provider 0.10.0 + liboqs 0.15 on Linux (April 2026). Composite signature spec: `draft-ietf-lamps-pq-composite-sigs-16`.

```
# -----
# Pre-req: OpenSSL 3.5+ with oqs-provider loaded
# -----
# Confirm the provider is visible:
openssl list -providers
# Should show: default, oqsprovider

# -----
# Step 1 – Generate a composite ML-DSA-65 + Ed25519 root CA key
# -----
openssl genpkey -algorithm mlrsa65_ed25519 \
  -out root-ca.key.pem \
  -provider oqsprovider -provider default

# -----
# Step 2 – Self-sign the root CA (composite signature)
# -----
openssl req -new -x509 -nodes \
  -key root-ca.key.pem \
```



```

    -out root-ca.crt.pem \
    -days 3650 \
    -subj "/CN=Acme PQ Root CA 2026/O=Acme/C=US" \
    -provider oqsprovider -provider default

# Inspect: should show signatureAlgorithm = id-MLDSA65-Ed25519
openssl x509 -in root-ca.crt.pem -text -noout | grep -A2 "Signature Algorithm"

# -----
# Step 3 – Issue a leaf cert (server)
# -----
openssl genpkey -algorithm mldsa65_ed25519 \
    -out leaf.key.pem \
    -provider oqsprovider -provider default

openssl req -new -nodes \
    -key leaf.key.pem \
    -out leaf.csr \
    -subj "/CN=api.example.com/O=Acme/C=US" \
    -addext "subjectAltName=DNS:api.example.com,DNS:*.api.example.com" \
    -addext "keyUsage=digitalSignature,keyEncipherment" \
    -addext "extendedKeyUsage=serverAuth,clientAuth" \
    -provider oqsprovider -provider default

openssl x509 -req -in leaf.csr \
    -CA root-ca.crt.pem -CAkey root-ca.key.pem \
    -CAcreateserial \
    -out leaf.crt.pem \
    -days 90 \
    -copy_extensions copy \
    -provider oqsprovider -provider default

# -----
# Step 4 – Build the chain bundle (server presents this)
# -----
cat leaf.crt.pem root-ca.crt.pem > leaf-fullchain.pem

# -----
# Step 5 – Verify the chain end-to-end
# -----
openssl verify -CAfile root-ca.crt.pem leaf.crt.pem
# Expected: leaf.crt.pem: OK

# -----
# Step 6 – Test in a TLS server
# -----
openssl s_server -accept 8443 \
    -cert leaf-fullchain.pem -key leaf.key.pem \
    -groups X25519MLKEM768:X25519:P-256 \
    -tls1_3 \
    -provider oqsprovider -provider default

# In another shell:
openssl s_client -connect localhost:8443 \
    -groups X25519MLKEM768 \
    -CAfile root-ca.crt.pem \
    -showcerts \

```

```
-provider oqsprovider -provider default 2>&1 | grep -E "Server Temp Key|Verification|
Cipher"
# Expected:
#   Server Temp Key: X25519MLKEM768
#   Verification: OK
#   Cipher: TLS_AES_256_GCM_SHA384
```

Common gotchas:

- **openssl req -x509 -addext** order matters in some 3.5.x builds — put **-addext** AFTER **-subj**.
- **mldsa65_ed25519** is the composite algorithm name; **mldsa65** alone is pure-PQ (no classical fallback). For HNDL-defense use composite; for FIPS-only environments use **mldsa65**.
- **OID conflicts** — early oqs-provider releases used pre-standard OIDs. Composite OIDs were finalized in [draft-ietf-lamps-pq-composite-sigs-16](#). If certs from 2024 don't validate, regenerate them with current oqs-provider.
- **-copy_extensions copy** is required when issuing the leaf — without it, SANs disappear silently.

To convert to formats other tools accept:

```
# DER
openssl x509 -in leaf.crt.pem -outform DER -out leaf.crt.der
# PKCS#12 bundle (with key)
openssl pkcs12 -export -in leaf-fullchain.pem -inkey leaf.key.pem -out leaf.p12
# JWK (for JWS verifiers – requires manual encode of the composite SPKI)
# (no native openssl tool yet; use a Python helper with cryptography + oqs)
```

Scenario L: PQC Interop Test Matrix (April 2026)

The single most common PQC migration failure mode is interop — version A's **X25519MLKEM768** doesn't talk to version B's. Use this matrix to set expectations before you flip flags in production.

SERVER ↓ / CLIENT →	CHROME 131+	FIREFOX 132+	SAFARI 18.4+	CURL 8.10+ (OQS- PROVIDER)	OPENSSH 10.0+	GO 1.25+ STDLIB	JAVA JDK 24+	.NET 10
OpenSSL 3.5+ (oqs- provider)	✓	✓	✓	✓	n/a	✓	✓	✓
nginx (OpenSSL 3.5+)	✓	✓	✓	✓	n/a	✓	✓	✓
Caddy 2.8+	✓	✓	✓	✓	n/a	✓	⚠ TLS 1.3 only path	⚠ TLS 1.3 only path
Cloudflare edge	✓	✓	✓	✓	n/a	✓	✓	✓
AWS ALB / NLB	✓	✓	✓	✓	n/a	✓	✓	✓
AWS API Gateway	✓	✓	✓	✓	n/a	✓	✓	✓
GCP HTTPS LB (external)	✓	✓	✓	✓	n/a	✓	✓	✓
Azure App Gateway / Front Door	✗ no PQ today	✗	✗	✗	n/a	✗	✗	✗
HAProxy 3.0+ (OpenSSL 3.5+)	✓	✓	✓	✓	n/a	✓	✓	✓
Envoy 1.32+	⚠ depends on BoringSSL build	⚠	⚠	⚠	n/a	⚠	⚠	⚠

SERVER ↓ / CLIENT →	CHROME 131+	FIREFOX 132+	SAFARI 18.4+	CURL 8.10+ (OQS- PROVIDER)	OPENSSH 10.0+	GO 1.25+ STDLIB	JAVA JDK 24+	.NET 10
OpenSSH 10.0 server	n/a	n/a	n/a	n/a	✓ default	n/a	n/a	n/a
Java tomcat (JDK 24 prim only)	△ no TLS hybrid in stdlib until JDK 27	△	△	△	n/a	△	n/a	△

Legend: ✓ negotiates `X25519MLKEM768` and completes handshake. △ works only with explicit configuration / partial support. ✗ no PQ support shipping today.

Common interop break patterns (matching the matrix gaps):

- Middlebox ClientHello fragmentation.** TLS 1.3 ClientHello with `X25519MLKEM768` is ~1.6 KB and exceeds the 1-record limit (16 KB is fine; some old middleboxes truncate at MTU). Symptom: classical-only fallback or hard fail after "received" but no ServerHello. Fix: bypass the middlebox or upgrade firmware.
- Azure Front Door / App Gateway** does not yet expose PQ groups — even if your origin runs OpenSSL 3.5 with oqs-provider, the AFD termination is classical. Track Azure roadmap for Phase 2 (later 2026).
- Envoy + BoringSSL** — BoringSSL has its own PQ shipping cadence; check Envoy build notes for whether your Envoy is BoringSSL or OpenSSL-based.
- Java stdlib pre-JDK 27** does ML-KEM as a primitive (JEP 496) but no TLS integration. Use Bouncy Castle 1.84+ with a custom `SSLContext` provider, or wait for JDK 27 GA in Sep 2026 (JEP 527).
- OpenSSH versions < 10.0** don't speak `mlkem768x25519-sha256`; fall back to `sntrup761x25519-sha512` or upgrade.

To run your own interop validation, the openquantumsafe.org test matrix publishes nightly builds against the major TLS implementations.

11.9.8 Key Rotation Cadences with PQC

NIST SP 800-57 Part 1 Rev. 6 (IPD, comments closed February 2026) reorganizes cryptoperiod guidance around NIST security categories (Cat 1/3/5) rather than key-size bits. Derived heuristics:

KEY ROLE	SUGGESTED CRYPTO PERIOD	RATIONALE
TLS leaf server cert	47-90 days	CA/B Forum target trajectory (47 days by 2029)
TLS client auth cert	12 months	Human distribution loop
Code signing (ML-DSA)	≤ 459 days	CA/B Forum ballot Feb 23, 2026 cap
Firmware signing (LMS/XMSS)	One-shot per tree; plan depth for product lifetime	Stateful — each leaf single-use
KMS/HSM data-key wrapping root	1-3 years	Hardware-backed, rarely rotated
Database column encryption DEKs	Re-wrap under new KEK rather than re-encrypt	Re-wrapping is cheap even at PQ sizes

Grace period for rollover (industry practice, not standardized): $2 \times \text{max_key_age} + \text{max_session_age} + \text{validation_delay}$. For 90-day certs with 24-hour sessions that's ~190-day windows during which both old and new KEM/signature algorithms must be accepted. PQ key distribution is slower than classical because OCSP/CRL responses and certificate chains are 4-10× larger — err on the longer side.

11.9.9 Downgrade Attack Prevention

Hybrid TLS gives *confidentiality* against a harvest-now attacker today, but the server certificate signature is still classical in most 2026 deployments, which means *authentication* against a future quantum adversary depends on the cert chain upgrading too. To ensure clients actually negotiate hybrid (not fall back silently):

1. **Strict policy enforcement** (recommended). OpenSSL client configuration:

```
SSL_CTX_set1_groups_list(ctx, "X25519MLKEM768"); # no classical fallback
```

Drop the classical group list entirely.

2. **HTTPS DNS records (RFC 9460)** can carry required-group metadata. Experimental.

3. **Service mesh mTLS policy** (Istio, Linkerd) can pin required suites per connection.

Cloudflare's pq.cloudflareresearch.com reports that as of April 2026 ~57% of browser-initiated HTTPS connections to Cloudflare carry an [X25519MLKEM768](#) key share (and the share is climbing quarter-over-quarter; this figure is the load-bearing one to cite). The remainder is older classical-only clients or middleboxes that strip PQ groups from the ClientHello.

Unit 11.9 Summary

Cloud KMS integration for PQC in April 2026 is a mosaic: AWS is furthest ahead with GA offerings across KMS/ACM/Private CA/Secrets Manager and AWS-LC-FIPS 3.0 being the first crypto library to bring ML-KEM inside a FIPS 140-3 module boundary. GCP leads on client adoption via Chrome's default hybrid TLS (~57% of browser-initiated traffic per pq.cloudflareresearch.com) and has Cloud KMS PQ signing/KEM in preview. Azure trails its hyperscaler peers — Microsoft has GA'd Windows/.NET PQC primitives but not the Azure data-plane services, with Phase 2 rollout targeted for later in 2026. HSM vendors (Luna, nShield, Utimaco, Fortanix) have firmware-level PQC shipping with CAVP validation, but completed FIPS 140-3 module certificates for the PQ firmware are still pending industry-wide.

The migration playbook is less about cryptography and more about inventory (CBOM), choice of hybrid vs composite certs, grace windows for rotation, and policy enforcement to prevent downgrade. Use HPKE (RFC 9180) for envelope encryption to keep KEM-swap as a configuration change. Use [cosign](#) + KMS-backed ML-DSA keys for artifact signing. Follow NIST IR 8547 timelines: deprecated by 2030, disallowed by 2035. The work to do is largely plumbing — but the plumbing is extensive.

Unit 11.10: PQC CVE Registry and Implementation Pitfalls

A growing catalog of real CVEs, academic attacks, and developer-bites from real PQC deployments. This registry is current as of April 2026; cross-reference CVE databases and library changelogs before taking a dependency.

11.10.1 Confirmed CVEs (by library)

liboqs and oqs-provider

IDENTIFIER	LIBRARY	SEVERITY	SUMMARY
CVE-2024-37305	oqs-provider < 0.6.1	8.2 High	Unchecked <code>DECODE_UINT32</code> lengths in hybrid key/signature deserialization → memory corruption on parsing user-supplied composite keys. Non-hybrid PQ keys unaffected. Fix: 0.6.1.
CVE-2024-37880	pq-crystals/kyber ref, liboqs < 0.10.1, PQClean, rustpq/pqcrypto	High (key recovery)	KyberSlash: Clang optimizer (versions through 18.x at <code>-Os</code> / <code>-O1</code> / etc.) converted a constant-time <code>cmov</code> into a secret-dependent branch in <code>poly_frommsg</code> . End-to-end ML-KEM-512 secret key recovery demonstrated in < 10 min on Raspberry Pi 2 and Cortex-M4 via decapsulation timing. Fix: move conditional into separate compilation unit so compiler can't optimize it. liboqs 0.10.1 (June 2024).
CVE-2024-54137	liboqs < 0.12.0, PQClean HQC	Moderate (NVD/Red Hat; information leakage, no concrete exploit demonstrated)	HQC reference bug: mishandled <code>sigma</code> (implicit-rejection) field — treated part of secret key as public and used broken comparison during FO check. Malformed ciphertexts returned secrets derived from key material instead of implicit-rejection values. Present since 2023-04-30 HQC spec; lurked 19 months. Fix: liboqs 0.12.0 (Dec 2024).
CVE-2025-48946	liboqs ≤ 0.12.x HQC	Low (design flaw)	Many malformed ciphertexts share the same implicit-rejection KDF input — HQC has weaker KEM binding than ML-KEM. No concrete attack known but enough for NIST to disable HQC by default in liboqs 0.13.0 (May 2025). Awaits HQC team spec update.
CVE-2025-52473	liboqs < 0.14.0 HQC	Moderate	Same class as CVE-2024-37880 but for HQC under Clang 17–20: secret-dependent branches in HQC reference. Full local key recovery PoC. Fix: liboqs 0.14.0 disables compiler optimizations for HQC sources.

OpenSSL native PQC

IDENTIFIER	LIBRARY	SEVERITY	SUMMARY
CVE-2025-15469	OpenSSL 3.5/3.6 <code>openssl dgst</code> CLI	Moderate	<code>dgst</code> buffer truncates input > 16 MB for one-shot algorithms (ML-DSA, Ed25519/448). No error reported; signature appears valid but covers only first 16 MB. Library callers unaffected — CLI-only. Fix: 3.5.6 / 3.6.2 (April 2026 security update).
CVE-2026-31790	OpenSSL 3.x RSA-KEM	Moderate	RSASVE encapsulation leaks uninitialized memory to peer. Not PQC-specific but frequently co-deployed in composite hybrids. Fix: 3.0.20, 3.3.7, 3.4.5, 3.5.6, 3.6.2.

Go / Rust / Microsoft

IDENTIFIER	LIBRARY	SEVERITY	SUMMARY
Go issue #75528	Go 1.24+ <code>crypto/tls</code>	High (panic)	<code>X25519MLKEM768</code> listed as FIPS-approved in <code>defaults_fips140.go</code> , but <code>crypto/ecdh/x25519.go</code> blocks all X25519 in <code>GODEBUG=fips140=only</code> . Hybrid KEX internally calls <code>ecdh.X25519()</code> → handshake panics. Any FIPS-only Go TLS fails. Fixed in Go 1.25-era patches.
RustSec 2026-01-27	A Rust ML-DSA crate	TBD	Timing leak in ML-DSA Decompose operation.

Vendors/libraries with no published PQC CVEs as of April 2026: AWS-LC/AWS-LC-FIPS (aws-lc-verification has held up), BoringSSL/Chrome PQC path, Microsoft SymCrypt/CNG (GA Nov 2025), Rustls ML-KEM (via aws-lc-rs), Go stdlib `crypto/mlkem`, Bouncy Castle, Sigstore cosign ML-DSA integration, Cloudflare CIRCL, pq-crystals/dilithium reference.

11.10.2 Microarchitectural and Fault Attacks (academic, practical)

ATTACK	TARGET	PUBLICATION	IMPACT
KyberSlash1/2	Kyber/ML-KEM Compress_d DIV	2024, CVE-2024-37880	Full key recovery in minutes (Pi 2) to hours (Cortex-M4)
GoFetch	Apple M1/M2/M3 Data Memory- dependent Prefetcher	March 2024	Kyber + Dilithium keys extracted by co-resident userspace code in 49min–15h. M1/M2 cannot disable DMP — effectively unmitigable without kernel patches. M3 has a user-space knob.
PQ-Hammer	Kyber, BIKE, Dilithium via Rowhammer	IEEE S&P 2025	Full key recovery on commodity DDR4/DDR5 by memory spray + hammer + massaging.
SLasH-DSA	SLH-DSA in OpenSSL 3.5.1	USENIX 2025	Universal forgery via software-only Rowhammer. Deterministic mode forged in ~1 hour, randomized in ~8 hours. OpenSSL declined fix: "fault attacks outside threat model."
Crowhammer	FALCON/FN-DSA reference (future FIPS 206)	CRYPTO 2025	Single targeted bit-flip in RCDT gaussian-sampler table → signing-key recovery after ~100M signatures. More flips lower signature requirement.
GPUHammer / cuPQC key leakage	NVIDIA GDDR6 GPUs	July 2025	First Rowhammer on NVIDIA GPUs; extends to cuPQC ML-KEM key leakage. Mitigation: enable ECC (<code>nvidia-smi -e 1</code>), 10% perf and 6.25% memory cost.
Hedged ML-DSA fault attack	Hedged ML-DSA on Cortex-M4	CHES 2024 (ePrint 2024/238)	Voltage glitch skips seed derivation → signature leaks <code>s1</code> directly. ~53% single-trace success rate . Takeaway: hedged mode ≠ automatically fault-resistant.
Correction Fault on ML-DSA-44	ML-DSA-44	CHES 2024 (Krahmer et al.)	Key recoverable with 1024 faulty signatures (512 with lattice reduction).

ATTACK	TARGET	PUBLICATION	IMPACT
Mind the Faulty Keccak	ML-KEM and ML-DSA — all phases	ePrint 2024/1522	EMFI loop-abort on Keccak achieves 89.5% fault success on Cortex-M0+/M3/M4/M33 → key recovery, forgery, verification bypass.
SPA chosen-CT on ML-KEM	pqm4 reference on Cortex-M4	ePrint 2024/2051 + 2025 refinement	Full key recovery in ~30 seconds; 2025 extension needs only 179 traces. Works against a shuffled countermeasure.
Rejection Side-Channel on ML-DSA	ML-DSA on Cortex-M4	ePrint 2025/582 + 2026/056	Novel attack family: leak from both accepted AND rejected signing trials. ~300 traces (10× fewer than prior art); 2026 variant uses public templates (non-profiling).
Single-trace ML-KEM decoding	ML-KEM-512 INTT+decoder	ACNS 2025 workshops	Single-trace chosen-CT SPA with ML-assisted classification; full long-term secret recovery.

11.10.3 Specification Ambiguities That Caused Real Interop Failures

1. **ML-KEM / ML-DSA private key format fragmentation** — FIPS defines expanded private keys, IETF prefers 64-byte seeds (d, z), PKCS#11 has its own expectations. OpenSSL 3.5, Bouncy Castle Java 1.81, C# 2.6.1 only aligned formats in mid-2025. Keys generated by one library frequently unimportable by another until then.
2. **ML-DSA pure vs. HashML-DSA mode confusion** — FIPS 204 §5.4 defines HashML-DSA with a different `Verify()` routine. **RFC 9882 (2025) explicitly forbids HashML-DSA in CMS / X.509 PKIX**. Early implementations confused the two; signers used pre-hash, verifiers used pure → silent verification failures.
3. **ML-DSA externalMu vs. internal mu** — FIPS 204 permits external μ computation but FIPS certification discourages it. Library APIs differed (compute- μ vs. accept- μ); interop failures on the pqc-forum.
4. **SampleNTT rejection sampling buffer bounds** — A malformed ML-KEM key can force > 575 bytes of XOF output (normally probability $\sim 2^{-38}$). Implementations with bounded XOF buffers fail on crafted keys.

5. **Non-constant `memcmp` / `strcmp` in FO ciphertext check** — Decapsulation compares re-encrypted ciphertext for implicit rejection. `strcmp` stops at first zero byte (accepts some malformed CTs); `memcmp` is non-constant-time. Both seen in early pedagogical implementations.
6. **TLS key-share misprediction round-trip cost** — Clients caching a classical named group and then trying hybrid resume advertise the hybrid group but send classical key share → HelloRetryRequest costs extra RTT. Addressed by `draft-ietf-tls-key-share-prediction`.

11.10.4 Recurring Developer Pitfalls (no CVE, but real bugs)

PITFALL	BITTEN BY	FIX
Secret-dependent integer division (<code>Compress_d</code>)	Kyber ref, pqm4, Rust pqcrypto, PQClean	Replace with multiplication-by-reciprocal
Compiler optimizes <code>cmov</code> → branch (Clang through 18.x for Kyber; 17-20 for HQC)	Any C code not using inline asm <code>cmov</code>	Segregate constant-time code to separate TU; compile with <code>-fno-tree-ch -fno-if-conversion</code> ; verify via <code>objdump -d</code>
Missing modulus check on imported ML-KEM public key	pq-crystals/kyber HEAD	Add bounds validation per FIPS 203 §7.2
<code>int32_t</code> polynomial multiplication overflows ($256 \times 3328^2 = 2.84\text{B} > \text{INT32_MAX}$)	Many custom educational implementations	Use <code>int64_t</code> accumulator
Missing <code>#include <stdlib.h></code> → implicit-declaration <code>malloc</code> returning <code>int</code> truncates pointer on 64-bit	Tutorial ports	Always include standard headers; use <code>-Werror=implicit-function-declaration</code>
Using <code>openssl dgst</code> CLI for ML-DSA over large files	CVE-2025-15469	Use library API or OpenSSL $\geq 3.5.6$ / 3.6.2
FIPS-only Go TLS with hybrid enabled	Go issue #75528	Update to Go 1.25+ patch release

11.10.5 Using This Registry

- **Before deployment:** grep your dependency tree for affected library versions and upgrade. All liboqs ≤ 0.14 is affected by at least one CVE in this table.

- **For risk assessment:** if your threat model includes co-resident adversaries (shared hypervisors, multi-tenant cloud), treat GoFetch-class DMP attacks as unmitigated on Apple M1/M2 hardware.
- **For hardware deployment:** assume Rowhammer fault-attack threat model for any PQ signature stored in DDR RAM for more than a few minutes. Enable ECC (`nvidia-smi -e 1` or server DIMM ECC); use refresh-rate enforcement where supported.
- **For compiler choice:** prefer GCC or rebuild-from-source Clang for PQC code paths, or audit with ctgrind/dudect after every compiler upgrade — constant-time properties are not portable across compiler versions.

Unit 11.10 Summary

The PQC standards are mathematically sound but the implementation surface is young. Real CVEs clustered in 2024–2026 are concentrated in (a) compiler-introduced timing leaks, (b) HQC reference-implementation bugs (three CVEs in 18 months — HQC gets a disproportionate share), (c) microarchitectural side channels (GoFetch, Rowhammer class), and (d) specification ambiguities bridging FIPS and IETF. **None of these broke the algorithms; all broke implementations of the algorithms.** Maintain a rolling watch of github.com/open-quantum-safe/liboqs/security, NIST pqc-forum, and major vendor security advisories; pair it with CBOM-driven dependency tracking so each CVE disclosure immediately surfaces which of your systems are exposed.

Unit 11.11: Formal Verification Lessons — "Verification Theatre"

Formal verification is the most powerful tool available to catch PQC implementation bugs, but it is not a substitute for code review. In February 2026, a paper exposed 13 bugs that had escaped formally verified cryptographic libraries shipped in production, including Signal's post-quantum ratchet.

11.11.1 The Verification Theatre Incident (February 2026)

Nadim Kobeissi's "Verification Theatre: False Assurance in Formally Verified Cryptographic Libraries" (ePrint 2026/192) analyzed Cryspen's libcrux and hpke-rs — both marketed as formally verified, both deployed in Signal's SPQR post-quantum ratchet. Key findings:

- **13 distinct vulnerabilities** inside the "verified" boundary
- **Cross-backend endianness bug** — produced real decryption failures in Signal's post-quantum ratchet
- **Missing mandatory X25519 validation** — RFC 9180 violation in HPKE
- **Nonce reuse via integer overflow** — classical crypto bug in a "verified" implementation
- **Two FIPS 204 specification violations in the ML-DSA verifier:**
 1. A norm check rendered dead code by a wrong constant
 2. A missing final bound check in hint deserialization
- **ML-KEM bugs in verified code:** wrong decompression constant, missing inverse NTT, false serialization proof
- **ML-DSA verified code:** wrong multiplication specification rendered axiomatized AVX2 proofs unsound

The structural issue: **only 58.4% of the ML-KEM deployment code proofs were actually checked** when the library was shipped. The NEON ARM64 backend was entirely "admitted" — zero proofs checked. "Formally verified" covered a much smaller surface than the marketing suggested.

11.11.2 What Formal Verification Does Well

Formal verification works well for:

- **Functional correctness** on specific inputs (e.g., "this function returns the correct low-bits of its argument")
- **Memory safety** within a defined language model (Rust's ownership, WhyML refinement types)
- **Constant-time properties** when the cost model accurately captures the target CPU (not always the case)
- **Protocol-level proofs** (ProVerif, Tamarin) for key-agreement and authentication properties

The **Formosa-Crypto project** (EasyCrypt/Jasmin) delivers formally verified ML-KEM and ML-DSA implementations with security, correctness, *and* constant-time proofs — and crucially, the proof coverage is documented and complete. Reference: the Jasmin-verified reference code and the accompanying EasyCrypt proof scripts.

11.11.3 What Formal Verification Misses

Based on the Kobeissi findings and broader experience:

1. **Cross-backend bugs:** Proofs typically cover one implementation path (reference C or one optimized backend). Endianness, alignment, and integer-width mismatches between backends are outside most proof scopes.
2. **Microarchitectural side channels:** A function can be constant-time in the C abstract machine and leaky on a specific CPU (GoFetch, PREFETCH, speculation). Proofs in the C model don't catch hardware-specific leaks.
3. **Specification transcription errors:** If the formal spec embeds a wrong constant, the code is provably correct against a wrong target. The FIPS 204 norm-check bug above is exactly this class.
4. **Incomplete proof coverage** when some lemmas are "admitted" (assumed without proof) — a human-review check that readers of the verification artifact rarely perform.
5. **Serialization boundaries** — proofs often cover in-memory operations but not byte encoding, where interop failures cluster.
6. **Fault attacks** — almost always outside the threat model.

11.11.4 Recommendations for Using Verified Libraries

1. **Read the verification artifact, not the marketing.** Ask: what proportion of the shipping code is covered by checked proofs? What assumptions are admitted? Which backends are verified?
2. **Treat verification as a *complement* to testing**, not a replacement. Continue to run KAT test vectors, fuzzing, and duedct timing tests against verified implementations.
3. **Track spec-to-code mappings** — at each FIPS spec revision, re-audit whether the formal specification in the proof still matches the current standard.
4. **Prefer verified implementations for the hardest-to-get-right primitives** — NTT, rejection sampling, decomposition — while accepting that higher-level protocol plumbing will need conventional assurance methods.

5. **Cross-verify** — interop-test the verified implementation against multiple reference implementations. A divergence signals an ambiguous spec or a missed proof obligation.

Unit 11.11 Summary

Formal verification is a genuine engineering advance over testing-only assurance, but it is not a silver bullet. The Verification Theatre paper is essential reading for any team adopting verified PQC libraries — it documents realistic failure modes (bugs in "verified" code that shipped in production messaging apps). Use verified implementations where they exist and match your needs, but supplement them with conventional review, testing, CBOM tracking, and timing analysis. The Formosa-Crypto project shows what rigorous PQC verification looks like when done end-to-end.

Module 11 Summary: Migrating Legacy Codebases to PQC

Key Takeaways

1. **Start Now:** The HNDL threat makes immediate action essential—data encrypted today may be decrypted by future quantum computers
2. **Inventory First:** You can't protect what you don't know about—comprehensive cryptographic discovery is the foundation
3. **Build for Agility:** Design systems that can swap algorithms without code changes—this is a permanent requirement
4. **Hybrid Bridge:** Use hybrid cryptography (X25519 + ML-KEM) during migration for defense in depth
5. **Test Thoroughly:** Validate with NIST test vectors, interoperability testing, and production monitoring
6. **Phase the Migration:** Follow global timelines (2030 deprecation, 2035 disallowance) with phased approach
7. **Legacy Systems Require Alternative Strategies:** When direct migration is impossible (dead languages, lost source code, regulatory constraints), use gateway/proxy approaches, HSM offloading, or network-level encapsulation

Migration Checklist:

- ☐ Cryptographic inventory completed
- ☐ Risk assessment performed (Y + Z > X analysis)
- ☐ Crypto-agile architecture designed
- ☐ PQC libraries selected and tested
- ☐ Hybrid configuration deployed
- ☐ Testing and validation complete
- ☐ Monitoring and alerting configured
- ☐ Rollback procedures documented
- ☐ Team trained on PQC operations
- ☐ Compliance timeline tracked

Connections to Other Modules:

- Volume 1, Module 4: ML-KEM for key encapsulation
- Volume 1, Module 6: ML-DSA for signatures
- Volume 1, Module 7: SLH-DSA as conservative alternative

- Module 8: Hybrid construction patterns
- Module 9: Protocol integration (TLS, SSH)
- Module 10: Future algorithms and agility

What's Next: The Course Conclusion summarizes all modules and provides guidance for continued learning in the evolving post-quantum landscape.

Course Conclusion

This comprehensive PQC learning plan has covered:

Foundations (Volume 1, Modules 1-3):

- Mathematical background for lattice and hash-based cryptography
- Quantum computing threat model

Core Algorithms (Volume 1, Modules 4-7):

- ML-KEM (FIPS 203) - Key encapsulation
- ML-DSA (FIPS 204) - Digital signatures
- SLH-DSA (FIPS 205) - Hash-based signatures

Deployment (Modules 8-9):

- Hybrid cryptography construction
- Protocol integration (TLS, SSH, S/MIME, VPN)

Future Directions (Module 10):

- Emerging algorithms
- Cryptographic agility
- Best practices

Migration (Module 11):

- Cryptographic inventory and discovery
- Risk assessment (HNDL threat model)
- Crypto-agile architecture design
- Library selection and integration
- Phased migration execution
- Testing and validation

The post-quantum transition is underway. Start today.

C/C++ Implementation Projects

Project 1: Toy LWE Implementation (Beginner)

Build a complete educational LWE encryption system with:

- Key generation with small parameters
- Encryption and decryption
- Visual demonstration of noise addition
- Attack simulation (showing why small parameters are insecure)

Project 2: Full NTT Implementation (Intermediate)

Implement a production-quality NTT for ML-KEM-768:

- Forward and inverse NTT
- Montgomery reduction
- Barrett reduction
- Benchmark against reference implementation
- Test with NIST test vectors

Project 3: ML-KEM from Reference (Advanced)

Starting from the pq-crystals/kyber reference:

1. Study and annotate every function
2. Implement identical functionality
3. Test against KAT vectors
4. Profile and optimize hot paths
5. Add constant-time verification

Project 4: Hybrid TLS Client (Advanced)

Build a TLS 1.3 client with:

- X25519+ML-KEM-768 key exchange
- Certificate validation with ML-DSA
- Session resumption

- Performance comparison with classical-only
-

Debugging and Testing Cryptographic Code

B.1 Common Bugs in PQC Implementations

B.1.1 Modular Reduction Errors

```
// BUG: Integer overflow before reduction
int16_t bad_mod_reduce(int32_t a) {
    return a % Q; // May give negative result!
}

// CORRECT: Handle negative values
int16_t good_mod_reduce(int32_t a) {
    int16_t r = a % Q;
    if (r < 0) r += Q;
    return r;
}

// BEST: Use Barrett/Montgomery reduction for performance
```

B.1.2 NTT/INTT Mismatch

```
// BUG: Forgetting to apply scaling in inverse NTT
void bad_invntt(int16_t *poly) {
    // ... inverse butterfly operations ...
    // Missing: multiply by scaling factor
}

// CORRECT: Include the normalization
void good_invntt(int16_t *poly) {
    // ... inverse butterfly operations ...

    // For ML-KEM's incomplete NTT (7 stages, 128 pairs):
    // 128^(-1) mod 3329 = 3303; fmul constant = R^2 * 3303 mod Q = 1441
    // For a full NTT (8 stages, 256 scalars):
    // 256^(-1) mod 3329 = 3316
    for (int i = 0; i < N; i++) {
        poly[i] = montgomery_reduce((int32_t)poly[i] * N_INV);
    }
}
```

B.1.3 Bit Packing/Unpacking Errors

```
// Test bit packing round-trip
void test_compression(void) {
    for (int i = 0; i < Q; i++) {
        uint16_t compressed = compress_d(i, 10); // d=10 bits
        int16_t decompressed = decompress_d(compressed, 10);

        int32_t error = abs(i - decompressed);
        if (error > Q / (1 << 11)) { // Max error for d=10
            printf("BUG: compress(%d) = %d, decompress = %d, error = %d\n",
                   i, compressed, decompressed, error);
        }
    }
}
```

B.2 Debugging Tools

B.2.1 GDB with Crypto Extensions

```
# Compile with debug symbols
gcc -g -O0 -fsanitize=address,undefined your_code.c -loqs -o debug_test

# Run with GDB
gdb ./debug_test

# Useful GDB commands for crypto debugging:
# (gdb) break poly_ntt
# (gdb) x/256hd poly          # Print 256 half-words as decimal
# (gdb) p/x *(uint8_t[32]*)key # Print 32 bytes as hex
```

B.2.2 Printing Polynomials

```
void print_poly(const char *name, const int16_t *poly, int n) {
    printf("%s = [", name);
    int nonzero = 0;
    for (int i = 0; i < n; i++) {
        if (poly[i] != 0) {
            if (nonzero > 0) printf(" + ");
            printf("%d·x^%d", poly[i], i);
            nonzero++;
            if (nonzero >= 10) {
                printf(" + ...");
                break;
            }
        }
    }
    printf("]\n");

    // Also print statistics
    int32_t sum = 0, max = 0;
```



```

    for (int i = 0; i < n; i++) {
        sum += abs(poly[i]);
        if (abs(poly[i]) > max) max = abs(poly[i]);
    }
    printf("  L1 norm: %d, L $\infty$  norm: %d\n", sum, max);
}

```

B.2.3 Randomness Debugging

```

// Deterministic randomness for debugging
#include <string.h>

static uint8_t debug_seed[32] = {
    0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0x07, 0x08,
    0x09, 0x0a, 0x0b, 0x0c, 0x0d, 0x0e, 0x0f, 0x10,
    0x11, 0x12, 0x13, 0x14, 0x15, 0x16, 0x17, 0x18,
    0x19, 0x1a, 0x1b, 0x1c, 0x1d, 0x1e, 0x1f, 0x20
};

void debug_randombytes(uint8_t *buf, size_t len) {
    // Use SHAKE256 as deterministic RNG
    shake256(buf, len, debug_seed, 32);

    // Update seed for next call
    shake256(debug_seed, 32, debug_seed, 32);
}

#ifdef DEBUG_BUILD
#define randombytes debug_randombytes
#endif

```

B.3 Test Vectors and Validation

B.3.1 Creating Test Harness

```

// test_harness.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

typedef struct {
    const char *name;
    int (*test_func)(void);
} test_case;

#define TEST(name) { #name, test_##name }

int test_ntt_roundtrip(void);
int test_compress_decompress(void);
int test_keygen_deterministic(void);
int test_encaps_decaps(void);
int test_invalid_ciphertext(void);

test_case tests[] = {
    TEST(ntt_roundtrip),

```

```

    TEST(compress_decompress),
    TEST(keygen_deterministic),
    TEST(encaps_decaps),
    TEST(invalid_ciphertext),
};

int main(void) {
    int passed = 0, failed = 0;

    for (size_t i = 0; i < sizeof(tests)/sizeof(tests[0]); i++) {
        printf("Running %s... ", tests[i].name);
        fflush(stdout);

        int result = tests[i].test_func();

        if (result == 0) {
            printf("PASS\n");
            passed++;
        } else {
            printf("FAIL (code %d)\n", result);
            failed++;
        }
    }

    printf("\n%d passed, %d failed\n", passed, failed);
    return failed > 0 ? 1 : 0;
}

```

Performance Benchmarks

Comprehensive Operation Timings (x86-64 Desktop Reference)

Key Encapsulation Mechanisms:

ALGORITHM	KEYGEN	ENCAPS	DECAPS	VS RSA-2048 ENCAPS
ML-KEM-512	~25 μ s	~30 μ s	~25 μ s	30 $\times$ faster
ML-KEM-768	~40 μ s	~50 μ s	~40 μ s	20 $\times$ faster
ML-KEM-1024	~60 μ s	~75 μ s	~60 μ s	13 $\times$ faster
X25519	~25 μ s	~25 μ s	~25 μ s	40 $\times$ faster
RSA-2048	~50 ms	~1 ms	~50 μ s	baseline

Digital Signatures:

ALGORITHM	KEYGEN	SIGN	VERIFY	VS RSA-2048 SIGN
ML-DSA-44	~80 μ s	~200 μ s (avg)	~80 μ s	5 $\times$ faster
ML-DSA-65	~130 μ s	~350 μ s (avg)	~130 μ s	3 $\times$ faster
ML-DSA-87	~200 μ s	~500 μ s (avg)	~200 μ s	2 $\times$ faster
SLH-DSA-128s	slow	~60 ms	~3 ms	60 $\times$ slower
SLH-DSA-128f	slow	~5 ms	~0.3 ms	5 $\times$ slower
SLH-DSA-256s	slow	~400 ms	~20 ms	400 $\times$ slower
ECDSA-P256	~50 μ s	~75 μ s	~100 μ s	13 $\times$ faster
Ed25519	~30 μ s	~60 μ s	~80 μ s	17 $\times$ faster
RSA-2048	~50 ms	~1 ms	~50 μ s	baseline

NOTE:

ML-DSA sign times vary due to rejection sampling (shown as average). SLH-DSA-128s is optimized for small signatures; SLH-DSA-128f is optimized for fast signing.

Cycle Counts (Reference C, scalar; per operation)

These are the **reference / scalar** cycle counts as published in the pq-crystals/kyber, pq-crystals/dilithium, and sphincsplus reference repositories on a contemporary x86_64 (Skylake-class) at fixed frequency. Use them as an upper bound — the AVX2/AVX-512 numbers in §"Performance Impact by Algorithm" below are 3–5× faster.

ALGORITHM	KEYGEN	SIGN/ENCAPS	VERIFY/DECAPS
ML-KEM-768	~200k cycles	~260k cycles	~230k cycles
ML-DSA-65	~400k cycles	~1.1M cycles (variable; rejection sampling)	~400k cycles
SLH-DSA-128f	~25M cycles	~15M cycles	~1M cycles

(These reconcile with the Scalar column of the AVX2/AVX-512 comparison table later in this section. Older snapshots quoted 150k cycles for ML-KEM-768 KeyGen — that reflected an earlier reference implementation; current pq-crystals master is 200k.)

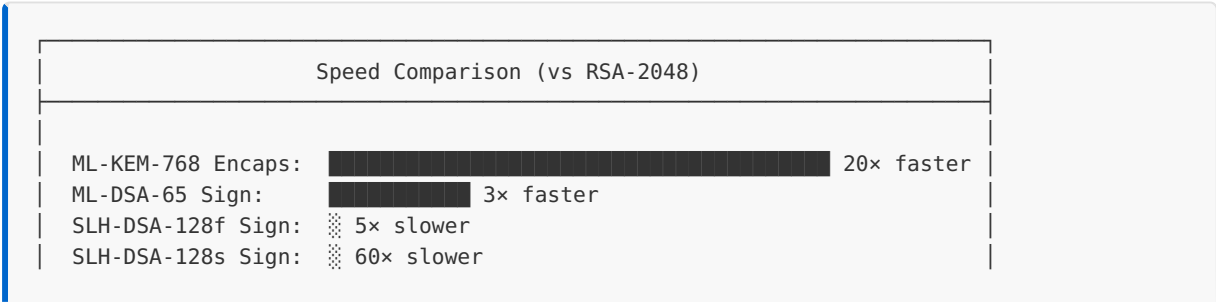
Embedded System Performance (ARM Cortex-M4 @ 168MHz)

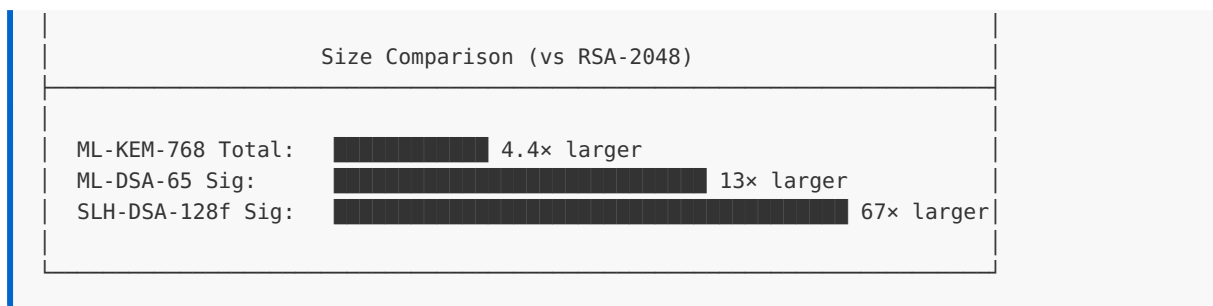
ALGORITHM	KEYGEN	ENCAPS/SIGN	DECAPS/VERIFY	STACK USAGE
ML-KEM-768	31 ms	11 ms	13 ms	~4 KB
ML-DSA-65	22 ms	32 ms	22 ms	~8 KB
SLH-DSA-128f	45 ms	900 ms	60 ms	~2 KB

Complete Size Comparison

ALGORITHM	SECURITY	PUBLIC KEY	SECRET KEY	CT/ SIGNATURE	TOTAL TRANSFER
KEMs					
X25519	128-bit	32 B	32 B	32 B	64 B
RSA-2048	112-bit	256 B	~1.2 KB	256 B	512 B
ML-KEM-512	128-bit	800 B	1,632 B	768 B	1,568 B
ML-KEM-768	192-bit	1,184 B	2,400 B	1,088 B	2,272 B
ML-KEM-1024	256-bit	1,568 B	3,168 B	1,568 B	3,136 B
X-Wing	192-bit	1,216 B	2,432 B	1,120 B	2,336 B
Signatures					
Ed25519	128-bit	32 B	64 B	64 B	96 B
ECDSA-P256	128-bit	64 B	32 B	64 B	128 B
RSA-2048	112-bit	256 B	~1.2 KB	256 B	512 B
ML-DSA-44	128-bit	1,312 B	2,560 B	2,420 B	3,732 B
ML-DSA-65	192-bit	1,952 B	4,032 B	3,309 B	5,261 B
ML-DSA-87	256-bit	2,592 B	4,896 B	4,627 B	7,219 B
SLH-DSA-128s	128-bit	32 B	64 B	7,856 B	7,888 B
SLH-DSA-128f	128-bit	32 B	64 B	17,088 B	17,120 B
SLH-DSA-256s	256-bit	64 B	128 B	29,792 B	29,856 B

Performance vs Classical Algorithms Summary





When to Use Which Algorithm

USE CASE	RECOMMENDED	RATIONALE
TLS key exchange	ML-KEM-768	Fast, reasonable size
Code signing	ML-DSA-65	Balance of size and speed
Firmware signing	SLH-DSA-128s	Small signatures, signing offline
Certificate chain	ML-DSA-44	Good size, sufficient security
Long-term archive	SLH-DSA-256s	Maximum security, size acceptable
Embedded systems	ML-KEM-512 + ML-DSA-44	Smaller, fits in constrained memory
Hybrid transition	X-Wing + Ed25519	Backward compatible, future-proof

Sources:

- NIST PQC Submission Documentation
- [liboqs Algorithm Datasheets](#)
- [CHERIoT PQC Benchmarks](#)
- [WolfSSL STM32 Benchmarks](#)

Memory Usage Analysis

Understanding memory requirements is critical for embedded systems, HSMs, and resource-constrained environments. PQC algorithms have significantly different memory profiles compared to classical cryptography.

Stack Requirements

Stack usage during cryptographic operations (peak values):

ALGORITHM	KEYGEN	ENCAPS/ SIGN	DECAPS/ VERIFY	NOTES
ML-KEM-512	~2.5 KB	~2.5 KB	~3.0 KB	NTT working space dominates
ML-KEM-768	~3.5 KB	~3.5 KB	~4.0 KB	Reference implementation
ML-KEM-1024	~4.5 KB	~4.5 KB	~5.0 KB	Can be reduced with optimization
ML-DSA-44	~6 KB	~8 KB	~6 KB	Signature expansion on stack
ML-DSA-65	~8 KB	~10 KB	~8 KB	Rejection sampling adds overhead
ML-DSA-87	~10 KB	~12 KB	~10 KB	Largest lattice-based
SLH-DSA-128s	~2 KB	~2.5 KB	~2.5 KB	Compact, iterative tree traversal
SLH-DSA-128f	~2 KB	~2.5 KB	~2.5 KB	Similar to 128s
SLH-DSA-256s	~3 KB	~3.5 KB	~3.5 KB	Larger hash outputs

NOTE:

Stack requirements can be reduced by 30-50% with stack-optimized implementations that reuse buffers.

Heap Requirements

Heap allocations (if any) during operations:

ALGORITHM	TYPICAL HEAP USAGE	HEAP-FREE POSSIBLE?
ML-KEM	0 (stack-only)	Yes, designed for it
ML-DSA	0-4 KB (impl. dependent)	Yes, with care
SLH-DSA	0-8 KB (tree buffers)	Yes, increases stack

Embedded-Friendly Design: All three FIPS PQC algorithms can be implemented without heap allocation, making them suitable for bare-metal and RTOS environments.

Working Memory During Operations

Breakdown of memory usage during key operations:

ML-KEM-768 Encapsulation:

Component	Size	Lifetime
Input public key	1,184 B	Full
Random seed	32 B	Start
NTT working buffer	1,024 B	NTT phase
Matrix row (one at a time)	768 B	Streaming
Intermediate polynomial	512 B	Compute
Output ciphertext	1,088 B	End
Shared secret	32 B	End
Peak total	~3.5 KB	

ML-DSA-65 Signing:

Component	Size	Lifetime
Secret key	4,032 B	Full
Message hash	64 B	Start
NTT working buffers	2,048 B	Compute
y vector (masking)	2,560 B	Attempt
w vector (commitment)	1,920 B	Attempt
Hint vector	640 B	End
Output signature	3,309 B	End
Peak total	~10 KB	

SLH-DSA-128f Signing:

Component	Size	Lifetime
Secret key	64 B	Full
Message digest	32 B	Start
WOTS+ chain state	544 B	Per leaf
Auth path buffer	320 B	Per tree
Tree traversal state	256 B	Iteration
Hash working buffer	512 B	Compute
Output signature	17,088 B	End
Peak total	~2.5 KB	

Memory Comparison Summary

ALGORITHM	STACK	HEAP	TOTAL WORKING	VS RSA-2048
ML-KEM-768	~4 KB	0	~4 KB	2× more
ML-DSA-65	~10 KB	0-4 KB	~12 KB	6× more
SLH-DSA-128f	~2.5 KB	0-8 KB	~8 KB	4× more
RSA-2048	~2 KB	0	~2 KB	baseline
ECDSA-P256	~1 KB	0	~1 KB	0.5×

EMBEDDED GUIDANCE:

For Cortex-M4 class devices with 32-64 KB RAM, all PQC algorithms are viable.
For Cortex-M0 with 8-16 KB RAM, prefer SLH-DSA or stack-optimized ML-KEM-512.

Benchmarking Methodology

Reproducible benchmarks require careful methodology. Poorly controlled measurements lead to incorrect algorithm comparisons and flawed deployment decisions.

Environment Setup

1. **CPU Frequency Pinning:** Disable dynamic frequency scaling (turbo boost, SpeedStep) to eliminate measurement variance:

```
# Linux: pin all cores to base frequency
sudo cpupower frequency-set -g performance
echo 1 | sudo tee /sys/devices/system/cpu/intel_pstate/no_turbo
# Verify: cat /proc/cpuinfo | grep MHz
```

2. **Process Isolation:** Bind the benchmark to a dedicated core and minimize interference:

```
# Isolate core 3 from scheduler, bind benchmark to it
taskset -c 3 ./benchmark
# Or use cgroups for stronger isolation
```

3. **Memory Configuration:** Disable ASLR for reproducible memory layout and lock pages:

```
echo 0 | sudo tee /proc/sys/kernel/randomize_va_space
```

Measurement Protocol

- **Warm-up:** Discard the first 1,000 iterations to fill instruction/data caches
- **Iterations:** Minimum 10,000 per operation for sub-microsecond timings
- **Timing Source:** Use `rdtsc` (x86) or `clock_gettime(CLOCK_MONOTONIC)`, never `gettimeofday()`
- **Statistical Reporting:** Report median and IQR (interquartile range), not mean/stddev — crypto timings are right-skewed due to cache misses and interrupts

SUPERCOP Framework

SUPERCOP (System for Unified Performance Evaluation Related to Cryptographic Operations and Primitives) is the standard framework for cryptographic benchmarking:

- Automates compiler flag exploration (finds optimal `-march`, `-O` combinations)
- Provides calibrated cycle counting across architectures
- Results are comparable across submissions (same methodology)
- Use `supercop/do` to build and run; results appear in `bench/` directory

Variance Reporting Checklist

When publishing PQC benchmarks, always report:

ITEM	EXAMPLE	WHY
CPU model + stepping	Intel i7-13700K (B0)	Microarch affects NTT speed
Frequency	3.4 GHz base, turbo disabled	Comparability
Compiler + flags	GCC 13.2 -O3 -mavx2	Codegen differences of 2-3x
OS + kernel	Ubuntu 22.04, 6.5.0	Scheduler, ASLR
Iterations + warm-up	100K iterations, 1K warm-up	Statistical validity
Metric	Median cycles (IQR: $\pm 2.1\%$)	Resist outlier bias

COMMON PITFALL:

Comparing ML-KEM benchmarks compiled with AVX2 against RSA benchmarks without hardware acceleration gives misleading results. Always compare at the same optimization level, or clearly label the comparison as "deployment-realistic."

Minimal Cycle-Counting Harness

Portable cycle-counter skeleton suitable for dropping into any PQC benchmark:

```
#include <stdio.h>
#include <stdint.h>

#if defined(__x86_64__) || defined(_M_X64)
#include <x86intrin.h>
static inline uint64_t cycles_now(void) { return __rdtsc(); }
#elif defined(__aarch64__)
static inline uint64_t cycles_now(void) {
    uint64_t v; __asm__ volatile("mrs %0, cntvct_el0" : "=r"(v));
    return v;
}
#else
#include <time.h>
static inline uint64_t cycles_now(void) {
    struct timespec ts; clock_gettime(CLOCK_MONOTONIC, &ts);
    return (uint64_t)ts.tv_sec * 1000000000ULL + (uint64_t)ts.tv_nsec;
}
#endif

static int cmp_u64(const void *a, const void *b) {
    uint64_t x = *(const uint64_t *)a, y = *(const uint64_t *)b;
    return (x > y) - (x < y);
}
```

```

/* Run `fn(ctx)` N times, return median cycles. */
uint64_t bench(void (*fn)(void *), void *ctx, size_t iters, size_t warmup) {
    for (size_t i = 0; i < warmup; i++) fn(ctx);
    uint64_t *samples = malloc(iters * sizeof(uint64_t));
    for (size_t i = 0; i < iters; i++) {
        uint64_t t0 = cycles_now();
        fn(ctx);
        samples[i] = cycles_now() - t0;
    }
    qsort(samples, iters, sizeof(uint64_t), cmp_u64);
    uint64_t median = samples[iters / 2];
    free(samples);
    return median;
}

```

Compile with `-O3 -march=native`. For ML-KEM encapsulation: 10,000 iterations × 1,000 warmup produces a stable median within $\pm 1\text{-}2\%$ on pinned cores.

Statistical Validation Tools

- **Google Benchmark** — well-tested C++ harness with automatic iteration tuning and statistical reporting
- **hyperfine** — shell-level wrapper for end-to-end latency (good for full handshake timing)
- `perf stat -r 10 --all-user ./bench` — Linux perf counters with repeated runs; reports cycles, cache-misses, IPC
- **dudect** — NOT a performance tool; detects *timing leaks* (t-test on two distributions)

Hardware Acceleration for PQC

Current CPU Support

x86-64 SIMD Extensions:

EXTENSION	SPEEDUP	OPERATIONS ACCELERATED
AVX2	2-4×	NTT butterfly, polynomial arithmetic
AVX-512	3-6×	Vectorized NTT, batch hashing
AES-NI	10×	SHAKE/SHA-3 (Keccak)
VPCLMULQDQ	5×	Polynomial multiplication in $GF(2^n)$

ARM Extensions:

EXTENSION	AVAILABILITY	SPEEDUP
NEON	ARMv7+, AArch64	2-3×
SVE/SVE2	ARMv8.2+	2-5× (variable vector length)
SHA3 instructions	ARMv8.2+	5-10× for Keccak
Crypto Extensions	ARMv8+	AES, SHA acceleration

NTT Optimization with SIMD

Standard NTT (scalar):	AVX2-Optimized NTT:
<pre>for i in 0..n/2: butterfly(a[i], a[i+n/2]) // 1 coeff/cycle</pre>	<pre>Load 8 coefficients at once ymm0 = _mm256_loadu_si256(a) ymm1 = _mm256_loadu_si256(b) 8 butterflies in parallel // 8 coeffs/cycle</pre>

Performance Impact by Algorithm:

ALGORITHM	SCALAR	AVX2	AVX-512	SPEEDUP
ML-KEM-768 KeyGen	200k cycles	75k cycles	55k cycles	3.6×
ML-KEM-768 Encaps	260k cycles	95k cycles	70k cycles	3.7×
ML-DSA-65 Sign	1.5M cycles	450k cycles	320k cycles	4.7×
SHAKE256 (1KB)	15k cycles	5k cycles	3k cycles	5×

Future PQC CPU Instructions

Expected in future processors:

1. Dedicated NTT instructions

- Single instruction for butterfly operation
- Hardware modular reduction for PQC primes (3329, 8380417)

2. Lattice-specific extensions

- Polynomial multiply-accumulate
- Modular reduction pipelines

3. NIST PQC acceleration

- Similar path to AES-NI adoption after AES standardization
- Expected timeline: 2027-2030 for mainstream CPUs

FPGA/ASIC Considerations

PLATFORM	LATENCY	THROUGHPUT	POWER	USE CASE
Software (CPU)	Low	Medium	High	General purpose
FPGA	Medium	High	Medium	Hardware security modules
ASIC	Very Low	Very High	Low	High-volume applications

FPGA Resources for ML-KEM-768:

Typical Xilinx Artix-7 Implementation:

Resource	Used	% of XC7A100T
LUTs	~15,000	24%
Registers	~8,000	6%
Block RAM (36Kb)	~20	15%
DSP Slices	~50	21%
Throughput	50,000 ops/sec	
Latency	~100 μ s	

Memory Bandwidth Considerations

ML-KEM-768 Memory Access Pattern:

NTT: Sequential access (cache-friendly)

- └ Forward NTT: $256 \times 12 \times \log(256) = 24,576$ reads
- └ Inverse NTT: Same

Matrix Multiply: Strided access (less cache-friendly)

- └ A.s: $3 \times 3 \times 256 \times 256 = 589,824$ multiplies
- └ Random access to A from seed (XOF-expandable)

Bottleneck: Matrix generation from seed dominates

Solution: Parallel XOF (SHAKE128) for A expansion

Implementation Recommendations

1. Always use available SIMD

- Check for AVX2/AVX-512 at runtime
- Fall back to NEON on ARM
- Use compiler intrinsics or assembly

2. Optimize critical paths

- NTT: Most operations per byte of output
- SHAKE: Bottleneck for deterministic sampling
- Modular reduction: Use Barrett or Montgomery

3. Consider memory layout

- Coefficient interleaving for SIMD
- Cache-aligned polynomial storage
- Prefetching for matrix operations

4. Library selection by platform

- x86-64: liboqs with AVX2 (compile with `-march=native`)
- ARM64: pqm4 reference or wolfSSL
- Embedded: pqm4 for Cortex-M series

Side-Channel Attacks on PQC Hardware Implementations

Physical attacks on PQC implementations are a critical concern, especially in embedded and HSM deployments where attackers may have physical access.

Differential Power Analysis (DPA) on NTT

The Number Theoretic Transform (NTT) is the performance-critical inner loop of lattice-based schemes. Its butterfly operations involve secret-dependent multiplications that leak through power consumption:

Attack Vector: An attacker measures power traces during NTT computation. Each butterfly multiplies a coefficient by a twiddle factor ζ^i . When the coefficient is secret (e.g., during decapsulation), the power consumed by the multiplication correlates with the Hamming weight of the intermediate value.

Countermeasure — Boolean/Arithmetic Masking: Split each secret coefficient s into shares $s = s_1 + s_2 \pmod{q}$. Perform NTT on each share independently. The power trace of each share is statistically independent of the secret, requiring $O(2^d)$ traces for d -th order masking.

```
// First-order masked NTT butterfly
// s = s1 + s2 (mod q), attacker sees only s1 or s2
ntt_butterfly(s1[i], s1[i+step], twiddle);
ntt_butterfly(s2[i], s2[i+step], twiddle);
// Refresh masks periodically to prevent higher-order combination
```

COST:

First-order masking roughly doubles NTT time; second-order roughly quadruples it.

Fault Injection on Fujisaki-Okeyama (FO) Transform

The FO transform's implicit rejection mechanism is a prime fault injection target:

Attack Vector: By inducing a computational fault (voltage glitch, laser) during re-encryption in ML-KEM's decapsulation, an attacker can bypass the ciphertext comparison check. If the faulty decapsulation returns the real shared secret instead of the pseudorandom rejection value, the attacker recovers the secret key one query at a time.

Countermeasure — Redundant Computation:

- Perform the re-encryption check twice with independent computations
- Use an instruction-level fault detection scheme (e.g., algorithm-level redundancy)
- Verify the return path: ensure the branch selecting between K and $\tilde{K}$ cannot be skipped

Electromagnetic (EM) Emanation Attacks

EM probes placed near a chip can capture localized signals from individual functional units, offering higher spatial resolution than power analysis:

Key Differences from Power Analysis:

- EM probes can target specific on-chip regions (e.g., NTT multiplier vs. hash unit)
- Near-field EM captures are effective even with on-chip power filtering
- Template attacks using EM traces have recovered ML-KEM secret keys in <1,000 traces in laboratory settings

Countermeasure — Shuffling + Masking:

- Randomize the order of NTT butterfly operations (shuffling) to decorrelate traces
- Combine with arithmetic masking for defense in depth
- Add random delays between operations to desynchronize traces

Summary of Hardware Countermeasures

ATTACK	TARGET	COUNTERMEASURE	PERFORMANCE COST
DPA	NTT butterflies	Arithmetic masking	2-4× slowdown
Fault injection	FO transform comparison	Redundant computation	1.5-2× slowdown
EM emanation	Localized operations	Shuffling + masking	2-3× slowdown
Timing	Branch on secret	Constant-time code	Minimal
Cache	Table lookups	Bitsliced implementations	1.2-1.5× slowdown

DESIGN PRINCIPLE:

Hardware countermeasures are layered. A production HSM implementation typically combines masking (against DPA/EM), shuffling (against template attacks), redundant computation (against faults), and constant-time code (against timing). Budget 3-5× performance overhead for a FIPS 140-3 Level 3+ target.

Alphabetical Index

Quick lookup of key terms and where they are defined or discussed in depth. Use browser find (Ctrl/Cmd+F) or PDF search to locate. Section references use unit numbers (e.g., §2.1 = Unit 2.1) or appendix names.

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ACVP (Automated Cryptographic Validation Protocol)	§11.5.5
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Azure Key Vault PQC (absence)	§11.9.4
Barrett reduction	§2.1, Appendix: Quick Reference
β (beta — ML-DSA rejection bound)	§6.1, Master Parameter Reference
BIKE	§10.1, Glossary
Bit-packing (ML-KEM/ML-DSA encoding)	§2.3, §6.6

C-D

TERM	LOCATION
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CBD (Centered Binomial Distribution)	§2.6, §4.2
č (c-tilde, ML-DSA challenge hash)	§6.4, §6.5
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Fujisaki-Okamoto transform	§4.3

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ζ (zeta — primitive root, ML-KEM NTT)	§2.5

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Resources Summary

Official Standards

- [FIPS 203 \(ML-KEM\)](#)
- [FIPS 204 \(ML-DSA\)](#)
- [FIPS 205 \(SLH-DSA\)](#)

C/C++ Libraries

- [liboqs](#)
- [pq-crystals/kyber](#)
- [pq-crystals/dilithium](#)
- [sphincs/sphincsplus](#)
- [mlkem-native](#)

Learning Resources

Books:

- "Serious Cryptography" by Jean-Philippe Aumasson - Excellent introduction to modern cryptography
- "An Introduction to Mathematical Cryptography" by Hoffstein, Pipher, Silverman - Strong mathematical foundations including lattices
- "Post-Quantum Cryptography" edited by Bernstein, Buchmann, Dahmen - Comprehensive academic treatment

Online Courses:

- [Boneh & Shoup Cryptography Course](#) - Free Stanford course
- [Introduction to Post-Quantum Cryptography](#) - Lecture series

Official Documentation:

- [NIST PQC Project](#) - Official standards and announcements
- [NIST SP 800-227](#) - Recommendations for Key-Encapsulation Mechanisms
- [NIST IR 8547](#) - Transition to Post-Quantum Cryptography Standards (migration timelines)

- [NIST SP 800-208](#) - Recommendation for Stateful Hash-Based Signature Schemes (LMS, XMSS)
- [CNSA 2.0](#) - NSA's Commercial National Security Algorithm Suite 2.0 transition guidance

IETF Drafts and RFCs (PQC in Protocols):

- [RFC 9180](#) - Hybrid Public Key Encryption (HPKE) — framework for hybrid KEM integration
- [RFC 9370](#) - Multiple Key Exchanges in IKEv2 — enables PQ+classical hybrid in IPsec/VPN
- [draft-ietf-tls-hybrid-design](#) - Hybrid key exchange in TLS 1.3 (ML-KEM + X25519)
- [draft-ietf-lamps-dilithium-certificates](#) - ML-DSA in X.509 certificates and CRLs
- [draft-ietf-lamps-kyber-certificates](#) - ML-KEM in X.509 and CMS
- [draft-ietf-pquip-pqc-engineers](#) - PQC for Engineers — excellent practical overview

Community & Support:

- [PQCrypto mailing list](#) - Academic community
- [Open Quantum Safe GitHub](#) - Libraries and discussions
- [IETF Post-Quantum Use in Protocols \(PQUIP\)](#) - Standardization efforts
- [IETF LAMPS Working Group](#) - PQC certificate and CMS standards

Papers (Essential Reading):

- Original Kyber paper: "CRYSTALS-Kyber: A CCA-Secure Module-Lattice-Based KEM"
 - Original Dilithium paper: "CRYSTALS-Dilithium: A Lattice-Based Digital Signature Scheme"
 - SPHINCS+ paper: "SPHINCS+: Submitting to NIST's Post-Quantum Cryptography Project"
 - Fujisaki-Okamoto Transform: "Secure Integration of Asymmetric and Symmetric Encryption Schemes"
-

Appendix: Quick Reference Cards

ML-KEM-768 Quick Reference

Key Sizes:

Encapsulation Key (public): 1,184 bytes
Decapsulation Key (private): 2,400 bytes
Ciphertext: 1,088 bytes
Shared Secret: 32 bytes

Parameters:

$n = 256$, $k = 3$, $q = 3329$
 $\eta_1 = 2$, $\eta_2 = 2$
 $d_u = 10$, $d_v = 4$

Security: NIST Level 3 (equivalent to AES-192)

ML-DSA-65 Quick Reference

Key Sizes:

Verification Key (public): 1,952 bytes
Signing Key (private): 4,032 bytes
Signature: 3,309 bytes

Parameters:

$n = 256$, $k = 6$, $l = 5$, $q = 8,380,417$
 $\eta = 4$, $\tau = 49$, $\gamma_1 = 2^{19}$, $\gamma_2 = (q-1)/32$

Security: NIST Level 3 (equivalent to AES-192)

ML-KEM Parameter Comparison

PARAMETER	ML-KEM-512	ML-KEM-768	ML-KEM-1024
NIST Level	1	3	5
Security	~AES-128	~AES-192	~AES-256
n	256	256	256
k	2	3	4
q	3329	3329	3329
η_1, η_2	3, 2	2, 2	2, 2
d_u, d_v	10, 4	10, 4	11, 5
Public Key	800 bytes	1,184 bytes	1,568 bytes
Private Key	1,632 bytes	2,400 bytes	3,168 bytes
Ciphertext	768 bytes	1,088 bytes	1,568 bytes
Shared Secret	32 bytes	32 bytes	32 bytes

ML-DSA Parameter Comparison

PARAMETER	ML-DSA-44	ML-DSA-65	ML-DSA-87
NIST Level	2	3	5
Security	~SHA-256 col.	~AES-192	~AES-256
n	256	256	256
k, l	4, 4	6, 5	8, 7
q	8,380,417	8,380,417	8,380,417
η	2	4	2
τ	39	49	60
Y₁	2 ¹⁷	2 ¹⁹	2 ¹⁹
Y₂	(q-1)/88	(q-1)/32	(q-1)/32
Public Key	1,312 bytes	1,952 bytes	2,592 bytes
Private Key	2,560 bytes	4,032 bytes	4,896 bytes
Signature	2,420 bytes	3,309 bytes	4,627 bytes

SLH-DSA Parameter Comparison (Selected Variants)

PARAMETER	128F	128S	192F	192S	256F	256S
NIST Level	1	1	3	3	5	5
Public Key	32 B	32 B	48 B	48 B	64 B	64 B
Private Key	64 B	64 B	96 B	96 B	128 B	128 B
Signature	17,088 B	7,856 B	35,664 B	16,224 B	49,856 B	29,792 B
Sign Speed	Fast	Slow	Fast	Slow	Fast	Slow
Verify Speed	Fast	Fast	Fast	Fast	Fast	Fast

Note: "f" = fast signing, larger signatures; "s" = small signatures, slower signing

SLH-DSA Hash Function Variants

Each parameter set has both SHAKE and SHA2 variants:

- **SHAKE variants:** SLH-DSA-SHAKE- $\{128,192,256\}\{f,s\}$
- **SHA2 variants:** SLH-DSA-SHA2- $\{128,192,256\}\{f,s\}$

Performance is similar; SHAKE is recommended for new deployments due to consistent security properties.

X-Wing Hybrid KEM Quick Reference

Components:

Classical: X25519 (Curve25519 ECDH)
Post-Quantum: ML-KEM-768

Key Sizes:

Public Key: 1,216 bytes (32 + 1,184)
Secret Key: 2,432 bytes (32 + 2,400)
Ciphertext: 1,120 bytes (32 + 1,088)
Shared Secret: 32 bytes

Security: NIST Level 3 + classical ECDH security
Combiner: HKDF-based key derivation from both secrets

FALCON Quick Reference (FIPS 206 Draft, FN-DSA)

FALCON-512:

Public Key: 897 bytes
Secret Key: 1,281 bytes
Signature: ~666 bytes (average)
Security: NIST Level 1

FALCON-1024:

Public Key: 1,793 bytes
Secret Key: 2,305 bytes
Signature: ~1,280 bytes (average)
Security: NIST Level 5

Key Features:

- Smallest signatures among lattice-based schemes
- Based on NTRU lattices
- Requires floating-point arithmetic or careful emulation
- Variable signature size (average values shown)

Algorithm Selection Guide

USE CASE	RECOMMENDED ALGORITHM	NOTES
General KEM	ML-KEM-768	Good balance of security/performance
High Security KEM	ML-KEM-1024	For long-term sensitive data
Resource-Constrained	ML-KEM-512	When space is critical
General Signatures	ML-DSA-65	Good balance, fastest lattice sig
High Security Sig	ML-DSA-87	For critical infrastructure
Conservative Security	SLH-DSA-128s	Hash-only assumption
Hybrid KEX (TLS 1.3)	X25519MLKEM768	TLS-specific draft (draft-ietf-tls-ecdhe-mlkem) — what Chrome/Cloudflare ship today
Hybrid KEM (general)	X-Wing	Standalone hybrid KEM (draft-connelly-cfrg-xwing-kem) — different binding/KDF; not interchangeable with X25519MLKEM768
Small Signatures	FALCON-512	When sig size is critical

Algorithm Decision Table (by Requirement Class)

Practitioner cheat sheet for on-the-spot algorithm selection. Cells: ✓ recommended / △ conditional / ✗ avoid. Read by row, applying your most-binding constraint first.

CONSTRAINT	ML-KEM-768	ML-KEM-1024	ML-DSA-65	ML-DSA-87	SLH-DSA-128S	SLH-DSA-128F	FALCON-512
Latency p99 < 5 ms (single op)	✓	✓	⚠ rejection tail	⚠	✗ slow sign	⚠	⚠ slow sign on FPU-less; verify <1 ms with FPU
Bandwidth < 5 KB / handshake	✓ (1.1 KB CT)	⚠ (1.5 KB CT)	⚠ (3.3 KB sig)	✗ (4.6 KB sig)	✗ (7.8 KB sig)	✗ (17 KB sig)	✓ (~666 B sig padded)
Memory < 16 KB stack (Cortex-M0)	⚠ tight (~14.2 KB ref; ~4 KB w/ mlkem-c-embedded)	✗	⚠ tight	✗	⚠ ok	✗	⚠ FP overhead
HNDL defense (long-term)	✓	✓ highest pure-PQ	N/A (sig)	N/A	N/A	N/A	N/A
FIPS 140-3 today	✓ FIPS 203	✓ FIPS 203	✓ FIPS 204	✓ FIPS 204	✓ FIPS 205	✓ FIPS 205	✗ FIPS 206 draft
Conservative crypto assumptions	⚠ MLWE	⚠ MLWE	⚠ MLWE	⚠ MLWE	✓ hash-only	✓ hash-only	⚠ NTRU
Code/ firmware signing (long-life)	N/A	N/A	⚠	⚠	✓	✓	⚠
TLS/protocol KEX	✓	✓	N/A	N/A	N/A	N/A	N/A
Cert chain < 10 KB	N/A	N/A	✓ leaf	⚠	✗	✗	✓

Worked examples:

- "Web TLS server, 5 KB chain budget, FIPS-required, defend HNDL today" → X-Wing (KEX) + ML-DSA-65 (cert) — only combo meeting all four.
- "Embedded firmware signing, 30-year retention, Cortex-M4 verifier" → SLH-DSA-128s (slow sign OK at build time, fast verify on device, hash-only is most conservative for 30-year horizon).
- "Internal mTLS, latency-sensitive, FIPS optional" → ML-KEM-768 + Ed25519 today; migrate signature to ML-DSA-65 once latency budget allows.
- "Resource-constrained IoT (8 KB RAM)" → SLH-DSA-128s for signing (small key, slow but tolerable for one-shot), ML-KEM-512 for KEM (smallest PQ KEM), accept reduced security category.

Master Parameter Reference (All Algorithms)

One-page lookup consolidating every parameter needed to implement or integrate the three FIPS PQC standards plus X-Wing.

ML-KEM (FIPS 203)

PARAMETER	ML-KEM-512	ML-KEM-768	ML-KEM-1024
NIST security level	1	3	5
Classical equivalent	~AES-128	~AES-192	~AES-256
n (ring degree)	256	256	256
q (modulus)	3329	3329	3329
k (module rank)	2	3	4
η_1 / η_2	3 / 2	2 / 2	2 / 2
d _u / d _v	10 / 4	10 / 4	11 / 5
Public key (bytes)	800	1,184	1,568
Secret key (bytes)	1,632	2,400	3,168
Ciphertext (bytes)	768	1,088	1,568
Shared secret (bytes)	32	32	32

ML-DSA (FIPS 204)

PARAMETER	ML-DSA-44	ML-DSA-65	ML-DSA-87
NIST security level	2	3	5
q (modulus)	8,380,417	8,380,417	8,380,417
k / l (matrix dims)	4 / 4	6 / 5	8 / 7
η (secret coeff bound)	2	4	2
τ (challenge weight)	39	49	60
β ($= \tau \cdot \eta$)	78	196	120
γ_1	2^{17}	2^{19}	2^{19}
γ_2	$(q-1)/88$	$(q-1)/32$	$(q-1)/32$
ω (hint bound)	80	55	75
$\tilde{c}$ bytes ($\lambda/4$)	32	48	64
Public key (bytes)	1,312	1,952	2,592
Secret key (bytes)	2,560	4,032	4,896
Signature (bytes)	2,420	3,309	4,627

SLH-DSA (FIPS 205) — 6 variants (f = fast signing, s = small signature)

PARAMETER	128F	128S	192F	192S	256F	256S
NIST security level	1	1	3	3	5	5
n (hash output bytes)	16	16	24	24	32	32
h (total tree height)	66	63	66	63	68	64
d (layers)	22	7	22	7	17	8
h' (= h/d)	3	9	3	9	4	8
a (FORS leaf height)	6	12	8	14	9	14
k (FORS trees)	33	14	33	17	35	22
w (WOTS+ Winternitz)	16	16	16	16	16	16
Public key (bytes)	32	32	48	48	64	64
Secret key (bytes)	64	64	96	96	128	128
Signature (bytes)	17,088	7,856	35,664	16,224	49,856	29,792

Each row has a SHA-2 variant (SLH-DSA-SHA2-) and a *SHAKE* variant (SLH-DSA-SHAKE-) with identical parameters but different underlying hash function.

X-Wing Hybrid KEM ([draft-connolly-cfrg-xwing-kem](#))

PARAMETER	VALUE
Classical component	X25519 (Curve25519 ECDH)
Post-quantum component	ML-KEM-768
Public key (bytes)	1,216 (32 + 1,184)
Secret key (bytes)	2,432 (32 + 2,400)
Ciphertext (bytes)	1,120 (32 + 1,088)
Shared secret (bytes)	32
KDF label	<code>\.//^\</code> (hex <code>5c 2e 2f 2f 5e 5c</code> , 6 bytes)
KDF	SHA3-256(label ss_m ss_x ek_x pk_x)

FALCON / FN-DSA (FIPS 206 Draft)

PARAMETER	FALCON-512	FALCON-1024
NIST security level	1	5
Public key (bytes)	897	1,793
Secret key (bytes)	1,281	2,305
Signature (bytes, average)	~666	~1,280
Signature (bytes, max)	752	1,462

Protocol Overhead Cheat Sheet (Classical vs Hybrid PQ)

Impact of adding PQC to common protocols. Numbers are approximate and depend on exact key/signature encoding. Useful for quick impact assessment during migration planning.

PROTOCOL	CLASSICAL	HYBRID (CLASSICAL + PQ)	MULTIPLIER
TLS 1.3 handshake (ECDHE-P256 + ECDSA-P256)	~670 B	X25519+ML-KEM-768 + Ed25519+ML-DSA-65: ~2,500 B	~3.7×
TLS 1.3 handshake (ECDHE-P384 + ECDSA-P384)	~900 B	X448+ML-KEM-1024 + Ed448+ML-DSA-87: ~4,800 B	~5.3×
SSH transport (per connection)	~1.5 KB	sntrup761+x25519 + Ed25519+ML-DSA-65: ~4.8 KB	~3.2×
S/MIME email (per recipient, RSA-2048)	~256 B	ML-KEM-768 wrap: ~1.2 KB; +ML-DSA-65 sig: ~4.5 KB	~4-18×
X.509 certificate (ECDSA-P256 leaf)	~1 KB	ML-DSA-65 leaf: ~4.5 KB; +hybrid composite: ~5.5 KB	~4.5-5.5×
TLS 1.3 cert chain (3 certs, ECDSA)	~3 KB	ML-DSA-65 chain: ~13.5 KB	~4.5×
Code signing (Authenticode, ECDSA)	~1.2 KB	ML-DSA-65: ~5 KB; SLH-DSA-128s: ~9 KB	~4-7.5×
DNSSEC DNSKEY record (ECDSA-P256)	~100 B	ML-DSA-44 DNSKEY: ~1.5 KB	~15×

Takeaways for capacity planning:

- Budget **3-5× bandwidth increase** for TLS/SSH under hybrid PQ
 - DNSSEC and cert chains see the largest multipliers — plan DNS UDP fallback and cert bundling carefully
 - SLH-DSA signatures are **~7× larger than ML-DSA-65** — reserve for non-bandwidth-sensitive uses
-

Document History

Live document. Reset for the public edition; ongoing changes will accumulate from here.

VERSION	DATE	CHANGES
1.0	Apr 2026	Initial public release.
1.1	Jun 2026	Currency re-verification against primary sources. Confirmed still-current: NIST IR 8610 Round-2 outcome (9 advance / 5 eliminate), FIPS 206/FN-DSA draft timeline (submitted Aug 2025, final late 2026/early 2027), FIPS 207/HQC (draft early 2026, final 2027), OpenSSH 10.0 default (<code>mlkem768x25519-sha256</code>), liboqs 0.15.0, Cloudflare ~57% browser-initiated PQ key shares, IBM/Google quantum-hardware roadmaps. Added: liboqs API churn note (Dilithium removed in 0.15, SPHINCS+ slated for removal in 0.16 → SLH-DSA). Point-in-time vendor/KMS/IETF snapshots retain their "as of April 2026 — verify before relying" markers and were not individually re-confirmed.

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Good luck on your post-quantum cryptography journey!